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CHEMISTRY

Grade 11

Government of Nepal

Ministry of Education, Science and Technology

Curriculum Development Centre

Sanothimi, Bhaktapur

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Preface

The curriculum is the central guide that decides what is essential for teaching and learning. A textbook is one of the main tools to deliver the intent and the content of the curriculum. Therefore, it should have such an attribute that it presents relevant, practical, and useful knowledge and the process of skills for the overall development of a learner in accordance with the curricular aims and intents. This textbook **'Chemistry Grade 11'** has been developed to address the main aims of the Secondary Level (Grade 11-12) Chemistry Curriculum and is aligned with the intent of the National Curriculum Framework for School Education, 2076.

This book was prepared by an academic team that includes Mr. Khem Raj Bhatta, Mr. Hari Prasad Regmi, Mr. Parmatma Mishra and Mr. Lav Dev Bhatta. Several people notably; the Director Generals Mr. Ima Narayan Shrestha and Mr. Baikuntha Prasad Aryal, Professor Dr. Krishna Bakta Maharjan, Dr. Kamal Prasad Achary, Mr. Umanath Lamsal, Mr. Keshar Bahadur Khulal, Mr. Heramba Raj Kandel, MS. Mina Shrestha, MS. Pramila Bakhati, Dr. Ram Lochan Aryal, Dr. Bhoj Raj Paudel, Dr. Bisnu Datta Bhatta, Mr. Naresh Prashad Bhatta and Mr. Yub Raj Adhikari also contributed significantly on the development of this book. The language of this book was edited by Nabin Kumar Khadka.

Effort has been made to make the book activity oriented and interesting to the learners. All the contents of each lesson in this textbook are equally important. However, teachers can adapt the contents and tasks to the need of their learners and classroom contexts. This textbook can be used as a major resource for classroom teaching but it is not all in all. The teachers are also encouraged to explore other resources in addition to this book and use them to supplement the curricular contents as provisioned in the curriculum. The Curriculum Development Centre always welcomes constructive feedback for the betterment of its publications.

Curriculum Development Centre
Sanothimi, Bhaktapur

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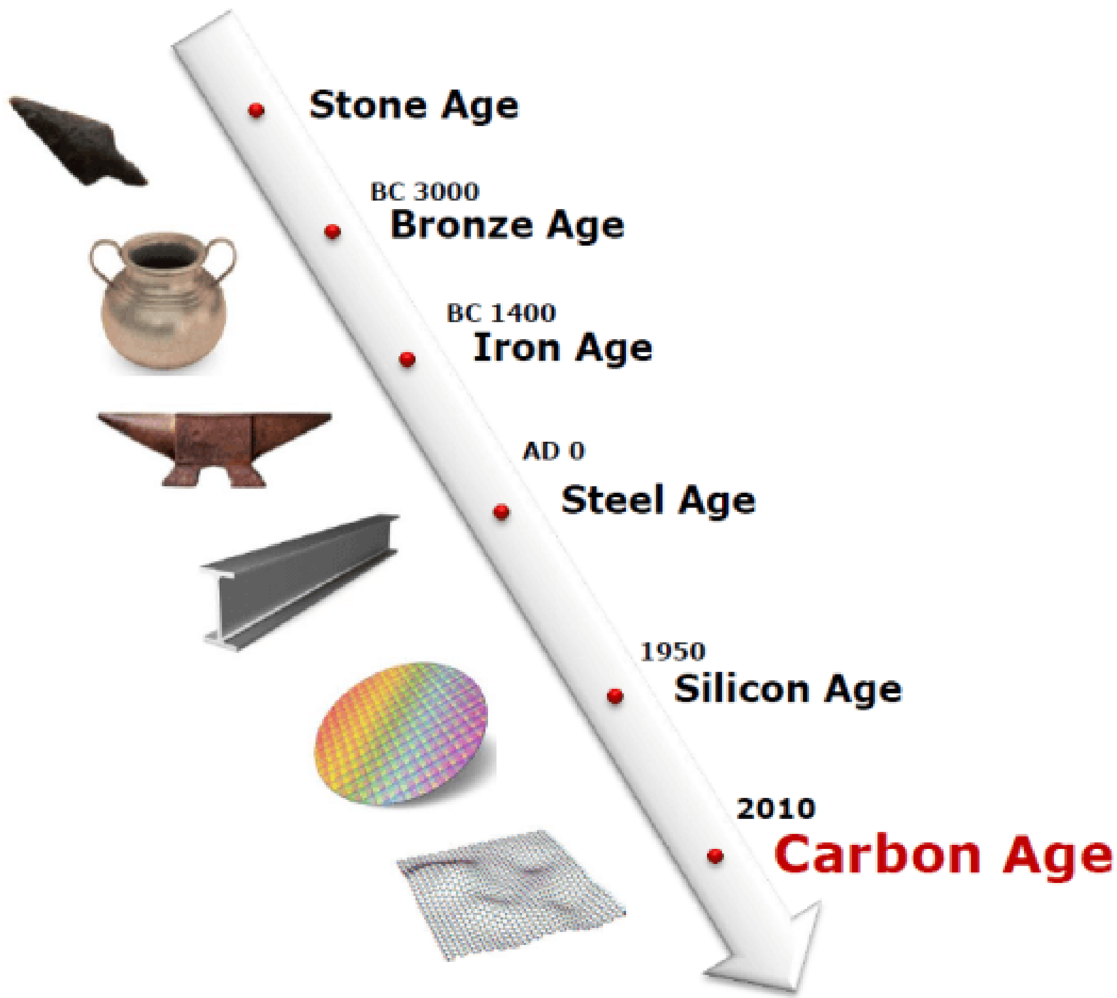
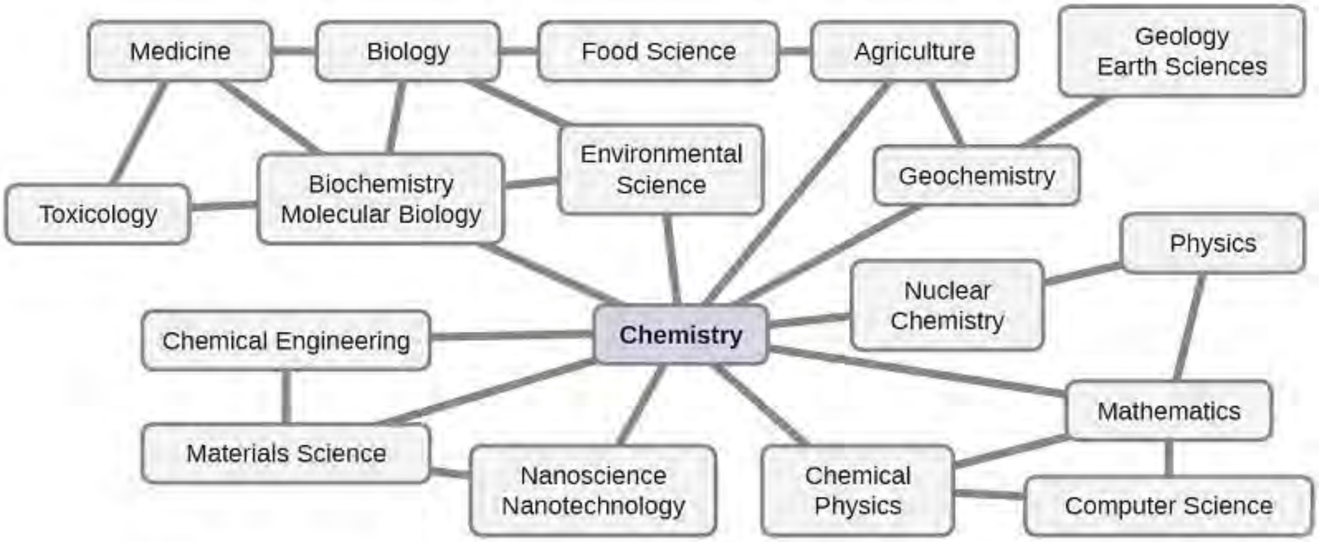
UNIT 1: FUNDAMENTALS OF CHEMISTRY

Activity 1

How does transition from the Stone Age to the Carbon Age represent human civilization? Discuss.



1.1 General introduction to chemistry

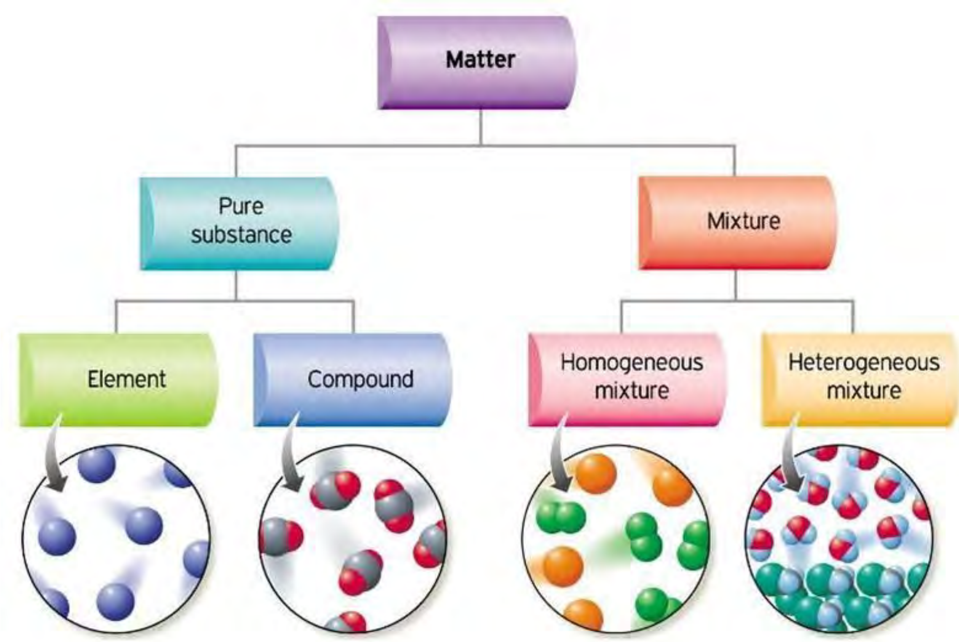


Chemistry is the branch of science that deals with the composition and structure of matter and the changes that matter undergoes. *Matter is anything that has mass and occupies space*. Your chemistry book, body and the air you breathe etc. are all examples of matter. Matter is simply the stuff that makes up our universe.

So far, we have noted that chemistry involves the composition and structure of matter. But beyond knowing what the composition of matter is and how the pieces fit together, chemistry is about how matter changes from one substance to another. When we refer to chemical change, we mean that the composition of matter has changed. A chemical change occurs when a substance changes into another substance. For example, when the carbon present in the paper burns in air, it changes into carbon dioxide gas. Chemical changes result in the production of something entirely different from the original substances.

Chemistry is often referred to as the "**central science"** because it serves as a necessary foundation for many other scientific disciplines. Regardless of which scientific field you are interested in, every single substance you will discuss or work with is made up of chemicals. Also, many processes related to those fields will be based on an understanding of chemistry. We also consider chemistry as a vital branch of science because of its crucial role in responding to the needs of society. Knowledge of chemistry helps to discover new processes, develop new sources of energy, produce new products and materials, provide more food, and ensure better health.

Chemistry and its language play vital roles in biology, medicine, materials science, forensics, environmental science, and many other fields. The basic principles of physics are essential for understanding many aspects of chemistry, and there is extensive overlap between many sub disciplines within the two fields such as chemical physics and nuclear chemistry.

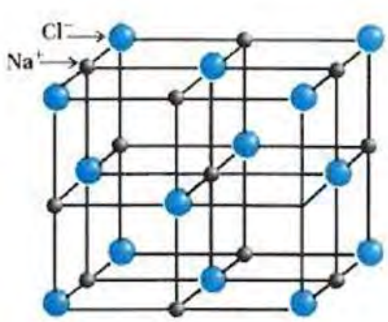


Classifications of matter

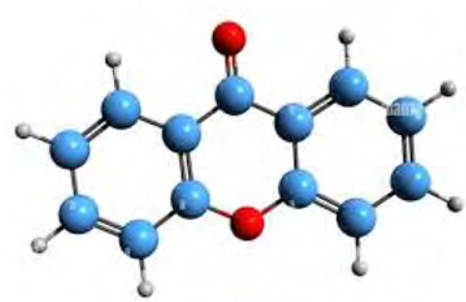
1.1.1 Divisions of chemistry

Physical chemistry

Biochemistry



Inorganic chemistry



Organic chemistry

Although all chemists are involved in the study of matter, the field is so broad that it helps to divide chemistry into divisions. These divisions characterize different aspects of the study of chemistry using some common features. Some of the major divisions of chemistry are:

a. Inorganic chemistry

It involves the study of the chemical elements, their compounds, and reactions excluding those that contain carbon as the principal element. Inorganic substances were thought to originate only from minerals, non-living sources.

b. Organic chemistry

Figure 1.4: Petroleum industry produces different types of petroleum products

It is the study of the composition and reaction of carbon-containing compounds and their derivatives called organic compounds. Generally, organic compounds are obtained from living sources (plants or animals).

c. Biochemistry

Figure 1.5: Analysis of blood sample in the pathology

Biochemistry is the study of chemical substances associated with living organisms. Biochemists study the nature of biological substances. The study of DNA, viruses, and immune systems are examples of biochemical study.

d. Physical chemistry

Physical chemistry deals with the structure of matter and how energy affects matters. Physical chemists are concerned with the physical characteristics of matters. Areas of research in physical chemistry might include how chemicals absorb light or how much energy is released or absorbed when a chemical reaction occurs.

e. Nuclear chemistry

Figure 1.6: A nuclear power plant producing electricity

Nuclear chemistry focuses on the study of atomic nuclei, nuclear fission reactions, and nuclear fusion reactions. It gives ideas about radioactive isotopes and their applications.

f. Analytical chemistry

Analytical chemistry deals with techniques used to identify and quantify the composition of matter. As the name implies, analytical chemists analyze substances for their content. An analytical chemist might perform tasks such as determining the sugar content of a fruit juice, the amount of pollutant in water or air and the purity of drugs etc.

g. Environmental chemistry

Figure: 1.7 Water Pollution

Environmental chemistry focuses on the occurrence of natural and synthetic substances and their impact on the environment. The study of the chemistry of pollution is a major area of concern for environmental chemists. Environmental chemistry studies the problems involving water pollution, air pollution, solid waste, hazardous materials, toxic substances etc.

h. Forensic chemistry

It deals with the analysis of evidence taken from the crime scenes to draw the conclusion from tests conducted on the evidence. CIB of Nepal applies this tool to find the evidence of the crimes.

Many branches of chemistry exist in combination with other areas of science. For example, the study of the chemistry of rocks and minerals falls under the category of geochemistry. The combination of medicine and chemistry leads to medicinal chemistry. Other divisions include agricultural chemistry, food chemistry, petroleum chemistry, soil chemistry, and polymer chemistry.

Test Yourself!

If chemists determine some arsenic content present in the drinking water supply of Kathmandu valley, name the branch of chemistry related to it and find its importance.

1.2 Importance and scope of chemistry

Activity 2

a. Observe the picture and identify the resulting chemical substance produced.
b. What are the various applications of the product obtained?

How do we get the paper and the ink that we are using in the classroom?

They are the products of modern chemical technology. In fact, it would be hard to find materials that are not related to the chemical industry. Plastics, medicines, paints, textiles, explosives, fertilizers, cosmetics, fuels, detergents, glass, electroplating, metallurgy, and photography etc. are examples of products and processes that illustrate the impact of the chemical industry on our daily lives.

Figure 1.9: Forensic examination of seized drugs

The careers of chemists and chemical laboratory technicians consist of study of composition, structure, properties, and changes of material substances. The daily used items like: batteries, plastics and clothing fibres, pharmaceuticals and medicines are the products of chemistry. Likewise, textiles, pesticides, fertilizers, soaps, shampoos, toothpaste perfumes etc. are the products of Chemistry.

Figure 1.10: products used in washing and cleaning

There are numerous careers in chemistry and chemistry related fields. Some of the chemistry related careers and their duties are briefly mentioned below.

Chemical engineers use engineering principles to solve problems of chemicals. They design chemical plants, develop chemical processes, and optimize production methods.

Chemical technicians assist chemists and chemical engineers to perform chemical analyses, test products, calibrate equipment, prepare solutions and reagents, and monitor quality control.

Hazardous materials managers supervise and monitor the removal, transportation, and disposal of harmful materials and substances.

Figure 1.11 Medicines

Pharmacists provide patient care that involves dispensing drugs, monitoring their use, and advising patient and health care workers about their effects.

Chemistry professionals can serve as teacher, instructor or professor in school, college or university.

Chemical sales and Marketing Representative (MR) are the professionals who are employed to deal with medicine and related accessories.

Forensic chemists combine the knowledge of chemistry with crime investigation in forensic chemistry to analyse samples collected from crime scenes.

Do you know?

Cell phones are made from numerous chemical substances. About 30% of the elements that are found in nature are found within a typical smartphone. The case, body, or frame consists of a combination of durable polymers composed primarily of carbon, hydrogen, oxygen, and nitrogen. Likewise, the metals such as aluminium, magnesium, and iron are used to make light and strong structures. The display screen is made from especially hard-edged glass (silica glass) and coated with a material to make it conductive. The circuit board uses a semiconductor material (usually silicon); commonly used metals like copper, tin, silver, and gold; and more unfamiliar elements such as yttrium (Yt), praseodymium (Pr), and gadolinium (Gd). The battery relies upon lithium ions and a variety of other materials, including iron, cobalt, copper.

Test yourself!

How does the study of chemistry help in agriculture? Explain.

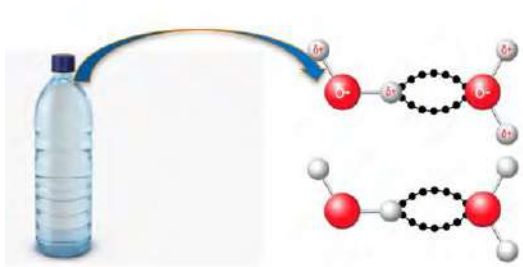
1.3 Basic concepts of chemistry

Activity: 3

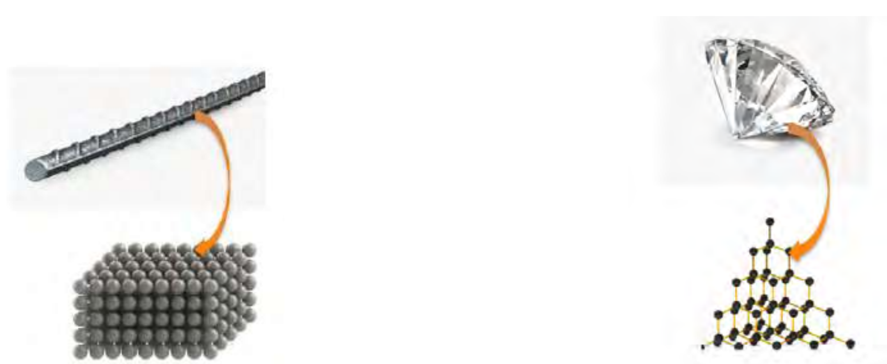
Observe the given figure:

a. What is the unit particle (atom, ion or molecule) by which the substance is made in each figure?
b. What is the difference in the properties of these two substances?

John Dalton (1766–1844) was an English chemist and physicist, best known for his pioneering work in the development of modern atomic theory. Born in Eaglesfield, England, Dalton came from a humble background and largely self-educated himself in science and mathematics. Apart from his work on atomic theory, Dalton also made contributions to meteorology. He formulated Dalton's Law of Partial Pressures, which describes the relationship between the pressure of a gas mixture and the partial pressures of its individual gases.



Atom and molecule



Iron atoms in iron rod   Diamond is made up of carbon atoms

Figure 1.12

An English scientist named **John Dalton** (1766–1844) proposed a theory of matter based on the original thoughts of Democritus. His ideas are now known as **atomic theory**. The major conclusions of atomic theory are as follows:

a. Matter is composed of small, indivisible particles called atoms.
b. Atoms of the same element are identical and have the same properties.
c. Chemical compounds are composed of atoms of different elements combined in small whole-number ratios.
d. Chemical reactions are merely the rearrangement of atoms into different combinations.

The atomic theory is now universally accepted as our current view of matter. Thus, we may define an ***atom as the smallest fundamental particle of an element that has the properties of that element.*** Atoms possess specific properties that define their behaviour in reactions and interactions with other atoms.

Just as the basic particles of most elements are atoms, the basic particles of a particular type of compound are known as molecules.

A molecule is formed by the chemical combination of two or more atoms.

Figure 1.13 Common salt in a bowl contains molecules of NaCl

The atoms in a molecule are joined together by a force called the **chemical bond**. The bond may be ionic or covalent depending on the nature of combining atoms. The number of atoms present in a molecule is called atomicity. Molecules containing only two atoms (atomicity 2) are called diatomic molecules. For example, H_2, O_2, N_2 are diatomic molecules. Likewise, molecules with more than two atoms are called polyatomic molecules, for example H_2O, CO2 etc. Each molecule of water is composed of two atoms of hydrogen joined by covalent bonds to one atom of oxygen. But in some molecules (the complex molecules on which life is based), like carbohydrate, protein, vitamin, etc. millions of atoms are combined.

Molecules have their own distinct chemical and physical properties that are different from those of individual atoms.

Test yourself!

What is the difference between these two molecules given in the figure?

Figure 1.14: H_2 molecule

Figure 1.15: HCl molecule

Relative atomic mass and relative molecular mass

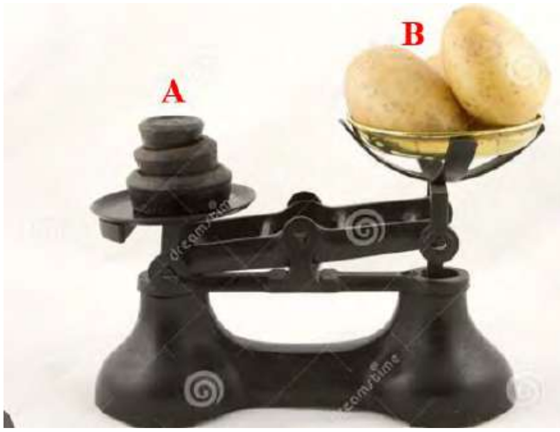
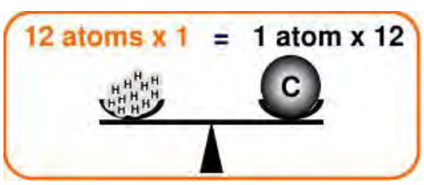


Figure 1.16

Activity: 4

Observe the balances given containing weights A and B.

Identify the standard and unknown substance in each figure.



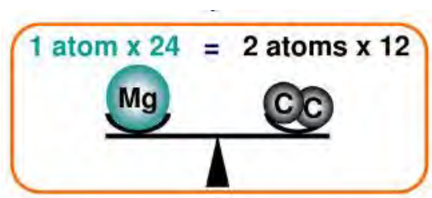


Figure 1.17: diameter of atom and nucleus

The atom is so small in size that it is a million times smaller than the thickest human hair. The diameter of an atom ranges from 0.1 to 0.5 nanometres (1.0×10^{-10} m to 5.0×10^{-10} m). Though the actual masses of the atoms could not be determined, their relative masses could be determined.

Like the standard weights used to measure the mass of unknown substances, carbon is used as a standard while measuring the mass of atoms or molecules.

Since, both the atomic mass and molecular mass are relative to carbon-12 isotopes, they are called relative atomic mass and relative molecular mass, respectively.

The relative atomic mass of an element is the average of the mass of the isotopes in a naturally occurring sample of the element relative to the mass of $\frac{1}{12}$ th of an atom of carbon-12.

Relative Atomic mass = $\frac{mass\ of\ an\ atom\ of\ element}{\frac{1}{12} of\ mass\ of\ an\ atom\ of\ c-12}$

Atomic mass of some of the common elements are given:

The relative atomic mass of silver is 107.87 means; the mass of a silver atom is 107.87 times the mass of one twelfth of a carbon-12 atom. No silver atoms actually exist with the mass of 107.87; this is just the average relative atomic mass of silver having isotopes ^{107}Ag and ^{109}Ag.

The relative atomic mass in fraction indicates that such an element occurs as a mixture of isotopes.

Many elements and compounds exist as a molecule. For example, hydrogen and water exist as molecules. Each molecule of hydrogen contains two hydrogen atoms and each molecule of water contains one oxygen and two hydrogen atoms. The mass of such a molecule is measured as relative molecular mass.

The relative molecular mass of a compound is a mass of a molecule of that compound relative to the mass of 1/12th of an atom of carbon-12

Relative Molecular mass = $\frac{mass\ of\ a\ molecule}{\frac{1}{12} th\ of\ mass\ of\ an\ atom\ of\ C-12}$

Molecular mass is calculated by summing up the atomic mass of all the atoms in a molecule.

For example, the molecular mass of chloroform ($CHCl_3$) = $1\times12+1\times1+3\times35.5$ =119.50

It means a molecule of chloroform ($CHCl_3$) is 119.50 times heavier than 1/12th of mass of an atom of carbon-12

If a compound is made up of ions, and therefore does not contain discrete molecules, the term ***relative formula mass*** or formula mass is used.

For example, formula mass of sodium chloride = 23+35.5=58.5

The problem discussing the masses of atom(s) or the subatomic particles is that they are so tiny , any numerical values that we state for the mass (in grams) would have little meaning to us. Scientists recognized this and decided to create a new unit for mass, the *atomic mass unit, or amu.*

The amu is defined as one-twelfth the mass of one atom of carbon-12 atom.

1 amu = 1/12th of mass of atom of carbon-12

$= \frac{1X1.99\times10-23g}{12}$ [mass of 1 atom of carbon $=1.99\times10^{-23}$g]

$= 6.67x10^{-24}$ g

Try yourself!

If you are given iron (III) oxide or ferric oxide.

a. Find the molecular mass of ferric oxide (Fe_2O_3) using the atomic mass given.

b. What could be the reason behind the atomic mass in fraction?

Ions and radicals

H, H^+, Cl^-, H_2,

a. From the list of the chemical species given, identify ion, molecule and atom.
b. Why are they called so?

Atoms and molecules are electrically neutral, they do not possess charges, whereas **ions** are charged species. Ions may have positive or negative charges. Radical is a chemical species that contains one or more unpaired electrons and may or may not have charge.

Remember: All radicals are ions but the reverse is not true. Free radicals are those chemical species having odd electrons without charge.

Cations (Basic radicals)

Metals have the tendency to lose electron(s) and form cations. So, they are also called **metallic ions or electropositive radicals or basic radicals.** Generally, the charge(s) of cation is equal to the valency of that ion. Radicals with one valency are called **monovalent ions**, with valency two are **bivalent** and with three are called **trivalent** ions.

Ion	***Classical Name of the ion***	***Name using roman numeral***	***valency***
Fe^{++}	Ferrous	Iron (II)	2
Fe^{+++}	Ferric	Iron (III)	3

Cu^{+} Cu^{++}	Cuprous Cupric	Copper (I) Copper (II)	1 2
Hg^{+} Hg^{2+}	Mercurous mercuric	Mercury (I) Mercury (II)	1 2
Mn^{2+} Mn^{4+}	Manganous Manganic	Manganese (II) Manganese (IV)	2 4
Sn^{2+} Sn^{4+}	Stannous Stannic	Tin(II) Tin(IV)	2 4
Pb^{++} Pb^{4+}	Plumbous Plumbic	Lead(II) Lead(IV)	2 4
Cr^{2+} Cr^{3+}	Chromous Chromic	Chromium(II) Chromium(III)	2 3
Co^{2+} Co^{3+}	Cobaltous Cobaltic	Cobalt(II) Cobalt(III)	2 3
Au^{+} Au^{3+}	Aurous Auric	Gold(I) Gold(III)	1 3

Most of the transition metals have variable valency. They can form more than one cation. Classically, while naming these ions, ***'ous'*** suffix is used for relatively lower valency state and '***ic***' suffix is used for relatively higher valency state. Nowadays, the charge on the metal ion follows the name of the metal in Roman numerals. This is referred to as the ***Stock method***. For example, the names of the iron ions are Iron (II) and Iron (III) and the names of the copper ions are Copper (I) and Copper (II). Some of the metal ions with variable valency is given below

Anions (Acid radicals)

Atom	ion	Name of ion	valency
Carbon	C^{4-}	Carbide	4
Nitorgen	N^{3-}	Nitride	3
Phosphorous	P^{3-}	Phosphide	3
Oxygen	O^{2-}	Oxide	2
Sulphur	S^{2-}	Sulfide	**2**

Selenium	Se^{2-}	Selenide	2
Tellurium	Te^{2-}	Telluride	2
Fluorine	F^-	Fluoride	1
Chlorine	Cl^-	Chloride	1
Bromine	Br^-	Bromide	1
Iodine	I^-	Iodide	1

Non-metals have a tendency to gain electrons and form anions. So, they are called **non-metallic ions,** electronegative ions or acid radicals**.** Radicals with one valency are called **monovalent ion**, with valency two are **bivalent** and with three are called **trivalant** ions.

The anions with single atom are called **monoatomic** anion, whereas ions with more than one atom are called **polyatomic ions** (compound radical). Monoatomic anions are named ending(suffix) with "**ide**" in the corresponding name of the atom. Some of the common anions with their symbol and valency are given.

ClO^-	hypochlorite	SO4- -	sulphate
ClO_2^-	chlorite	HSO4-	bisulphate
ClO_3^-	chlorate	SO3- -	sulphite
ClO_4^-	perchlorate	S- -	sulphide

In most cases, the anions are composed of oxygen and one other element. Thus, these anions are called ***oxyanions***. When there are two oxyanions of the same element, they of course, have different names. The anion with the smaller number of oxygens uses the *root* of the element plus *–ite as* suffix. The one with the higher number uses the *root* plus *–ate* suffix. For the case of anions with even more number of oxygens prefix - ***hypo*** is used before the name indicating least oxygen and prefix- ***per*** is used to indicate the highest number of oxygens. Examples are given in the following table.

Certain anions are composed of more than one atom but behave similarly to monatomic anions in many of their chemical reactions. They also have -*ide* endings similar to the monatomic anions. For example, cyanide ion (CN^-) and hydroxide ion (OH^-) are named as chloride ion. The ions with single atom are called **simple ions**. Examples Cl^-, H^-, I^- etc., Ions with more than one atom are called compound **ions.**
Examples: CrO_4^-, HCO_3^-, PO_4^{3-} etc. Some of the simple and compound ions with their formula are given:

Formula	Name of the ion	Formula	Name of the ion
F^-	Fluoride	CrO_4^-	Chromate
Cl^-	Chloride	$Cr_2O_7^{--}$	Dichromate
Br^-	Bromide	CO_3^{2-}	Carbonate
I^-	Iodide	HCO_3^-	Hydrogen carbonate(bicarbonate)
H^-	Hydride	NO_2^-	Nitrite
O^{2-}	Oxide	NO_3^-	Nitrate
S^{2-}	Sulfide	SiO_4^{4-}	Silicate
O_2^{2-}	Peroxide	PO_4^{3-}	Phosphate
O_2^-	Superoxide	HPO_4^{--}	Hydrogen phosphate
OH^-	Hydroxide	$H_2PO_4^-$	Dihydrogen phosphate
CN^-	Cyanide	$C_2O_4^{--}$	Oxalate
CNO^-	Cyanate	SCN^-	Thiocyanate
MnO_4^-	Permanganate	MnO_4^{--}	Manganate
$S_2O_3^{--}$	Thiosulphate	SO_4^{--}	Sulphate
HSO_4^-	Hydrogen sulphate (Bisulphate)	SO_3^{--}	Sulphite

Do you know?

Ions play an important role in biological systems.

For example;

Na^+ and K^+: Critical for nerve impulse transmission.

Ca^{2+}: Involved in muscle contraction and blood clotting.

Cl^-: Maintains electrolyte balance and helps in digestion

Molecular formula

Activity: 6

Packets of commonly used chemical substances are given.

a. Identify their applications in everyday life.

b. Name the elements present in these substances.
c. Write the molecular formula for the provided substances.

Vinegar is a liquid which is commonly used as a food preservative and contains mainly acetic acid. The molecular formula of acetic acid is CH_3COOH. It indicates that vinegar contains carbon, hydrogen and oxygen atoms.

Compound is represented by the symbols of the elements of which it is composed. This is called the **molecular formula** of the compound. **Simply, a molecular formula is the symbolic representation of a molecule**.

The molecular formula of water is H_2O indicating two atoms of hydrogen and one atom of oxygen in a molecule. Each chemical compound has a unique formula or arrangement of atoms in its molecules. For example, hydrogen peroxide (H_2O_2) is distinctly different from water (H_2O). The formulae of other well-known compounds are $C_{12}H_{22}O_{11}$ (sucrose also known as table sugar), $C_9H_8O_4$ (aspirin), NH_3 (ammonia), and CH_4 (methane).

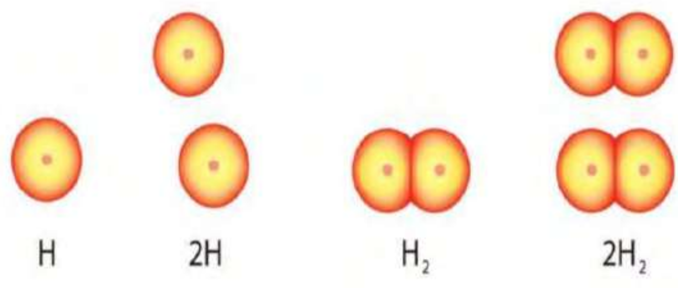
We can write the molecular formula of ionic compounds by knowing the radicals(ions) and their valency. For example, different compounds formed by the combination of cation and anion are given in the table below.

	Cl^-	CO_3^{2-}	OH^-	So_4^{2-}	PO_4^{3-}
Na^+	$NaCl$	Na_2CO_3	$NaOH$	Na_2SO_4	Na_3PO_4
Ca^{2+}	$CaCl_2$	$(NH_4)_2CO_3$	$Ca(OH)_2$	$CaSO_4$	$Ca_3(PO_4)_2$
Mg^{2+}	$MgCl_2$	$MgCO_3$	$Mg(OH)_2$	$MgSO_4$	$Mg_3(PO_4)_2$
Zn^{2+}	$ZnCl_2$	$ZnCO_3$	$Zn(OH)_2$	$ZnSO_4$	$Zn_3(PO_4)_2$
K^+	KCl	K_2CO_3	KOH	K_2SO_4	K_3PO_4
Fe^{3+}	$FeCl_3$	$Fe_2(CO_3)_3$	$Fe(OH)_3$	$Fe(SO_4)_3$	$FePO_4$
Al^{3+}	$AlCl_3$	$Al_2(CO_3)_3$	$Al(OH)_3$	$Al_2(SO_4)_3$	$AlPO_4$
Co^{3+}	$CoCl_3$	$Co_2(CO_3)_3$	$Co(OH)_3$	$Co_2(SO_4)_3$	$CoPO_4$
Fe^{2+}	$FeCl_2$	$FeCO_3$	$Fe(OH)_2$	$FeSO_4$	$Fe_3(PO_4)_2$
H^+	HCl	H_2CO_3	HOH	H_2SO_4	H_3PO_4

Most of the household chemical compounds are known by their common name rather than their systematic names. The given table summarizes some of these common chemicals with their common name, systematic names and molecular formula.

Common name	Systematic name	Formula
Baking soda	Sodium bicarbonate	$NaHCO_3$
Bleach	Sodium hypochlorite	$NaClO$
Bleaching powder	Calcium chlorohypochlorite	$CaOCl_2$
Borax	Sodium tetraborate decahydrate	$Na_2B_4O_7.10H_2O$
Brimstone	Sulphur	S_8
Chalk	Calcium carbonate	$CaCO_3$
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4.7H_2O$
Gypsum	Calcium sulphate dihydrate	$CaSO_4.2H_2O$
Lime (quick)	Calcium Oxide	CaO
Lime (slaked)	Calcium hydroxide	$Ca(OH)_2$
Lime stone	Calcium Carbonate	$CaCO_3$
Lye (Caustic soda)	Sodium hydroxide	$NaOH$

Magnesia	Magnesium oxide	MgO
Milk of magnesia	Magnesium hydroxide	$Mg(OH)_2$
Muriatic acid	Hydrochloric acid	HCl
Potash	Potassium carbonate	K_2CO_3
Saltpetre	Potassium nitrate	KNO_3
Talc (Talcum)	Magnesium silicate	$Mg_3Si_4O_{10}(OH)_2$
Vinegar	Acetic acid	CH_3COOH
Water	Dihydrogen oxide	H_2O
Ammonia	Nitrogen trihydride	NH_3



It is important to note that a subscript following a symbol and a number in front of a symbol do not represent the same thing; for example, H_2 and 2H represent distinctly different species. H_2 is a molecular formula; it represents a diatomic molecule of hydrogen, consisting of two atoms of the element that are chemically bonded together. The expression 2H indicates two separate hydrogen atoms. The expression $2H_2$ represents two molecules of hydrogen.

Test yourself!

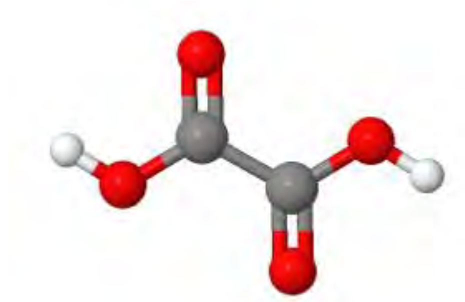
Write the molecular formula of the given compounds.

- **a.** Sodium thiosulphate
- **b.** Calcium oxalate
- **c.** Ferric ferrocyanide
- **d.** Sodium nitroprusside
- **e.** Ammonium Acetate

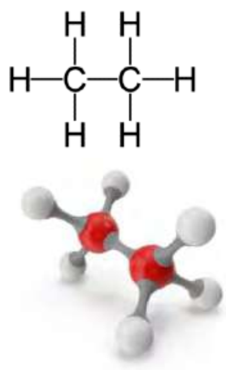
Empirical formula

Activity:

Ball and stick model of one of the organic compounds containing carbon (grey), hydrogen(white) and oxygen(red) is given.



a. Find the number of each atom in the structure.
b. What is the molecular formula of the compound?
c. Determine the simplest whole number ratio of atoms from molecular formula.



structural formula *ball and stick model*

C_2H_6 *molecular formula* **CH_3** empirical formula

Figure 1.17: Different formula of ethane molecule

The molecular formula of a substance specifies the number of atoms of each element in one molecule of that substance. Thus, the molecular formula of carbon dioxide CO_2 indicates, each molecule of carbon dioxide contains 1 atom of carbon and 2 atoms of oxygen. The exact ratio of the number of atoms of carbon to oxygen is 1:2. The simplest ratio of number of atoms is also 1:2. It means its empirical formula is the same as molecular formula.

The molecular formula of glucose is $C_6H_{12}O_6$; each glucose molecule contains 6 atoms of carbon, 6 atoms of oxygen, and 12 atoms of hydrogen. These atoms are in an exact ratio of 6:12:6. But the simplest whole number ratio of the number of atoms is 1:2:1. It means the empirical formula for glucose is CH_2O.

The empirical formula of a compound is the simplest formula that gives the simplest ratio of numbers of atoms of each kind in a compound.

Relationship between molecular and empirical formula

If we know the simple whole number ratio of the atoms present in a molecule, it is possible to calculate the exact whole number ratio of atoms present using the molar mass

Molecular formula = empirical formula × 'n'

Where 'n' = molecular weight /empirical weight

For example, the empirical formula of glucose is CH_2O and its molecular mass is 180

Then empirical weight =30

n = 180/30=6

Molecular formula =CH_2O×6

=$C_6H_{12}O_6$

Test yourself!

What is the molecular formula of lactic acid having empirical formula CH_2O and molecular weight is 9?

1.4 Percentage composition from molecular formula

Activity:

a. What do you mean by the difference in the percentage of the two elements given?
b. Find the mass of hydrogen and oxygen present in 18 g of water (H_2O).

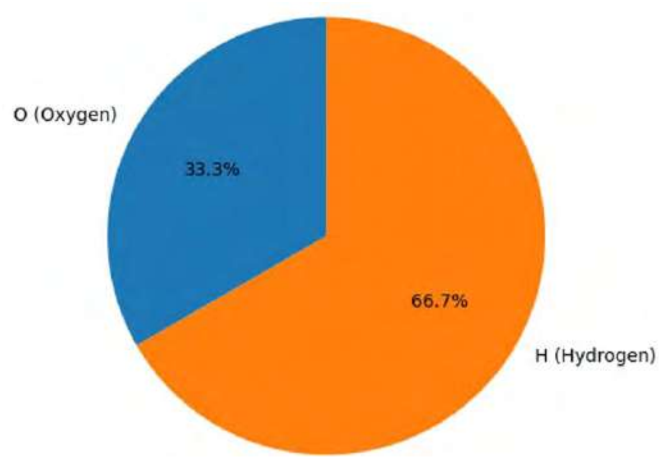


Fig: % composition of ingredients in the food

Compound	OPC(%)	PPC(%)
CaO	64	40
SiO_2	19	28
Al_2O_3	4.6	8.3
Fe_2O_3	3.7	3.8
MgO	0.7	1.0

Table: %composition of OPC and PPC cement

Have you noticed the percentage composition of the ingredients present in the packaging food or medicine?

Percentage composition is the amount of that particular constituent in the 100 gram of the substance. It gives the idea about the amount of each component present in the substance. We can calculate the percentage composition of the constituent elements present in the given compound, if the molecular formula is known. The calculation of percentage composition is based on the law of constant proportion.

Percentage composition = $\frac{amount\ of\ that\ element}{total\ mass} \times 100$

For example: The percentage composition of carbon and oxygen in carbon dioxide is calculated as

percentage composition of carbon in CO_2 = $\frac{1\ x\ atomic\ mass\ of\ carbon}{molecular\ mass\ of\ CO2} \times 100$

$= \frac{1\times 12}{44} \times 100$

= 27.27%

Percentage composition of oxygen in CO_2 = $\frac{2\times\ atomic\ mass\ of\ oxygen}{molecular\ mass\ of\ CO2} \times 100$

$= \frac{2\times 16}{44} \times 100$

=72.73%

Urea (NH_2 -CO-NH_2) is the most important nitrogenous fertilizer in the market, with the highest Nitrogen content which supports the growth of plants and vegetables. Percentage compositions is calculated as

Mass of C =1 × 12 = 12

Mass of O = 1 × 16 = 16

Mass of N =2 × 14 = 28

Mass of H= $4 \times 1 = 4$

Molecular mass of urea: $12 + 16 + 28 + 4 = 60$

Percentage of C = $(12 / 60) \times 100\% = 20\ \%$

Percentage of O: $(16 / 60) \times 100\% = 26.66\%$

Percentage of N: $(28 / 60) \times 100\% = 46.67\%$

Percentage of H : $(4 / 60) \times 100\% = 6.67\%$

Similarly, ethanol (C_2H_5OH), the commonly used alcohol, percentage composition of each element is calculated as:
Mass of C = $2 \times 12= 24$
Mass of H = $6 \times 1=6$
Mass of O = $1 \times 16=16$
Molecular mass of C_2H_5OH = 46

% composition of carbon=$24/46\times100\%$
=52.14%

% composition of hydrogen =$6/46\times100\%$
=13.04%

% composition of oxygen =$16/46\times100\%$
=34.78%

Applications of percentage composition

It is a fundamental concept that aids in understanding and manipulating chemical substances. Knowing the percentage composition of the molecule is helpful in;

Determination of empirical formula: Knowing the percentage composition helps to know the simplest whole number ratio of atom. It means empirical formula can be determined.

Determination of molecular formula: Combined with molecular mass data, percentage composition helps in determining the molecular formula of a compound, which represents the actual number of atoms of each element in a molecule.

Quality Control: The industries which produce chemical compounds, maintaining a specific percentage composition is crucial for ensuring the quality and consistency of products. Deviations from the expected composition can indicate impurities or variations in the production process.

Stoichiometry calculations: Percentage composition is essential for stoichiometric calculations, which involve the quantitative relationships between reactants and products in a chemical reaction. It helps to determine the amounts of substances involved in a reaction.

Predicting properties: The percentage composition provides understanding into the properties of a compound. Different compounds with the same elements but different percentage compositions may exhibit distinct physical and chemical properties.

Test yourself!

Write the molecular formula and find the percentage composition of each element in the given compound.

a. Citric acid
b. Sulphuric acid
c. Paracetamol
d. Ammonia

Project work

1. Make a comprehensive list of all Nepalese government and non-governmental offices and departments employing chemistry personnel for various chemical observations, analysis, and identification purposes. Conduct a visit to or utilize website resources to expose the specific roles and responsibilities of chemists within these organizations:

S.N	Name of organizations /office	Role of chemist
1.	Nepal oil corporation	refining, quality control, research, environmental issues, product development, and safety management.
2.		
3.		
4.		
5.		
6.		
7.		

2. Prepare a pictorial chart book about the scope and applications of chemistry.

Exercises

A. Multiple choice questions

1. Which of the following determines the major portion of atomic mass?
a. neutrons and protons
b. electrons and protons
c. electrons and neutrons
d. neutrons and positrons

2. There are two isotopes of the same element. Which of the following is common?
a. atomic weight
b. chemical reactivity
c. the number of neutrons
d. the number of protons

3. Which of the following is true about compounds?
a. Compounds are pure substances that are composed of two or more elements in a fixed proportion.
b. Compounds cannot be broken down chemically to produce their constituent elements or other compounds.
c. All the compounds are alike in structure and properties.
d. Compounds are also called mixtures.

4. Which of the following is true about the symbol $_{17}Cl^{35}$?
a. 17 protons, 17 electrons, and 17 neutrons.
b. 17 protons, 17 electrons, and 18 neutrons.
c. 17 protons, 18 electrons, and 17 neutrons.
d. 35 protons, 35 electrons, and 17 neutrons

5. Which of the following molecular formula contains the highest number of atoms?
a. $KClO$
b. $KClO_2$
c. $KClO_3$
d. $KClO_4$

6. Which of the following radicals has maximum valency?
a. Nitrate
b. Nitrite
c. Cyanide
d. Sulphate

7. What is the approximate mass of 1 atomic mass unit?
a. Mass of 1 atom of carbon
b. Mass of 1 atom of oxygen
c. Mass of 1 atom of hydrogen
d. Mass of 1 atom of chlorine

8. Which of the following molecules is tetra atomic?
   a. Ozone
   b. Ammonium chloride
   c. Glucose
   d. Ammonia
9. Calcium compounds are used to repair the fractured bones. Which of the following compounds is used for this purpose?
   a. $CaSO_4$
   b. $CaSO_4.H_2O$
   c. $CaSO_4.2H_2O$
   d. $CaSO_4.1/2\ H_2O$
10. Which of the following compounds has the same empirical formula and molecular formula?
   a. Glucose
   b. Sucrose
   c. Ethanol
   d. Carbon tetrachloride
11. What is the percentage composition of water in $CuSO_4.5H_2O$? [Atomic mass of Cu is 63.5 amu)
   a. 7.21%
   b. 14.42%
   c. 21.63%
   d. 36.07%
12. Which of the following is heaviest?
   a. 100 atoms of nitrogen
   b. 30 molecules of ammonia
   c. 20 molecule of calcium carbonate
   d. 10 molecules of oxygen

B. Short answer questions

1. Define chemistry? How does study of chemistry help in the development of society? Write any four points.
2. Why is chemistry called central science? Write the role of chemistry in agriculture and medicine in two points for each.
3. Gypsum is a one of the minerals having the molecular formula $CaSO_4$. $2H_2O$ is commonly used in the preparation of plaster of Paris.

   a. How many atoms are there in a molecule of gypsum?
   b. Calculate the molecular mass of gypsum.
   c. Why is it used in making plaster of Paris?
   d. Calculate the percentage composition of sulphate in the given compound.

4. Ammonia and ammonium nitrate are commonly used agricultural fertilizers.

  a. Write the molecular formula of these compounds

  b. Calculate the percentage composition of nitrogen in each compound

  c. Which one do you think is the best fertilizer, why?

5. Fill the given table with correct molecular formula.

Radicals	K^+	Zn^{++}	Fe^{+++}	Cu^+
MnO_4^-	$KMnO_4$			
MnO_4^{--}				
$Cr_2O_7^{--}$				
O^{2-}				
N^{3-}				
PO_4^{3-}				

6. Write any two differences in each of the following:

a. Atom and molecule
b. Atomic mass and molecular mass
c. Molecular formula and empirical formula
d. Atom and radical
e. Acid and basic radicals

7. The mass of an atom and molecules are measured using different types of scale

a. Why is it difficult to measure the mass of an atom and molecule?
b. Define the term 1 amu. Calculate the value of 1 amu in gram.

8. Write the correct formula of the ion and place a checkmark in the given boxes where necessary.

Ion	**Formula /symbol**	**Monatomic Ion**	**Polyatomic ion**	**Cation**	**anion**
Phosphate					
Iodide					
Fluoride					
Cyanide					
Carbonate					
Nitride					
Oxalate					

Nitrate					
Sodium					
Aluminum					
Ammonium					

9. If the red one indicates oxygen atom and the grey colour indicates hydrogen atom in the figure.

a. Find out the total number of oxygen and hydrogen atoms.

b. Write the molecular formula of the molecules shown.

10. Chemistry plays very important role in the different areas of science

a. How does chemistry contribute to the development of such new things in this field?

b. What role does chemistry play in environmental protection and sustainable practices?

c. How is chemistry involved in the design and manufacturing of electronic devices and materials?

11. Penicillin ($C_{14}H_{20}N_2SO_4$) is the first antibiotics (antibacterial agents), discovered by Alexander Fleming.

a. Why is penicillin called compound?

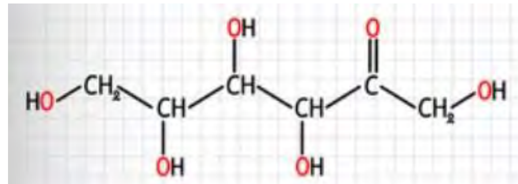
b. How many atoms are there in a molecule of penicillin?

c. Calculate the molecular weight of Penicillin

d. Calculate the mass percent of each element present in the penicillin.

12. Structural formula of organic compounds is given below.

a. Write the molecular and empirical formula of the compound.
b. Find the empirical and molecular weight.
c. Calculate the percentage composition of carbon and hydrogen in the molecule.



13. The percentage composition of Nicotine, a stimulant that's found in tobacco, is given in the table. Determine its molecular formula if its molecular weight is 162.

C	74.02%
N	17.27%
H	8.710%

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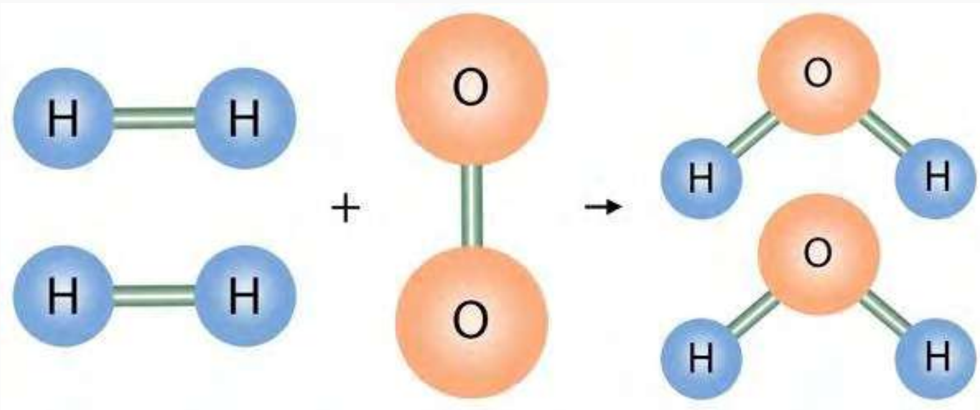
UNIT 2: STOICHIOMETRY

Activity:



a. What is the total number of atoms in the reactants and total number of atoms in the product?
b. Find the total mass before the chemical reaction and total mass after the reaction.

2.1 Dalton's atomic theory and its postulates

In1808 John Dalton, an Englishman, published a theory based on certain facts and experimental evidence that were known at the time. Based on these evidences he proposed a theory to describe the existence of the atom called Dalton's atomic theory. The theory consists of the following postulates.

a. All matter is made up of tiny indivisible particles called atoms, and the atom is the smallest structural unit of an element.
b. In terms of their mass and chemical and physical properties, atoms of the same element are all very similar and atoms of different elements are very different.
c. Atoms chemically combine with each other in simple whole number ratios to form molecules.
d. In chemical reactions, atoms of elements retain their identity, but rear- range, reorganize, and reunite in the formation of other substances.

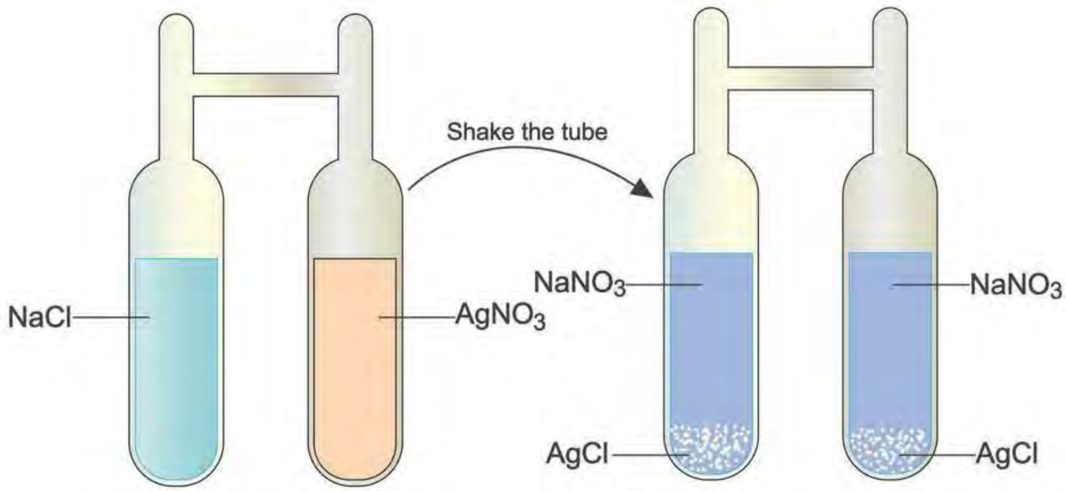
2.2 Laws of stoichiometry

Stoichiometry is derived from the Greek word ***stoikhein*** means element and ***metron*** means measure. So, stoichiometry means measure of an element. While cooking, we use the right amount of salt or sugar to minimize the waste and maximize the taste, similarly stoichiometry ensures that chemical reactions happen with the right amounts of reactants to maximize efficiency and avoid waste. It's a fundamental concept in chemistry, bridging the gap between the microscopic world of atoms and molecules and the macroscopic world of measurable quantities.

Law of conservation of mass /energy

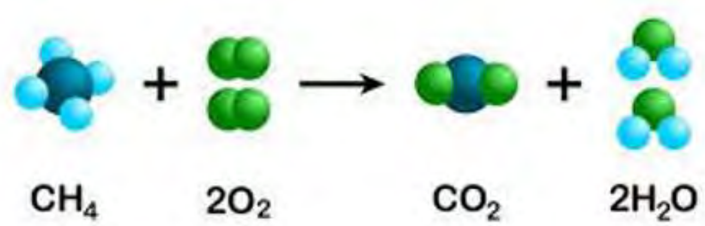
Activity:

Two chemicals are kept in the sealed H shaped. After shaking the tube, the products formed are shown.



Write the chemical equation for the above reaction.

a. Is there any change in mass of the tube before and after the reaction? why?



In the above reaction of combustion of methane gas, there are 9 atoms in the reactant before the reaction and again there are 9 atoms in the product after the reaction. It means the number of atoms remains constant, so the total mass of the substance also remains constant in the chemical reaction.

For constant mass, the total number of atoms in the reactant should be equal to that of products. Therefore, the chemical equation should be balanced.

If ‘x’ g be the mass of A and ‘y’ g be the mass of B, then the total mass of AB after the chemical reaction will be x + y.

$$A + B \longrightarrow AB$$

For example:

$$\mathbf{x \quad y \quad = \quad x + y}$$

where x is the mass of A, y is the mass of B and x+y is the mass of AB.

$2C_2H_6 + 7O_2 \longrightarrow 4CO_2 + 6H_2O$
$2\times30 + 224 \qquad 4\times44 + 6\times18$
284 g $\qquad 284$ g

Law of conservation of mass/energy states that
“**when *a chemical reaction occurs in a closed system, there is no gain or loss of mass and energy during the* reaction process i.e. mass is conserved”.**
It means mass /energy can neither be created nor be destroyed however it can be transformed from one form to another *i.e.* matter cannot be destroyed. So, it is also called law of indestructibility of matter.

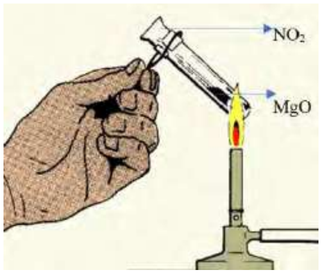
Test yourself!

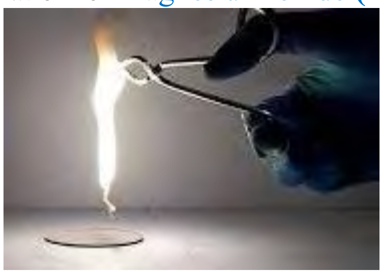
10 g of calcium carbonate produces 6.2 g of calcium oxide upon heating along with carbon dioxide gas.

a. Write the balanced chemical equation for the reaction given.
b. Find the number of atoms before and after the reaction of each element.
c. Calculate the mass of carbon dioxide gas produced.

Law of definite proportion

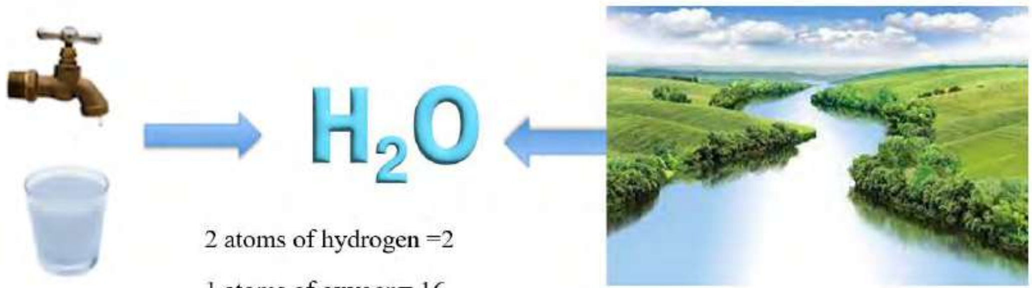
Activity:
Observe the two methods for the formation of magnesium oxide **(MgO)**





$Mg + O_2 \longrightarrow$ **MgO** $\qquad Mg(NO_3)_2 \longrightarrow$ **MgO** $+ 2NO_2 + 1/2O_2$

a. Calculate the ratio of mass of magnesium and oxygen in MgO obtained in each case.
b. Find the difference in these ratios.



2 atoms of hydrogen =2

1 atoms of oxygen= 16

Molecular weight of water =18

% of hydrogen $=\frac{2}{18}\times 100=11.11\%$

% of oxygen$=\frac{16}{18}\times 100=88.88\%$

Water obtained both from tap and river contains hydrogen and oxygen in the ratio of 1:8 by mass.

The law of constant proportion, also known as the **law of definite proportions**, is a principle in chemistry that was first formulated by **Joseph Proust** in the late 18th century. The essence of the law lies in the constant composition of pure chemical compounds. It states that

"Regardless of the source of the compound or the method used to prepare it, the ratio of the masses of the elements in the compound remains constant."

Let us consider, compound AB is formed by two different methods:

Method **1**, if p gram of A combines with q gram of B

ratio of mass of A and B = $\frac{p}{q}$

Method **2**, if x gram of A combines with y gram of B,

ratio of mass of A and B = $\frac{x}{y}$

then according to this law,

$\frac{p}{q}=\frac{x}{y}$

For example, carbon dioxide gas can be obtained either by burning carbon or by burning methane gas, the ratio of mass of carbon and oxygen in each case will be 12: 32 or 3:8

$C + O_2 \longrightarrow CO_2$ **(12:32)**

$CH_4 + 2O_2 \longrightarrow CO_2$ **(12:32)** $+ 2H_2O$

Sample of calcium carbonate contains 28% of calcium, 15% of carbon, and 57% of oxygen. Applying the Law of Constant Proportion, we can find the mass of calcium, carbon, and oxygen in 20.0g of calcium carbonate prepared by another method.

In $CaCO_3$, Ca =28%, C= 15%, O= 57%

Mass of the second sample of CaCO3 =20.0g

On the application of the Law of Constant proportion
Mass of the calcium = 20.0 × 28/100 =5.6g
Mass of the carbon = 20.0×15/100 =3g
Mass of the oxygen = 20 - (5.6+3) = 11.4g

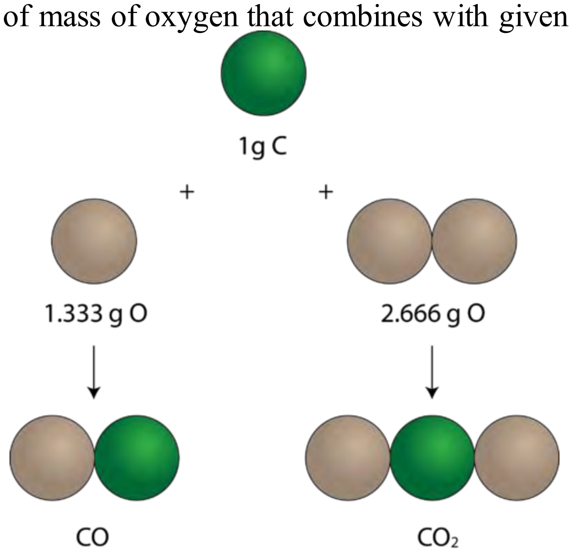
Although this law is easily understandable today, it was of great use in the late 18th century when chemical compounds did not have any proper definition. The law of definite proportions also contributed to the development of Dalton's atomic theory. There are few limitations of this law:

a. This law of definite proportion is applicable for simple compounds; in case of complex molecules it may not accurately represent the fixed ratio of all elements.

b. This law is not applicable for non-stoichiometric compounds having various compositions of elements in different samples due to crystallographic vacancies. For example, in wusite (oxide of iron), the ratio of iron and oxygen can range from 0.83 to 0.95

Test yourself!
Prove the law of definite proportion in the given sets of experiment.
2.8g of calcium oxide prepared by heating limestone was found to contain 0.8g of oxygen. When 1.0 g of oxygen is treated with calcium,3.5 g of calcium oxide was obtained.

Law of multiple proportion
Activity:
Fixed mass of carbon reacts with different mass of oxygen to form two different compounds as given:
a. Name the two compounds formed.
b. What is the ratio of mass of oxygen that combines with given mass of carbon?

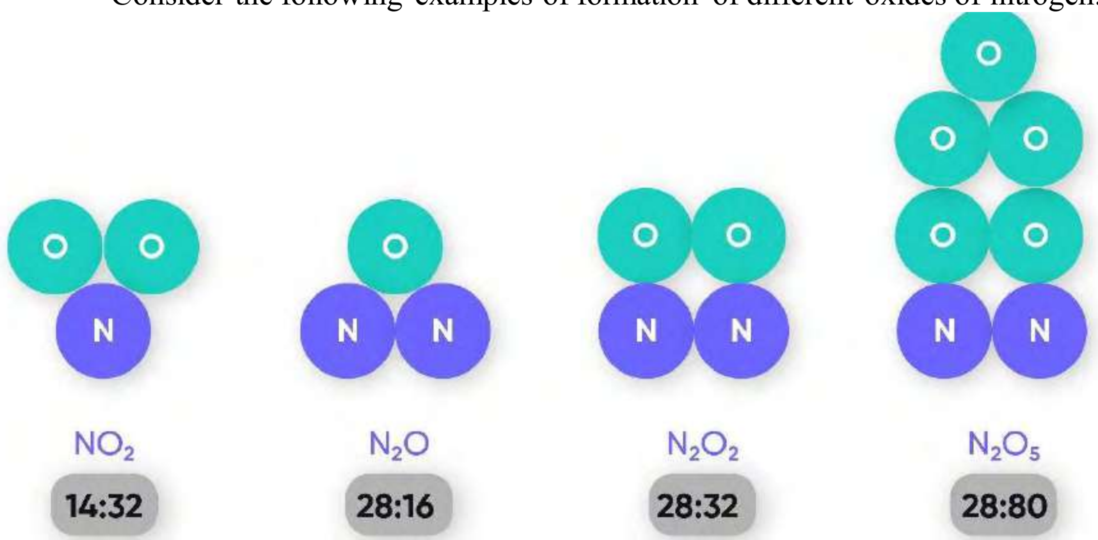


The law of multiple proportions is a fundamental principle in chemistry that was formulated by **John Dalton** in the early 19th century. This law deals with the relationship between the masses of elements that combine to form different compounds.

Let's take two elements A and B combined to form two different compounds. If the mass of element B that combines with the fixed mass of element A in two different compounds be w_1 and w_2 respectively. Then, from the law of multiple proportion

$\mathbf{w_1 : w_2}$ will be a simple whole number.

Consider the following examples of formation of different oxides of nitrogen.



14:32 **14:8** **14:16** **14:40**

When mass of nitrogen is kept constant (14), the ratio of mass of oxygen will be

32 :8 :16 :40

4:1:2:5

Or **1:2:4:5** which is a simple whole number ratio.

If we consider the formation of only two oxides of nitrogen,

The ratio of mass of oxygen that combines with constant mass of nitrogen(14g) is 16:32 or 1:2 which is a simple whole number ratio.

Law of multiple proportions states that "***When two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in simple, whole-number ratios".***

The law of multiple proportions is the extension of Dalton's Atomic Theory, which proposed that matter is composed of indivisible atoms and that atoms of different elements combine in simple, whole-number ratios to form compounds. It helped chemists understand the stoichiometry of chemical reactions and the consistent ratios in which elements combine to form compounds.

Example:
Let us consider three compounds of hydrogen and nitrogen has the following composition:

Compound	Mass of hydrogen (g)	Mass of nitrogen (g)
A	0.108	0.50
B	0.072	1.00
C	0.108	0.75

Mass of nitrogen combined by 1.0 g of hydrogen in each compound is given as:

Compound	Mass of hydrogen (g)	Mass of nitrogen (g)
A	1.0	4.6
B	1.0	13.88
C	1.0	6.94

The ratio of mass of nitrogen combining with 1.0 g of hydrogen will be

4.6 /4.6: 13.88/4.6 : 6.94 4.6

=1: 3: 0.5

=2:6:3 (simple whole number ratio)

Test yourself!

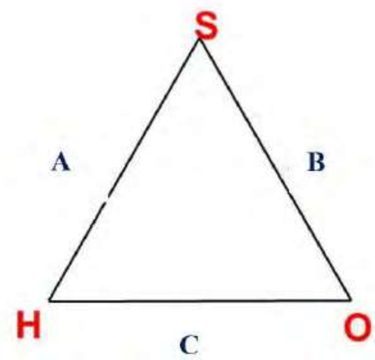
Two common oxides of iron (FeO and Fe_2O_3) and their composition are given below. Use this data to prove the law of multiple proportion.

Complete the table given

Law of reciprocal proportion

Activity:

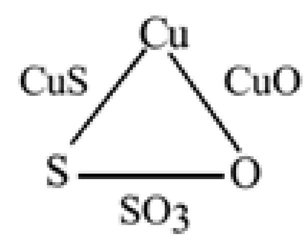
Name the compounds A, B and C formed by the given elements as shown in the figure.



The law of reciprocal proportions was proposed by **Jeremias Richter** in 1792. It deals with mass ratio relationships among three different compounds which can form three different compounds.

Oxygen and sulphur react with copper to give copper oxide (CuO) and copper sulfide (CuS), respectively. Sulphur and oxygen also react with each other to form Sulphur trioxide (SO_3)

Element	Compound	Ratio of mass of each element
Copper and Sulphur	CuS	Cu:S = **63.5**:32
Copper and Oxygen	CuO	Cu:O =**63.5**:16
Sulphur and Oxygen	SO_3	S:O = 32:16×3 = 32:48



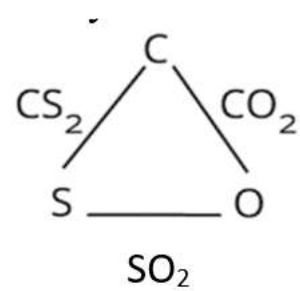
The ratio of mass of Sulphur and oxygen that combines with constant mass of copper (63.5) =**32:16 i.e 2:1**

The ratio of mass of Sulphur and oxygen when they combine themselves in SO_3 = 32:16×3 i.e **2:3**

2:3 and 2:1 are simple multiple of each other.

Likewise,

CO_2, SO_2, and CS_2 are compounds of carbon, Sulphur and oxygen



CO_2 contains **12** parts by mass of carbon and **32** parts by mass of oxygen,
SO_2 contains **32** parts by mass of Sulphur and 32 parts by mass of oxygen.
the mass ratio of carbon and Sulphur when combined with a fixed quantity of oxygen is **12:32 or 3:8**

CS_2 contains **12** parts by mass of carbon and **64** parts by mass of Sulphur, resulting in a ratio of **12:64 or 3:16**.

Therefore, **3:8** and **3:16** are simple multiple of each other.

Law of reciprocal proportion states that, "***If two different elements combine separately with the same weight of a third element, the ratio of the masses in which they do so are either the same or a simple multiple of the mass ratio in which they combine***."

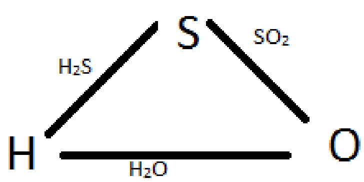
This law is also called **the law of equivalent proportions** where the combined mass ratio of elements is in their equivalent masses.
There are few elements which will combine with the third element and also combine with each other. Thus, the law of reciprocal proportion is applicable to limited elements only.

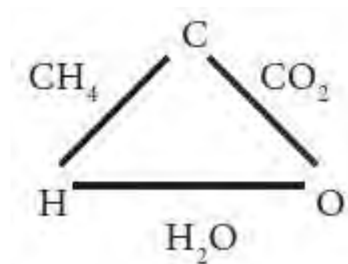
How does the law of reciprocal proportion support Dalton's atomic theory?

According to Dalton's atomic theory, all the atoms of the same element are identical and the compounds are formed by the combination of atoms of different elements in a simple ratio of whole numbers. Therefore, the weights of the elements combining with a fixed weight of another element should bear a simple ratio to the ratio of the weights of the elements when they combine with each other. This explains that the law of reciprocal proportion is on the basis of Dalton's atomic theory

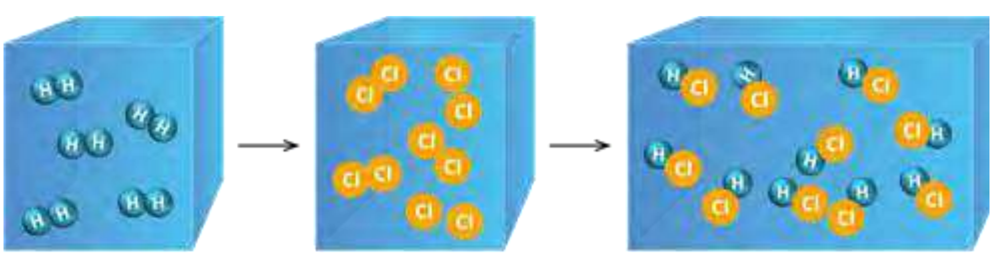
Test yourself!

Prove the law of reciprocal proportion in the given:





Law of gaseous volume
Activity: Observe the container containing gaseous reactants which gives gaseous product.



a. Write the balanced chemical equation for the given reaction.
b. Find the ratio of the number of mol of gas in the reactants and products.

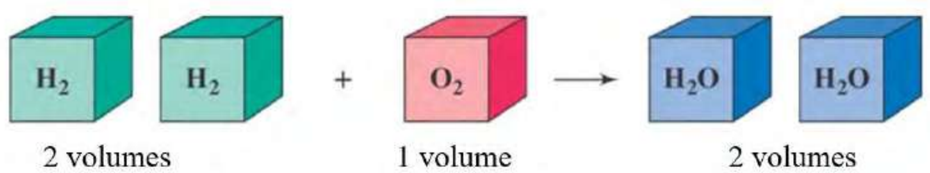
Joseph Louis Gay-Lussac's Law of Gaseous Volumes describes the relationship between the volumes of reactants and products in a chemical reaction involving gaseous substances, assuming the pressure and temperature are constant.

Let us consider a chemical reaction involving gaseous molecules having the volume V_1, V_2 and V_3, then the law can be represented as

$\mathbf{V_1 : V_2 : V_3 = a : b : c}$

Where a, b and c are the coefficients in the balanced chemical equation which represents the molar ratio of the gas.

In the reaction of hydrogen and oxygen, the law of gaseous volume can be applied to calculate the volume of water vapor formed under constant temperature and pressure.



The ratio of volume for the formation of water vapour from hydrogen and oxygen is

2:1:2 which is a simple whole number ratio.

If 100 mL of hydrogen reacts with excess oxygen, then the volume of water vapour formed will be 100 mL.

The law is often stated as follows:

"The **volumes of gases involved in a chemical reaction are in simple whole-number ratios, provided the volumes are measured under the same conditions of temperature and pressure."**

Test yourself!

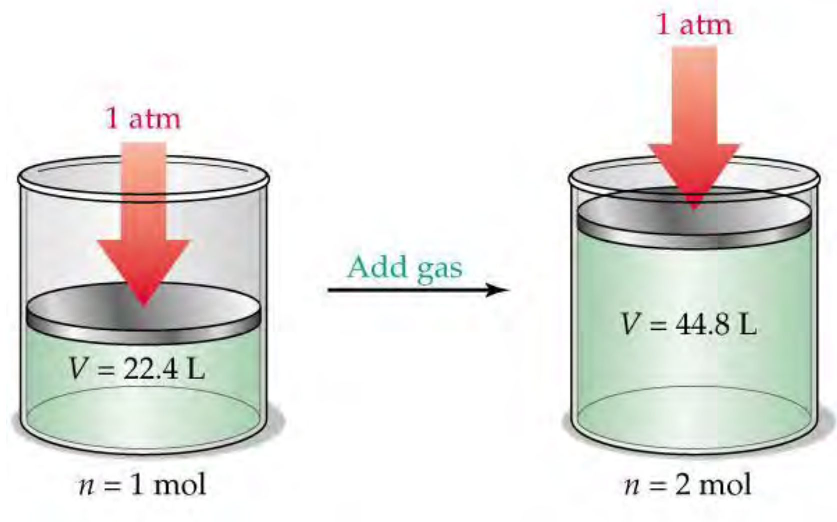
Nitrogen and hydrogen combine under certain conditions to form ammonia gas.

Write the balanced chemical equation for the ammonia gas formation.
Find the ratio of volume of Nitrogen, hydrogen and ammonia gas.
Calculate the volume of ammonia gas produced by the reaction of 20 ml of hydrogen with excess of oxygen gas.

2.3 Avogadro's law and some deductions

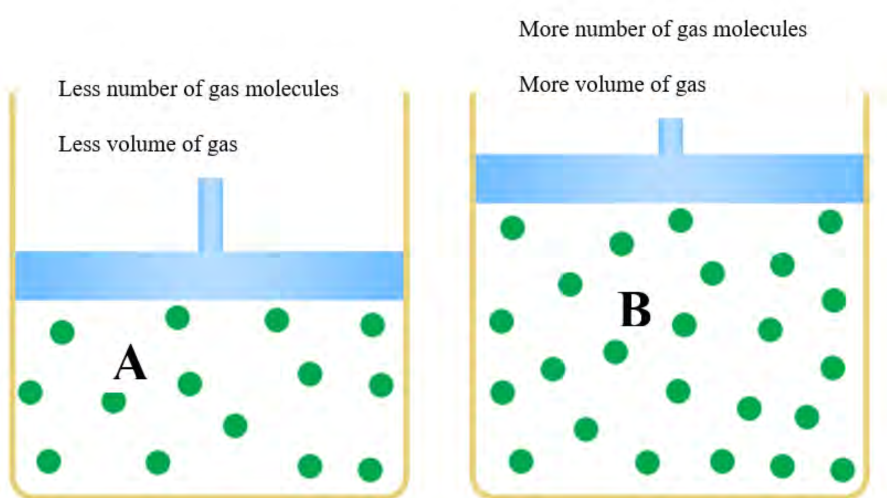
Activity:

Which of the given containers contain a greater number of gas molecules and why?



The space occupied by the gas in the container is called its volume. The volume of gas is equal to the volume of the container. More the gas means more the number of gas molecules which increases the volume of gas. The Italian chemist **Amedeo Avogadro**, in early 19^{th} century established a relationship between the volume of a gas and the corresponding number of molecules under a given set of conditions of temperature and pressure. This law is called **Avogadro's law**.

In the figure A and B, the volume of container B is more than A means it has a greater number of gas molecules than A



Do you know?

In 1909, **Jean Baptise perrin** provided experimental support for Avogadro's hypothesis using the technique like Brownian motion and the concept of Avogadro's Number became firmly established. Today Avogadro's number (N_A) is defined as the number of atoms, ions or molecule in one mol of a substance and its value if approximately 6.022×10^{23}

1 mol is defined as the amount of a substance that contains the same number of particles (atoms, insolences) as there are atoms in 12.00 grams of isotope of carbon 12.

Whatever is the gas in the container, the number of gas molecules will be the same if they have the same volume at the same condition of temperature and pressure. For example, the shape and mass of NH_3, Cl_2 or O_2 molecules are different but there is same number of molecules in same volume of gas.

22.4 L of gas at STP (standard temperature Pressure) or NTP (normal temperature pressure) contains 6.022×10^{23} molecules =one mol of gas

44.8 L of gas at STP contains 1.24×10^{23} molecules = two mol of gas

Room temperature and pressure (RTP)

Temperature =25^0C =298K

Pressure =1 atm=760 mmHg

Normal temperature pressure (NTP) or Standard temperature pressure (STP)

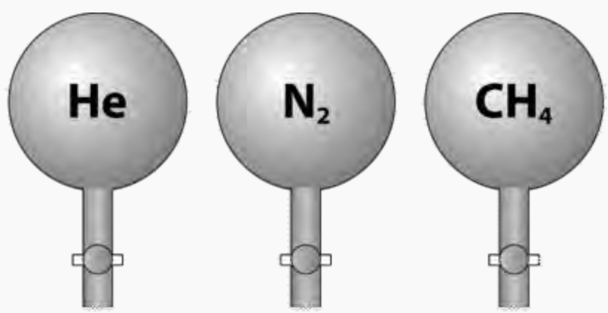
Temperature =0^0C =273K

Pressure =1 atm=760 mmHg

Avogadro's law states that, "***under similar conditions of temperature and pressure, an equal volume of all gases contain an equal number of molecules".***

$$V \propto N$$

Where V is the volume of the gas and N is the number of molecules.



Volume	22.4 L	22.4 L	22.4 L
Pressure	1 atm	1 atm	1 atm
Temperature	0^oC	0^oC	0^oC
Mass of gas	4.00 g	28.0 g	16.0 g
Number of gas molecules	6.02×10^{23}	6.02×10^{23}	6.02×10^{23}

Bezelius hypothesis

"Equal volume of all gases under similar conditions of temperature and pressure contains equal number of atoms" this hypothesis contradicts the Dalton's atomic theory. So, this theory was rejected.

Avogadro's hypothesis can be used to derive various types of relationship for the gas.

1. **The relationship between molecular mass and vapor density of gas.**

   We know that vapour density is the density of a certain gas in comparison to the density of hydrogen gas under the same condition of temperature and pressure.

   Vapour Density $= \frac{density\ of\ gas}{density\ of\ hydrogen\ gas}$

   $= \frac{certain\ mass\ of\ gas \times volume\ of\ hydrogen\ gas}{volume\ of\ vapour \times certain\ mass\ of\ hydrogen\ gas}$ [∵density $= \frac{mass}{volume}$]

   $= \frac{certain\ mass\ of\ gas}{certain\ mass\ of\ hydrogen\ gas}$ [∵volume is equal for both]

   $= \frac{mass\ of\ 1\ molecule\ of\ gas}{mass\ of\ 1\ molecule\ of\ hydrogen\ gas}$

   $= \frac{mass\ of\ 1\ molecule\ of\ gas}{2\times mass\ of\ 1\ atom\ of\ hydrogen\ gas}$ [∵hydrogen is diatomic gas contains 2 atoms per molecule]

   $= \frac{molecular\ weight}{2}$ [∵molecular weight$= \frac{mass\ of\ 1\ molecule\ of\ gas}{mass\ of\ 1\ atom\ of\ hydrogen\ gas}$]

   Molecular weight =2× vapor density

where molecular mass is the sum of the mass of all the atoms in a molecule whereas vapor density is the density of vapor compared to the density of vapor of hydrogen under same temperature and pressure.

2. **The relationship between molar mass and molar volume**

   We know that

   Molecular weight =2× vapor density

   $$=2 \times \frac{mass\ of\ certain\ volume\ of\ gas}{mass\ of\ same\ volume\ of\ hydrogen\ gas}$$

   $$= \frac{mass\ of\ 1\ L\ of\ \ of\ gas\ at\ STP}{mass\ of\ 1\ L\ \ of\ hydrogen\ gas\ at\ STP}$$

   $$=2 \times \frac{mass\ of\ 1\ L\ of\ gas}{0.9} [\because mass\ of\ 1\ L\ \ of\ hydrogen\ gas\ at\ STP = 0.9g]$$

   Molecular weight $=22.4 \times mass\ of\ 1\ L\ of\ gas$

   Molecular weight $=mass\ of\ 22.4\ L\ of\ gas$

The mass of one mol of substance is called **molar mass** whereas the volume occupied by one mol of gas is called **molar volume**. It means 1 mol of gas occupies 22.4 L at STP. When temperature increases volume of gas also increases therefore 1 mol of gas at room temperature and pressure (RTP) occupies 24.0 L.

Molar volume =22.4 L at STP

Molar volume =24.0 L at RTP.

3. **The relationship between molecular mass and number of particles**

   We know that molar volume of any gas at STP =22.4 L

   Or 1 mol of gas = 22.4 L

   From Avogadro's hypothesis, equal volume of all gas under constant conditions contains the same number of molecules. Using different experimental measurements, and the gas laws it was found that 22.4 L of gas contains 6.022×10^{23} molecules at STP.

   Therefore 1 mol of gas = 6.022×10^{23} molecules

   The number 6.022×10^{23} is constant for all substance so called Avogadro's constant (N_A)

For example; 2 gram of hydrogen (1 mol) contains 6.022×10^{23} molecules of hydrogen

100 g of $CaCO_3$(1 mol) contains 6.022×10^{23} molecules of calcium carbonate.

4. **Determination of molecular formula of the gas molecule from volumetric composition vapor density**

   For example, if the oxide of nitrogen contains same volume of nitrogen and vapor density of the oxide is 38,

   volume of oxide = volume of nitrogen

   then using Avogadro's hypothesis, they have same number of molecules

   1 molecule of oxide =1 molecule of nitrogen (2 atoms of nitrogen)

   Therefore, molecular formula of oxide $=N_2O_y$

   Since vapour density =38,

   molecular weight $=2\times38=76$

   So, $N_2O_y=76$

   Or, $2\times14+y\times16=76$

   Or, $y=3$

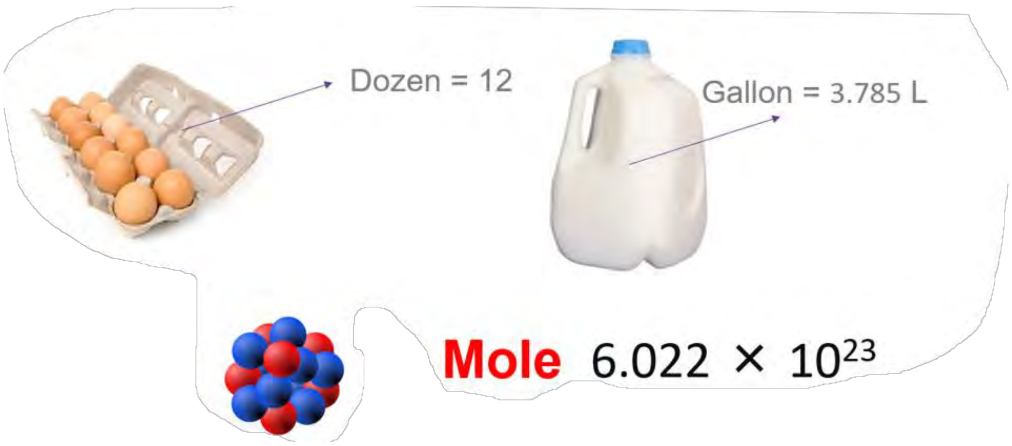
   Therefore, molecular formula of oxide $=N_2O_3$

> **Try yourself!**
>
> An oxide of nitrogen contains its half volume of nitrogen. Find the molecular formula of the oxide if vapour density of oxide is 15.

2.4 Mol and its relation with mass, volume and number of particles

Activity:

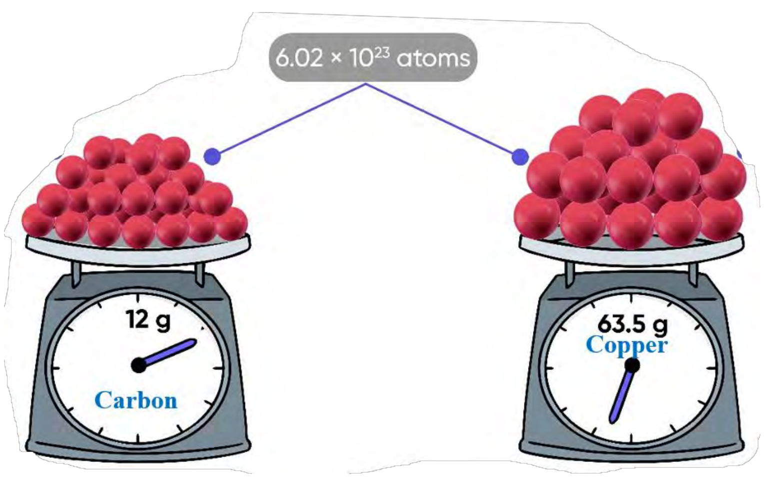
Identify the measurements and respective units in the above figure.



Have you ever tried to count the number of sugar particles in a packet of sugar? It is very difficult because the particles are very small and numerous. Chemists face the similar challenge while counting the atoms, ions or molecules because they are again smaller than the size of sugar. Here, we will discuss the way of counting such microscopic particles in a convenient way.

Mol in terms of number

Find the similarity and differences between the given amount of copper and carbon in terms of number of particles and mass.



We use different units in our daily life to measure the number of substances. For example, buying 1 dozen bananas means 12 pieces of banana and buying 1 ream of paper means 500 sheets. These units like dozen or a ream are inappropriate to a chemist while dealing with microscopic particles like atoms, molecules, electrons or ions. So, they use mol to express the number of such minute particles.

Figure: 1 mol of carbon=12 g.

Different ways of expressing numbers

1 dozen =12 pieces

1 gross =144 pieces

1 ream =500 pieces

1 mol = 6.022×10^{23} particles

One mol is defined as the number of atoms as there are in exactly 12 g of carbon -12.

One mol contains 6.022×10^{23} particles**.** This number (6.022×10^{23}) is referred to as Avogadro's number (N_A) named in honor of Amedeo Avogadro, a pioneer investigator of the quantitative aspects of chemistry.

1 mol = 6.022×10^{23} (atoms, ions or molecules)

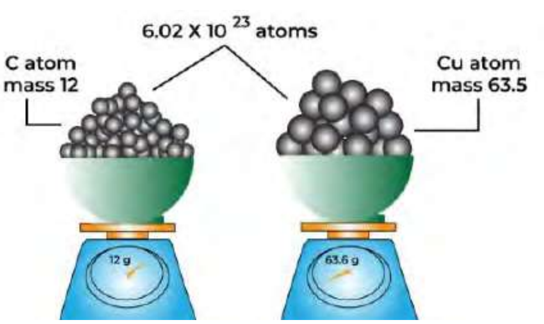
Mol of substance can be converted in to number of particles (molecules, atoms, ions or electron) by multiplying with Avogadro's number (N_A) whereas number of particles can be converted into mol by dividing with Avogadro's number (N_A).

Number of mol = $\frac{number\ of\ particles}{6.022\, x\, 10^{23}}$

Do you know how big is Avogadro's number?

If you started contains now at the rate of 10 billion atoms per second it will take 2 billion years to count the Avogadro's number

Equal number of mol contains an equal number of particles. Therefore, 12 g of carbon and 63.5 g of copper contain same number of particles (6.022×10^{23}) .



Do you know?

The number of molcules in a single droplet of water is roughly 100 billion times greater than the number of people on earth.

Solved problem

Calculate the number of molecules, number of atoms and number of electrons in 0.5 mol of water.

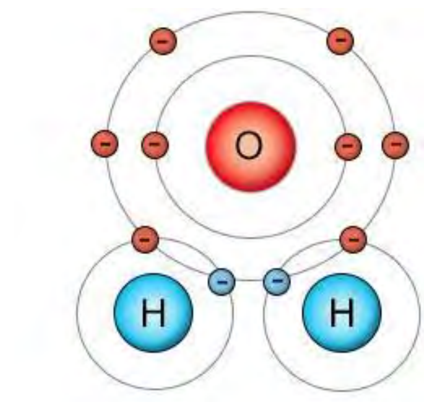
Solution,

Number of molecules of water = number of mol× N_A

$=0.5\ 6.022x10^{23}$

$=3.011 \times 10^{23}$

Since one molecule of water contains three atoms



Number of atoms =number of molecules × 3

$= 3.011\ x10^{23}\times 3 = 9.033\times 10^{23}$

Likewise, one molecule of water contains 10 electrons

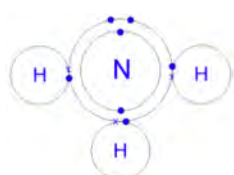
Number of electrons in 0.5 mol of water = number of molecules ×10

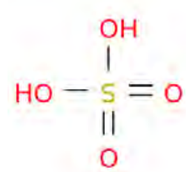
$= 3.011 \times 10^{23}\times 10$

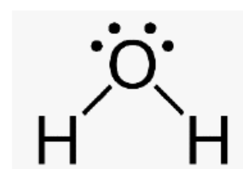
$= 3.011 \times 10^{24}$

Test yourself!

Using the structures of the given molecules answer the following questions:







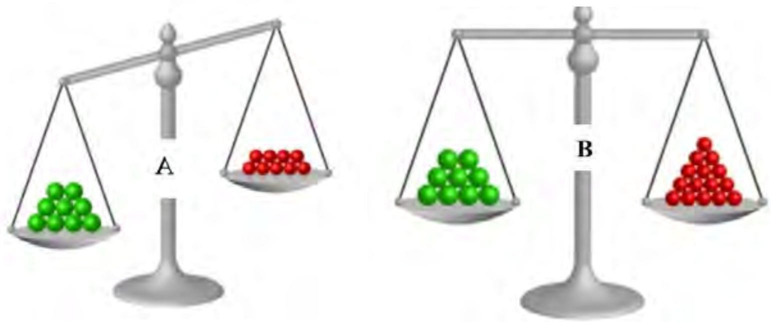
a. Find out the number of molecules, number of atoms and number of electrons in 0.1 mol of ammonia gas.
b. How many hydrogen atoms are there in a mixture of 1 mol of water and 2 mol of Sulphur acid?

Mol in terms of mass

Activity: 2.4.2

Two different substances with different colour and number are kept in the balance in the figure A and B.

What is the difference between the figure in terms of number of particles and mass?



gnesium=24 amu

of magnesium=24 g

f carbon dioxide =44 amu

nolecule) of carbon dioxide =44 g

1 mol of any substance is numerically equal to the atomic mass or molecular mass of the substance in grams.

12 g of carbon =1 mol [atomic mass of carbon =12]

55.8 g of iron =1 mol [atomic mass of iron = 55.8]

Likewise, 44 g of carbon dioxide =1 mol [molecular mass of carbon dioxide = 44]

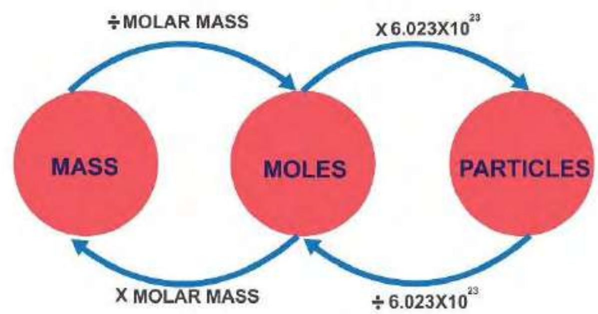
$\therefore$ Number of mol = $\frac{given\ mass\ (g)}{atomic\ mass\ or\ molecular\ mass}$

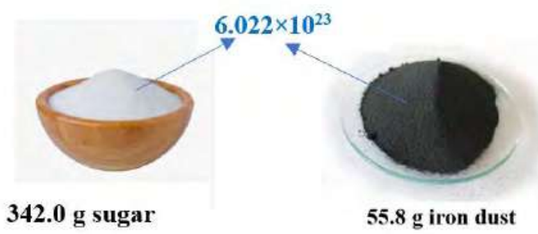
If atomic mass is used in the above relation, it is also called the number **of grams of atom**. When molecular mass is used it is called the number **of gram molecules.**

Number of gram atom = $\frac{given\ mass\ (g)}{atomic\ mass}$,

Number of gram molecule = $\frac{given\ mass\ (g)}{molecular\ mass}$

Now, you can convert the mol into mass and number using the relationship as shown below





Solved problem:

If oxygen gas cylinder contains 0.16 kg of oxygen gas at room temperature and pressure.

Calculate

Number of gam atom of oxygen
Number of gram molecule of oxygen
Number of mol of oxygen

Solution,

Mass of oxygen gas =0.16 kg =160 g

Number of gram atom = = = =10

Number of gram molecule = = 5

Number of mol =

Therefore, number of mol =5

Test yourself!

What is the number of gram atom, number of gram molecule and number of mol of nitrogen in 140 milligrams of it?

Mol in terms of volume

Activity:

n = 1 mol	n = 1 mol	n = 1 mol
P = 1 atm (760 torr)	P = 1 atm (760 torr)	P = 1 atm (760 torr)
T = 0°C (273 K)	T = 0°C (273 K)	T = 0°C (273 K)
V = 22.4 L	V = 22.4 L	V = 22.4 L
Number of gas particles = 6.022×10^{23}	Number of gas particles = 6.022×10^{23}	Number of gas particles = 6.022×10^{23}
Mass = 4.003 g	Mass = 28.02 g	Mass = 32.00 g

a. Which one is the heaviest gas among the ones given?
b. What is the number of mol of each gas?
c. What is the relationship between mol and volume at given conditions?

At STP 6.022×10^{23} molecules (1mol) of any gas occupies 22.4 L called molar volume.

It means 1 mol of gas at STP=22.4 L

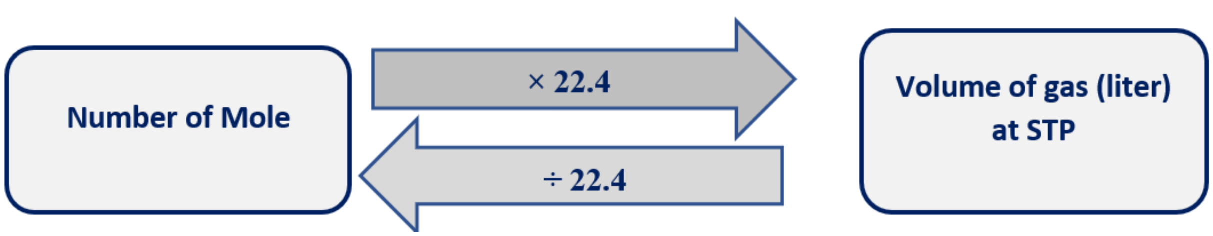
0.5 mol of gas at STP= 11.2L

But at room temperature and pressure (RTP) volume of 1 mol of gas increases due to increase in temperature.

1 mol of gas at RTP =24.0 L

Number of mol = $\frac{given\ volume\ of\ gas\ (L)}{22.4}$ at STP.

Like mass and number of particles, volume of gas can also be converted to mol and vice versa using 22.4 at STP.



Solved problem1:

Calculate the volume of 2 mol of SO_2 gas at STP?

Number mol =

Therefore, volume of gas $=2\times22.4=44.8$ L

Solved problem 2;

Find the number of mol of CO_2 gas in 22.4 mL at STP.

Number mol =

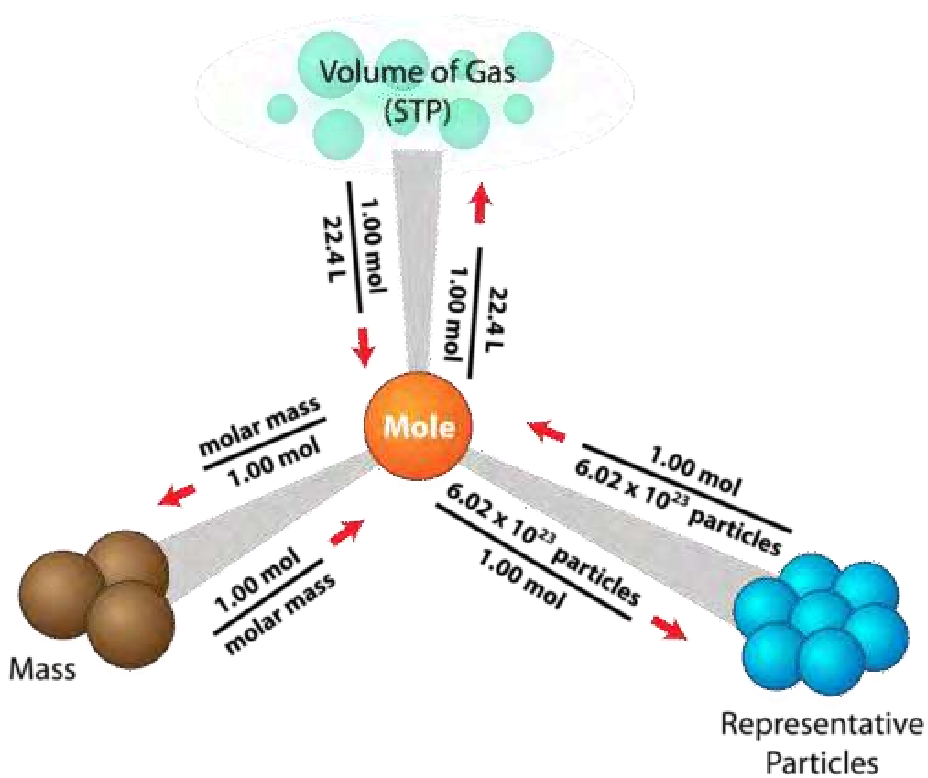
Or, Number mol =

= 0.001

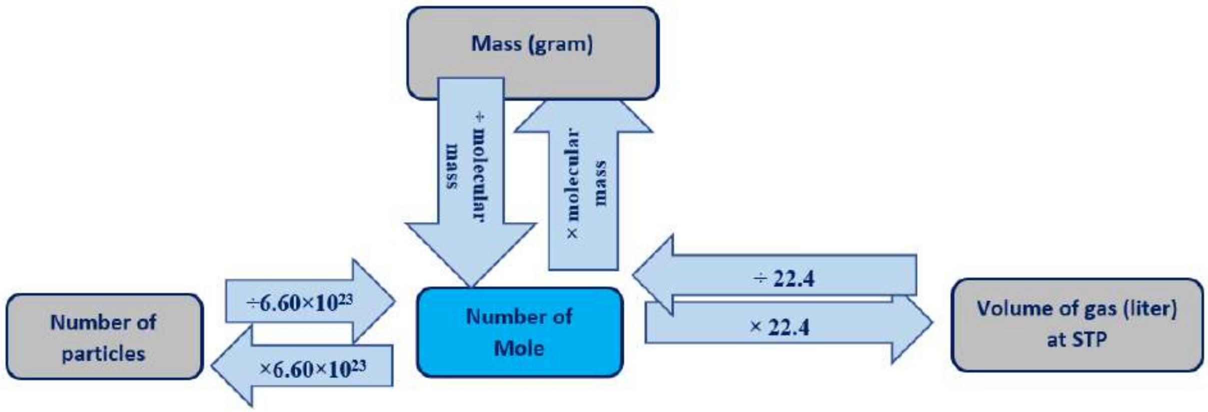
Solve yourself!

Calculate the number of mol of H_2S gas in 1120 mL at NTP. Also, find the mass and number of molecules.

2.5 Calculations based on mole concept



The mole concept is a fundamental part of chemistry. This idea about the inter relationship among mol, mass, volume and number is applicable in various areas of chemistry, providing a standardized and convenient way to quantify and analyze chemical reactions. The relationship among the mol with the number of particles, mass of substance and volume of gas can be summarized in the diagram below:



Number of mol $= \frac{given\ mass\ (g)}{atomic\ mass\ or\ molecular\ mass} = \frac{given\ volume\ of\ gas\ (L)}{22.4} = \frac{number\ of\ particles}{6.022\ x\ 10^{23}}$

Solved problem

Fire extinguisher contains liquid CO_2 gas as an extinguishing agent. When you activate the fire extinguisher, the liquid CO_2 is released and expands into a gas. Use the information given in the figure and calculate;

a. number of mol of carbon dioxide gas in the extinguisher
b. number of CO_2 molecules in the container
c. number of gram atom of Oxygen
d. Number of carbon atoms
e. the volume of CO_2 gas released at STP (assume no gas is left in the vessel)

Solution,

a. Number of mol $= \frac{given\ mass\ (g)}{molecular\ mass} = \frac{4.4x1000}{44} = 100$
b. Number of CO_2 molecules =number of mol x 6.022×10^{23}
$=100 \times 6.022\times10^{23} = 6.022\times10^{25}$
c. number of gram atom of Oxygen = no. of mol of $CO_2 \times 2$
[1 molecule of CO_2= 2 atoms of oxygen]
$=100 \times 2$
$= 200$

d. number of carbon atoms =number of molecules of $CO_2 \times 1$
[1 molecule of CO_2=1 atom of carbon]
$=6.022\times10^{25} \times 1$
$=6.022\times10^{25}$.

e. Volume of CO_2 released = number of mol of $CO_2 \times 22.4$
$=100 \times 22.4$
$=2240$ L.

Test yourself!

1. Solve the following.
   a. Determine the mass of 6.022×10^{23} number of N_2 molecules.
   b. Determine the number of bromide ion in 0.2 mol of $MgBr_2$.
   c. Calculate the ratio of molecules present in 16 g of methane and 16 g of oxygen.
   d. Calculate mass of Nitrogen (N_2) which contains the same number of molecules as are present in 4.4 g of Carbon dioxide (CO_2).
   e. Calculate the mass of 100 mL of ammonia gas at STP.
2. You have 0.5 mol of formic acid (HCOOH). Calculate the;
   a. Mass of formic acid
   b. Number of formic acid molecules
   c. Mass of oxygen atom
   d. Number of hydrogen ions in solution.
   e. Gram atom of carbon.

Chemical calculations based on chemical equations

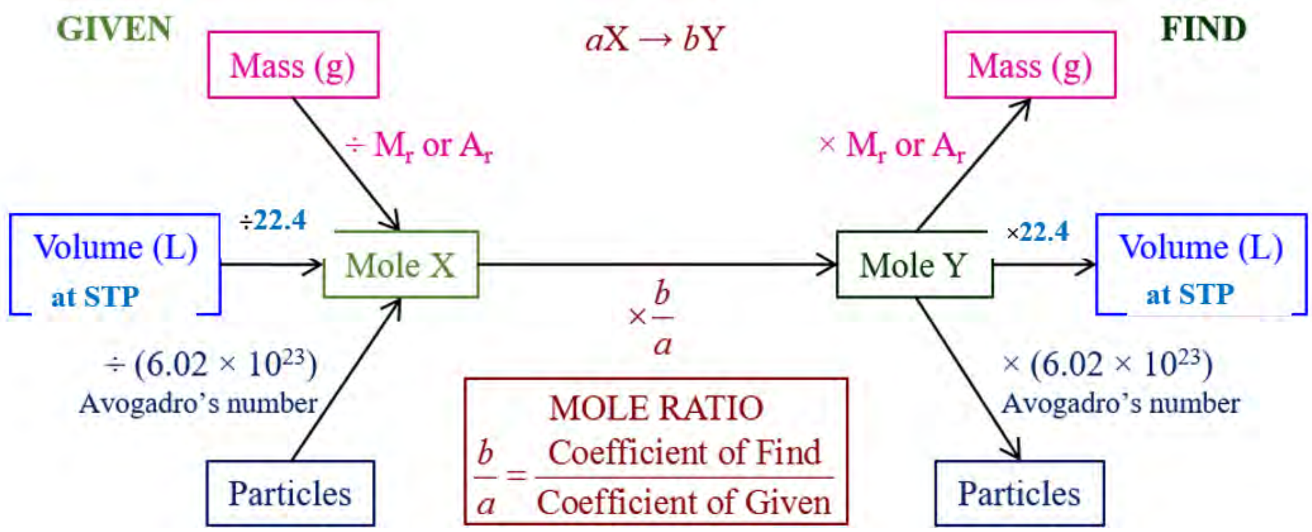
Activity:

Chemical equation for the reaction of iron oxide and carbon monoxide is given.

$$Fe_2O_3 + 3\,CO \xrightarrow{\Delta} 2\,Fe + 3\,CO_2$$

reactants		yields	products	
1 formula unit	3 molecules		2 atoms	3 molecules
1 mol	3 mol		2 mol	3 mol
159.7 g	84.0 g		111.7 g	132 g

a. Find the mol of iron produced by 4 mol of iron oxide with excess of carbon monoxide.
b. What is the mass of carbon dioxide produced by 8.4 g of carbon monoxide with excess of iron oxide?



Right amount of reactant maintains the quantity and quality of the product and saves the expensive ingredients from unwanted wastage. So, the knowledge of mol concept is essential for chemists in the industries. Chemical calculations are a fundamental aspect of chemistry and are used to determine various properties and quantities related to chemical reactions and substances. Stoichiometry involves calculating the relative quantities of reactants and products in a chemical reaction. We can calculate the mass, mol, or volume of substances involved in a reaction. This is typically done using balanced chemical equations. In a balanced chemical equation, the ratio of mol of reactant and products is equal to the ratio of stoichiometric coefficient of these substances.

$$A_2 + B_2 = 2AB$$

Number of mol of A_2: number of mol of B_2: number of mol of AB = 1:1:2

Or, Number of mol of A_2: number of mol of AB $= \frac{1}{2}$

Or, Number of mol of B_2: number of mol of AB $= \frac{1}{2}$

For example, the given cylinder contains 3.2 kg of methane. Can you calculate the mass of oxygen required to burn completely? What will be the amount of carbon dioxide and water produced during combustion?

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

Now, Number of mol of CH_4: number of mol of =1: 2

$$\frac{Number\ of\ mol\ of\ CH_4}{number\ of\ mol\ of\ O_2} = \frac{1}{2}$$

$$\frac{3.2 \times 1000 \times 32}{16\ \times X} = \frac{1}{2} [X\ is\ the\ unknown\ mass\ of\ oxygen\ gas]$$

Or, X= 12800 g

Or, X = 12.8 kg

It means 12.8 kg of oxygen gas is required for the complete combustion of 3.2 kg of methane gas.

Likewise, volume of CO_2 gas produced at STP can be calculated as

$$\frac{Number\ of\ mol\ of\ CH_4}{number\ of\ mol\ of\ CO_2} = \frac{1}{1}$$

$$\frac{3.2 \times 1000 \times 22.4}{16\ \times X} = \frac{1}{1} [X\ is\ the\ unknown\ volume\ of\ of CO_2\ gas]$$

Or, X=4480 L.

The mass of water produced can be calculated as

$$\frac{Number\ of\ mol\ of\ CH_4}{number\ of\ mol\ of\ H_2O} = \frac{1}{2}$$

$$\frac{3.2 \times 1000 \times 18}{16\ \times X} = \frac{1}{2} \qquad [X\ is\ the\ unknown\ mass\ of\ water]$$

Or, X= 7200 g =7.2kg

Test yourself!

a. Calculate the volume of nitrous oxide (N_2O) gas produced when 5 grams of ammonium nitrate (NH_4NO_3) decomposes at normal temperature and pressure (NTP).

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

b. Calculate the mass of hydrogen gas (H_2) evolved when 25 grams of zinc (Zn) reacts with excess hydrochloric acid (HCl). [At. Mass of Zn = 65.5 amu]

c. Calculate the volume of carbon dioxide produced when 1 Kg of 90% pure limestone decomposes at NTP.

2.6 Limiting reactant and excess reactant

Activity 2.6.1

Sugar and tea leaves are the primary ingredients to make tea. Use the amounts given and discuss to find the answer to the following questions.

a. Which one will run out first when making tea?

b. How many cups of tea can you prepare using the provided quantities of sugar and tea leaves? (Assume one cup of tea requires 20 grams of sugar and 5 grams of tea leaves)

Activity2.6.2

How many bicycles can be made from the parts given? Discuss in pairs and find the reason behind the answer.

During the chemical reaction it is not necessary to completely finish all the reactants used. One of the reactants may be left over in the reacting vessel even after the complete chemical reaction called **excess reagent**. The reactant that gets completely consumed or "limits" the amount of product that can be formed is called **limiting reactant**. It's the reactant that is used up first, and once it's finished, the reaction can't proceed any further. The limiting reactant is a fundamental concept in stoichiometry and plays a central role in various aspects of chemistry and industry.

- **a.** It tells us how much product we can make at most in a chemical reaction.
- **b.** It helps us avoid wasting chemicals by using them efficiently.
- **c.** It prevents the formation of unnecessary byproducts and waste.
- **d.** It helps in making economical and environmentally friendly choices in chemical reactions.

For example, if we have **2 mol** of A_2 and **3 mol** of B_2 react to form AB.

Then the balanced chemical equation is

A_2 + B_2 ▼ 2AB

Solved problem 1

Potassium superoxide, KO_2, is used in rebreathing masks to generate oxygen according to the reaction below. If the mask contains 0.150 mol KO_2 and 0.100 mol water, what is the mass of oxygen that can be produced?

$4KO_2(s) + 2H_2O(\ell) \rightarrow 4KOH(s) + 3O_2(g)$

4 mol of KO_2=2 mol of H_2O

0.150 mol of KO_2= 0.075 mol of H_2O **(required)**

Given mol of H_2O is 0.1 mol

Since required amount is less than given amount, H_2O is an excess reagent and KO_2 is a limiting reagent.

Now,

= (Where X is mass of oxygen gas in gram)

Mass of oxygen =3.6g

1 mol 1 mol

2 mol (given) **2 mol (required)**

But the given number of mol of B_2 is 3 mol.

Since the given number of mol of $\mathbf{B_2}$ (3 mol) is greater than the required number of mol (2mol) it is an excess reagent and $\mathbf{A_2}$ is a limiting reagent. It can be directly calculated from the given relationship.

$$\frac{number\ of\ mol\ given}{Stoichiometric\ coefficient} = \frac{2}{1} \text{ for } A_2$$

$$\frac{number\ of\ mol\ given}{Stoichiometric\ coefficient} = \frac{3}{1} \text{ for } B_2$$

The ratio is smaller for A_2 so is a limiting reactant and B_2 is an excess reactant.

Amount of excess reagent left depends upon the given amount and amount consumed by limiting reagent. It is calculated as;

Number of mol of excess reagent left over = given number of mol – number of mol consumed.

= 3 mol -2 mol

= 1 mol

Solved problem 2:

If 4 mol of nitrogen reacts with 9 mol of hydrogen gas to form ammonia gas, then answer the given questions:

Which one is limiting reactant?
What is the mass of ammonia gas produced in the reaction?
Find the mass of excess reagent left over after the complete reaction.

Solution,

$\mathbf{N_2(g) + 3H_2(g) \qquad 2NH_3(g)}$

The balanced equation tells us that the stoichiometric coefficient for nitrogen and hydrogen is 1 and 3, respectively.

It means 1 mol of N_2=3 mol of H_2

4 mol of N_2 = **12 mol of H_2** (required)

But the given amount of **H_2 is 9 mol**

Since given amount is less than required amount, **H_2 is a limiting reagent**

Since limiting reactant is nitrogen, it is used to determine the unknown amount of ammonia,

[X is the mass of NH_3]

Or, x = 714 g

$N_2(g) + 3H_2(g) \quad 2NH_3(g)$

1mol 3 mol

4 mol 9 mol

number of mol of excess reagent left over = given number of mol – number of mol consumed

= 4mol -3 mol

= 1 mol

=28 grams

Try yourself!

When 10.0 grams of calcium carbonate reacts with 15.0 grams of hydrochloric acid according to the following equation:

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

a. Which reactant is the limiting reagent?
b. Calculate the amount of excess reagent remaining after the reaction is complete.
c. Find the volume of CO_2 gas produced at STP.

2.7 Theoretical yield, experimental yield and percentage yield

When lime stone ($CaCO_3$) is burnt at high temperature in the lime kiln, it is decomposed to quick lime and carbon dioxide.
If a lime kiln operator used 1 kg of limestone in the kiln and obtained **500 g** of quick lime.
But using the chemical equation, the amount of lime produced is **560 g** by burning 1 kg of limestone as given:

$\mathbf{CaCO_3 \rightarrow CaO + CO_2}$

100 g   56 g

1000g   **560g**

Fig: Lime kiln

500 g is the theoretical or experimental yield of the process.

Theoretical yield is an idealized value and may not always be achievable in practice. Theoretical yield represents the ideal amount of product you would obtain if the reaction proceeded perfectly, with 100% conversion of reactants to products.
The theoretical yield in a chemical reaction is the maximum amount of product that can be produced based on stoichiometric calculations.

The amount of product formed in the chemical process will be different from the calculated product obtained theoretically. Due to various factors, including the loss of product during handling, incomplete reactions, and the presence of impurities, the amount of product obtained in the chemical process is always less than theoretical yield.
That amount of product obtained in the actual chemical process is called experimental yield or actual yield.

Percentage yield is a measure of the efficiency of the reaction. Higher the percentage yield means higher the efficiency of the process.

$$\%yield = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

For example, in the above case of decomposition of limestone if only 500 g of quick lime is produced from 1 kg of limestone, the percentage yield is calculated as

Percentage yield$=\frac{500}{560}\times 100$

[Where 500g=experimental yield and 560g= theoretical yield]

$= 89.28\%$

Test yourself!

In Haber's Process, Nitrogen and hydrogen are converted to ammonia at low temperature and high pressure. When 400 g of H_2 is added to an excess amount of N_2, 104 g of NH_3 is formed.
Write the balanced chemical equation for the reaction involved.
What is the theoretical yield of ammonia?
Calculate the percentage yield.
Washing soda is a hydrate of sodium carbonate. Its formula is $Na_2CO_3.xH_2O$.
2.714 g sample of washing soda is heated until a constant mass of 1.006 g of Na_2CO_3 is reached. What is value of x? Calculate the mass percent of water in the hydrate.

Solved problem1:

0.1234 g of curdy white precipitate of silver chloride (AgCl) was formed by the reaction of silver ion (Ag^+) with excess of chloride ion (Cl^-). Assuming the percentage yield to be 98.7%, how many grams of silver ions were present in the solution?

Solution,

$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

108 g 143.5g

0.0928g 0.1234g

Experimental yield of Ag^+ = 0.0928g

Percentage yield=×100

98.7 =×100

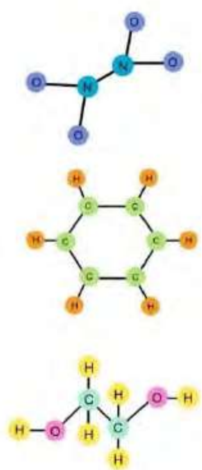
Therefore, theoretical yield = 0.094g

The ratio of actual yield to theoretical yield expressed in percentage is called the percentage yield.

Calculation of empirical and molecular formula from percentage composition

Activity:

Find the ratio of mole of atoms of given compounds in both molecular and empirical formulas.



Molecular	Empirical
N_2O_4	NO_2
C_6H_6	CH
$C_2H_6O_2$	CH_3O

Do you know?

Green chemistry is a philosophical approach to maximize the efficiency of processes for synthesizing chemical products. Such efficiency is measured in terms of **atom economy.**

We have already discussed the empirical formula, molecular formula and calculation of percentage composition from molecular formula in unit 1.3. In this chapter we will discuss the calculation of empirical formula and molecular formula from percentage composition. We use the idea about the mol for this conversion.

Empirical formula is the simplest formula of the compound obtained from the simplest whole number ratio of the atoms involved in the compound. We can deduce the empirical formula using these simple steps.

a. Write the percentage composition of each element in the given compound.

b. Convert the percentage in to mol (for example :20% means 20 gram in 100 gram)

c. Determine the mol ratio of constituent elements.

d. Find the simplest whole number ratio of mol by dividing each mol value by the smallest mol value obtained. Round to the nearest whole number.
e. Write the empirical formula.
f. Find molecular formula using the given relationship:
   Molecular formula =empirical formula × n [$n = \frac{molcualr\ weight}{empirical\ weight}$]

For example;

Percentage composition of an organic compound is given as

Carbon =40%

Hydrogen=6.7%

oxygen= 53.3%

Assume 100 g of the compound, so you have 40 g of carbon, 6.7 g of hydrogen, and 53.3 g of oxygen.

Number of mol of carbon = $\frac{40}{12}$=3.33

∵ Number of mol =

Number of mol of hydrogen = $\frac{6.7}{1}$ =6.7

Number of mol of oxygen = $\frac{53.3}{16}$=3.33

The smallest value is 3.33, so divide each by 3.33.

Carbon: 3.33/3.33=1

Hydrogen: 6.7/3.3 ≈2

Oxygen: 3.33/33.33 =1

∵ simplest whole number ratio of number of mol of carbon, hydrogen and oxygen=1:2:1

So, the empirical formula of the compound is $\mathbf{CH_2O}$

if vapor density of the given compound is 90, we can calculate the molecular formula as

Molecular formula =empirical formula × n

$= CH_2O \times \frac{molcualr\ weight}{empirical\ weight}$

$= CH_2O \times \frac{2\times\ Vapour\ density}{empirical\ weight}$

$= CH_2O \times \frac{2\times\ 90}{30}$ [where 30 is the empirical weight of CH_2O]

$= CH_2O \times 6$

$= \mathbf{C_6H_{12}O_6}$

Test yourself!

1. Find the empirical formula of compound X, Y and Z from the composition given.

Compound X	Compound Y	Compound Z
Carbon=50% Hydrogen=10% Oxygen=40%	Nitrogen=25% Hydrogen=3.3% Oxygen=71.7%	Phosphorus = 40% Hydrogen = 6.7% Oxygen = 53.3%

2. What is the molecular formula of the organic compound having molecular weight 46 containing 52.17% carbon ,13.04% hydrogen and 34.78% oxygen?

Erxercises

A. Multiple choice questions

1. Two oxides of a metal contain 27.6% and 30% of oxygen respectively. If the formula of the first oxide is M_3O_4, then what is the formula for the second oxide?
   a. MO
   b. M_2O_3
   c. MO_2
   d. M_2O

2. What is the definition of a mol?
   a. The number of atoms in 12 g of carbon-12.
   b. The number of molecules in 1 gram of water.
   c. The number of protons in one atom of hydrogen.
   d. The mass of 1 L of a substance.

3. How many mol of oxygen atoms are there in 0.5 mol of oxygen gas?
   a. 0.5 mol
   b. 1.0 mol
   c. 2.0 mol
   d. 0.25 mol

4. How many molecules are there in 88 g of carbon dioxide?
   a. 12 x 10^{23}molecules
   b. 6.02 x 10^{23} molecules

c. 3.01×10^{23}molecules
d. 1.2×10^{23}molecules

5. How many mol of water are present in 1.8 mL of water?
a. 1.8 mol
b. 0. 1 mol
c. 0.08 mol
d. 8.03×10^{-5} mol

6. Which of the following defines a limiting reactant?
a. that is left over after the reaction has gone to completion
b. that has the lowest coefficient in the balanced equation
c. for which you have the lowest mass in g
d. that has the lowest molar mass

7. The atomic masses of two elements (A and B) are 20 and 40 respectively. x g of A contains y atoms, how many atoms are present in 2x g of B?
a. x
b. y
c. 2y
d. y/2

8. What is the ratio of the volumes occupied by 1 mol of O_2 and 1 mol of O_3 in identical conditions?
a. 2:1
b. 1:3
c. 1:1
d. 2:3

9. How many mol of oxygen are needed to react completely with 0.5 mol of methane to form water and carbon dioxide?
a. 2 mol
b. 1 mol
c. 0.5mol
d. 0.05mol

10. When 4.8 g of magnesium reacts with excess hydrochloric acid, what is the volume of hydrogen gas produced at STP?
a. 22.4 L
b. 2.24 L
c. 4.48 L
d. 44.8 L

11. In a chemical reaction, 10g of a substance were expected to produce 12g of a product. If the actual yield was found to be 9g, what is the percent yield?
a. 100%
b. 90%
c. 80%
d. 75%

12. In the chemical reaction given, how many mol of D are produced from 16 mol of A and 18 mol of B?

$4A+6B \rightarrow 2C+3D$

a. 54 mol
b. 48 mol
c. 16 mol
d. 9.5 mol

B. Short answer questions

1. State the given laws of stoichiometry with one example of each:

a. Law of multiple proportion
b. Law of reciprocal proportion
c. Law of gaseous volume

2. State Avogadro's law. Write any four applications of this law.

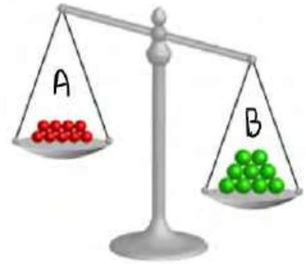
3. Differentiate the following

a. limiting reactant and excess reactant.
b. law of conservation of mass and law of constant composition
c. experimental and theoretical yield.

4. Identify the law of stoichiometry from the given data:

Experiment no.	Compounds	Elements	Law of stoichiometry
1.	CO_2 CS_2 SO_2	27.27% of carbon 15.79% of carbon 50% of Sulphur.	
2.	CuS CuO SO_3	66.49% of copper 79.50 % of copper 40 % of sulphur	
3.	1st oxide 2nd oxide	42.9 % of carbon 27.3 %. of carbon	

5.



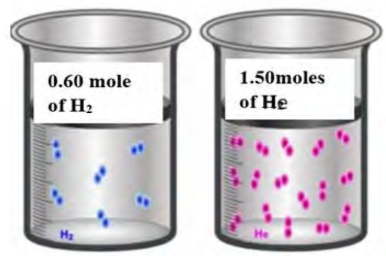
a. What is atomic mass? Why is atomic mass in fraction?
b. Equal atoms of sodium and aluminum are weighed in the balance. Identify A and B with reason.

6. For the reaction $2S(s) + 3O_2(g) \rightarrow 2SO_3(g)$ If 6.3 g of S is reacted with 10.0 g of O_2

a. Calculate the number of mol of each reactant.
b. Which one will be the limiting reactant?
c. Find the mass of SO_3 produced.

7. Hydrogen and helium gas are kept in the container given.

a. Calculate the number of molecules and number of atoms in the container.
b. Which of the gas occupies more volume at STP? Why?



8. Combustion of methane is shown below.

a. Write the balanced chemical equation based on the molecular model given.

b. Find the ratio of the number of mol of methane and oxygen gas in the equation.

c. Calculate the number of mol of O2 required for the complete combustion of 0.5 mol of methane.

9. Carbon dioxide is produced in the laboratory using limestone and hydrochloric acid If 68.1 g solid $CaCO_3$ is mixed with 51.6 g HCl.

a. Write the balanced chemical equation for the given reaction.
b. Identify the limiting reactant.
c. What is the mass and volume of CO_2 produced at STP?

10. In the following reaction: **3** NaBr + H_3PO_4 → **3** HBr + Na_3PO_4

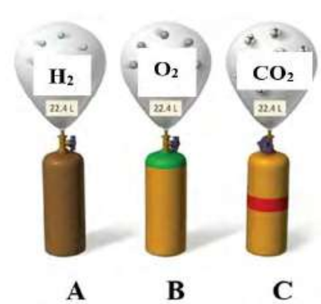
If 200.0 g of each reactant is used, determine the following:

a. Which is the limiting reagent?
b. What is the theoretical yield of each product (in g)?
c. How much excess reactant will remain unreacted, assuming 100% yield?
d. If the percent yield of this reaction is only 70.0%, what would be the actual yield of each product?

11. When 28g N_2 reacts with excess H_2, only 30g of ammonia is produced. What is the percentage yield? How can we increase the percentage yield of the product?
12. Sodium reacts violently with water to produce hydrogen and sodium hydroxide. How many gram of hydrogen are produced by the reaction of 400 mg of sodium with water?
13. Magnesium hydroxide Mg $(OH)_2$ is commonly used as an antacid, if 0.5 mol of it reacts with 0.2 mol of HCl as
Mg $(OH)_2$+ 2HCl → $MgCl_2$+ $2H_2O$
a. Find the limiting reactant.
b. What is the mass of each reactant used?
c. Calculate the mass of magnesium chloride produced.
d. Find the mass of H_2SO_4 to react completely with the given amount of Mg $(OH)_2$.
14. Methanol is used as a fuel for racing cars. It burns in the engine according to the equation If 40.0 g of methanol is mixed with 46.0 g of O_2,
$2CH_3OH$ (l) + $3O_2$(g) → $2CO_2$(g) + $4H_2O$(g)
a. What is the mass of CO_2 produced?
b. Why is methanol used as fuel?
c. Calculate the mass of methanol that can be burnt completely with 0.5 Kg of oxygen gas.

C. Long answer questions

1. A chemist conducts an experiment to determine the composition of a compound containing carbon and oxygen. After analysis, it is found that 1.33 grams of carbon combines with 2.66 grams of oxygen to form one compound, while another compound contains 1.33 grams of carbon combined with 5.32 grams of oxygen.



a. Calculate the number of mol of each atom in both compounds.
b. Determine the empirical formulas of the two compounds.
c. How do the results support the Law of Multiple Proportions?
2. A, B and C represents the container containing three different gases at STP.
a. Find the number of mol of gas in A.
b. Calculate the mass of gas in container B.

c. Find the number of gas molecules in container C.
d. Find the number of atoms in A.
e. Does it explain Avogadro's law? Explain.

3. The first step in the Ostwald process for manufacturing nitric acid is the reaction of ammonia (NH_3), with oxygen (O_2), to produce nitric oxide (NO), and water.
   a. Write the balanced chemical equation for the given reaction.
   b. Find the mass of water produced if 1 kg of ammonia gas is consumed in the reaction.
   c. Which one is limiting the reactant if 0.1 mol of each reacts together?
   d. Calculate the volume of NO gas produced at NTP if 0.1 mol of each reactant reacts.

4. During the combustion process, nitrogen in the air reacts with oxygen at high temperatures, forming nitrogen monoxide (NO) and it can react with oxygen to form gaseous nitrogen dioxide (NO_2). If 3.13 g of nitrogen monoxide reacts with 4.16 g of oxygen,
   a. Write the balanced chemical equation of the reaction.
   b. Which one is the limiting reagent? Why?
   c. How much nitrogen dioxide will be formed?
   d. Which reactant remains in excess, and finds the mass?
   e. What is the environmental impact of this chemical reaction?

5. 56.6 g of calcium and 30.5 g of nitrogen undergo a reaction to form calcium nitride that has a 93.0% yield,
   a. Write the balanced chemical equation for the reaction given.
   b. Find the number of mol of each reactant in the given reaction.
   c. What do you mean that percentage yield of the given reaction is 93%?
   d. Find the mass of solid calcium nitride formed in the reaction.

6. A chemical equation between aluminum sulphide (48.0 g) and water (32.0 g) is given

$$Al_2S_3(s) + H_2O\ (l) \rightarrow Al\ (OH)_3(s) + H_2S(g)$$

   a. Balance the given chemical equation
   b. Define the limiting reactant and identify it in the given equation.
   c. How much Al $(OH)_3$ will be produced?
   d. How much of the excess reactant remains after the reaction is complete?

7. Hydrogen sulfide decomposed to hydrogen and Sulphur at 2000 ^{0}C. It was found that 13 mg of H_2 were produced.

$$H_2S \rightarrow H_2 + S$$

   a. Why is 13 mg called experimental yield of the process? Convert this amount into volume at STP.
   b. Calculate the theoretical yield using the equation given.
   c. Find the percentage yield of the process
   d. Write any two ways by which percentage yield of the reaction can be increased.

8. Oxygen gas is prepared in the laboratory by heating potassium chlorate ($KClO_3$) in a hard glass test tube as shown in the figure.

a. Write the balanced chemical equation for the decomposition of $KClO_3$ in the presence of heat.

b. Find the number of mol of oxygen produced in the gas jar at STP.

c. What is the value of X to produce such an amount of oxygen gas?

d. If 13 g of $KClO_3$ produces the same volume of oxygen gas under the same condition, calculate the percentage yield of the reaction.

e. Calculate the mass of magnesium required to react completely with the mass of oxygen produced above.

9. Solve the given numerical problems

a. Calculate the mass of 6.022×10^{23} molecule of NH_4Cl.

b. An atom of some element Y weighs 6.644×10^{-23} g. Calculate the number of gram atoms in 40 kg of it.

c. If a mol were to contain 1×10^{24} particles, what would be the mass of
(i) one mol of oxygen
(ii) a single oxygen molecule?

d. A chemist weighs a steel cylinder of compressed oxygen, and finds that it has a mass of 1027.8 g. After using some of the oxygen in an experiment, the cylinder has a mass of 1023.2 g. How many mol of oxygen gas are used in the experiment?

e. Calculate the mass of aluminum that would have the same number of atoms as 6.35 g of copper.

f. The weight of a diamond is given in carats. One carat is equivalent to 200 mg. A pure diamond is made up entirely of carbon atoms. How many carbon atoms make up a 1.00 carat diamond?

g. Element X combines with element Y to form two compounds. In the first compound, the mass of X is 12 g, and in the second compound, the mass of X is 18 g. If the mass of Y in the first compound is 8 g, find the mass of Y in the second compound.

h. CO, CO_2 and C_xO_y are the oxides of carbon. If a 2.5 g sample of C_xO_y contains 1.32 g of Carbon and 1.18 g of Oxygen, using the law of multiple proportions, find the value of x and y in the formula.

i. In an experiment 4.70 g of H_2 is allowed to react with an excess of N_2,12.5 g of NH_3 is formed. What is the percent yield of this experiment?

j. The equation for a reaction occurring in the synthesis of methanol is:
$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$
What is the maximum amount of methanol that can be formed from 2 mol of carbon dioxide and 3 mol of hydrogen?

k. Methyl salicylate is made up of carbon, hydrogen, an oxygen atom. When a 5.287 g sample of methyl salicylate is burned in excess oxygen, 12.24 g of carbon dioxide and 2.522 g of water are formed. What is the empirical formula for methyl salicylate?

Project work

Use chart paper and complete the chart as given below.

S. N	Product	Composition /reactants
1.	Cement 	• Limestone ($CaCO_3$) • Clay • Silica (SiO_2) • alumina (Al_2O_3) • Iron ore • Gypsum ($CaSO_4 \cdot 2H_2O$)
2.	Beverage	
3.	Fertilizer	
4.	Juice	
5.	Noodles	
6.	Paracetamol	

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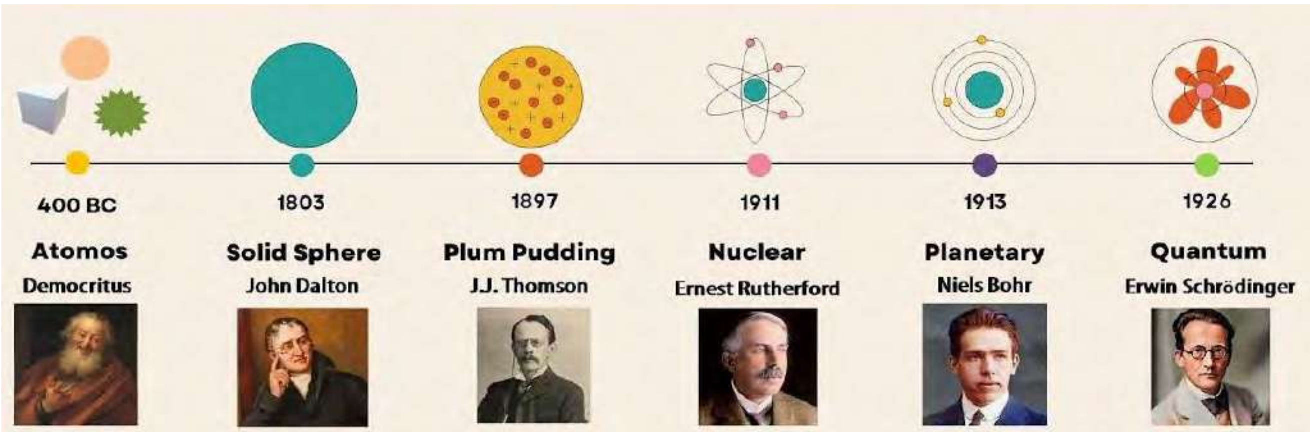
UNIT 3: ATOMIC STRUCTURE

Activity:

Observe the picture about the development of atomic model and identify the model that:

a. discovers electrons.
b. proposes the orbits.



We have discussed the atom and Dalton's atomic theory and its postulates in unit-2.1. In this unit we will discuss the structure of atoms in detail. Dalton postulated atom as a fundamental unit of matter which is indivisible. Later on, various experiments were conducted and concluded that the atom consists of a nucleus as a central core and negatively charged **electron** around the nucleus.

In 1897 the British Physicist J.J Thomson discovered negatively charged **electrons** in the atom from his famous experiment using cathode ray tubes. Likewise, Rutherford discovered positively charged **proton** in 1911 from an alpha rays scattering experiment. In 1932 James Chadwick discovered a neutral particle in the atom called a **neutron**, the same mass as that of proton and much heavier than electron.

3.1 Rutherford's atomic model and its limitations

Activity:

a. Create a circle with a center on the whiteboard.
b. Soak 4-5 tennis balls in ink and throw each ball one by one toward the center of the circle and aim to hit it.
c. Observe and record the number of balls that successfully hit the center and those that do not.
d. Examine the bouncing behavior of the balls and identify the causes of bouncing.

Ernest Rutherford (1871-1937) was a New Zealand physicist has been known as 'father of nuclear physics". He made a significant contribution in the development of atomic model and proposing the concept of radioactivity and radioactive decay. He was awarded the Nobel prize in chemistry in 1908 for his investigation into disintegration of the elements.

Hans Geiger and Ernest Marsden conducted the experiment called Alpha particles scattering experiment in 1909 under the guidance of Ernest Rutherford. Based on these experiments Rutherford developed an atomic model called Ruther ford's atomic model or nuclear model or planetary model.

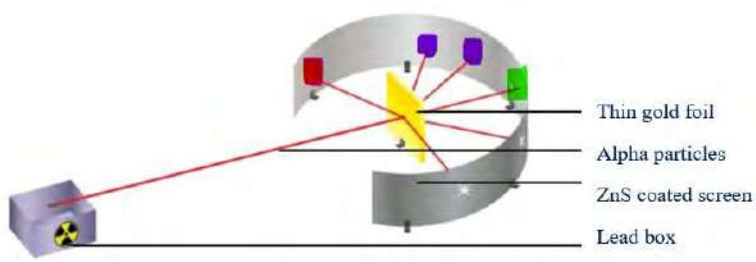


Fig: Alpha particles scattering experiment

These scientists observed the effect of bombarding thin gold foil with positively charged alpha particles (helium nuclei) on the zinc sulfide coated screen. They found that:

a. Most of the particles (99 %) passed through the foil without significant deflection as shown by green dots in the figure.
b. A small fraction of alpha particles experienced small angle deflections as in purple dots.
c. Very few of them bounced back in the same direction of the source.

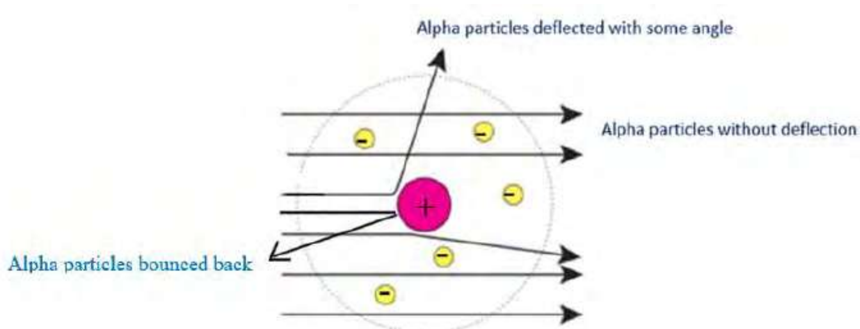
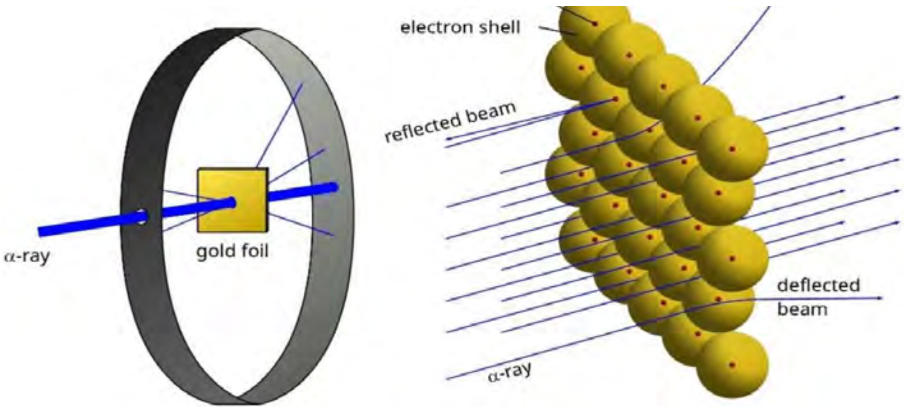


Figure: structure of gold atom according to rutherford

Conclusions of the Rutherford experiment:



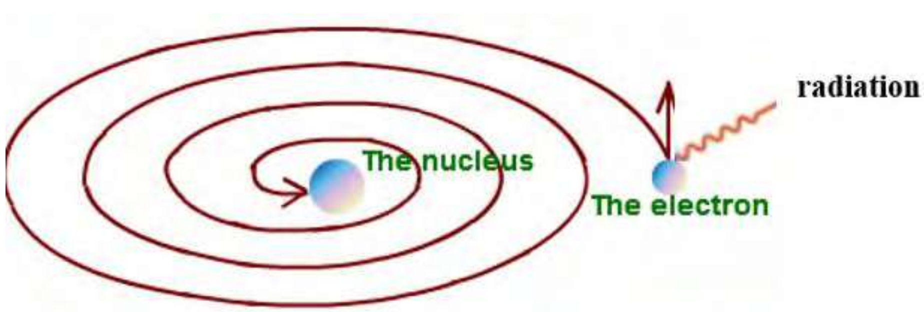
a. Most of the alpha particles passes through the atom which indicates most of the space of the atom is empty.
b. Some of the particles deflected with some angle means there is a positively charged mass inside the atom.
c. Very few alpha particles bounced back or deflected with almost 180 ^{0} angles meaning there is a dense mass concentrated in the small region inside an atom.

Test yourself!

What is the cause of deflection of alpha particles while passing through the gold foil?

3.2 Limitation of Rutherford atomic model

The Rutherford atomic model provided a foundation in the atomic theory by giving the existence of a central nucleus surrounded by electrons. But it could not explain the stability of the atom. According to classical physics, accelerated electrons should emit electromagnetic radiation and lose energy, eventually fall into the nucleus. If so the electron would collide with the nucleus. It means Rutherford atomic model fails to explain the stability of atom.



Test yourself!

What could be the cause Rutherford chose gold foil for his experiment?

3.3 Postulates of Bohr's Atomic Model and its applications

Activity:

a. Take a chart paper, colored marker and 7 beads or marbles.
b. Prepare circular orbits using colored markers around the center and assign number 1,2,3 for each circle.
c. Distribute beads or marbles, to represent electrons on each circle in the order of 2,5 in the first and second circle respectively.
d. Identify the atom with reference to this model.

Niels Bohr(1885-1962) a Danish physicist Renowned for his ground breaking contributions to quantum theory. He received Nobel prize in physics for the development of atomic model which revolutionized out understanding of the atom.

Neil's Bohr in 1931 proposed an atomic model combining the principles from classical physics as well as quantum theory. The major postulates of Bohr's atomic model are:

$$E_1<E_2<E_3<E_4$$

where

energy of K shell (n=1) $=E_1$

energy of L shell (n=2) $=E_2$

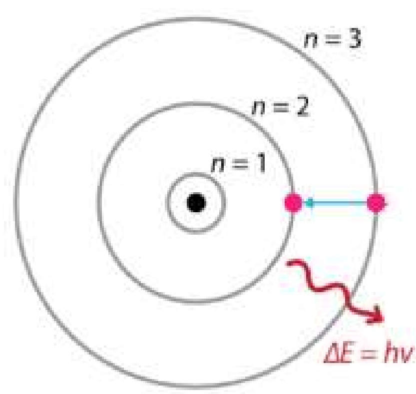
energy of M shell(n=3) $= E_3$

energy of N shell (n=4) $=E_4$

a. Electrons revolve around the nucleus in well-defined circular paths called orbits or shell. Each shell resembles specific energy so are also called energy levels.

b. The distance as well as energy of the shells increases as moving away from the nucleus. As long as electrons remain in a particular shell, electrons do not absorb or emit energy.

c. Electrons when they absorb energy move to higher energy levels and emit energy as radiation when they fall back to the lower energy level. The wavelength of radiation emitted depends up on the difference in energy between the two shells



$$\Delta E= E_2- E_1 = h\nu$$

Where, E_2= higher energy h=Planck's constant

E_1=lower energy ν=frequency of radiation

d. The angular momentum of an electron in a particular orbit is an integral multiple of $h/2\pi$

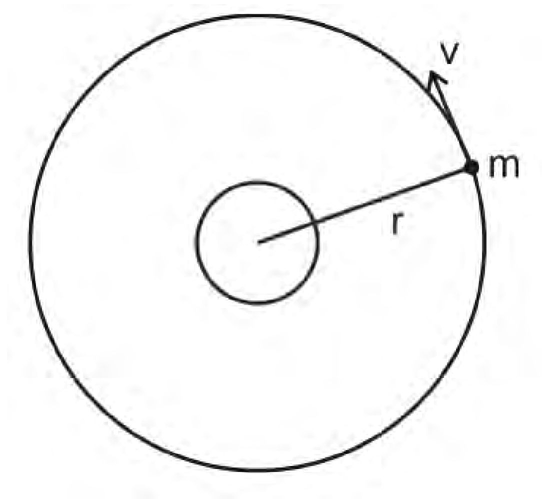
i.e $mvr = n\times\frac{h}{2\pi}$

where n is a principal quantum number =1,2,3 ….and h= Planck's constant

m= mass of an electron

v= velocity of an electron

r= radius of the orbit



How small is the nucleus in the atom?

The marble in the center of football stadium will be equivalent to nucleus in the center of atom.

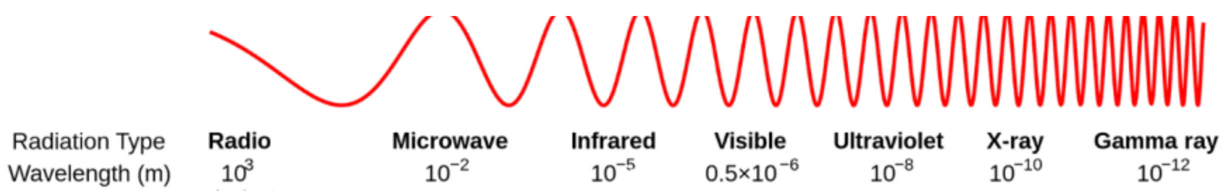
Test yourself!

How many orbits are possible around the nucleus in an atom?

3.4 Spectrum of Hydrogen Atom

Activity 3.4: Observe the spectrum given below

a. Identify the radiation having minimum and maximum frequency.
b. Which of the radiation is most harmful among?



When gaseous atoms are heated, the electrons in the particular shells are excited to higher energy levels. The same case is observed when the electrons absorb specific frequencies of light. By using the instrument like a spectrophotometer, patterns of absorption of light by a substance are recorded called **absorption spectra**.

In the same way when electrons jump from higher energy level to lower energy level, the patterns of emitted light at different wavelengths by a substance is called **emission spectra**. Emission spectra may be **continuous spectra** (broad range of wavelength without distinct lines) or **line spectra** (sharp lines corresponding to specific wavelength).

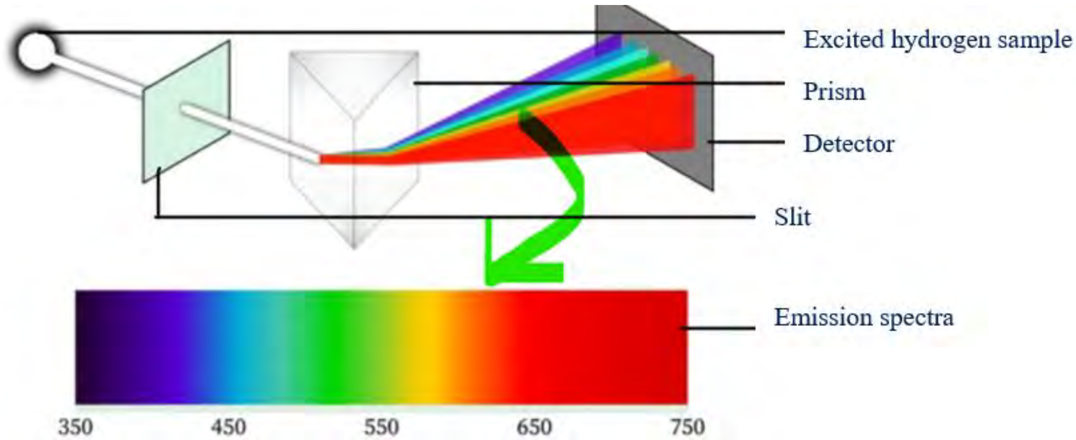


Fig: emission spectra of hydrogen

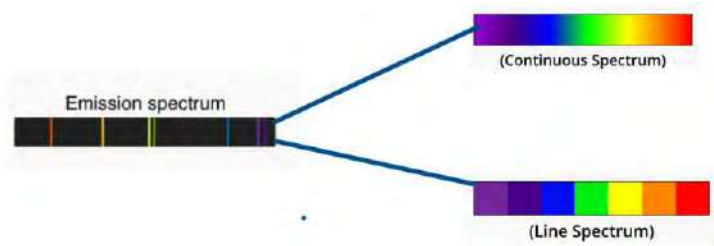


Figure: Like the bar code, which gives the identity of the product, emission spectra gives the identity of elements.

According to Bohr's atomic model, the electrons in hydrogen atoms revolve the nucleus in discrete, quantized orbit having specific energy. When high voltage is supplied across electrode in the hydrogen gas discharge tube, and direct the emitted light from the tube through a prism series of lines are observed. Depending upon the wavelength of radiation, these lines lie in visible, UV and IR regions of the electromagnetic spectrum. These lines are called **hydrogen spectra or hydrogen emission spectra**.

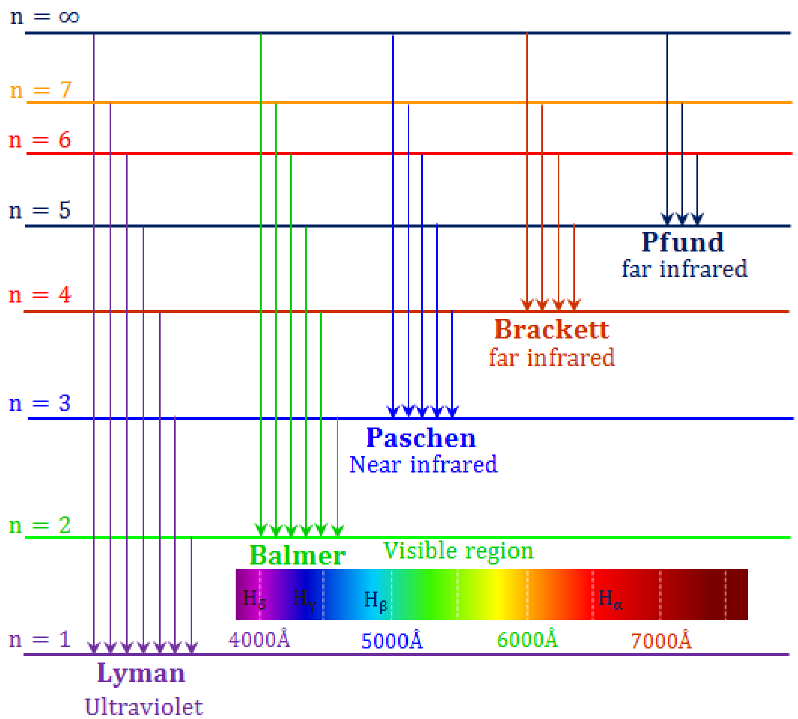


Figure: spectral series of hydrogen

The wavelength of spectral lines in the hydrogen spectrum depends upon the energy level involved in the transition. The relationship between the wavelength and the energy level was formulated by Johannes Rydberg in 1888 called Rydberg formula:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Where λ= wavelength of radiation

R_H= Rydberg constant for hydrogen= 1.09×10^7per meter

n_1=principal quantum number of lower energy level

n_2= principal quantum number of higher energy level

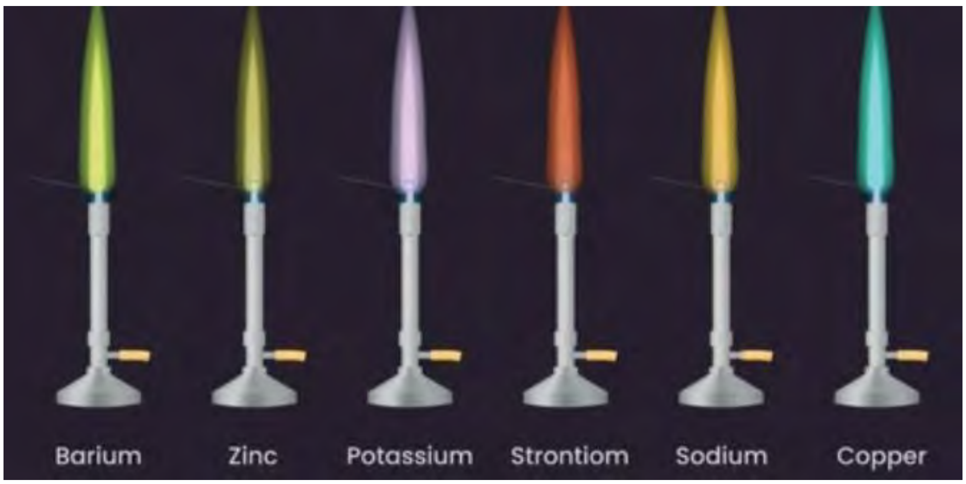
different series of spectral lines observed in the hydrogen spectrum are

a. **Lyman series**: It involves the transition from higher energy level to the 1st shell. It is observed in the UV region.
b. **Balmer series**: It involves the transition from higher energy level to the 2nd shell. It is observed in visible regions.

c. **Panchen series**: It involves the transition from higher energy level to the 3rd shell. It is observed in the IR region.
d. **Brackett series:** It involves the transition from higher energy level to the 4th shell. It is observed in the IR region.
e. **Pfund series:** It involves the transition from higher energy level to the 5th shell. It is observed in the IR region.

Cause of different colored flame

Salts on heating give a unique colored flame. The color of flame is used to identify the particular metal in the salt. The colors are due to the excitation of electrons in the metals by the application of heat. As the electrons come back to the lower energy level, it emits the photons of light.



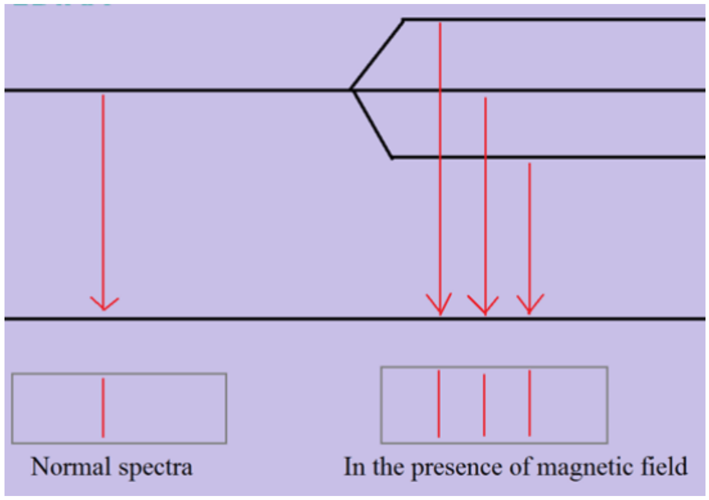
Test yourself!

Although hydrogen atom contains only one electron, why are various spectral series observed?

3.5 Defects of Bohr's theory

Though Bohr's atomic model plays a significant role in understanding the structure of atoms, there are some defects of this model.:

a. It could not explain the spectrum of atoms or ions having more than one electron like Helium, lithium ions etc.
b. It could not explain further splitting of spectral lines in the presence of magnetic field (**Zeeman effect**) and electric field (**stark effect**).



c. It could not explain wave particle duality of electrons, the concept emerged with the development of quantum mechanics. (**duality of electron** means acting as wave and particle)

d. It could not explain the theoretical explanation of **quantization of angular momentum.**

3.6 Elementary idea of quantum mechanical model

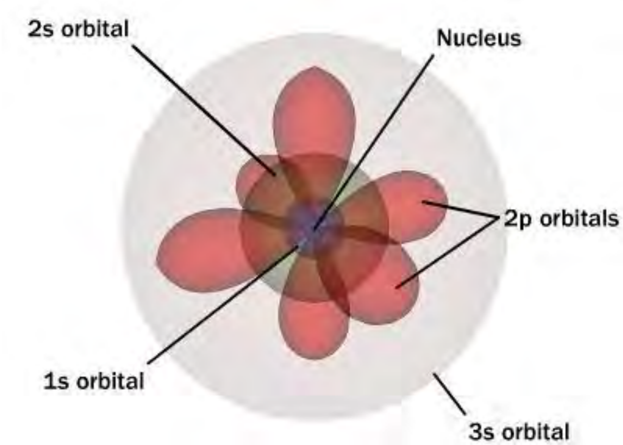
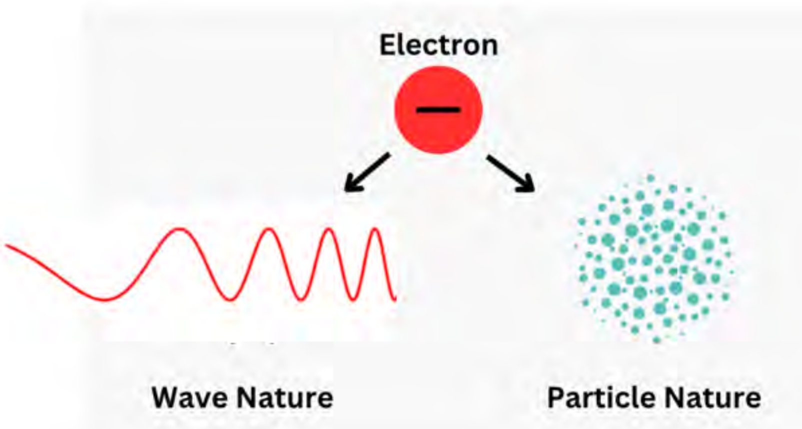


Fig: quantum mechanical model

Louis de Broglie (1892-1987) French physicist, who made ground breaking contribution to the field of quantum mechanics. He was awarded the Nobel prize in physics in 1929 for his contribution revolutionary idea of wave particle duality.

The behavior of electrons within the atom can be described by the quantum mechanical model. Unlike the earlier model, where electrons orbit the nucleus in fixed paths, the quantum mechanical model defines electrons by wave functions. There were some developments which led to the formulation of the quantum mechanical model.

de-Broglie’s wave equation



In 1900 a German physicist Max Planck studied the emission spectra of some hot material objects and concluded that light was capable of behaving both as a wave and as a particle. He proposed that the energy, E, of a single quantum (fixed amount) equals a constant time the frequency of the radiation) as

E=h ν ………..(1)

Where **h** is a Planck’s constant = $6.62607015 \times 10^{-34}$ joule-seconds

Einstein’s mass (m) energy(E) relationship can be applied to the photon when it behaves as a particle

$E=mc^2$ [c is the velocity of light]……(2)

French physicist de-Broglie in 1924 proposed a hypothesis regarding the dual nature of particle

called wave particle duality. It states that:

“a material particle whether macroscopic or microscopic in motion is also associated with wave character”

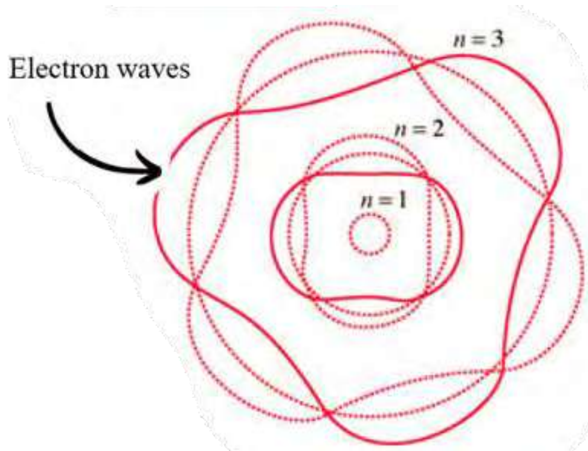
using equation 1 and 2

or, $mc^2= h\nu$

or, $mc^2=hc/\lambda$ [$\nu=c/\lambda$]

or, $\lambda = h/mc$ [c = velocity of light]

Therefore, the wavelength of moving particle (λ) is given by



$\lambda = h/mv$ or $\lambda = h/p$(momentum)

This equation is called **de Broglie's wave equation**.

The momentum and wavelength are inversely related to each other, indicating that wavelength is measurable for microscopic particles like electrons having very small mass. Very large particles have large momentum and negligible wavelength so this equation is not applicable for macroscopic objects.

Test yourself!

1. Calculate the wavelength of the electron assuming that it moves with the speed of light.
2. Can we apply the de-Broglie wave equation for the motion of planets?

3.7 Heisenberg Uncertainty Principle

Activity 3.7

a. Draw a circle on the chart paper using a marker and place two blue marbles within the circle.
b. Use a black marble to strike one of the blue marbles and try to determine the exact location of the blue marble.

Werner Heisenberg (1901-1976)
German physicist ,known for his immense contribution in the development of quantum mechanics. He was awarded to Novel prize in 1932 for the creation of quantum mechanics.

The Heisenberg Uncertainty Principle, formulated by Werner Heisenberg, it states that:

"we cannot precisely know both the position and momentum of an electron simultaneously".

Mathematically,

$$\Delta x \ \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta x \ \Delta p \geq \frac{\hbar}{2}$$

Where, Δx is the uncertainty in position

Δp is the uncertainty in momentum.

h is the Planck's constant

$\hbar$ (h-bar) is the reduced Planck's constant($h/2\pi$)

It means that the more accurately we know the position of a particle, the less accurately we can know its momentum, and vice versa. Therefore, the absolute determination of momentum and position of an electron is impossible.

Let us consider an electron moving in its own path. When the incident photon is reflected to the microscope after interacting with the electron then only the exact position of that electron can be detected. Since photons and electrons have nearly the same energy, any attempt to locate an electron with a photon will knock the electron off course, resulting in uncertainty about where the electron is located.

This principle is applicable only to microscopic particles like electrons but not for macroscopic particles because for large objects, the uncertainty in position and momentum are very small.

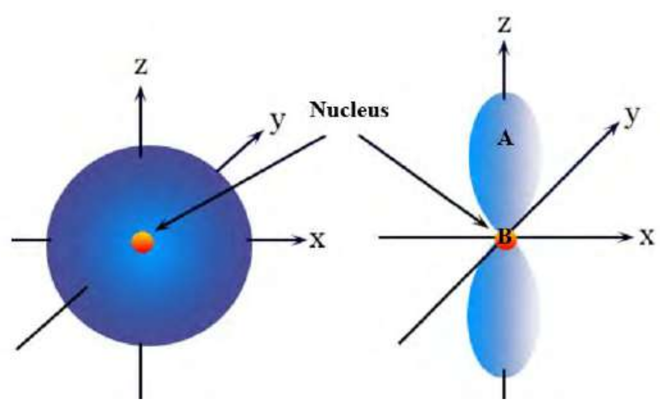
Test yourself!

Find the error in the determination of position if the momentum of the electron is measured with absolute accuracy i.e. $\Delta p=0$

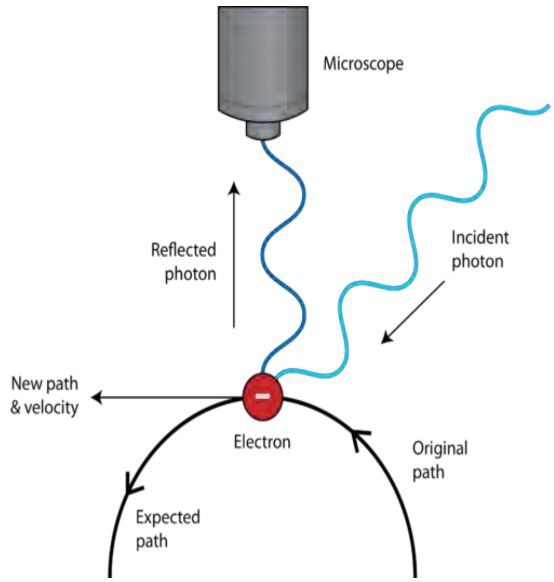
3.8 Concept of Probability

Activity:

a. Identify the orbitals given in the figure.
b. In which region (A or B) the probability of finding an electron is maximum?

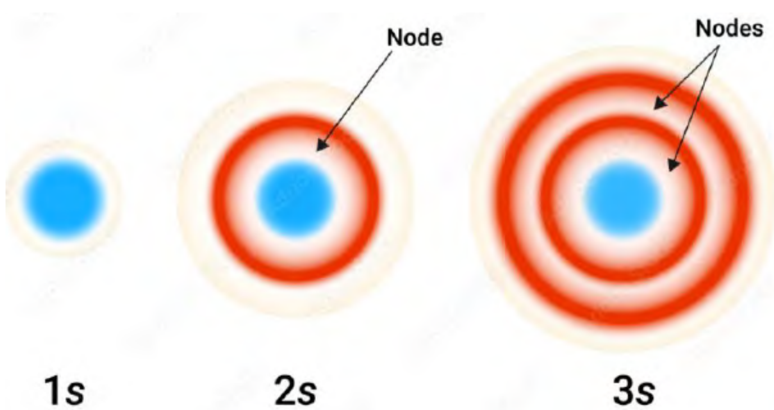


According to Bohr's atomic model, an electron moves around the nucleus in a fixed path where both the position and momentum of the moving electron can be exactly determined. But the Heisenberg Uncertainty principle challenges Bohr's model by introducing a fundamental limit in the determination of these pairs of properties (position and momentum). It means electrons are not found in specific orbit instead there is a probability distribution of electrons in a particular region.



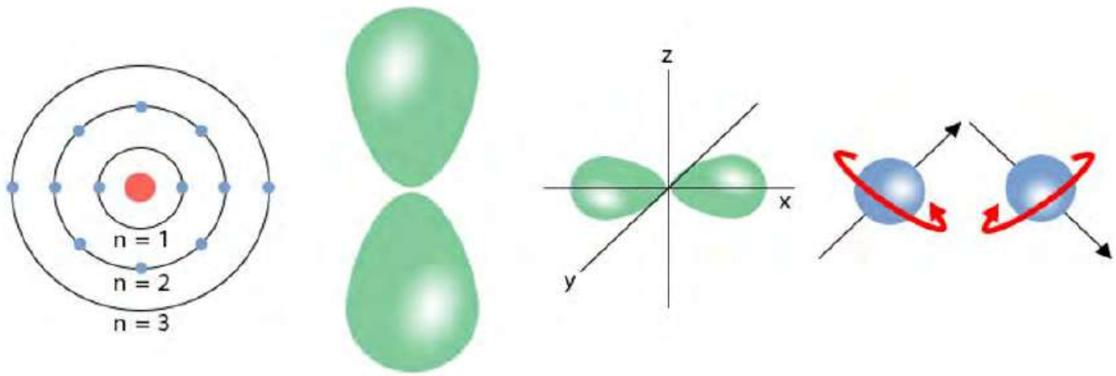
The concept of probability is fundamental and it plays a central role in understanding particles like electrons at the quantum level. In quantum mechanics, the state of a particle is described by a mathematical function called wave function (Ψ). The square of the wave function (Ψ^2) gives the probability density function, which represents the probability of finding a particle in a particular region of space called **orbitals**. Let us take an example of electron distribution in 1s, 2s and 3s orbitals in the given diagram. Node indicates the region where electron finding probability is zero.

Comparing these probabilities for 1s,2s and 3 s orbitals it is found that electron density becomes more spread out with increasing number of shell (n)



3.9 Four Quantum Numbers

Activity:



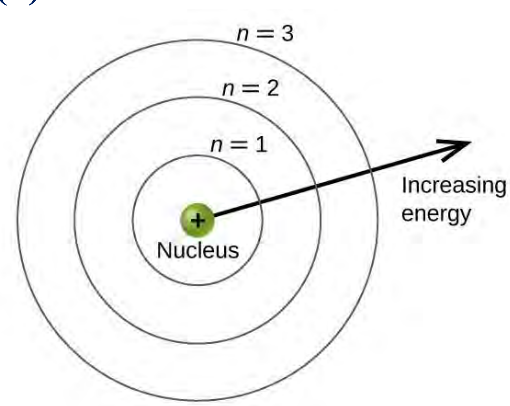
A **B** **C** **D**

Observe the figures A, B, C and D given above and identify the figure which is related to:

a. Size of the shell
b. Shape of the subshell
c. Spin of the electron

The term quantum came from the Latin word ***quantus*** meaning:" how much "or "how great". It refers to the fundamental theory to understand the behavior of matter and energy on a microscopic scale. The set of values that are used to describe the state of a quantum system, such as an atom or subatomic particles are called **quantum numbers**. The values of quantum numbers are useful not only to understand the distribution of electrons in the atom but also to define the energy, orientation and spin of electrons. There are four quantum numbers, they are:

a. Principal quantum number(n)



Observe the structure of atom given above and:

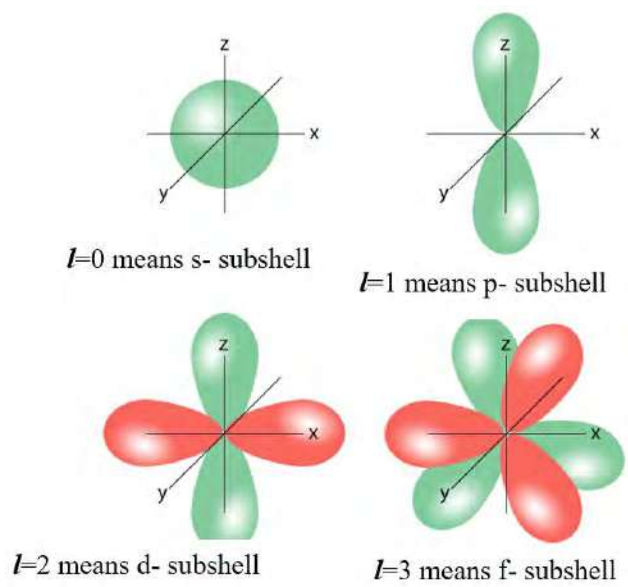
a. Identify the shell that is farthest to the nucleus.
b. Identify the shell having highest energy.

Principal quantum number was introduced by Wolfgang Pauli and developed further by others, including **Niels Bohr** in 1913.Principal quantum number denoted by '**n**' represents the major energy level or shell of an electron. It determines the **overall energy** of the electron and **the distance** of the electron from the nucleus. It can have positive integer (1,2,3 …) values. As the value of principal quantum number (n) increases, the energy of the electron and the distance from the nucleus also increases.
The maximum number of electrons in a particular shell is given by $2n^2$, where n is a principal quantum number.
For example, n=1 means K shell (1^{st} shell) can hold maximum 2 electrons

n=2 means L shell (2^{nd} shell) can hold a maximum of 8 electrons and so on.

The elements of the same period have the same value of principal quantum number. As moving down the group, the size of atoms increases due to increase in principle quantum number.

b. Azimuthal quantum number (*l*)



Azimuthal quantum number or subsidiary quantum number denoted by '*l*' was introduced by **Arnold Sommerfeld** in 1915. It determines the **shape** of the orbital in which an electron is likely to be found. The value of azimuthal quantum numbers is limited by the value of the principal quantum number. It means its value ranges from 0 to (n-1) where n is a principal quantum number. This quantum number relates to the magnitude of the electron's angular momentum within an atom so is also called **angular momentum quantum number**.
Different values of '*l*' corresponds to different orbitals with different shapes as given:

Azimuthal quantum number(*l*)	Type of orbital	Shape of orbital
0	s	Spherical
1	p	dumbbell
2	d	Complex (looks like double dumb bell)
3	f	complex

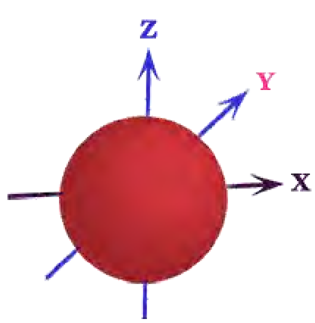
The values of '*l*' in a particular shell gives the information about the number of subshells in that shell:

Energy level	n	*l*	Number of subshells	Type of subshell
1^{st} (k shell)	1	0,	1(s -subshell)	1s
2^{nd} (L shell)	2	0,1	2(s and p subshell)	2s and 2p
3^{rd} (M shell)	3	0,1,2	3(s, p and d subshell	3s,3p and 3d

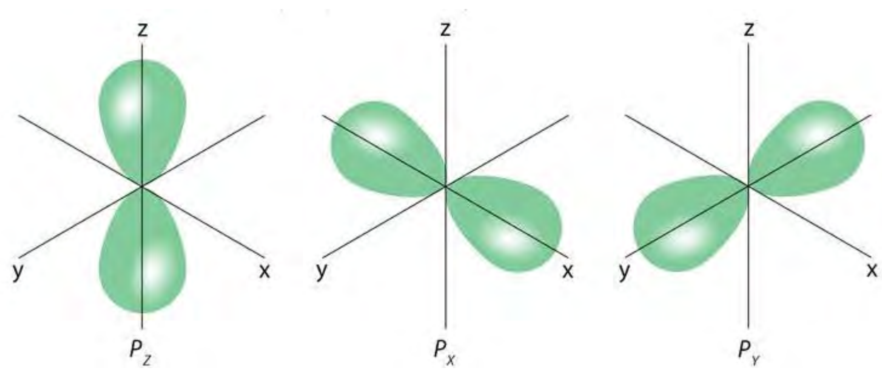
Test yourself!

If the total number of electrons in the given subshell is determined using $2(2l+1)$, find the number of electrons in s, p, d and f subshells.

c. Magnetic Quantum Number(m)



m=0 means no specific orientation of s orbital



Three values of m i.e (0,+1,- 1) means three different orientations of p-orbitals

Magnetic quantum number was also proposed by **Arnold Sommerfeld.** It is denoted by 'm' which primarily determines the number of **orbital** and **orientation** of orbitals in a given subshell. Value of the magnetic quantum number depends on the value of azimuthal quantum number(l). It means it can have integer values ranging from ***-l* to *+l***. The total number of allowed orientations(orbitals) in a given subshell is given by **2*l*+1**.

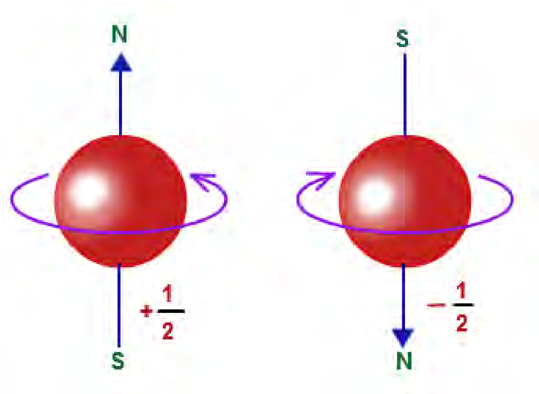
The only one value of magnetic quantum number for the s -subshell indicates the electron cloud is uniformly distributed in all directions and has no specific orientation in space. The three values of **m** for the p -subshell indicates there are three orbitals in the p-subshell and have three different orientations in the x, y and z axis.

Test yourself!

Write down the values of magnetic quantum numbers(m) for the f -subshell and find the total number of orbitals present in the given subshell.

d. Spin quantum number(s)

Activity:



Subshell	Azimuthal quantum number (*l*)	Magnetic quantum number(m)	Orientation in space or total number of orbitals(2*l*+1)
s	0	0	1
p	1	-1,0,+1	3
d	2	-2,-1,0,+1,+2	5

Identify the clockwise and anticlockwise spin in the given figure.

The spin quantum number was introduced by **George Uhlenbeck** and **Samuel Goudsmit** in 1925. Spin quantum number denoted by **'s'** describes the spin of the electron. The spin quantum number has only two values allowed: **+1/2** and **-1/2** for clockwise and anticlockwise spin respectively. In electronic configuration these quantum numbers are often represented by the arrows ↑ and ↓ to indicate the two possible spin i.e. clockwise or anticlockwise respectively. The only two values of **s (**$+1/2$ **and -1/2**) implies that there are only two electrons in each orbital and they are distinct to each other.

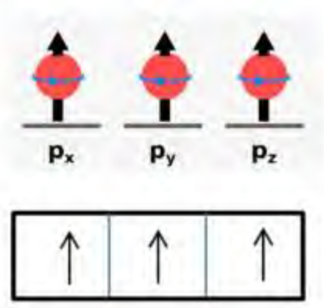


Fig: Paramagnetic

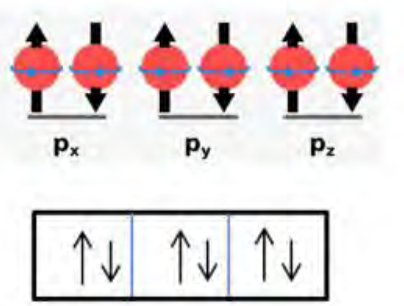


Fig: diamagnetic

The intrinsic spin of electrons contributes to the overall magnetic properties of materials. Paired electrons with opposite spins in the same orbital result in a canceling effect on magnetic moments called **diamagnetic** (repelled by external magnetic field) while unpaired electrons contribute to the overall magnetic behavior called **paramagnetic** (attracted by external magnetic field).

Solved problem

What is the total number of orbitals associated with the principal quantum number n =3?

Solutions,

For n = 3, the possible values of l are 0(s), 1(p) , and 2(d)

Values of m= 0, -1,0,+1 and -2,-1,0,+1,+2

Total values of m indicate number of orbitals in the given shell, therefore there are total 9 orbitals associated with principal quantum number(n)=3

Project work

Take a chart paper or A_4 size paper and make a column as shown below and complete the table. Use different colors for different sets of quantum numbers.

Principal quantum no.(n)	Azimuthal quantum no.(*l*)	Magnetic quantum no.(m)	Spin quantum no.(s)	No. of orbitals	Name of orbitals	Electrons in the subshell=2(2*l* +1)	Total no.of electrons in the shell ($2n^2$)
1(1^{st} shell)	0	0	+½ -½	1	1s	2	2
2(2^{nd} shell)	0	0	+½ -½	1	2s	2	

	1	-1	+½ -½	3	2p	6	8
		0	+½ -½				
		+1	+½ -½				
3(3^{rd} shell)							
4(4^{th} shell)							

The foundation of MRI (magnetic resonance imaging) is based on the observation that, like electrons, the nuclei of many elements possess an intrinsic spin. Like electron spin, nuclear spin is quantized. For example, the nucleus has two possible magnetic nuclear spin quantum numbers +1/2 and -1/2.

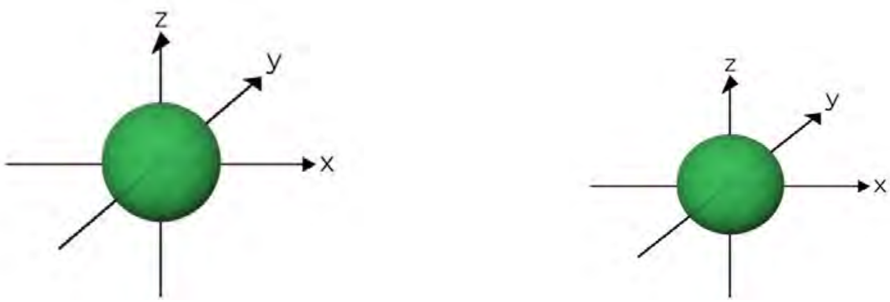
3.10 Orbitals and shape of s and p orbitals

Activity:

Identify the objects and name the shapes given in the figure.

From the above explanation of quantum number, we know that orbitals are present in the subshell where probability of finding the electron is maximum.

The **s** orbital is **spherically symmetrical**; it means the probability of finding electrons from the nucleus is equal in all directions.



1s orbital **2s orbital**

Figure: Spherical shape of s-orbital

The single orbital in the first shell is called 1s orbital and the s-orbital in the second shell is called 2s orbital. 2s orbital is larger and has more energy than 1s orbital.

The three orbitals pointing towards three axes (x. y and z) mutually perpendicular to each other are called $2p_x$, $2p_y$ or $2p_z$ orbitals, respectively.

p_x orbital has highest probability of finding electrons on or near x axis and zero probability of finding in y and z axis. Likewise, the py orbital has the highest probability of finding electrons in the y-axis and zero probability in the x and z axis.

Because of the unsymmetrical distribution of electrons in p orbitals they are like a long spongy solid cylinder having two lobes called **dumbbell shape**. The region of zero probability of finding the p-electron at the center of the atom is called **node**.

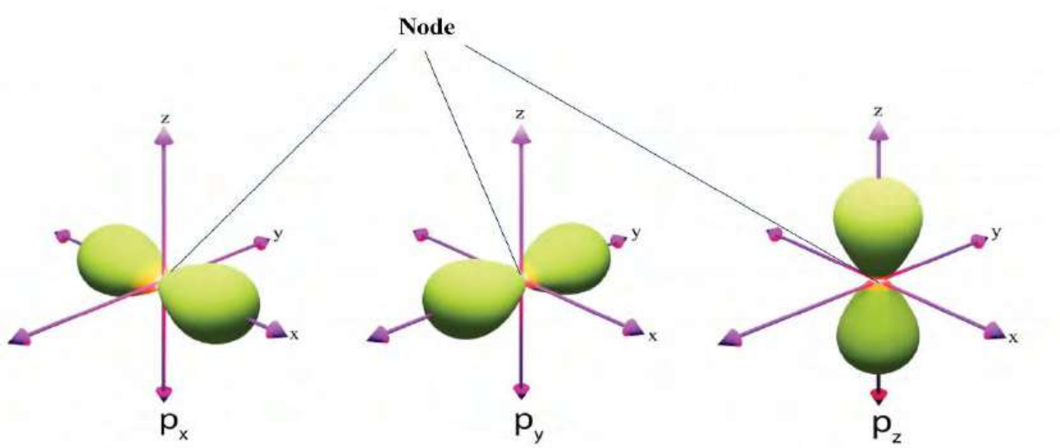


Figure: dumbbell shape of p- orbitals

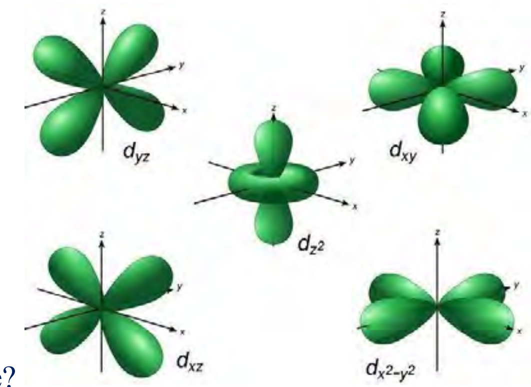
Though p_x, p_y and p_z orbitals occupy different regions of space, the electrons in these orbitals have exactly the same amount of potential energy. So, these orbitals are called **degenerate orbitals** and the phenomenon is called **degeneracy**.

What is there behind the name of s, p, d and f-orbitals?

While studying the emission spectra of atom, scientists found that some of the lines were sharp **(s)**, some were very strong and hence referred to as **principal (p)** and some of lines were rather **diffuse (d)**. However, **fundamental (f)** designation follows alphabetic order after letter **d**.

Test yourself:

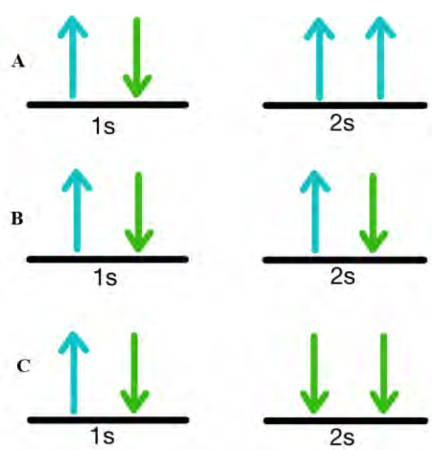
Identify the orbitals given in the figure?



3.11 Pauli's Exclusion Principle

Activity:

a. Compare the filling of electrons in the figure and find out the difference.
b. Which of the figures shows least repulsion among the electrons? Why?



From the study of quantum numbers, we know that electrons in the same orbit have the same value of principal quantum number(n) and electrons in the same subshell have the same value of azimuthal quantum number(l). Likewise, two electrons in the same orbital can have the same value of magnetic quantum number(m) but not the same value of spin quantum number(s).

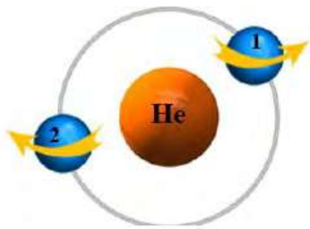
From the study of these quantum numbers, Austrian physicist Wolfgang Pauli in 1925 proposed the principle called Pauli's exclusion principle. It states that:

"**no two electrons in an atom can have the same set of all four quantum numbers**".

The word **exclusion** means there is a restriction of the same values of all four quantum numbers in the same orbitals.

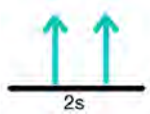
It means that two electrons in the same atom cannot occupy the same space. Two electrons in an orbital can have the same energy level, same orbital shape and orientation but have at least different spin. It means **each orbital can accommodate a maximum of two electrons with opposite spins.**

Let us consider two electrons (1 and 2) of helium atom, the four quantum numbers for these electrons are:



Electron	n	l	m	s
1	1	0	0	+1/2
2	1	0	0	-1/2

This principle ensures that electrons are distributed in the way to maintain the stability of the matter.



Same spin (unstable)

opposite spin (stable)

Test yourself!

How many electrons are there in Lithium ions? Write down all sets of quantum numbers for each electron.

3.12 Aufbau Principle

Activity:

Observe the staircase given below:

a. What is the purpose of using a staircase?
b. Why is it discomfort when climbing to the third step of the staircase directly from the ground?

Fig: orbitals with increasing order of energy

When climbing the staircase, we always start with the first step on each floor because it has the lowest energy. The same is true while filling the electrons in the orbitals. Each step represents an orbital. Lower the step, lower the energy of the orbital. Electrons require more energy to occupy orbitals on higher energy levels. **Aufbau principle**, also called building up principle, provides a systematic way to fill the electrons in the atomic orbitals of an atom. It states that:

"electrons are filled in the atomic orbitals with increasing order of their energies".

According to this principle, among the available atomic orbitals, those with lower energy are occupied before higher energy orbitals. The energy of the orbitals can be determined with the help of the sum of the principal and azimuthal quantum numbers also called the **(n+*l*) rule**.

i. Lower the value of **(n+*l*)** corresponds to lower energy.
ii. For the same value of **(n+*l*),** an orbital with lower value of n is said to have lower energy.

The (n+*l*) value of different orbitals is calculated in the table to know the order of energy as given:

Orbitals	**n**	***l***	**(n+*l*)**
1s	1	0	1
2s	2	0	2
3s	3	0	3
3p	3	1	4
3d	3	2	**5**
4s	4	0	4
4p	4	1	5

Therefore, the increasing order of energy is **1s<2s<2p<3s<3p<4s<3d<4p**…

Test yourself!

Find the values of (n+*l*) for the given orbitals and arrange them in increasing order of energy.

4d, 5s, 6s, 6p

3.13 Hund’s rule and Electronic Configurations of Atoms and Ions

Activity:

Take a look inside the bus.

a. How are the seats occupied in the bus?
b. What happens when a new passenger steps on board?

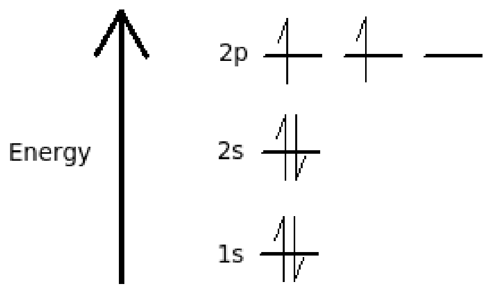


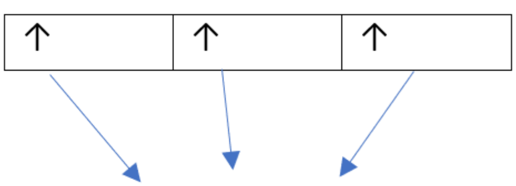
Figure: Filling of electrons in degenerate orbitals

According to the Aufbau principle, electrons are filled in the orbitals in increasing order of energy of orbitals. But it does not explain about the filling of electrons in the orbitals of equivalent energy (called **degenerate orbitals).** German physicist Friedrich Hund in 1925 put forward a rule regarding the filling of degenerate orbitals called Hund’s rule. It states that:

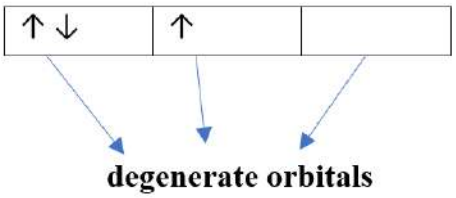
“**electrons will occupy degenerate orbitals singly before pairing up**”.

In other words, when electrons are single in the orbital their spin is parallel and becomes antiparallel when there is a pairing of electrons.

Maximum multiplicity (number of unpaired electrons =3)
Less repulsion
More stable
More symmetrical



degenerate orbitals



Less multiplicity (number of unpaired electrons =1)
More repulsion
Less stable
Less symmetrical

The number of unpaired electrons in a degenerate orbital is called **multiplicity**. According to Hund's Rule, filling of electrons singly maximizes the number of unpaired electrons. So, this rule is also called **Hund's rule of maximum multiplicity.** When electrons are unpaired, the repulsion between the negatively charged electrons decreases and stability increases**.**

For example, consider the p orbitals at a given energy level. There are three **degenerate p orbitals (p_x, p_y, p_z).** According to Hund's rule, each orbital is singly occupied by electrons with parallel spins so that it has maximum multiplicity and higher stability. Filling of electrons in the orbital using Hund's rule and the number of unpaired electrons in the atoms of some elements are given in the table:

Element	Symbol and configuration (Hund's rule)		1s	2s	2p	No. of unpaired electron (multiplicity)
Lithium	Li	$1s^2 2s^1$	↑↓	↑	[ ] [ ] [ ]	1
Beryllium	Be	$1s^2 2s^2$	↑↓	↑↓	[ ] [ ] [ ]	0
Boron	B	$1s^2 2s^2 2p^1$	↑↓	↑↓	[↑] [ ] [ ]	1
Carbon	C	$1s^2 2s^2 2p^2$	↑↓	↑↓	[↑] [↑] [ ]	2
Nitrogen	N	$1s^2 2s^2 2p^3$	↑↓	↑↓	[↑] [↑] [↑]	3
Oxygen	O	$1s^2 2s^2 2p^4$	↑↓	↑↓	[↑↓] [↑] [↑]	2
Fluorine	F	$1s^2 2s^2 2p^5$	↑↓	↑↓	[↑↓] [↑↓] [↑]	1
Neon	Ne	$1s^2 2s^2 2p^6$	↑↓	↑↓	[↑↓] [↑↓] [↑↓]	0

Test yourself!

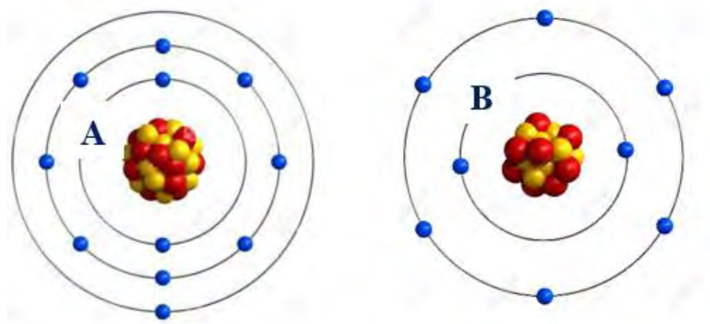
Find the multiplicity in sodium and chlorine atom

Electronic configuration

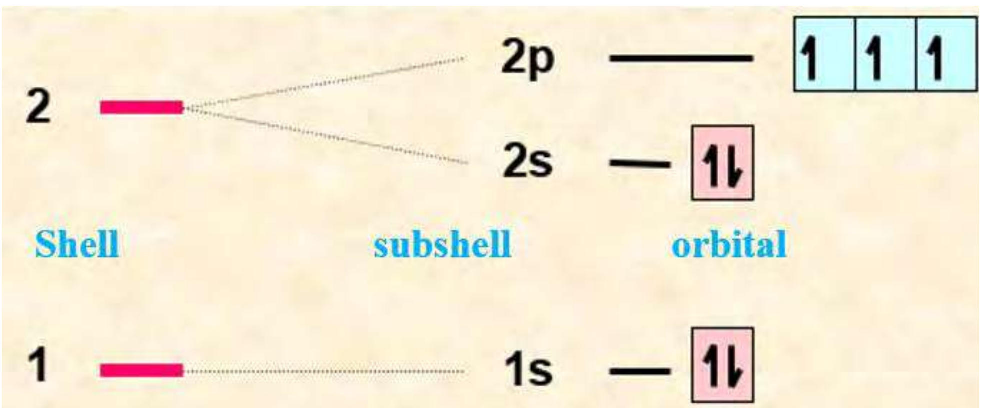
Activity:

Take a look at the atomic models of two atoms, **A** and **B**, provided.

a. Identify these atoms with reason.
b. What is the reason behind the distribution of electrons separately in the shell?



Identify the shell and subshell that has the highest energy in the figure.



According to Bohr's atomic model, atom consists of positively charged nucleus surrounded by negative electrons in a fixed energy level. The farther these electrons are from the nucleus; the more energy they have. Electrons can be distributed in these energy levels, or shells, using $2n^2$ rule (n is a principal quantum number). However, this rule is not enough for the distribution of electrons in subshells. This is explained by Aufbau principle using n+*l* rule.

The distribution of electrons in different subshells of an atom is called electronic configuration.

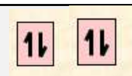
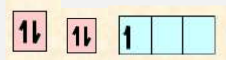
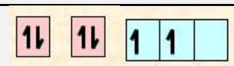
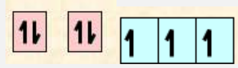
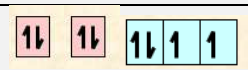
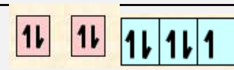
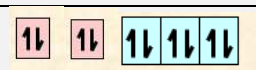
It follows the **Aufbau principle,** which says **"electrons go to the lowest energy level first**" According to the Aufbau principle, the increasing order of energy is **1s < 2s < 2p < 3s < 3p < 4s < 3d and so on.**

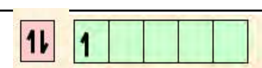
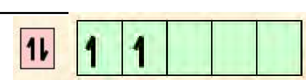
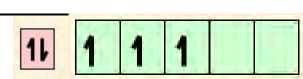
The electronic configurations of the elements correspond to their locations in the periodic table. Thus, elements in the same column of the table have related outer-shell (valence) electron configurations as shown above.

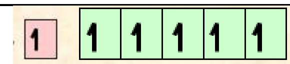
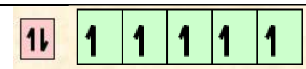
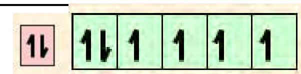
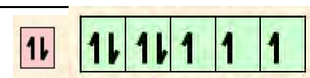
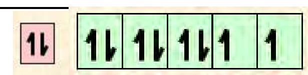
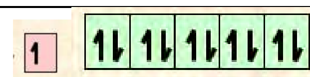
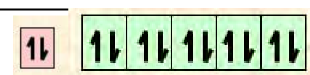
When filling orbitals with the same energy, it follows **Hund's rule**. It means putting one electron in each orbital before pairing them up. Take nitrogen, for example. Its electronic configuration is $1s^2\ 2s^2\ 2p^3$.

Instead of writing out the entire electronic configuration for **sodium ($1s^2\ 2s^2\ 2p^6\ 3s^1$),** we can use the noble gas notation and write it as **[Ne] $3s^1$.** This is called **shorthand notation** for electronic configuration or **condensed electronic configuration**. Electronic configuration of some elements is given in the table:

Element	**Electronic configuration**
Hydrogen	$\boxed{\uparrow}$ 1s1
Helium	$\boxed{\uparrow\downarrow}$ 1s2
Lithium	$\boxed{\uparrow\downarrow}$ $\boxed{\uparrow}$ 1s2 2s1

Beryllium	 1s2 2s2
Boron	 1s2 2s2 2p1
carbon	 1s2 2s2 2p2
Nitrogen	 1s2 2s2 2p3
Oxygen	 1s2 2s2 2p4
Fluorine	 1s2 2s2 2p5
Neon	 $1s^2$ $2s^2$ $2p^6$

Elements	**Electronic configuration (subshell)**	**Electronic configuration (orbital)**
Scandium (21)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$	 [Ar] $4s^2$ $3d^1$
Titanium (22)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$	 [Ar] $4s^2$ $3d^2$
Vanadium (23)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$	

		[Ar] $4s^2$ 3d3
Chromium (24)	**$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$**	 **[Ar] 4s1 3d5**
Manganese (25)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^5$	 [Ar] 4s2 3d5
Iron (26)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$	 [Ar] 4s2 3d6
Cobalt(27)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^7$	 [Ar] 4s2 3d7
Nickel(28)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$	 [Ar] 4s2 3d8
Copper (29)	**$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$**	 **[Ar] 4s1 3d10**
Zinc (30)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}$	 [Ar] 4s2 3d10

Some of the elements do not follow the Aufbau principle. For example, electronic configuration of **Chromium** and **Copper** are different from expected configuration. In these cases, one of the 4s electrons is promoted into the 3d to achieve a more stable arrangement of lower energy.

Electronic configuration of ions

Electronic Configuration of ion is similar to that of atom**.** Electrons are always lost by an atom from the highest energy level.

Examples: sodium atom (Na) $1s^2\ 2s^2\ 2p^6\ 3s^1$

sodium ion (Na^+) $1s^2\ 2s^2\ 2p^6$ [1 electron is removed from the 3s]

chlorine atom (Cl) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

chloride ion (Cl^-) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ [1 electron is added to the 3p] d-block Transition metals usually exhibits multiple oxidation states. The reason is their electronic configuration and tendency to lose or gain electrons to achieve a stable configuration. For example, Different oxidation states of Titanium are possible as:

Ti $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$

Ti^+ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^2$

Ti^{3+} $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1$

Ti^{4+} $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

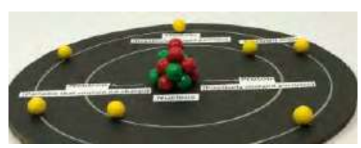
Though 4s has lower energy than 3d, electrons in the 4s orbital are removed before any electrons in the 3d orbitals due to less ionization energy of 4s electron than 3d.

Test yourself!

a. Write the electronic configuration of cuprous ion (Cu^{+1}) and cupric ion (Cu^{++}).
b. Find the number of unpaired electrons in each ion.

Project work

1. Use the materials such as (thread, cardboard, rubber, wire, table tennis ball, clay) and make a model as shown in the diagram.
    a. Rutherford atomic model
    b. Bohr's atomic model



History of development of atomic model

Leucippus and Democritus: The concept of atoms dates back to ancient Greece. Leucippus and his student Democritus proposed that everything is composed of indivisible particles called "atomos," laying the groundwork for the atomic theory.

Dalton's Atomic Theory: In the early 19th century, John Dalton formulated the atomic theory, stating that elements are composed of indivisible atoms, each with a specific mass. While some aspects of Dalton's theory were refined, the idea of indivisible atoms persisted.

J.J. Thomson's Discovery of the Electron: In 1897, J.J. Thomson discovered the electron through experiments with cathode rays. This discovery challenged the idea of indivisible atoms and led to the development of the "plum pudding" model.

Rutherford's Nuclear Model: In 1911, Ernest Rutherford proposed that an atom has a small, dense nucleus surrounded by electrons. His gold foil experiment demonstrated the existence of a concentrated, positively charged nucleus.

Bohr's Quantized Orbits: Niels Bohr introduced the idea of quantized electron orbits in 1913. This explained the stability of certain electron orbits and successfully accounted for the spectral lines of hydrogen.

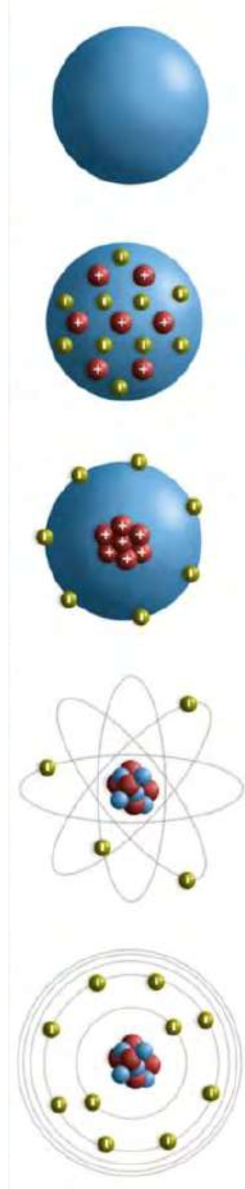
Quantum Mechanics: In the 1920s, quantum mechanics emerged as a more comprehensive theory to describe the behavior of particles at the atomic and subatomic levels. Wave-particle duality, Heisenberg's uncertainty principle, and the Schrödinger equation became fundamental concepts.

Electron Cloud Model: The modern atomic model is often represented as an electron cloud surrounding a nucleus. This model, based on quantum mechanics, describes the probability of finding electrons in specific regions around the nucleus.

Discovery of Neutrons: James Chadwick discovered neutrons in 1932, completing the picture of the atomic nucleus. Neutrons, along with protons, form the nucleus, and electrons orbit the nucleus.

Standard Model of Particle Physics: The Standard Model, developed in the 20th century, is a comprehensive theory describing the electromagnetic, weak, and strong nuclear interactions among subatomic particles.

Advancements in Imaging: With technological advancements, scientists can now visualize individual atoms using techniques like scanning tunneling microscopy (STM) and transmission electron microscopy (TEM).



Exercise

A. Multiple choice questions

1. Which of the following statements best describes the Pauli Exclusion Principle?

a. Electrons in the same atom must have opposite charges.

b. No two electrons in an atom can have the same set of quantum numbers.

c. Electrons in the same energy level must have the same spin.

d. Electrons in the outermost shell determine an element's chemical properties.

2. Of the following pairs of species, which one will have the same electronic configuration?

a) Li^+ and Na^+

b) He and Ne^+

c) H and Li

d) C and N

3. Which of the following principles was used by Bohr to explain his atomic model?

a) Conservation of linear momentum

b) The quantization of angular momentum

c) Conservation of quantum frequency

d) Conservation of mass

4. Assuming equal velocity, which of the following has the highest de-Broglie's wavelength?

a. NH_3

b. CO_2

c. H_2S

d. SO_2

5. Half-filled and fully filled orbitals are extra stable due to

a. Greater repulsion

b. Greater attraction

c. Greater symmetry

d. Greater energy

6. Which of the following orbital is filled last in the electronic configuration?

a. 3p

b. 4p

c. 5s

d. 3d

7. An electronic configuration representing an atom in the excited state is

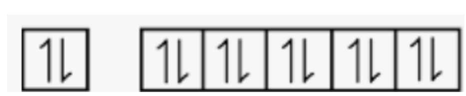
a. $1s^2\ 2s^2\ 2p^1$

b. $1s^2\ 2s^2\ 2p$

c. $1s^2\ 2s^2\ 2p^5$

d. $1s^2\ 2s^2\ 2p^2\ 3s^1$

8. The valence shell electronic configuration of one of the elements of d- block metal is given, identify the metal

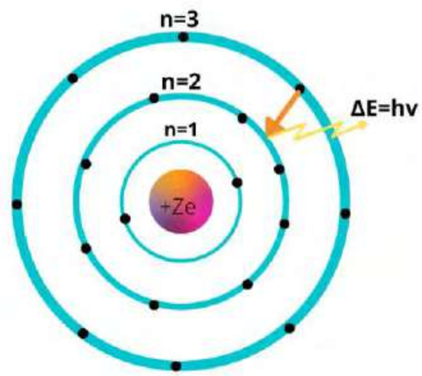


a. Fe

b. Co

c. Cr
d. Zn

9. In which region, the emission spectra are observed in the figure.



a. UV
b. IR
c. Visible
d. Far IR

10. What does the magnetic quantum number (m) indicate?
a. The size of the orbital
b. The shape of the orbital
c. The orientation of the orbital
d. The spin of the electron

11. In the quantum number set (n=3, *l*=2, m=-1), which orbital does it represent?
a. 3p
b. 3s
c. 3d
d. 3f

12. According to Bohr, what happens when an electron transition occurs from a higher energy level to a lower one?
a. It absorbs energy
b. It emits energy
c. It remains at the same energy level
d. Its mass increases

13. What is the total number of electrons in the given configuration having azimuthal quantum number (l) =0

a. 10
b. 8
c. 6
d. 4

B. Short answer questions

14. Rutherford atomic model is based on his famous experiment called alpha ray scattering experiment.
a. Why is this experiment called the alpha ray scattering experiment?
b. What are the observations made in this experiment?
c. Write any three conclusions of these observations.

15. Write the answer to the following questions.
a. What was the major limitation of the Rutherford atomic model?
b. Write any three postulates of Bohr's atomic model.
c. How did Bohr explain the stability of atom?

16. Write the answer to the following questions.

a. Define orbit. How many orbits are possible around the nucleus of an atom?

b. What is the relationship between the distance and energy of the orbit?

c. Why are orbits called stationary energy levels?

17. One of the success of Bohr's atomic model was the explanation of hydrogen spectra which could not be explained by Rutherford atomic model

a. What are similarities and dissimilarities in these two models? Write any two of each.

b. Write the postulate of Bohr's atomic model which explains the cause of hydrogen spectra.

c. Hydrogen spectra are called emission spectra, why?

d. Name the spectral series of hydrogen and the region they are observed.

18. Give reasons:

a. Electrons do not jump to the nucleus.

b. 4p has greater energy than 3s.

c. Bohr's atomic model is against the uncertainty principle.

d. Dual nature of particles is valid only for microscopic objects.

e. Ground state configuration of the nitrogen atom has 3 unpaired electrons.

19. State and explain the following:

a. Hund's rule of maximum multiplicity

b. Heisenberg Uncertainty principle

c. de-Broglie's wave equation

d. Dual nature of electrons

20. Differentiate the following:

a. Orbit and orbital

b. Principal and azimuthal quantum number

c. Lyman series and Balmer series

d. s and p orbitals

21. Electronic configuration of an element X is given

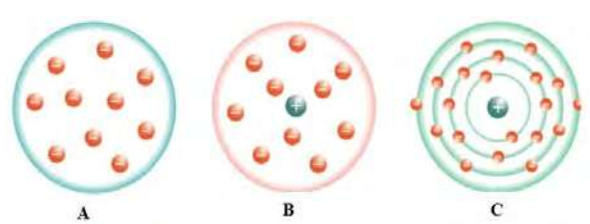
[Ar] $3d^{10}$ $4s^2$

a. Identify the number of protons and electrons in the given atom.

b. Name the group, block and period of the element X in the periodic table.

c. Write the electronic configuration of X^{++} ion and compare the size of X^{++} with X.

22. Different models of atoms proposed by different scientists are given below in figure A,B and C.



a. Identify the model proposed by Rutherford and give any two reasons behind it.

b. What is the limitation of the Rutherford atomic model?

23. The maximum number of electrons in the orbit can be calculated by $2n^2$ rule where n is a principal quantum number

a. What are the three pieces of information that can be obtained from the value of principal quantum number (n)?
b. What are the possible values of n in an atom?
c. Write the values of all quantum numbers for the electron in 3p orbital.

24. Magnetic orientation of orbitals is also explained by one of the quantum numbers.

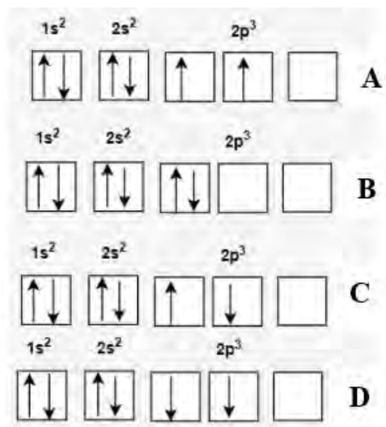
a. Name the quantum number which explains the orientations of orbitals in space.
b. Write down the value of azimuthal(l) and magnetic quantum number (m) for the p subshell and prove that p- orbitals have three different orientations
c. Draw the shape of all these three orbitals
d. Which of these orbitals have more energy and why?

25. One of the relationships which explains the quantum mechanical model of an atom is given

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

a. Name and state the principle that shows the given relationship.
b. Write the meaning of each term in the above relationship.
c. Does this principle apply for the macroscopic objects? why?

26. Hund's rule of maximum multiplicity explained the distribution of electrons in the degenerate orbitals. The four possible ways of distributing the electrons of an element is given.



a. What do you mean by degenerate orbitals? give one example of it.
b. Which of the given arrangements is the correct way of distributing electrons and why?
c. How does Pauli's exclusion principle support Hund's rule?
d. Is there any difference in the value of spin quantum number(s) for the electrons in the p subshell between C and D? Why?

C. Long answer questions

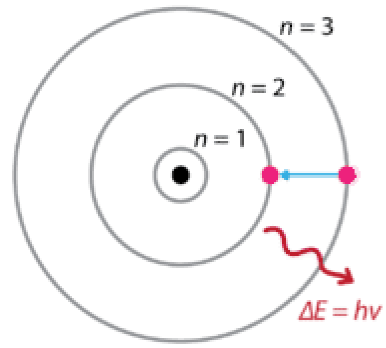
27. Electronic configuration is used to distribute the electrons in various orbitals of an atom and is based on Aufbau Principle.

a. State Aufbau principle.
b. What do you mean by (n+*l*) rule in the Aufbau principle?
c. Write the electronic configuration of chromium atoms (24) and prove that 4s orbitals have lower energy than 3d.

d. Write the values of all four sets of quantum numbers for the 15th electron of chromium atom.
e. Write the configuration of chromium (III) ion and find the number of unpaired electrons in it.

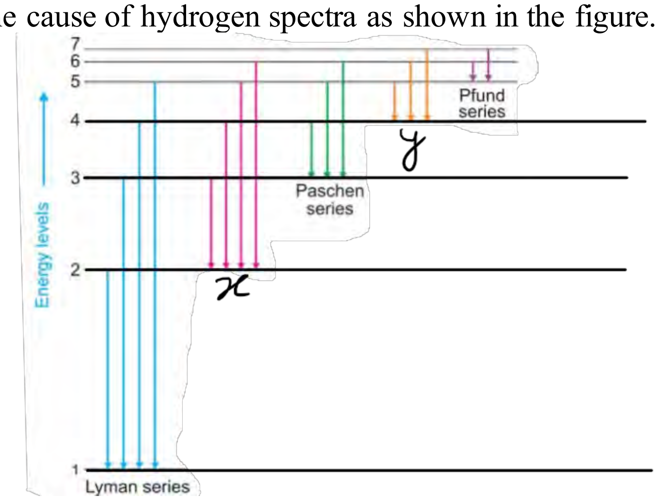
28. Bohr's atomic model defined orbit to explain the stability of an atom as shown in the figure.

a. Define the term orbit. How does it explain the stability of an atom?
b. Arrange the given orbits in figure in the increasing order of energy.
c. Name the spectral series that is observed if an electron jumps as in the figure.
d. How does this model explain the stability of an atom?

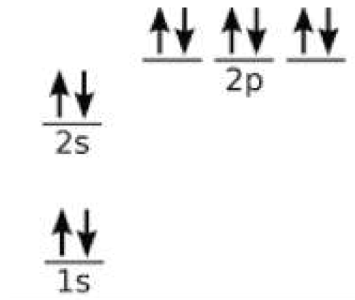


29. Bohr's atomic model explains the cause of hydrogen spectra as shown in the figure.

a. What do you mean by hydrogen spectra? How are such spectra originated?
b. Hydrogen spectra are emission spectra. why?
c. Identify x and y in the figure.
d. Name the region where x and y are observed.
e. What happens if an electron jumps from 1st orbit to 2nd orbit?



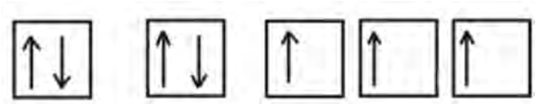
30. Electronic configuration of one of the atoms in the subshell is given



a. Name the subshell that is farthest from the nucleus.
b. Write the value of **n** and ***l*** for the second shell in the given configuration.
c. Arrange the given orbital in increasing order of energy.
d. Find the number of unpaired electrons in the given atom.
e. It is difficult to remove the valence shell electron for this atom to convert into cation. Why?

31. Some of the d- block transition metals like copper and chromium do not follow the Aufbau principle while writing the electronic configuration of an atom.
a. State Aufbau principle and give one example to prove this principle.
b. Write the electronic configuration of copper and chromium atoms. How can you justify these configurations?
c. Write the configuration of copper (II) ion and find the number of unpaired electrons in it.

32. The complete ground state electronic configuration of an atom as explained by Hund's rule is given.



a. State Hund's rule? Why is this rule called the rule of maximum multiplicity?
b. Identify the atom from the configuration given and write the value of azimuthal (*l*) and magnetic quantum number (m) for the p subshell of the given atom.
c. Write the value of spin quantum number for the electrons in 1s orbital?
d. What happens to the stability of the atom if one of the orbitals in the p subshell is paired?

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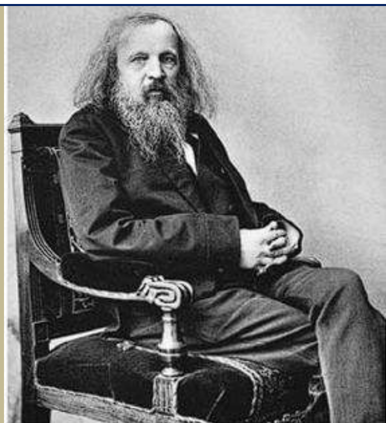
UNIT 4: CLASSIFICATION OF ELEMENTS

Activity:

Take a look at the provided periodic table and discuss:

a. Name the person who developed the table and find the elemental property that was the foundation of this table.
b. How was this table a cornerstone in chemistry?
c. What are the limitations or drawbacks associated with the table?

I	II	III	IV	V	VI	VII	VIII		
H 1.01									
Li 6.94	Be 9.01	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0			
Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5			
K 39.1	Ca 40.1		Ti 47.9	V 50.9	Cr 52.0	Mn 54.9	Fe 55.9	Co 58.9	Ni 58.7
Cu 63.5	Zn 65.4			As 74.9	Se 79.0	Br 79.9			
Rb 85.5	Sr 87.6	Y 88.9	Zr 91.2	Nb 92.9	Mo 95.9		Ru 101	Rh 103	Pd 106
Ag 108	Cd 112	In 115	Sn 119	Sb 122	Te 128	I 127			
Ce 133	Ba 137	La 139		Ta 181	W 184		Os 194	Ir 192	Pt 195
Au 197	Hg 201	Ti 204	Pb 207	Bi 209					
			Th 232		U 238				

 **Dmitri Ivanovich Mendeleev (1834–1907)** constructed a periodic table as part of his effort to systematize chemistry. He received many international honors for his work.	 In 1913 the British physicist **Henry Moseley (1887–1915)** arranged the elements in the periodic table in order of atomic number, Z, instead of atomic weight. This is the basis for the modern periodic table of elements.

In 1869 the Russian chemist Dmitri Mendeleev (1834–1907) and the German chemist J. Lothar Meyer (1830–1895), working independently, made similar discoveries. They found that when elements are arranged in order of atomic mass in horizontal rows, the elements of each vertical column have similar properties.

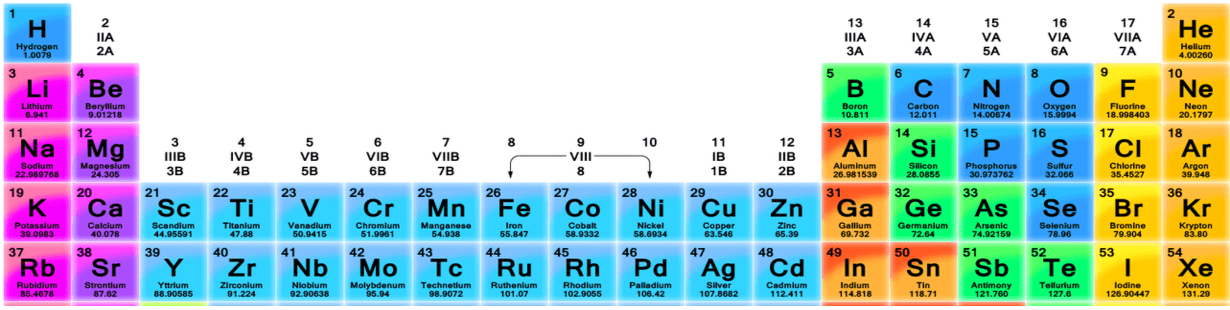
Mendeleev's periodic table, published in the late 19th century, was a significant milestone in understanding of the elements and their properties. Mendeleev's periodic table was a key step for arranging the elements which led to the development of modern chemistry. It served as a foundation upon which later scientists, such as Henry Moseley, built more accurate, systematic and comprehensive periodic tables. In addition, it further reflects our understanding of atomic structure and properties.

4.1 Modern periodic law and modern periodic table

Activity:

Observe the part of periodic table given below.

a. Name any five commonly used elements in our daily life.
b. Identify any one metal and any one non-metal from the table.



In 1913, the English physicist Henry Moseley observed and studied the X-ray spectra of various elements and concluded that atomic number is a more fundamental property of an element than atomic mass. Then, he modified the Mendeleev periodic law into a new law called modern periodic law. Modern periodic law states that;

"The physical and chemical properties of the elements are periodic functions of their atomic numbers".

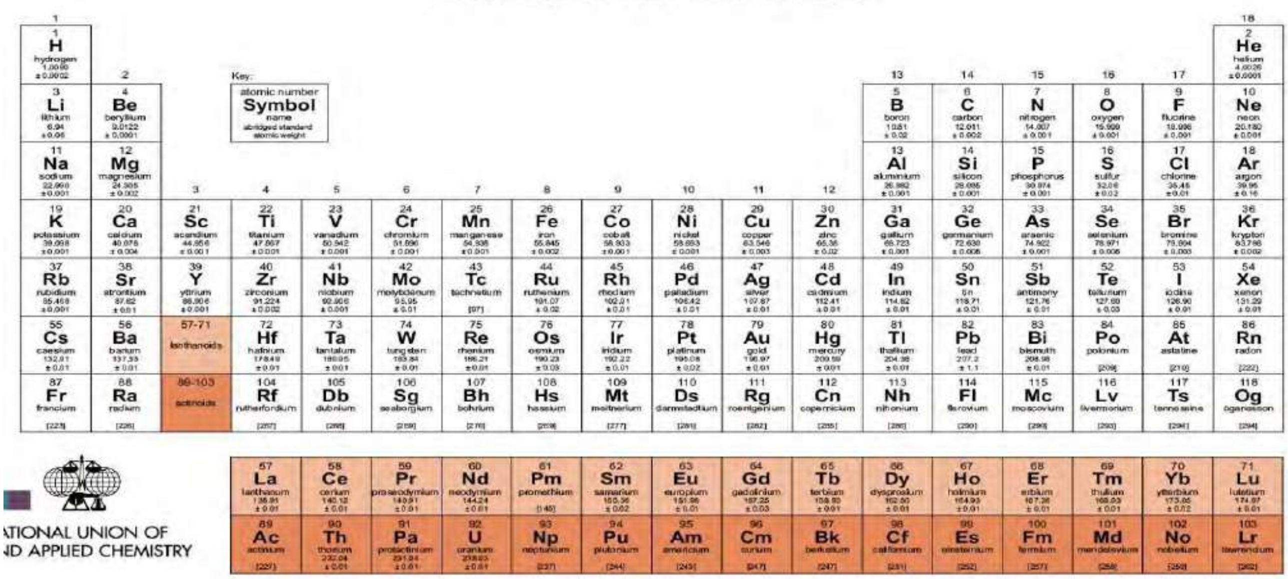
This means that when elements are arranged in order of increasing atomic number, their properties repeat in a periodic manner.

4.1.1 Classification of elements in period, group and block

Activity 4.3:

Count the number of rows and columns in the periodic table given.

IUPAC Periodic Table of the Elements



Vertical columns in the table are called **Groups**. There are 18 groups in the modern periodic table. Initially, the groups were numbered using Roman numerals I to VIII. However, the International Union of Pure and Applied Chemists (IUPAC) recommended the Arabic system of numbering from **1 to 18**. Group **1, 2, 13-17** are called **main group or representative** elements. Likewise, Group **18** elements are called Noble or Inert Gases and the elements of Group **3 to 12** are called **Transition Metals**. Main group elements are further divided into **alkali metals** (group 1), alkaline **earth metals** (group **2**), pnictogens (group 15) and chalcogens **(**group 16). Elements in the same group have the same number of valence electrons, therefore elements in the same group have similar chemical properties.

Horizontal rows in the Table are called **Periods**. There are **seven** periods in the modern periodic table which are numbered from **1 to 7**. Elements of the same period have same number of shells. The nature and properties of the elements change in a similar manner across each period.

Li [He]$2S^1$	Be [He]$2S^2$
Na [Ne]$3S^1$	Mg [Ne]$3S^2$
K [Ar] 4 S^1	Ca [Ar] $4S^2$
Rb [Kr] $5S^1$	Sr [Kr] $5S^2$
Cs [Xe]$6S^1$	Ba [Xe]$6S^2$

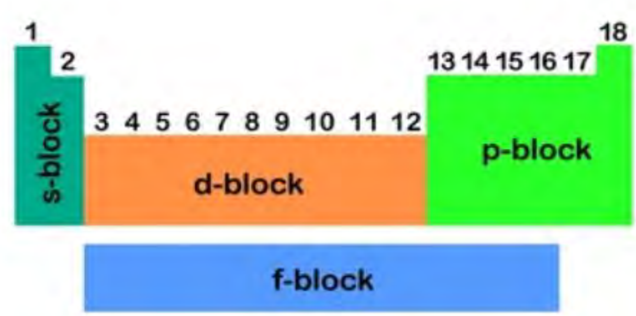


Figure*: Valence shell configuration of elements*

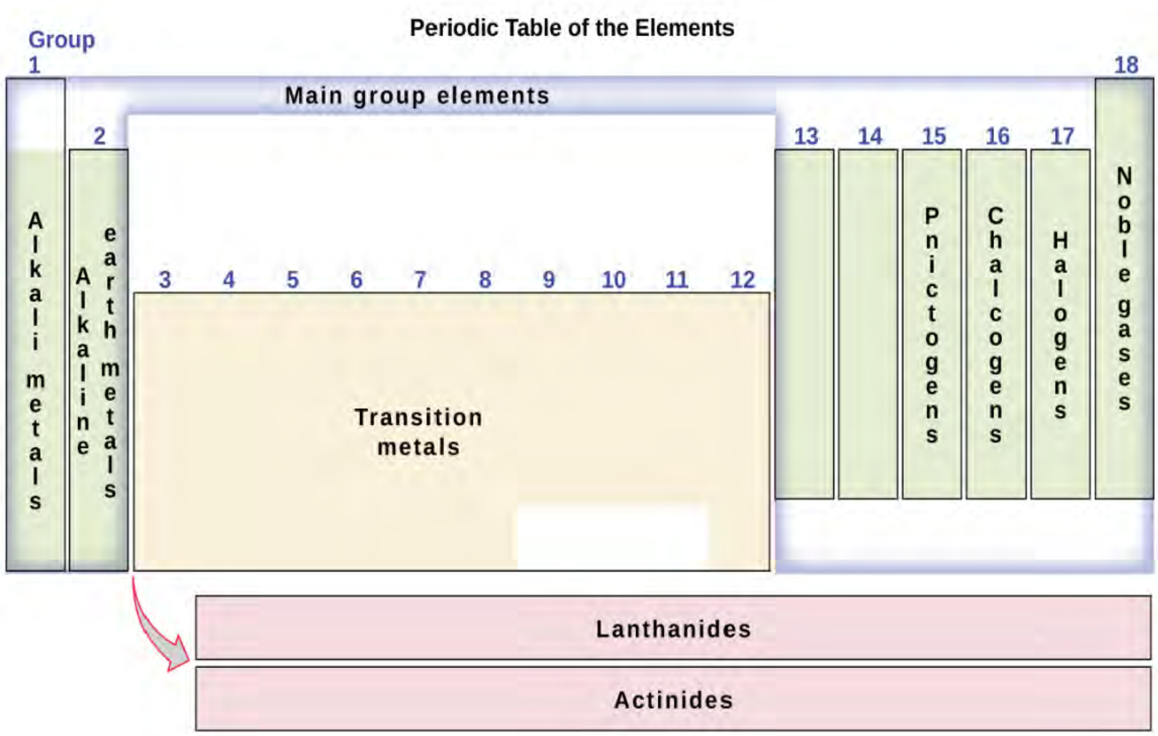
According to the type of subshell being filled, elements are further classified into four different categories called **blocks (**s, p, d and f). When a valence shell electron goes to **s** subshell, then these elements are called ***s- block elements*.** Elements of group 1(alkali metals) and 2(alkaline earth metals) are s-block elements. The general valence shell electronic configuration of these elements is ns^1 (group 1) and ns^2(group 2), where n is the principal quantum number for the valence shell.

s-block elements are on the extreme left side of the periodic table. They are very active, soft and shiny metals which can lose electrons easily and become electropositive. The general valence shell electronic configuration of s- block element is $\mathbf{ns^{1\text{-}2}}$.

p-block elements are on the right side of the periodic table. The general valence shell electronic configuration is **ns ^{2}np$^{1-6}$**. They are usually nonmetals having a tendency to gain electrons and become electronegative. s and p -block elements are collectively called ***representative or main group*** elements.

Potassium dichromate($K_2Cr_2O_7$)

potassium permanganate ($KMnO4$)



The elements in between s and p -block are called d-block elements, and are also called ***transition metals***. The general valence shell electronic configuration is ns^2 $(n\text{-}1)$ $d^{1\text{-}10}$. Metals like Fe, Cu, Ag, Au etc. are d-block transition metals which are hard, have high melting point, malleable, ductile and good conductor of heat and electricity. So, they are used in daily life for various purposes. These elements and their compounds are usually colored and are used as a catalyst.

The elements which are at the bottom of the periodic table (lanthanides and actinides) have valence electrons in the f- subshell called ***f-bloc transition metals or inner transition metals***. They are also called rare earth metals because they are rarely found on earth.

> **Do you know?**
>
> The heaviest element officially recognized on the periodic table is Oganesson (Og), with an atomic number of 118. Oganesson is highly unstable and has a very short half-life, making it extremely challenging to study.

4.1.2 Classification of elements in metals, nonmetals and metalloids

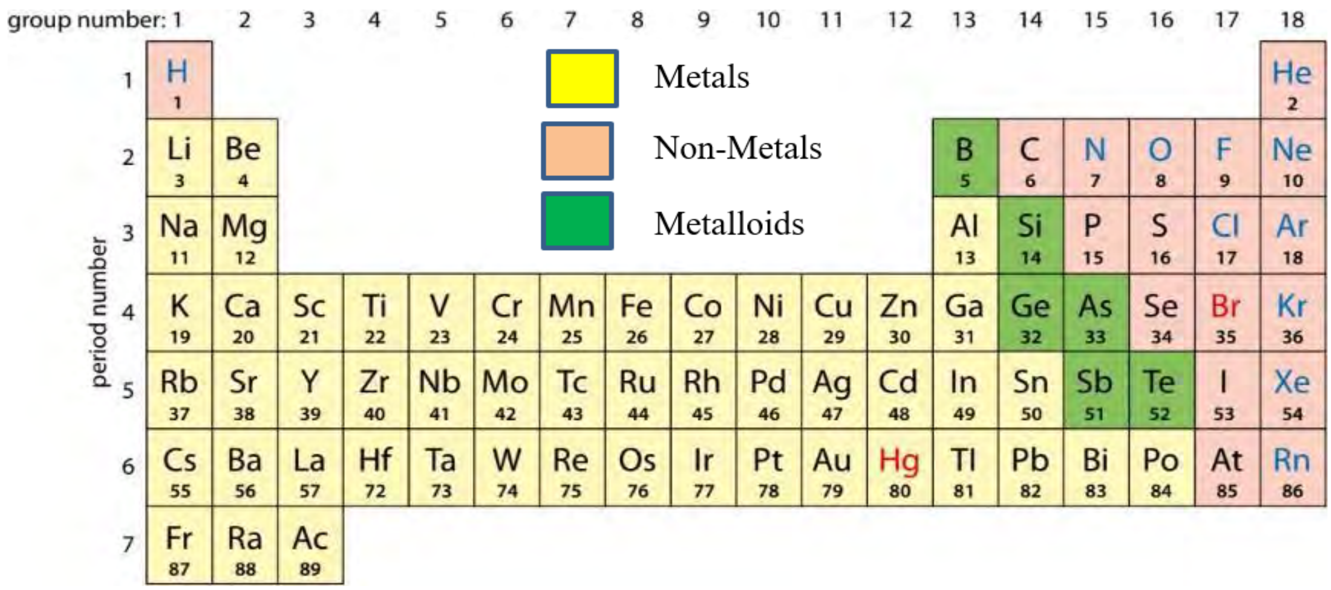
Activity: 3

a. Name any five metals and any five non-metals that are used in your daily life.
b. Locate the position of these elements in the periodic table given above.

Most of the elements in the periodic table are metals which are kept in the left and bottom of the table as shown in table below. Metals are hard, malleable, ductile and are good conductors of heat and electricity. They are good reducing agents because they can easily lose the electrons and become electropositive.

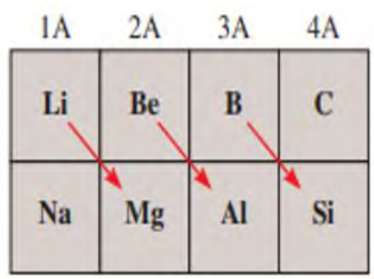
Elements in the right of the periodic table are non-metals. Non-metals exist in solid, liquid or gas states. They tend to gain electrons and become electronegative so are called oxidizing agents. There are some elements which exhibit the properties of both metals and non-metals called semi **metals or metalloids**. Metalloids like silicon (Si) and Germanium (**Ge**) are used as semiconductors in electronic devices.

> **Useful resources**
>
> http://www.periodicvideos.com/



Lithium **(Li),** a member of the alkali metal group, shares some similarities in behavior with magnesium (**Mg**), an alkaline earth metal. This similarity arises due to comparable **charge-to-size ratios** and the presence of similar outer electron configurations. This relationship is called a **diagonal relationship**. It refers to the similarity in properties between elements that are diagonally adjacent to each other.

Similarly, beryllium (**Be**) and aluminum (**Al**) also show a diagonal relationship.



Do you know?

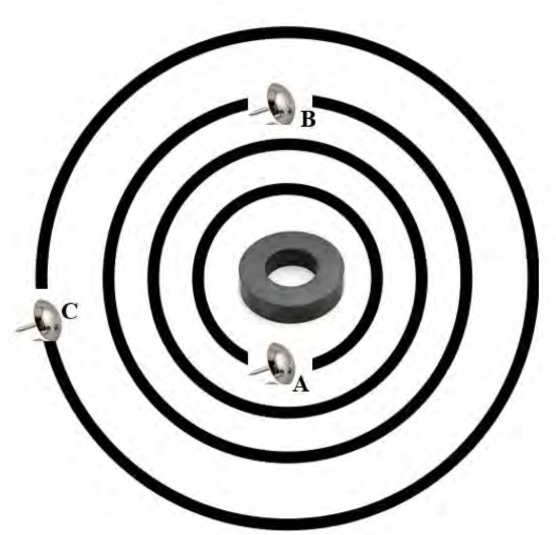
At room temperature and 1 atm pressure, 11 elements are gas (hydrogen, nitrogen, oxygen, fluorine, chlorine and inert gases), 2 elements are liquid (bromine and mercury) and the rest are solids.

Test yourself!

1. Name each of the following elements and predict whether they are metals, nonmetal or metalloid:

a. Element X is in period 4 and group 3.

b. Element Y is in period 3 and group 2.
c. Element Z is in period 5 and group 6.

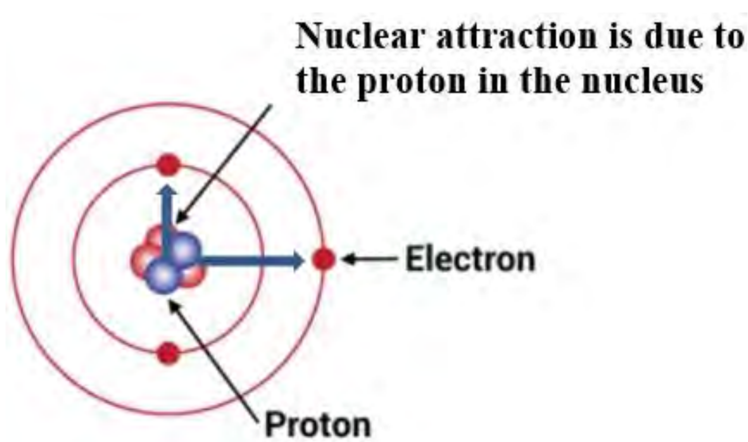


4.2 Nuclear charge and effective nuclear charge

Activity:

Consider the diagram with a circular magnet at the center and iron pins A, B, and C placed at various distances around the magnet.

Which pin experiences the least attraction from the magnet, and why?



The negatively charged electrons in the atoms are held in the orbit due to the attraction of the positively charged nucleus at the center.

The charge of the nucleus is called **nuclear charge(z).** It depends upon the number of protons present in the nucleus. As the number of protons increases in the nucleus, the nuclear charge increases. Therefore, nuclear charge increases both in group and period. The electrons in the inner shells of the atom decrease the nuclear charge experienced by a valence electron. This effect of reducing the nuclear charge experienced by the valence electron is termed **shielding or Screening effect.** Due to the screening effect of inner shell electrons, the overall nuclear charge decreases which is called effective ***nuclear charge (Z_{eff}).***

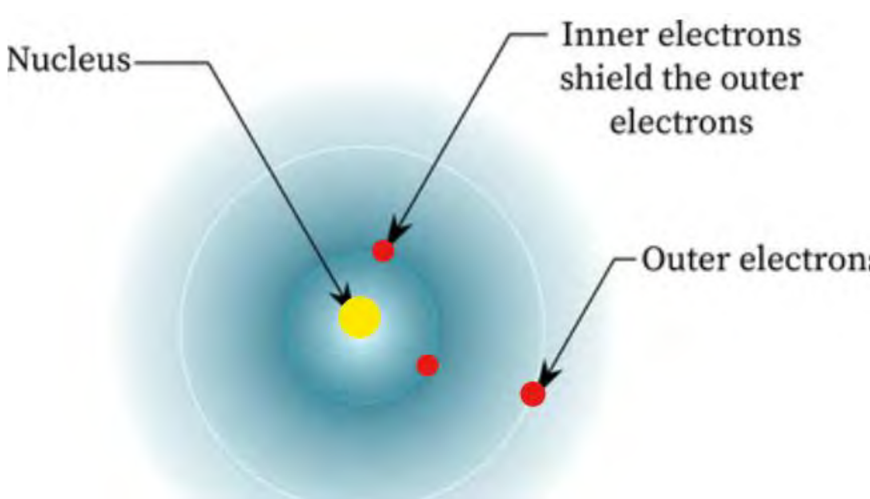
*The **effective nuclear charge is the net positive charge that an electron experiences from the nucleus.***

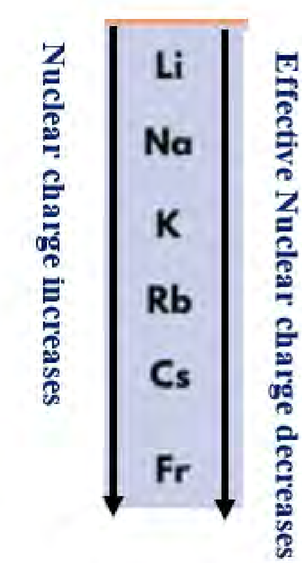
It equals to the nuclear charge but is reduced by any shielding or screening from inner shell electrons.

Z_{eff} = Z – S or σ

where Z = actual nuclear charge (atomic number) and S =screening or shielding constant

In the group moving from top to bottom nuclear charge as well as shielding increases due to increase in number of protons and number of shells respectively. But the shielding effect dominates the nuclear charge and lowers the effective nuclear charge. Therefore, the effective nuclear charge (net charge) decreases in groups. While moving from left to right in the period there is increase in nuclear charge but shielding remains nearly constant. It causes an increase in effective nuclear charge. Increasing the effective nuclear charge reduces the size of the atom by pulling the electrons inward.



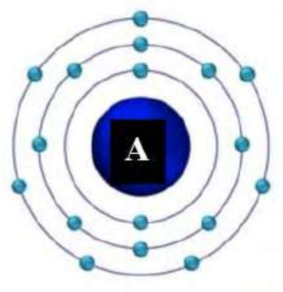


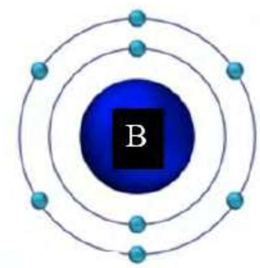
Test yourself!

Identify the element with highest nuclear charge and highest effective nuclear charge in the table given and give reason for your choice.

C 6	N 7	O 8
Si 14	P 15	S 16

4.3 Periodic trend and periodicity





Activity 4.3.2

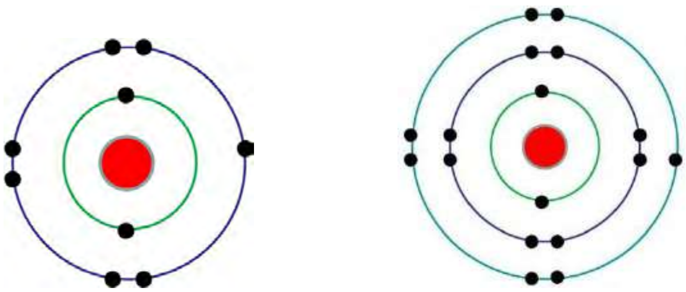
Observe the atoms A and B in the figure

a. Find the number of shells and the number of protons in each.
b. Compare the radius of these atoms.

An understanding of physical properties of a substance helps chemists identify its structure and bonding. When the elements are arranged by atomic number, their physical properties vary periodically. ***The repeating pattern or trend of properties on the periodic table is called periodicity.***

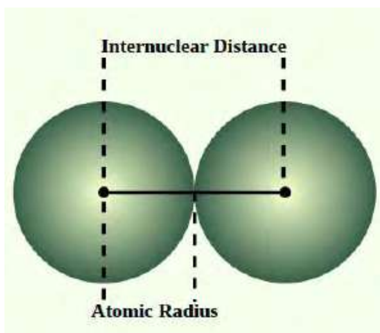
4.3.1 Atomic radius

Activity:

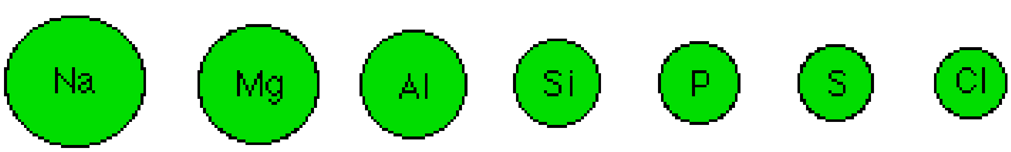


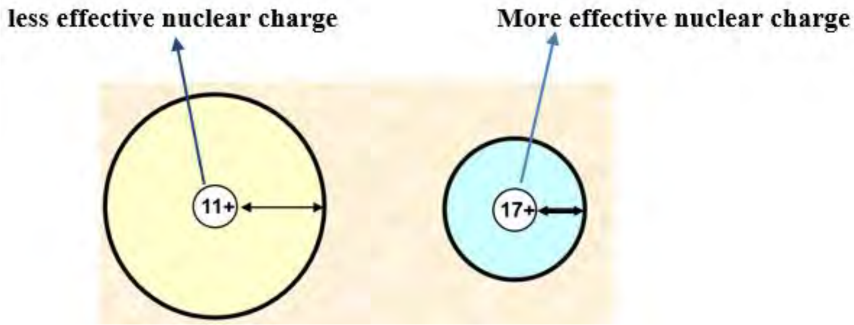
Identify the atoms given and compare their atomic radius.

The atomic radius is basically used to describe the size of an atom. The larger the atomic radius, the larger the atom. The atomic radius is usually taken to be half the internuclear distance in a molecule of the element.

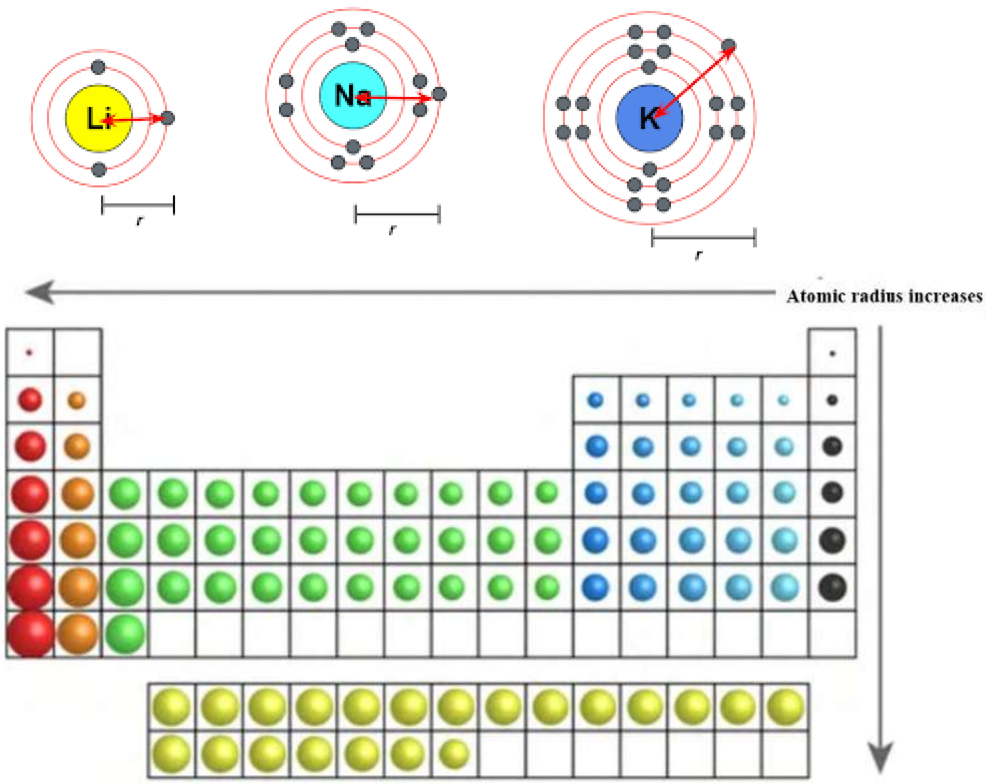


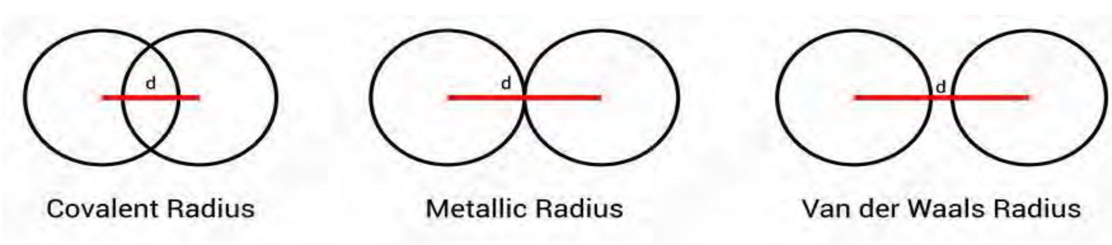
Atomic radius depends on the number of shells and effective nuclear charge of the nucleus. As increasing the ***number of shells, atomic*** radius increases, therefore chlorine atom is bigger in size than that of fluorine atom. As increasing the ***effective nuclear charge*** nuclear attraction increases which reduces the radius of the atom. Atomic radius of Chlorine is smaller than sodium because the nuclear attraction by 17 protons in chlorine is greater than 11 protons in sodium.





As moving from top to bottom in the group *atomic radius increases* due to increase in the number of shells. Whereas as moving from left to right in the period, atomic *radius decreases* due to increase in effective nuclear charge.





Depending on the nature of the bond formed by an atom, atomic radius may be covalent, metallic or Vander Waal's radius. Covalent radius is defined as half the distance between the nuclei of two atoms of the same element that are bonded together by a single covalent bond. Metallic radius is half the distance between the nuclei of two adjacent, identical metal atoms. Ionic radius is the radius of ions, it is defined as the distance from the nucleus of an ion to its outermost electron shell. In the same way van der Waals radius is defined as half the distance between the nuclei of two non-bonded atoms of the same element when they are closest to each other.

Remember: Atomic radius is usually expressed in Angstrom units(A°). It is equivalent to 1×10^{-10} meters. The atomic radius of hydrogen atom is 0.53 Å(angstrom) = 37 pm (picometre)

Inert gases (Ne, Ar etc.) do not form covalent bonds and the internuclear distance therefore cannot be measured so it is not possible to measure atomic radius for these elements but it is possible to measure a value for the Vander Waals' radius

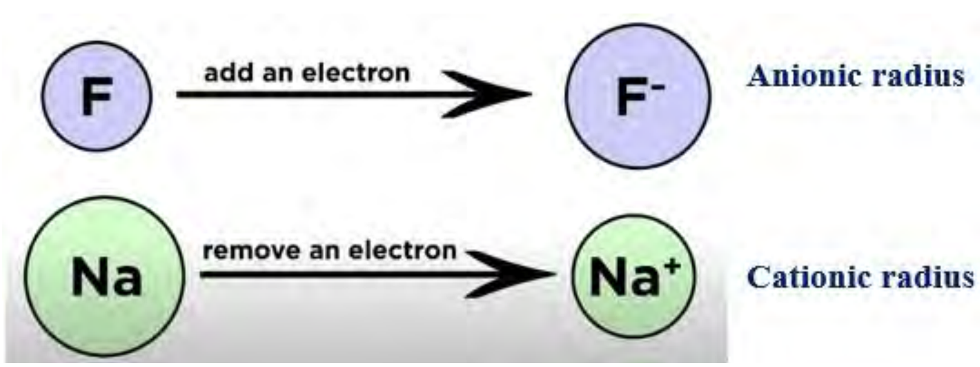
Test yourself!

Arrange the following elements in order of increasing atomic radius:
Na, Be, Mg, Cl, O, S

Atoms of oxygen, nitrogen and hydrogen are given as A, B and C in the figure
Identify the atoms and give reason for your answer.

4.3.2 Ionic radius

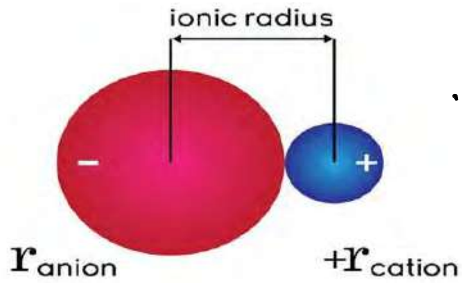
Activity:



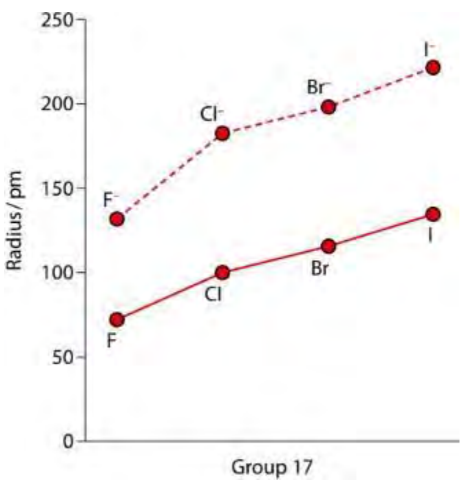
Compare the radius of:

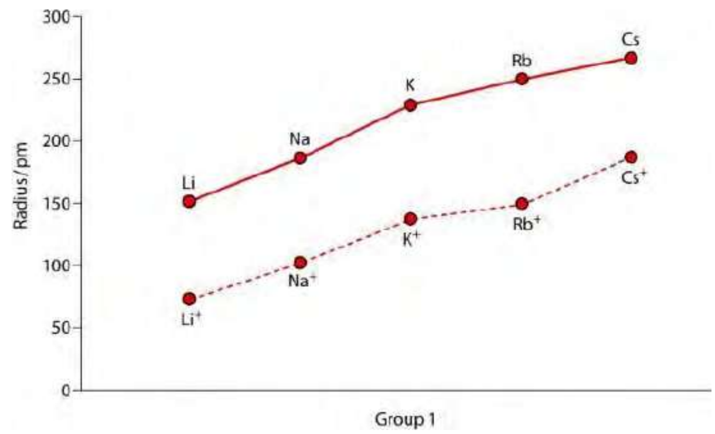
a. F and F-
b. Na and Na+

The size of an ion is measured in terms of ionic radius. Ionic radius of cation is smaller than the atomic radius due to decrease in the number of shells or increasing the effective nuclear charge of the nucleus. Likewise, the ionic radius of anion is greater than the atomic radius due to decrease in effective nuclear charge by adding extra electrons. The periodic trend of ionic radii is the same as that of atomic radii.

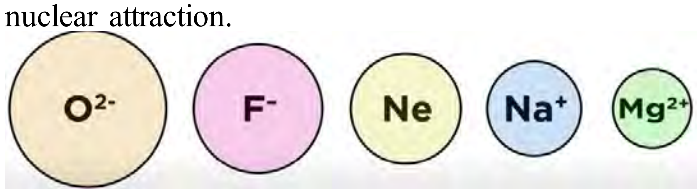


Ionic radius is the distance from the nucleus of an ion up to which it has an influence on its electron cloud.





In the case of ***isoelectronic*** species (having the same number of electrons) the size depends upon the number of protons (nuclear charge). Moreover, the number of protons in the nucleus causes stronger nuclear attraction and decreases the size of atom or ion. For example, the increasing order of size of Na^+, Ne and F^- is **Na^+ < Ne < F^-.**
They all have the same number of electrons but the number of protons in Na^+ is greater than Ne and F^- which causes the electrons are pulled in more strongly and decreases the size.
It means in isoelectronic species, as increasing the number of protons, size decreases due to increase in nuclear attraction.



8P⁺ 9 P⁺ 10 P⁺ 11 P⁺ 12 P⁺

Size decreases

Test yourself!

Fill the given table and arrange the given species in increasing order of size.

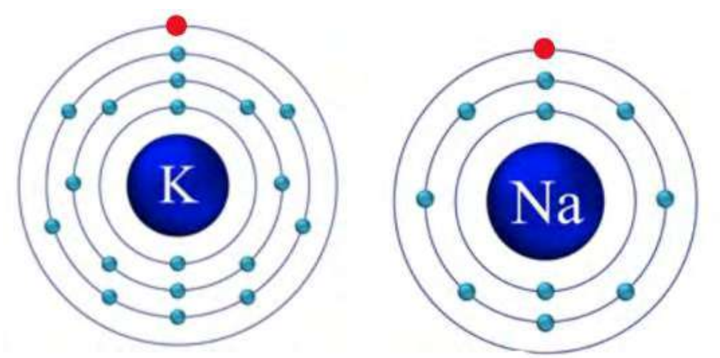
Species	Cl^-	Ar	K^+	P^{---}
No. of protons				15
No. of positive charge	17			
No. of electrons		18		

4.3.3 Ionization energy

Activity:

Observe the atomic structures of two atoms in the figure.

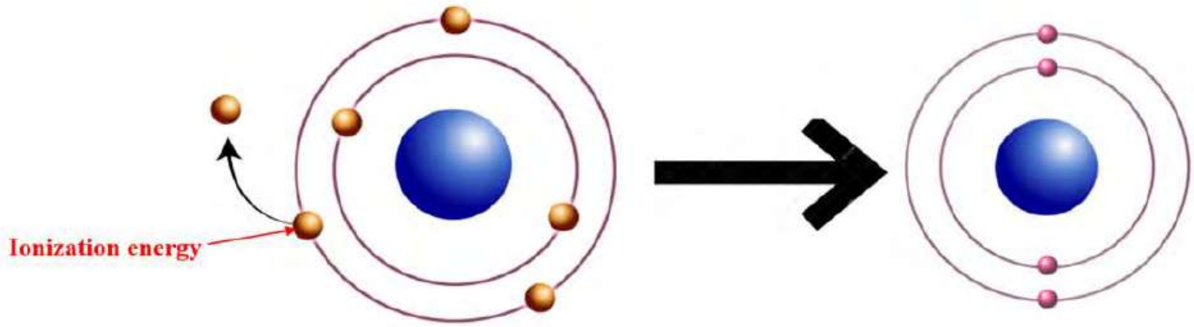
a. Find the number of protons in each.
b. If you are trying to remove the valence electron (red colour) from each atom, which one is easier to remove?



The electrons of the atom are strongly pulled by the nucleus so we need to supply energy to remove the outermost electron from the atom to make it cation (+ve ion). ***Ionization energy is the minimum amount of energy required to remove one mole of electrons from one mole of atoms in the gaseous state.***

$$Energy + Na_{(g)} \rightarrow Na^{+}_{(g)} + e^{-}$$

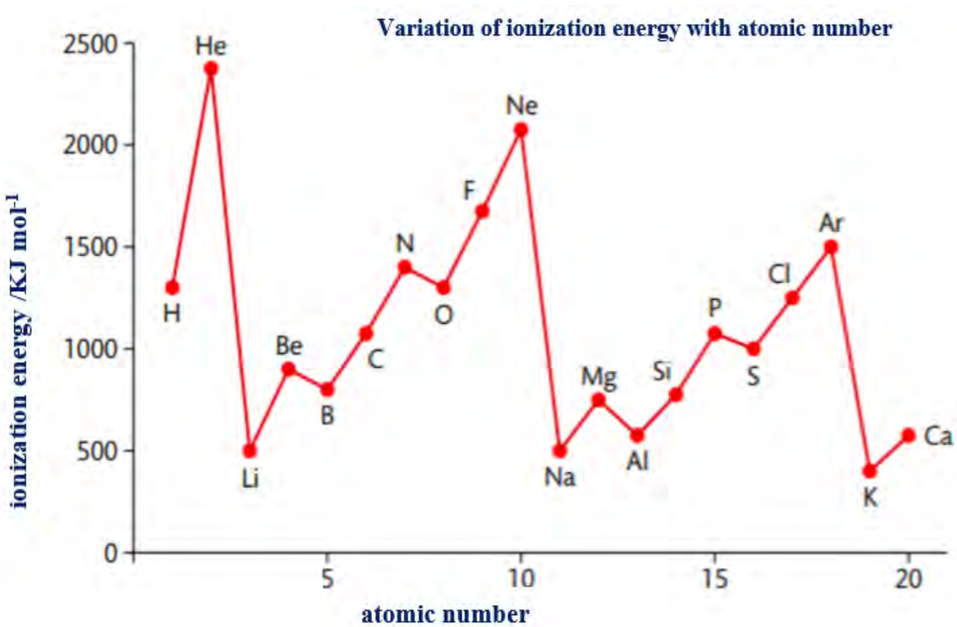
Note: Atom in the gas state is uninfluenced by its neighbors and so there are no intermolecular forces.



Ionization energy mainly depends upon the ***distance from the nucleus*** and ***effective nuclear charge*** of the nucleus.

When the nuclear dissonance increases, nuclear attraction decreases and it is easy to lose the outermost electron. It means ionization energy decreases. When effective nuclear charge increases nuclear attraction increases and it is difficult to remove the outermost electron. It means ionization energy increases.

As moving from top to bottom in the group atomic size increases, nuclear attraction decreases so ionization energy decreases. In the period moving from left to right in the table effective nuclear charge increases which makes it difficult to remove the electron and ionization energy increases.



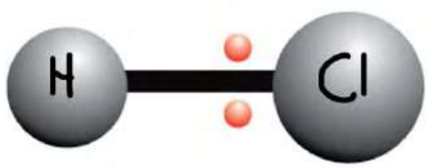
Unusual trend of ionization energy from **Be** to **B** and

N to **O** is due to the change in *electronic configuration* from stable to unstable.

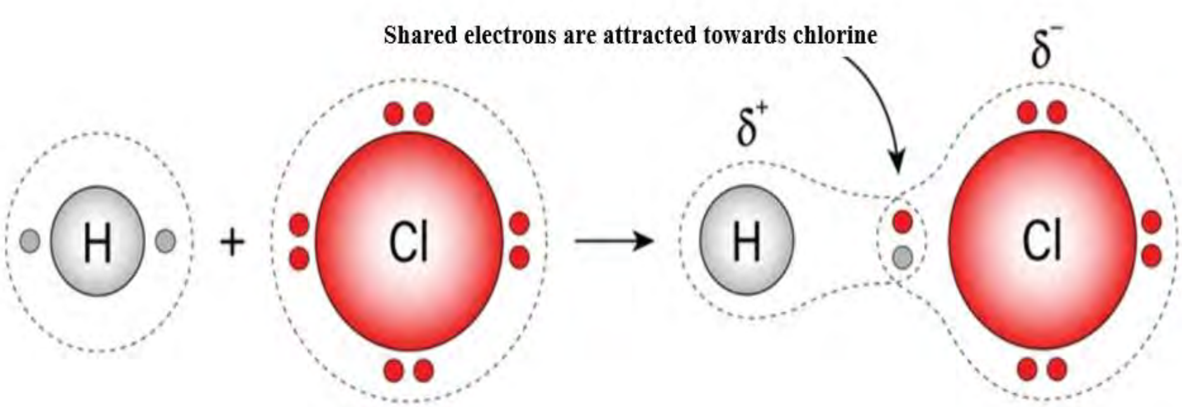
4.3.4 Electronegativity

Activity: Observe the figure given:

a. Name the game shown.
b. What is the major factor that affects the pulling?
c. Why are the electron pairs closer to chlorine than hydrogen in the HCl molecule given?

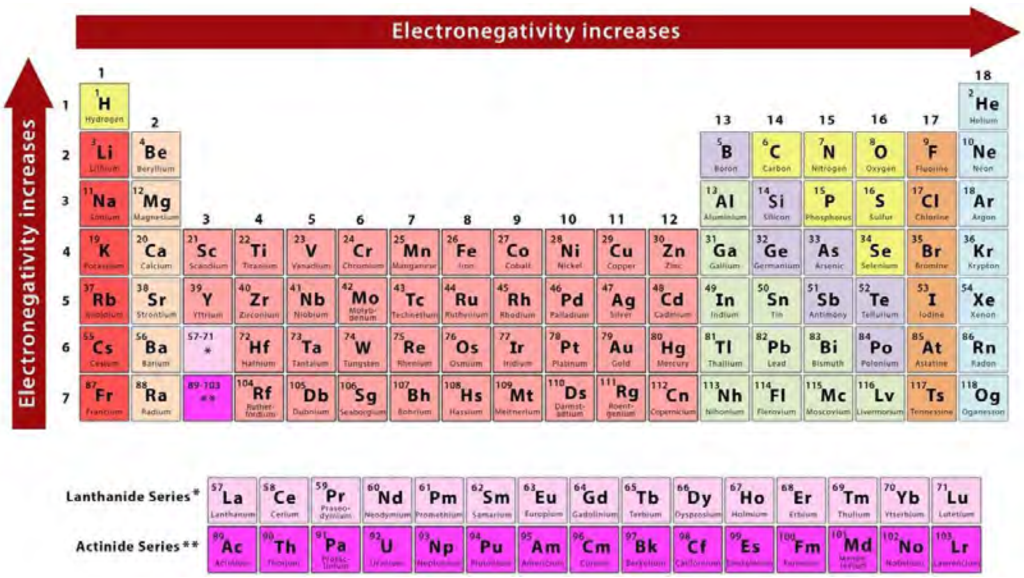


We know that a covalent bond is formed by sharing of electrons between the atom. In the case of hetero diatomic molecules, one of the atoms involved in bonding may have more affinity towards the electrons than the other. It means electrons are shifted toward this atom having higher power to pull the electrons. For example, in an HCl bond, the shared pair electrons are pulled towards the chlorine atom rather than the hydrogen. It means Chlorine is more electronegative than hydrogen. Therefore, it pulls the electron cloud towards itself and becomes partially negative.



Electronegativity is a measure of the attraction of an atom for a shared pair of electrons when it is covalently bonded to another atom.

Electronegativity of an atom depends upon the size of the atom. Smaller the size of an atom higher the effective nuclear charge and more the electronegativity. As moving from top to bottom of the periodic table, atomic size increases and tendency to pull the electrons decreases so electronegativity decreases. Whereas as moving from left to right, effective nuclear charge increases and the nuclear attraction increases which increases the electronegativity.



Electronegativity is typically measured on the Pauling scale, with values ranging from **0.7 (Francium) to 4.0 (Fluorine)**. Electronegativity plays a crucial role in various chemical processes and has several practical applications in chemistry. Here are some key applications of electronegativity:

a. **Chemical Bonding**: Electronegativity helps to predict the type of chemical bond. When the electronegativity difference between two atoms is large, an ionic bond is formed. If the difference is small, then the bond becomes covalent.

b. **Predicting Bond Polarity**: Electronegativity also helps to determine the polarity of a covalent bond. If two atoms in a covalent bond have different electronegativity, the bond will be polar. If the bonded atoms have the same electronegativity, then the bond is nonpolar.

Test yourself!

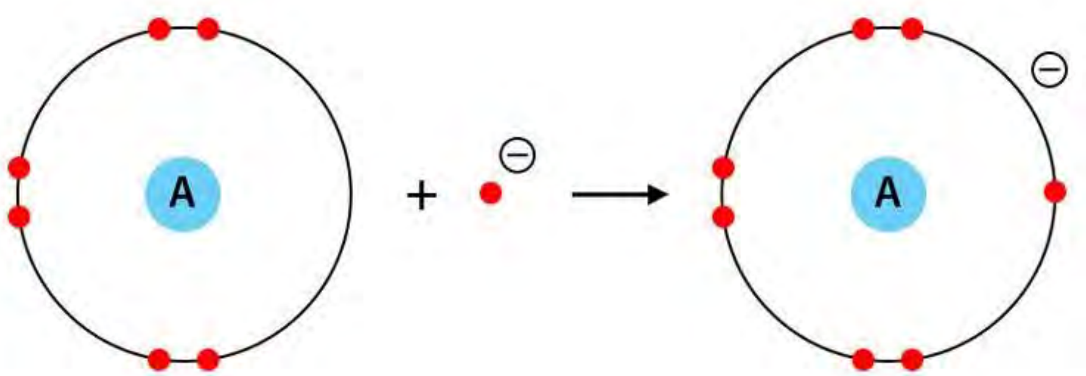
Arrange the given molecule with the bond polarity and give reason for your answer.

HF, HCl, HBr , HI

Bond type	Electronegative difference	Examples
Pure covalent(diatomic)	0	H_2
Non-polar covalent	<0.4	CH_4
Polar covalent	0.4-1.7	HF
Ionic	>1.7	NaF

4.3.5 Electron Affinity (EA)

Activity:



a. What is the difference in the number of electrons in A and A^-?

b. Identify whether energy is absorbed or released in the given process.

When an electron is added to a neutral atom, a negatively charged ion is formed. During the process, the energy is generally released so the process is an exothermic process. More the negative value of electron affinity, higher the tendency of atoms to gain electrons to form anion. It is measured in kJ/mole or eV (electron volt). Electron affinity of inert gas is positive suggests that the anion is not stable, so it cannot be formed in the gas phase.

Electron affinity depends upon nuclear charge and size of the atom. When nuclear charge increases, nuclear attraction increases and electron affinity also increases. Likewise, with the increase in size of atom nuclear attraction decreases and electron affinity decreases.

Electron affinity is the energy released when 1 mol of electrons is added to 1 mol of neutral atoms in the gas phase

Li -60	**Be** > 0	**B** -27	**C** -122	**N** > 0	**O** -141	**F** -328	**Ne** > 0
Na -53	**Mg** > 0	**Al** -43	**Si** -134	**P** -72	**S** -200	**Cl** -349	**Ar** > 0

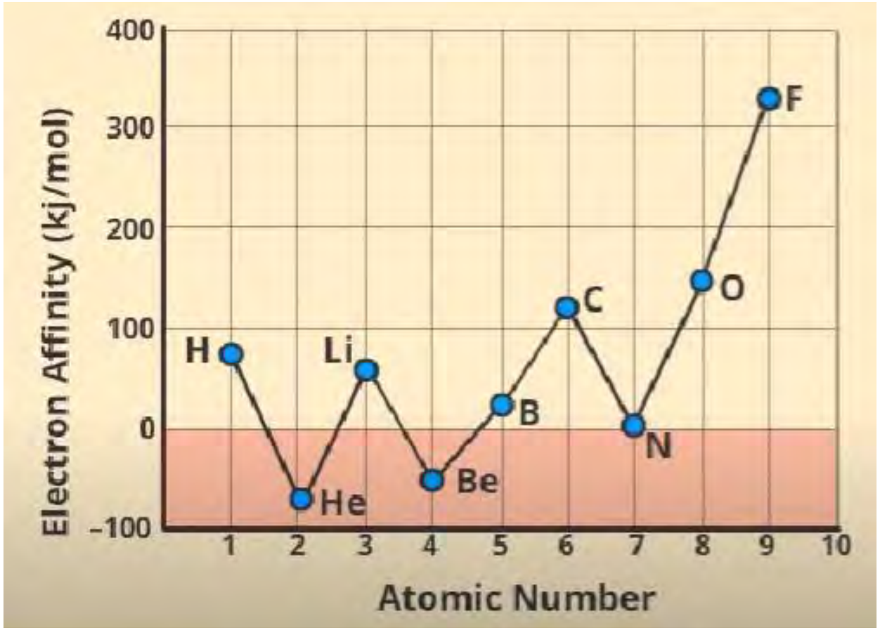
Table: electron affinity (in eV) of some elements

In general, electron affinity increases as moving from left to right due to decrease in atomic size whereas in group, from top to bottom electron affinity decreases due to increase in atomic size.

Electron affinity of elements (p-block) of 3rd period is higher than that of corresponding elements of 2nd period. Atomic size of elements of period 3 is larger than that of period 2 elements. Therefore, there is less repulsion for the incoming electron in the valence shell. For example, electron affinity of chlorine is abnormally higher than that of fluorine atom.

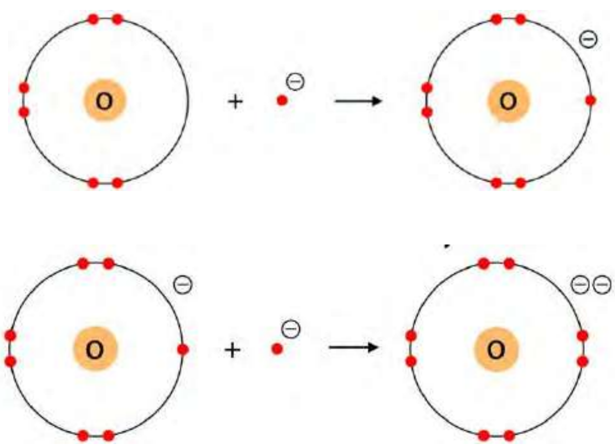
Atoms with higher value of electron affinity means they easily accepts electrons.

Atoms with lower value of electron affinity means they do not accept electrons easily or they have comparatively stable electronic configuration.



Test Yourself!

Compare the energy E_1 and E_2 in the given figure and give reason.



+ (first electron affinity) E_1

+(second electron affinity) E_2

4.3.6 Metallic character

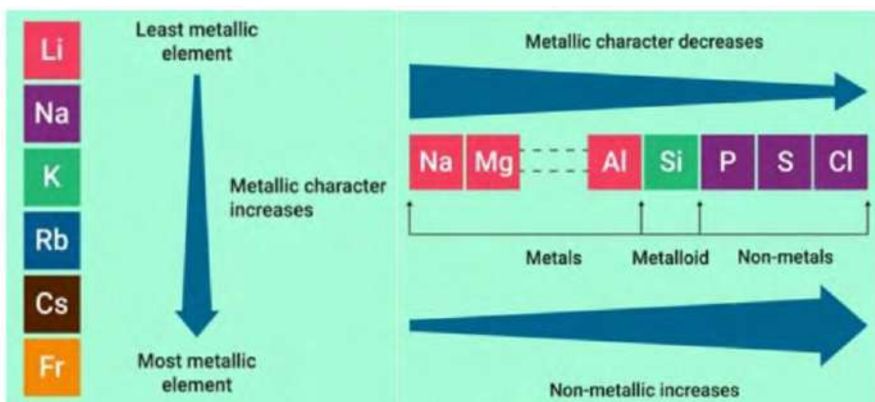
Activity: 6

a. Identify the metals in the figure and write one use of each.
b. Which property is responsible for such shapes?

The metallic character of an atom refers to its ability to exhibit properties of metals. These properties include:

Electrical Conductivity, thermal Conductivity, malleability, ductility, luster etc. Presence of free electrons in the metal are responsible for these properties of metals. Metals tend to lose electrons and form positively charged ions (cations) in chemical reactions This tendency of losing electrons increases as increasing the size of the atom.

Metallic character tends to decrease as moving from left to right across a period of the periodic table. This is because the number of protons in the nucleus increases across a period, leading to a stronger attraction between the positively charged nucleus and the negatively charged electrons in the outermost shell. As a result, the outermost electrons are held more tightly, making it harder for them to move freely and conduct electricity.



Metallic character generally increases moving down in the group of the periodic table. This is because as moving down a group, there are a greater number of shells of electrons, and the outermost electrons are farther from the nucleus These outermost electrons are shielded by the inner electrons. So, they are less strongly attracted to the nucleus and can move more freely, leading to better metallic properties. Transition metals, found in the middle of the periodic table, exhibit variable metallic character depending on their position within a period.

More about periodic table

Metalloids, such as silicon and germanium, have properties intermediate between metals and nonmetals. They are often used in semiconductor technology.
The noble gases in Group 18 of the periodic table are often used in lighting (neon lights)
Helium is the only element that was discovered in space before it was found on Earth. It was first observed in the Sun's spectrum during a solar eclipse in 1868.
Electron affinity of chlorine is greater than that of fluorine because of large atomic size of chlorine there is less electronic repulsion in the valence shell than that of fluorine.
First ionization energy of nitrogen is greater than oxygen but second ionization energy of oxygen is greater than nitrogen.
The noble gases (helium, neon, argon, krypton, xenon, and radon) were once called "inert gases" because they were believed to be completely non-reactive. However, it was later discovered that they could form compounds under specific conditions, challenging this perception.

Figure: Periodic trend of metallic characters

Use of transition metals
Transition metals are used in daily life for wide range of applications due to their unique properties and versatility.
Titanium **(Ti)**is used in MRI as well as in the production of parts and engines of aircraft.
Platinum **(Pt)** is used in cancer treatment.
Cobalt (**Co)** is used in paints and pigments.

Test yourself!

Give reasons for the following.

a. Sodium metal is not used for making household utensils.
b. Copper is used for making electrical wires.
c. Gold is used in jewelry and ornaments.

Exercise

A. Multiple choice questions

1. Which of the following elements has the same number of groups and period?

a. Li
b. Be
c. B
d. Mg

2. An element is in group 3 and period 2. How many electrons are present in its outer shell?

a. 2
b. 3
c. 5
d. 6

3. Which of the following atoms, designated by their electronic configurations, has the highest ionization energy?

a. $[Ne]3s^2\ 3p^2$
b. $[Ne]\ 3s^2\ 3p^3$
c. $[Ar]3d^{10}\ 4s^2\ 4p^3$
d. $[Kr]4d^{10}\ 5s^2\ 5p^3$

4. Which of the following elements shows the most metallic behavior?

a. Li
b. Be
c. Si
d. Ca

5. Which of the following is true about the elements of the same group?

a. Same number of valence electrons
b. Same valence shell electronic configuration
c. Similar physical properties
d. All are true

6. What is the correct order of electronegativity among the following options?

a. Li<Na<K<Rb<Cs
b. Li<K<Na<Rb<Cs
c. Li>Na>K>Cs>Rb
d. Li>Na>K>Rb>Cs

7. The relationship between Li/ Mg and Be/ Al is

a. Diagonal b. periodic c. Triangular d. Regular

8. Which of the following has more electron affinity than that of fluorine?

a. sodium

b. potassium

c. chlorine

d. oxygen

B. What happens when an electron is added to the valence shell of an atom?

a. Energy is absorbed

b. Energy is released

c. Energy remains same

d. Force of attraction increases

10. What is the name of the effect which decreases the force of attraction between valence electrons and nucleus?

a. screening effect

b. shielding effect

c. photoelectric effect

d. doppler effect

11. Element B lies above element A in the periodic table, and element C lies to the left of element A. What is the correct ascending order of their electronegativities?

a. C<A<B
b. A<B<C
c. B<A<C
d. C<B<A

12. Which of the following gives the information about the period in the modern periodic table?

a. atomic number

b. atomic mass

c. principal quantum number
d. azimuthal quantum number.

C. Short answer questions

1. Differentiate the following:

a. Period and periodicity

b. Nuclear charge and effective nuclear charge

c. Atomic radii and ionic radii

d. Electronegativity and electron affinity

2. Give reason for the given statements:

a. The atomic radius of carbon is smaller than that of lithium.

b. It is more difficult to ionize magnesium than that of calcium.

c. Nitrogen and oxygen are both p- block elements.

3.

a. State modern periodic law.
b. How does this law overcome the limitations of Mendeleev periodic law?
c. Identify the elements with the smallest and largest atomic radius from the table given.

5 **B** boron 10.81 ± 0.02	6 **C** carbon 12.011 ± 0.002	7 **N** nitrogen 14.007 ± 0.001	8 **O** oxygen 15.999 ± 0.001	9 **F** fluorine 18.998 ± 0.001
13 **Al** aluminium 26.982 ± 0.001	14 **Si** silicon 28.085 ± 0.001	15 **P** phosphorus 30.974 ± 0.001	16 **S** sulfur 32.06 ± 0.02	17 **Cl** chlorine 35.45 ± 0.01

4.

a. Define the term periodicity. What is the cause of periodicity of atomic properties?
b. Name the periodic properties that generally decrease from left to right in the period and give reason.

5. As moving down the group nuclear charge increases but effective nuclear charge decreases.

a. Give the meaning of the term nuclear charge and effective nuclear charge.

b. What is the cause of increasing nuclear charge and decreasing effective nuclear charge in the group?

c. How is effective nuclear charge used to explain the trend in atomic radii and electronegativity?

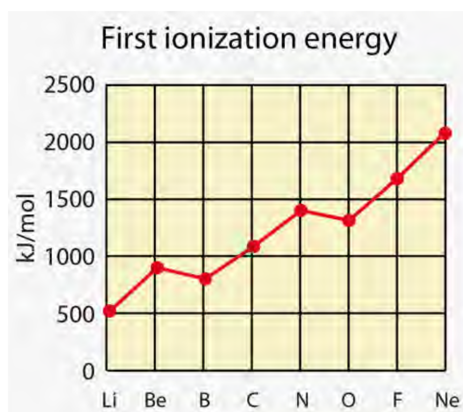
6. Atomic radius of the atom plays vital role to determine the various properties of the atom

a. Define the term atomic radius. What are the general periodic trends of atomic radii in group and period? Give reason for the trend.

b. Why is cation smaller than the atom?

7. First ionization energy of some elements is given in the graph

a. Define first ionization energy.
b. What is the general periodic trend in ionization energy in the graph moving from Li to Ne?
c. Identify the element having highest first ionization energy and give reason for the choice.



8. Some isoelectronic species are given

O^{2-}, F^{-}, Na^{+}, Mg^{2+}, Al^{3+}

a. What do you mean by isoelectronic species?

b. Write the number of protons and electrons in each.

c. Arrange the given atoms and ions in ascending order of size and give reason.

9. Electron affinity values (kJ/mole) of some elements of the periodic table are given in the table

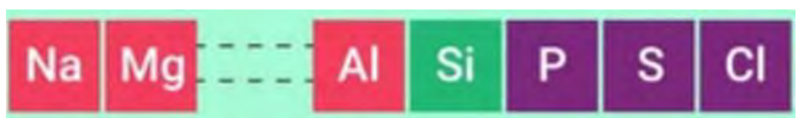
B	C	N	O
-27	-122	> 0	-141
Al	**Si**	**P**	**S**
-43	-134	-72	-200

a. Define electron affinity. What is the general trend in electron affinity across the given period and group?

b. What is the meaning of negative sign in the values given?

c. Which element has the highest electron affinity in the table? why?

10. Elements of same period are given below:



a. Identify the period.
Why are these elements kept in the same period?

b. What is the meaning of the different colors of the elements given?

c. Why is there a gap between magnesium and aluminum?

d. Name the elements that have metallic and covalent radius.

11. Energy change while adding electrons in the fluorine atom is given.

$F(g) + e \rightarrow F^{-}(g)$ Energy = -328 kJ /mol

a. Name the energy and define it.

b. What do you mean by the negative value of energy in the given equation?

c. Will the value of energy be higher or lower than fluorine if the above case is applied for nitrogen?

Long answer questions

17. Consider these elements of period 3

Al	Cl	Mg	Si	P	S

a. Identify the elements which act as metalloid. Write one property of that element.

b. List these elements in the order of increasing electronegativity and give reason for such order.

c. Which element do you think is easy to make cation? Why?

d. What do you mean by metallic character? Which of the given elements has the highest metallic character and why?

18. Effective nuclear charge is the net resultant charge of the nucleus which is a very useful parameter to explain the periodic trend of the elements.

a. How is effective nuclear charge related to the atomic number of an element?

b. Explain how shielding affects effective nuclear charge.

c. In which direction (left to right or top to bottom in the periodic table) does effective nuclear charge generally increase and why?

d. How does effective nuclear charge affect the size of the atom, ionization potential and electronegativity of the element?

19. Elements in the periodic table are primarily divided into metals, metalloid and nonmetals on the basis of metallic properties of the elements.

a. What is the trend in metallic character in group and period in the periodic table?

b. Why do alkali metals and alkaline earth metals exhibit high metallic character?

c. Which groups of elements in the periodic table are known for their non-metallic character, and why?

d. How does the number of valence electrons influence the metallic character of an element?

e. Transition metals are used for making wire. why?

20. Electronic configuration of one of an element (X) is given

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$

a. What is the atomic number of that element?

b. Identify the period, group and block of this element.

c. Write any two properties of the element which lies in this block

d. Compare the size of X with X^+ and X^-

21. Some of the elements from the same block are given :

a. Write down the electronic configuration of calcium.

b. Identify the block and write any two properties of this block.

Li	Be
Na	Mg
K	Ca

c. Why are these metals very soft in nature?

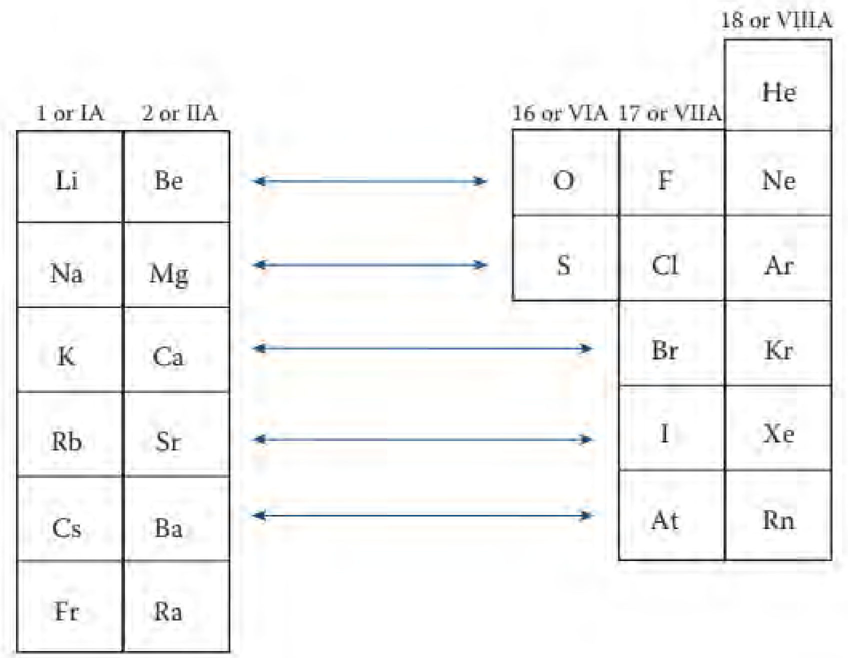
d. Arrange these elements in increasing order of atomic size.

e. Name the elements that are in a diagonal relationship?

22. A. Part of the periodic table is given. Use the table and find the element having:

a. Largest atomic radius
b. Highest ionization energy
c. Smallest electron affinity
d. Most metallic
e. Most difficult to ionize

B. Write the group name having given properties:



a. Forms a strong base when dissolved in water.
b. Forms ores
c. Highest electronegativity

Project work

Make a modern periodic table in a chart paper. Use different color to distinguish block, and state of elements at room temperature.
Take a chart paper and draw a various graph to show the periodic trend of different properties that you have studied in the class.
The smart phones which we are using today contain more than thirty elements for various purposes. These elements are used for making casing, batteries, circuit board or lenses. Make a table and categorize the elements as metal, nonmetal and metalloid.
Mention the special property of each element so that it is used in the smart phone.

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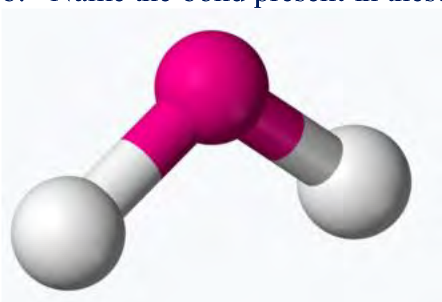
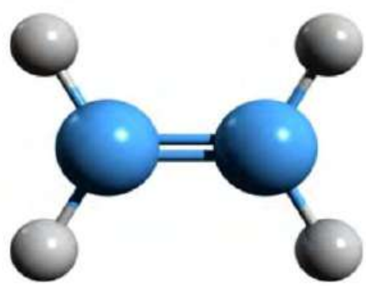
UNIT 5: CHEMICAL BONDING AND SHAPES OF MOLECULES

Activity:

a. Find the numbers of atoms in each molecule given.
b. Name the bond present in these molecules.





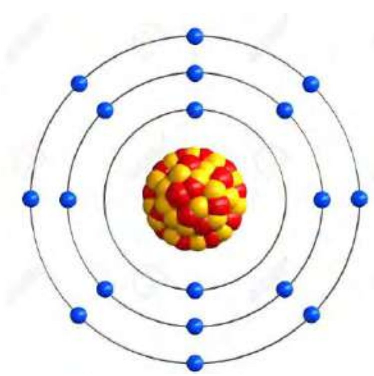
Gilbert N. Lewis (1875–1946) was an American physical chemist known for his significant contributions to the understanding of chemical bonding and the concept of electron pairs.

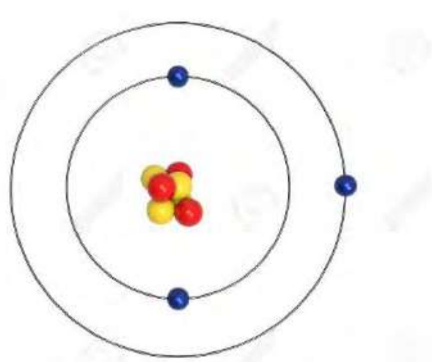
5.1 Valence shell, valence electron and octet theory

Activity:

Electron arrangements of two atoms are given in the figure.

a. Identify these atoms given and find the number of shells in each.
b. Which shell is the valence shell in these atoms?
c. How many valence electrons are there in each atom?

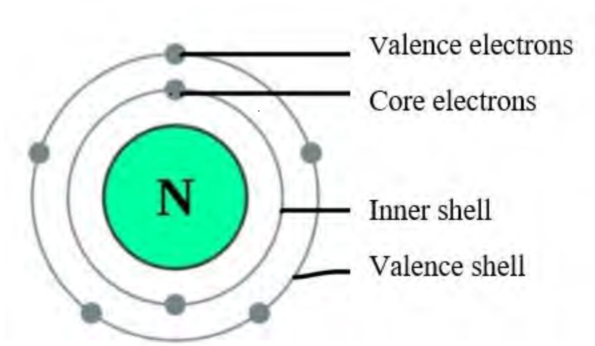




The combining capacity of an atom depends on the number of electrons present in the outermost shell. Electrons present in the outermost shell are called **valence electrons** and such shells are called **valence shells**. Valence electrons are important in determining the element's reactivity and its ability to form chemical bonds.

The number of valence electrons often equals the group number of an element in the periodic table.

The electrons present in the inner shell are called **inner electrons or core electrons**. Core electrons are not involved in chemical reactions because they are tightly bound to the nucleus.



Group 1 elements (e.g., Li, Na) have 1 valence electron.

Group 2 elements (e.g., Be, Mg) have 2 valence electrons.

Group 17 elements (e.g., Cl) have 7 valence electrons.

Group 18 elements (e.g., He, Ne) have either 2 or 8 valence shells.

The octet rule, also known as the octet theory was primarily formulated by **Gilbert Lewis**, an American chemist.is a fundamental principle in chemistry. It states that

"atoms tend to gain, lose, or share electrons in order to achieve a stable electron configuration with eight electrons in the valence shell".

Do you know?

For hydrogen and helium, achieving a duet (two electrons in the outer shell) is sufficient for stability.
This rule is particularly relevant for elements in the s- and p-blocks of the periodic table.

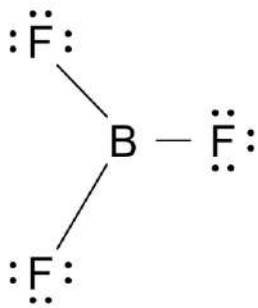
The octet rule is the fundamental principle in understanding chemical bonding and explaining the stability of certain molecules and ions. It provides a simple framework for predicting the behavior of atoms in chemical reactions.

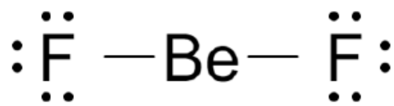
Exceptions of octet rule

There are exceptions and modifications to the octet rule, especially for elements in the transition metals and those with incomplete d or f orbitals. These exceptions are due to the specific electronic configurations and bonding behaviors of these elements.

a. Incomplete Octet

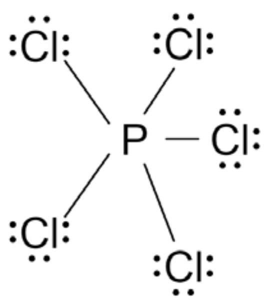
Some molecules and ions have fewer than eight electrons in their valence shells of the central atom. Examples include BeH_2, $BeCl_2$, BeF_2, BF_3 BCl_3, etc.

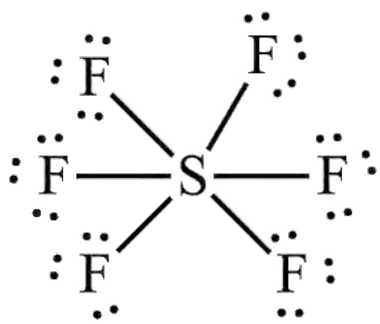




b. Expanded Octet

Elements of group 15 and above can accommodate more than eight electrons in their valence shells, expanding their octet. For example, SF_6, PCl_5, etc.

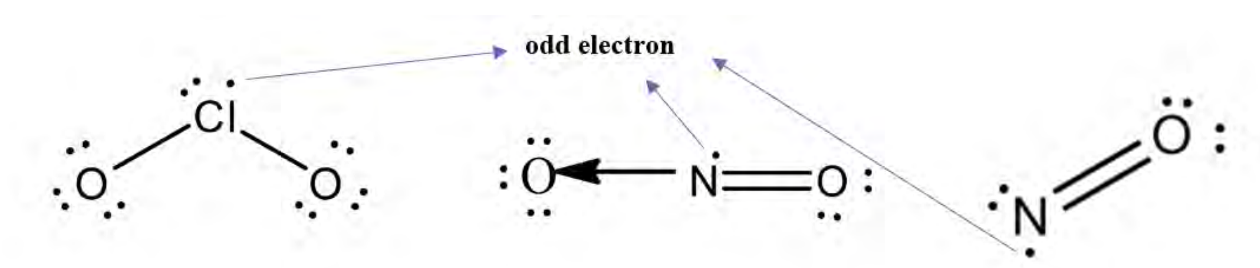




These molecules containing more than eight electrons in the central atom are also called hypervalent or super octet molecules.

c. Odd Electron Molecules

Molecules with an odd number of electrons cannot satisfy the octet rule for all atoms. Examples include nitrogen dioxide (NO_2), nitric oxide (NO), chlorine oxide (ClO_2), etc.

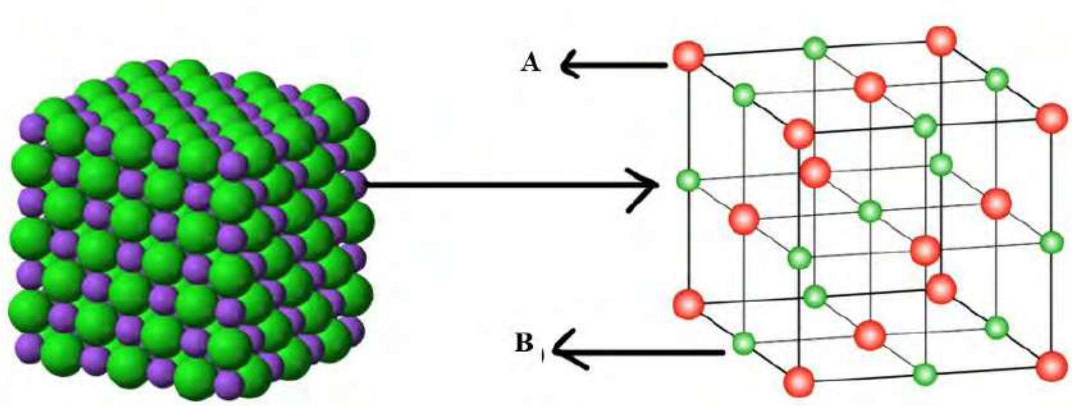


Test yourself!

Identify whether the given species follows octet rule or not.
XeF_4, H_2, H_2O, NH_3, $AlCl_3$

2 5.2 Ionic bond and its properties

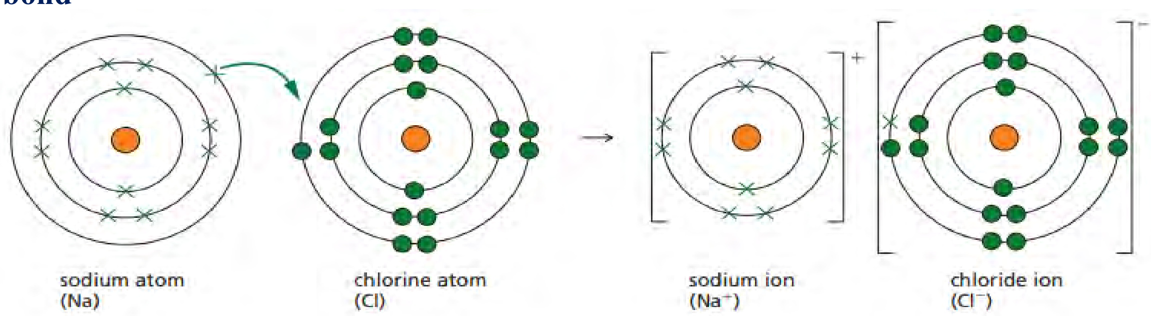
Activity:



a. Identify the ions (A and B) in the figure if it shows the crystals of NaCl.
b. Discuss the reason for the different sizes of these ions.
c. Name the force that holds these ions together.

Bond is a type of relationship between two atoms, ions or molecules which hold these species together. When valence electrons of different atoms come close enough to each other, there is a formation of chemical bond. When chemical bond is formed, energy is released called **bond energy** which makes the bond stable. The nature and strength of bond depends on the nature of combining elements and mode of formation of bond. If the bond is formed between metal and nonmetal then the bond is ***ionic***, if it is within the metal, it is called ***metallic***, likewise ***covalent*** bond is formed if the bond is between nonmetal and nonmetal.

Ionic bond



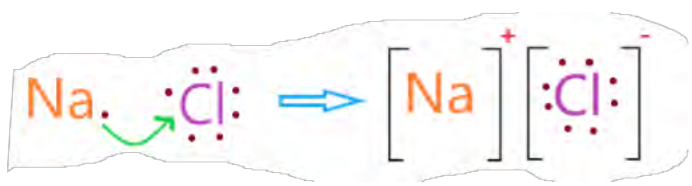
The metallic elements have a tendency to lose the valence electrons to make the valence shell octet. In the same way nonmetals have the tendency to accept the electrons to maintain a stable octet**.** So, there is a bond between metal and nonmetal**.**

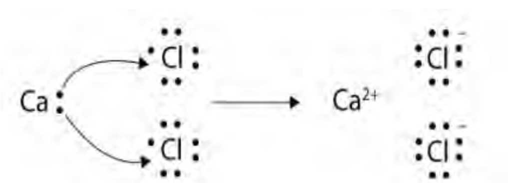
As the metal atom loses electrons, it becomes a positively charged ion (**cation**). Similarly, the non-metal atom accepts electrons, becomes a negatively charged ion (**anion**). The positive and negative ions are attracted to each other by electrostatic force of attraction. **The electrostatic force of attraction existing between the oppositely charged ions is called ionic bond**. Since there are electrical charges on the atoms involved in the bonding ionic bond is also called electrovalent bond. The compounds with ionic bonds are called ionic compounds. For example, KCl, MgSO4, K_3PO_4 NaCl, etc.

When these ions approach each other to form ionic compounds a certain amount of energy is released. The **amount of energy released when the oppositely charged gaseous ions are brought from an infinite distance to form one mole of ionic crystal is called lattice energy.** Higher the lattice energy means higher the stability of the compound and lower the solubility in water**.**

Ionic bond in NaCl

When very active sodium metal loses its one valence electron, it becomes sodium ion (Na^+). Chlorine accepts one electron easily and becomes chloride ion (Cl^-). It results in the formation of ionic compound NaCl with ionic bond.





Likewise, two valence electrons of calcium are transferred to two chlorine atoms to form calcium chloride molecules.

Ionic bonds involve the electrostatic attraction between positively and negatively charged ions without a well-defined directional arrangement or specific orientation, it is called **non-directional nature** of ionic bond.

The conditions for the formation of ionic bond are given below:

a. A high electronegativity difference between the two atoms increases the possibility of electron transfer and formation of ionic bonds.
b. Metals with low ionization energy can lose electrons easily and form the ionic bond.
c. Non-metals with high electron affinity can accept electrons and form the ionic bond,

d. The presence of polar solvents can facilitate the dissociation of ions and the formation of an ionic bond.

Test yourself!
Draw diagrams to represent the bonding in each of the following ionic compounds:
Potassium fluoride (KF)
Lithium chloride (LiCl)
Magnesium fluoride (MgF_2)
Calcium oxide (CaO).

Properties of ionic compounds

Activity:

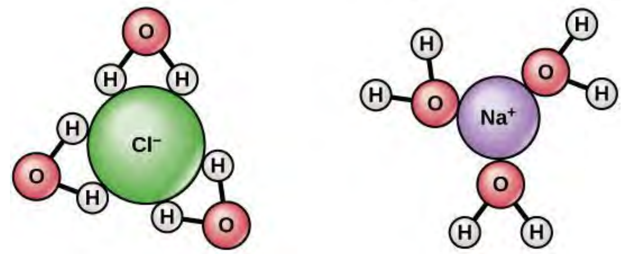
a. Take piece of potash alum ***(fItkiri)*** and observe the color and nature of salt.
b. Hammer(hit) the alum and observe the change.
c. Pour the alum in a glass of water and observe the solubility.

Figure: Potassium chloride(KCl)

Figure: Ferrous sulphate (FeSO4)

Due to the strong electrostatic force of attraction between the oppositely charged ions, ionic compounds tend to form regular crystalline structure in solid state with high density and melting point. Some of the important characteristics of ionic compounds are given below.

a. Ionic compounds are generally soluble in polar solvent like water. The polarity of water helps water molecules to surround and interact with the ions called **hydration**. These interactions pull the ions away from the crystal lattice then the solid dissolves into the solution.



b. In the solid state, ionic compounds do not conduct electricity because the ions are held in a fixed position and cannot move. In the molten state or aqueous state (when dissolved in water), ionic compounds conduct electricity as the ions are free to move and carry an electric charge.

c. Ionic compounds are often hard and brittle due to the arrangement of ions in a crystal lattice structure. Applied force causes the repulsion within the similar charged ions then the layers of ions slide make the salt brittle(breakable).

Figure: hydration

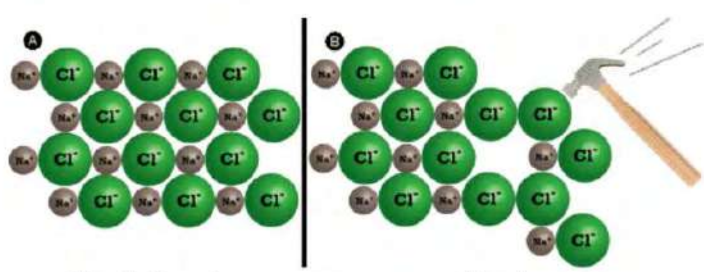


Fig: before stress *after stress*

An ionic compound forms by the arrangement of oppositely charged ions in a rigid, closely packed structure called a **unit cell**. It repeats itself to produce a crystal.

Unit cell of KCl **unit cell of CaO** **unit cell of NaCl**

Test yourself!

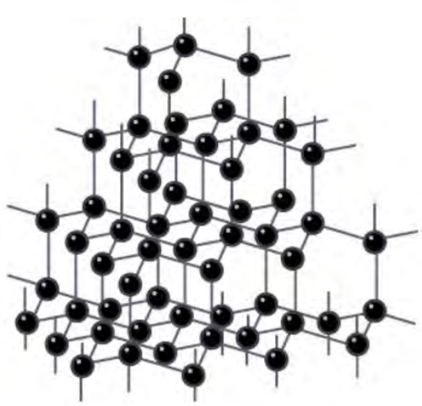
Silver chloride is an ionic compound but is insoluble in water, why?

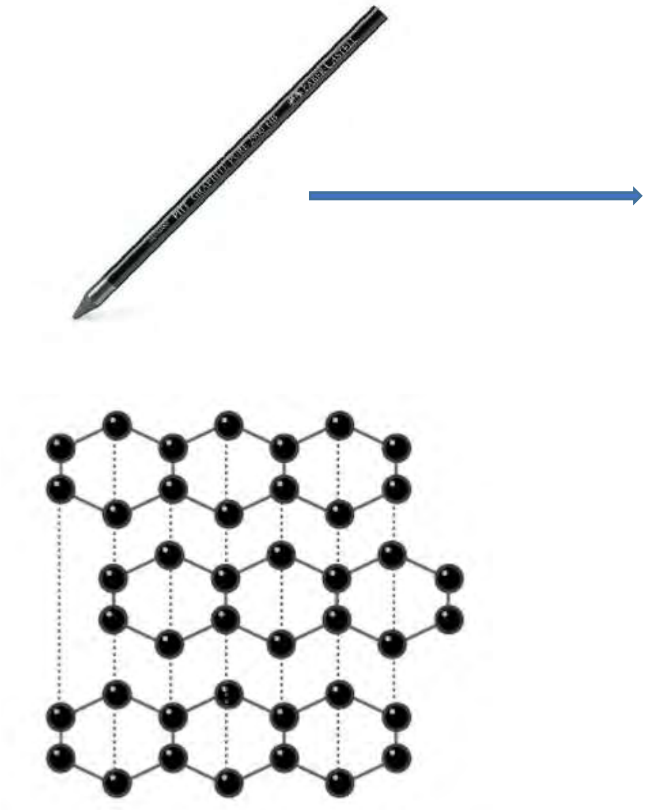
5.3 Covalent bond and coordinate covalent bond

Activity:

Two substances along with the bonding between the atoms are given in the figure.

a. Identify the materials given and find the variation in physical properties.
b. Name the bond present between the atoms.





How is covalent bond formed between the atoms?

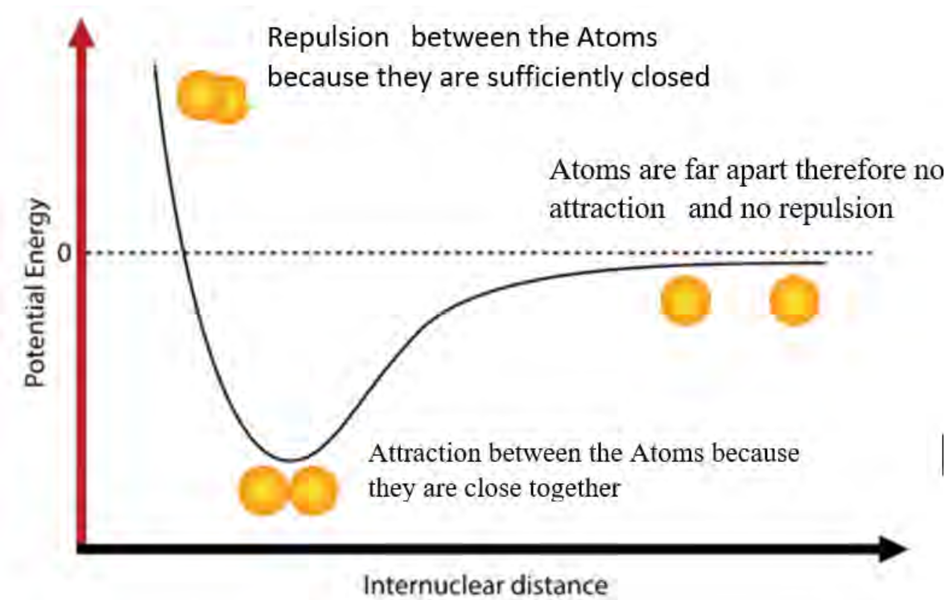


Fig: Potential energy against internuclear distance for the hydrogen molecule.

Let us consider two hydrogen atoms, initially far from each other. There is no overlap, no attraction and no repulsion.

Therefore, potential energy is nearly zero. When they come closer, there is a repulsion between the negatively charged electron as well as attraction between the electrons and the nucleus. It causes decrease in internuclear distance and potential energy as well. When these atoms overlap with each other, there is a balance between the attraction and repulsion forces. It means at this point:

The force of attraction between the electrons and the nucleus = repulsion between the electrons + repulsion between the nucleus.

The bond formed between these atoms is **covalent bond**. The distance between the nucleus of the bonded atoms is called **bond length** and the decrease in potential energy at this point is called **bond energy**. Further overlapping between these atoms increases the repulsion force and becomes unstable. Then the potential energy increases.

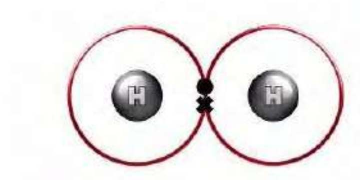
Representation of covalent bond

Formation of covalent bond in some simple molecules can be represented as:

H_2

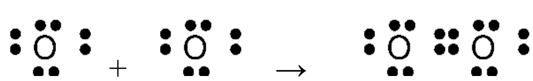
H• + •H → H •• H

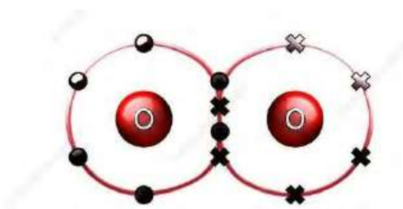
H─H (single covalent bond)

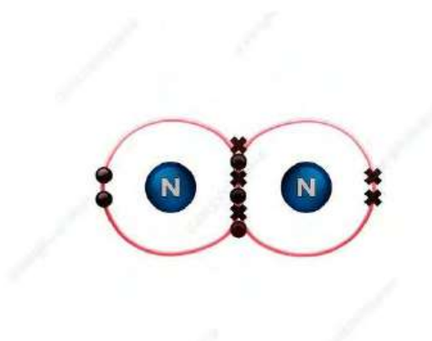


Likewise, oxygen shares two electrons and nitrogen shares 3 electrons respectively to maintain stable electronic configuration as shown.

O_2

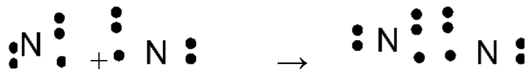






O ═O (double covalent bond)

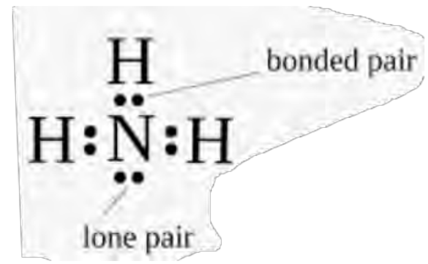
N_2



N≡N (triple covalent bond)

Those pairs of electrons involved in the bonding are called **bonding pair or bond pair.** The pair of valence-shell electrons that do not actually take part in the bond are known as **non-bonding pairs** (**lone pairs**). These non-bonding pairs of electrons can have a significant effect on the properties of substance and shape of the molecule.

For example, in ammonia there are three bond pairs and one lone pair of electrons.



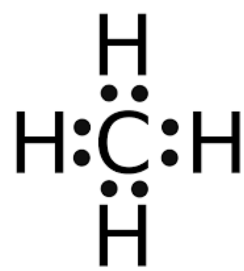
Covalent bond is defined as the bond formed by mutual sharing of valence electrons between the combining atoms. The number of electrons shared is equal to the number of covalent bonds formed.

Conditions for the formation of covalent bond

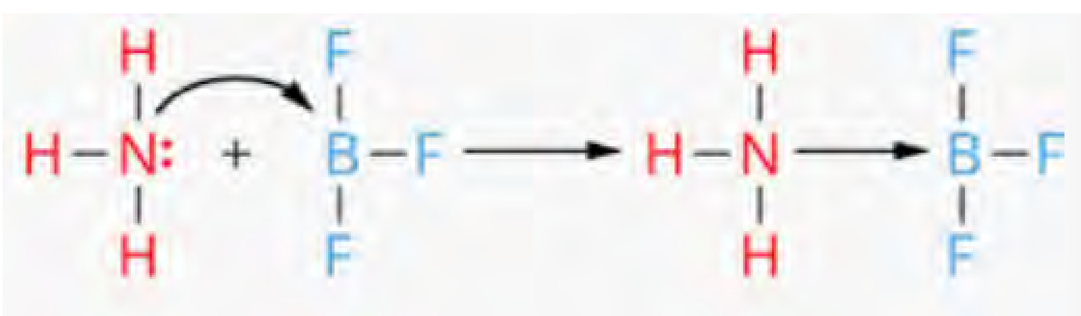
a. Covalent bonds usually form between atoms with similar electronegativities.
b. The participating atoms should have high ionization potential and high electron affinity.

Test yourself!

a. How many electrons are shared by carbon with hydrogen atoms? Why?
b. Find the number of bond pairs of electrons in a carbon atom.
c. Is there any nonbonding pair of electrons in the carbon? Why?
d. Draw the covalent bond around the carbon atom.

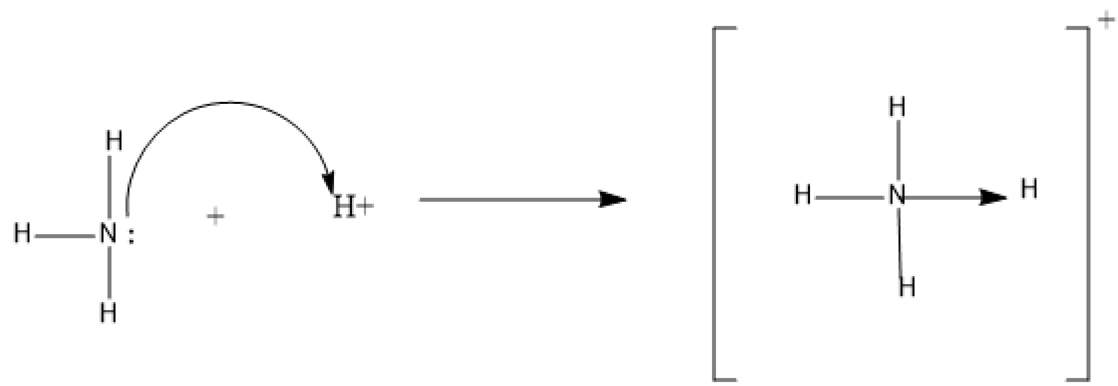


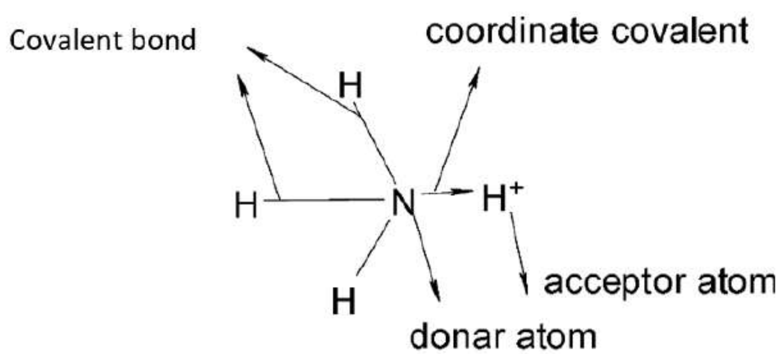
Coordinate covalent bond (dative bond)



The bonding between ammonia and Boron trifluoride is shown.

a. Which of the above molecules follows octet rule and which does not?
b. How is the B-F bond and N-B bond different from each other?





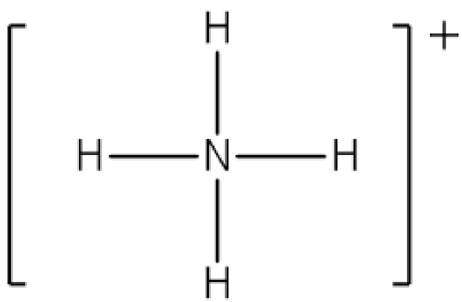
Nitrogen atom in the NH3 molecule donates a pair of electrons to the hydrogen ion and forms a bond between nitrogen and hydrogen called coordinate covalent bond.

In **coordinate covalent bonding**, the shared pair of electrons come from ***only one of the two atoms***. The atom which contributes shared pair of electrons is called *donor(nitrogen)* and the atom which accepts these electrons is called ***acceptor(hydrogen).*** For the coordinate bond to be formed, the donor atom must be in octet condition before donating the electrons.

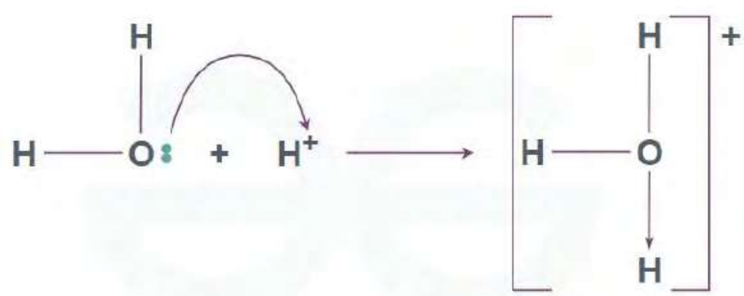
Coordinate covalent bond is also called dative bond and is often represented by an arrow pointing from the donor atom to the acceptor atom. The arrow indicates the direction of the electron pair transfer.

Though the process of formation of coordinate covalent bonds is different from the ordinary covalent bond, they are similar in nature and properties. Therefore, co-ordinate bonds can be represented by usual covalent bonds (except complexes).

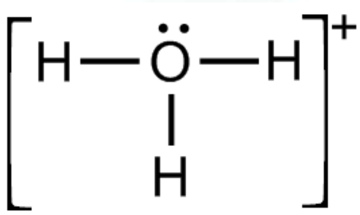
Ammonium ion, therefore can also be represented as



Likewise, formation of hydronium ion(H_3O^+) is represented as:



or



Do you know?

Coordinate covalent bonds play a vital role in many chemical reactions mainly in the formation of complex molecules and ions.

Non-polar and polar covalent bonds

a. Identify the given molecule as homonuclear and heteronuclear.
b. Why is the location of electrons pairs different in these two molecules?

Fig: tug of war

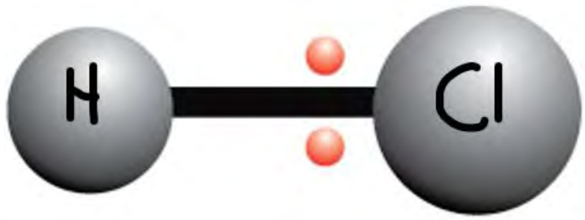
In figure **A,** the pulling power of both the teams are the same, so, the red ribbon is in the center. Whereas, in figure **B,** the pulling power of one of the teams is comparatively more, so the ribbon is shifted towards that team. The same is true in the case of covalent bonds.

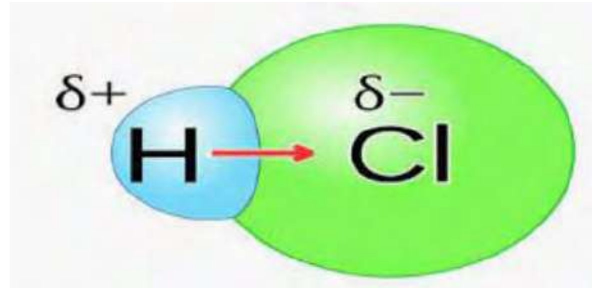
If the electrons are shared equally, the shared electron pair will be in the exact middle of the atoms. Hence there is no formation of charge. Such covalent bond is called **non-polar covalent bond or pure covalent bond**. In nonpolar bonds, the electronegativities of the bonded atoms are the same. All homo diatomic molecules like H_2, O_2, Cl_2 etc. contain nonpolar covalent bonds.

If the covalent bond is formed between two unlike atoms having different electronegativity, then the bond is called **polar covalent bond**. In this bond, there is uneven distribution of electrons between the bonded atoms.

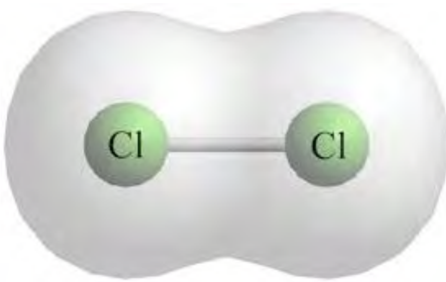
Like a tug-of-war where the bigger and stronger team wins, the atom with higher electronegativity pulls the electrons closer to itself.

This unequal sharing results in partial charges on the atoms. The atom which pulls the shared pair of electrons becomes partially negative and another atom becomes partially positive. All hetero diatomic molecules contain polar covalent bonds. For example, HCl, HF, N-H, etc.

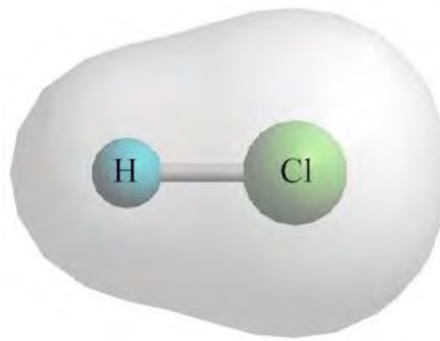




The equal and unequal distribution of electron clouds between the bonded atoms make the bond nonpolar and polar, respectively.



Non-polar covalent bond



polar covalent bond

Do you know?

Unlike covalent bonds, where electrons are shared between atoms in specific directions, ionic bonds do not have a defined direction. The electrostatic attraction between the positive and negative ions occurs uniformly in all directions, leading to a three-dimensional arrangement of ion. So ionic bonds are **non-directional** whereas covalent bonds are **directional** in nature.

Note:

All homoatomic molecules are non-polar and all heteroatomic molecules are polar but poly atomic molecules may be polar or non-polar depends up on the geometry of molecule. It will be discussed later in VSEPR theory polar bond does not mean polar molecule. The molecule with polar bond may be nonpolar depends up on the shape of the molecule.

Test yourself!

Which of the following molecule is more polar and why?

$H-F$ $\qquad\qquad$ $H-Br$

Properties of covalent compounds

Activity:

a. Fill a beaker with water.
b. Add a small amount of either kerosene or petrol into the water.
c. Use a rod and gently stir the mixture.
d. Observe any changes, with a specific focus on solubility.

A **B**

a. The cylinder given in the figure A contains cooking gas, Name the gas. What type of molecules are present in this gas?
b. There are two substances in the figure B. Are they soluble? why?

Covalent compounds can exist in all three states of matter (solid, liquid, and gas) depending on temperature and pressure. The major properties of covalent compounds are:

a. The intermolecular forces in covalent compounds are generally weaker than the electrostatic forces between ions in ionic compounds. Therefore, these compounds are usually soft and have lower melting and boiling points compared to ionic compounds.
b. Many covalent compounds are soluble in nonpolar solvents but may not be soluble in polar solvents like water. This is because like dissolves like, and covalent compounds often have nonpolar characteristics.

c. In general, covalent compounds do not conduct electricity in their pure form. This is because they do not contain free ions or charged particles.

Do you know?

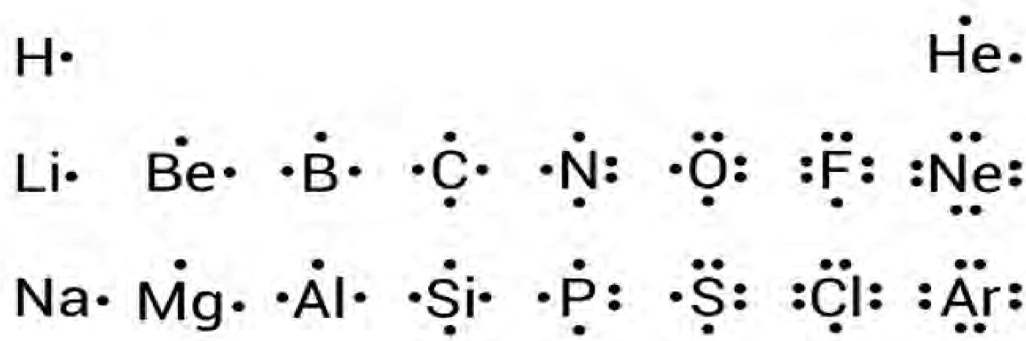
Toluene, ether, CCl_4 and benzene are commonly called nonpolar solvents which are usually used to dissolve the nonpolar substance like grease, fat, etc.

Test yourself!

Compound	Melting point/^{0}c	Boiling point/^{0}c
NaCl	801	1413
MgO	2852	3600
CO	-199	-191
C_6H_{14}	-95	69

a. Identify the covalent and ionic compounds from the table above.
b. Why does MgO have a higher melting point than NaCl?
c. Name the compound that has the highest rate of evaporation.

5.5 Lewis structures or Lewis dot diagrams



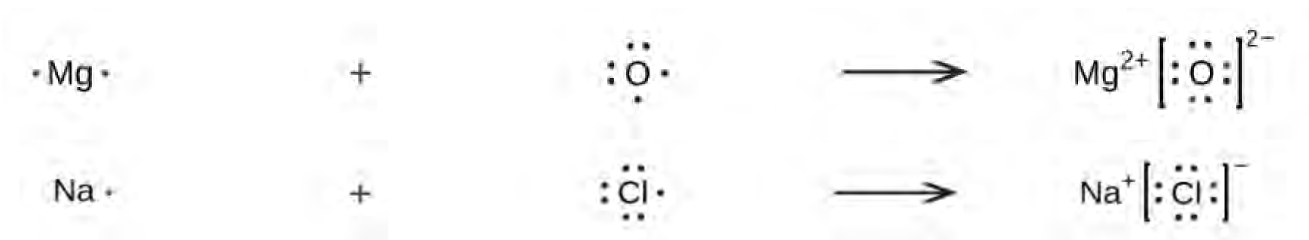
a. What is the meaning of dots around the symbol of the elements?
b. Why are the number of dots different in each symbol?

A Lewis dot diagram, also known as a Lewis structure or electron dot diagram, is a representation of the valence electrons in an atom. It was introduced by **Gilbert N. Lewis** in 1916. The main purpose of a Lewis dot diagram is to show the arrangement of valence electrons around the atomic symbol of an element.

Using the **Lewis dot symbol,** we can explain the formation of ionic and covalent bonds in a molecule. The goal of the Lewis structure is to predict the ***structural formula*** for a molecule whose molecular formula is already known.

In a Lewis structure, valence electrons are represented by dots or crosses.

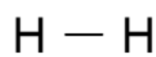
Lewis dot structures of some ionic compounds are:



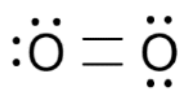
Covalent bond consists of a pair of bonding electrons between the two atoms involved, a dot and a cross between the atoms represent a bond. Likewise, a pair of dots or crosses only within the central atom represent a non-bonding pair of electrons.

Lewis dot structure of some common molecules of s and p block elements are:

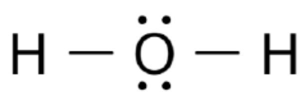
a. H_2



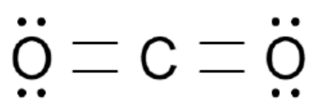
b. O_2



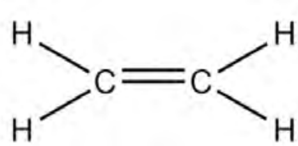
c. H_2O



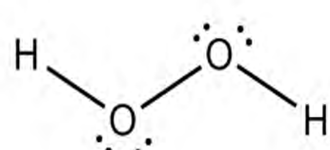
d. CO_2



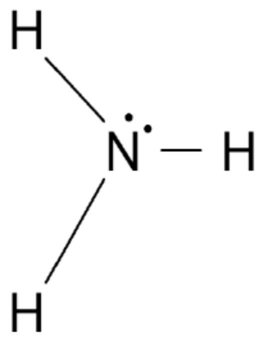
5.C_2H_4

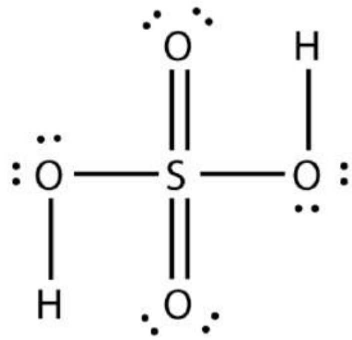


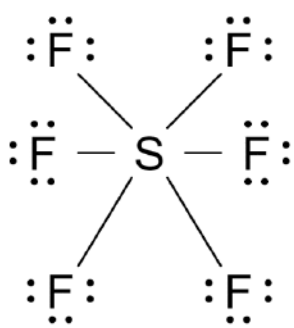
e. H_2O_2



Likewise, Lewis structure of NH_3, H_2SO_4 and SF_6 are:







Lewis dot structure of ions

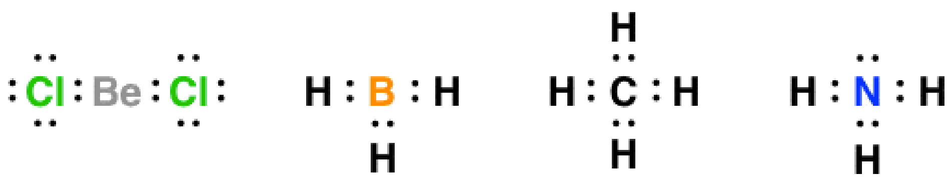
In a positively charged ion, electrons can be removed from **central or peripheral atoms. The number of electrons lost is equal to the number of positive charges in the ion. In the same way** an electron is added to either central atom or peripheral atoms in case of a negatively charged ion.

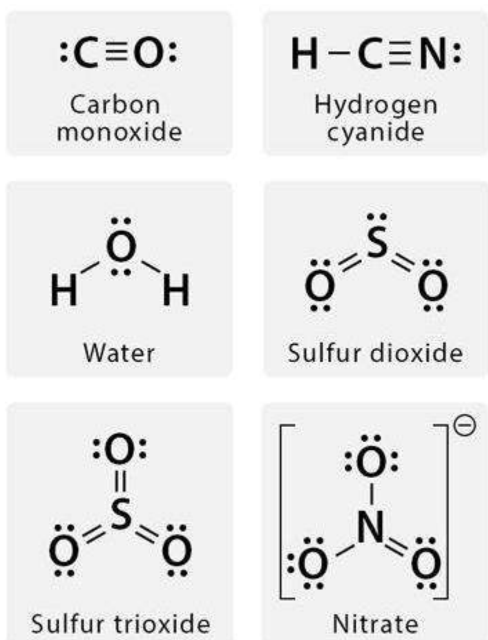
Lewis dot structures of some common ions

https://idealcalculator.com/lewis-structure-calculator-lewis-structure-generator/

Test yourself!

1. Draw the Draw the Lewis dot diagrams of following molecules and ions:
   a. SO_3 a. SO_3 b. SO_2 c. PO_4^{---} d. N_2O_3 e. NO_2^-
2. Draw the bond between the atoms in the given molecules.

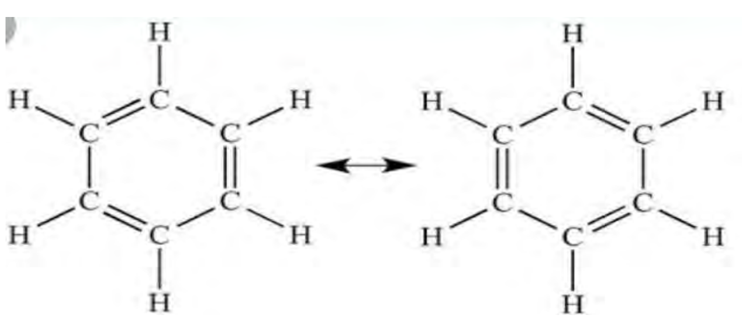




5.6 Resonance

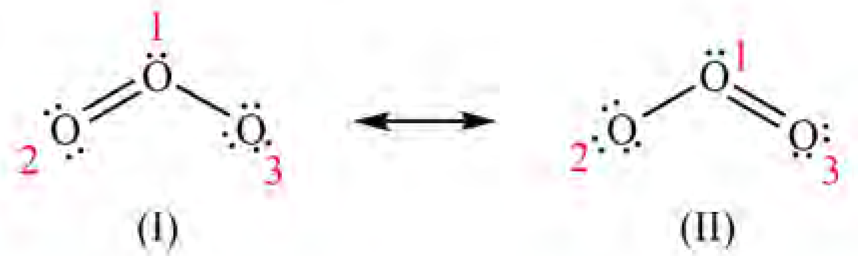
Activity:

Two structures of the same molecule are given below.



a. What is the molecular formula of the molecule given?
b. What is the difference between these two structures?

Let us consider an ozone molecule(O_3). The three atoms of oxygen are covalently bonded with each other as shown in the Lewis structure.



Ozone Hybrid

The dot lines indicate the delocalization of *pi* bonding electrons

These two structures have the same number of oxygen atoms and the same position of atoms (as indicated 1,2 and3) but the position of the double bond is different. In structure I double bond is in between atoms 1 and 2 whereas in structure II double bond is in between atoms 1 and 3.

These structures I and II are called **resonance structures or canonical forms** and the phenomenon is called **resonance**. Resonance is a concept in chemistry that describes the delocalization (electrons are not bound to a single atom) of electrons within molecules. It is often used to explain certain properties and behaviors of molecules that cannot be effectively represented by a single Lewis structure.

Do you know?
The more resonating structures a molecule has, the more stable it is considered to be. This is because resonance delocalizes electrons, spreading them out over multiple atoms and reducing electron repulsion. Example, in Niro benzene.

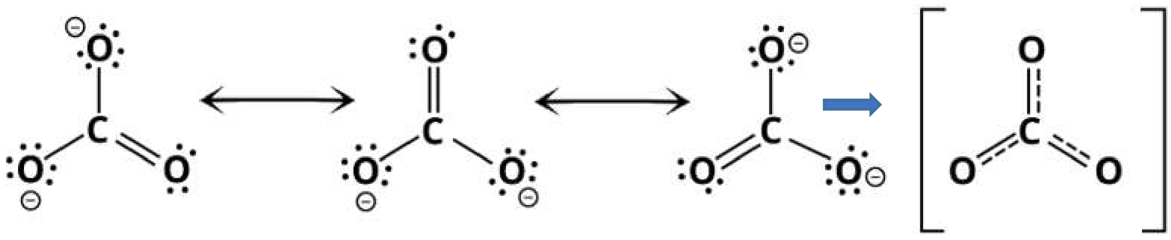
The stability is due to the delocalization of electrons, which lowers the overall energy of the molecule called **resonance energy**. The resonance structures are equivalent in energy and stability. The actual structure of the molecule is considered a **resonance hybrid**, representing the average of all contributing resonance structures. The hybrid is more stable than any individual resonance structure.

Importance's of resonance

Resonance plays a crucial role in explaining the stability of molecular species.
Resonance can affect bond lengths and strengths. Bonds in a molecule with resonance tend to have intermediate characteristics between those of single and double bonds.

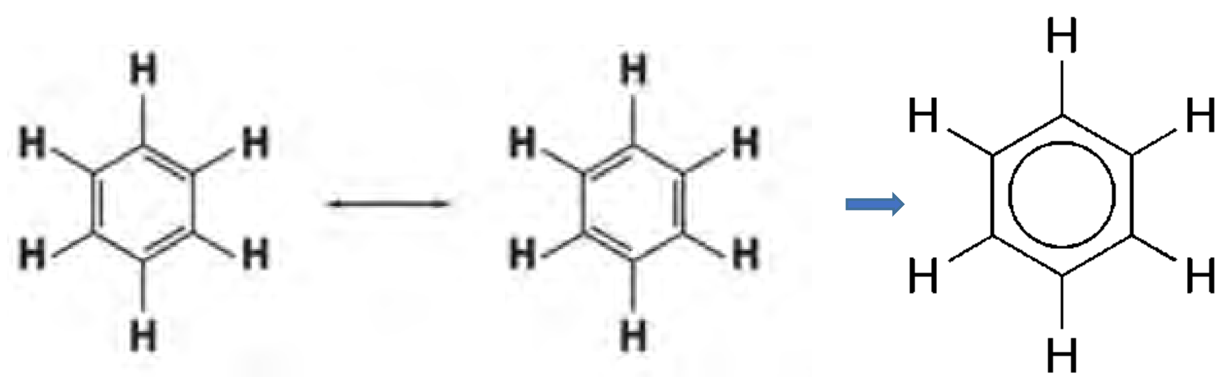
Examples:

1. **Carbonate ion (CO_3^{--})**



Resonance structures resonance hybrid

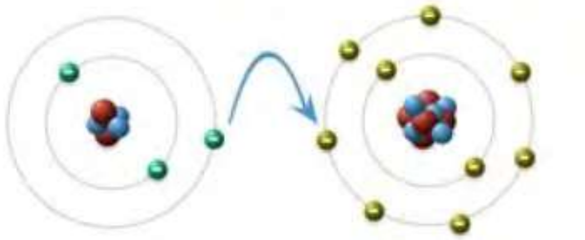
2. **Benzene**



Resonance structures **resonance hybrid**

Test yourself!

1. Identify the compound formed and draw the Lewis structure of the compound. Write any two properties of that compound.

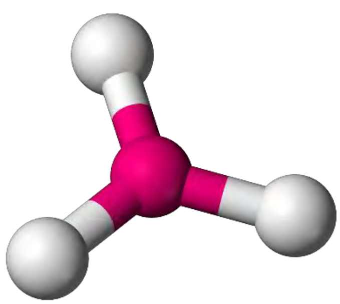


2. Draw the resonating structure of following:
   a. So_4^{--}
   b. H_2o_2
   c. Po_4^{3-}
   d. No_2^{-}

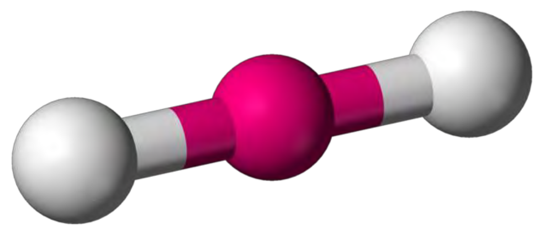
5. 7Valence Shell electron Pair repulsion (VSEPR) Theory and Shapes of some molecules

Activity:

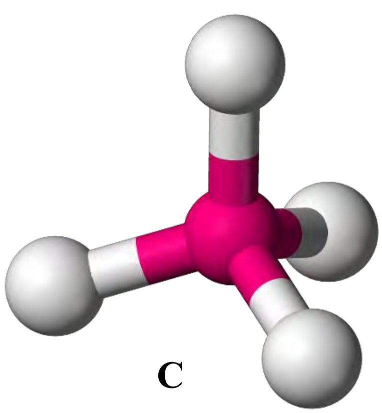
Identify the name of shapes (linear, angular, trigonal planar or tetrahedral) given below.



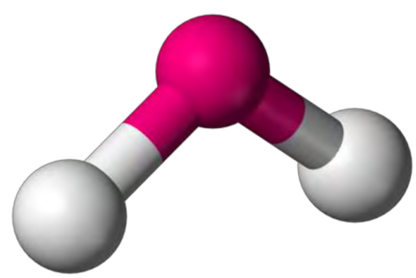
A



B



C



D

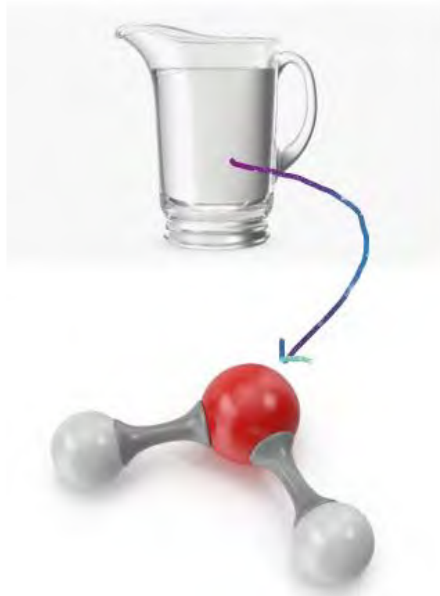


Figure: Glass of water and shape of water molecule

a. Identify the atoms present in the water molecule.
b. Name the type of bond present in the given molecule.
c. Relate the geometrical shape of water molecule with one of the shapes (linear, trigonal, angular or pyramidal).

Water in a container has no fixed shape but the molecules of water are bent or angular in shape. In this unit we will discuss the cause of the shape of the molecule.
The Lewis dot formula of a molecule is not enough to explain the geometry or shape of the molecule. Various experimental techniques like X- ray crystallography, nuclear magnetic resonance (NMR) or electron microscopy help scientists to determine the shapes of molecules and understand how their atoms are arranged.

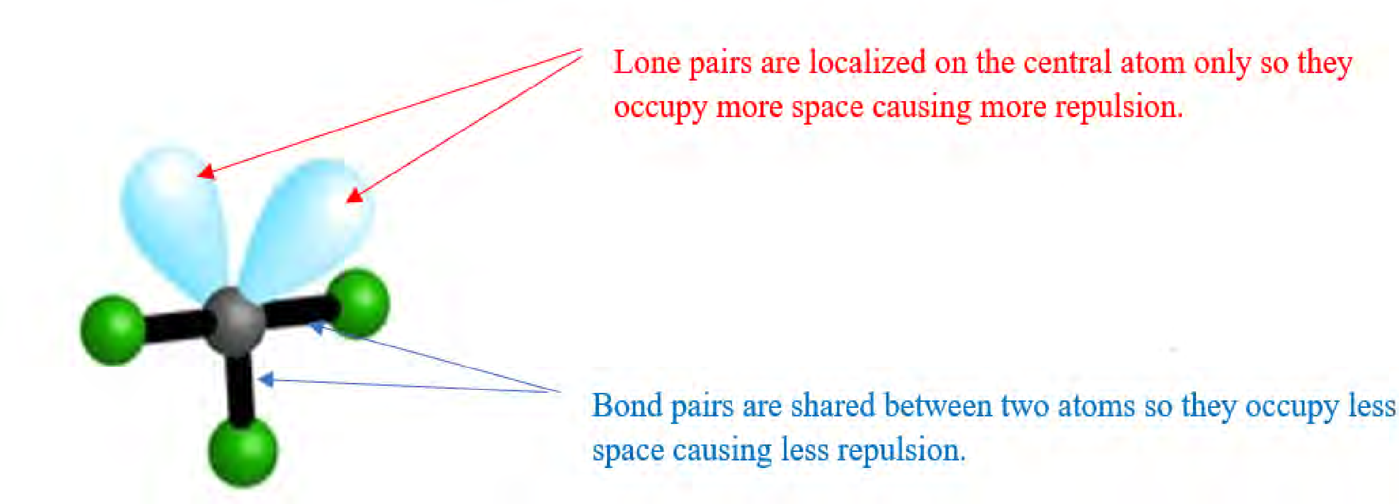
The valence-shell electron-pair repulsion (VSEPR) model predicts the shapes of molecules and ions by assuming that:

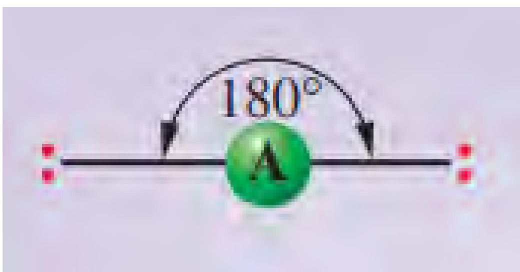
a. There are two types of electron pair around the central atom in a covalent molecule. They are bond pair (which form the bond) and lone pair (nonbonding pair)
b. These pair of electrons undergo mutual electrostatic repulsion in the order,

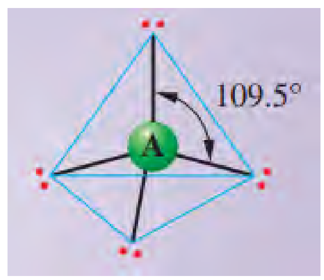
   Lone pair-lone pair > lone pair -bond pair>bond pair- bond pair

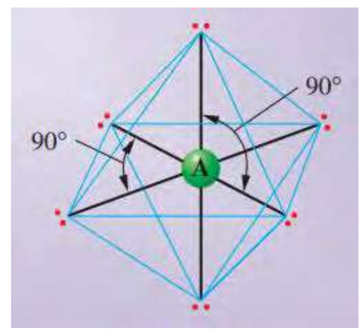
c. These electron pairs arrange themselves in space to minimize the repulsion. Hence, the shape of the molecule is determined.

   Why is repulsion between lone pair -lone pair greater than bond pair-pair?

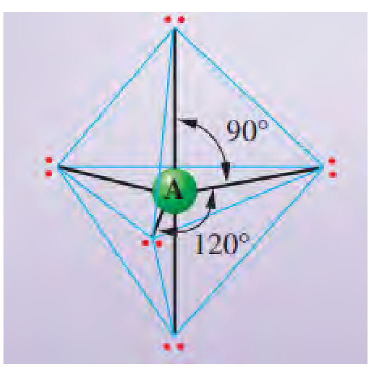


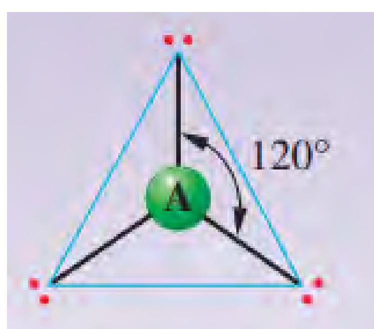






Let us consider the various polyatomic molecules of atom **A** having different bond pair electrons without lone pairs. The possible arrangements of electrons pair around the central atom "**A**" are shown. Identify the shapes of the given molecules (linear, Tetrahedral, Trigonal bipyramidal, trigonal planar or octahedral)

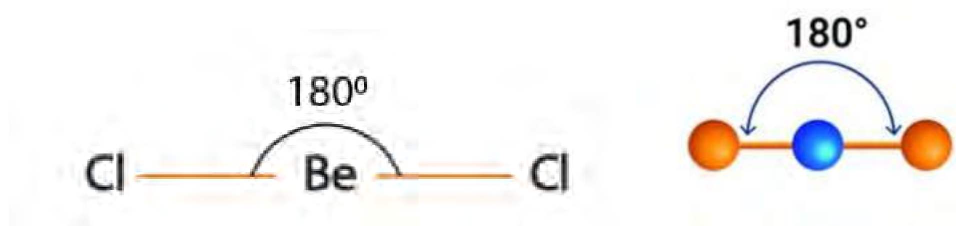




Shape of some common molecules

a. Molecule with two bond pair (AB_2 type)

These types of molecules have only two bond pairs of electrons around the central atom and the repulsion will be minimum when these pairs are in linear arrangement with bond angle 180 ^{0}. Therefore, these molecules have linear geometry. Examples $BeCl_2$.BeF_2 CO_2, etc.



Linear geometry

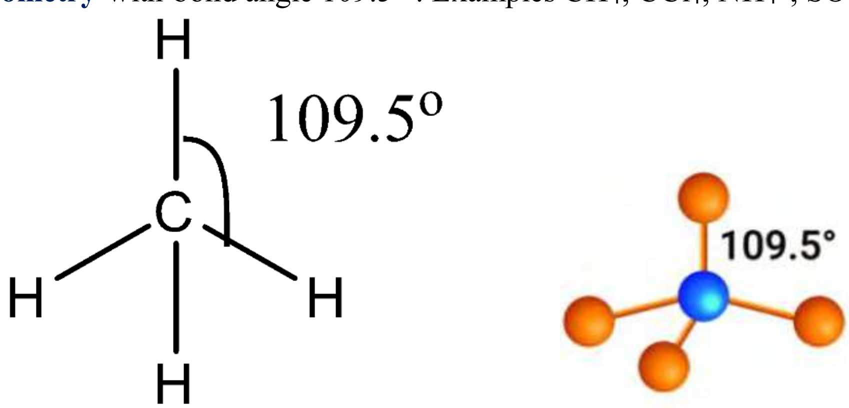
b. Molecules with three bond pair (AB_3 type)

If there are three bond pairs of electrons around the central atom of a molecule, the repulsion among these pairs will be minimum when they are 120^0 apart from each other in the same plane. So, the molecule has trigonal planar geometry**.** Examples, BF_3, BCl_3, $AlCl_3$

Trigonal planar geometry

c. Molecules with four bond pair electrons (AB_4 type)

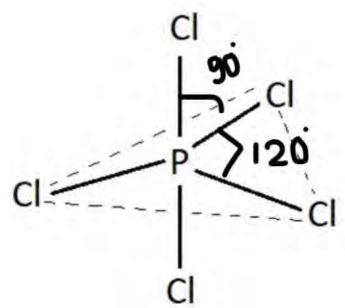
When there are four bond pairs of electrons around the central atom, the molecule has **tetrahedral geometry** with bond angle 109.5 ^{0}. Examples CH_4, CCl_4, NH_4^+, SO_4^{--}

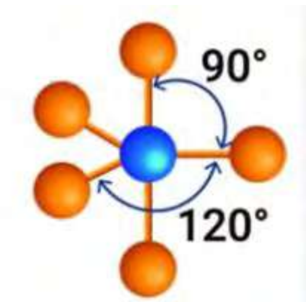


Tetrahedral geometry

d. Molecules with five bond pair electrons (AB_5 type)

When there are five bond pairs of electrons around the central atom, three pairs are located in a plane (**equatorial positions**), and two are situated along the axis perpendicular to that plane **(axial positions)**. This type of geometry is called **trigonal bipyramidal**. The bond angles between the equatorial atoms are 120 degrees, while the bond angles between an axial atom and the three equatorial atoms are 90 degrees. Examples, phosphorus pentafluoride (PF_5), phosphorus pentachloride (PCl_5)

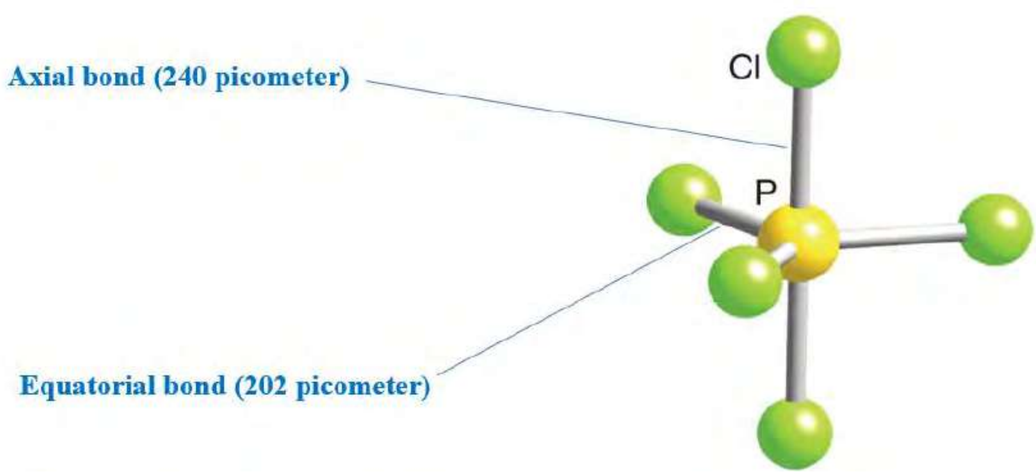




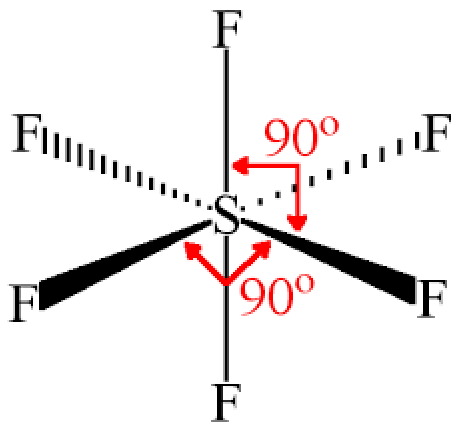
Trigonal bipyramidal geometry

Do you know?

In trigonal bipyramidal molecules, **the axial bonds** tend to be longer than the **equatorial bonds** because the axial positions experience greater repulsion than the equatorial positions.



Likewise, the molecules with six bond pair electrons have octahedral geometry with bond angle 90^0. Example SF_6.

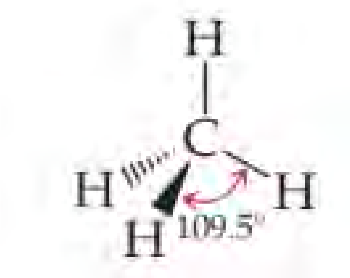


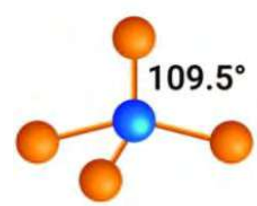
Octahedral geometry

Molecules with distorted geometry (with lone pair electrons)

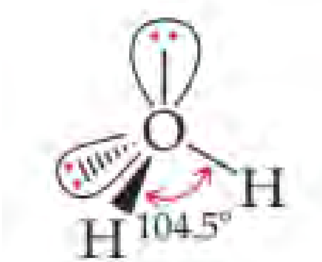
Activity:

Structures of methane and water molecules are given:





Tetrahedral shape of methane molecule



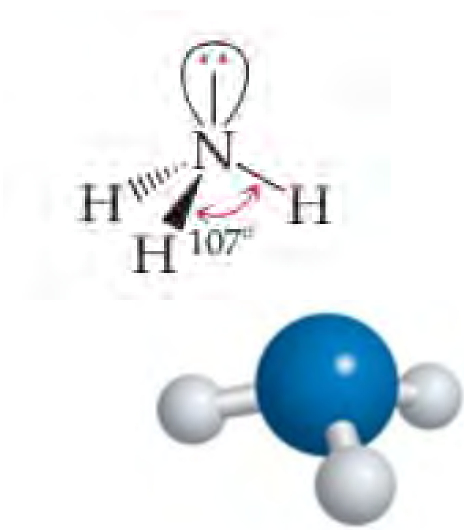
Bent shape of water molecule

a. Find the number of bond pair and lone pair electrons in the given molecules.
b. Why is the shape of these molecules different?

The number of electron pairs around the central atom in CH_4, H_2O and NH_3 are the same i.e. four. But, the shapes of these molecules are different. In CH_4, all the electrons are **bond pairs,** so the repulsion is equal (**bond pair-bond pair)** making the molecular shape **tetrahedral**.

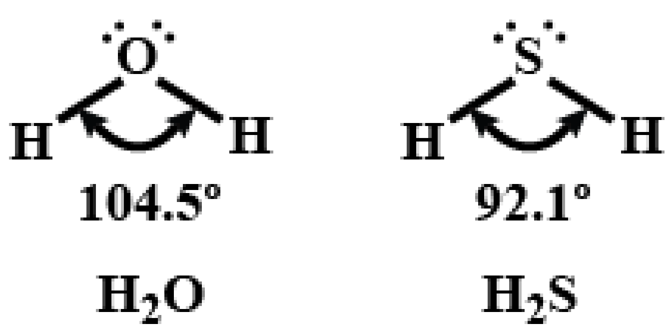
In the case of water molecules among the four-electron pair two are bonding (bond pair) and two pairs are nonbonding (lone pairs). Lone pair -lone pair repulsion is greater than lone pair -bond pair. It causes the molecule to be **bent or angular** in shape with bond angle 104.5^0.

Likewise, in NH_3 molecules only three pairs are bonding and one pair is nonbonding. The repulsion is minimum when these three bonding pairs are at the trigonal base of pyramid resulting trigonal pyramidal shape of the molecule with bond angle 107^0.



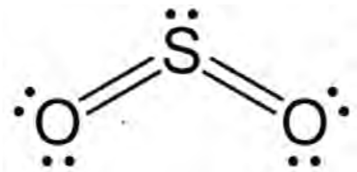
Trigonal pyramidal shape of ammonia molecule

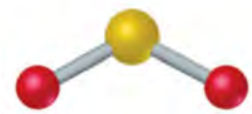
The shape of H_2O and H_2S molecules are the same (Bent) but have different bond angle. Oxygen atom in water molecule is more **electronegative** than sulphur atom in Hydrogen sulphide molecule. So oxygen atom attracts the bond pair electron towards it more causing more repulsion and increases the bond angle in H_2O ($104.5\ ^0$) than H_2S (92.1^0).



When there is a double bond in a molecule (4 electrons), it is considered as one bond pair electrons. For example, in CO_2, there are two double bonds between carbon and oxygen atoms which are considered as two bond pairs resulting in a linear shape.
Similarly, SO_2 has two double bonds between carbon and oxygen atoms. But due to the presence of one lone pair of electrons in the Sulphur, the molecule has a bent shape.

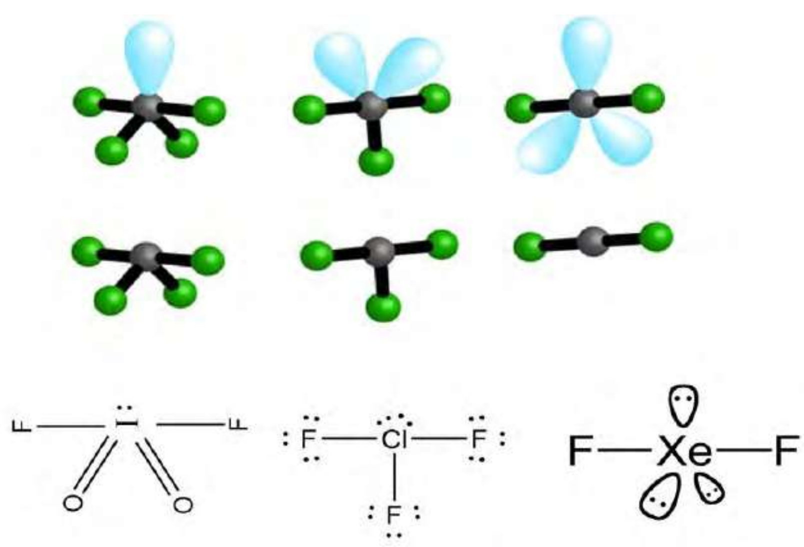
linear shape of CO_2





Bent shape of SO_2

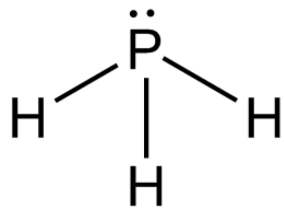
Do you know?
Molecules with the same atomicity (number of atoms) may have different shapes due to different number of bond pairs and lone pairs.

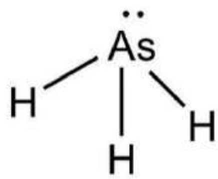


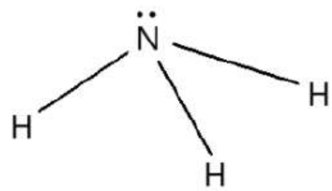
Sea saw shaped T-shaped linear

Test yourself!
Some of the Hydrides of group 15 elements are given. They have similar shapes but different bond angles.

a. Name the shape of these molecules and give reason for the same shape.
b. Why is the bond angle different?
c. Arrange these molecules in increasing order of bond angle.



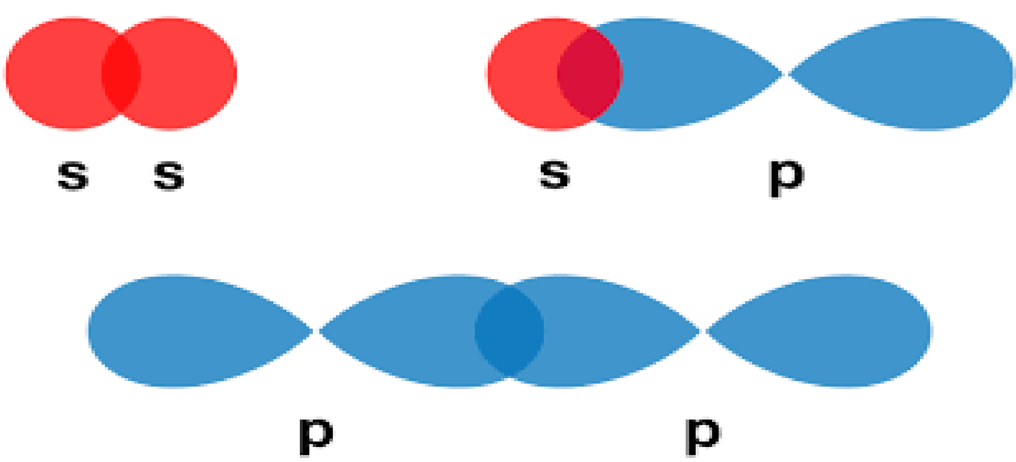




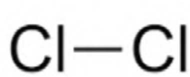
Linus Pauling (1901-1994) Valence Bond Theory (VBT) was developed by Linus Pauling in the 1930s. Pauling was awarded the Nobel Prize in Chemistry in 1954 for his research on the nature of the chemical bond, which included his work on VBT.Pauling was known not only for his scientific contributions but also for his strong stance on social and political issues.

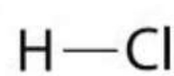
e.8 Elementary idea of valence bond theory (VBT)

Activity: Observe the overlapping of atomic orbitals in the figure given.



a. What is the shape of s and p orbitals given?
b. Identify the overlapping given that can form the molecules below.

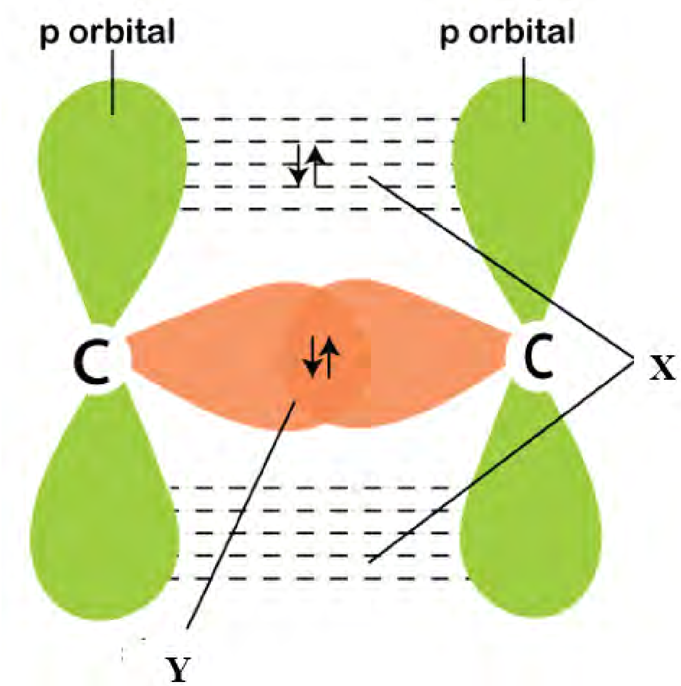




Valence-shell electron-pair repulsion (VSEPR) model usually predicts the correct general shape of a molecule. It does not explain the formation of chemical bonds between the atoms. **Valence bond theory (VBT)** developed and improved by Linus Pauling is based on quantum mechanics. It provides further understanding about the reason for formation of covalent bonds. It also explains that bonds have definite directions in space, so that the molecules have particular molecular geometry. It is based on certain fundamental postulates.

a. All atoms have atomic orbitals which are responsible for holding electrons and each orbital cannot contain more than two electrons.
b. Atoms form covalent bonds by the overlapping of their only half filled atomic orbitals.
c. Strength of bonding depends on the amount of overlap. When overlapping increases, the strength of the bond increases.
d. When there is a **head-on overlapping** of atomic orbitals, a **sigma bond** is formed and when there is a **lateral** overlapping, a **pi bond** is formed.
e. When atomic orbitals overlap to form a bond, the overall energy of the system decreases. As a result, the molecule is stable.

Sigma bond and pi bond.

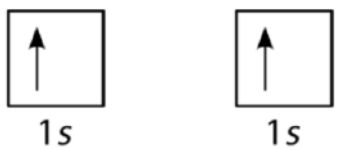


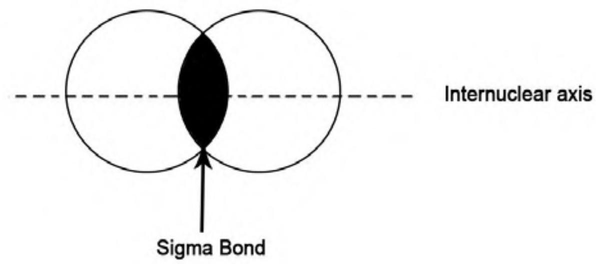
a. What is the difference in overlapping X and Y?
b. Which overlapping will result in a strong bond?

Sharing of electrons in the outermost shell involves overlapping of atomic orbitals (s, p, d or f). Depending upon the nature of overlapping of atomic orbitals, covalent bonds may be sigma(σ) bond or pi(π) bond.

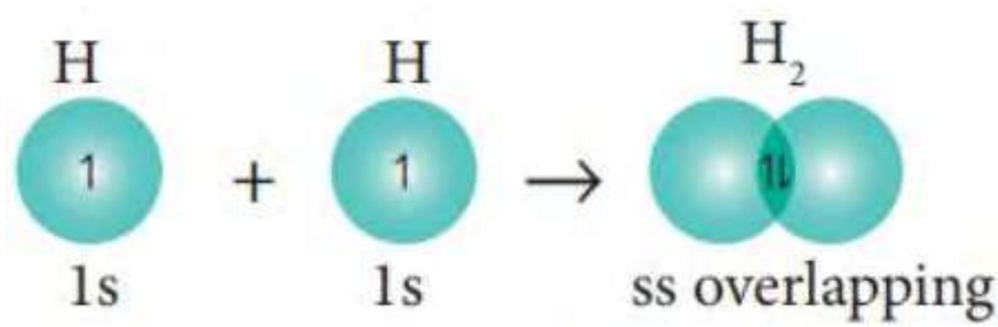
Let us explain the formation of hydrogen molecules. Two half-filled atomic orbitals (s -orbital) of each hydrogen atom overlap with each other and a single bond (sigma bond) is formed between the atoms.

H $1S^1$ + H $1S^1$





s- s orbital overlapping



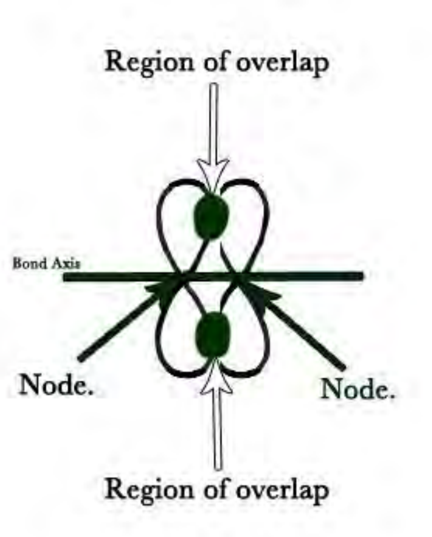
The type of covalent bond which is formed by head on overlapping of atomic orbitals is called sigma bond.

There are three types of overlapping which result in sigma bonds; they are s-s overlapping, s-p overlapping and p-p overlapping.

Sigma bond is a strong bond because there is a direct overlap and maximum interaction between the atomic orbitals and their electrons. The Sigma bond passes through the internuclear axis. There is a free rotation of atoms without breaking the bond.

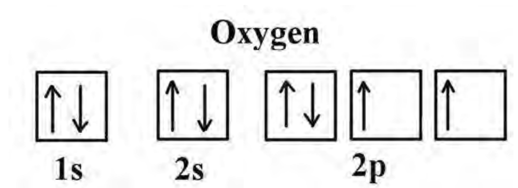
Note: hybridized orbitals (will be discussed later) always have head on overlapping and forms sigma bond

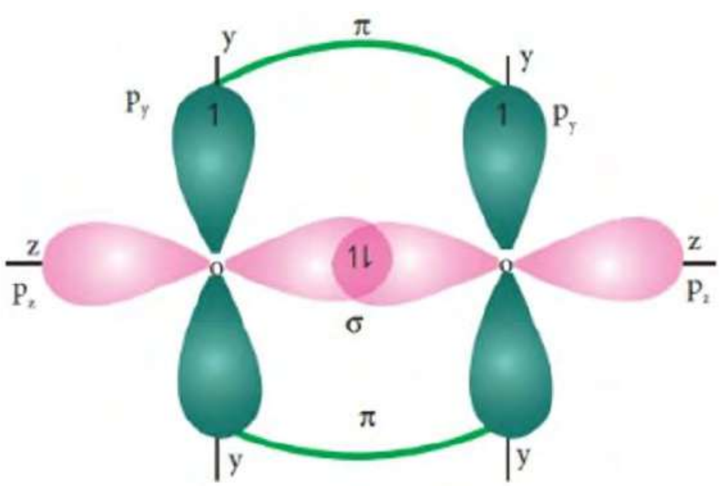
When p-orbitals overlap laterally (side to side) then the bond formed is called Pi- bond. Since the extent of overlapping is less than head-on overlapping, the pi bond is weaker than sigma bond. This bond passes perpendicular to the internuclear axis. Unlike sigma bonds, pi bonds do not allow free rotation around the bond axis. Sigma bond is essential for the formation of pi bonds.

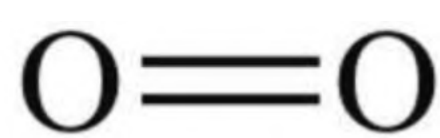


The sideways overlaping results the more electron density above and below the plane of bonding atoms

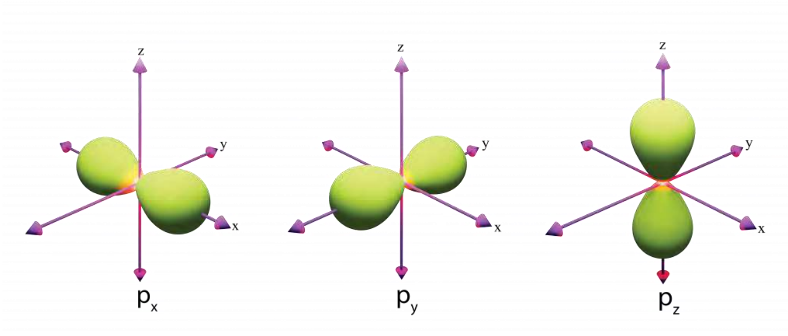
In the case of oxygen molecules there are two half-filled atomic orbitals, one of these orbitals overlap by head on overlapping forming **sigma bond** while other overlap laterally (side to side) to form **pi bond.**

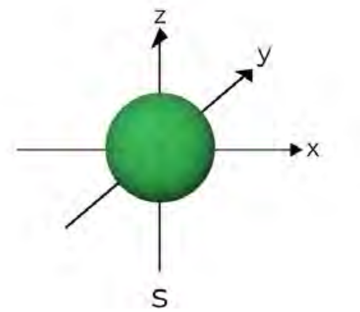






s- orbital is spherical in shape having equal distribution of electrons in all directions so there is only head on overlapping of these orbital and can form only sigma bonds. Whereas p – orbitals are dumb bell in shape so the overlapping of electron clouds can occur laterally or head on forming both sigma and pi bonds.

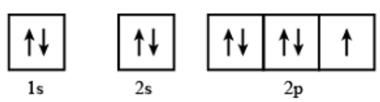


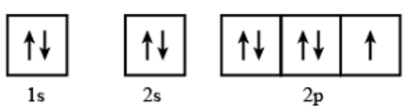


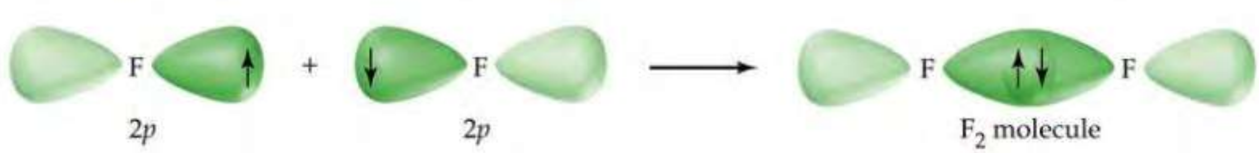
Let us explain the formation of some covalent molecules based on valence bond theory(VBT).

Fluorine molecule(F_2)

F $1s^2\ 2s^2\ 2p^5$ **F $1s^2\ 2s^2$ 2p5**

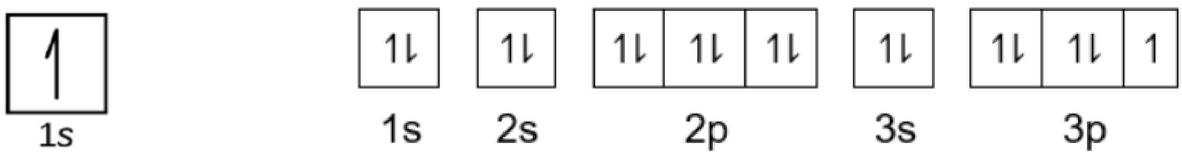


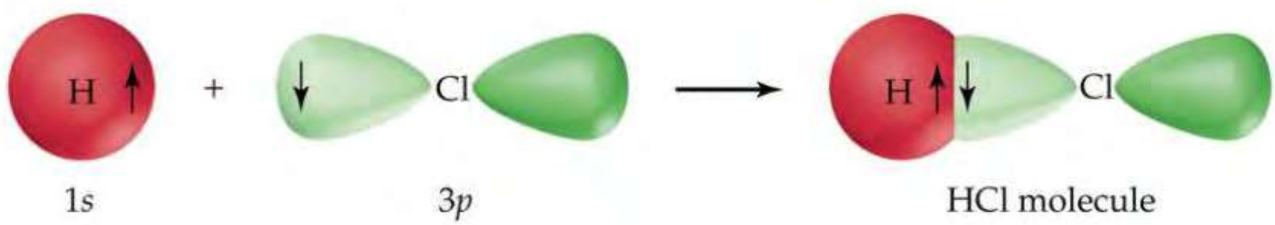


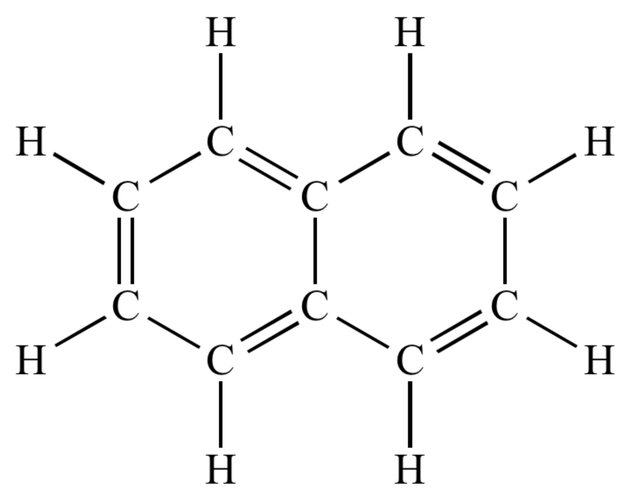


Formation of HCl molecules involves the overlapping of half-filled s-orbital of hydrogen atom and Half-filled p-orbital of chlorine atom as shown.

H $1S^1$ **Cl $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$**







Test yourself!

a. Write the molecular formula of the given compound
b. Find the number of sigma and pi bonds in the molecule.

e.9 Concept of Hybridization

Activity:

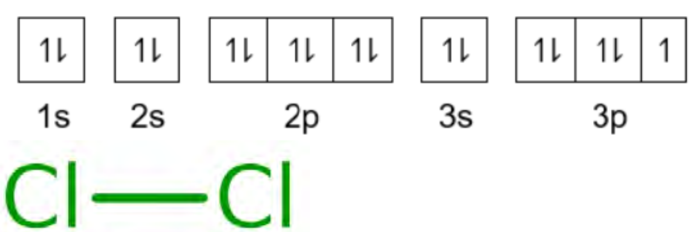
a. Take two measuring cylinders.
b. Fill one cylinder with 10 ml of red ink and the other with 10 ml of blue ink.
c. Carefully pour both inks into a separate container and mix them thoroughly.
d. Observe the resulting color of the mixture and discuss the reason why it is different from the previous color?

A **B**

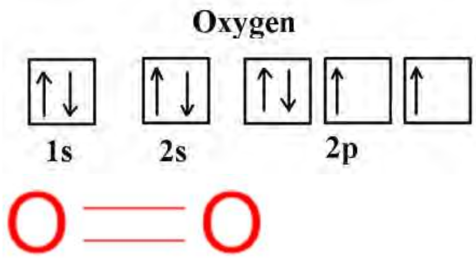
What happens to the taste and color of the mixture:

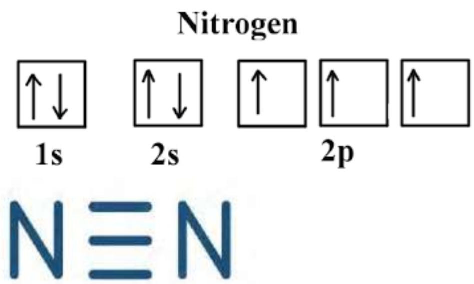
a. When 20 ml of each juice A is mixed with 20 ml of B?
b. When 20 ml of A is mixed with 60 ml of B?

From the discussion of VBT we know that the number of half-filled atomic orbitals is equal to the number of bonds in the molecule. For example, a chlorine atom has one half-filled (singly filled) p-orbital, resulting in a single bond in the Cl_2 molecule.

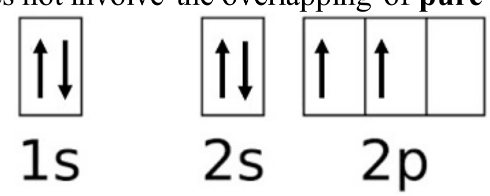


Similarly, oxygen and nitrogen atoms have two and three half-filled atomic orbitals, respectively, leading to double and triple bonds in these molecules

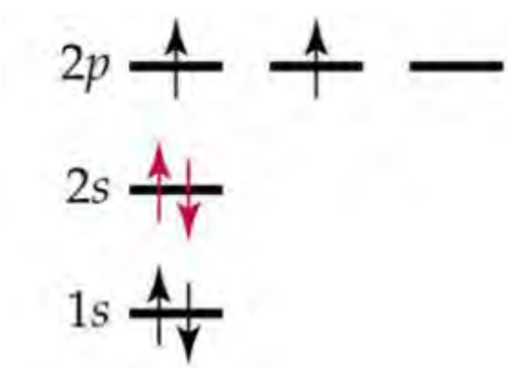




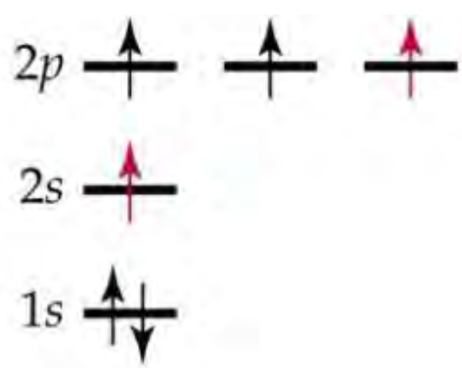
However, considering the case of formation of methane molecules, the number of half-filled atomic orbitals is not equal to the number of bonds formed. Carbon has only two half-filled p- orbitals but in CH_4 carbon forms four bonds with four hydrogen atoms. It means formation of methane does not involve the overlapping of **pure p- orbitals of carbon**.



One of the electrons from 2s- orbital is excited to higher (2p) orbital and forms four half-filled orbitals.

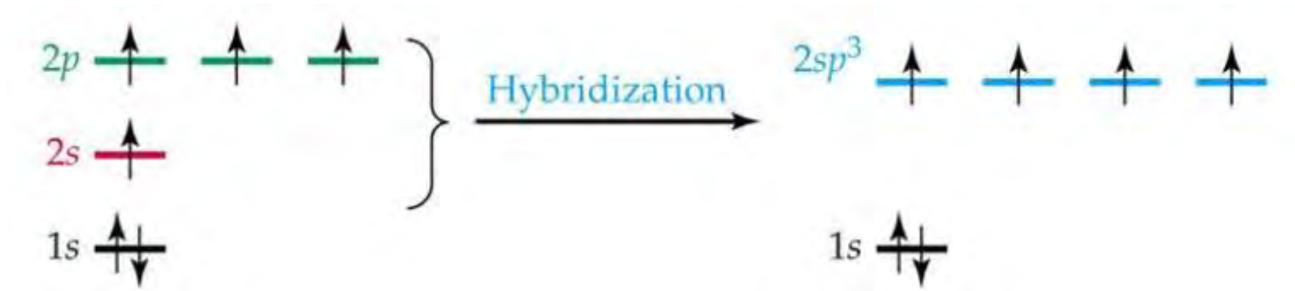


Ground state electronic configuration of carbon



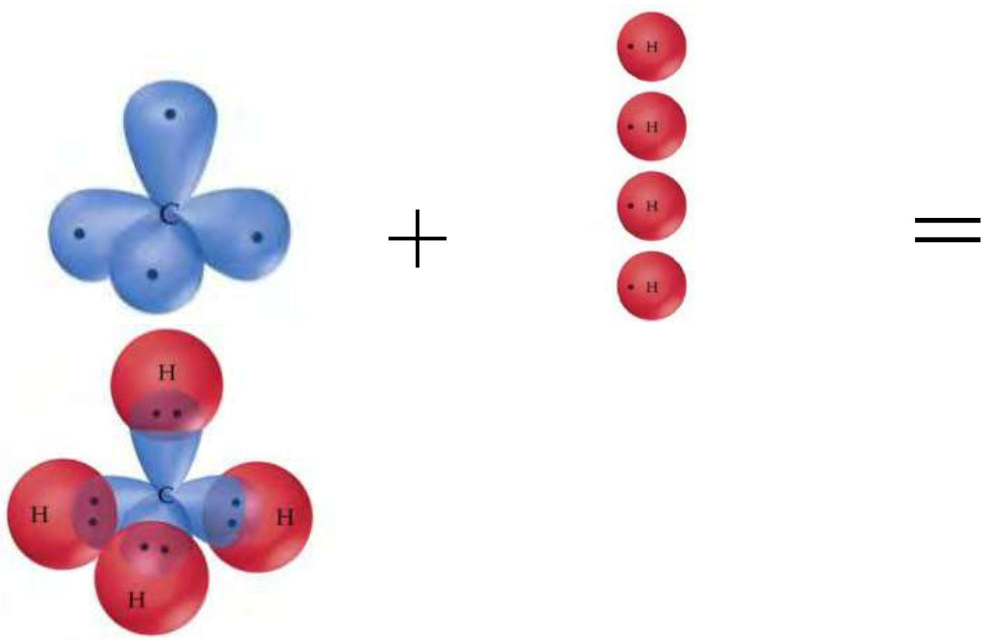
excited state electronic configuration of carbon

But these pure orbitals are not used for bonding, instead these orbitals combine to form four new and equivalent orbitals called **hybrid orbitals**. **The process of formation of hybrid orbitals from pure atomic orbitals is called hybridization.** Since one s- orbital mix with three p-orbitals, the new orbitals formed are called sp3 hybrid orbitals and the phenomenon is called sp3 hybridization.



Excited state electronic configuration of carbon ***sp3 hybridized configuration of carbon***

These four sp3 hybrid orbitals overlap with four s – orbitals of hydrogen atom to form tetrahedral methane molecules.



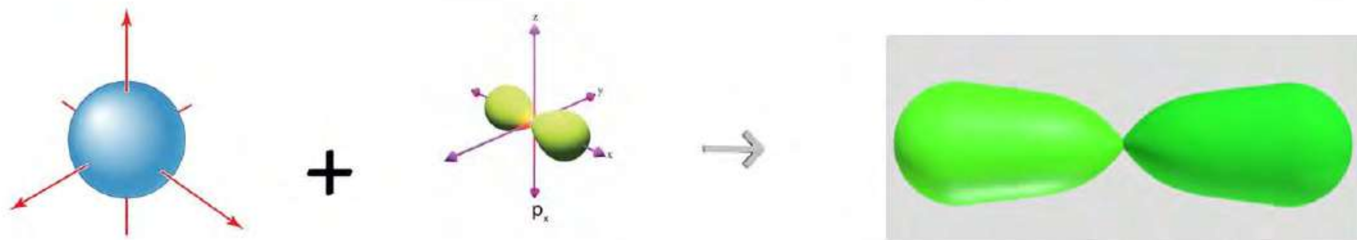
Hybridized orbitals formed have certain characteristics, they are:

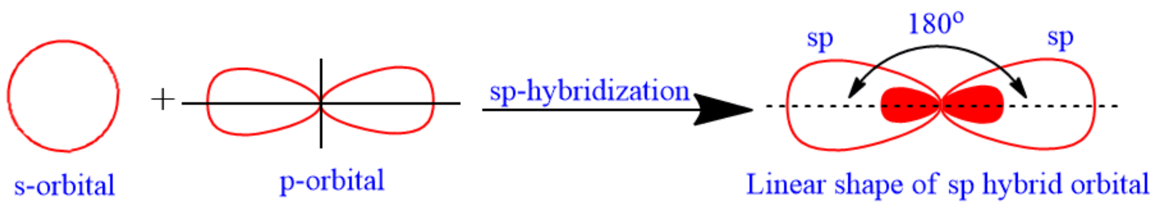
a. The number of hybrid orbitals formed always equals the number of atomic orbitals used.
b. A set of hybrid orbitals always have definite directional characteristics.
c. Hybrid orbitals are equivalent in energy also called degenerate hybrid orbitals.

Types of hybridizations involving s and p orbitals

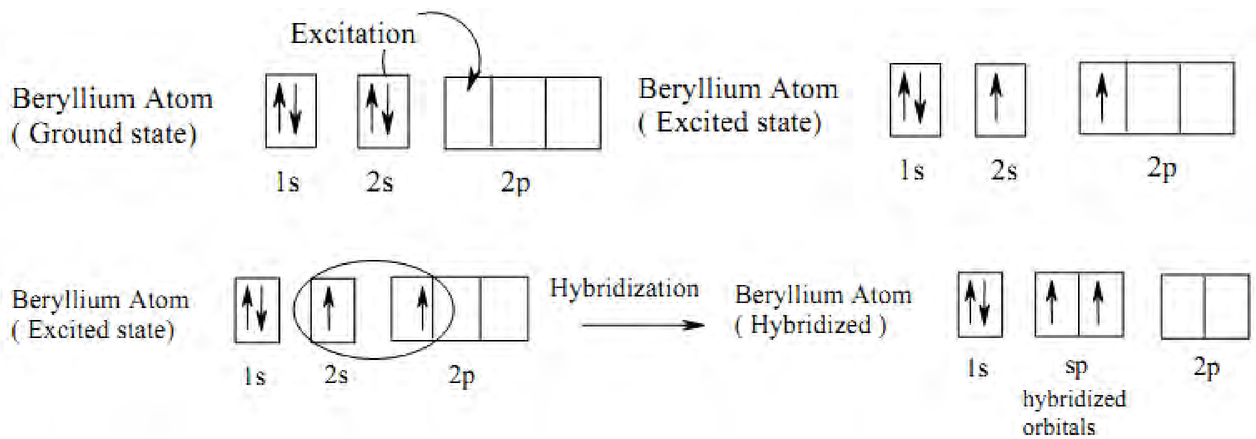
a. sp hybridization

When one s- orbital mixes with one p-orbital to form two identical hybrid orbitals, it is called **sp hybridization or linear or diagonal hybridization**. The resulting two sp hybrid orbitals remain in opposite directions in a straight line at an angle of 180^0 angle making a linear shape.
sp hybridization forms a sigma bond but leaves the space for the creation of pi bonds. Examples $BeCl_2$, CO_2, C_2H_2, etc.

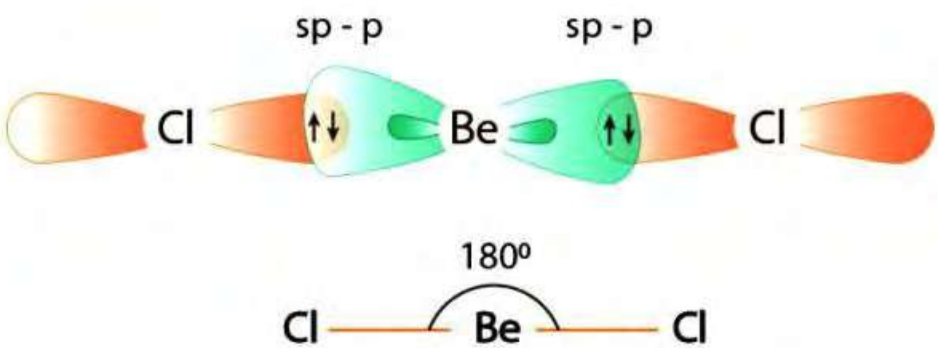




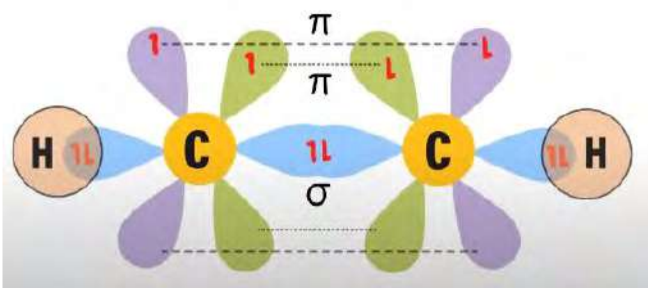
The formation of $BeCl_2$ with sp hybridization is shown as:



These **sp** hybridized orbitals overlap with the two p-orbitals of chlorine atoms to form a linear $BeCl_2$ molecule.



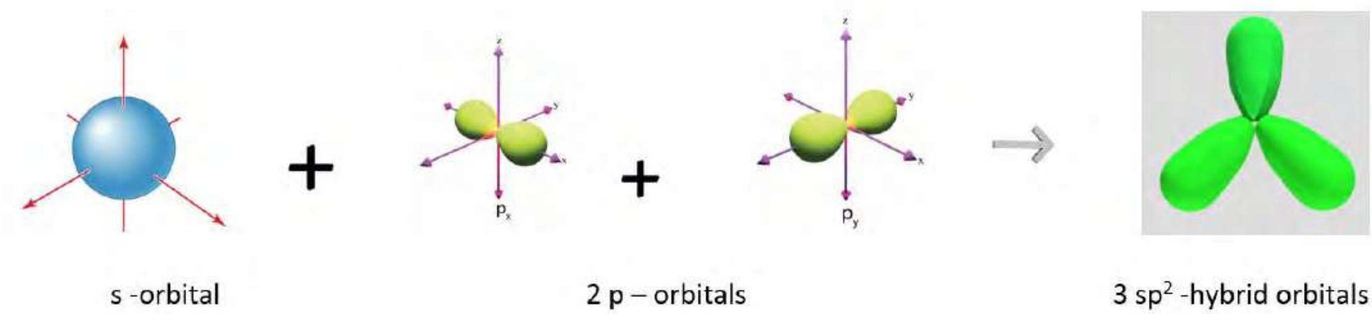
In the case of ethyne, among the four half-filled atomic orbitals of excited carbon atom, only one is used to mix with s -orbital forming two **sp -** hybrid orbitals and the remaining two unhybridized p- orbitals overlap laterally forming two pi bonds.

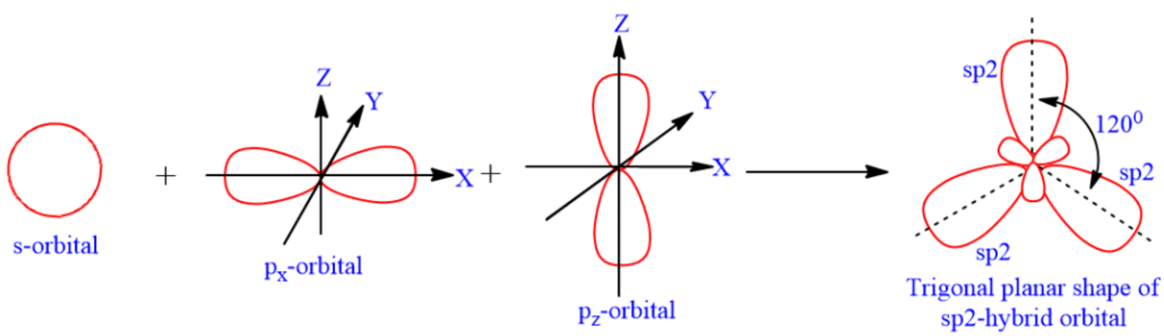


$H - C \equiv C - H$

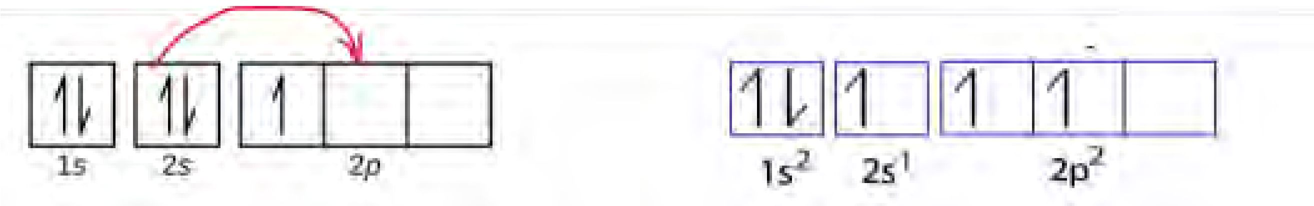
b. Sp^2 hybridization

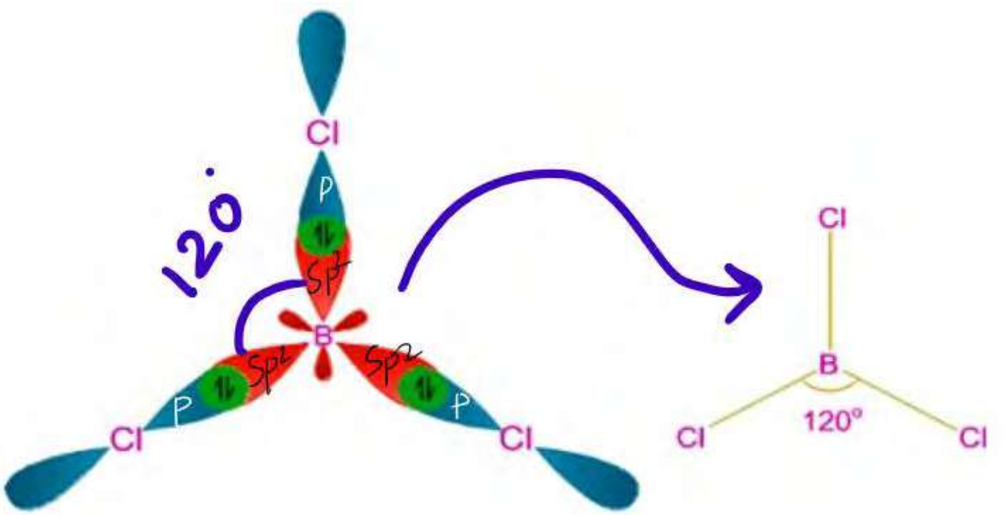
When one s- orbital mixes with 2 p-orbitals forming 3 new hybrid orbitals, it is called sp^2 hybridization. These 3 sp^2 hybrid orbitals lie in the same plane and directed at 120^0 angles to each other making a trigonal planar shape. BF_3, $AlCl_3$, C_2H_4, benzene, etc.



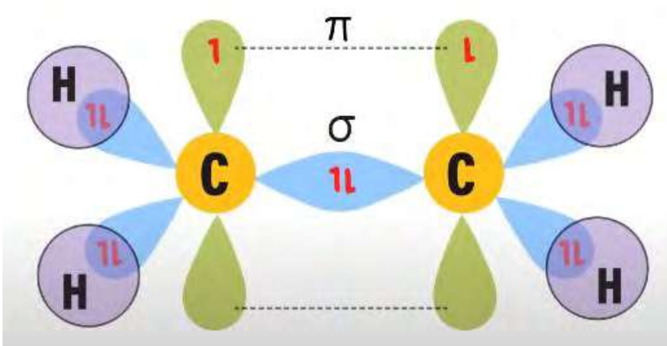


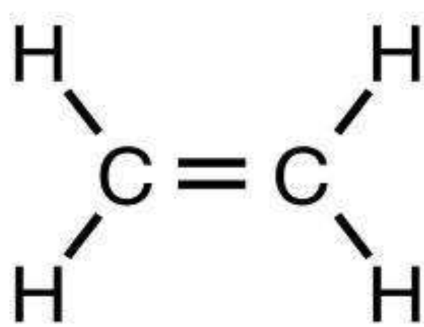
In the formation of BCl_3, the three sp^2 hybrid orbitals of boron overlap with the three pure p-orbitals of chlorine to form three sigma bonds which are at 120^0 forming trigonal planar geometry.





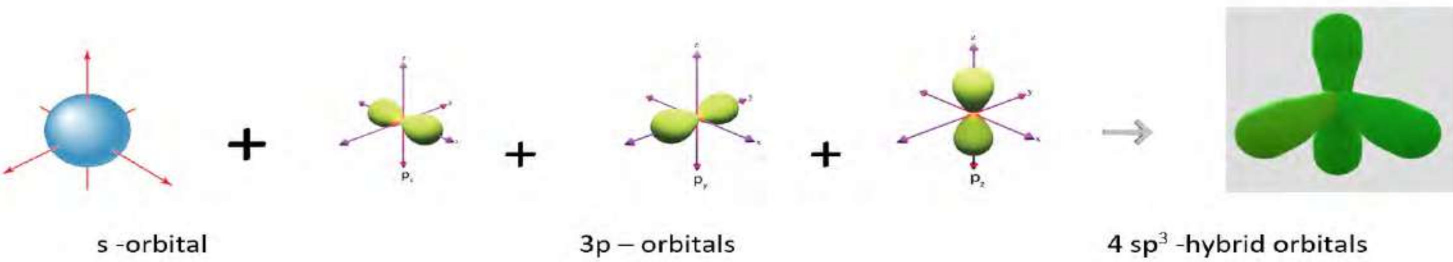
In case of ethene among the three p- orbitals of excited carbon atoms, only two are used to form sp2 hybrid orbitals. The unhybridized p orbital is perpendicular to the plane of the sp2 hybrid orbitals. Three sp2 hybrid orbitals overlap with each other or with the s- orbital of the hydrogen atom and form three sigma bonds. This unhybridized p – orbitals of two carbon overlap with each other and form pi bond as shown:

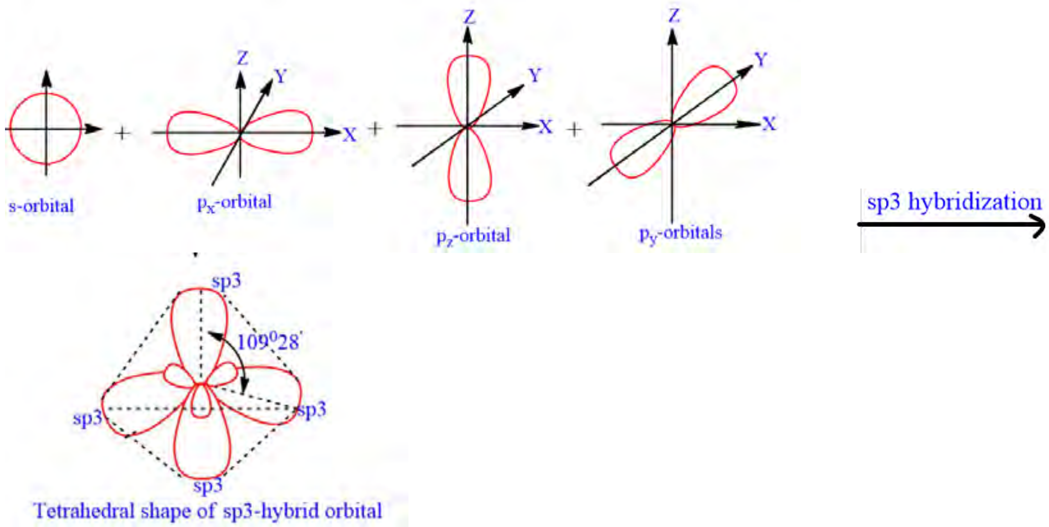




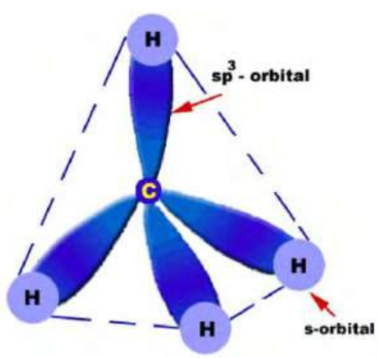
c. sp^3 hybridization

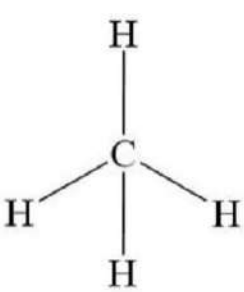
When one s-orbital combines with three p-orbitals to form four hybrid orbitals, it is called sp^3 hybridization. These four sp^3 hybrid orbitals point in tetrahedral directions with bond angle 109.5^0. Examples include $H_2O.NH_3$, CH_4, methyl radical, etc.





Formation of sp^3 hybridization in the carbon atom in CH_4 is discussed earlier. These four hybrid orbitals overlap with the s- orbitals of four hydrogen atoms to form tetrahedral molecules of methane.



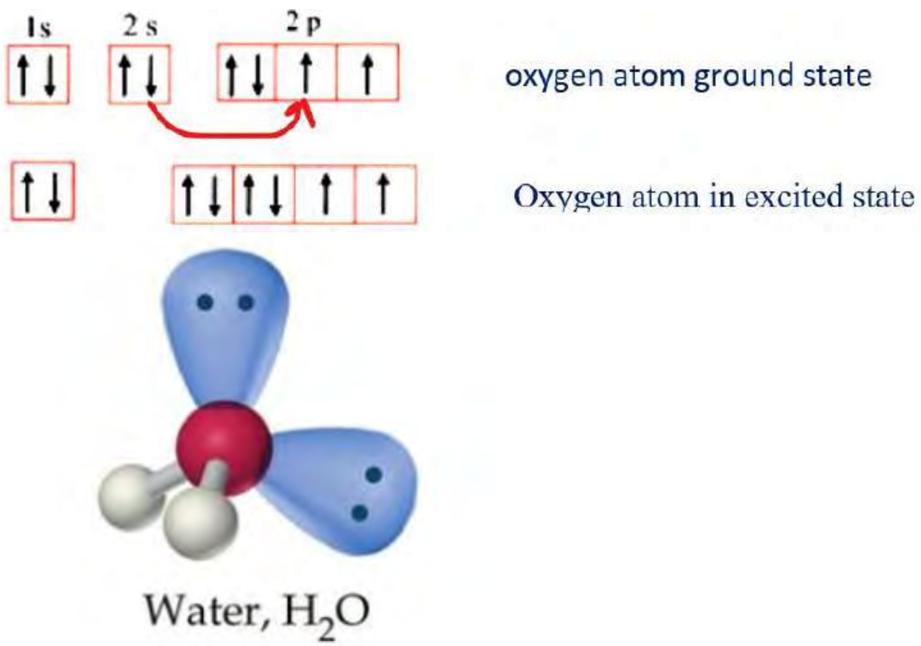


Do you know?

Size of hybridized orbitals increases in the order $sp<sp^2<sp^3$
Percentage of s- character increases in the order

$$sp > sp^2 > sp^3$$

In the case of water, among the four hybrid orbitals of oxygen atoms, two are fulfilled and two are half filled. Only half filled hybrid orbitals are used to overlap with s-orbitals of hydrogen atoms. The unused full filled hybrid orbitals will remain as lone pair. Because of these lone pair - lone pair repulsion, the shape of H_2O molecule becomes bent instead of tetrahedral as expected from sp^3 hybridization.

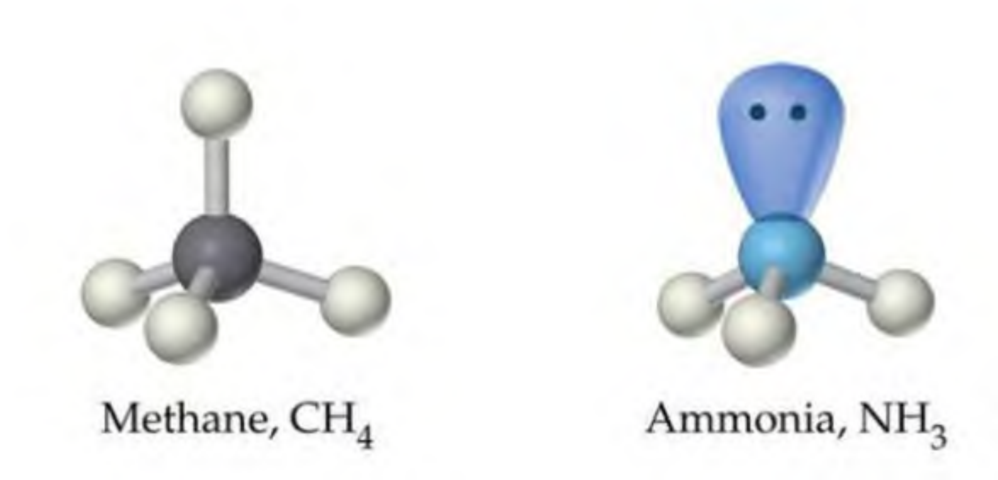


Test yourself!

Shape of two molecules having same hybridization are given.

a. What is the hybridization of these molecules?
b. Name the shape of these molecules.

c. Explain the reason for having different shapes of these molecules though they have same hybridization.

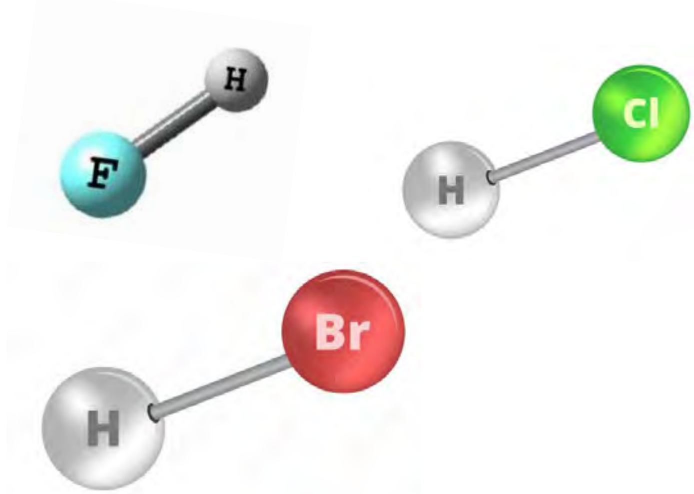


e.10 Bond Characteristics

Activity:

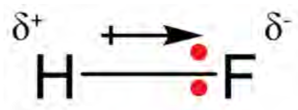
Observe the molecules of hydrogen halides given.

a. Arrange the halogen atoms in increasing order of size.
b. Arrange these molecules in increasing order of bond length and bond strength.



We have discussed the formation of covalent bonds by the overlapping of atomic orbitals. In this chapter we will discuss the different properties of bond whether it is long or short or whether it is strong or weak.

5.10.1 Dipole moment and polarity of bond



a. Identify whether the bond in the above molecule is ionic or covalent and why?
b. Which one is polar and which one is non-polar? Give a reason.

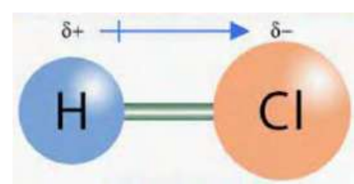
We have discussed the development of polarity in a covalent molecule previously. Dipole moments measure the polarity of a molecule. **The separation of charge due to shifting of bonded electrons towards more electronegative atoms is called dipole**.
Dipole moment(μ) is a vector quantity and can be calculated mathematically by using the relationship

μ=charge × distance

Dipole moment is usually measured in Debye unit (D) after the name of chemist Peter Debye.

1 Debye = 3.336×10^{-30} coulomb-meters

All homo diatomic molecules (H_2, O_2, Cl_2) are nonpolar because the electronegativity difference between the bonded atoms is zero and there is no formation of dipole (μ=0). But, all hetero diatomic molecules (HCl, HF) have dipole moment > 0 means they are polar. because shared pairs of electrons shift towards more electronegative atoms.

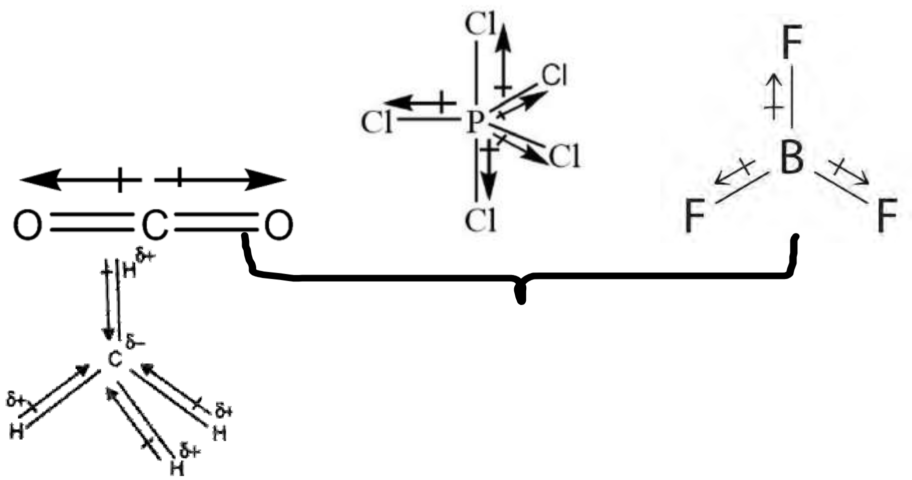


Nonpolar molecule (μ)=0

polar molecule (μ)>0

The individual bond dipoles depend on the relative electronegativity of the bonded atom whereas the overall dipole moment of the molecule depends on the molecular shape.

If the bond dipoles cancel each other, the molecule will have no dipole making the molecule nonpolar. For example, CO_2 and BeF_2 are nonpolar molecules because individual bond dipoles cancel each other due to their linear shapes.

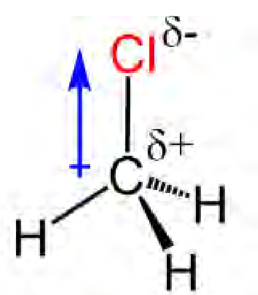


Nonpolar molecules Polar molecule

Likewise, CH_4, CCl_4 are nonpolar though the C-H and C-Cl bond is polar. It is because of their tetrahedral geometry; bond dipoles cancel each other making the net dipole moment zero. Similarly, BF_3, PCl_5, SF_6 are non-polar molecules.

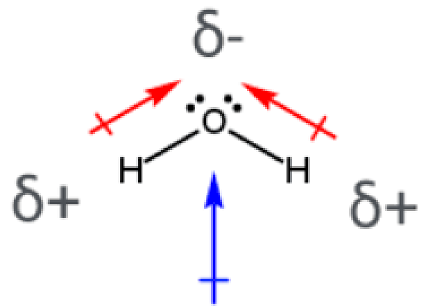
> **Test yourself!**
>
> Draw the direction of dipole in each bond and predict the polarity of molecule given
>
> Which of the following molecule is most polar? Why?

When individual bond dipoles do not cancel each other, the molecule has a very large net dipole. It makes the molecule polar. For example, chloromethane (CH_3Cl) is a polar molecule because of net dipole moment greater than zero.



(μ)=1.87 D (polar)

Some of the neutral molecules like water and ammonia are polar due to the lone pair of electrons in the central atom. The lone pair develops a slight negative charge(δ^-) making the other end slightly positive(δ^+) resulting in a highly polar molecule.



(μ)=1.84 D (polar)

5.10.2 bond length

Bond	H—F	H—Cl	H—Br	H—I
Length	0.917 Å	1.275 Å	1.415 Å	1.609 Å

a. Identify the molecule in which the halogen atom is largest in size.

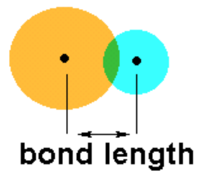
b.

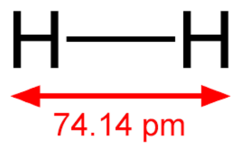
molecule	Bond length (Pm)
H-H	74.17
F-F	144
Cl-Cl	199
Br-Br	228
H-F	92
H-Cl	127
H-Br	141
H-I	160

Table: bond lengths of some diatomic molecules

c. Which of the given molecules has the longest bond length and why?

Bond length refers to the distance between the nuclei of two bonded atoms. Bond length is typically measured in picometers (pm) or angstroms (Å). One picometer $=10^{-12}$m and one angstrom $=10^{-10}$m. Bond lengths are determined experimentally by using x-ray diffraction or molecular spectroscopy.





There are various factors that affect the bond length.

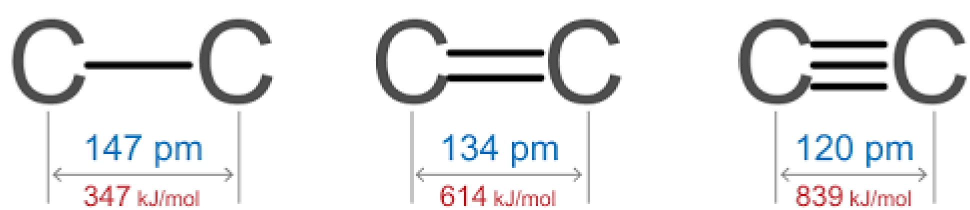
a. When atomic size is large the electrons are far from the nucleus. It makes the atomic radius larger and increases the bond length. For example, the difference in bond length among halogen molecules is mainly due to differences in their atomic sizes. Fluorine molecules have shorter bond lengths(144pm) compared to chlorine molecule(199pm) and bromine molecule(228pm).

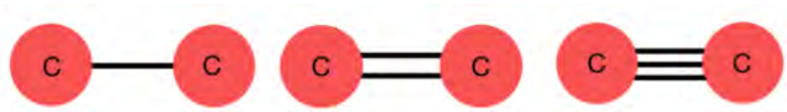
b. If one atom in the bond is more electronegative, it pulls the shared electrons closer, making the bond shorter. For example, H-F bond is shorter than H-Cl bond and H-Br due to decrease in electronegativity of halogen atoms.

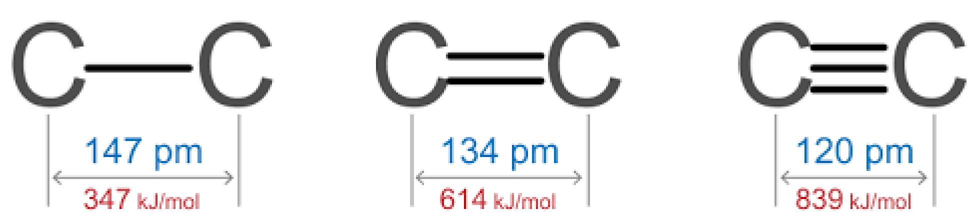
c. When the number of bonds between the atoms increases, the electron pair also increases which increases the force of attraction. Therefore, double and triple bonds are shorter than single bonds between the carbon atoms.

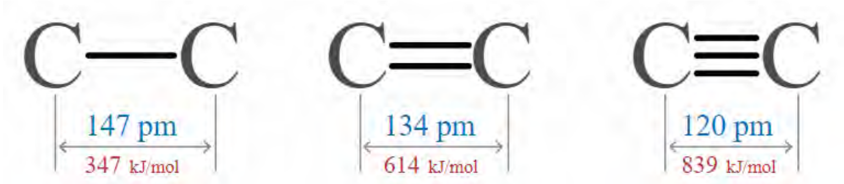
Remember:

Bond length is inversely proportional to bond strength. Large the bond length means weaker the force of attraction between the bonded atoms and has low **bond energy (kJ/mole)** (energy required or release to break/form 1 mole of bond)









5.10.3 Top of Form

Ionic character in covalent bond

Element	Li	Be	B	H	C	N	O	F
Electronegativity in Pauling scale(χ)	1.0	1.6	2.0	2.1	2.5	3.0	3.4	4.0

a. Which of the atoms is most electronegative? Why?
b. Identify two elements having the highest electronegativity difference.
c. Li F is ionic but HF is covalent. Why?

If the electronegativity difference between the bonded atoms increases, the ionic character also increases in the covalent bond.

More the electronegativity difference means more the percentage of ionic character in the covalent bond. The percentage ionic character in a covalent bond can be calculated by using the given relationship.

$$\%\text{ ionic character} = \frac{observed\ dipole\ moment}{maximum\ posible\ dipole\ moment} \times 100$$

Where, observed dipole moment= actual dipole moment of the molecule
$Maximum\ possible\ dipole\ moment$= dipole moment that would exist if the bond were completely ionic.

Molecule	Li-F	Be-F	B-F	C-F	N-F	O-F	F-F
Electronegativity difference ($\Delta\chi$)	3.0	2.4	2.0	1.5	1.0	0.6	0
%ionic character	89	76	63	40	22	9	0

Covalent character increases →

← **Ionic character increases**

Table: ionic and covalent character in some bonds

Whether the bond is pure covalent, polar covalent or ionic depends on the electronegativity difference between the bonded atoms.

Pure Covalent Bond: $\Delta\chi \approx 0$	Polar Covalent Bond: $0.5 < \Delta\chi < 1.7$	Ionic Bond: $\Delta\chi > 1.7$
		

Figure: nature of covalent bond and electronegativity difference

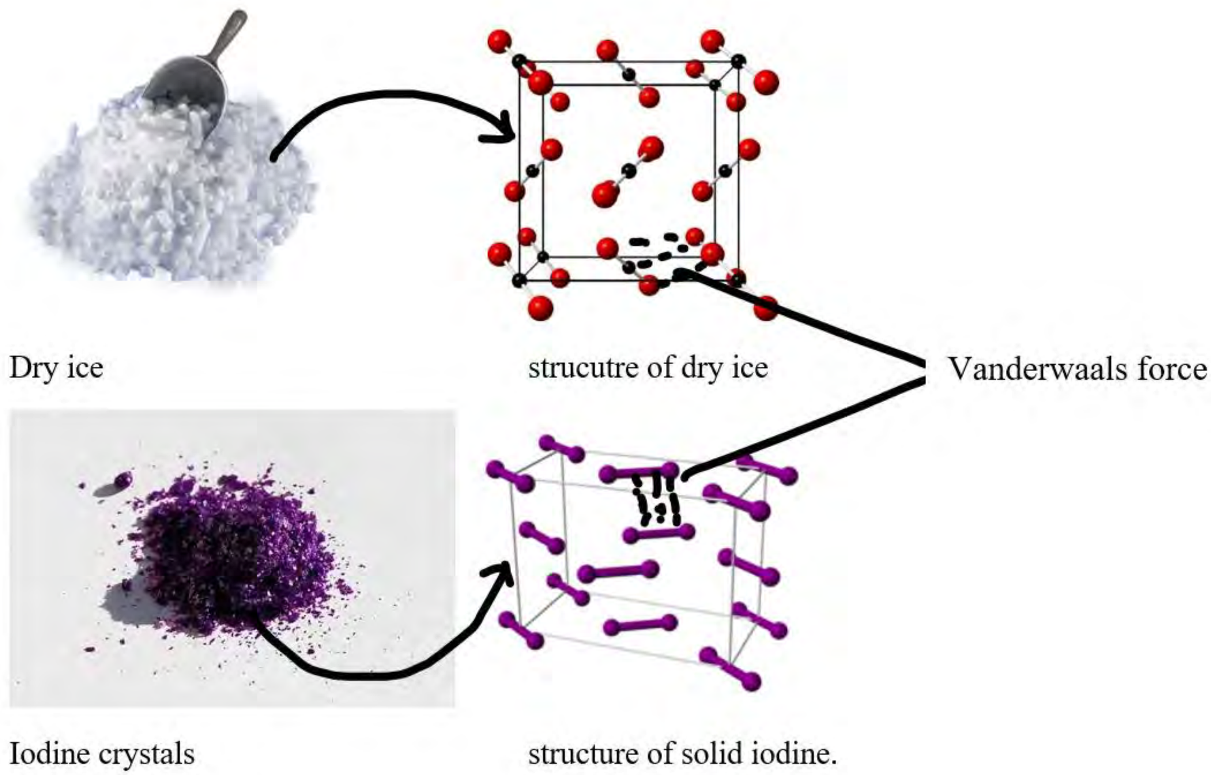
Test yourself!
Use the tables given above and identify whether the bond given is pure covalent, polar covalent or ionic

H- H, H-Cl, Cl-Cl , F-F, N-O, Li-O C-O

e.11 Vander Waal's force and Molecular solids

Activity:

a. Take some camphor or naphthalene bulb in a dry test tube.
b. Warm the test tube gently.
c. Observe the changes.
d. What is the chemistry behind the change?

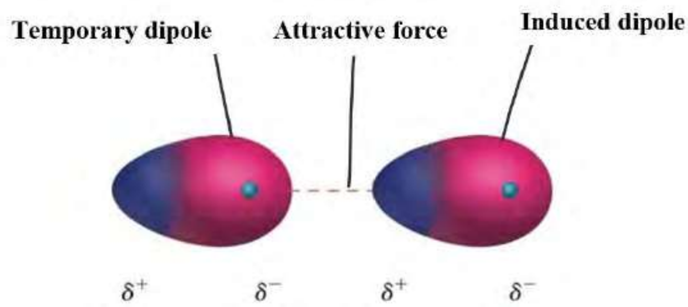


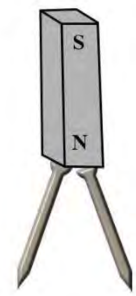
Dry ice (solid CO_2) has a surface temperature of -78.5^0C is used in the food industry as a refrigerant. It sublimes efficiently at room temperature without leaving any residue. The unit particles in dry ice are CO_2 molecules. These molecules are held together by a very weak force of attraction called **Vander Waals force** named after Dutch scientist **J.D Vander Waal**. Vander Waals force, which exists between atoms or molecules, is relatively weaker compared to covalent and ionic bonds.

All these solids that consist of individual molecules held together by weak Vander Waals force of attraction are called **molecular solids**. For example, dry ice, solid iodine, solid ammonium salts.

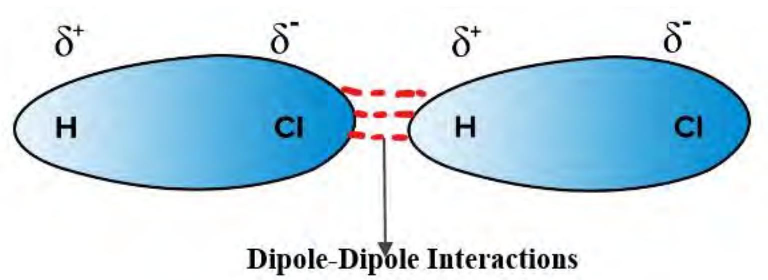
Depending upon the nature of interactions, there are two main types of van der Waals forces:

a. **London Dispersion Forces (or London Forces):** These are the weakest of the van der Waals forces and occur due to temporary asymmetrical distribution of electrons around atoms. It creates temporary dipoles, leading to an induced dipole in neighboring atoms or molecules, resulting in an attractive force. This attraction is similar to a magnet picking up an iron nail by inducing a magnetic moment in the nail. The attractions in non-polar molecules like H_2, Cl_2, Inert gases etc. are London Dispersion Forces.





b. **Dipole-Dipole Interactions**: These occur between molecules that have permanent dipoles. A molecule with a positive end and a negative end can attract another molecule with an opposite orientation, leading to dipole-dipole interactions. The interaction in polar covalent molecules like HCl, CH_3Cl, SO_2 etc. are dipole-dipole Interactions



Vander Waals force mainly depends upon the shape and size of molecules. Larger molecules generally have stronger van der Waals forces due to the large surface area of contact between molecules. Substances with stronger Vander Waals force typically have high melting point, boiling point and viscosity.

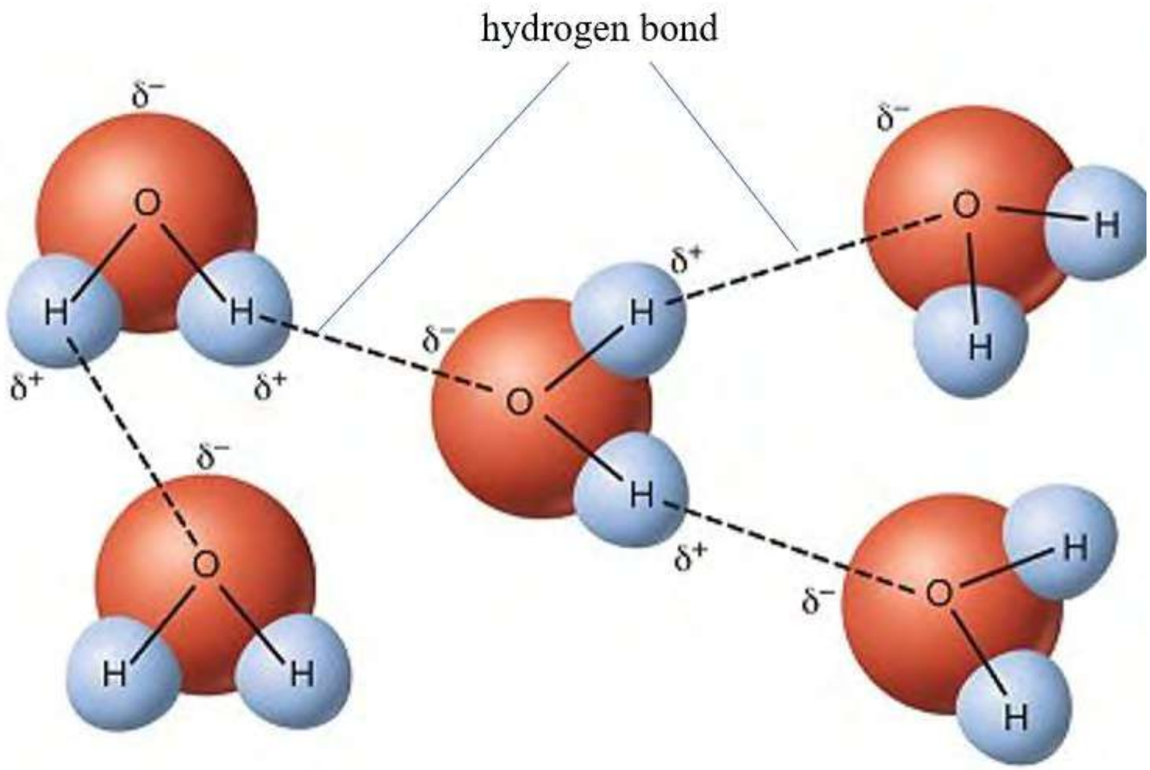
Test yourself!
Which of the given alkane has strongest Vander Waals force and why?

Halogens	**State**	**Boiling point(^{0}C)**
F_2	Gas	-188
Cl_2	Gas	-35
Br_2	Liquid	58.8
I_2	Solid	184
At_2	Solid	337

5.12 Hydrogen bonding and its applications

Activity 5.12:

Observe the figure and discuss the reason of floating ice in water though ice is solid and water is liquid.



You know that alcohol is an organic compound but is soluble in water. Likewise, water has some unique properties such as high boiling point, high specific heat capacity and high surface tension. All these phenomena are due to hydrogen bonding between the molecules.

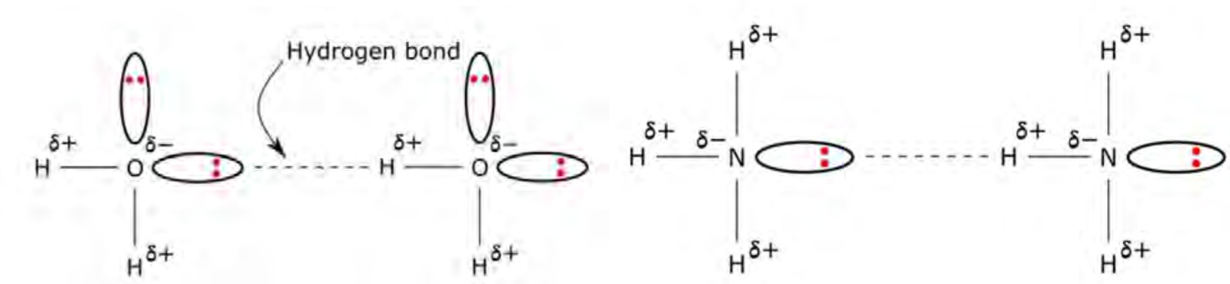
This interaction involves a strong electrostatic attraction between the positively charged hydrogen atom and the negatively charged atom it is bonded to or attracted to. It is a strong electrostatic force of attraction between highly <u>δ+ hydrogen and lone pair of electrons on adjacent molecule. It is roughly 10 times stronger than dipole -dipole interactions.</u>

Hydrogen bonding is a special type of dipole-dipole interaction that occurs between a hydrogen atom that is covalently bonded to a highly electronegative atom (usually nitrogen,

oxygen, or fluorine) and another electronegative atom in a different molecule or within the same molecule

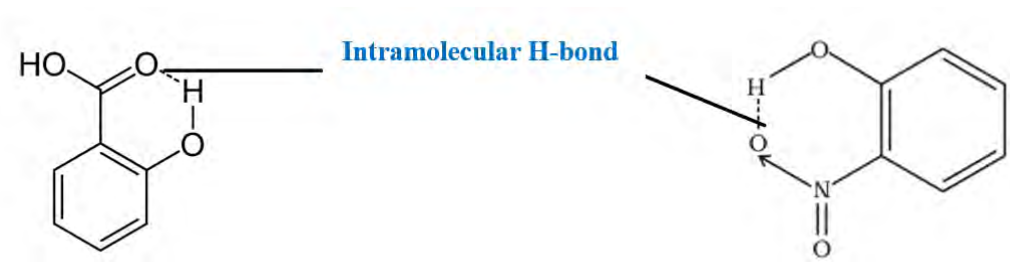
Depending upon the nature of molecules involved and the context of formation of bond, Hydrogen bonding are of two types:

a. Intermolecular hydrogen bonding occurs when the hydrogen atom of one molecule is bonded to an electronegative atom of another molecule. Intermolecular bonding in water, alcohol or ammonia are examples of intermolecular H-bonding



Do you know?
Each H_2O molecule can make four H-bonds, two with lone pair and two with H-atom.

b. Intramolecular Hydrogen bonding also called self-hydrogen bonding occurs within a single molecule where a hydrogen atom is bonded to an electronegative atom within the same molecule.



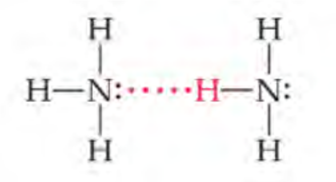
o- salycylic acid **o**-nitrophenol

Hydrogen bonding has several significant consequences and influences various properties and behaviors of substances. Some of the key consequences of hydrogen bonding include:

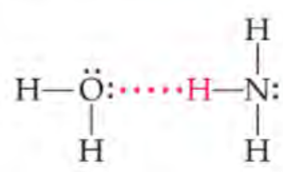
Hydrogen bonding between the water molecule and base pairs contribute to the stability, structure and function of biological molecule

like DNA

Higher Boiling and Melting Points
Substances that exhibit hydrogen bonding generally have higher boiling and melting points compared to similar substances without hydrogen bonding. Ammonia has higher boiling point than phosphine (PH_3), HF has more boiling point than HCl.

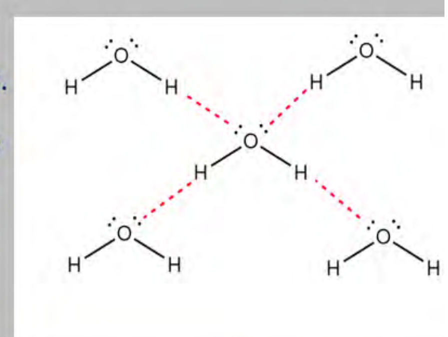


Increased Solubility in Water: Hydrogen bonding enhances the solubility of certain substances in water. Solubility of alcohol, acid or ammonia in water is due to Hydrogen bonding.



Higher Surface Tension and viscosity: Hydrogen bonding develops the strong cohesive forces between molecules at the surface of a liquid which increases the surface tension and viscosity of the liquid.

In the liquid state, water molecules are in constant motion, forming and breaking hydrogen bonds with neighboring molecules. When water freezes into ice, the molecules slow down and arrange themselves in a hexagonal lattice structure. Each water molecule in ice forms **hydrogen bonds** with four neighboring water molecules, creating a stable and open structure.

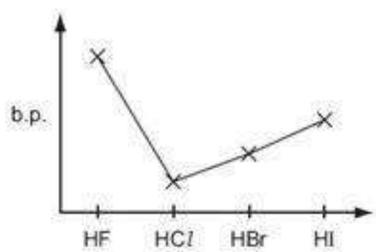


Due to 3-dimensional network with open space density of ice decreases than water then it floats.

Test yourself!

Use the graph given:

Arrange the hydrogen halides in increasing order of boiling point. Give reason for the trend of such boiling point.



5.13 Metallic bonding and properties of metallic solids

Activity 5.13:

Discuss the properties of metals that are responsible for the given use.

There are approximately 3.57×10^{22} gold atoms in one *tola* (11.66g) of gold. These huge numbers of gold atoms are held together by a strong force of attraction called **metallic bond**. The bonding in metal is explained by the electron sea model. According to this model, metal atoms donate their electrons and become positively charged metal ions. Electrons lost from metal atoms are delocalized and form a sea or pool of electrons.

The Electrostatic force of attraction between positively charged metal ions and sea of electrons is called metallic bond.

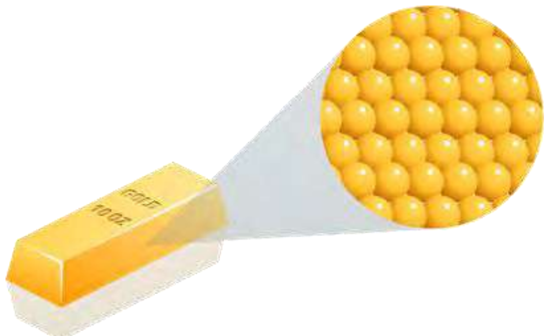


Fig: Gold atoms in gold bar

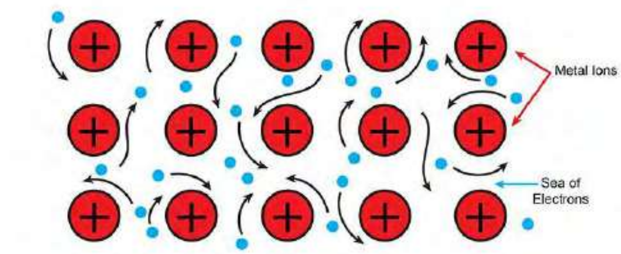


Fig: Electron sea model of metallic bonding

Strength of metallic bond depends upon:

a. **Charge of the metal ion**: Metals with higher positive charges on their ions increase the electrostatic attraction between the positive ions and the negatively charged delocalized electrons and form stronger metallic bonds.

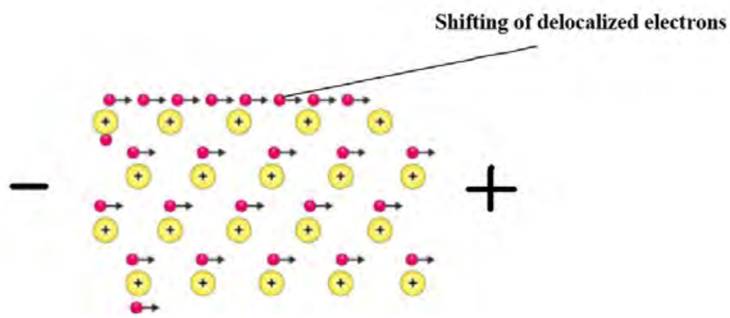
c. **Size of the metal ion:** Smaller ions allow for a higher charge density, resulting in stronger attractions between the positively charged metal ions and the delocalized electrons making the bond stronger.

d. **Number of delocalized electrons:** The greater the number of delocalized electrons per atom, more the attraction with the positively charged metal ions and stronger the metallic bond.

The solid substances with metallic bonds are called **metallic solids**. All metals are examples of metallic solids.

The special properties of metallic solid are due to its delocalized electrons which are spread over a number of atoms. Most of the metals are hard with high melting point and boiling point is due to strong electrostatic force of attraction between positively charged metal ion (**kernel**) and negatively charged sea of electrons.

When the metal is connected to the electric source, delocalized electrons move easily away from the negative to the positive side as shown in the figure. Such mobility of valence electrons throughout the entire metal lattice are responsible for making the metal a **good conductor of electricity**. Similarly, shifting of delocalized electrons from hot end to cold when connected to heat source causes metals to be the **good conductor of heat**.



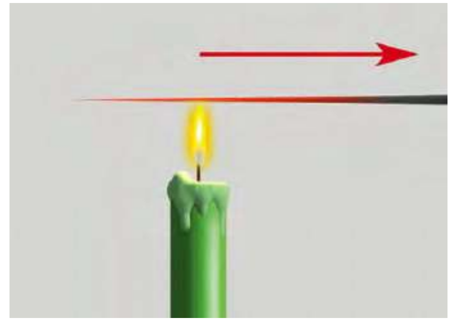


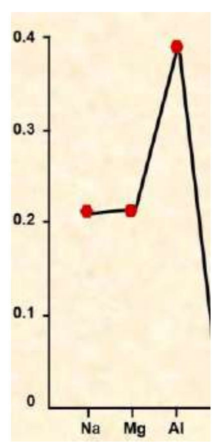
Fig: Flow of current from negative to positive pole *fig: Flow of heat from hot to cold end*

Metal atoms are arranged in a layer making a closely packed structure. Upon hammering or stressing, layers of atoms slide each other without breaking the bond. Therefore, metals are **malleable** (rolled into thin sheets) and **ductile** (drawn into thin wires).

Do you know why are metals heated before deforming?
Heating metals can enhance their malleability and ductility by providing energy to the metal lattice, weaken the metallic bond so it is easier for layers of atoms to slide over each other.

The free electrons in the metal interact with the light source and absorb photons and re- emit, making the metal surface **shiny and reflective**.
Alkali metals have larger atomic size and only one loosely held valence electron makes them soft and have low melting point. In contrast, transition metals have strong metallic bonding due to smaller atomic size and presence of multiple valence electrons. That is the reason why transition metals (Fe, Ag, Cu etc.) are essentials for various household and industrial applications.

Test yourself!
The electrical conductivity of Sodium, Magnesium and Aluminum are given in the graph.
Arrange the metals in increasing order of electrical conductivity.
Give reason for such trend.



Exercise

A. Multiple choice questions

1. Which statement about hydrogen bonding is true?
   a. It is stronger than covalent bonds.
   b. It only occurs in liquids and gases.
   c. It exists in all molecules containing Hydrogen atoms.
   d. It is responsible for the high boiling point of water.

2. The shape of a molecule with a central atom surrounded by four bonding pair and no lone pairs:
   a. Linear b. Trigonal Planar c. Tetrahedral d. Octahedral

3. What is the effect of increasing the number of delocalized electrons on the strength of metallic bonds?
   a. Weakens the bonds
   b. No effect on the bonds
   c. Strengthens the bonds
   d. Converts them into covalent bonds

4. Which molecule has the highest bond angle according to VSEPR theory?
   a. CO_2 b. NH_3 c. CH_4 d. H_2O

5. Which statement is true regarding the effect of hydrogen bonding on the physical properties of substances?
   a. It decreases boiling points.
   b. It increases solubility in non-polar solvents.
   c. It weakens intermolecular forces.
   d. It results in higher melting and boiling points.

6. What does Valence Bond Theory (VBT) describe regarding the formation of chemical bonds?
   a. The sharing of electron pairs in overlapping orbitals.
   b. The transfer of electrons between atoms.
   c. The arrangement of atoms in a crystal lattice.
   d. The repulsion between electron pairs in the valence shell.

7. Name the molecule with lowest boiling point:
   a. HCl
   b. HBr
   c. HI
   d. HF

8. Which of the following compounds is soluble in water?
   a. Formic acid
   b. Acetone
   c. Ether
   d. Carbon tetrachloride

9. Which of the following molecules is most likely to have strong London dispersion forces?

a. H_2O

b. CO_2

c. HF

d. CH_4

10. How many sigma and pi bonds are there in a molecule of benzene?

a. 6 sigma 6 pi

b. 6 sigma 3 pi

c. 12 sigma 6 pi

d. 12 sigma 3 pi

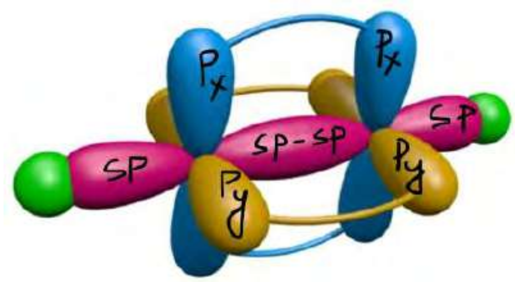
11. Orbital picture of one of the hydrocarbons is given. What is the number of atoms that lie in the internuclear axis in the given molecule?

a. 1

b. 2

c. 3

d. 4



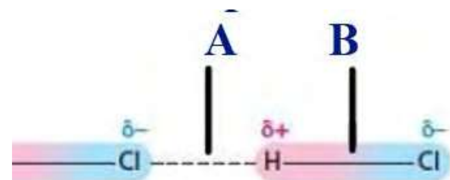
12. The bond A and B are ……………. respectively.

a. ionic and covalent

b. covalent and ionic

c. h-bond and covalent

d. vander walls force and covalent



13. The correct combination of shape and hybridization of the ion given is

a.	Trigonal	SP^2
b.	Tetrahedral	Sp^2
c.	Pyramidal	Sp^3
d.	Tetrahedral	Sp^3

B. Short answer questions

14. Write any two differences between:

a. Ionic solid and covalent (network solid)

b. Sp^2 and sp^3 hybridization

c. Sigma bond and pi bond

15. Give reasons for the following statements:

a. Copper sulphate is brittle whereas copper is malleable

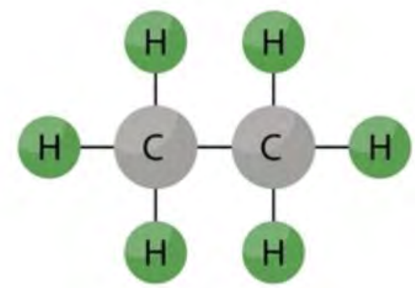
b. Solid iodine sublimes easily to vapor

c. Glycerol is more viscous than ether
d. CO_2 is linear whereas SO_2 is bent in shape
e. NH_3 is polar molecule but BH_3 is not
f. Sigma bond is stronger than pi bond
a. Petrol and diesel are kept in a sealed container.

16. What do you mean by octet rule? Give any three compounds that follow octet rule and any three that do not follow octet.

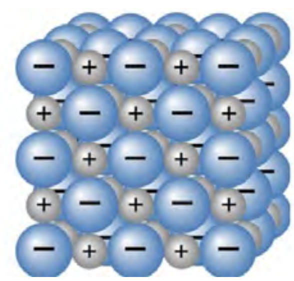
17. Structure formula of one of the hydrocarbons is given:

a. Write the Lewis dot structure to show the bond pair electrons.
b. What is the hybridization and bond angle in each carbon atom?
c. The compound is insoluble in water. Why?



18. Structure of one of the solid substances is given in the figure.

a. Name the bond present in the solid and find the unit particles present
b. What happens when this solid is hammered?
c. What is the reason that it is good conductor of electricity only in molten or aqueous state?
d. Give any two examples of such solid.



19. Resonance helps explain the stability of certain molecular structures that cannot be sufficiently represented by a single Lewis structure.

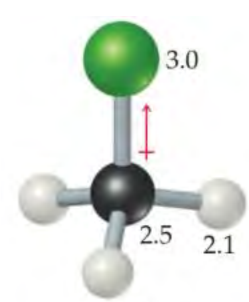
a. Define the term resonance and resonating structures with one example.
b. What is the cause of resonance?
c. How does resonance explain the stability of species?

20. Dipole moment is the important parameter which explains various property of that molecule

a. Define dipole moment and write its unit.
b. How does dipole moment predict the polarity of the molecule?
c. Polar bond does not indicate polar molecule. Prove with one example.

21. Electronegativity value of each atom along with the shape of CH_3Cl molecule is given:

a. Which bond (C-H or C-Cl) is the most polar bond? Why?
b. Is the molecule polar? Give reason.
c. Name the shape and hybridization of the molecule.



CH_3Cl

22. Draw the Lewis diagrams of following molecule or ions:

a. H_2O
b. CCl_4

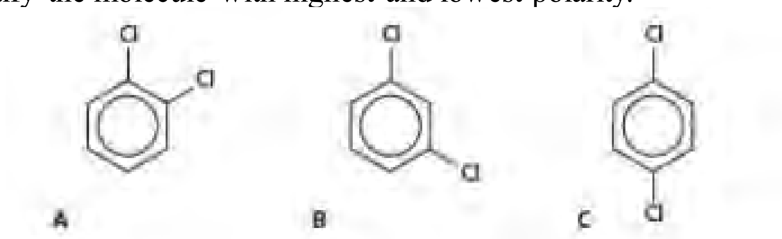
c. PCl_5
d. NO_3^-
e. SO_4^{--}
f. O_3
g. I_3^- (tri-iodide ion)
h. ClO_3^+
i. SiH_4
j. IF_5
k. HNO_3
l. NO_2^-

23. Find the number of lone pair and bond pair electrons around the central atom of a given molecule and draw the shape of the molecules.

b. **c.**

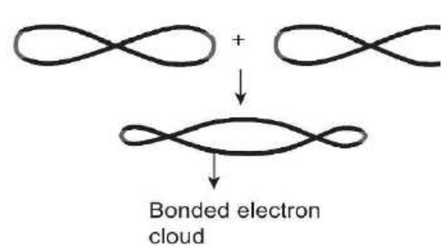
d. **e.**

24. Draw the direction of the bond dipole in the given molecules and find whether the net dipole is zero or not. Identify the molecule with highest and lowest polarity.

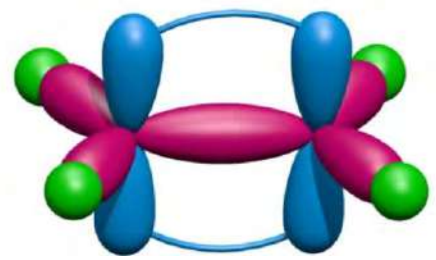


25. Valence bond theory explains the formation of bond by the overlapping of atomic orbital as shown in the figure.

a. Name the type of covalent bond formed by the overlapping of atomic orbitals.
b. Write any three properties of the bond formed by such overlapping of atomic orbitals.

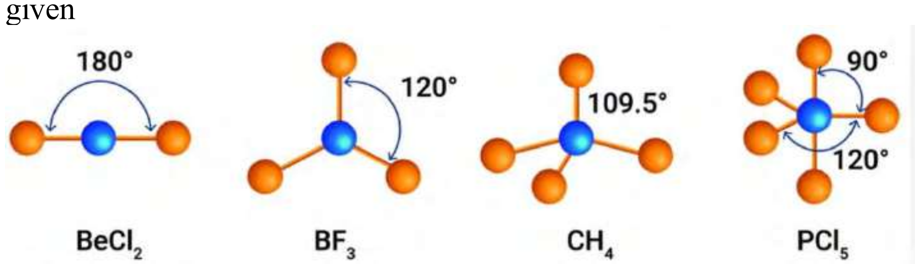


26. Orbital overlapping between the atoms during the formation of ethene molecules is shown in the diagram.



a. Find the number of heads on and sidewise overlapping in ethene molecule.
b. Define the term hybridization and name the hybridization and bond angle in the carbon atom.
c. Draw the orbital overlapping in ethyne molecules showing sigma and pi bonds and write the hybridization in carbon atoms.

27. Fill the table below using the structures of the molecule given

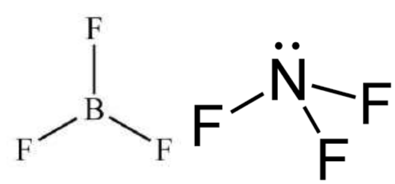


Formula of the molecule	Shape	Hybridization

28. Copper, dry ice and sodium chloride are all solids but have different physical as well as chemical properties.

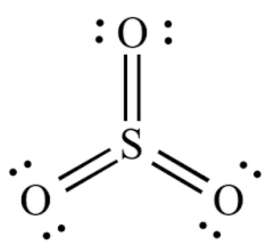
a. What is the reason for varying properties of these solids?
b. Name the solid that is good conductor of heat and electricity and give reason for that property
c. Why is sodium chloride brittle in nature?
d. Dry ice is converted to vapor directly while heating. What is the reason behind this?

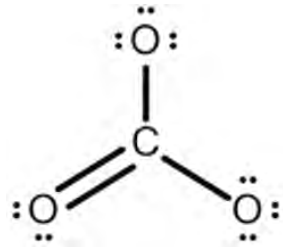
29. Dipole moment helps in determining the polarity as well as shape of the molecule. Structure of two molecules is given. Give reason for the following:

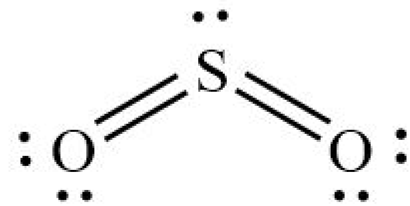


a. Different shapes of these molecules.
b. NF_3 is polar molecule.
c. BF_3 is nonpolar molecule.
d. B-F bond is polar.

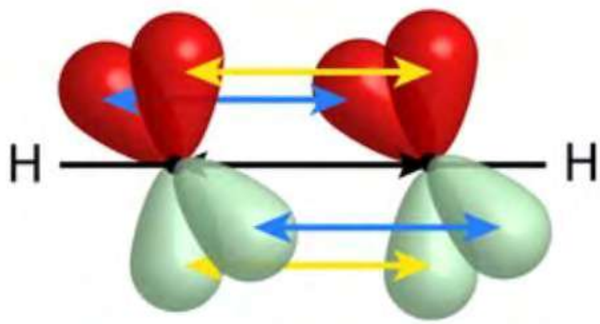
30. Define the term resonance. Write the resonance and resonance hybrid structure of given species:





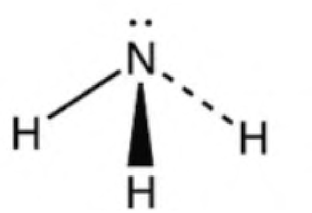


31. Orbital overlapping of one of the hydrocarbons is given in the figure:



a. What is the structural formula of this hydrocarbon?
b. Name the bond that passes through the internuclear axis and write any two characteristics of this bond.
c. Draw the orbital picture of ethane.

32. Ammonia is highly soluble in water and has a pungent smell. Shape of ammonia molecule is given:



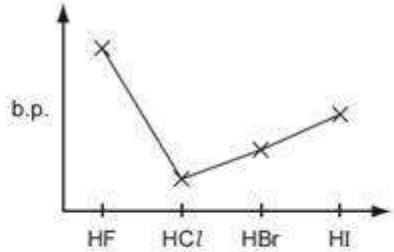
a. Name the given shape. What is the cause of this shape of the molecule?
b. What happens to the bond angle if hydrogen atoms are replaced by fluorine atoms?
c. Why is ammonia Highly soluble in water?

33. Hydrogen halides are the molecules of hydrogen with halogens. Three of the halides are given

HCl
HBr
HI

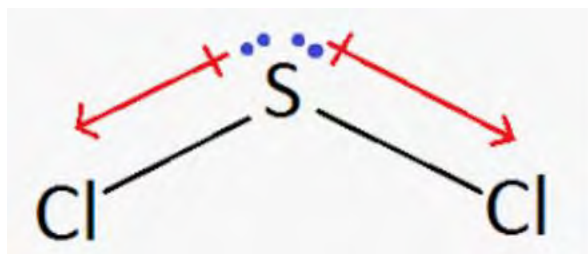
a. Arrange the given halide molecules in the increasing order of
   i. Bond dipole
   ii. Bond length
   iii. bond strength
b. Give a reason for such an order.

34. Boiling point of different hydrogen halides are given in the graph given:



a. Name of halide with highest boiling point
b. Give a reason for the highest boiling point for that halide.
c. Which of the halides is highly soluble in water? Give a reason.

35.



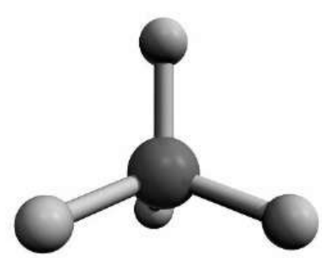
a. Write the formula of given molecule.
b. Name the shape and give reason for such shape.
c. Identify whether the molecule is polar or non-polar and why?

C. Long answer questions

36. Consider two molecules: boron trifluoride (BF_3), and phosphorus trifluoride (PF_3), both have the same general formula AX_3, but their molecular structures are different as explained by VSEPR theory.

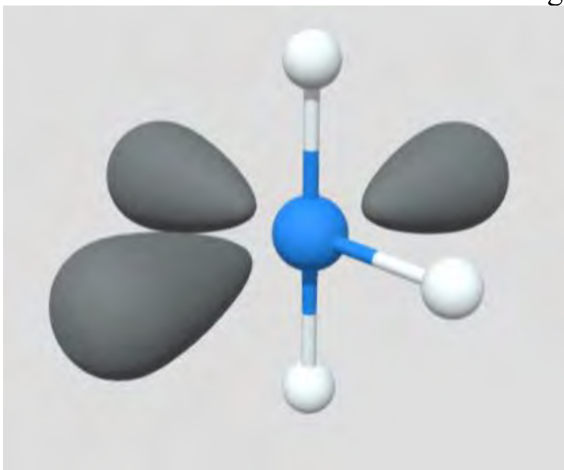
a. Write any three postulates of VSEPR theory.
b. Draw the structure of BF_3 and PF_3 and write the name of their shape.
c. What is the reason for the different shape of these molecules?
d. What is the hybridization of Boron and Phosphorus in these molecules?
e. Compare the bond angle in these molecules.
f. Which one is more polar molecule and why?

37. Methane is the main component of natural gas. It is used primarily as fuel to make heat and light. Jar of natural gas contains a huge number of methane molecules. The structure of the methane molecule is given.



a. How many bond pair and lone pair electrons are there in the given molecule?
b. Name the shape of the molecule and write bond angle in the molecule.
c. Show the bond dipole in each bond and write the polarity of the molecule with reason.
d. What will be the common name of the molecule when all the hydrogen atoms are replaced by chlorine atoms? Draw the shape of this molecule and compare the bond angle with methane.

38. Molecular model of unknown molecule is given:

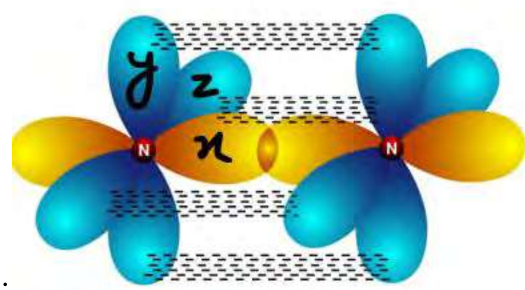


a. How many lone pair and bond pair electrons are around the central atom?
b. Predict the shape and polarity of the molecule.
c. Give one example of such molecule.

39. Benzene is an example of aromatic hydrocarbon containing both sigma and pi bonds between the carbon atoms.

a. How many sigma and pi bonds are there in benzene?
b. All C-C bonds in benzene are of same length (shorter than single bond but longer than double bond), why?
c. What is hybridization and bond angle of each carbon in benzene?

40. Nitrogen molecule is formed by the overlapping of three atomic orbitals (x, and z) to form



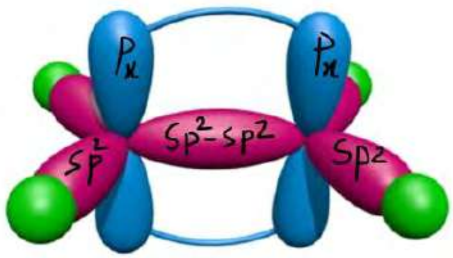
as shown in the diagram:

a. Write the electronic configuration of atoms to show the number of half-filled atomic orbitals.
b. Name the orbitals which overlap laterally.
c. Why is the sigma bond stronger than the pi bond?
d. How many sigma and pi bonds are there in a Nitrogen molecule?
e. Why is nitrogen called inert gas?

41. Some of the molecules with odd electrons are given below which do not follow octet rule.

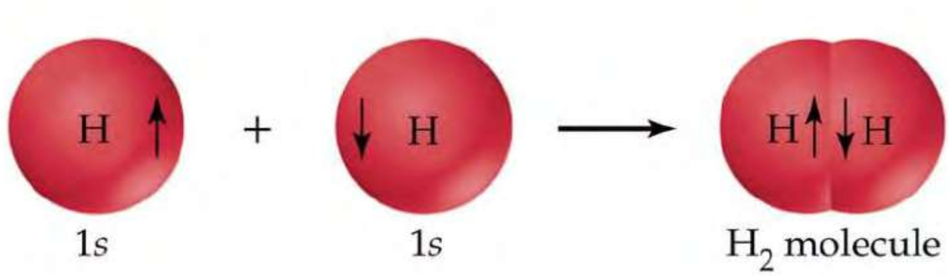
a. Define the term odd electron and octet rule.
b. Give any three examples of other compounds which do not follow octet rule.

42. Orbital picture of hydrocarbon is given where central atom has sp^2 hybridization.



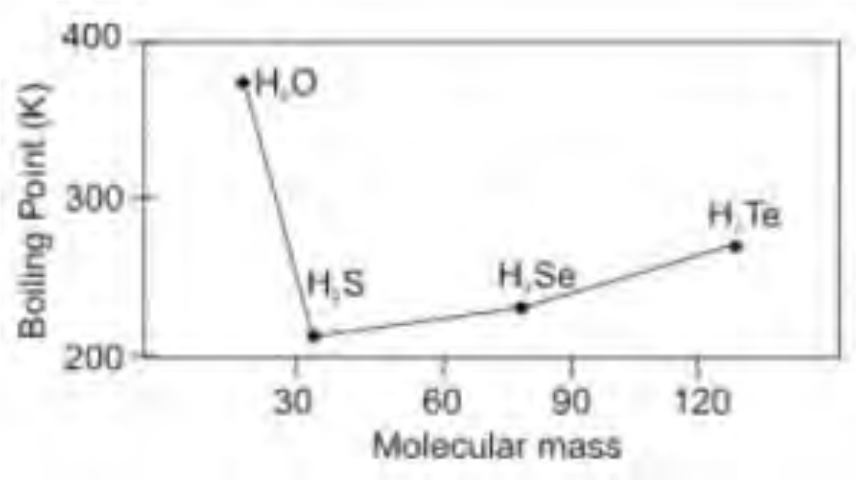
a. Define term hybridization.
b. How are sP^2 hybridized orbitals formed?
c. What is the shape and bond angle of Sp2 hybridization?
d. Find the number of sigma and pi bonds.
e. Write the structural formula of the given molecules.

43. Valence bond theory (VBT) explains the formation of Hydrogen molecules as shown in the figure.



a. What are the postulates of VBT?
b. What types of overlapping is there in the hydrogen molecule?
c. Draw the orbital overlapping in the HF molecule.
d. Why is the He2 molecule not possible like H_2?

44. Boiling points of hydrides of group 16 elements are given in the graph.
a. Define the term boiling point.
b. Water has the highest boiling point though its molecular mass is smallest among the compounds. Why?
c. The boiling point increases from H_2S to H_2Te. Why?
d. Which of the molecules has the highest bond angle? Why?



45. Valence bond theory describes the formation of sigma and pi bonds (covalent bond) by orbital overlapping.

a. Write any two conditions for the overlapping of orbitals in valence bond theory.
b. Give any two differences between sigma and pi bond.
c. Draw the orbital picture of an oxygen molecule showing the sigma and pi bond.

46. Draw the shapes of the given molecule and write the types of hybridization on the center atom.

H_2S, PH_3, CCl_4, $BeCl_2$

47. Electronic configuration of element A and B are given.

A

B

a. Draw the Lewis dot structure of compound formed by A and B.
b. Name the bond present in AB and write any two conditions for the formation of this bond.
c. Write the electronic configuration of A^{++} and B^{---} and compare their size.

48. Bohr's atomic theory was able to explain the spectra of hydrogen, which was not explained by Rutherford atomic model.

a. Write any three postulates of Bohr's atomic model
b. What causes the origin of hydrogen spectra?
c. Draw the figure to show the different spectral series of hydrogen.
d. Write any two limitations of Bohr's atomic model,

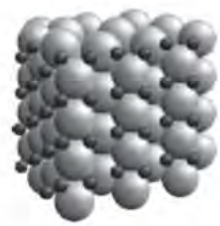
49. Depending upon the nature of bonding solid substances are divided into different categories (ionic, molecular metallic and covalent) as shown below:

a. Identify A, B, C and D and write the name of the bond present.
b. Which of the solid substances is soluble in water?
c. Which of the solid is a good conductor is solid state, and why?
d. Which one is called network solid?
e. Name the solid that has lowest melting point and give reason.

A

B

C



D

50. Use the table given and answer the following questions.

a. Identify the molecules that have tetrahedral and pyramidal shape.

b. Name the molecule that is most polar and least polar. Why?

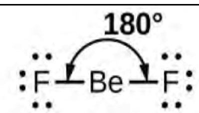
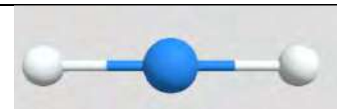
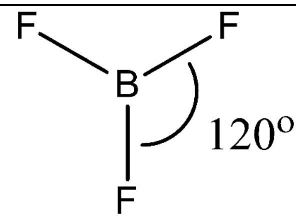
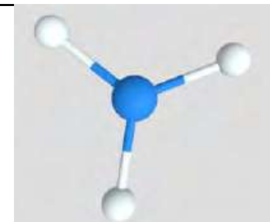
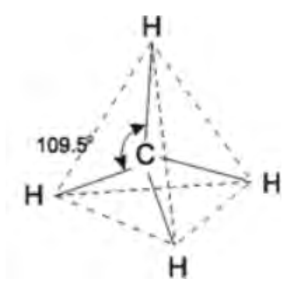
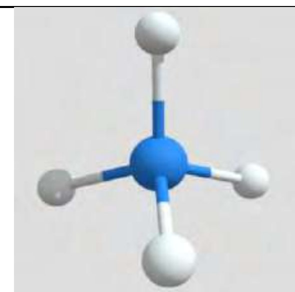
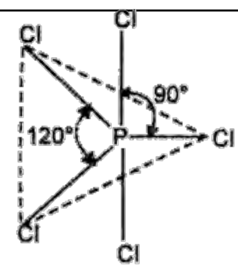
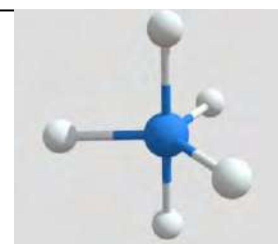
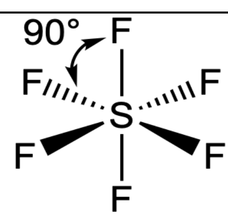
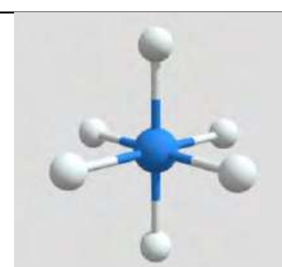
c. Which of the halide molecules has the highest bond length and why?

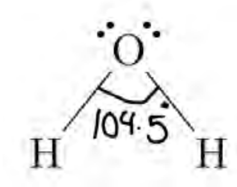
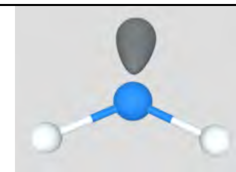
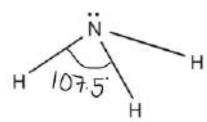
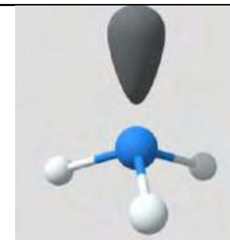
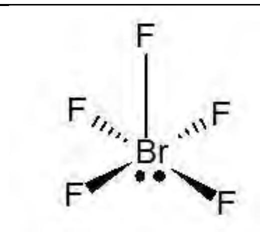
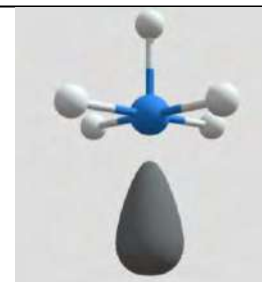
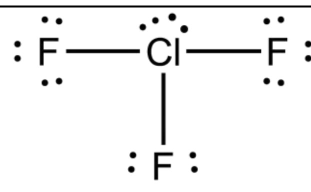
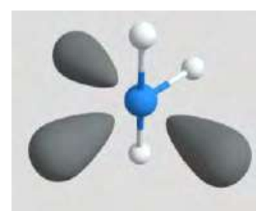
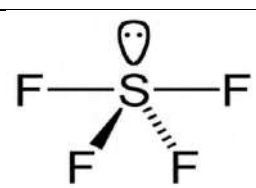
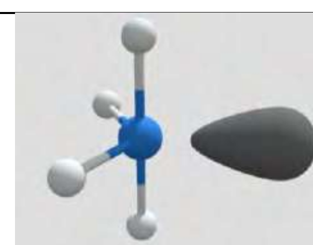
d. $CHCl_3$ is tetrahedral but polar. Why?

Molecule	Dipole moment Debye(D)
HI	0.42
HBr	0.80
HCl	1.03
$CHCl_3$	1.04
NH_3	1.48
H_2O	1.84
HF	1.91

Project work

1. Use clay and tooth pin and make a shape of different molecules: (H_2O, NH_3, CH_4, BeF_2, BF_3, PCl_5, SF_6)
2. Take a chart paper and draw the column and fill the corresponding rows and column as given:

molecule	Bond pair	Lone pair	Shapes and bond angle	3-d shape
BeF_2	2	0	 Linear	
BF_3	3	0	 Trigonal plannar	
CH4	4	0	 Tetrahedral	
PCl_5	5	0	 Trigonal bipyramidal	
SF_6	6	0	 Octahedrtal	

H_2O	**2**	**2**	 **Bent or V shaped**		
NH_3	**3**	**1**	 **Trigonal pyramidal**		
BrF_5	**5**	**1**	 **Square pyramidal**		
ClF_3	**3**	**2**	 **T -shaped**		
SF_4	**4**	**1**	 Sea saw shaped		

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UNIT 6:
OXIDATION AND REDUCTION

Activity:

a. Take a fresh apple and cut it in half.

b. Expose one of the halves to the air and keep the other half in an airtight container.

c. After 10 minutes, observe both pieces and note down the changes.

6.1 General and electronic concept of oxidation and reduction

A. General concept of oxidation and reduction

Activity:

a. $PbO + C \rightarrow Pb + CO$
b. $H_2 + Cl_2 \rightarrow 2HCl$
c. $H_2S + Br_2 \rightarrow 2HBr + S$

From the given chemical reactions, identify the reactants that

i. gains oxygen.
ii. loses oxygen.
iii. gains hydrogen.
iv. loses hydrogen.

Have you observed the rusted iron nails? Do you know the reason for the formation of rust? In this unit we will discuss the chemistry behind such a chemical phenomenon. Rust is the oxide of iron ,it is formed when iron reacts with oxygen in the presence of water.

$$Fe(s)+3O_2(g)\rightarrow 2Fe_2O_3(s)$$

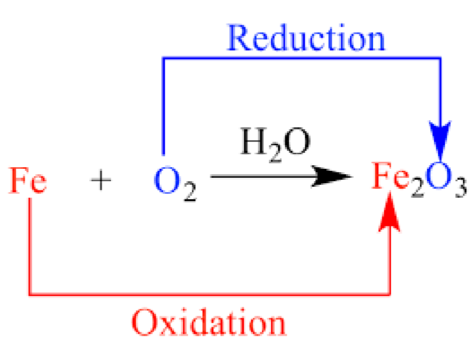


Fig: Rusting of machinery part is due to the oxidation of iron

Fig: magnesium ribbon is oxidized to magnesium oxide.

Fig: magnesium undergoes oxidation when it burns with oxygen

In the same way, when magnesium ribbon burns in air, bright white light is observed. There is a reaction between magnesium metal and oxygen to form magnesium oxide.

$$2Mg(s)+O_2(g)\rightarrow 2MgO(s)$$

Both of these reactions involve gain of oxygen.

The reaction involving gain of oxygen is called oxidation and the substance that gains oxygen is said to be oxidised.

There are some chemical reactions where oxygen is released. For example, when zinc oxide is heated with hydrogen, zinc and oxygen are formed.

$$ZnO+ H_2 \rightarrow Zn +H_2O$$

Likewise, while heating copper oxide with carbon copper oxide loses oxygen to form carbon and carbon dioxide.

$$CuO + C \rightarrow Cu + CO_2$$

The reaction where oxygen is lost is called reduction and the substance that loses oxygen is said to be reduced.

Oxidation involves the addition of any electronegative species (usually nonmetals) besides oxygen and reduction involves the addition of any electropositive species (usually metals) besides hydrogen. This early concept of oxidation and reduction is limited because it couldn't explain all the reactions (e.g. ionic reactions) where there are no electronegative or electropositive elements involved.

$Fe+S \rightarrow 2$ (Fe is oxidised to FeS)

$KI+Cl_2 \rightarrow KCl+I_2$(KI is oxidised to KCl)

$FeCl_3+H_2 \rightarrow FeCl_2+HCl$ ($FeCl_3$ is reduced to $FeCl_2$)

General or classical concept

Addition of oxygen or any electronegative species or removal of hydrogen or any electropositive species is called oxidation.

Addition of hydrogen or any electropositive species or removal of oxygen or any electronegative species is called reduction.

Test yourself!

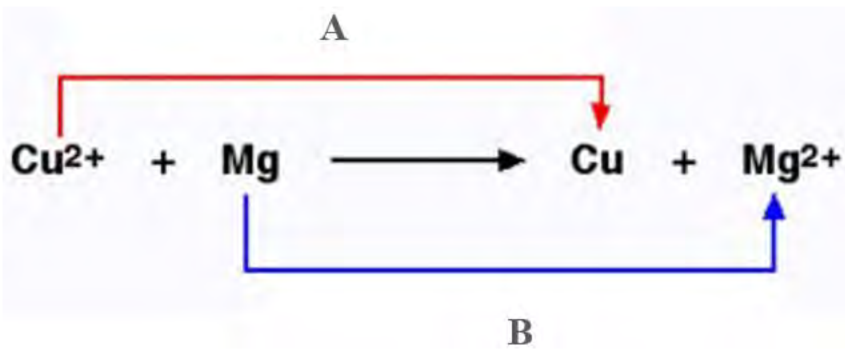
Identify the following biochemical reaction as oxidation or reduction and give reason for your choice

a. $C_6H_{12}O_6(a)+6O_2(g)\rightarrow 6CO_2(g)+6H_2O(l)+$ energy

b. $6CO_2(g)+6H_2O(l)+$light energy$\rightarrow C_6H_{12}O_6(a)+6O_2(g)$

B. Electronic concept of oxidation –reduction

Activity: Observe the given ionic reaction which involves both loss and gain of electrons.



a. Identify the process (A or B) which proceeds by loss or gain of electrons.

b. Find out the number of electrons lost and gained in each process A and B.

Figure: reaction between zinc plate and copper sulphate solution

When a piece of zinc rod is kept in the copper sulphate solution, the blue color slowly becomes colorless. Zinc atoms lose electrons, becoming positively charged zinc ions (Zn^{2+}). These electrons are transferred to copper ions (Cu^{2+}) in the solution and Cu^{++} changed into metallic copper (Cu).

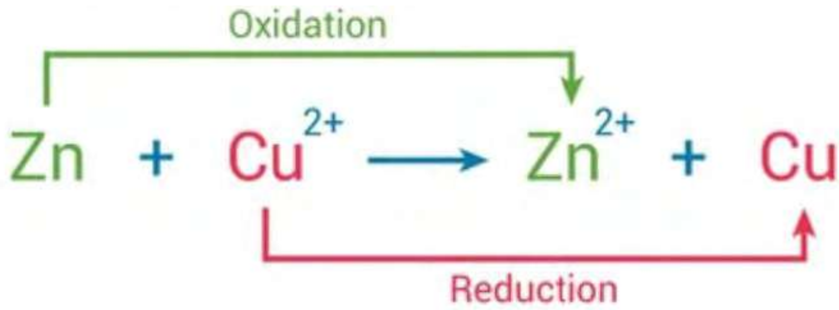
$Zn \longrightarrow Zn^{++} + 2e^-$ [Zinc is oxidised to Zinc ion]

$Cu^{++} + 2e^- \longrightarrow Cu$ [copper ion is reduced to copper]

The net ionic equation is the sum of above two reactions

$Zn + Cu^{++} \longrightarrow Zn^{++} + Cu$(ionic equation)

In this reaction, Zinc loses electrons and oxidized whereas copper ion gains electrons and reduced. It means both oxidation and reduction are there in this reaction called **redox reaction**.



In Electronic concept**,**

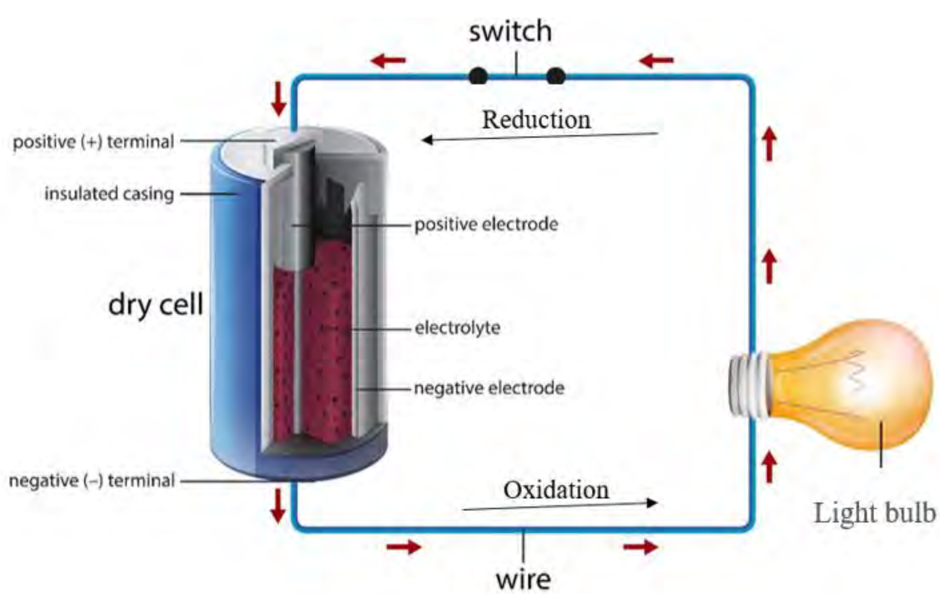
Oxidation is the loss of electrons or de-electronation. Reduction is the gain of electron or electronation. The reaction involving both loss and gain of electrons is called redox reaction.

Redox reactions are the fundamental processes in batteries, where one half-cell undergoes oxidation, and the other undergoes reduction. For example, dry cells used in toys generate electrical energy due to the spontaneous redox reaction.

$2NH_4^+(aq) + 2MnO_2 + 2e^- \rightarrow Mn_2O_3(s) + 2NH_3(aq) + H_2O$ (reduction)

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ (oxidation)

$2NH_4^+(aq) + 2MnO_2 + Zn \rightarrow Mn_2O_3(s) + 2NH_3(aq) + H_2O + Zn^{2+}(aq)$ (redox reaction)



Alcohol (ethanol) on a driver's breath is detected using Breathalyzer. The device contains potassium dichromate (VI). Alcohol on the breath causes a color change of dichromate from orange to green. This is an example of a redox reaction.

$$2K_2Cr_2O_7 - + 8H_2SO_4 + 3C_2H_5OH \longrightarrow 2Cr_2(SO_4)_3 + + 3CH_3COOH + 11H_2O$$

Test yourself!

The reaction in the lead acid battery is given:

Identify oxidation and reduction using elecectronic concept and add these reactions to make a redox reaction.

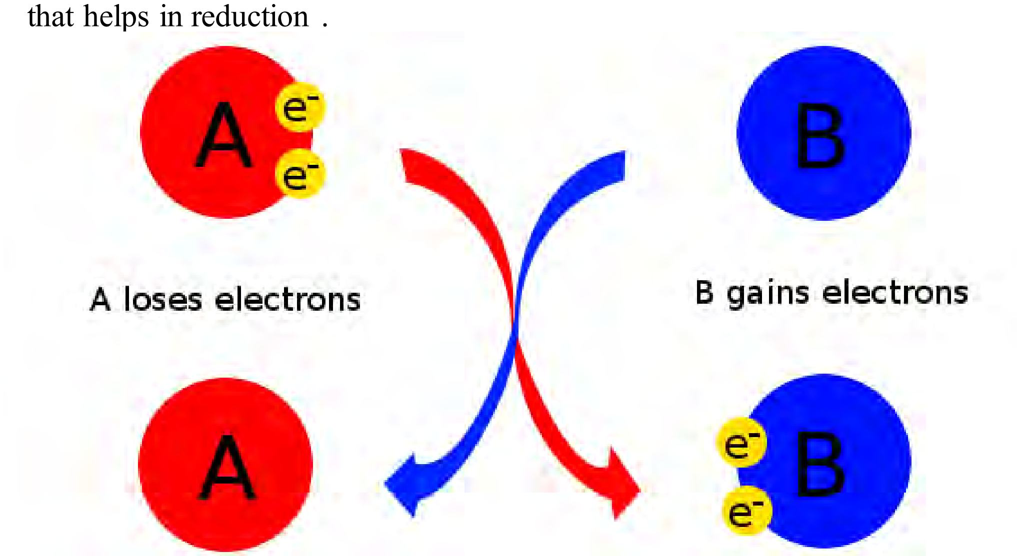
Oxidising and Reducing agent

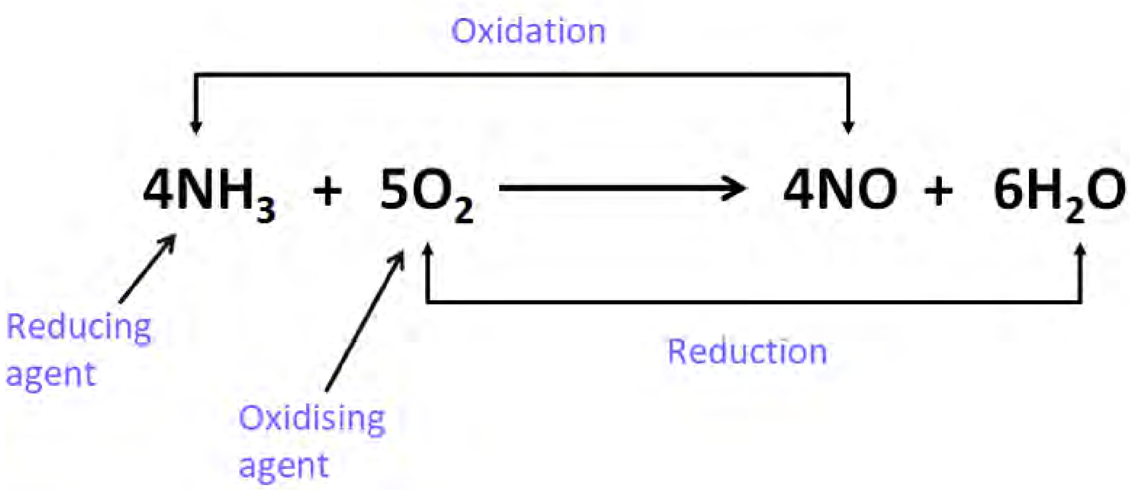
Activity:

Observe the given picture and identify the substance:

a. that helps in the oxidation.

b. that helps in reduction .





A small piece of sodium metal glows white when placed in a flask filled with chlorine gas. The molecular equation for this reaction is;

$Na + Cl_2 \longrightarrow NaCl$

Thus, the half-reaction is;

$Na \longrightarrow Na^+ + e^-$ [Sodium metal oxidised to sodium ion]

$2e^- + Cl_2 \longrightarrow 2Cl^-$ [chlorine molecule is reduced to chloride ion]

In this reaction, sodium oxidizes chlorine, so it is a reducing agent whereas chlorine reduces sodium, so is an oxidising agent.

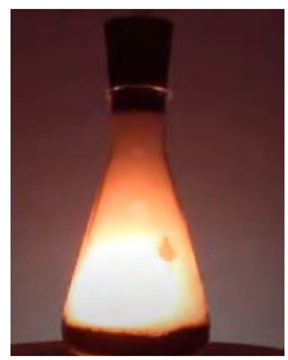


Fig: the redox reaction between sodium and chlorine is highly exothermic(releases energy

The substance that itself is oxidised helps in the reduction of others is called reducing agent or reductant. The substance that itself is reduced and helps in the oxidation of others is called oxidising agent or oxidant.

The species that easily gains electrons are comparatively stronger oxidising agents. Similarly, species that easily lose electrons are stronger reducing agents. The comparative strength of some chemical species is given.

Strongest oxidizing agent	$F_2 + 2e^- \rightleftharpoons 2F^-$	Weakest reducing agent
	$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	
	$\boxed{O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O}$	
	$Br_2 + 2e^- \rightleftharpoons 2Br^-$	
	$Ag^+ + e^- \rightleftharpoons Ag$	
	$Cu^{2+} + 2e^- \rightleftharpoons Cu$	
	$\boxed{2H^+ + 2e^- \rightleftharpoons H_2}$	
Increasing strength of oxidizing agent	$Pb^{2+} + 2e^- \rightleftharpoons Pb$	Increasing strength of reducing agent
	$Sn^{2+} + 2e^- \rightleftharpoons Sn$	
	$Ni^{2+} + 2e^- \rightleftharpoons Ni$	
	$Fe^{2+} + 2e^- \rightleftharpoons Fe$	
	$Cr^{3+} + 3e^- \rightleftharpoons Cr$	
	$Zn^{2+} + 2e^- \rightleftharpoons Zn$	
	$\boxed{2H_2O + 2e^- \rightleftharpoons H_2 + 2OH^-}$	
	$Al^{3+} + 3e^- \rightleftharpoons Al$	
Weakest oxidizing agent	$Mg^{2+} + 2e^- \rightleftharpoons Mg$	Strongest reducing agent
	$Na^+ + e^- \rightleftharpoons Na$	

How does antioxidant work in our body?

Strong oxidising agent hydrogen peroxide(H_2O_2) oxidizes the normal DNA in our body and converts to altered DNA. Substances called antioxidants that are found in some of the food react with oxidising agents (such as hydrogen peroxide) and thus remove them from our system. It causes the repairing of DNA faster and helps to slow down the problems that come with aging.

Figure: antioxidants are found naturally in fruits and vegetables

Project work:

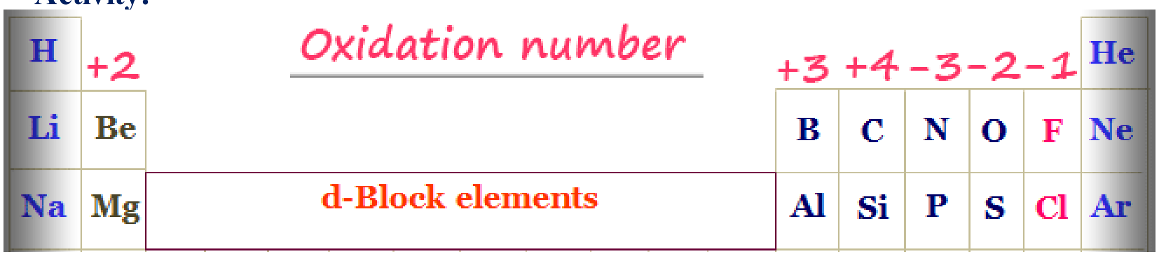
a. Take 10 ml of silver nitrate solution on the beaker.

b. Insert the piece of copper wire or copper coin into the solution.

c. Observe the change in the metal as well as solution.

6.2 Oxidation number and rules for assigning oxidation number

Activity:



Observe the part of the periodic table where certain numbers are given in each group except for group 1 and 18. Identify the numbers for these groups.

During the redox reactions, electrons are lost as well as gained by the atoms. Certain numbers are used to describe the degree of loss or gain of electrons in such reactions. These numbers areoxidation numbers.

The oxidation number of an atom in a molecule or ion is the charge that an atom would have if all electrons in its bonds were assigned to the more electronegative atom.

The following simple conventions are useful to calculate the oxidation number of each atom in a molecule or ion:

a. Oxidation number of atoms in free state or elemental form is zero.

For example, the oxidation number of chlorines in Cl_2, phosphorous in P_4 or iron in Fe is 0.

b. Certain elements have same oxidation number in almost all their compounds:

i. Fluorine always has oxidation number of -1.

ii. Group 1 elements forming +1 ions (Li^+, Na^+, K^+) oxidation number =0.

iii. Group 2 always forms +2 ions: Mg^{2+}, Ca^{2+}, etc. oxidation number =+2.

c. Oxidation number of hydrogens is +1 in covalent compounds but -1 in ionic compounds(hydrides).

For examples: covalent compounds (H_2O, NH_3, CH_4) oxidation number of H =+1

ionic compounds (NaH, LiH, CaH_2) oxidation number of H =-1

d. Oxidation number of oxygens is -2 in normal oxides, but is -1 and -1/2 is peroxides and superoxides respectively.

Examples:

Normal oxides (NO, CO_2, P_2O_5), oxidation number of oxygen =-2

Peroxides (H_2O_2, Na_2O_2) oxidation number of oxygen =-1
Super oxides (KO_2, NaO_2) oxidation number of oxygen =-1/2

e. Sum of the oxidation number must be zero for the neutral compound and equal to charge for ions.

For example, the oxidation number of **carbons** in glucose and carbonate ion is calculated as:

i. $C_6H_{12}O_6$

$$6 \times x + 12 \times (+1) + 6 \times (-2) = 0$$
$$\therefore x = 0$$

ii. CO_3^{--}

$$1 \times x + 3 \times (-2) = -2$$
$$\therefore x = +4$$

Where, x is the oxidation number of carbons

The same atom may have variable oxidation number depending upon the nature of neighboring atoms and type of bonding. The range of oxidation number of particular atoms depends on the group of the atom in the periodic table.

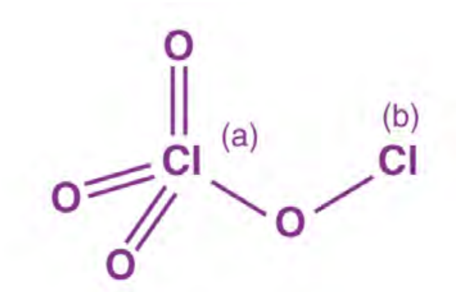
	Group 1	Group 2	Group 13	Group 14	Group 15	Group 16	Group 17
Highest oxidation number	+1	+2	+3	+4	+5	+6	+7
Lowest oxidation number	0	0	-5	-4	-3	-2	-1

In the case of molecule of nonmetals, more electronegative atom carries negative oxidation number whereas less electronegative atom carries positive oxidation number.

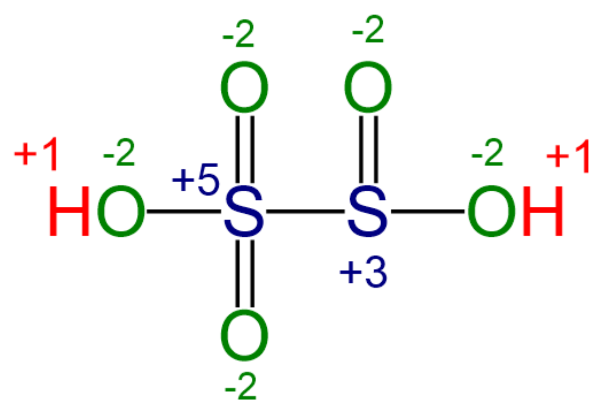
ClO_2 oxidation number of oxygen is -2 and that of chlorine is +7 because Oxygen is more electronegative than chlorine.

ICl_3 oxidation number of -1 and that of Iodine is +3 because chlorine is more electronegative than iodine.

All the atoms in a same molecule have the same oxidation number only when they have the same environment in the molecule. When the same atom in the molecule have different environment or bond structure, then they have different oxidation number, for example: Oxidation number of chlorine atom(**a**) is +7 whereas oxidation number of chlorine atom (**b**) is +1 in Cl_2O_4 molecule.



Likewise, the different oxidation number of Sulphur atoms in pyro sulphorous acid is calculated as:



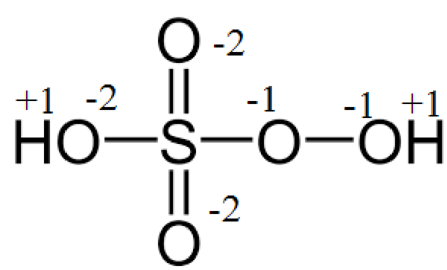
The oxidation number of an element is denoted by the Roman numeral inside the bracket while naming the compound. Examples:

H_2SO_4 Sulphur (VI) acid, H_2SO_3 Sulphur (IV) acid

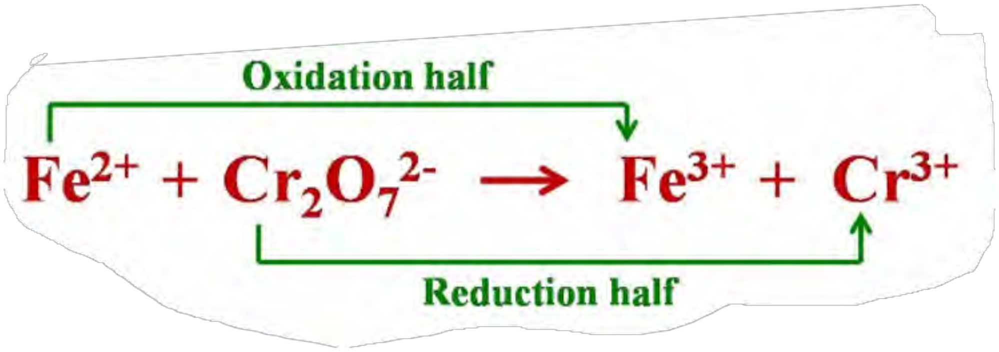
$FeCl_3$ iron (III) chloride , $FeCl_2$ iron (II) chloride

Test yourself!

Find the oxidation number of Sulphur atom in the given molecule



Oxidation and reduction in terms of oxidation numbers
Activity:

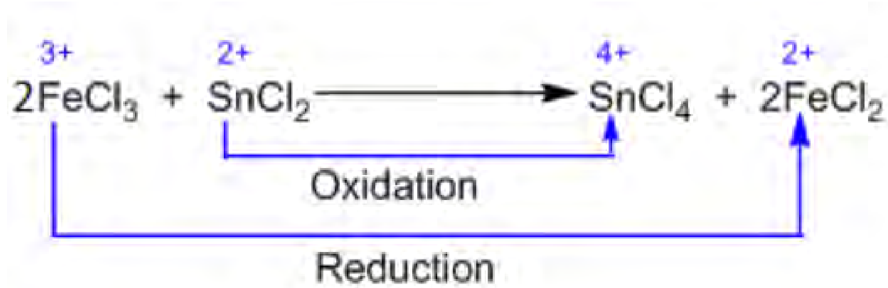


a. Find the oxidation number of **Fe** and **Cr** in both reactants and products.

b. Identify whether there is increase or decrease in oxidation number in these atoms during the chemical reaction.

If an atom gains electrons, the oxidation number decreases. Similarly, if an atom loses electrons, the oxidation number increases.

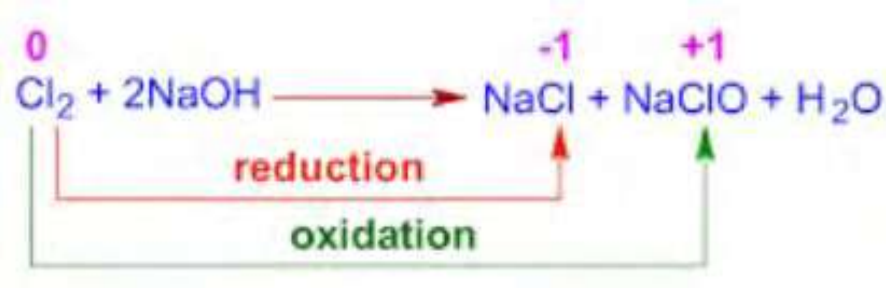
Oxidation is the increase in oxidation number and reduction is the decrease in oxidation number.

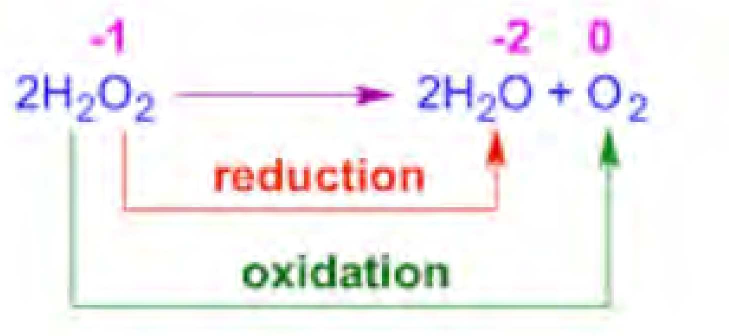


Oxidation number of iron decreases from +3 to +2. It means iron (III) chloride is reduced to iron (II) chloride so $FeCl_3$ is an oxidising agent.

Likewise, the oxidation number of tin increases from +2 to +4. It means tin (II) chloride is oxidised to tin (IV) chloride so it is called a reducing agent.

The redox reaction which involves the same element undergoing both oxidation and reduction, is called a disproportionation reaction. Examples:





Test yourself!

Assign the oxidation number to each species and complete the table given.

a. $MnO_2 + H^+ + Br^- \rightarrow Mn^{2+} + Br_2 + H_2O$

b. $CH_4 + O_2 \rightarrow CO_2 + H_2O$

c. $Fe^{2+} + MnO_4{}^- + H+ \rightarrow Fe^{3+} + Mn^{2+} + H_2O$

Reaction	Reactant oxidised	Product of oxidation	Reactant reduced	Product of reduction	Oxidising agent	Reducing agent
a.						
b.						
c.						

Test yourself!

Calculate the oxidation number of Sulphur and Chromium in the following species:

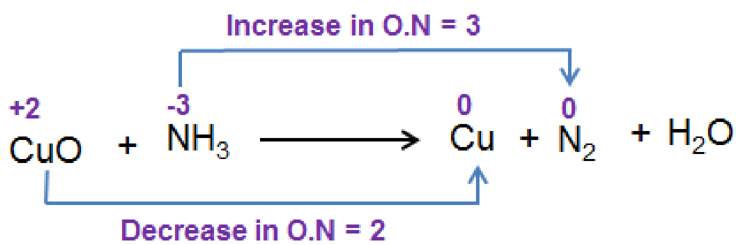
$SO_4{}^{--}$ b. $\mathbf{Cr}_2O_7{}^{--}$
c. $K_2\mathbf{Cr}O4$ d. $H_2\mathbf{S}O_3$

6.3 Balancing Redox reaction by Oxidation number and Ion electron method

Balancing redox reaction by these methods is based on the principle that the number of electrons lost in the oxidation is equal to the number of electrons gained in the reduction.

Oxidation number method

Activity:



Identify the atom that loses the electrons and the atom that gains the electrons.

Calculate the number of electrons lost or gained by each atom while moving to product.

The different steps to balance the equation by this method involves.

a. Assign the oxidation number for each atom involved in the reaction.

b. Identify oxidant and reductant on the basis of changes in oxidation number in reactant and product of the same species.

c. Write the total change in oxidation number per molecule or ion given.

d. Cross-multiply the obtained numbers so that loss of electrons equals to the gain of electrons per atom.

e. Balance other atoms, including metals, non-metals, hydrogen, and oxygen, by hit and trial method.

Solved example 1:

$$\mathbf{HNO_3(\mathit{aq}) + H_2S(\mathit{aq}) \longrightarrow NO(\mathit{g}) + S(\mathit{s}) + H_2O\ (\mathit{l})}$$

Step 1: Assigning the oxidation number of each atom

$$\overset{+5}{HN}O_3(aq) + H_2\overset{-2}{S}(aq) \longrightarrow \overset{+2}{N}O(g) + \overset{0}{S}(s) + H_2O\ (l)$$

Step 2: Identifying oxidant and reductant. In this equation HNO_3 is oxidant and H_2S is reductant

$$\overset{+5}{HNO_3}(aq) + \overset{-2}{H_2S}(aq) \longrightarrow \overset{+2}{NO}(g) + \overset{0}{S}(s) + H_2O\,(l)$$

$3e^-$ $2e^-$

Step 3: Writing the change in oxidation number of nitrogens is 3 (3 electrons are gained) and change in oxidation number of Sulphur is 2(2 electrons are lost)

$$\overset{+5}{HNO_3}(aq) + \overset{-2}{H_2S}(aq) \longrightarrow \overset{+2}{NO}(g) + \overset{0}{S}(s) + H_2O\,(l)$$

$3e^-$ $2e^-$

Step 4: Multiplying by criss-cross method. The number of electrons lost should be equal to the number of electrons gained i.e. there should be per atom.

$$\overset{+5}{HNO_3}(aq) + \overset{-2}{H_2S}(aq) \longrightarrow \overset{+2}{NO}(g) + \overset{0}{S}(s) + H_2O\,(l)$$

$3e^-$ $2e^-$

$$2HNO_3(aq) + 3H_2S(aq) \longrightarrow NO(g) + S(s) + H_2O(l)$$

Step 5: Balancing chemical equation by hit and trial method.

$$2HNO_3(aq) + 3H_2S(aq) \longrightarrow 2NO(g) + 3S(s) + 4H_2O(l)$$

Solved example 2:

Balance the given reaction in acidic medium.

$$\mathbf{Cr_2O_7{}^{2-}(aq) + Cl^-(aq) \rightarrow Cr^{3+}(aq) + Cl_2(g)}$$

$$\overset{+6}{Cr_2O_7}{}^{2-}(aq) + \overset{-1}{Cl^-}(aq) \rightarrow \overset{+3}{Cr^{3+}}(aq) + \overset{0}{Cl_2}(g)$$

The total change in oxidation number of chromium in $Cr_2O_7{}^{--}$ is 3×2 and total change in oxidation number of chlorine is 1. Balancing the number of electrons lost equals the number of electrons gained .

$$Cr_2O_7{}^{2-}(aq) + Cl^-(aq) \rightarrow Cr3^+(aq) + Cl_2(g)$$

$3\times2=6$ 1

$Cr_2O_7{}^{2-}(aq) + 6Cl^-(aq) \rightarrow 2Cr^{3+}(aq) + 3\,Cl_2(g) + 7H_2O$

$Cr_2O_7{}^{2-}(aq) + 6Cl^-(aq) + 14H^+ \rightarrow 2Cr^{3+}(aq) + Cl_2(g) + 7H_2O$

Solved example -3:

Balance the given equation by oxidation number method.

$NO_2 + H_2O \longrightarrow HNO_3 + HNO_2$

+4 +5 +3

$NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

In the given reaction nitrogen atom is both oxidised and reduced, so is called disproportionation reaction.

+4 +4 +5 +3

$NO_2 + NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

Change in oxidation of nitrogen =1

Change in oxidation number of another nitrogen =1

$NO_2 + NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

$1e^-$ $1e^-$

Loss of **1** electron (oxidation) gain of **1** electron (reduction)

Criss crossing the number so that loss of electrons equals to the gain of electrons.

$NO_2 + NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

$1e^-$ 1e

Now the equation is balanced.

Oxidizing agent $= NO_2$

Reducing agent$= NO_2$

Try yourself!

Balance the given reaction by oxidation number method

a. $FeSO_4 + K_2Cr_2O_7 + H_2SO_4 \longrightarrow Fe_2(SO4)_3 + K_2SO_4 + Cr_2(SO_4)_3 + H_2O$
b. $H_2O_2 + ClO_2 \longrightarrow ClO_2^- + O_2$ (in basic solution)
c. $Cl_2 + NaOH$ (Cold & dil.) $\longrightarrow NaCl + NaOCl + H_2O$
d. $Br^- + Cl_2 \longrightarrow Cl^- + Br_2$
e. $Cr_2O_7{}^{2-}(aq) + Fe^{2+}(aq) \longrightarrow Cr^{3+}(aq) + Fe^{3+}(aq) + H_2O(l)$ (in acidic medium)

B. Ion electron (half -reaction) method

Activity:

Observe the given ionic equation.

$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$

a. Identify the species that loses and gains electrons.

b. What is the total charge of reactant and total charge of the product?

The systematic process for balancing by this method is outlined below:

1. Assign oxidation number of each atom of the skeleton of given chemical equation.
2. Identify oxidant and reductant on the basis of changes in oxidation number of the same species in reactant and product.
3. Separate the given equation into two half-equations: oxidation half and reduction half.
4. Balance each half-equation separately. Always balance metal atoms first followed by nonmetal, hydrogen, oxygen and charge respectively in both halves.
5. Balance oxygen by adding the required number of H2O molecules in the deficient side in both halves.
6. Balance hydrogen by adding H + in the deficient side in both halves.
7. If reaction is carried in alkaline medium add –OH to neutralize H+ ions in both halves.
8. Balance charge by adding e- in the required side in both halves.
9. Add both halves reactions to make a single and add any required radicals or groups or atoms in equal number in both halves.

Solved Example1:

Balance the given redox reaction by ion electron (half reaction) method.

$Cu + Ag^{+} \longrightarrow Cu^{++} + Ag$

Solution;

$Cu \longrightarrow Cu^{++} + 2e^{-}$ (oxidation- half)

$Ag^{+} + e^{-} \longrightarrow Ag$ (reduction- half)

Multiplying reduction half by 2 so that number of electrons lost equals number of electron gain

$Cu \longrightarrow Cu^{++} + 2e^{-}$] × 1

$Ag^{+} + e^{-} \longrightarrow Ag$] ×2

$Cu + 2Ag^{+} \longrightarrow Cu^{++} + 2Ag$ (Adding the balanced half equation)

This is the balanced redox equation in neutral medium.

Oxidant: Ag^{+}

reductant: Cu

Fig: Reaction between copper metal and silver nitrate solution

Fireflies produce light by the oxidation of complex chemicals (luciferin) in air. This process is called **bioluminescence**.

When cut pieces of apple are exposed to air, they turn brown due to the oxidation of phenolic compounds present in the apple.

Solved example 2:

Balance the given equation in acidic medium by ion electron (Half -reaction) method.

$MnO_4^{-}(aq) + Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + H_2O(l) + Fe^{3+}(aq)$

Solution;

$MnO_4^{-}(aq) + 8H^{+}(aq) + 5e^{-} \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$ [reduction - half]

$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^{-}$ [oxidation -half]

Multiplying oxidation half by 5

$MnO_4^{-}(aq) + 8H^{+}(aq) + 5e^{-} \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$

$[Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^{-}\ ]\times 5$

$MnO_4^{-}(aq) + H^{+}(aq) + Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 4H_2O(l) + Fe^{3+}(aq)$

This is the balanced redox reaction in acidic medium.

Oxidising agent: MnO_4^{-}

reducing agent: Fe^{2+}

Fig: Redox titration between KMnO4 and ferrous sulphate in acidic medium

Solved example 3:

Balance the given equation in alkaline medium by ion electron method.

$Cl^{-}(aq) + Cr_2O_7{}^{2-}(aq) \longrightarrow Cr^{3+(aq)} + Cl_2(g) + H_2O$

$6e\text{-} + 14H^{+} + Cr_2O_7{}^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$ (reduction half)

$2Cl^{-} \longrightarrow Cl_2 + 2e\text{-}$(oxidation half)

Multiplying the oxidation half by 3 so that loss of electrons equals to gain of electrons

$6e^{-} + 14H^{+} + Cr_2O_7{}^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$ (reduction half)

$2Cl^{-} \longrightarrow Cl_2 + 2e\text{-}$(oxidation half)] $\times 3$

$14H^{+}(aq) + 6Cl^{-}(aq) + Cr_2O_7{}^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 3Cl_2(g) + 7H_2O\ (l)$

This is the balanced equation in acidic medium

$14\ OH^{-}(aq) + 14H^{+}(aq) + 6Cl^{-}(aq) + Cr_2O_7{}^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 3Cl_2(g) + 7H_2O\ (l) + 14\ OH^{-}$ (adding OH- in both sides)

$14H_2O(l) + 6Cl^{-}(aq) + Cr_2O_7{}^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 3Cl_2(g) + 7H_2O(l) + 14\ OH^{-}$

$7H_2O(l) + 6Cl^{-}(aq) + Cr_2O_7{}^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 3Cl_2(g) + 14\ OH^{-}\ (aq)$

This is the balanced redox equation in alkaline medium.

Test yourself!

Balance the given equation by ion electron (half- equation) method and identify the oxidant and reductant.

a. $MnO_4{}^{-}(aq) + C_2O_4{}^{2-}(aq) \longrightarrow CO_2(g) + Mn^{2+}(aq)$ (in acidic solution)
b. $Mg(s) + N_2(g) \longrightarrow Mg_3N_2(s)$ (in neutral solution)
c. $Cr_2O_7{}^{2-}(aq) + Br^{-} \longrightarrow Cr^{3+}(aq) + Br_2(aq)$ (in acidic solution)
d. $C_2O_4{}^{2-} + MnO_4{}^{-} \longrightarrow CO_3{}^{2-} + H_2O + MnO_2$ (in alkaline medium)

6.4 Electrolysis

Nepal has been successful in using hydrogen as a cooking fuel. More than a dozen scientists from Nepal, the UK, and Switzerland have been conducting an 18-month experiment, according to the Kathmandu University Hydrogen Lab.

Activity:

Read the news article published in the newspaper:

a. Why is hydrogen regarded as a cooking fuel? Why is this fuel called green fuel?

b. Name the substance used and the process carried out to produce hydrogen using electricity.

Fig: electroplating of copper on iron spoon

Copper plating on an iron spoon is an application of electrolysis which is shown in the figure above.

a. Identify the terminal of the battery in which the copper plate and iron spoon are connected.

b. What is the solution kept in the beaker?

Have you ever observed gold-plated ornaments? Gold metal is plated onto copper(usually) using a solution containing gold salt (such as gold chloride or gold cyanide). This technique, known as **electroplating,** involves the deposition of one metal onto another.

The electroplating of metal is due to the decomposition salt solution into its constituent ions. The salt solution is either molten or aqueous (with water) and is a good conductor of electricity.

The aqueous or in molten form of the substance which is used in electrolysis is called electrolyte. Electrolyte conducts electricity by the migration of its ions.

Electrolytes may be weak or strong. For example, H_2SO_4 solution is a strong electrolyte whereas acetic acid (CH_3COOH) is a weak electrolyte. Solutions of cane sugar, glycerin, alcohol etc., are examples of non-electrolytes which do not decompose to produce ions.

This process, in which a chemical compound breaks down into its ions with the help of external electrical energy, is termed electrolysis.

In electrolysis, electrical energy is used to carry out a non-spontaneous chemical change. A cell arranged for such a purpose is called an **electrolytic cell.**
The good conductor solids dipped in electrolyte are called **electrodes**. Electrode connected with the positive terminal of the battery is called **Anode** and connected with the negative one is called Cathode**.**
When electrical energy is passed through the electrolyte, the electrolyte dissociates into its corresponding ions i.e. electrolysis occurs.

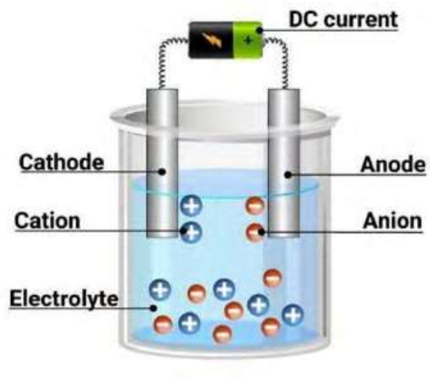


Fig: electrolysis

Let us consider an electrolyte AB is dissociated into A^+ and B^- ions. A^+ being the positively charged ion moves towards the cathode called **cation** and negatively charged ion (B^-) moves towards the anode called **anion**.

$$\mathbf{AB \longrightarrow A^+ + B^-}$$

Cation gains electrons and gets reduced to neutral atoms at cathode. Likewise, at anode anion loses its electron and oxidised to parent atom.

At **a**node

$B^- \longrightarrow B + e^-$ (**o**xidation)

Examples :

$2Cl^- \longrightarrow Cl_2 + 2e^-$

$4OH^- \longrightarrow O_2 + 2H_2O + 4e^-$

At cathode

$A^+ + e^- \longrightarrow A$ (reduction)

Examples

$Na^+ + e^- \longrightarrow Na$

$Cu^{++} + 2e^- \longrightarrow Cu$

6.4.1 Qualitative aspects of electrolysis

Activity:

Name the electrodes A and B.

Identify the gas produced at A and B.

Hydrogen gas is a potential clean energy source. It is mainly produced from the electrolysis of water.
In the presence of acid, water ionizes as

$2H_2O \rightarrow H^+ + OH^-$

These ions formed move towards the opposite electrodes.

At the cathode (negative electrode):

$2H^+ + 2e^- \rightarrow H_2(g)$

At the anode (positive electrode):

$4OH^- \rightarrow O_2(g) + 2H_2O + 4e^-$

The overall reaction for the electrolysis of water: $2H_2O \rightarrow 2H_2 + O_2$

The product of electrolysis depends upon

a. nature of electrolyte (whether it is molten or aqueous)
b. nature of electrodes (whether it is active or inert)

Why is acid or salt added to pure water for the electrolysis of water?

Pure water has only 10^{-7} M H^+ ions and 10^{-7} M OH^- ions. These very small quantities of ions are not enough to conduct the current. Acid or salt added produces sufficient number of ions to conduct electricity.

A. Electrolysis of molten NaCl using Pt electrodes.

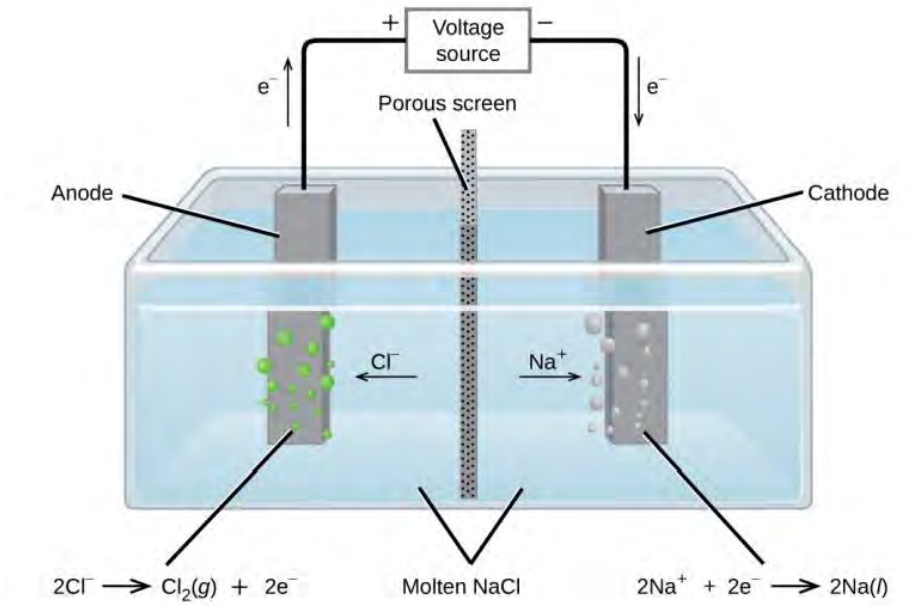


Fig: electrolysis of molten NaCl

A Down's cell is a commercial electrochemical cell used to obtain sodium metal. This process involves the electrolysis of molten sodium chloride using Pt (inert) electrodes. The reaction at anode and cathode are at cathode.

$Na^+ + e^- \rightarrow Na$ (reduction)

Sodium metal is deposited at cathode

at anode

$Cl^- \rightarrow Cl + e^-$ (oxidation)

$Cl + Cl \rightarrow Cl_2$

chlorine gas is liberated at anode.

B. Electrolysis of aqueous NaCl solution using Pt electrodes

When aqueous solution (salt dissolved in water) is used as an electrolyte, both the salt and the water take part in the reaction and form different types of ions. If there are different kinds of ions

near an electrode, the ones that need the least energy (lowest discharge potential) will be changed into products.
Consider the electrolysis of aqueous solution of NaCl.

Both NaCl and water ionizes,

$NaCl \rightarrow Na^{+} + Cl^{-}$

$H_2O \rightarrow H^{+} + OH^{-}$

At the anode, both Cl^{-} and OH^{-} ions move but discharge potential of Cl^{-} ion is less than that of OH^{-}. It means it is easier for Cl^{-} to change into Cl_2 than the OH^{-} ion to O_2.

The potential at which the ion is discharged or deposited on the appropriate electrode is termed the discharge or deposition potential.

The decreasing order of discharge potential of some of the ions is given below,

For cations:

$Li^{+} > K^{+} > Na^{+} > Ca^{++} > Mg^{++} > Al^{+++} > Zn^{++} > Fe^{++} > Ni^{++} > H^{+} > Cu^{++} > Hg^{+} > Au^{+++}$

For anions:

$SO_4^{--} > NO_3^{-} > OH^{-} > Cl^{-} > Br^{-} > I^{-}$

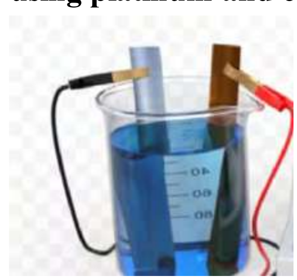
$2Cl^{-} \rightarrow Cl_2(g) + 2e^{-}$

At cathode, both H^{+} and Na^{+} ion move but the discharge potential of H^{+} is lower than that of Na^{+}. So, it is easier for H^{+} to become H_2 than Na^{+}.
$2H_2O + 2e^{-} \rightarrow H_2(g) + 2OH^{-}$

So, the products of electrolysis of aqueous solution of NaCl is H_2 and Cl_2 respectively at cathode and anode.

C. Electrolysis of Copper sulphate solution using platinum and copper electrodes.

I



II

Fig: electrolysis of CuSO4 solution using (I)graphite (inert)electrodes and (II)Copper(active) electrodes

In both cases, the ionization of copper sulphate and water occur as follows:

$CuSO_4 \rightarrow Cu^{2+}$ and SO_4^{2-}

$H_2O \rightarrow H^{+}$ and OH^{-} ions

The reaction at cathode is same for both the cases.

$Cu^{2+}+2e^- \rightarrow Cu$ [As Cu^{2+} has lower discharge potential than H^+, it is reduced first]

Using platinum (inert) electrodes	**Using copper electrodes (active)**
At anode, As OH^- has lower discharge potential than SO_4^{2-}, it gets preferentially oxidised. $OH^- \rightarrow O_2(g) + H_2O + 4e^-$ Hence, Cu at cathode and O_2 at anode are produced by electrolysis.	At anode, Cu oxidizes prior to the oxidation of OH^- and SO_4^{--} $Cu \rightarrow Cu^{++}$ At the end, copper at anode becomes thin whereas copper at cathode becomes thick in size.

The reaction at anode is different due to different nature of the electrodes.

The product of electrolysis of some common electrolytes using inert (non -attackable) or active electrodes(attackable) are summarized in the table below.

Electrolyte	Electrode	Product at anode	Product at cathode
Molten NaCl	Pt or graphite	$2Cl^- \longrightarrow Cl_2 + e^-$	$Na^+ + e^- \longrightarrow Na$
Aqueous NaCl	Pt or graphite	$2Cl^- \longrightarrow Cl_2 + e^-$	$2H^+ + e^- \longrightarrow H_2$
NaOH(molten)	Pt or graphite	$4OH^- \longrightarrow O_2+2H_2O +4e^-$	$Na^++e^- \longrightarrow Na$
NaOH(molten)	Pt or graphite	$4OH^- \longrightarrow O_2+2H_2O +4e^-$	$2H^+ + e^- \longrightarrow H_2$
$CuSO_4$ solution	Pt or graphite	$4OH^- \longrightarrow O_2+2H_2O +4e^-$	$Cu^{++}+2e^- \longrightarrow Cu$
$CuSO_4$ solution	Cu electrode	$Cu \longrightarrow Cu^{++}+2e^-$	$Cu^{++}+2e^- \longrightarrow Cu$
Conc.H_2SO_4	Pt or graphite	$HSO_4^- \longrightarrow H_2S_2O_8 +2e^-$	$2H^+ + e^- \longrightarrow H_2$
Dill. H_2SO_4	Pt or graphite	$4OH^- \longrightarrow O_2+2H_2O +4e^-$	$2H^+ + e^- \longrightarrow H_2$
$AgNO_3$ solution	Pt or graphite	$4OH^- \longrightarrow O_2+2H_2O +4e^-$	$Ag^++ e^- \longrightarrow Ag$
$AgNO_3$ solution	Ag electrode	$Ag \longrightarrow Ag^+ + e^-$	$Ag^++ e^- \longrightarrow Ag$

Test yourself!

What will be the product of electrolysis of $MgCl_2$ (molten) and aqueous using graphite electrodes?

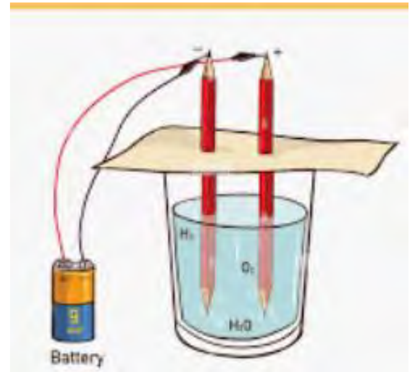
Project work

a. Fill the beaker with water and dissolve a small amount of dilute acid or salt to make the water conductive.

b. Immerse two pencils (graphite rod) in the water.

c. Connect one pencil to the positive terminal and the other pencil to the negative terminal of the battery as shown in the diag.

d. Observe any changes that occur inside the beaker.

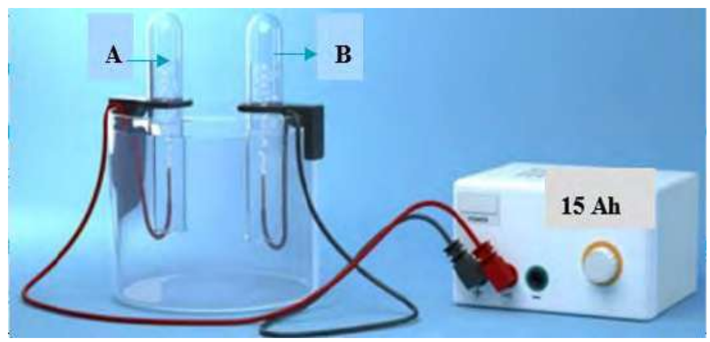


6.4.2 Quantitative Aspects of Electrolysis

Activity:

The figure illustrates the electrolysis of water.

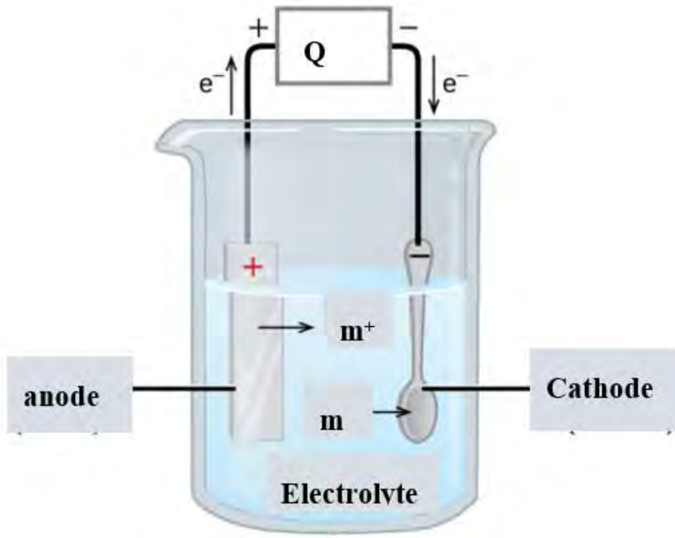
What will be the change in the volume of gas A and B, if 10 Ah (ampere hour) battery is used by replacing the given battery for the same time.



Michael Faraday demonstrated the quantitative relationships between the amount of charge used and the quantities of materials produced or consumed during electrolysis.

These relationships he discovered are known as **Faraday's laws.** When the amount of current or time to conduct the electrolysis increases, the amount of substance formed in the respective electrodes also increases.

Michael Faraday (1791-1867) an English scientist, made significant contributions to the fields of electromagnetism and electrochemistry. His groundbreaking work included the formulation of the law of electrolysis, which advanced our understanding of the complicated relationship between electricity and chemical reactions.



If m is the mass of a substance deposited at cathode by using Q amount of charge,

$m \propto Q$

or $m \propto I \times t$

or $m = z\,Q = z \times I \times t$

where I is the current in amperes (A), 't' is the time is second (s) and z is a constant called **electrochemical equivalent.**

if Q = 1Coulomb, then m = z

Hence, electrochemical equivalent(z) of a substance is defined as**;**

"the mass of the substance liberated or deposited in electrolysis by the passage of 1 coulomb of charge"

SI unit of z is Kg/C

The quantity of charge is usually expressed in Faraday (F).

1 Faraday is the quantity of electric charge carried by 1 mol of electrons.

1F = charge of 1 mol of electrons

= charge of 1 electron × 1 mol electron

$= 1.6\times10^{-19}\times6.022\times10^{23}$

= 96500 coulombs

For a general reaction of metal ion

$M^{n+}(aq) + ne^{-} \longrightarrow M(s)$

Charge (Q) = nF

Where F is a faraday's constant =96500 coulomb.

Examples:

$Pb^{2+} + 2e^{-} \rightarrow Pb$

1 mol 2 mol 1 mol [2F of charge is required to deposit 1mol of lead]

$2Cl^- \rightarrow$ Cl_2 + $2e^-$

2 mol 1 mol 2 mol [2F of charge is required to produce 1 mol of chlorine gas]

1st law of electrolysis:

In electrolysis, the mass of an element deposited on the cathode, or liberated on anode during passage of current through a solution is directly proportional to the charge which passed through the electrolytic solution.

Solved problem:

Calculate the mass of aluminum deposited at cathode when a current of 25.0 A is passed through the Aluminum solution for 15.0 minutes.

Solution,

Charge (Q) = It

$= 25A \times 15 \times 60$ second

$= 22{,}500$coulomb

$= 0.23F$

The reaction at cathode $Al^{3+} + 3\,e^- \longrightarrow Al(s)$

3 mol of electrons = 1 mol of Al

3 F = 27g

0.23F= 2.09g

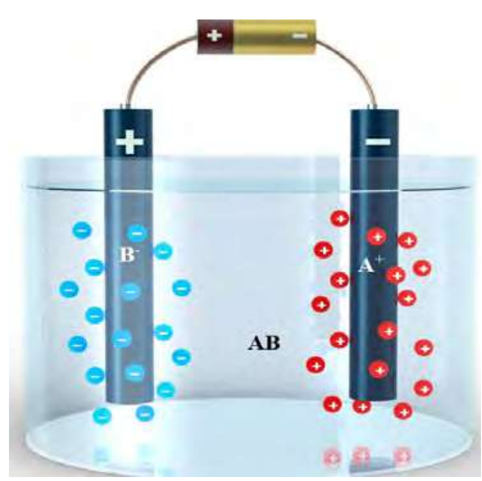
Test yourself!

In the given figure, electrodes A and B are kept in an electrolyte (A^{++}).

a. Identify the electrode (A or B) where oxidation and reduction take place.

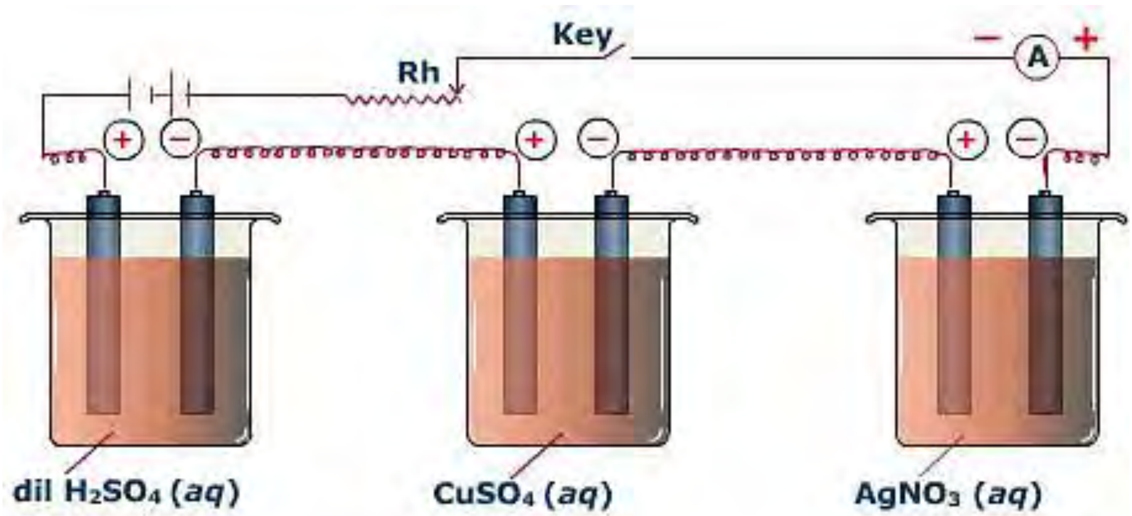
b. Outline the reaction occurring at the cathode, and anode.

c. Identify the electrode that increases in size after the reaction.



Second law of electrolysis

Activity:



a. Write the balanced equation for the ionization of each electrolyte given.

b. Identify the substances produced at cathode in each beaker.

In the case of different electrolytes connected in series, the amount of charge is the same for all the electrolytes. In such cases, the amount of substance at the electrodes only depends upon the equivalent weight of that substance. Higher the equivalent weight more the mass of the substance formed and vice versa.

Mathematically, **m∝ E**

$$\text{or, } \frac{m1}{E1} = \frac{m2}{E2} = \frac{m3}{E3}$$

where E is the chemical equivalent or equivalent weight of an element which is numerically equal to atomic mass divided by its valency. **E=** $\frac{A}{V}$

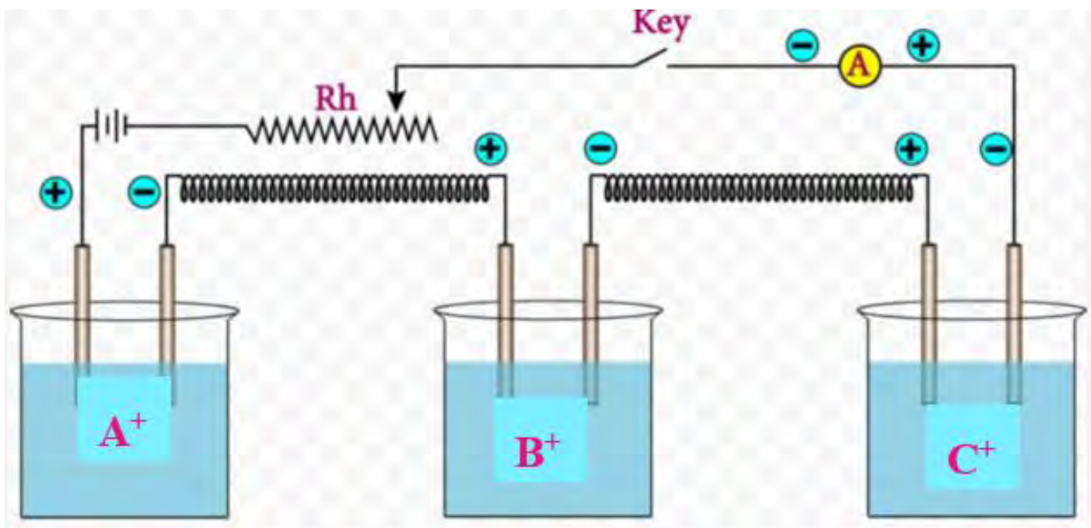


Fig: electrolysis of different electrolyte connected in series

When same charge is passed through three aqueous solutions of A^+, B^+ and C^+, then the mass of substance deposited at the respective cathode is:

$$\frac{mA}{EA} = \frac{mB}{EB} = \frac{mC}{EC}$$

Relationship among electrochemical equivalent (Z), equivalent weight(E) and faraday constant(F)

We know that electrochemical equivalent(z) is the mass of a substance when 1 coulomb of charge is passed.

If Q = 1C   m = z

If Q= 96500 C(1F) m=96 500 × z = equivalent weight (E)

Where E is mass deposited by 1 faraday of charge

E = 96 500 × z = z × F

or z= E/F

therefore, the first law of electrolysis can be written as

Mass deposited or liberated (m) = $\frac{E}{F} \times \mathbf{I} \times \mathbf{t}$

2nd law of electrolysis states that

"When the same quantity of electricity is passed through different electrolytes, connected in series, the masses of different substances produced at electrodes are proportional to their chemical equivalents (equivalent weight)"

Solved example 1:

Constant current was passed for 5 hours through the solution of copper sulphate using copper electrodes. Calculate the amount of current passed to deposit 6.3 g of copper at cathode.

Solution

The mass of Cu deposited in cathode(m) = 0.404 g.

Atomic mass of Cu = 63.5 amu

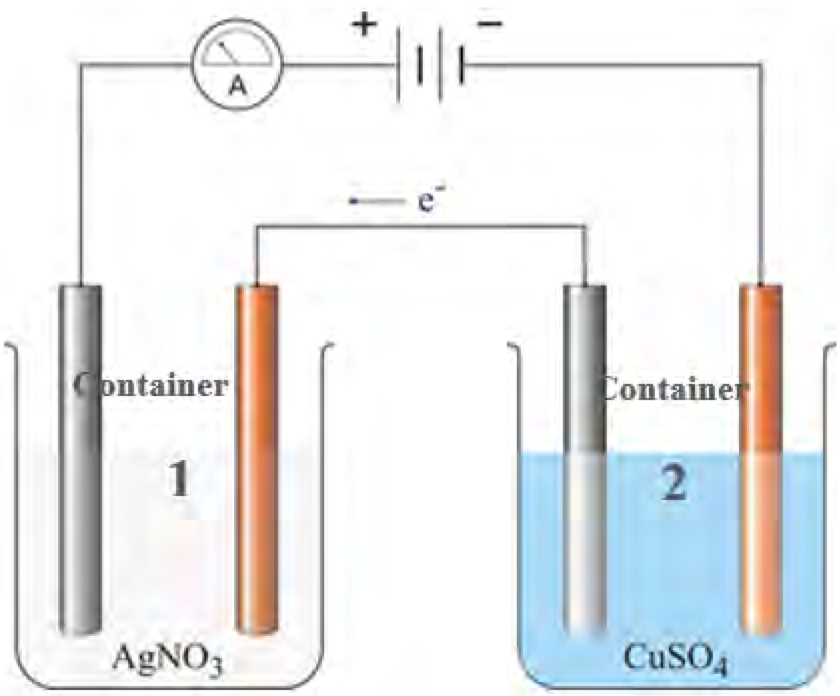
M=z It

$M = \frac{E}{F} \times I \times 5 \times 60 \times 60$

$0.404 = \frac{63.5}{96500 \times 2} \times I \times 5 \times 60 \times 60$

I =0.06A

Solved example 2:



Determine the mass of metal deposited in container 2, given that 1.08 g of silver is deposited in the first container.

Atomic mass of Ag=108 amu and Cu=63.5 amu

Solution,

Using second law of electrolysis

$$\frac{mass\ of\ silver}{Equivalent\ mass\ of\ silver} = \frac{mass\ of\ copper}{Equivalent\ mass\ of\ copper}$$

1.08

$$\frac{1.08\times1}{108} = \frac{mass\ of\ copper\times2}{63.5} \quad [E=\frac{A}{V}]$$

∴ Mass of copper deposited =0.635 g

Test yourself!

Consider the electrolysis of water, represented by the equation:

$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

a. Calculate the amount of hydrogen gas (in mol) liberated when a current of 5.0 A is passed through water for 2 hours.

b. Determine the volume of oxygen gas (in L) evolved at STP (molar volume = 22.4 Lat STP)

Exercises

A. Multiple choice questions

1. Which of the following statements is true about oxidation?

a. Gain of electrons and increase in positive charge

b. Loss of electrons and decrease in positive charge

c. Loss of electrons and increase in positive charge

d. Gain of electrons and decrease in positive charge

2. What is the substance that undergoes reduction in the redox reaction?

a. reducing agent because it reduces other substance

b. reducing agent because it oxidizes other substance

c. oxidising agent because it oxidizes other substance

d. reducing agent because it reduces other substance

3. In the given redox reaction:

$2Fe^{3+} + Sn^{2+} \rightarrow 2Fe^{2+} + Sn^{4+}$

What is the total number of electrons exchanged (lost and gained)?

a. 1

b. 2

c. 3

d. 4

4. Electrolysis of water involves the decomposition of water into hydrogen and oxygen gases. According to Faraday's laws, if 1 mol of oxygen is liberated, how many mol of hydrogen will be liberated?

a. 0.5 mol

b. 1 mol

c. 2 mol

d. 3 mol

5. Which of the following is a disproportionation reaction?

a. $N_2+O_2 \rightarrow NO$

b. $Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$

c. $NO_2+OH^- \rightarrow NO_2^- + NO_3^- + H_2O$

d. $Cu + Zn^{++} \rightarrow Cu^{++} + Zn$

6. Electrolysis of molten aluminum oxide (Al_2O_3) is used in the production of aluminum. If 108 g of aluminum is deposited, calculate the amount of electricity (in coulombs) required (atomic mass of Al=27).

a. 96,500 C

b. 193,000 C

c. 48,250 C

d. 24,125C

7. Which of the following is true in the reduction of 2 mol of permanganate ion (MnO_4^-) to Mn^{++} in acidic medium?

   a. 5 mol of electrons are lost.

   b. 10 mol of electrons are gained.

   c. 5 mol of electrons are gained.

   d. 10 mol of electrons are lost.

8. If the reaction at anode and cathode are:

   $Al \longrightarrow Al^{+++} + 3e^-$ (at anode)

   $H^+ + e^- \longrightarrow H_2$ (At cathode)

   Which one of the following is the overall redox reaction?

   a. $Al + Al^{+++} \longrightarrow H^+ + H_2$

   b. $H^+ + H_2 \longrightarrow Al + Al^{+++}$

   c. $H_2 + Al^{+++} \longrightarrow Al + H^+$

   d. $Al + H^+ \longrightarrow Al^{+++} + H_2$

9. During the electrolysis of copper chloride ($CuCl_2$), a current of 2.0 A is passed for 30 minutes. If the atomic mass of copper is 63.5 g/mol, calculate the mass of copper deposited on the cathode.

   (Given: Faraday's constant =96,500C/mol)

   a. 0.25 g

   b. 0.50 g

   c. 1.00 g

   d. 2.00 g

10. What are the products of electrolysis of aqueous solution of KCl upon using platinum electrodes?

    a. H_2 at cathode and O_2 at anode

    b. K at cathode and O_2 at anode

    c. H_2 at cathode and O_2 at anode

    d. K at cathode and Cl_2 at anode

11. In the reaction

$2Fe^{3+}+Sn^{2+} \rightarrow 2Fe^{2+}+Sn^{4+}$ which one is true?

a. Fe^{3+} is oxidised

b. Fe^{2+} is oxidised

c. Sn^{4+} is oxidised

d. Sn^{2+} is oxidised

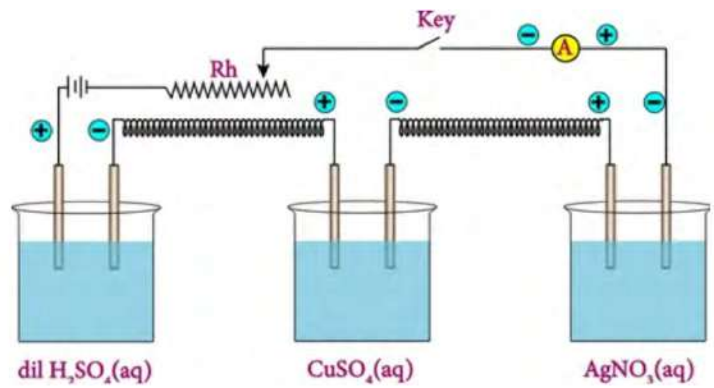
12. Which of the following elements has the highest mass in the cathode after 10 minutes of electrolysis in the figure given?

a. H_2

b. Cu

c. Ag

d. SO_2



B. Short answer questions

1. Define the following terms with one example of each:
   a. Oxidation in electronic concept
   b. Reduction in electronic concept
   c. Redox reaction
   d. Oxidation in terms of oxidation number
   e. Oxidising and reducing agent

2. Differentiate the following:
   a. Oxidising and reducing agent
   b. Oxidation and reduction
   c. Electrochemical equivalent and chemical equivalent
   d. 1^{st} law and 2^{nd} law of electrolysis

3. State and explain the following:
   a. Electrolysis
   b. Faraday's 1^{st} law of electrolysis
   c. Faraday's 2^{nd} law of electrolysis
   d. Oxidation number

4. Give reason with a suitable chemical equation.
   a. Combustion of ethane is a redox reaction
   b. Oxidation and reduction occur simultaneously
   c. Oxidising agent itself is reduced

5. The lead-acid battery, commonly known as the lead storage battery, undergoes a redox reaction during both discharge and recharge. Here's the basic reaction during discharge:

$$PbO_2 + Pb + H_2SO_4 \rightarrow PbSO_4 + H_2O$$

a. Write the oxidation number of leads in the reaction.

b. Write oxidation and reduction half of the given reaction.

c. Balance each half and find the balanced redox reaction.

d. What happens to the above reaction when the battery is recharged?

6. Assign oxidation numbers to the underlined atoms in the following molecules and ions:

$Na\underline{Cl}$, $\underline{Cl}O_2$, $Fe_2(\underline{S}O_4)_3$, $\underline{S}O_2$, $\underline{I}_2$, $K\underline{Mn}O_4$, $\underline{Ca}H_2$, $\underline{P}O_4^{---}$

7. Define the term oxidation number and find out oxidation numbers for nitrogen in the following species.

NF_3, N_2O, N_2H_4, NO, N_2O_4, NO_2^-, Ba_3N_2

8. Chlorine and compounds of chlorine are used as oxidising agents. Different compounds of chlorine are given:

Cl_2O, $HClO_3$, HCl, ClF_3

a. Define oxidising agent.
b. Calculate the oxidation number of chlorines in the given compounds.
c. Using oxidation numbers, which one is the strongest oxidising agent among and why?

9. The redox reaction that occurs in the dry cell (Zinc -carbon) is given.

$$Zn(s) + 2NH_4^+(aq) + 2MnO_2(s) \rightarrow Zn^{2+}(aq) + Mn_2O_3(s) + H_2O(l) + 2NH_3(aq)$$

a. Assign oxidation numbers to each atom in the given reaction and prove that it is a redox reaction.

b. Determine the oxidising and reducing agents based on changes in oxidation numbers.

10. What do you mean by redox reaction?

$$S + HNO_3 \longrightarrow H_2O + H_2SO_4 + NO_2$$

a. How can you prove that the above reaction is a redox reaction?

b. Balance the reaction by oxidation number method.

11. Balance the given redox reaction by oxidation number method and identify the oxidising and reducing agent.

a. $Zn + AgNO_3 \longrightarrow Zn(NO_3)_2 + Ag$

b. $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + H_2O + NO_2$

c. $O_2 + HI \longrightarrow H_2O + I_2$

d. $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

e. $Cl_2 + Br^- \longrightarrow Cl^- + Br_2$

f. $I_2O_5 + CO \longrightarrow CO_2 + I_2$

g. $S + HNO_3 \longrightarrow H_2O + H_2SO_4 + NO_2$

h. $Na_2S_2O3 + I_2 \longrightarrow Na_2S_4O_6 + NaI$

i. $KMnO_4 + Na_2C_2O_4 + H_2SO_4 \longrightarrow MnSO_4 + CO_2 + K_2SO_4 + Na_2SO_4 + H_2O$

j. $FeSO_4 + K_2Cr_2O_7 + H_2SO_4 \longrightarrow Fe_2(SO4)_3 + K_2SO_4 + Cr_2(SO_4)_3 + H_2O$

k. $CuO + NH_3 \rightarrow Cu + N_2 + H_2O$

l. $Cl_2 + 2KBr \rightarrow Br_2 + 2KCl$

m. $Mg + HBr \rightarrow MgBr_2 + H_2$

n. $Fe + O_2 \longrightarrow Fe_2O_3$

o. $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + NO + H_2O$

p. $Sn + HNO_3 \longrightarrow SnO_2 + NO_2 + H_2O$

q. $Fe^{2+} + MnO_4^- \longrightarrow Fe^{3+} + Mn^{2+} + H_2O$

r. $I^-(aq) + Cr_2O_7^{2-}(aq) \longrightarrow Cr^{+++}(aq) + I_2(s)$

s. $Sn + MnO_4^- \longrightarrow Sn^{2+} + Mn^{2+}$(in acidic solution)

t. $H_2O_2 + ClO_2 \longrightarrow ClO_2^- + O_2$ (in basic solution)

12. The redox reaction between chlorate ion and nitride ion is
$ClO_3^- + N^{3-} \longrightarrow Cl^- + NO_3^-$

a. Write the oxidation and reduction half of the given equation.

b. Balance each half.

c. Write the balanced redox reaction.

13. Balance the given redox reactions by ion electron method and identify oxidant and reductant.

a. $MnO_4^{-\,-} + Fe^{2+} \longrightarrow Fe^{3+} + Mn^{2+}$

b. $I^- + Cr_2O_7^{\,2-} \longrightarrow I_2 + Cr^{3+}$

c. $Cl_2 + NO^{2-} \longrightarrow Cl^- + NO^{3-}$

d. $S + NO \longrightarrow S^{2-} + NO_3^-$

e. $MnO_4^- + H_2C_2O_4 \longrightarrow Mn^{2+} + CO_2$

f. $IO_2^- + N_2 \longrightarrow I^- + NO_3^-$

g. $P + IO_4^- \longrightarrow PO_4^{3-} + I^-$

h. $SO_2 + BrO_3^- \longrightarrow SO_4^{2-} + Br^-$

i. $ClO_4^- + S \longrightarrow Cl^- + SO_4^{2-}$

j. $ClO_3^- + N^{3-} \longrightarrow Cl^- + NO_3^-$

k. $ClO^- + SO_2 \longrightarrow Cl^- + SO_4^{2-}$

l. $Ca(s) + 2\ H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$

m. $CN^-(aq) + MnO_4^-(aq) \longrightarrow CNO^-(aq) + MnO_2(s)$ (In basic solution)

n. $MnO_2(s) + BrO_3^-(aq) \longrightarrow MnO_4aq) + Br^-(aq)$ (In basic solution)

o. $ClO\text{-}(aq) + Cr(OH)_4^- \longrightarrow CrO_4^{2-}(aq) + Cl^-(aq)$ (In basic solution)

p. $NO_2^-(aq) + Al(s) \longrightarrow NH_3(g) + Al(OH)_4^-$ (in acidic solution)

q. $Cr_2O_7^{2-}(aq) + Br^- \longrightarrow Cr^{3+}(aq) + Br_2(aq)$ (in acidic solution)

r. $Sn^{4+}(aq) + H_2\ (g) \longrightarrow Sn^{2+}(aq) + H^+(aq)$ (in acidic solution)

s. $SO_3^{2-}(aq) + MnO_4^- \longrightarrow SO_4^{--} + Mn^{++}$ (in acidic solution)

C. Long answer questions

1. In an experiment of electrolysis, a certain amount of current was passed for five hours through two cells connected in series. The amount of gold deposited in one of the cells containing $AuCl_3$ solution was 9.85 g. [atomic mass of gold =196]

a. State 1st law of electrolysis.

b. Find the amount of current passed through in the given experiment.

c. Calculate the amount of copper deposited in the next cell containing copper sulphate solution. [atomic mass of Cu = 63.5)

d. What is the application of electrolysis?

2. In an electrolytic solution, 10 gm of Ag needs to be deposited on the Cu surface.

a. Identify anode and cathode for the process.

b. Write the reaction at anode and cathode.

c. Calculate the time taken, if current passed is 2×10^{-3} Ampere to deposit 10 g of Ag.

d. Name the electrolyte that can be used.

3. The given diagram illustrates Faraday's second law of electrolysis

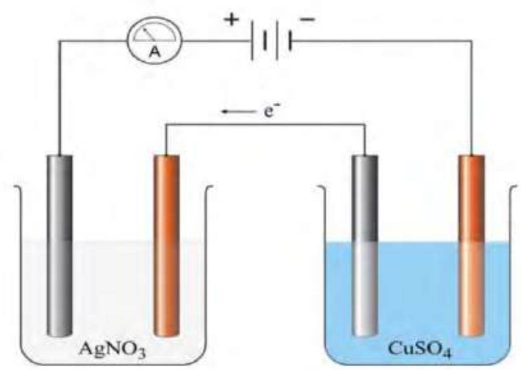
a. State second law of electrolysis.

b. Write the reaction that occurs at cathode in both beakers.

c. Calculate the mass of silver deposited at cathode if 9650 coulombs of charge are passed through the circuit.

d. Balance the given redox reaction by oxidation number or ion electron method.

$Cu^{++} + Ag \longrightarrow Ag^{+} + Cu$



4. Consider a vessel containing silver nitrate solution. If copper wire is dipped in the solution of silver nitrate, there is a change in the color of copper wire.

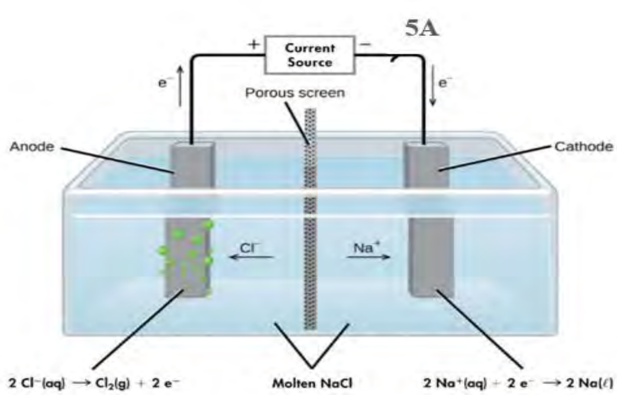
a. Write the reaction between copper and silver nitrate.

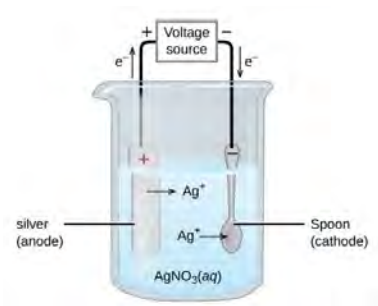
b. Assign the oxidation number in each atom and identify the oxidant and reductant.

c. Balance the given equation using the oxidation number.

d. What is the application of this type of reaction?

5. One of the applications of electrolysis is electroplating as given in the figure:





a. Define the term electrolysis and electroplating.

b. Write the reaction at anode and cathode.

c. What happens to the product if anode is replaced by platinum?

d. Find the mass of silver deposited over iron spoon if 2 amperes of current are passed for half an hour [Atomic mass of Ag=107.9]

e. Is there any change in the amount of silver nitrate in the solution? Why?

6. The redox reaction between chlorine and hot and conc. Solution of caustic soda is disproportionation or auto redox reaction:

$$Cl_2 + NaOH \rightarrow NaCl + NaClO_3 + H_2O$$

a. What do you mean by redox reaction?

b. Assign the oxidation number to chlorine atoms in reactants and products.

c. Why is this reaction called an auto redox reaction?

d. Identify the reductant and oxidant in the reaction.

e. Balance the given equation by oxidation number method.

7. Electrolysis of molten sodium chloride is given in the cell below using platinum electrodes.

a. Define the term electrolysis.

b. Write the balanced redox reaction for the cell.

c. Find the mass of sodium and volume of chlorine produced at STP at respective electrodes when the process is carried for 1 hour.

d. What is the change in the product of electrolysis if water is added on the electrolyte?

e. Give one application of this process.

Project work

A. Make a lemon battery.

a. Take a fresh lemon and insert the zinc piece into one side of the lemon and the copper piece into the opposite side.

b. Connect the galvanometer with these metal pieces using wires.

c. Observer the deflection in the galvanometer.

B. Make an electrolytic cell and perform electroplating.

a. Take a beaker, $CuSO_4$ solution, a copper piece, an iron nail, wires, and batteries.

b. Measure the weights of the copper and iron nail before beginning the experiment.

c. Connect the positive terminal of the cell to copper and the negative terminal to the iron nail.

d. Submerge these metals in a $CuSO_4$ solution and activate the circuit.

e. After a 5-minute interval, remove the metal pieces, reweigh them, and calculate the change in weight before and after the electrolysis process.

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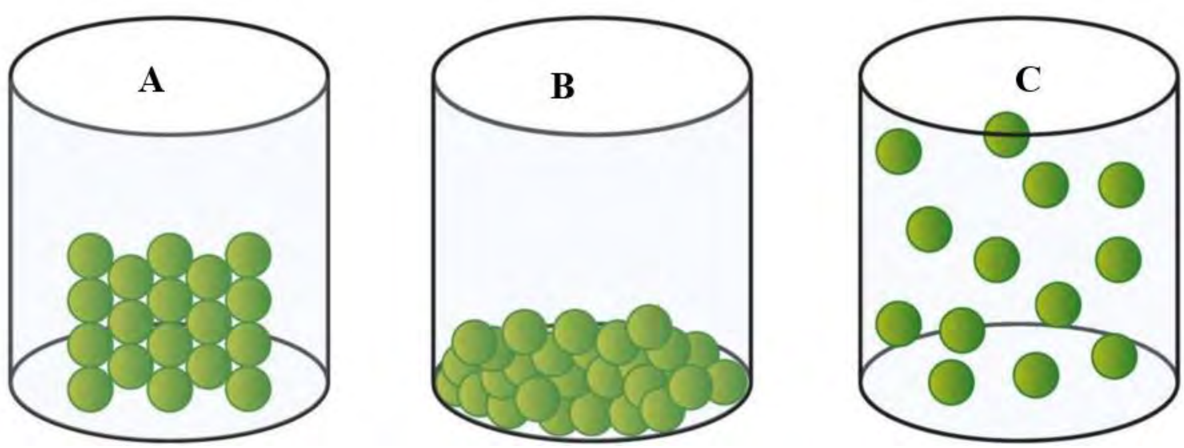
UNIT 7: STATES OF MATTER

Activity:

a. Observe the different matter given above and identify the corresponding molecular arrangement from the given figure A, B and C below.
b. Compare the intermolecular force of attraction and intermolecular space among A, B and C



Whatever you see around, you will find matter. The term "**matter"** refers to anything that has mass and occupies space. It includes all the elements, compounds, and substances that exist, whether in solid, liquid, or gaseous form. The study of matter involves understanding its composition, properties, structure, and the changes it undergoes during chemical reactions. The three states of matter differ in properties and structure due to the different molecular force of attraction between the particles.

7.1 Gaseous State

Activity:

a. Why is the balloon called a hot air balloon?
b. What is the fundamental principle behind the working of a hot air balloon?

The state of matter in which the molecular forces of attraction between the particles of matter are minimum, is known as **gaseous state.** It is the simplest state of matter in which particles are uniformly distributed.

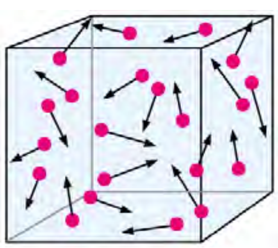


Fig: Gas exerts pressure

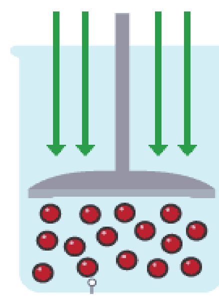


Fig: Compression of gas molecules

There is greatest intermolecular space and least intermolecular force of attraction in the gas states. Therefore, gas possesses some unique properties which are not observed in liquids and solids.

Gases do not have definite shape and volume like liquids.

Gases have very low density due to negligible intermolecular forces.
Gases have infinite expansibility and high compressibility.
Gases exert pressure.
Gases possess high infusibility. (property to diffuse)

Depending upon the temperature and pressure, the same substance can be solid, liquid or gas. As the change in temperature and pressure the intermolecular arrangements among the molecules changes and there is a change in the state of the matter.

Do you know?

1. Only 11 elements in the periodic table are gas at room temperature and pressure, they are H_2, N_2, O_2, F_2, and Cl_2 and inert gases (He, Ne, Ar, Kr, Xe, Rn).
2. LPG gas cylinders primarily contain propane and butane gas. These gases are compressed and cooled to liquid form, making them easier to store and transport.

Measurable properties of gases

Fig: measuring volume of gas by eudiometer tube

1. Volume

Since gases occupy the entire space available to them, the measurement of volume of a gas only requires a measurement of the container confining the gas.

Volume is expressed in L (L), milliliter (mL) or cubic centimeter(cc)

1L = 1 decimeter cube (dm^3) =1000 mL

2. Temperature

Gases expand on increasing the temperature and contract up on cooling and become liquid or solid at very low temperature.

Temperature is measured in centigrade degree (^{0}C) with the help of thermometers. Temperature is also measured in Fahrenheit (^{0}F).

S.I. unit of temperature is kelvin (K) or absolute degree. K = ^{0}C + 273.15

3. Pressure

Pressure is one of the most readily measurable properties of a gas. Pressure of the gas is the force exerted by the gas per unit area of the walls of the container in all directions. The pressure given by the atmospheric gas is called atmospheric pressure which helps in the upward movement of liquid through capillary.

Fig: Barometer

Pressure of a pure gas is measured by manometer while that of a mixture of gases by barometer.

The S.I. unit of pressure, the pascal (Pa),

1 Pa= 1 N/m^2

The pressure exerted by earth's atmosphere is expressed in atm unit

1 atm=101325 pascal= 760mmHg =760 torr at sea level at 0^0C

1bar =100,000 pascal

Atmospheric pressure is commonly measured by **barometer** whereas **Manometer** is used to measure the pressure of gas other than the atmosphere.

Note:

NTP (Normal Temperature and Pressure) and **STP** (Standard Temperature and Pressure) are two different sets of conditions used as reference points for measuring and comparing the properties of gases. STP is more commonly used in modern scientific Literature and research than NTP.

At STP temperature =0 ^0C (273.15K)

Pressure =1 atm=760mmHg

7.1.1 Kinetic theory of gas and its postulates

Activity:

Place some marbles or table tennis balls inside the glass container with the lid.
Shake the container and observe how the balls move more rapidly after shaking.
Allow the container to sit undisturbed for a while and observe how the motion of particles slows down.

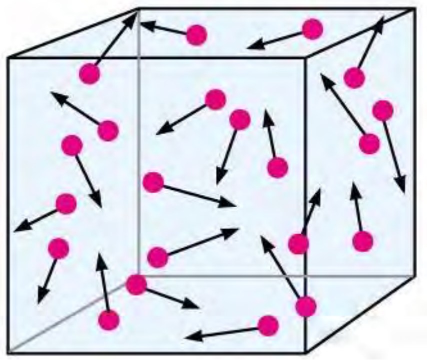


fig: Random motion of gas particle in the container

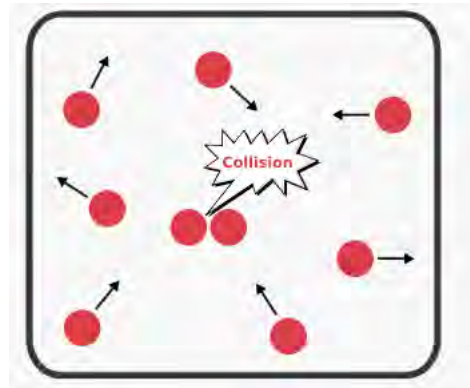


fig: Elastic collision in the gas molecules

The kinetic theory of gases provides a theoretical framework for understanding the properties of gas. This theory is useful in explaining various gas properties, such as pressure, temperature, and volume. The key assumptions of kinetic theory are:

a. Gases are composed of tiny particles (atoms or molecules) which are in constant random motion.
b. The size of individual gas particles is negligible compared to the size of the container in which gas is kept.
c. There are negligible interactions among the gas particles.
d. Due to the random motion, gas particles undergo collisions with each other and with the walls of the container.
e. Collisions between gas particles and with the walls of the container are perfectly elastic, i.e. there is no net loss of kinetic energy.
f. The pressure exerted by a gas is the result of the constant collisions of its particles with the walls of the container.
g. The temperature of a gas is directly related to the average kinetic energy of its particles. Higher temperature means higher kinetic energy of the gas molecules. It means $K.E \propto T$

Test yourself!

a. What factors contribute to the spherical shape of a tire as shown?
b. In colder winter conditions, why is there a need to inflate the tire with more air compared to the summer?

7.1.2 Gas laws

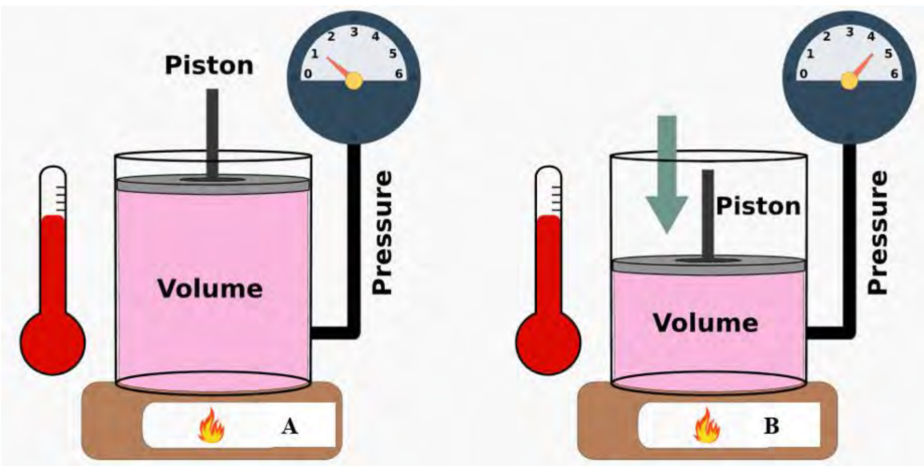
Activity:

a. Fill a balloon with air.
b. Observe any changes in the volume of the balloon.
c. Gently squeeze one end of the balloon and observe how its shape changes.

If you've ever punctured your bicycle tire or a football, you may have noticed changes in their shape. This change in the shape is due to a decrease in the volume of gas. Similarly, air balloons might burst on a sunny day. These examples represent the physical properties of gas and their changes under the influence of temperature or pressure. These everyday activities are based on a set of principles known as **gas laws**. Gas laws describe the behavior of gases under various conditions. These laws provide an experimental framework to understand how gases respond to changes in pressure, temperature, or volume.

7.1.2.1 Boyle's law and Charles' law

Activity:



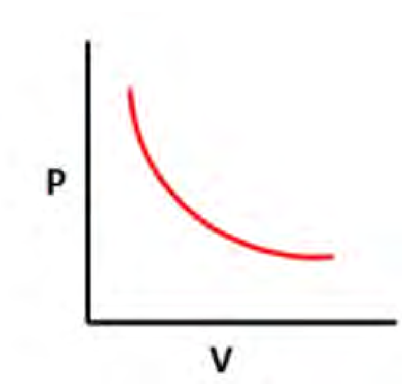
Robert Boyle (1627-1691) was an Irish philosopher and chemist and was pioneer in promoting scientific methods and experimentation. The Boyles law published in 1662 was the foundation for understanding the behavior of gases. He made significant contributions to various scientific disciplines, including physics, medicine, and theology.

a. Compare the volume and pressure of gas in the given containers A and B.
b. What is the relationship between pressure and volume of gas at a given temperature?

Robert Boyle, the British chemist studied the behavior of gases systematically in the 17th century. From his series of experiments on the gas sample, He derived the pressure -volume relationship in the form of Boyle's law.

We can write a mathematical expression showing the inverse relationship between pressure and volume:

$P \propto 1/V$ [temperature is constant] or $P = k/V$ [where k is a proportionality constant]



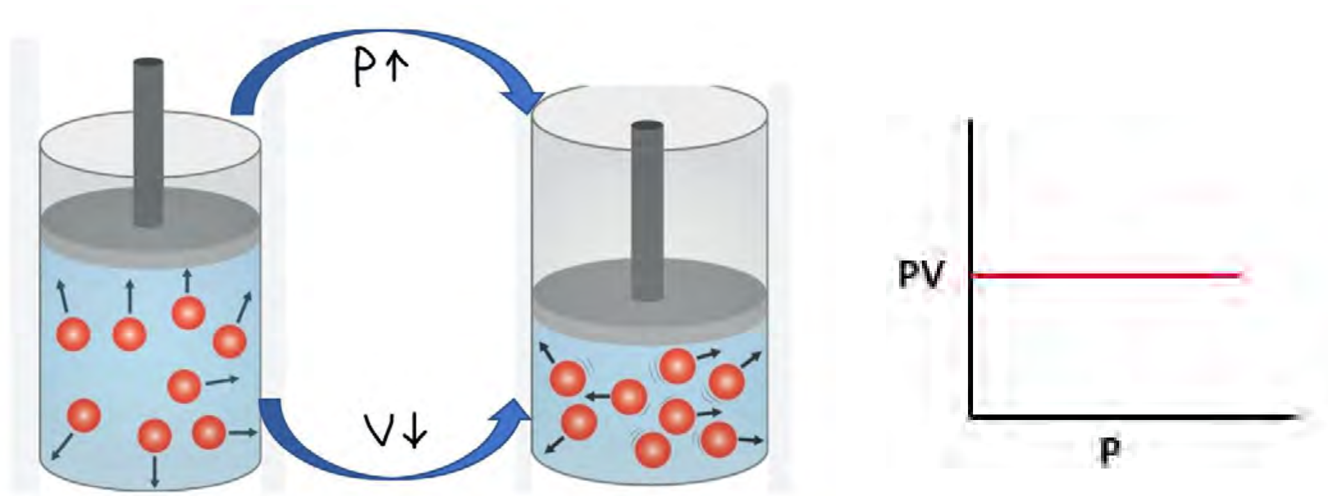


Fig: Effect of pressure on the volume of gas

Above relation can also be written as

Or $PV=k$

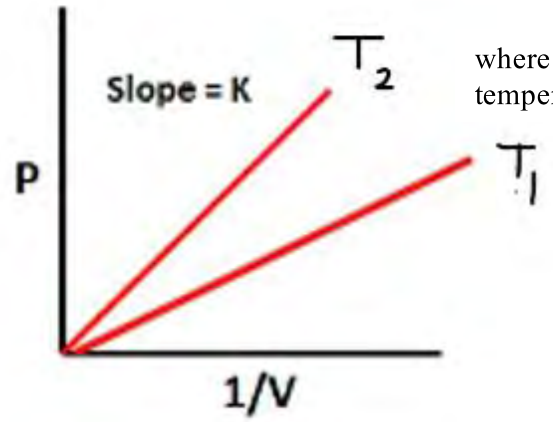
It means the product of pressure and volume of a gas at constant temperature is constant.

For two different set of conditions of pressure and temperature at constant temperature, the above relation can be written as

$P_1V_1=P_2V_2$ where P_1= initial pressure V_1=initial volume

P_2= final pressure and V_2= final volume

If temperature increases the volume of gas increases at constant pressure, therefore the relationship between pressure and volume of gas at various temperature can be plotted as



where T_2 is higher temperature than T_1

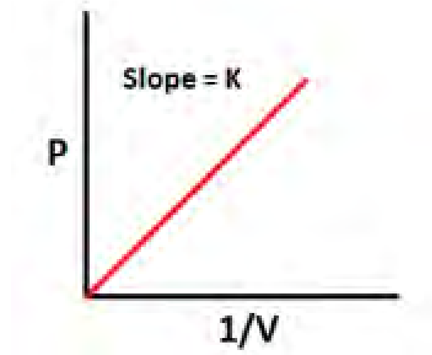


Fig: Plot of P vs 1/V at different temperatures

The relationship between pressure and reciprocal of volume(1/V) can be plotted at constant temperature as shown.

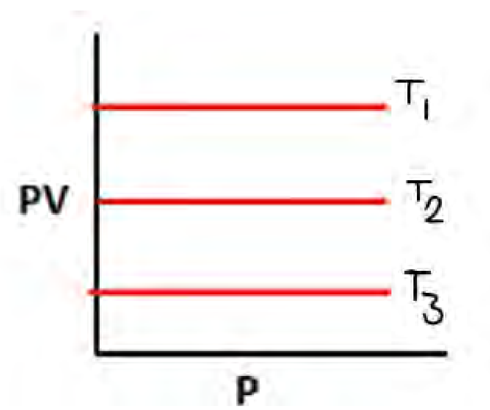
Boyle's law states that **"the pressure of a fixed amount of gas at a constant temperature is inversely proportional to the volume of gas**".

Applications of Boyle's law:

a. Paint sprays or aerosol sprays.
b. Breathing of humans and animals.
c. Packing of soda bottles.
d. Storage of LPG gas in the cylinders.

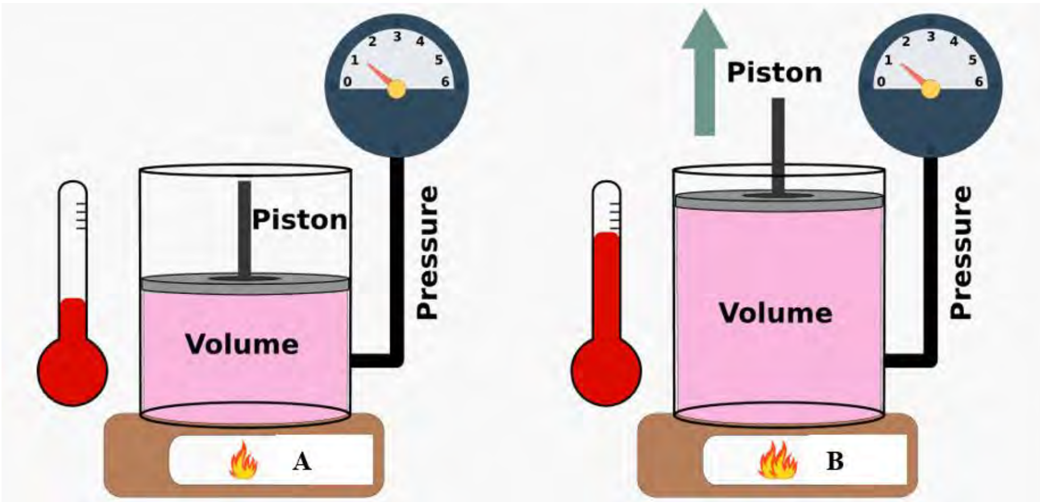
Test yourself!

Arrange the temperature in increasing order in the given relation between PV and P at various temperatures and give reason for the trend.



Charles's law

Activity:



a. Compare the temperature pressure and volume in the given figures A and B.
b. Find the relationship between temperature and volume at a given pressure.

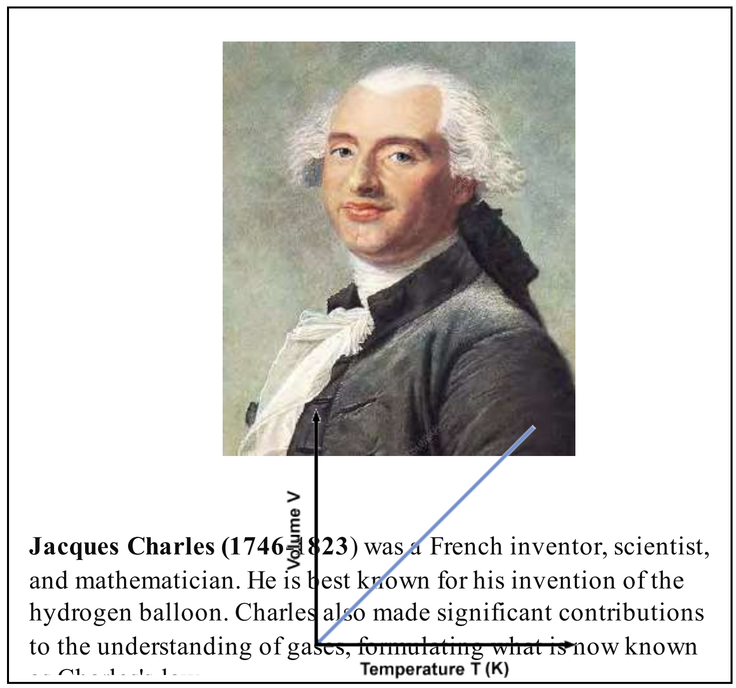


Fig: plot of volume vs temperature (K) at constant pressure

Jacques Charles and Joseph Gay-Lussac French scientist studied the effect of temperature on the volume of gas and found that gas expanded on heating and contracted on cooling. The relationship between *temperature and volume was named as Charles's and* Gay-Lussac's law, or simply **Charles's law.** It states **that**

"the volume of a fixed amount of gas at constant pressure is directly proportional to the absolute temperature of the gas,"

Mathematically volume of gas(V) ∝ absolute temperature (T)

Or $V = KT$ or $V/T =$ constant (K)

For two different sets of conditions

$V_1/T_1 = V_2/T_2$

where, V_1=initial volume T_1 =initial temperature

V_2= final volume T_2=final temperature

Upon plotting the relationship between volume and temperature, a straight line as shown is obtained:

According to Charles's law,

For every 1-degree change (rise or fall) in the temperature of a gas, its volume changes (increases or decreases) by 1/273 of its volume at 0 degrees Celsius. This new relation was given by:

$$V_t = V_0 + (1/273 \times V_0) \times t$$

Or

$$V_t = V_0 - (1/273 \times V_0) \times t$$

Where.

V_t is the volume of gas at temperature 't' and V_0 is the volume of gas at 0^oC temperature

Therefore, the volume of gas at (-273^0C) can be calculated as

$$V = V_0 - (1/273 \times V_0) \times (273^0C)$$

$$=0$$

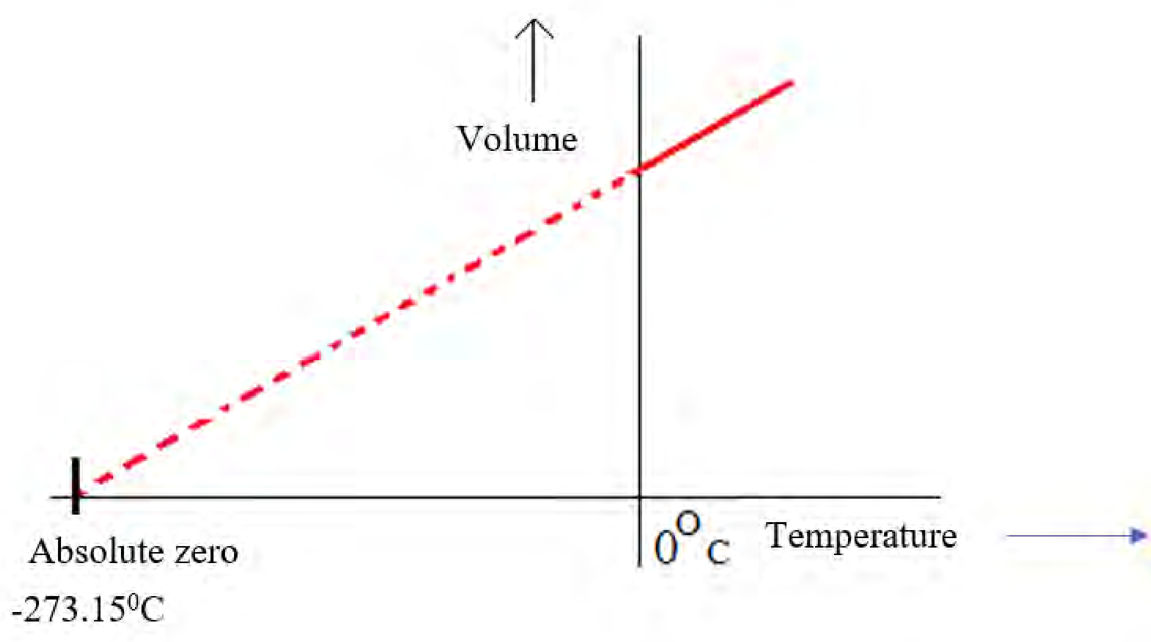


Fig: Plot of volume of gas vs temperature (^0C)

The graph indicates that the volume of gas is zero at -273^0C =0 kelvin. In 1848 Lord Kelvin identified the temperature (-273^0C) (more exact value $-273.15\ ^0C$) as theoretically the lowest attainable temperature called as **absolute zero**. Then a new scale called an **absolute temperature** scale was developed with absolute zero as the starting point.

Absolute zero $=-273.15\ ^0C \approx -273\ ^0C$ is a theoretical temperature because volume of gas can be measured practically only over a certain range of temperature. At very low temperature gas condenses to liquid.

where $P_1>P_2>P_3$

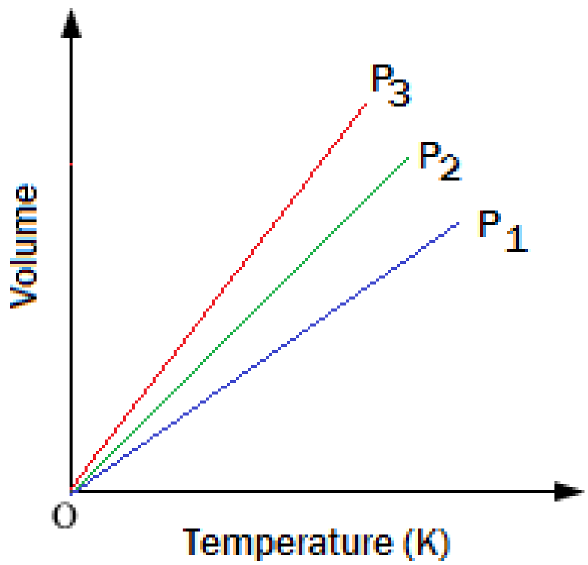


Fig: Plot of volume vs temperature at different pressures

By convention, we use **T** to denote absolute (kelvin) temperature and **t** to indicate temperature on the Celsius scale. They are interconvertible to each other using the relationship:

$T=t^0C + 273.15$

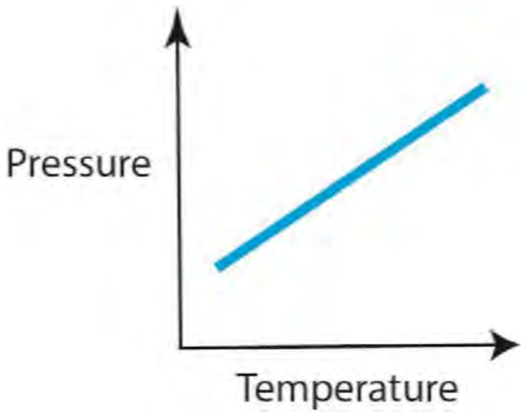
It means the boiling point of water (100^0C) = (100+273.15)=373.15K in Kelvin scale.

If the volume of a given amount of gas is constant, pressure is directly proportional to temperature.

$P \propto T$ or $P=KT$

Or $P_1/T_1=P_2/T_2$

This relationship is another form of Charle's law which is also called **Gay-Lussac's law**



Applications of Charles law

a. A flying hot air balloon
b. Shrinking of a car tire in the winter season
c. Reduction of the gap between railway lines on cold days
d. Blasting of balloons on hot days

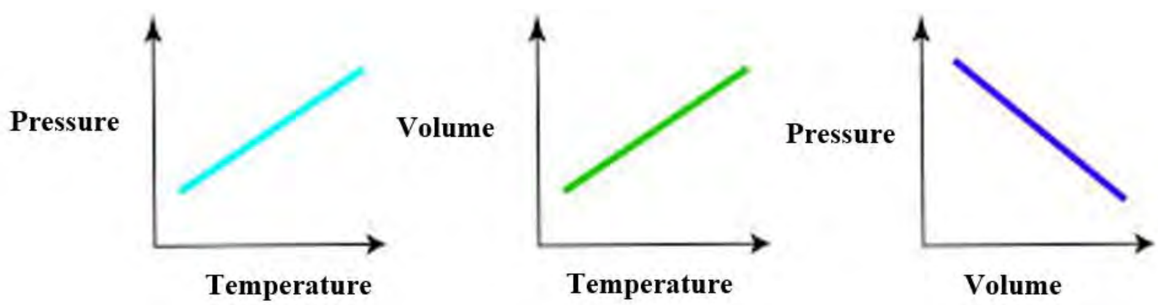
Do you know?

When gas is cooled to nearly absolute zero (-273.15^{0}C) atoms lose their kinetic energy and condenses to form a new form of matter called **super cooled atom** named as **Bose -Einstein condensate (BEC).** These are useful in the development of technology like quantum computers.

Test yourself!

What is the cause of the pressure cooker explosion?

Combined gas equation



From Boyles law, the relationship between the volume and pressure at constant temperature is

$V \propto 1/P$ [temperature is constant].

Or, V=K/P………. (I)

From Charles's law, the relationship between volume and temperature at constant pressure is

$V \propto T$ [pressure is constant].

Or, V=KT…………(II)

Combining these equations (I and II) V= K T/P or PV=KT

For two different sets of conditions, it can be written as

$$\frac{P1V1}{T1} = \frac{P2V2}{T2}$$

Where, P_1=initial pressure V_1=initial volume

P_2= final pressure V_2=final volume

This equation is called the combined gas equation. This equation is a useful tool in understanding the behavior of gases under different conditions and allows for calculations involving pressure, volume and temperature.

Solved problem

A gas initially has a pressure of 200 torr in a container of 300ml capacity at 26^0C. The volume is decreased to 0.25 L by increasing the pressure to 0. 60atm. Calculate the final temperature of the vessel.

Solution,

Given, initial pressure(P_1) = 200 torr =200/760atm =0.26atm

Initial volume (V_1) = 300ml=0.3L

Initial temperature (T_1) = 26+273.15=299.15K

Final volume (V_2) = 0.25 L

Final pressure(P_2) = 0.60atm

Final temperature (T_2) =?

Using combined gas equation,

$$\frac{P1V1}{T1} = \frac{P2V2}{T2}$$

$$or, T2 = \frac{P2V2T1}{P1V1}$$

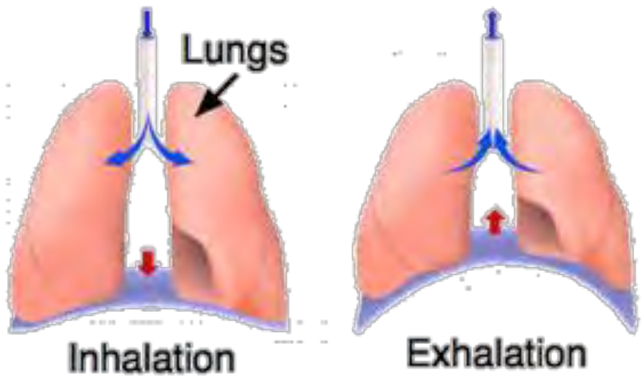
or, $T_2 = \frac{0.6\times0.25\times299.15}{0.26\times0.3} = 575.28K$

Test yourself!

6 L of gas at 20^0C temperature and 2.5atm pressure is compressed to 3 L by decreasing the temperature to 10 ^{0}C. Find out the final pressure of the gas.

7.1.2.2 Avogadro's law

Activity: Observe the given figure and find the difference in volume and number of gas molecules in the lungs during these two processes.



When we inhale, the lungs expand because they get filled with air. Similarly, while exhaling, the lungs let the air out and shrink in size. The change in volume is due to a change in the number of gas molecules in the lungs. It means when we inhale, the number of gas molecules increases and when we exhale the number of gas molecules decreases. Likewise, when we press the bicycle pump, the number of gas molecules in the bicycle tire increases. So, there is a change in shape and volume of the tire.

In 1811 Italian scientist, Amedeo Avogadro published the hypothesis relating volume of gas with number of gas molecules called **Avogadro's hypothesis**. We have discussed this hypothesis in unit 2.3 in detail.

If you have a container containing **n** mol of gas at volume **V**, and the number of mol increased to 2n, the volume also becomes twice the original (2V) as shown in the figure

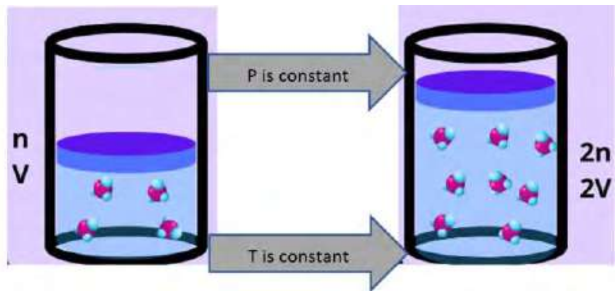


Fig: relationship between number of mol and volume of gas

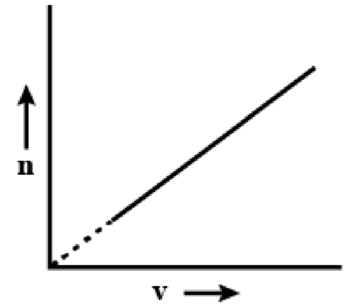


fig: Plot of mol vs volume of gas

The relationship between the number of mol and volume can be expressed mathematically as:

volume of gas (V) ∝ number of mol of gas(n) [at constant temperature and pressure]

Or, V =k n

k is the proportionality constant.

For two different sets of conditions, the above equation can be written as

$$\frac{V1}{n1} = \frac{V2}{n1}$$

Where, V_1=initial volume of gas n_1= initial number of mol of gas

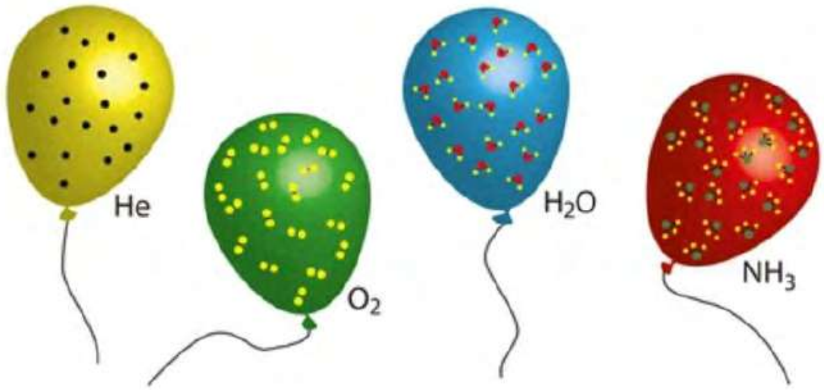
V_2= final volume of gas n_2= final number of mol of gas

Avogadro's law is a specific case derived from Avogadro's hypothesis. It states that **"the volume of a gas is directly proportional to the number of mol of gas keeping temperature and pressure constant."**

Test yourself!

The given balloon contains equal volume of different gases.

a. Which of them contain the greatest number of gas molecules?
b. Which of the balloons weighs the highest mass?

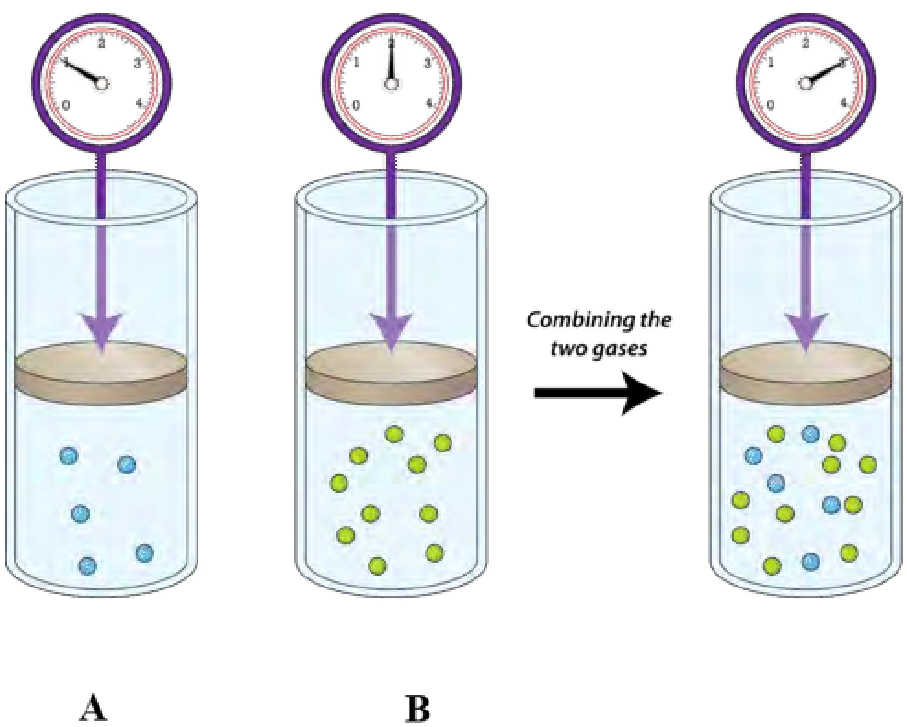


7.1.2.6 Dalton's law of partial pressure

Activity:

Under same conditions of temperature:

a. Compare the number of gas molecules in container A, B and C.
b. Which of the containers possess the highest pressure of gas? Why?



Gas exerts pressure due to the continuous collision of gas molecules on the wall of the container. What happens to the pressure of mixture, when more than one non-reacting gas is mixed together? That was studied by **John Dalton** and formulated a law in 1801 called **Dalton's law of partial pressure**.

Partial pressure is the pressure exerted by each individual gas if it were present alone in the container.

Consider a case in which 'n'$_1$ and 'n_2'moltwo gases are mixed together in a container of volume V to get 'n' mol of mixture. If P_1 and P_2 be the partial pressure of each gas in the mixture,

The total pressure of the mixture of gas (P_t) = P_1+P_2 ……….

ideal gas equation can be applied to each gas

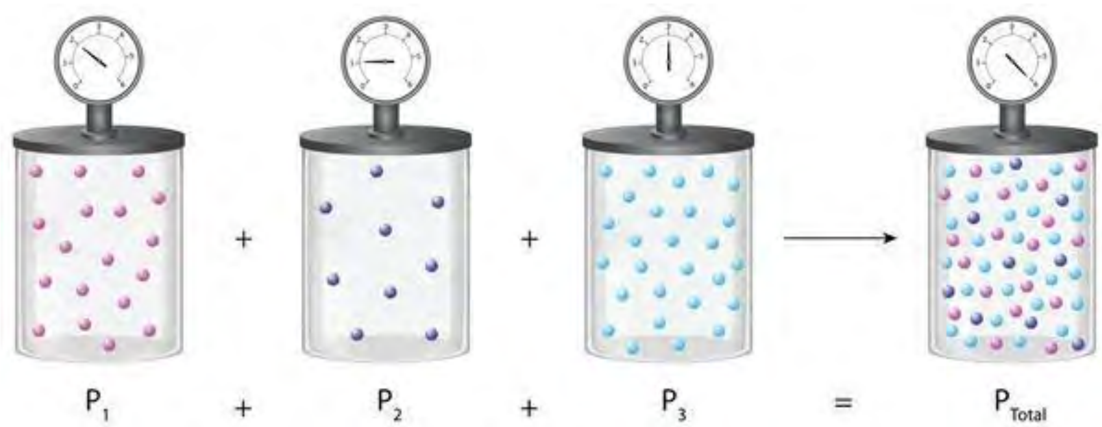
$P_1 = \frac{n1RT}{V}$ and $P_2 = \frac{n2\ RT}{V}$

The above equation (i) becomes

$P_t = \frac{n1RT}{V} + \frac{n2\ RT}{V}$

$P_t = (n_1+n_2)\frac{RT}{V}$ [All gas is at constant temperature and occupies the same volume V]

$P_t = n\frac{RT}{V}$ [$n=n_1+n_2$]



Dalton's law of partial pressure states that "**the total pressure of a mixture of non-reacting gases is just the sum of partial pressure of individual gases**".

Partial pressure and mol fraction

We know that partial pressure of each gas $P_1 = \frac{n1RT}{V}$

And total prerssure of mixture of gas $Pt = \frac{nRT}{V}$

Or, $\frac{P1}{Pt} = \frac{n1RT}{V} / \frac{nRT}{V}$

Or $\frac{P1}{Pt} = \frac{n1}{n} = \chi_1$ (mol fraction of gas)

P_1 (partial pressure of 1st gas) = $P_t \times \chi_1$

Likewise, P_2 = partial pressure of second gas = $P_t \times \chi_2$

Where, χ_1 is the mol fraction of 1st gas $= \frac{n1}{n1+n2}$

χ_2 is the mol fraction of 1st gas $= \frac{n2}{n1+n2}$

The pressure exerted by each gas in a mixture is independent of the presence of other gases. The behavior of one gas is not affected by the behavior of the other gases in the mixture. This law is mainly applicable for the gases in the closed spaces, such as in gas cylinders.

Dalton's Law is commonly applied in analytical chemistry to separate and analyze the components in the gas mixture. Besides, this law is applicable in processing and separation of natural gas, which is a mixture of various hydrocarbons.

How do you calculate the pressure of dry gas?

The pressure exerted by a gas collected over water is the sum of pressure of dry gas and the pressure of moisture (aqueous tension).

The pressure of dry gas is calculated as

$P_{dry} = P_{moist}$ - aqueous tension (f)

Solved problem:1

A mixture of gases contains 3.25 mol of helium (He), 1.50 mol of nitrogen (N_2), and 2.80 mol of carbon dioxide (CO_2). Calculate the partial pressures of each gas if the total pressure is 3.50 atm at a certain temperature.

Solution,

Mol of helium (nHe) = 3.25 mol

Mol of carbon dioxide (n_{CO_2}) = 2.80 mols

Mol of nitrogen (n_{N_2}) = 1.50 mol

Total pressure (P_T) = 3.50 atm

We know that the partial pressure of each gas in a mixture = P_T ×mol fraction of that gas

$$P_{He} = P_T \times \chi_{He} = 3.50 \times \frac{3.25}{1.5+3.25+2.80} = 1.51 \text{ atm}$$

$$P_{N2} = P_T \times \chi_{N2} = 3.50 \times \frac{1.50}{1.5+3.25+2.80} = 0.69 \text{ atm}$$

$$P_{CO2} = P_T \times \chi_{CO2} = 3.50 \times \frac{2.80}{1.5+3.25+2.80} = 1.29 \text{ atm}$$

Solved problem:2

Calculate the total pressure of the gas mixture in 10-L container in the figure given, when both the valves are opened at 27 ^{0}C. Also calculate the partial pressure of each gas.

Solution,

From the ideal gas equation, PV = nRT

Number of mol of gas (n_A) = (1×5)/0.0821×300=0.2mol

Number of mol of gas (n_B) = (2×7)/ 0.0821×300=0.5 mol

Total number of mol in the gaseous mixture = 0.7 mol

Total pressure inside the 10L container = P_{tot} = nRT/V

P_{tot} = 1.72atm

Partial pressure of A (P_A) = mol fraction of A × total pressure (P_{total})

$=\frac{0.2}{0.2+0.5}\times 1.72$

=0.49atm

Partial pressure of B(P_B)= total pressure (P_{total}) -P_A

=1.72-0.49

=1.22atm

Test yourself!

A freshly prepared oxygen is collected by the downward displacement of water from an inverted bottle. The total pressure shown on the pressure gauge is 747 mm Hg, and the temperature of the room is 27^0C. Calculate the partial pressure of oxygen? Vapour pressure of H_2O at 27^0C is 26.739 mmHg.

7.1.2.7 Graham's law of Diffusion

Activity:

Consider that the volume of gas is equal and is released from the same size opening at the same time in the figure given.

a. Which gas will be smelled first?

b. Which gas cylinder empties first?

Thomas Graham (1805–1869) was a Scottish chemist renowned for his groundbreaking contributions to the field of chemistry, particularly in the study of diffusion of gases and colloids.

Fig: diffusion of ink in water

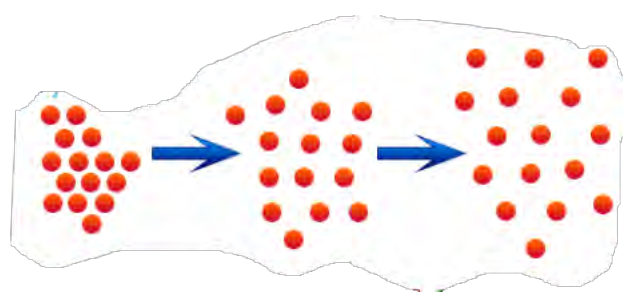


Fig: diffusion of gas molecules

When a drop of ink is added into clear water, it gradually disperses, a phenomenon known as **diffusion**. This process involves the movement of particles from higher concentration to regions of lower concentration. Like liquids, gases also show diffusion due to the random motion of gas molecules. You may have smelled the pungent smell of ammonia, it is due to the diffusion of the gas in air. Various factors, such as temperature, pressure, and the nature of the gas, influence the rate of diffusion.

In the 19th century, Scottish chemist **Thomas Graham** studied the relationship between the rates of diffusion of two gases and their respective molar masses, leading to the formulation of **Graham's law of diffusion**.

Mathematically, it can be expressed,

$$Rate\ of\ diffusion\ of\ gas \propto 1/\sqrt{molar\ mass\ of\ gas}$$

For two different gases 1 and 2 having different rate of diffusion the law can be written as

$$\frac{Rate\ of\ diffusion\ of\ gas\ 1}{rate\ of\ diffusion\ of\ gas\ 2} = \frac{\sqrt{molar\ mass\ of\ gas\ 2}}{\sqrt{molar\ mass\ of\ gas\ 1}}$$

Or, $\frac{r1}{r2} = \frac{\sqrt{M2}}{\sqrt{m\ 1}}$

Or $\frac{r1}{r2} = \frac{\sqrt{d2}}{\sqrt{d\ 1}}$ [∵ density of gas = $\frac{Molecular\ mass}{2}$]

Let us consider the gas X and Y having density dx and dy respectively are diffused at the same time from the opposite ends of the glass tube as shown in the figure. These gases meet together near to the gas X indicates that the rate of diffusion of gas Y is faster than that of gas X.

Fig: relationship between density and rate of diffusion of gas

Graham's law of diffusion states that" **the rate of diffusion of a gas is inversely proportional to the square root of its molar mass"**.

Rate of diffusion can be expressed in different ways.

Rate of diffusion is the volume of gas diffused per unit time.

Rate of diffusion (r) = $\frac{volume\ of\ gas\ diffused\ (v)}{time\ taken\ for\ diffusion(t)}$

If two different gases are diffused from the same volume of container at different time t_1 and t_2, then

$$\frac{r1}{r2}=\frac{t2}{t1}$$

likewise, if two different gases diffuse at same time the equation becomes,

$$\frac{r1}{r2}=\frac{V1}{V2} \text{ or } \frac{r1}{r2}=\frac{l1\times t1}{t1\times l2} \quad [\because \text{velocity} = \frac{distance\ travelled\ (l)}{time\ taken(t)}]$$

$$\frac{r1}{r2}=\frac{l1}{l2}$$

From Avogadro's law, volume of gas is directly proportional to the number of mol of gas

v $\propto$ n or $\frac{V1}{V2}=\frac{n1}{n2}$

Therefore, this law can be summarized as

$$\frac{Rate\ of\ diffusion\ of\ gas\ 1(r1)}{rate\ of\ diffusion\ of\ gas\ 2(r2)}=\frac{\sqrt{M2}}{\sqrt{M1}}=\frac{\sqrt{d2}}{\sqrt{d\,1}}=\frac{t2}{t1}=\frac{V1}{V2}=\frac{l1}{l2}=\frac{n1}{n2}$$

Where, M= molecular mass of gas

d= density of gas

v=velocity of gas

t=time taken by gas

l=length travelled by gas

n=number of mol of gas

Applications of Graham's law of diffusion in:

a. separation techniques such as **gas chromatography**, where various components of a gas mixture require separation for analysis.
b. predicting the speed at which different pollutants disperse and mix in the atmosphere.
c. calculating the rates at which hydrogen and oxygen gases can diffuse and react thereby influencing the **efficiency of fuel cells.**

When you make a small pore in the balloon filled with air, it starts to escape the air from the balloon. **This process by which a gas under pressure escapes from the container through a small opening is called effusion.** The rate of effusion depends upon the molecular mass similar to that of rate of diffusion.

$Rate\ of\ effusion\ of\ gas \propto 1/\sqrt{molar\ mass\ of\ gas}$

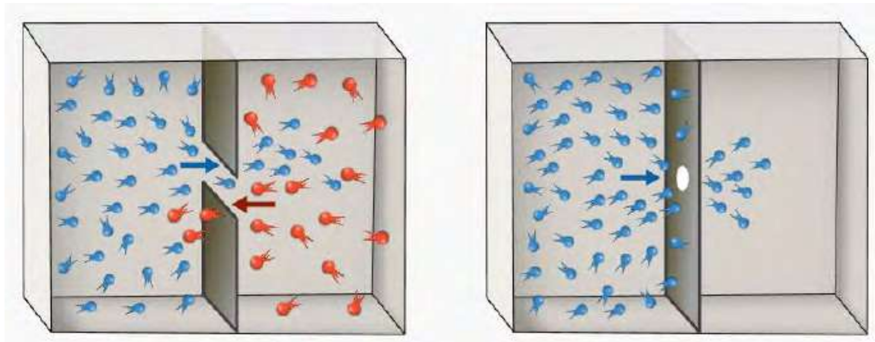
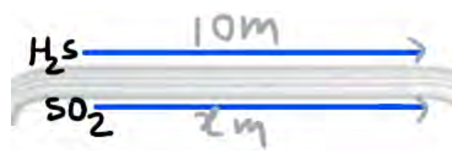


Fig: diffusion vs effusion

Solved problem: 1



Hydrogen sulfide and Sulphur dioxide gas are released at the same time from the same end of the glass tube as shown in the figure.

a. What is the value of x after 1 minute?
b. If 3.4 grams of H_2S takes 5 minutes to diffuse, what is the mass SO_2 that can be diffused in the same time?

Solution,

Molecular mass of H_2S=34

Molecular mass of SO_2=64

Distance travelled by H_2S=10 meter

Distance travelled by SO_2(x)=?

$$\frac{Rate\ of\ diffusion\ of\ gas\ 1\,(r1)}{rate\ of\ diffusion\ of\ gas\ 2\,(r2)} = \frac{\sqrt{M2}}{\sqrt{M1}} = \frac{l1}{l2}$$

Or, $\frac{\sqrt{M2}}{\sqrt{M1}} = \frac{l1}{l2}$

$$\frac{\sqrt{M(H2S)}}{\sqrt{M(SO2)}} = \frac{x}{10}$$

$$\frac{\sqrt{34}}{\sqrt{64}} = \frac{x}{10}$$

∴ x=7.2m

Again.

$$\frac{Rate\ of\ diffusion\ of\ gas\ 1\,(r1)}{rate\ of\ diffusion\ of\ gas\ 2(r2)} = \frac{\sqrt{M2}}{\sqrt{M1}} = \frac{n1}{n2}$$ [n is the number of mol of gas $= \frac{given\ mass}{molecular\ mass}$]

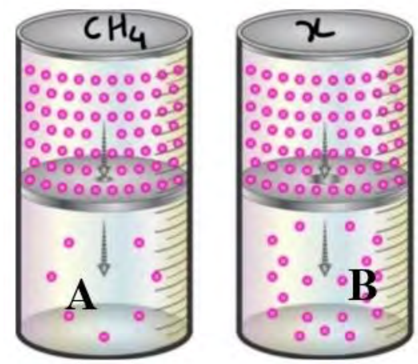
Or, $\frac{\sqrt{MH2S}}{\sqrt{MSO2}} = \frac{nSO2}{nH2S}$

Or $\frac{\sqrt{34}}{\sqrt{64}} = \frac{x\,/64}{3.4/34}$

X= 4.66g

Solved problem: 2

The gas in container B diffuses 2 times faster than the gas in container A. If 6×10^{20} molecules of gas in container **A** are effused at 10 minutes, find the number of unknown gas molecules that effuse at the same time in container **B**.



(hint: 1 mol of gas $=6.022\times10^{23}$ gas molecules)

Solution,

$$\frac{Rate\ of\ diffusion\ of\ unknown\ gas\ (rx)}{rate\ of\ diffusion\ of\ methane\ (rCH4)} = \frac{\sqrt{MCH4}}{\sqrt{Mx}}$$

Or, $\frac{2}{1} = \frac{\sqrt{16}}{\sqrt{Mx}}$

$\therefore M_x=4$

Again,

$$\frac{\sqrt{MCH4}}{\sqrt{Mx}} = \frac{nx}{nCH4}$$

$$\frac{\sqrt{16}}{\sqrt{4}} = \frac{nx\times6.022\times10^{23}}{6.022\times10^{23}}$$

Or number of mol of x$=2/10^3$

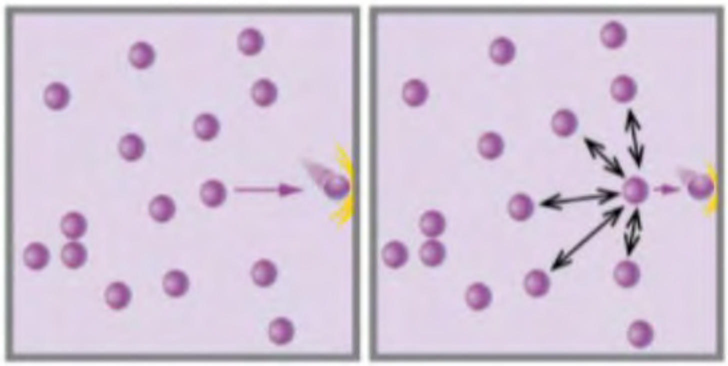
Number of molecules of x$= 2/10^3\times 6.022\times10^{23}$

$= 1.2\times10^{21}$

Test yourself!

10 ml of Sulphur dioxide gas diffuses in 18 seconds through the porous plug. What is the time taken by the same volume of helium to diffuse from the same plug under the same temperature?

7.1.3 Ideal gas and ideal gas equation



Ideal gas **real gas**

The ideal gas equation, also known as ideal gas law, describes the behavior of ideal gas (perfect gas) under various conditions. This equation is derived from Boyle's law, Charle's law and Avogadro's law.

Boyle's law $v \propto 1/P$ (at constant n and T)

Charles's law $V \propto T$ (at constant n and P)

Avogadro's law $V \propto n$ (at constant T and P)

Combining these relationships

$$V \propto \frac{nT}{P}$$

Or $$V=R\frac{nT}{P}$$

$$\boxed{PV = nRT}$$ [where, R is a proportionality constant called gas constant and n is the number of mol of gas]

This equation is called the ideal gas equation because this equation is followed **by ideal gas** at all conditions of temperature and pressure. **An ideal gas is a theoretical concept that serves as a simplified model for the behavior of gases under certain conditions**. The key assumptions of an ideal gas are based on the kinetic theory of gases. It means, Ideal gas is a hypothetical gas in which the volume and interactions of the gas molecule is completely negligible. There is no ideal gas in nature. All-natural gases (H_2, He, CO_2, NH_3 etc.) are real.

Ideal gas equation can also be written as

$$PV=\frac{w\,RT}{M} \quad [n= \frac{w(\text{given weight})}{M(\text{molecular weight})}]$$

Or, $P= \frac{w\,RT}{V.M}$ or $P=\frac{d\,RT}{M}$ where d (density) is the mass per unit volume of gas

$$d= \frac{P.M}{RT}$$

7.1.4 Gas constant and its significance

In the ideal gas equation, PV=n RT, the constant R is called **Universal gas constant** or simply gas constant. It helps in connecting various gas laws and helps in understanding the behavior of gas in different situations.

R$=\frac{PV}{nT} = \frac{work\ done}{nT}$

[PV refers to the work done by a gas during a change in volume under constant pressure.]

Therefore, R is the measure of work done per mol of gas per kelvin temperature. It provides a link between macroscopic observables (like pressure and volume) and the microscopic behavior of gas molecules.

R = 0.0821 L atm/mol.K is often used when pressure is expressed in atm, volume in liters, and temperature in Kelvin.

R=**8.324J/mol. K** is used when pressure is in pascals (Pa), volume in cubic meters (m^3), and temperature in Kelvin.

The unit and value of constant R depends upon the units used for pressure, volume and temperature.

At **STP** (standard temperature and pressure) volume of 1 mol of gas is 22.4 L

$$R=\frac{PV}{nT} = \frac{1\ atm \times 224 L}{1\ mole \times 273.15K}$$

$$= 0.0821 \quad L\ atm/mol.\ K$$

Common Values for R
8.3145 kPa · L / mol · K
8.3145 N · m / mol · K
8.3145 J / mol · K
8.3145 kg · m2/s2 mol · K
0.08206 atm · L / mol · K

Likewise, the other units and values of R = 8.324J/mol K

$R = 8.324 \times 10^7$ erg/mol K

Solved problem 1:

Calculate the number of mol of the gas kept in the given cylinder under given conditions.

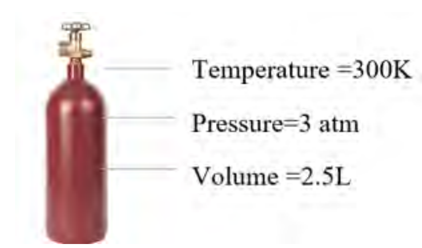
Solution,

Volume of gas(V)=2.5 L

Temperature(T)=300 K

Pressure(P)=3 atm

R=0.0821L·atm/mol K



Using ideal gas equation, PV=nRT

$$n = \frac{PV}{RT}$$

$$= \frac{3 \times 2.5}{0.0821 \times 300} = 0.305 \text{ mol}$$

Test yourself!

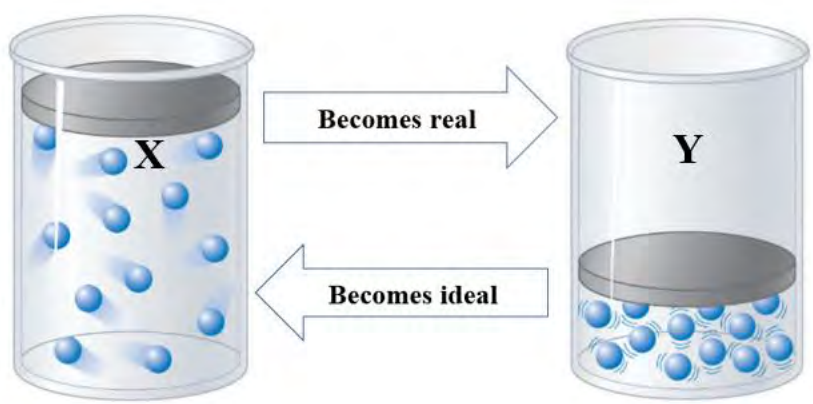
1. Calculate the volume (in litres) occupied by 4.4 g of CO_2 gas at STP.
2. How many mol of gas will a 1.25 litre flask can hold at 35^0C and a pressure of 640 mmHg?

7.1.5 Deviations of real gas from ideality

Activity:

Observe the cylinders X and Y containing the same gas under different conditions.

a. Which container has more interactions among the gas molecules?
b. What should be the change in temperature and pressure to convert Y to X?



The concept of an ideal gas and its corresponding equation (PV=nRT) is based on gas laws and the kinetic theory of gases. This theory assumes that **gas molecules have negligible volume** and **interact negligibly with each other.** When volume and interactions are ignored, the effect of temperature and pressure on gas behavior is negligible. This is called ideal **behavior or ideality**.

When temperature is increased and pressure is decreased real gas molecules begin to separate and move farther apart. In such conditions, the above-mentioned assumptions of negligible volume and interaction can be considered, and real gases exhibit ideal behavior.

In contrast when pressure increases and temperature decreases, gas molecules come closer and show certain volume and interactions called **deviation from ideality.**

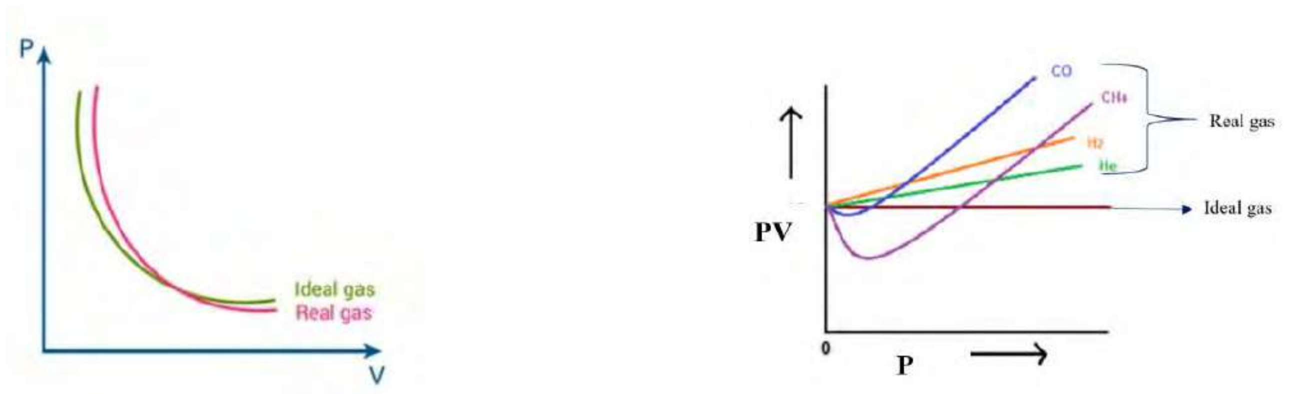


Figure: deviations of real gas from ideal behavior

Cause of deviation

For a gas to exhibit ideal behavior, both the **volume occupied by gas molecules and intermolecular forces should be ignored**, as suggested by the kinetic theory of gases. These assumptions remain valid mainly under conditions of high temperature and low pressure.

However, under conditions of high pressure and low temperature, gas molecules tend to approach each other closely. In such conditions, the volume occupied by the molecules and the interactions between them cannot be neglected. It means, real gases deviate from ideal behavior under these conditions.

Consider the example of carbon dioxide, which can be condensed into dry ice by reducing the temperature and increasing the pressure. This phenomenon indicates that molecules of real gases possess some volume and interactions at low temperature and high pressure.

Upon plotting the graph of PV vs P, it gives a straight line for ideal gas, indicates PV= constant at all pressure. But for real gas this plot is not the same with ideal gas except at very low pressure. It means real gas possesses ideal behaviour at very low pressure.

To predict the behavior of real gases one useful equation was proposed by the Dutch scientist **Johannes van der Waals** by modifying the ideal gas equation. This equation is called Vander Waal's equation:

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

where a and b are constants called Vander Waal's constants.

Test yourself!

The plots of Hydrogen and helium are different from other gases in the PV vs P graph. Why?

Exercises

A. Multiple choice questions

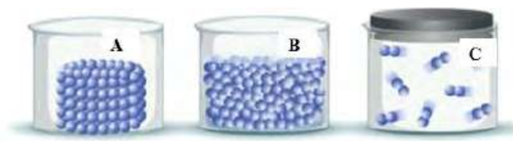
1. At constant temperature, what happens to the volume of a gas if the pressure is doubled?

a. It becomes half.
b. It doubles.
c. It remains the same.
d. It quadruples

2. When collecting gases over water, the partial pressure of the collected gas is equal to:

a. the atmospheric pressure.
b. the vapor pressure of water.
c. the sum of the atmospheric pressure and the vapor pressure of water.
d. the difference between the atmospheric pressure and the vapor pressure of water.

3. Which of the given figures shows least intermolecular force of attraction and most intermolecular distance among the particles?

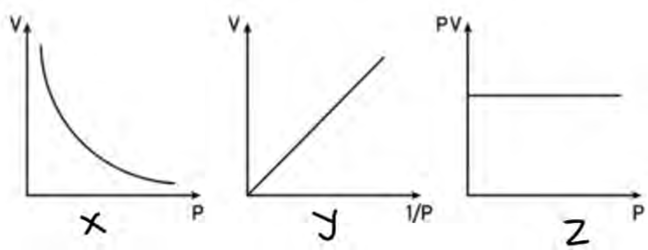


a. A
b. B
c. C
d. Both A and C

4. What is the pressure (in atmospheres) exerted by a mixture of 12.0 g of N_2 and 12.0 g of O_2 in a 2.5 L container at 25°C? Assuming ideal behaviour, which of the following gases will have the greatest volume at STP?

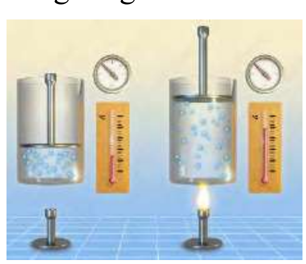
a. 1 mol of He.
b. 14 g of N_2.
c. 5.0 3 10 23 molecules of Cl_2
d. 0.2 mol of CO2 gas

5. Which of the given graph shows the Boyle's law?



a. X
b. Y
c. Z
d. All are related to Boyle's law

6. Which gas law is expressed in the figure given?



a. Boyle's law
b. Charle's law
c. Dalton's law
d. Grahm's law

7. Which of the following gas has the highest rate of diffusion?

a. He
b. O_3
c. O_2
d. N_2

8. How much does the volume of the gas increase if we increase the temperature by 1 Degree?

a. 273 Ls
b. 1 by 273^{rd} of the original volume
c. 1 L
d. Hundred L

9. Hydrogen sulphide (H_2S) has a very strong rotten egg odour. Methyl salicylate($C_8H_8O_3$) has a strong "minty" odour, and benzaldehyde(C_7H_6O) has a pleasant almond odour. If the vapours for these three substances were released at the same time from across a room, which odour would you smell first?

a. rotten egg odour
b. minty" odour,
c. pleasant almond odour
d. all odour at the same time

10. A mixture of carbon dioxide and an unknown gas was allowed to effuse from a container. The carbon dioxide took 1.25 times as long to escape as the unknown gas. Which one could be the unknown gas?

a. Cl_2
b. CO
c. HCl
d. H_2

11. What is the partial pressure (in mmHg) of neon in a 4.00 L vessel that contains 0.838 mol of methane, 0.184 mol of ethane, and 0.755 mol of neon at a total pressure of 928 mm Hg?

a. 234 mm
b.394 mm
c.400 mm
d.294 mm

12. A real gas will behave most like an ideal gas under conditions of

a. high temperature and high pressure
b. high temperature and low pressure
c. low temperature and high pressure
d. low temperature and low pressure

13. The volume of an ideal gas is zero at

a. –45°F
b. -273 K
c. -363 K
d. –273°C

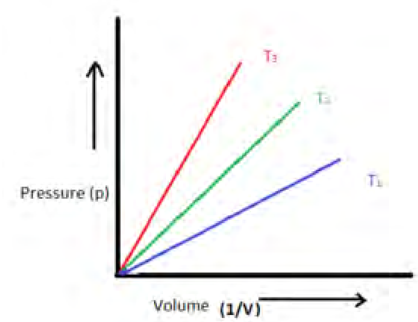
14. When a sample of gas in a closed container at constant volume is heated until its absolute temperature is doubled, which of the following is also doubled?

   a. the density of the gas
   b. the average velocity of the gas molecules
   c. the pressure of the gas
   d. the potential energy of the molecules

B. Short answer questions

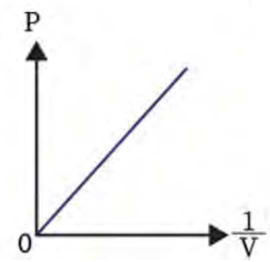
1. What are the postulates of kinetic theory of gas? Which postulate is responsible for the cause of gas pressure?
2. State Boyle's law with mathematical relation and graph.
3. Give the statement of Charles's law and show the relationship between volume and temperature graphically.
4. State Dalton's law of partial pressure and give two applications of Dalton's law of partial pressure.

5. The relationship between pressure and volume is plotted.



   a. State the law shown in the graph.
   b. Identify the highest temperature and give reason.
   c. A gas is confined to a cylinder with an initial volume of 4.0 L at a pressure of 2.0 atm. If the gas is compressed to a volume of 2.0 L, what will be the new pressure if the temperature remains constant?
6. Differentiate between:
   a. Diffusion and effusion
   b. Real gas and ideal gas
7. Give reasons:
   a. Hot air balloon goes up in the air.
   b. Gas is released from LPG cylinders when the regulator is turned on.
   c. Ammonia gas diffuses faster than oxygen gas.
   d. Gas escapes from LPG cylinders when the regulator is turned on.
   e. Bicycle tires are more likely to puncture during summer compared to winter.
8. Volume of gas changes with change in pressure at constant temperature.
   a. Why does the volume of gas decrease upon increasing pressure?
   b. Find the change in volume of gas if pressure is increased from 4.0 atm to 12 atm.
   c. Does the volume of the gas reach zero if the pressure is infinitely increased? Why?

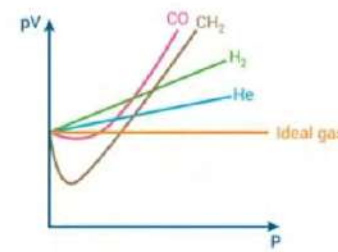
9. Plot of Pressure versus reciprocal of volume at constant temperature is given in the graph.



   a. What do you think about the relationship between pressure and volume from this graph?
   b. Identify the gas laws which show this relation.
   c. What is the application of this law?

10. Pv=nRT is called ideal gas equation where R is constant.
   a. Define ideal gas and write the meaning of each term in the given equation.
   b. Calculate the unit and value of R at STP for 1 mol of gas. Also, write its significance.
   c. 3.20 g sample of gas occupies 2.0 L at 17°C and 380 mmHg. What is the molar mass of the gas?

11. Pv versus pressure at constant temperature is shown in the graph.



   a. Why does ideal gas show a straight line at all pressure?
   b. Real gas shows deviation at high pressure but not at very low pressure. Why?
   c. Identify the real gas that shows maximum deviation.

12. A gaseous mixture consists of 11.0 g of carbon dioxide and 48.0 g of oxygen gas. The volume of the container is 222.4 L and the temperature is 273 ^{o}C. Calculate the following:
   a. the mol of each gas
   b. total mol of all gases
   c. the mol fraction of each gas
   d. the partial pressure of each gas
   e. the total pressure of the mixture

13. Two gas cylinders each contain hydrogen and helium gas, respectively. The gases from these cylinders are then mixed in a third container.
   a. Which cylinder is likely to contain a greater number of gas molecules if they have the same volume at same conditions?
   b. Which cylinder will empty first if opened simultaneously through pores of the same size, and why?
   c. What happens to the total number of gas molecules and total pressure when these gases mixed together keeping the volume constant and why?

14. The volume of moist oxygen gas collected in the figure is 15 cc at 25^{0}C temperature and 645mmHg pressure. is 15 ml.
   a. What would be the volume of gas at STP?
   b. Calculate the pressure of dry oxygen gas. Vapor pressure of water at 25^{0}C =23.76mm of Hg

15. State Graham's law of diffusion and write the relationship between the rate of diffusion of two gases with their molecular mass, length of the tube and volume of container.

16. Using Boyle's law, Charles law and Avogadro's law, derive an ideal gas equation and write the significance of this equation.

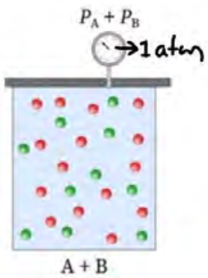
17. Charles's law defines the term absolute zero while giving the relationship between volume and temperature of gas at constant pressure

a. Define the term absolute zero. Why is it called a theoretical concept?

b. Show the absolute zero temperature in the graph of volume and temperature (^0C)

C. Long answer questions

18. O.5 mol of A and 1.5 mol of B are mixed in the given container. The total pressure is measured by pressure gauze as shown.

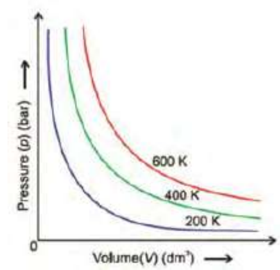


a. What is the cause that gas exerts pressure?

b. What do you mean by partial pressure?

c. Calculate the partial pressure of A and B.

19. Pressure vs volume of gas is plotted at different temperatures as shown in the graph.



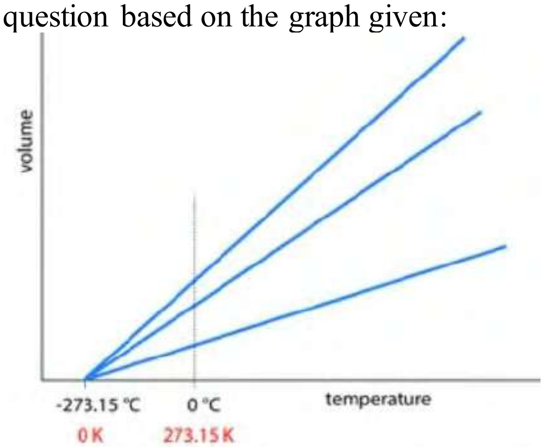
a. Which laws of gas are shown by the graph? State the law.

b. What will be the nature of the graph for the real gas?

c. What is the meaning that the plot goes up with increasing temperature?

d. Helium balloon with a volume of 0.2 L at sea level is allowed to rise to a height where the pressure is about 0.40 atm. Assuming that the temperature remains constant, what is the final volume of the balloon?

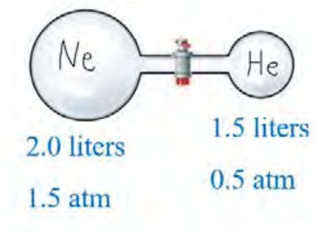
20. Answer the question based on the graph given:



a. What is the relationship between volume of gas and temperature at constant pressure?
b. What is the cause of such a relationship?
c. Name the temperature O K. Why is this temperature called theoretical temperature?
d. A gas occupies a volume of 2.0 L at STP. If the temperature is increased to 100^0C, what will be the new volume assuming constant pressure?

21. Two different gases at different pressures are connected through a gas tube fitted with a stopcock in the figure.

a. When a stopcock is opened there is a change in the pressure of each gas. Why?
b. Calculate the pressure of each gas and the total pressure after opening the stopcock.



22. Nitrogen gas is collected over the water by downward displacement of water.

a. Calculate the pressure of dry gas on the vessel at 25 ^0C if total pressure of gas is found to be 645 mmHg. (vapour pressure of water at 25 ^0C =23.8 mmHg)
b. Find the mass of nitrogen gas collected in the test tube occupying 100 ml volume.

23. State Graham's law of diffusion. Write one application of such law. Consider two gases, A and B, with molar masses of 16 g/mol and 64 g/mol, respectively. If the rate of diffusion of gas A is 2 meters per second, what is the rate of diffusion of gas B under the same conditions?

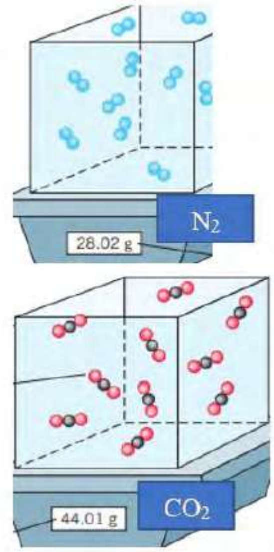
24. You have three gases: X, Y, and Z having molar mass 20, 30, and 40 respectively. The initial rate of diffusion of X is 3 m/s.

a. Identify the gas that can travel fastest among. What is the cause of having different rate of diffusion of these gases?
b. Calculate the rate of diffusion for Y and Z in comparison to X.

25. Two gases are released at once from the opposite end of a 24 cm long glass tube as shown in the diagram.

a. Which gas do you think travels faster and why?
b. Find the distance from A at which these gases meet together.
c. Name the molecular formula of compounds formed when these gases meet together.

26. Certain amounts of nitrogen and carbon dioxide are kept separately in the given container shown in the diagram.

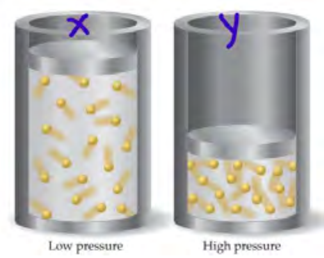


Calculate:

a. Number of mol of each.
b. Number of molecules of each.
c. Volume occupied by each gas at STP.
d. The total pressure of gas, if they are mixed together in a container of 1.5 Ls capacity at 15^0C.

27. The effect of pressure on the volume of gas at constant temperature is shown in the figure.

a. What is the reason for the compressibility of gas?
b. What is the change in the volume of gas when pressure is 4 times increased?



c. Which vessel shows more ideal behaviour of gas at the given temperature and why?

Project work

a. Take a needle-free plastic syringe, fill it with air and note the initial volume reading.
b. Compress the piston while blocking the nozzle with a rubber tip or your hand, then record the new air volume.
c. Fill the syringe with water and record the volume of water.
d. Once more, press the piston with the sealed nozzle and note the final volume of water
e. Calculate the difference in volume between these two trials.

Analyse and draw conclusions regarding the distinct properties of liquids and gases based on your observations

7.2 Liquid State

Unlike solids, liquids have no definite shape because liquid molecules are in constant motion. So, the molecules flow and take the shape of a container. The intermolecular forces are stronger than in gases which causes higher density compared to gas. Unlike gas, liquids are nearly incompressible due to the limited space between the molecules.

7.2.1 Physical properties of liquid

a. Evaporation and condensation

After 4 hours

a. Name the physical process by which the water level decreases in the glass.
b. What happens if water is replaced by alcohol in the same container and exposed to air for the same time?

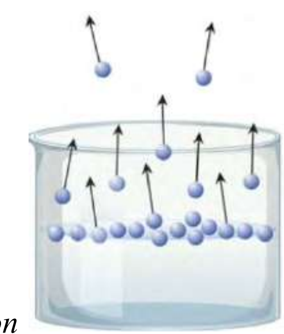
When a glass of water is kept in the sunlight, water level in the glass decreases. It is due to the evaporation of water. Heat energy supplied to the liquid weakens the intermolecular force and increases the movement of particles in the liquid. It causes the liquid molecules to escape from the surface and become gas.

Evaporation is a surface phenomenon because surface molecules have more kinetic energy than the inner molecules. Evaporation happens at all temperatures below the boiling point.

The energy required to change one mol of liquid to one mol of gas is called the enthalpy (heat) change of evaporation (ΔH_{eve}).

What happens when we heat the water in a closed container?

Fig: evaporation and condensation



The water molecules from the surface evaporate and form vapor molecules. As the number of vapor molecules increases, they lose their kinetic energy and come closer to become a liquid. This reverse process of evaporation which involves the conversion of a substance from its gaseous state to a liquid state is called **condensation**. The formation of clouds in the atmosphere, dew on the grass are examples of condensation. Condensation is an exothermic process, it means it releases heat.

Remember:

a. Evaporation occurs at all temperatures.
b. Evaporation occurs only at the surface.
c. Evaporation causes cooling of liquid.
d. Evaporation depends upon temperature and nature of liquid.

Upon increasing the temperature kinetic energy increases which increases the rate of evaporation. Liquids with strong intermolecular force of attraction have a lower rate of evaporation in compared to liquids with weak intermolecular force of attraction.

Test yourself!

a. Does water inside a sealed bottle undergo evaporation?
b. Evaporation causes cooling of liquid. why?
c. Which one has a higher rate of evaporation ethyl alcohol or water?

b. Vapour Pressure and Boiling point

Activity: Observe the given beakers containing two different liquids**:**

a. Which liquid will have more rate of evaporation at a given temperature and why?
b. Which one will boil first?

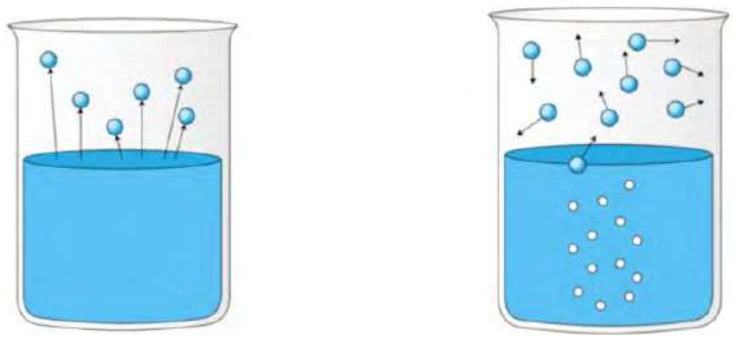


Fig: difference between evaporation and boiling

When a liquid is heated, its surface molecules gain energy and transform into vapor. These vapor molecules start to collide with each other and condense to form liquid. Initially, the rate of evaporation is more than the rate of condensation because the number of molecules on the surface is greater than the vapor molecules. Gradually, the rate of condensation also increases and becomes equal to the rate of evaporation, establishing equilibrium.

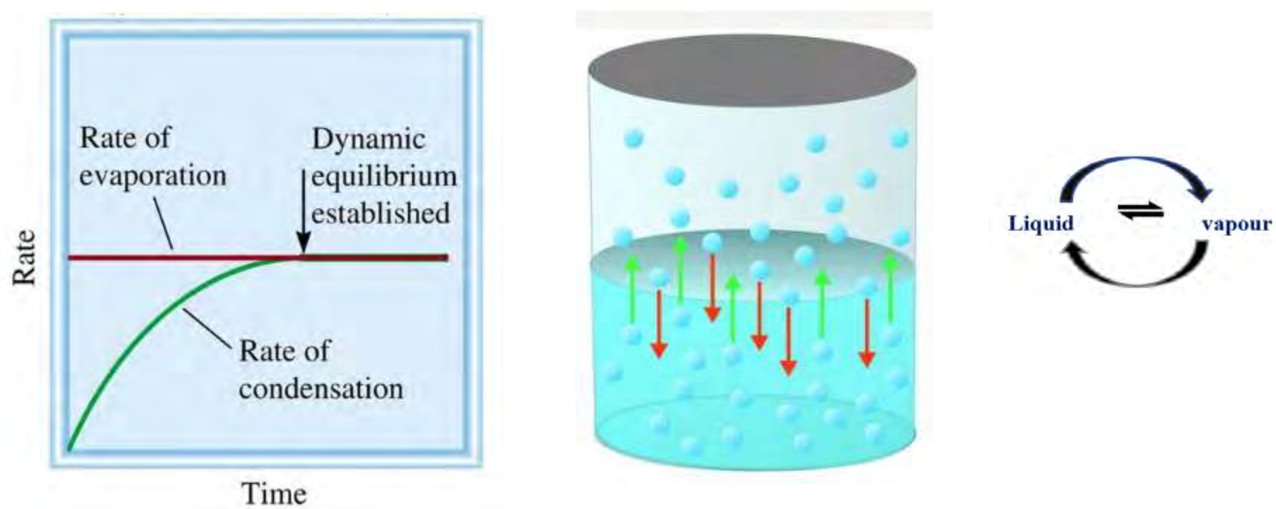


Fig: equal rate of evaporation and condensation at equilibrium

At equilibrium, the concentration of water molecules in the vapor remains constant.
The pressure exerted by vapor on the liquid surface at equilibrium is called vapor pressure. Vapor pressure increases with temperature. As the temperature rises, the molecules in the liquid gain more energy. So, the number of water molecules changing into the vapor phase increases. The vapor pressure also depends on the strength of intermolecular force in the liquid. Liquid with strong intermolecular force where molecules are strongly held together have lower vapor pressure because it does not allow the molecules to move freely and change into vapor.

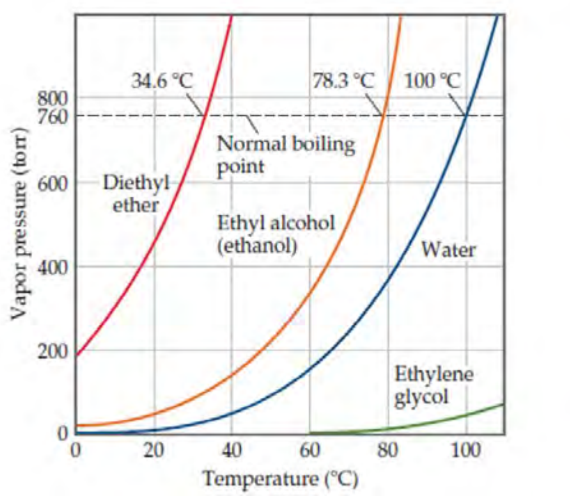
After reaching the equilibrium between evaporation and condensation, if liquid is further heated, the vapor pressure of liquid starts to increase and at one point becomes equal to the atmospheric pressure of that place.

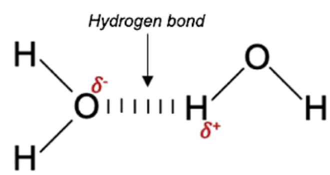
This phenomenon is called boiling and the temperature at which vapor pressure of liquid equals that of atmospheric pressure is called boiling point.

Unlike evaporation, boiling is throughout the entire bulk of the liquid.

Vapour pressure = atmospheric pressure (boiling point)

Boiling point of liquid depends up on the vapor pressure of the liquid. It means liquid with high vapor pressure (less intermolecular force) boils earlier and has low boiling point compared to liquid with low vapor pressure (strong intermolecular force). Diethyl ether has more vapour pressure compared to alcohol and water so has least boiling point(34.6^0C). Due to strong intermolecular force of attraction (H-bonding) in the water, it has less vapor pressure than the ether.





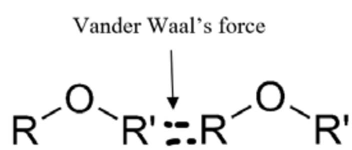


Figure: Variation of vapour pressure and boiling point of some liquids

Liquid	Vapour pressure at 25^0C/atm	Boiling point /^{0}C
Diethyl ether	0.7	34.6
Bromine	0.3	58.8
Ethyl alcohol	0.08	78.5
Water	0.03	100

At sea level, where atmospheric pressure is typically around 1 atmosphere (101.3 kPa), water boils at 100 degrees Celsius. As you climb in altitude, atmospheric pressure decreases and it becomes easier for the vapour pressure of the liquid to meet the atmospheric pressure. It means the liquid can boil at a lower temperature. This is why water boils at a lower temperature in mountainous regions compared to sea level.

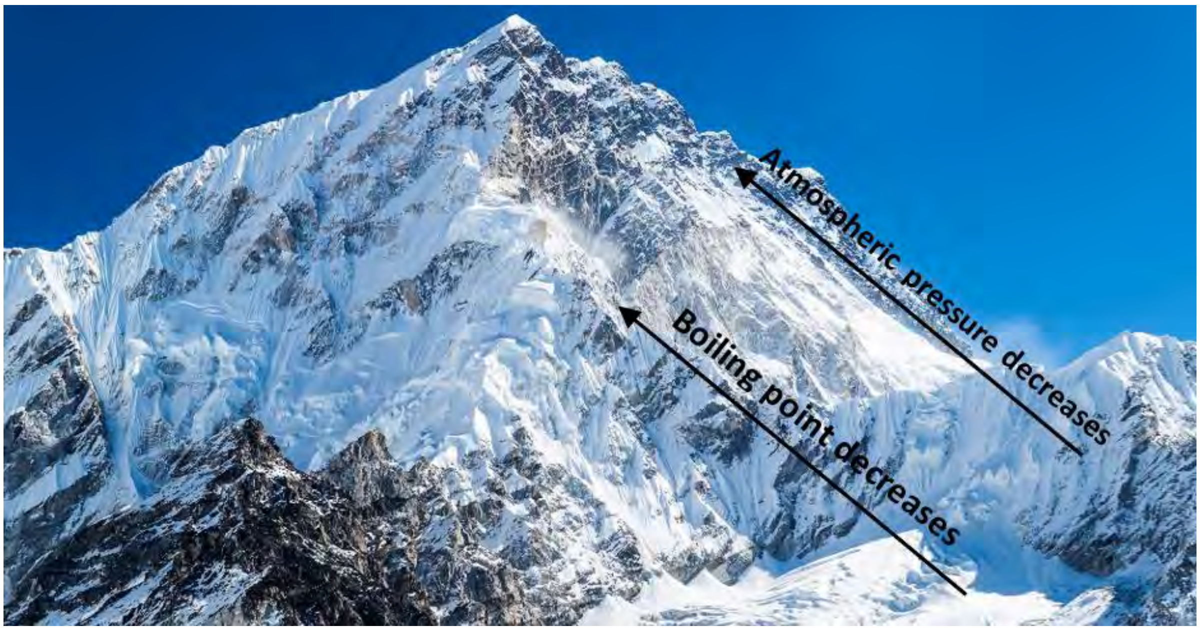


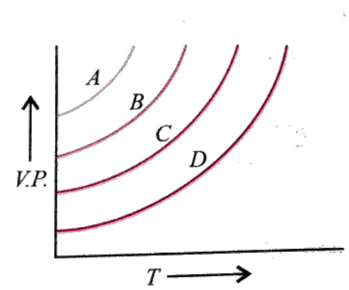
Fig: variation of atmospheric pressure and boiling point of liquid with altitude

Do you know?

A pressure cooker is designed to trap steam and increase the pressure within the cooking chamber. It means the atmospheric pressure inside the cooker is more than outside. This higher pressure raises the boiling point and allows for more effective and faster cooking.

Test yourself!

The plots of vapour pressure of four liquids A, B, C and D with temperature are given.

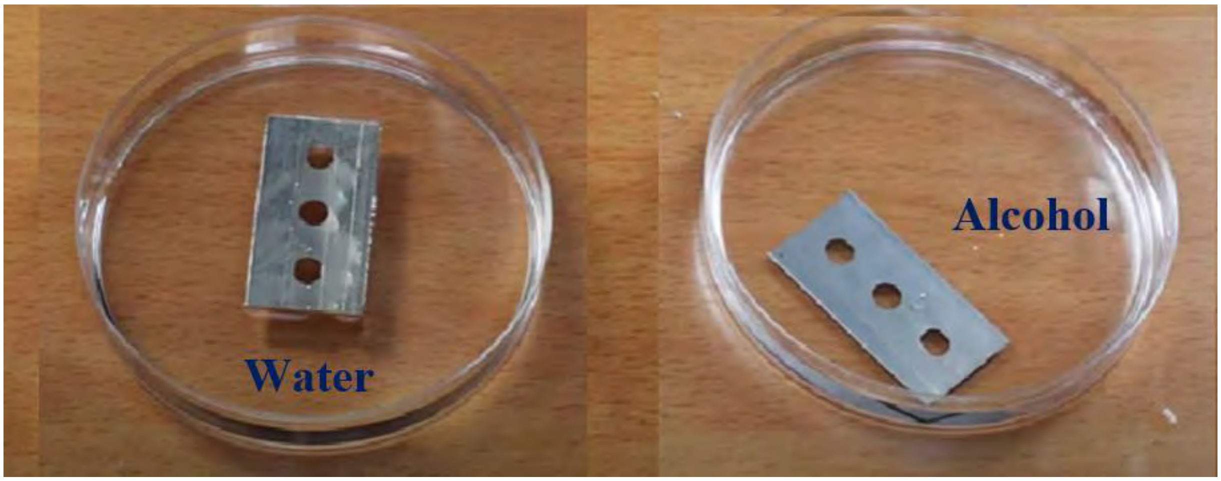


a. Arrange the liquids in increasing order of vapour pressure.
b. Which of the liquid will have the lowest boiling point? why?

C. Surface Tension

Activity: Observe the figures below containing two liquids with steel blade in each:

a. What is the difference in the figure?
b. Which liquid has more tension on the surface?



Have you ever noticed raindrops carefully resting on the surface of a green leaf? Perhaps you've observed flies walking on the water surface easily, challenging gravity. Now, we'll explore the reasons behind these fascinating phenomena.

Fig: The surface tension of the water helps in the formation of droplets and allows the insect to walk on the water without sinking

The intermolecular force of attraction between the molecules of the same substance is termed as cohesive force. The cohesive forces among molecules at the liquid's surface create an unbalanced force on those molecules. This force is responsible for the development of tension on the surface.

The molecules in the bulk of the liquid are attracted equally in all directions, while those at the surface are attracted more strongly to the molecules below than to those above and beyond. This results in a net inward force, causing the liquid surface to behave as a stretched rubber. Surface tension is the net inward force per unit length of the surface

Therefore, surface tension(γ)= $\frac{Force(F)}{Length\ (L)}$

Surface tension is measured in **Nm^{-1}(SI) or dyne cm^{-1}(CGS)** units**.**

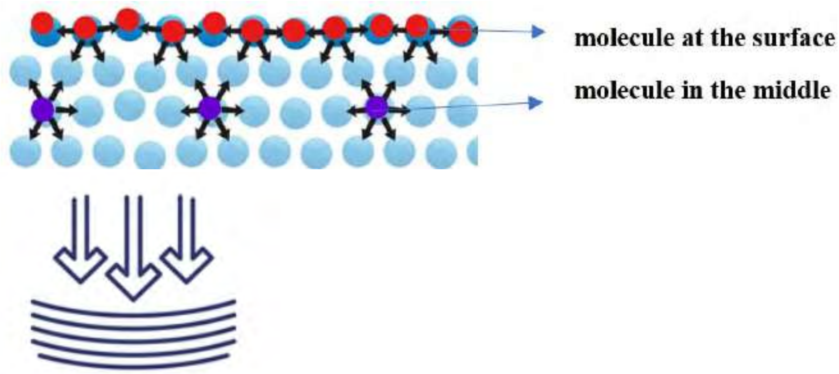


Fig: unbalanced force at the surface results the inward pull

Surface tension is defined as "**the property of a liquid that arises from cohesive forces among surface molecules, so as to minimize surface area and behave like a stretched elastic rubber**.

Do you know?

Because of the relatively high attraction of water molecules for each other through hydrogen bonds, water has a higher surface tension compared to that of most other liquids like alcohol and ether. Therefore, water drops are more spherical in shape and bigger in size compared to alcohol and ether.

Generally surface tension of liquid decreases with temperature. When liquid is heated, kinetic energy increases that makes the liquid molecules moving faster. It weakens the cohesive force to lower the surface tension.

The temperature at which surface tension of liquid becomes zero is called critical temperature of the liquid.

Critical temperature of water is 3744K. Impurities present in the liquid lower the surface tension of the liquid by weakening the intermolecular force in the liquid molecules.

Applications of surface tension

a. Soap and detergent contain surfactants which reduces the surface tension of water so that water can spread over large areas and wash the clothes more effectively.

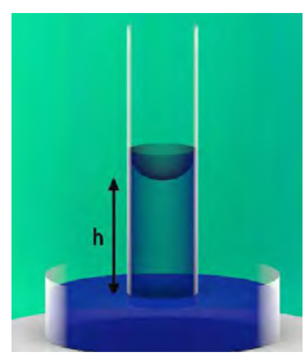


Figure: capillary action

b. A duck is able to float on water because its feathers secrete oil that lowers the surface tension of water.
c. Plants draw the water from the ground to their leaves due to capillary action which is the result of surface tension.
d. Surface tension tries to minimize the surface area of liquid so that liquid drops are spherical in shape.
e. Inkjet printers create images or text by precisely squeezing tiny droplets of ink. Surface tension helps control the size and placement of these droplets on paper.

Measurement of surface tension by drop count method

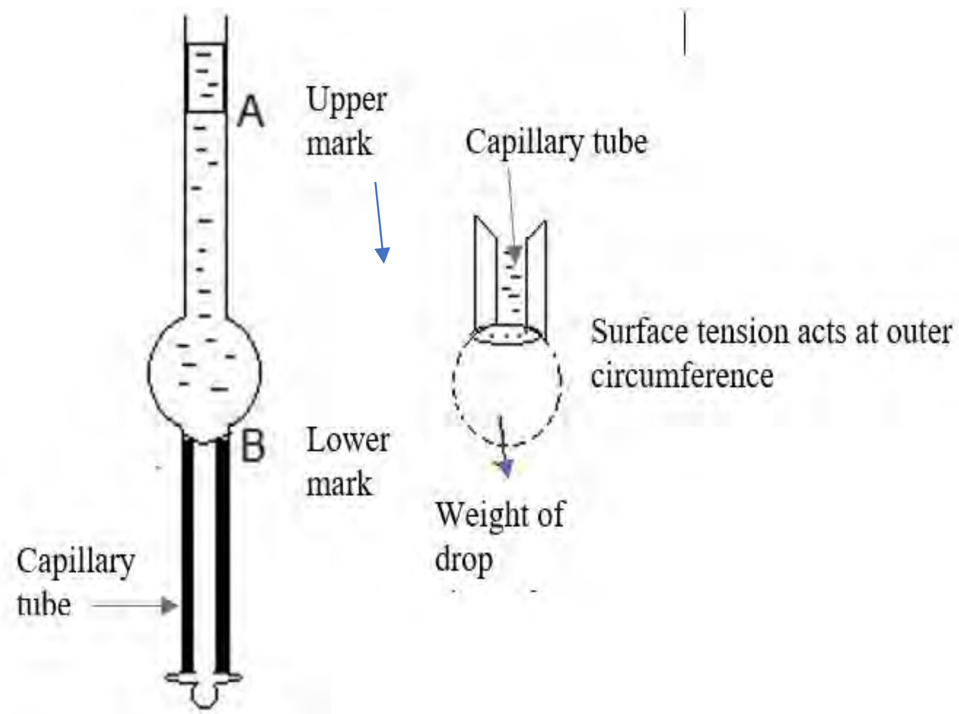


Figure: Stalagmometer and drop of liquid

In the drop count method, surface tension of liquid is measured using a stalagmometer. When a drop of liquid is detached from the circular end of a vertical stalagmometer, the upward and downward force must balance each other. Upward force of surface tension holds the drop whereas weight of drop pulls it downward.

At the point detached of drop

Upward force =downward force (figure D)

$2\pi r\gamma = mg$

where, γ = surface tension

m= mass of the drop and r= radius of the capillary for two different liquids the relation $2\pi r\gamma = mg$

becomes

$$2\pi r\gamma_1 = m_1 g \qquad 2\pi r\gamma_2 = m_2 g$$

$$\frac{\gamma_1}{\gamma_2} = \frac{m_1}{m_2}$$

$$\frac{\gamma_1}{\gamma_2} = \frac{M_1 \times n_2}{M_2 \times n_1} \left[\text{mass of each drop} = \frac{\text{total mass}}{\text{number of drops formed}} \right]$$

$$\text{Or } \frac{\gamma_1}{\gamma_2} = \frac{d_1 \times V \times \ldots}{d_2 \times V \times n_1} \text{ or } \frac{\gamma_1}{\gamma_2} = \frac{d_1 \times V \times n_2}{d_2 \times V \times n_1}$$

Where γ_1 = surface tension of known liquid

γ_2 = surface tension of unknown liquid

d_1= density of known liquid

d_2= density of unknown liquid

v = volume of liquid

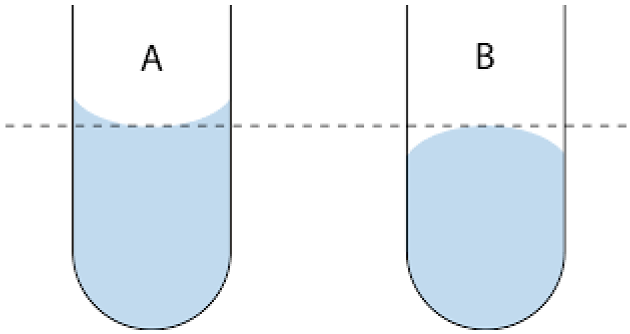
n_1 = number of drops of known liquid

n_2= number of drops of unknown liquid

Test yourself!

Two liquids, water and mercury, are kept in test tubes given, showing different meniscus at the surface.

a. Name the meniscus in A and B.
b. Identify the liquid with the help of the meniscus and provide the reason behind your identification.



Activity:

a. Collect some liquids that are available to you (e.g., water, honey, syrup, vegetable oil, glycerine, kerosene, etc.).
b. Pour 10 ml of each liquid onto an inclined surface (preferably glass) and note the time it takes for each liquid to reach the bottom, one by one.
c. Identify the liquid that reaches the bottom first and the one that takes the most time.
d. Find the possible causes behind the varying speeds of the liquids.

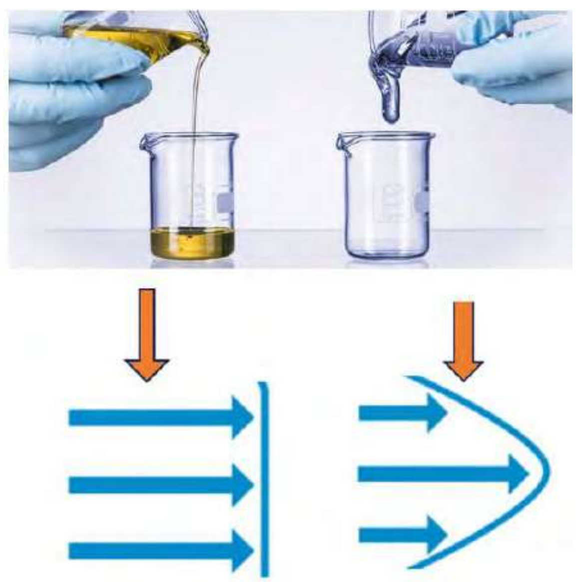


Fig: honey is a viscous liquid Fig: high viscosity means high resistance force between the layers

During the winter season, we use glycerin to moisturize our skin, while mechanics apply lubricants to machinery parts. Have you ever questioned why such liquids are used for these purposes? Now, we will explore the properties of liquids that will answer these questions.

We have already discussed why liquids flow, but it's important to note that not all liquids flow at the same speed. Certain liquids, such as petrol or ether, flow easily, while others, like honey and glycerin, flow more slowly. This variation in movement is credited to a property known as **viscosity**.

Viscosity is defined as the fluid's property that resists flow due to internal frictional forces between different layers.

The internal frictional force is termed **viscous force,** and liquids with higher viscosity are referred to as **viscous liquids**.

Viscous forces are intermolecular forces acting between molecules in different layers of liquid moving with different velocities.

Consider a liquid moving through a pipe in different layers of cross-sectional area **A** moving at varying velocities. The velocity of the layer is zero at the boundary called **stationary or fixed layer**. Moving away from the boundary towards the center of the tube, the velocity of the liquid increases. Let v be the velocity of the layer(B) next to the central layer(A) at a distance **dx**, then the velocity of the central layer should be **v + dv.**

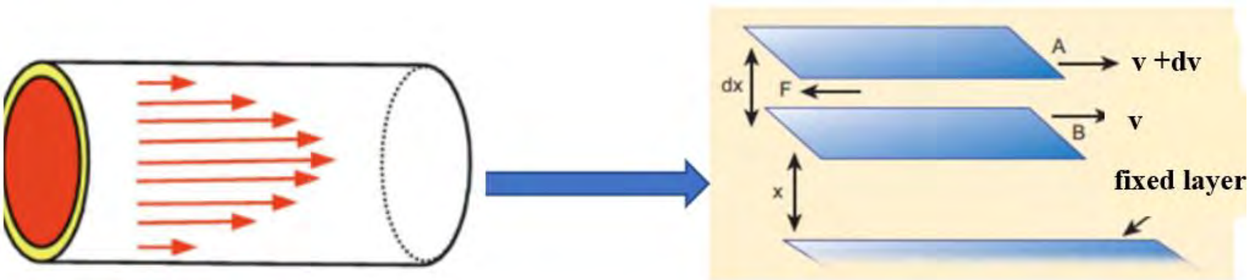


Fig: laminar flow of liquid in a pipe

The resistive force **(F)** offered by one layer to the next adjacent layer depends on the cross-sectional area of the layer, the difference in velocity **(dv),** and the difference in distance **(dx).**

Mathematically,

$$\mathbf{F} \propto \mathbf{A}\ \frac{dv}{dx}$$

where **A** is the cross-sectional area of each layer

$\frac{dv}{dx}$ is the rate of change of velocity with respect to the distance traveled called velocity gradient.

This relationship can be expressed as

$$\mathbf{F} = \boldsymbol{\eta}\ \mathbf{A}\ \frac{dv}{dx}$$

where η is a constant called the coefficient of viscosity.

The coefficient of viscosity(η) is defined as the resistive force required to maintain a unit velocity gradient between two parallel layers of liquid of unit area.

SI unit of coefficient of viscosity (η) is Newton-second per square meter **($Ns \cdot m^{-2}$)** or Pascal-seconds **(Pa·s)**. In the CGS unit system, the coefficient of viscosity is expressed as dyne-sec/cm^2, which is equivalent to Poise,

one Poise =1 dyne.sec cm^{-2} = 0.1 Pa·s.

The coefficient of viscosity of different substances at room temperature are given in the table below.

Substance	**η (Pa.s)**
Air	10^{-5}
Water	10^{-3}
Ethyl alcohol	1.2×10^{-3}
Mercury	1.5×10^{-3}
Ethylene glycol	20×10^{-3}
Olive oil	0.1
100% Glycerol	1.5
Honey	10
Corn syrup	100
Bitumen	10^{8}

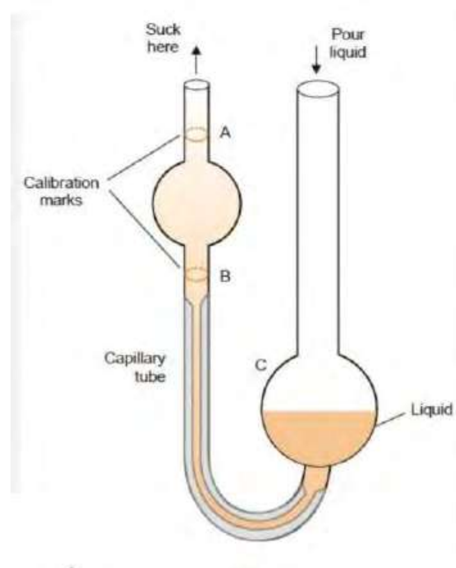


Fig: Ostwald's viscometer

Note: It's important to distinguish between viscosity and density, viscosity refers to the thickness of a fluid, while density relates to the space between its particles.

Viscosity of liquid is measured by using Ostwald's viscometer.

The viscosity of a liquid is influenced by several factors:

a. **Temperature** has a significant impact on viscosity. In general, as the temperature increases, the viscosity of most liquids decreases. This is because higher temperatures lead to greater molecular motion and weaker intermolecular forces, making it easier for the liquid to flow

b. **The size and shape of molecules** in the liquid impact viscosity. Long and complex molecular structures often result in higher viscosity. Polymer molecules, which are long chains of repeating units, can significantly impact the viscosity of a liquid.

c. **The molecular structure and composition** of the liquid play a crucial role in determining its viscosity. Liquid with strong intermolecular force has more viscosity than the substance with weak intermolecular force. For example, glycerol is more viscous due to the strong hydrogen bonding between the molecules.

Applications of viscosity

a. Viscosity is vital in the formulation of lubricants used in machinery. Proper viscosity ensures that the lubricant effectively reduces friction between moving parts.

b. The viscosity of paint and coatings impacts their application properties. Controlling viscosity ensures even spreading and adherence to surfaces during painting.

c. Viscosity is considered in the formulation of cosmetic products, such as lotions and creams, to achieve the desired uniformity for application on the skin.

d. In medicine, viscosity is important in understanding blood flow, as well as in the design of drug delivery systems and diagnostic devices.

e. Monitoring viscosity is often part of quality control processes in industries producing paints, adhesives, and various liquid products to ensure reliability and performance.

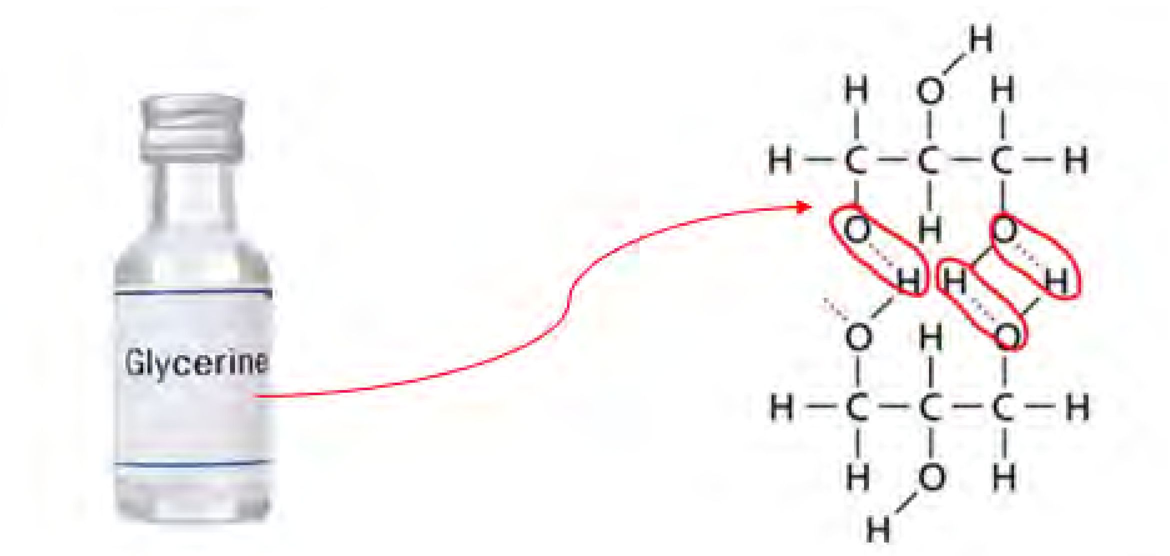
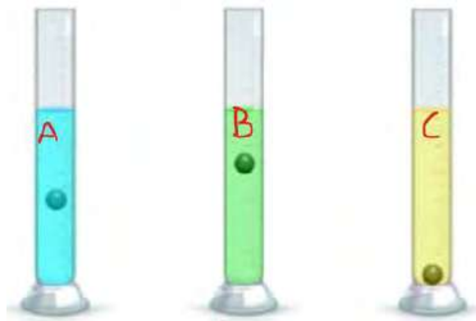


Fig: Multiple H-bonding in glycerol increases the viscosity of glycerol, the key component of glycerin

Test yourself!

1. Each measuring cylinder holds a single marble submerged in an equal volume of various liquids. Determine the container with the highest viscous liquid and give reason.



7.2.2 Liquid crystals and their applications

Activity:

Name and compare these monitors given below in terms of technology, image quality, energy efficiency, and pricing.

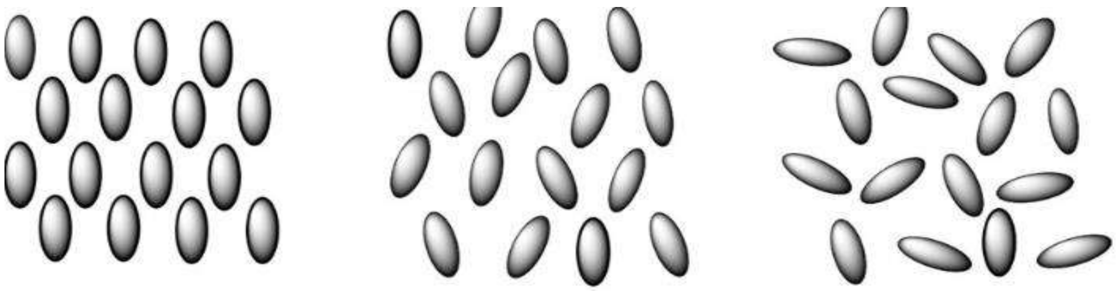
Have you observed the changes while pressing the display of a digital wrist watch? Applying pressure might cause a temporary change in the colors displayed. The distorted image persists for a short duration and then gradually returns to normal. This distortion occurs because the pressure can affect the arrangement of crystal molecules called **liquid crystals.**

Friedrich Reinitzer (1857–1927) was an Austrian botanist and chemist best known for his discovery of liquid crystals. His discovery of liquid crystals laid the foundation for the field of liquid crystal research, which has since led to numerous technological advancements, particularly in the development of liquid crystal displays (LCDs) used in various electronic devices.

Fig: Liquid crystal displays (LCDs) watches

In the late 19th century Austrian botanist **Friedrich Reinitzer** discovered a peculiar substance while studying cholesteryl benzoate. Later on, **Otto Lehmann**, a German physicist expanded the understanding about these substances and coined the term "**liquid crystal**" in 1922.

In the liquid phase, these molecules are oriented randomly and in solid phase molecules are in perfect order. In the liquid crystalline phase, by contrast, the molecules are arranged in specific patterns that differ from solid and liquid as shown in the figure.



solid **liquid crystal** **liquid**

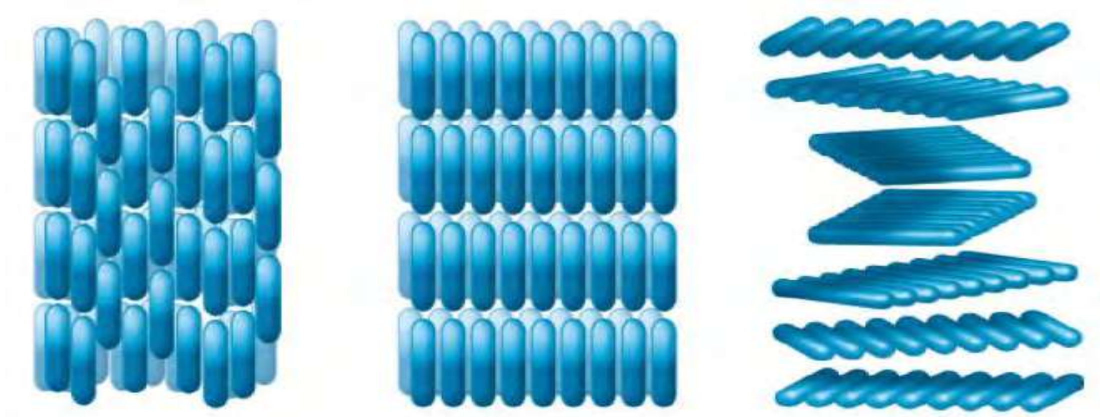
Fig: Molecular arrangements in solid, liquid and liquid crystal

Liquid crystals are defined as the intermediated state of matter having the properties of solid but can flow like a liquid.

The unique property of liquid crystals is their sensitivity to external stimuli, such temperature, pressure, or an applied electric field. When exposed to these stimuli, the molecular alignment of liquid crystals can change and hence changes their optical properties. These crystals have the ability to curve and rotate the light in different directions.

Liquid crystals are anisotropic, meaning their properties vary with direction. This anisotropy is a result of the alignment of the long, rod-shaped molecules within the liquid crystal structure.

Types of liquid crystals



Nematic **Smectic** **Cholesteric**

Fig: Different types of liquid crystals

Depending on the nature of the ordering of the rod-shaped molecules, liquid crystals are classified as noematic, smectic and cholesteric.

In a **nematic liquid crystal**, the molecules are aligned so that their long axes tend to point in the same direction but the ends are not aligned with one another.

In **smectic liquid crystals**, the molecules maintain the long-axis alignment as well as they pack into layers.

In a **cholesteric liquid crystal**, the molecules are arranged in layers, with their long axes parallel to the other molecules within the same layer. Upon moving from one layer to the next, the orientation of the molecules rotates. It causes the spiral pattern, so it is also called spiral nematic.

Do you know?

a. Fireflies have layers of light-emitting cells in their abdomen that regulate the direction and intensity of light. These cells are filled with a gel-like substance containing **liquid crystals**.
b. The scales of certain fish incorporate **liquid crystals**, allowing them to display varying colours when viewed from different angles. This feature helps these fish observe and exhibit different types.

Applications of liquid crystals

fig: Polarized light microscope

Fig: Electro-Optical camera

Fig Liquid crystal thermometer

a. **Liquid Crystal Displays (LCDs)**
   The most widespread application of liquid crystals is in the manufacturing of LCDs. The ability of liquid crystals to change orientation in response to an electric field created high-quality, flat-panel displays used in televisions, computer monitors, smartphones, and other electronic devices.

b. **Electro-Optical Devices**

Liquid crystals are employed in various electro-optical devices, including light modulators and optical shutters. These devices utilize the response of liquid crystals to electric fields to control the passage of light, making them valuable in optics and photonics applications

c. **Biomedical Imaging**

Liquid crystal materials are utilized in certain biomedical imaging techniques, such as polarized light microscopy. The birefringent properties of liquid crystals enhance contrast and enable the visualization of structures in biological samples.

d. **Liquid Crystal Polymer (LCP)**

Liquid crystal polymers, a type of liquid crystal, are used in the development of antennas for wireless communication systems.

e. **Holographic Displays**

Liquid crystals play a role in the development of holographic displays. The manipulation of liquid crystal properties contributes to the creation of dynamic holographic images for applications in entertainment and visualization.

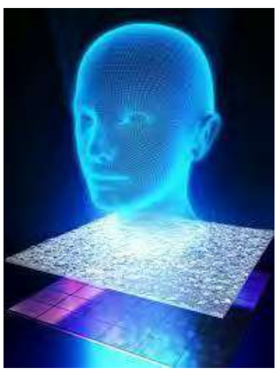


Fig: Holographic display *fig: Liquid Crystal Polymer* ***(LCP)***

Test yourself!

Despite the widespread applications of LCD, what encouraged the development of new LED technology?

Exercises

A. Multiple choice questions

1. Evaporation is a surface phenomenon which occurs at all temperatures. The rate of evaporation increases with
   a. increasing the pressure
   b. increasing the surface area
   c. viscosity of liquid
   d. decreasing temperature
2. When vapour pressure of liquid equals the atmospheric pressure, liquid boils. At which place the boiling point of liquid is minimum?
   a. In the pressure cooker in the mount Everest
   b. In the open vessel in the mount Everest
   c. In the pressure cooker in the sea level
   d. In the open vessel in the sea level
3. Which of the following liquids is most viscous?
   a. Water
   b. Ethyl Alcohol
   c. Diethyl ether
   d. Methyl alcohol
4. Which of the following is responsible for the capillary action in the narrow tube?
   a. Viscosity
   b. Evaporation
   c. Surface tension
   d. Gravity
5. Characteristic property of liquid crystal is
   a. intermediate phase between liquid and gas.
   b. intermediate phase between solid and gas.
   c. intermediate phase between liquid and solid.
   d. intermediate phase between liquid and plasma.
6. What happens to the boiling point of the water when some amount of common salt is added to it?
   a. No change in boiling point
   b. Boiling point increases.
   c. Boiling point decreases.
   d. Boiling point first decreases then increases.
7. Which of the following points explains that beans are cooked faster in a pressure cooker?
   a. Low boiling point of water inside the pressure cooker
   b. More heat is produced inside the pressure cooker
   c. High boiling point of water inside the pressure cooker.
   d. More heat is produced outside the pressure cooker
8. Which of the following liquid has the most spherical shape of its drop?
   a. Water

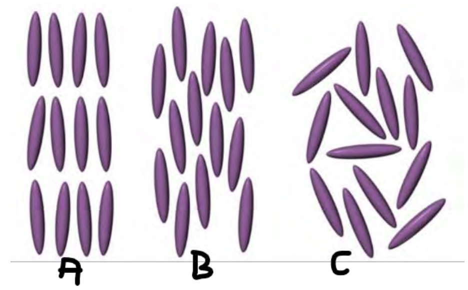
b. Ether
c. Ethyl alcohol
d. Methyl alcohol

9. What is the primary characteristic of liquid crystals?
   a. Rigidity
   b. Fluidity
   c. Transparency
   d. Opacity
10. Grease and other lubricants are commonly used in the machinery parts. Which of the following properties is responsible for such liquids?
    a. High rate of evaporation and low viscosity
    b. High viscosity and low boiling points.
    c. High viscosity and low rate of evaporation
    d. High boiling points and low viscosity

B. Short answer questions

11. Define the term evaporation and vapour pressure. How does temperature and nature of bonding affect the rate of evaporation of the liquid?
12. Boiling point of liquid is influenced by nature of liquid and atmospheric pressure.
    a. What do you mean by boiling point of liquid?
    b. How does boiling point depend upon the intermolecular force of attraction and atmospheric pressure?
13.
    a. What is surface tension?
    b. How does intermolecular force of attraction affect the surface tension of liquid?
    c. Write any two applications of surface tension.
14. Differentiate the following:
    a. Evaporation and boiling
    b. Surface tension and viscosity
    c. Vapor pressure and atmospheric pressure
15. Give reasons for the following statements:
    a. Boiling point of water is less in Archaizer than in Bharatpur.
    b. Liquid drops are spherical.
    c. Water makes concave meniscus on the surface.
    d. Drops of water are bigger in size than that of alcohol when they are dropped from the same tube.
    e. It is difficult to transfer the organic solvent like ether using a glass tube.
    f. Glycerine is used in skin care.
    g. For a given substance, the liquid crystalline phase tends to be more viscous than the liquid phase.

16. The alignment of molecules in the liquid crystals depends on temperature. Three different types of alignment (A, B and C) at three different temperature is given:



a. Which one does not belong to liquid crystals and why?

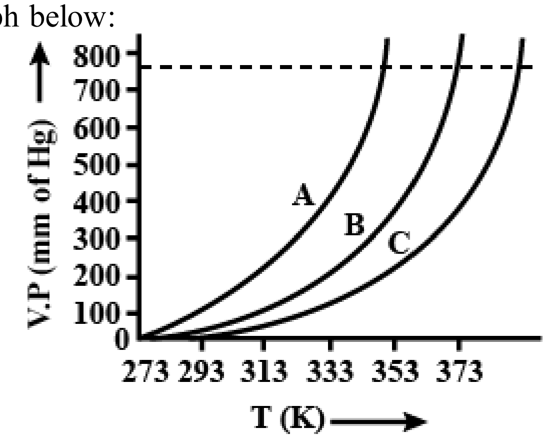
b. Identify the type that is at lowest temperature and give reason.

c. Which one is more viscous and why?

C. Long answer questions

17. LCDs are commonly used in electronic devices due to their unique and special features. LCD contains liquid crystals.
    - **a.** What do you mean by liquid crystal?
    - **b.** How do liquid crystals differ from traditional solid and liquid states of matter?
    - **c.** What unique properties make them suitable for various technological applications?
    - d. How is LCD superior to CRT?
    - **e.** What are the different phases in liquid crystals and how does liquid crystal respond to external stimuli?

18. The relationship between vapour pressure and temperature for different liquids A, B and C are given in the graph below:

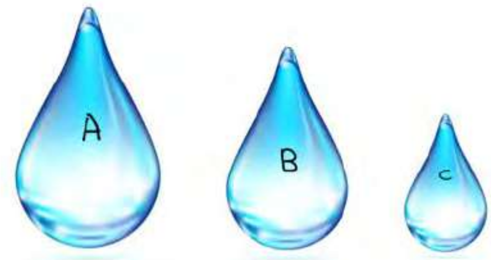


a. Define the term vapour pressure. How does the nature of bond affect the vapour pressure of the liquid?

b. Identify the liquids having the highest rate of evaporation and highest boiling point from the graph.

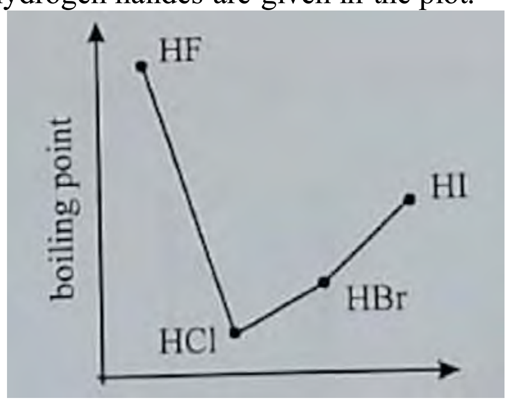
c. If atmospheric pressure is 300 mmHg instead of 760 mmHg at the sea level, which liquid would boil first and why?

19. Liquid forms drop from the open end of capillary due to surface tension. Drops of water, ether and ethyl alcohol formed from the same end of capillary are given in the figure.



   a. Define surface tension and write its unit.
   b. Why do liquid drops try to be spherical?
   c. Name the liquid that forms the drops A, B and C and give reason for your choice.
   d. After an injection, healthcare professionals might use a cotton ball or swab with rubbing alcohol to the injection site to provide a cooling sensation. Why?

20. Boiling points of hydrogen halides are given in the plot.



   a. Define boiling point. How does boiling differ from evaporation?
   b. In what factors does the boiling point of liquid depend?
   c. Which hydrogen halide has the highest boiling point? What could be the reason for such a high boiling point?
   d. Boiling point of HBr is less than HI. Why?
   e. Identify the halides having least vapour pressure.

21. The table below shows the boiling point and surface tension of different liquids.
   i. Define the terms boiling point and surface tension.
   ii. Boiling point of water is the highest among the three. Why?
   iii. Which of the given liquid forms the largest drop when poured from the same capillary? Why?
   iv. Which of the liquid boils first in the pressure cooker?

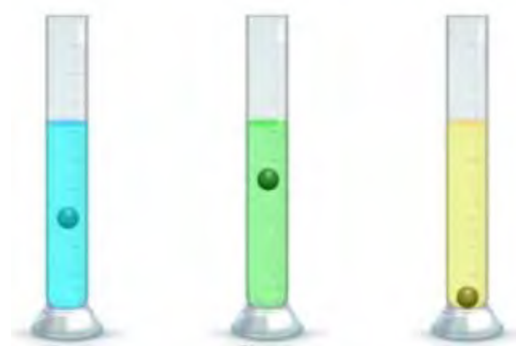
	Boiling point (^0C)	Surface tension(J/m^2)
Water	100	7.3×10^{-2}
Ethanol	78	2.3×10^{-2}
propanol	97	2.4×10^{-2}

22. Marble is placed in a separate measuring cylinder, each filled with liquids ((water, ether, glycerol) of different viscosity in the figure.

a. Define the term viscosity. How does temperature affect viscosity of liquid?

b. Identify the liquid from the figure and give reason for your identification.

c. Write any two applications of viscosity.



7.3 Solid State

Activity:

a. Collect some solid substances around you (common salt, rubber, plastic, iron nails, etc.)
b. Touch and examine the physical structure and surface smoothness of each.
c. Discuss the variations in properties of these substances (colour, hardness, solubility, etc.)

We have discussed the gas and liquid state in earlier sections. In this section we will discuss the different types of solids and the cause of variation of their properties. Solid state consists of molecules in fixed position held together by the strongest intermolecular force of attraction. Like liquid, the molecules of solid also have kinetic energy and have motion. But the motion of particles is restricted to fixed position so there is no freedom of movement. There are basically two general categories of solids: amorphous and crystalline.

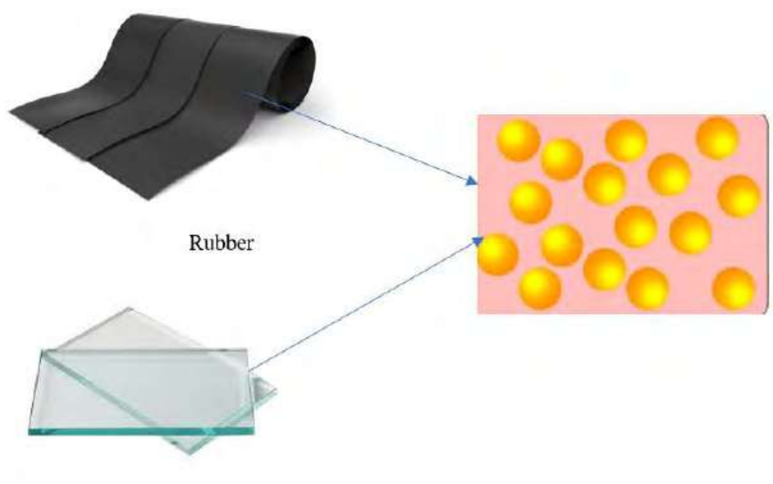


Fig: group of protestors are chaotic and random

Fig: amorphous solids

Amorphous solids have no defined shape or are shapeless. As a group of chaotic protestors, the particles in amorphous solids are not located in any particular positions. They are called isotropic in nature. It means they have uniform properties (mechanical, thermal or electrical) in all directions. Examples of amorphous solids are glass, rubber, plastics, etc.

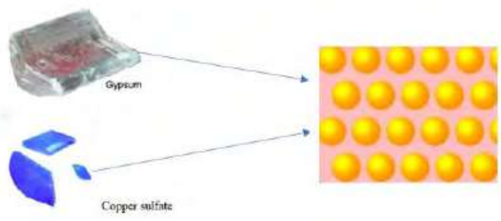


Fig: crystalline solids

Fig: the soldiers ready for march pass are in definite position

Like a perfectly arranged army, the particles in crystalline solid are arranged in a regular, symmetrical structure called a **crystal lattice**. They are anisotropic in nature means, show different properties (mechanical, thermal, electrical) in different directions. They have sharp melting points and are generally soluble in water. Crystals of salt, pieces of quartz or natural minerals etc. are crystalline solids.

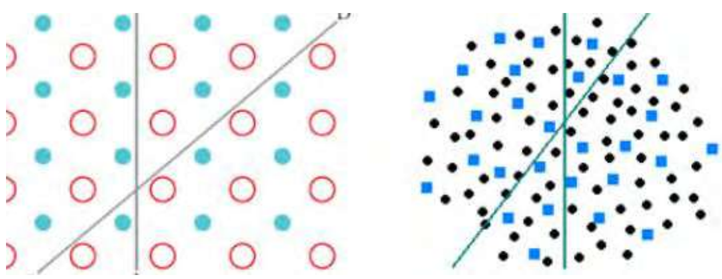


Fig: anisotropy and isotropy

An isotropic solid has uniform properties in all directions. This means its mechanical and thermal properties, like elasticity and conductivity, are the same regardless of the direction you measure them. On the other hand, anisotropic solids exhibit different properties when measured in different directions.

Test yourself!

Identify the given substances as crystalline or amorphous and place them in the table below. Write any three differences in their properties.

NaCl, plastic, rubber, lamp black, $CuSO_4.5H_2O$, gelatin, waxes, glass, snowflakes, quartz, copper

Crystalline solids	*Amorphous solids*

7.3.3 Efflorescent, Hygroscopic and Deliquescent solids

Activity:

a. Take some amount of copper sulphate crystals in a porcelain basin and gently heat them over a burner.
b. Observe the gradual change in the colour of the crystals.
c. Take some pellets of caustic soda (NaOH) in an open vessel for a few minutes.
d. Observe the change in the shape of the pellets.

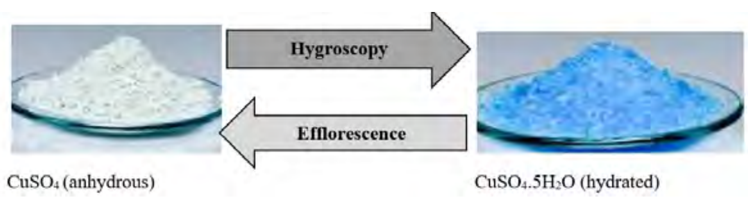


Fig: heating copper sulfate crystals

When you heat the crystals of blue vitriol(($CuSO_4.5H_2O$) the blue color slightly fades away and becomes white. It is the result of loss of water molecules present in the crystal. The substance formed after losing water molecules is called **anhydrous salt**. Efflorescent substances lose water molecules either partially or completely and become anhydrous. For example, Glauber's salt ($Na_2SO_4.10H_2O$), green vitriol ($FeSO_4.7H_2O$), blue vitriol ($CuSO_4.5H_2O$), washing soda ($Na_2CO_3.10H_2O$) etc. are efflorescent substances.

$CuSO_4.5\ H_2O$ (hydrated) → $CuSO_4$(anhydrous) + $5H_2O$ (at high temperature)

$Na_2CO_3.10H_2O$ (hydrated) → Na_2CO_3 (anhydrous) + $10H_2O$ (at high temperature)

Those substances that can lose water molecules that are already present in their molecular structure are called when exposed to air are called efflorescent substance and this phenomenon is called efflorescence.

Do you know we need to water a seed in the soil to germinate? It is because the outer coating of seeds tends to absorb moisture required for germination. The substances that absorb moisture from the atmosphere at ordinary temperatures are called **hygroscopic substances**, and the property is known as **hygroscopy**. Hygroscopic substances after absorbing water results in physical changes like an increase in volume, boiling point, temperature, and change in viscosity. Hygroscopic substances also called **humectants** are commonly used in desiccators. These substances are also used in food, cosmetics, and drugs to maintain the moisture level and preserve them. For example, CaO, $NaNO_3$, sucrose, CuO. conc. H_2SO_4 and conc. HCl are common hygroscopic substances. Hygroscopy and efflorescence are reversible to each other.

$$CuSO_4(\text{anhydrous}) + 5H_2O \rightarrow CuSO_4.5\,H_2O\ (\text{hydrated})$$

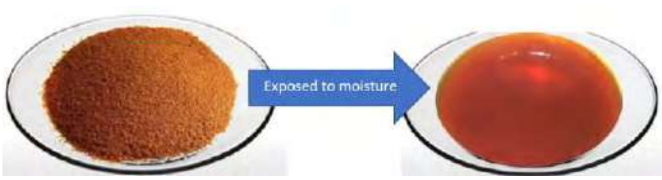


*Fig: Ferric chloride is **a deliquescent substance** which absorbs moisture and forms saturated solution*

*fig: Desiccator used in laboratory contains fused $CaCl_2$ as a **desiccant***

Like the hygroscopic, there are some solid substances which absorb moisture from the atmosphere and get dissolved to form an aqueous solution. These substances are called **deliquescent substance** and the phenomenon is called **deliquescence**. These deliquescent substances have a high affinity to water therefore they lose their crystalline shape and form saturated solution. Deliquescence is found to be maximum at low temperature and humid atmosphere. Example, solid NaOH, $CaCl_2$, P_4O_{10}, KOH, $MgCl_2$. These substances are used to remove moisture called **desiccants.**

Do you know?

A species of lizard commonly called the **thorny dragon** has **hygroscopic** grooves between its spines. Water condenses on the backbones at night is then distributed across its skin by means of capillary action.

Try yourself!

You've probably seen these little packets that come with new shoes or electronics. What is the role of these materials in those products?

7.3.4 Crystallization and Crystal growth

Activity:

a. Put 4 teaspoonful of copper sulphate powder into 1/2 cup of hot water and stir thoroughly until the solid dissolves completely.
b. Filter the impurities present using filter paper and allow the filtrate to stand until it undergoes complete evaporation (may be overnight)
c. Observe the colour and shape of the resulting crystals.

Note: You can do the same activity using common salt (NaCl) instead of Copper sulphate.

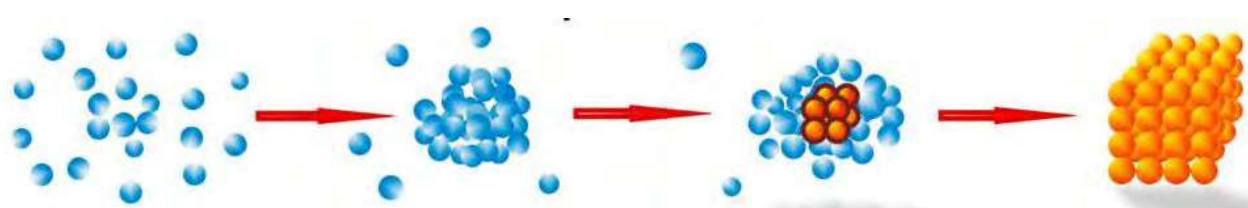
Remember that copper sulfate is poisonous.

Fig: Different types of crystals

Under suitable conditions, some solid substances form fixed geometrical shapes called **crystals**. A substance that is made up of crystals is called a **crystalline substance**. Sometimes the word polycrystalline is used to indicate a substance made up of many crystals. Crystals can be found in various forms, including minerals, metals, salts, and even biological molecules. Crystals of different substances have different properties like color, shapes, cleavage, etc. Crystals grow bigger by adding more layers of solid around their outsides. There are various ways by which crystals can be made. For example: precipitation, evaporation, sublimation, cooling, etc.

The process of formation of crystals is termed as crystallization.

Crystallization is only physical change because no new substances are formed. During crystallization solid crystal forms from a liquid, gas, or a solution. The general overview of the crystallization process:



Nucleation: The first step is nucleation, where tiny clusters of particles begin to form in the liquid or solution.

Crystal Growth: Once nucleation occurs, the crystal continues to grow as more atoms, ions, or molecules join the crystal lattice. This growth is typically in a specific direction, creating the characteristic shape of the crystal. In some crystals, growth occurs **layer by layer.** New particles attach to the existing crystal surface, creating concentric layers. Some crystals grow in a **spiral pattern**, with new particles attaching to the crystal surface in a helical arrangement.

The conditions under which crystallization occurs, play a crucial role in crystal growth (size and quality of the crystals):

a. The **temperature** of the system can affect the rate of crystal growth. Higher temperatures generally lead to faster growth.
b. In solutions, **supersaturation** (an excess of solute) helps in crystal growth.
c. The **concentration** of solute in a solution can affect crystal growth. Higher concentrations generally lead to faster growth of the crystal.

Applications of crystallization

Fig: NaCl obtained from the crystallization of sea water

a. Crystallization is widely used in the **chemical industry** to purify substances and produce high-quality products.
b. Many **pharmaceutical compounds** are obtained in crystalline form for stability.
c. The study of crystal structures is essential in **materials science** for understanding the properties and behavior of materials.
d. Crystals form the basis of many minerals, and studying their crystal structures helps in understanding **geological processes.**

Test yourself!

How does cooling contribute to crystal formation in the process of crystallization?

7.3.5 Water of Crystallization

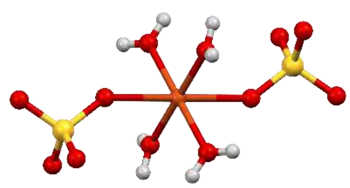
Activity 7.3.6

a. Take a spatula full of copper sulphate crystals.
b. Observe the initial colour of the crystals, then heat them directly over a flame.

c. Observe any changes in colour during the heating process.

Fig: Anhydrous coppe sulphate *Fig: Copper* sulphate *crystals(hydrated)*

When an anhydrous salt is exposed to moisture, it absorbs water molecules and becomes hydrated. These water molecules significantly affect the physical and chemical properties of the compound. For instance, anhydrous copper sulfate appears white, while its hydrated crystal is blue.



structure of $CuSO_4 .5H_2O$

The water molecules that are associated with the crystal structure of the compound are called water of crystallization and the crystals are called hydrated crystals.

Each unit of hydrated crystals contains a specific number of water of crystallization. For example, $CuSO_4 \cdot 5H_2O$, known as pentahydrate copper sulfate, indicates the presence of five water molecules per formula unit. Common hydrated salts and their water of crystallization is given:

Chemical name	Common name	Chemical formula
Copper(II) sulphate pentahydrate	Blue vitriol	$CuSO_4 \cdot 5H_2O$
Magnesium sulphate heptahydrate	Epsom salt	$MgSO_4.7H_2O$
Calcium sulphate dihydrate	Gypsum	$CaSO_4.2H_2O$
Iron (II) sulphate heptahydrate	Green vitriol	$FeSO_4.7H_2O$
Zinc sulphate heptahydrate	White vitriol	$ZnSO_4.7H_2O$
Soidum carbonate decahydrate	Washing soda	$Na_2CO_3.10H_2O$
Calcium sulphate semi hydrate	Plaster of Paris	$CaSO_4.1/2H_2O$

Aluminium potassium sulfate dodecahydrate	Potash alum	$KAl(SO_4)_2\cdot 12H_2O$
Sodium sulphate decahydrate	Glauber's salt	$Na_2SO_4.10\ H_2O$

The water of crystallization can be removed through heating, this process is called dehydration. This method is applicable in determining the water of crystallization in hydrated salts.

The working formula to calculate the water of crystallization in a hydrated salt is as follows:

Water of Crystallization (%) $= \frac{Mass\,of\,water\,lost\,during\,heating\,(g)}{mass\,of\,hydrated\,salt\,taken(g)} \times 100$

$$The\ number\ of\ water\ of\ crystallization$$

$$= \frac{Mass\ of\ water\ lost\ during\ heating(g)}{mass\ of\ hydrated\ salt\ taken(g) \times 18} \times molecular\ mass\ of\ hydrated\ salt$$

Note: anhydrous salt is different from dehydrate salt.

Anhydrous refers to a substance that does not contain water molecules in its crystalline structure. It indicates the absence of water. For example, anhydrous copper sulfate ($CuSO_4$) is copper sulfate without any water molecules.

Dehydrate specifically refers to a substance that has had water removed from it. For instance, if you take a hydrated salt and remove the water through heating, the resulting substance is said to be dehydrated.

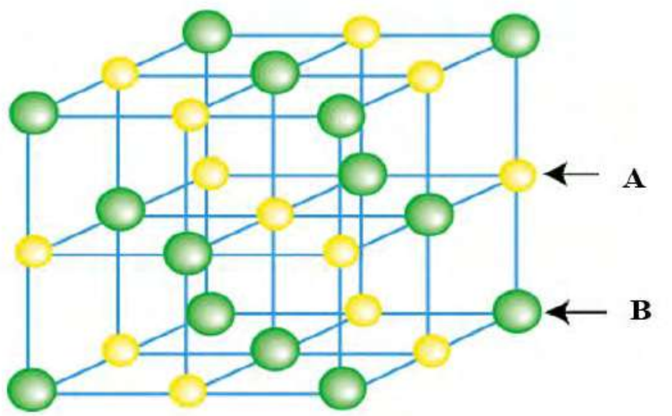
Test yourself!

An evaporating dish weighing 120.41 g is selected for the experiment. The salt is thoroughly heated, and the experimental outcomes concerning the heating process are detailed in the table.

Substance	Mass /g
evaporating dish + salt before heating	122.41
evaporating dish + salt after heating	121.36

a. Calculate the mass of water present in the given mass of hydrated salt.
b. Find the percentage of water in the salt.

7.3.6 Introduction to Crystal lattice and Unit cell



Observe the structure of NaCl and identify the ions A and B.

Crystal lattice

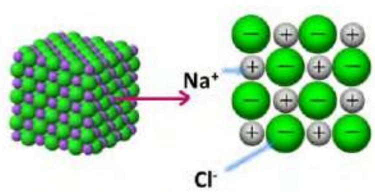


Fig: ionic lattice

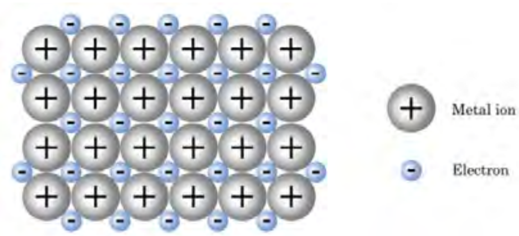


Fig: metallic lattice

In each solid substance, there is a regular pattern of some repeating units. Repeating units may be ions, atoms or molecules. In NaCl repeating units are positive and negative ions. Likewise, in metallic solid the repeating units are positive ions.

The regular repeating arrangement of unit particles (atoms, ions or molecules) called crystal lattice. Crystal lattice helps to maintain the **regular** structure of crystals**.**

Each unit particle (atom, ion or molecule) is represented by one point in a crystal lattice called **lattice point** or **lattice site**.

Crystal lattice may be ionic lattice, metallic lattice or molecular lattice depending upon the unit particles present. In an **ionic lattice**, the positive and negative ions arrange themselves in a repeating three-dimensional pattern to maximize the electrostatic force of attraction between the ions. Compounds with ionic lattices are sometimes called **giant ionic structures**. In an ionic lattice of sodium chloride, each sodium ion is surrounded by six chloride ions and each chloride ion is surrounded by six sodium ions. The properties of ionic compounds like high melting point, brittleness and conductivity are due to the strong electrostatic force of attraction between the ions in the ionic lattice.

In **metallic lattice**, the metal cations are arranged in a regular pattern, forming a **crystal lattice**. In metallic lattice, each metal ion is surrounded by negatively charged mobile electrons called **sea of electrons**. The properties of metals like high tensile strength, hardness is due to the strong attractive force between the metal ion and the delocalized electrons in the metallic lattice.

Fig: Molecular lattice of iodine and ice

In a **molecular lattice**, the basic building blocks are individual molecules, and they come together in an ordered, repeating pattern to form a solid structure. Molecular lattices are held together by intermolecular forces. These forces can include hydrogen bonding, van der Waals forces, and dipole-dipole interaction. The crystals of ice and iodine are examples of molecular lattice. Molecular lattices often have lower melting and boiling points compared to ionic or metallic lattices because the intermolecular forces are generally weaker than the electrostatic forces.

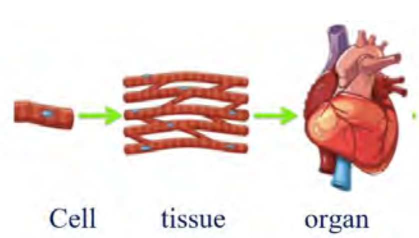
Test yourself!

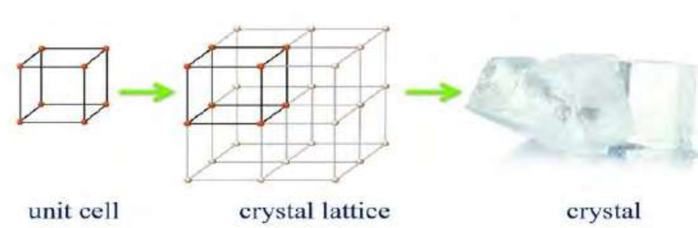
A structure of lattice is given in the figure

a. Identify the lattice point (atom, ion or molecule) and the bond that holds these particles together in the lattice.
b. Which properties of the substance are expected from such lattice?

Unit Cell

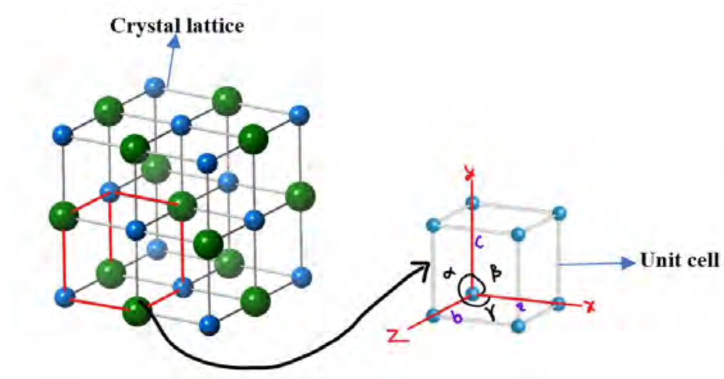
Activity:





Compare the figure above and draw the conclusion.

Like the cells in our body, unit cells are the building block of the crystals. The shapes and other physical properties of crystalline solid are determined by the geometry and arrangement of these unit cells. Different parameters like density of crystals, coordination number and number of atoms per unit cell can be calculated from the unit cell. These parameters are useful to explain and predict properties of crystals in crystallography.



Parameters of a Unit Cell:

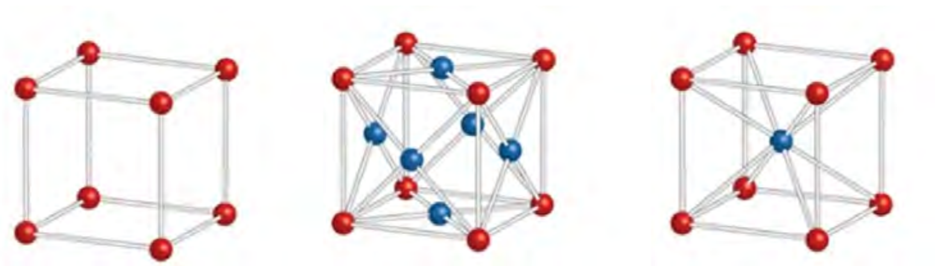
Lattice Constants: These are the lengths of the edges of the unit cell i.e. a, b and c

Angles Between Axes: The angles between the edges of the unit cell are represented by alpha (α), beta (β), and gamma (γ).

Fig: Crystal lattice and unit cell of sodium chloride crystal

Types of cubic unit cells

If lattice constants are equal (a=b=c) and the angle between the axes are equally 90 0 (**α**= B =γ 90^0, then the unit cell will be cubic.



Simple cubic Face centered cubic Body centered cubic

Fig: Different types of cubic unit cells

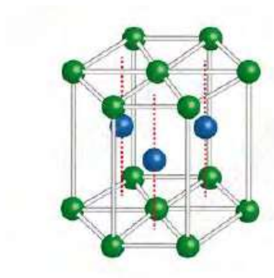
Depending up on the position of lattice points (atoms,ions or molecules) in the unit cells ,there are three types of unit cells:

a. **Primitive cubic or simple cubic (SC)**: Lattice points are located only at the corner of the cube. Crystals of Polonium(Po), Cesium(Cs) and Rubidium(Rb) have simple cubic unit cells.
b. **Body centered cubic (BCC):** One lattice point is at the center besides others at the corner of the cube. Crystals of iron, tungsten and potassium have BCC unit cells.

c. **Face centered cubic (FCC):** lattice points are at the center of each face besides the corner of the cube. Crystals of copper, aluminium, NaCl, KCl have FCC unit cells.

Test yourself!

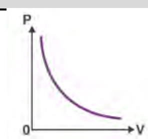
If a=b=c and **α**= B =γ 90^0, then the unit cell will be cubic, Identify the given unit cell based on geometry.



For animation https://youtu.be/BjVTdZ_htu8?si=i8mXHOa3DZ5DO7LO

Project work

1. Take a chart paper and make a column and write the information's about all the gas laws (where possible) as shown below.

Gas Law	Relationship	Formula	Constant conditions	Graph
Boyle's law	$V \propto \frac{1}{p}$	$P_1V_1=P_2V_2$	No. of mol and temperature	
Charle's law				

2. Make different types of unit cell and crystal lattice using clay and stick.

Exercises

A. Multiple choice questions

1. Which of the following is an example of a molecular solid?
   a. Sodium chloride
   b. Diamond
   c. Iodine
   d. Copper

2. What does efflorescence substances do?
   a. Absorb moisture from the air
   b. Lose water of crystallization on exposure to air
   c. Dissolve readily in water
   d. Form a crystal lattice structure

3. In the process of crystal growth, what is the term for the addition of new particles to the surface of a crystal?
   a. Nucleation
   b. Sublimation
   c. Lattice formation
   d. Deposition

4. Which type of solid allows for high electrical conductivity in molten and aqueous form?
   a. Molecular solid
   b. Ionic solid
   c. Metallic solid
   d. Covalent network solid
   a. The smallest portion of the lattice is known as
      a) Lattice structure
      b) Lattice point
      c) Bravais crystal
      d) Unit cell

5. Diamond is an example of
   a. solid with hydrogen bonding
   b) electrovalent solid
   c) covalent solid
   d) glass

6. Which property of the following is shown by crystals with water of crystallizations?
   a. Hygroscopic
   b. Anhydrous
   c. Deliquescent
   d. Efflorescent

7. Which of the following factor is not responsible maintain the size and quality of crystal during crystal growth?
   a. Pressure
   b. Concentration of solution

c. Temperature
d. Nature of crystal

8. Number of water of crystallization in copper sulphate, gypsum and sodium carbonate are respectively
a. 5,10,2
b. 2,5,10
c. 10,2,5
d. 5,2,10

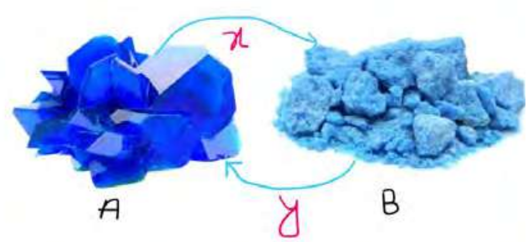
Short answer questions

1. Give reasons:
   a. Anhydrous copper sulphate changes its colour after exposing to the moisture.
   b. Calcium chloride is used as desiccant in the desiccators.
   c. Pellets of caustic soda change their shape up on exposed to air.

2. Differentiate between:
   a. Crystal and crystallization
   b. Hygroscopic and hydrated salt
   c. Deliquescent and hygroscopic
   d. Crystal lattice and unit cell

3. On the basis of arrangements of particles, solid substances are divided into crystalline and amorphous.
   a. Define and differentiate between amorphous and crystalline solids.
   b. Provide examples of materials that exhibit amorphous and crystalline characteristics.
   c. Imagine you are designing a new material for a specific application (e.g., electronic devices). Would you prefer to use an amorphous or crystalline material, and why?

4. In the figure given, two forms of copper sulphate A and B, are presented. Through certain physical processes, denoted as x and y, these substances can transform into each other.
a. Define the term hygroscopic and efflorescent and identify the substance.
b. Name the process x and y and the corresponding equation to represent these processes.
c. What is the reason for having different colors of A and B?



d. In laboratory settings, why is it important to be aware of whether a substance is efflorescent, deliquescent, or hygroscopic?

5. Discuss the environmental implications of efflorescent, deliquescent, and hygroscopic solids. How might these behaviours influence storage, transportation, or usage of certain materials in different climates?

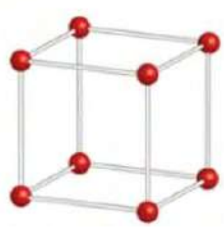
6. Crystallization is one of the techniques of separation of mixture commonly used into laboratory.
   a. Define the term crystal and crystallization.
   b. Name the major two steps of crystallization
   c. How does crystallization help in purification of substances? What types of mixture can be separated through crystallization?

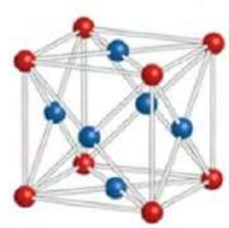
7. A sample of copper sulphate pentahydrate ($CuSO_4 \cdot 5H_2O$) has a mass of 250 grams. After heating the mass of salt is found to be 160 grams.
   a. $CuSO_4 \cdot 5H_2O$ is called efflorescent. Why?
   b. Why is there a loss of mass after heating?
   c. What is the change in colour of salt before and after heating and why?
   d. Determine the percentage of water of crystallization in the original sample.

8. Crystal lattice of one of the crystalline solid substance is given in the figure

   a. Define the term "crystal lattice" and identify the type of lattice in the figure and give any two properties of such lattice
   b. What are other types of crystal lattice?

9. Crystal lattice is made up of unit cells. Unit cells of some crystals are given:





   a. What do you mean by unit cell?
   b. Name the unit cell given and give reason for such naming

c. How does knowledge of crystal structures and unit cell contribute to the development of new materials?

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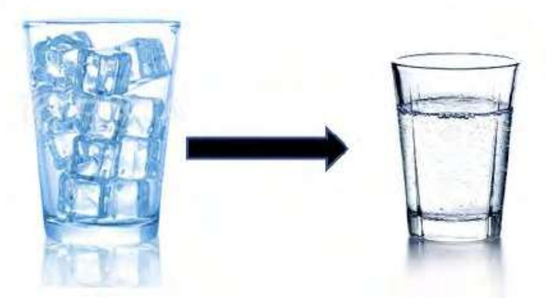
UNIT 8: CHEMICAL EQUILIBRIUM

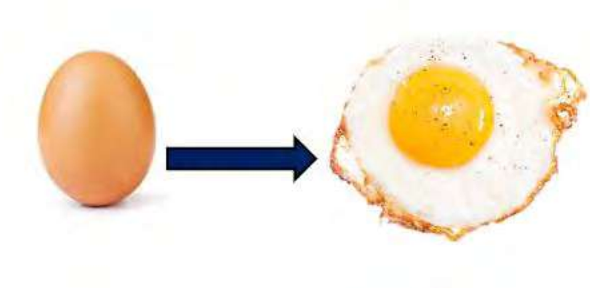
8.1: Reversible and Irreversible Reactions

Activity:

Which of the given processes (A and B) is reversible and which one is irreversible? Why?



A



B

$$A + B \rightleftharpoons C + D \quad \text{.............................} \quad (1)$$

$$A + B \rightarrow C + D \quad \text{.................................} \quad (2)$$

In equation **(1)**, the reactants and products are separated by double headed arrow (⇌) whereas in equation **(2)**, the reactants and products are separated by single headed arrow. The first equation represents that the reactants and products are inter convertible but in the second equation reactants and products are not inter convertible. Here equation **(1)** is reversible reaction and equation **(2)** is irreversible reaction.

A **reversible reaction** is that in which the products formed react back to give reactants under certain conditions. Such reaction involves two reactions i.e., the forward reaction and the backward reaction separated by a double headed arrow. For example,

$$CuSO_4.5H_2O \rightleftharpoons CuSO_4$$

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

$$PCl_{5(s)} \rightleftharpoons PCl_{3(s)} + Cl_{2(g)}$$

$$H_{2(g)} + Cl_{2(g)} \rightleftharpoons 2HCl_{(g)}$$

An **irreversible reaction** is that which proceeds in one direction only such that entire reactants get converted into products at given set of conditions. For example,

$$HCl_{(aq)} + KOH_{(aq)} \rightarrow KCl_{(aq)} + H_2O_{(l)}$$

$$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$

$$AgNO_{3(aq)} + NaCl_{(aq)} \rightarrow AgCl\downarrow_{(s)} + NaNO_{3(aq)}$$

$$Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$$

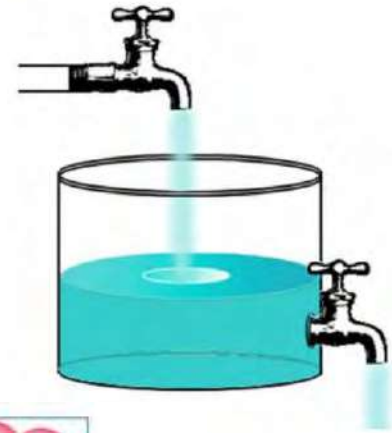
Most of the natural processes are irreversible in nature. Irreversible reaction proceeds faster than that of reversible. So that reversible reaction takes infinite time for the completion whereas irreversible reaction completes in finite time.

Test yourself:

Identify whether the given process is reversible or irreversible.

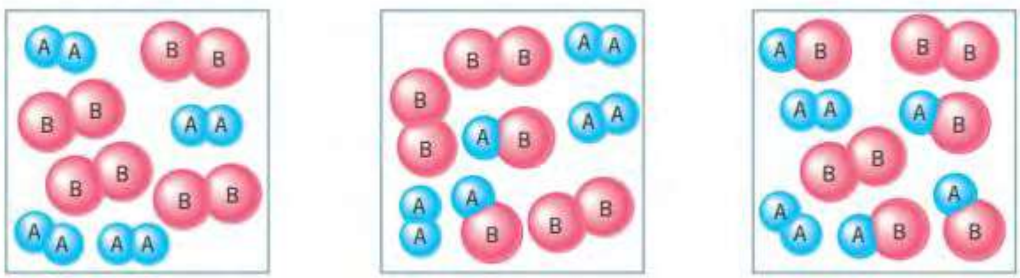
Process	Reversible	Irreversible
Evaporation of water		
Formation of ammonia		
Dissolution of salt		
Burning of petrol		

8.2 Chemical Equilibrium



Activity:

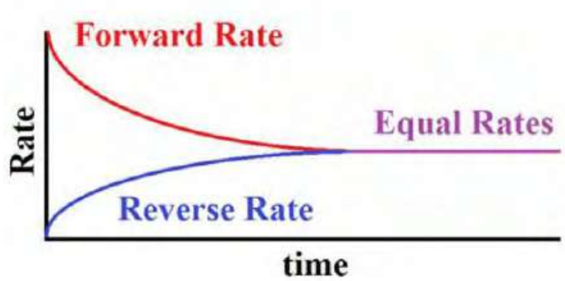
What happens to the amount of water in the given container after 1 hour if both the taps are opened? Why?

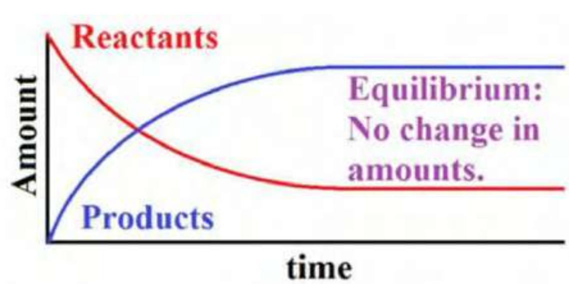


A **B** **C**

Let us consider a reversible reaction between $\mathbf{A_2}$ and $\mathbf{B_2}$ to form **AB**. Initially, there is no formation of product as in picture A. Concentration of reactants is maximum so rate of forward reaction is more. Slowly reactants start to change into products AB. Rate of backward reaction also increases as in figure **B.** After some time, the rate of forward reaction and rate of backward reaction are equal as shown in figure **C**. This is a state of chemical equilibrium.

$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$

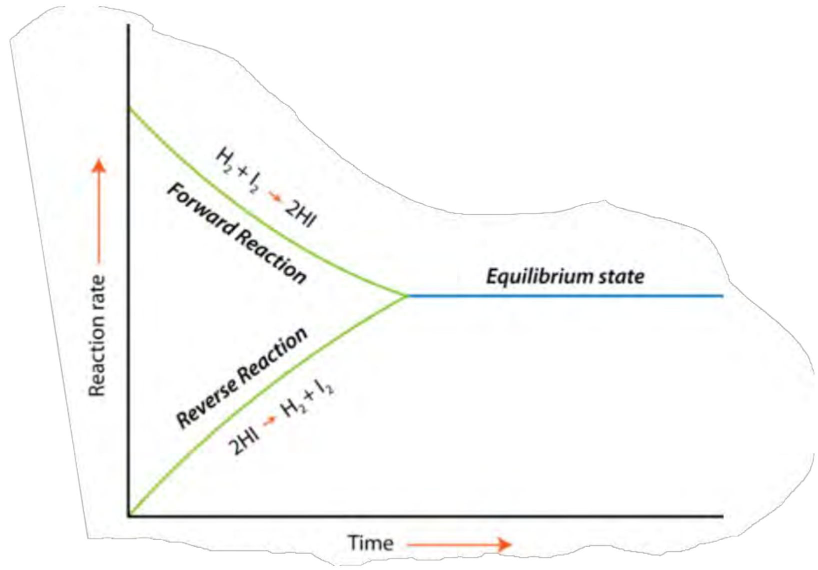




At equilibrium, the reaction continuously proceeds in both directions but there is no net change in the concentration of reactants and products as shown in the graph. Therefore, it is also known as **dynamic chemical equilibrium**. Equilibrium is attained only in a reversible reaction occurring in a closed vessel.

Chemical equilibrium is a state of chemical reaction in which the rate of forward reaction is equal to the rate of backward reaction. For example, in the following reversible reaction, the rate of formation of HI is equal to the rate of decomposition of HI at equilibrium condition.

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$



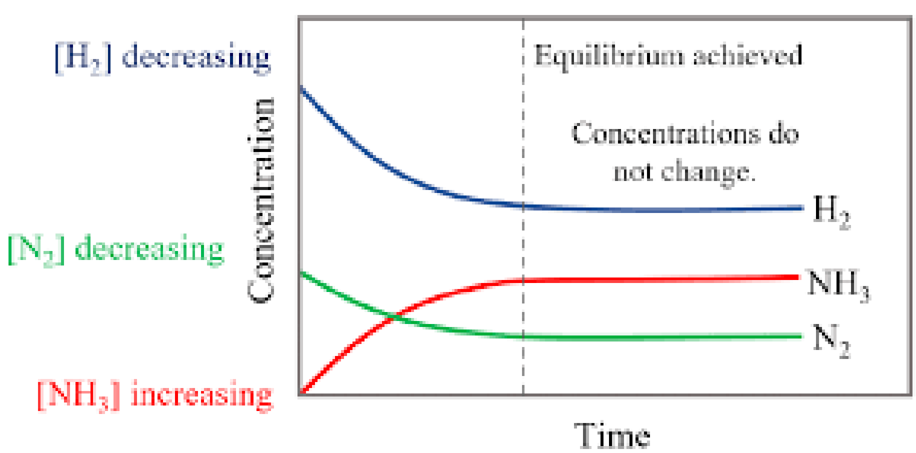
Characteristics of chemical equilibrium

The important features of chemical equilibrium are:

i. It is established in a closed container.
ii. It is dynamic in nature. It means reactants and products are reacting continuously.
iii. The concentrations of reactants and products remain constant.
iv. Rate of forward reaction is equal to the rate of backward reaction.

Test yourself!

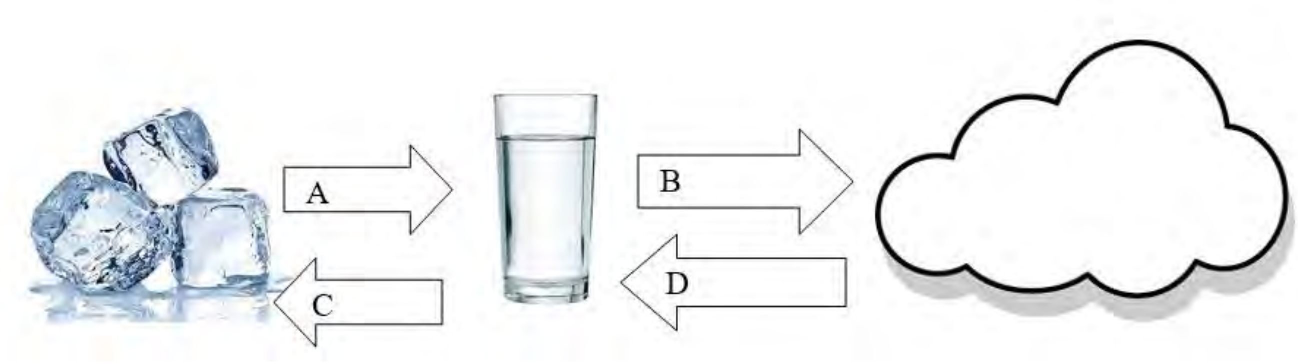
Why is concentration of ammonia zero at zero time? Which two chemicals have equal concentration at a point, during the reaction? Write the equation at equilibrium for the given reaction.



8.3: Equilibrium in a physical process

Activity:

a. Name the process A, B, C and D in the given figure.
b. Identify the process having equal rate at equilibrium.



A book on a table, water enclosed in a container, saturated solution, etc. are some real-life examples of physical equilibrium. In this process the chemical composition of the substance remains the same.

Types of physical equilibrium

There are five types of physical equilibrium. They are:

a. Solid-liquid equilibrium

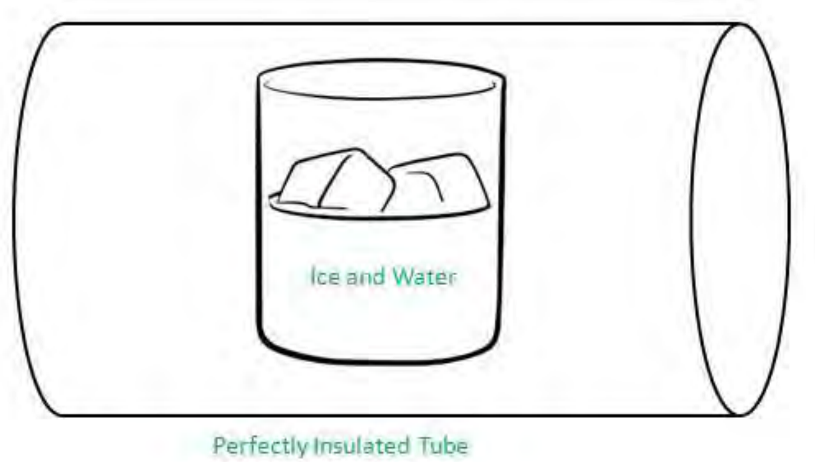


FIGURE: ILLUSTRATION OF SOLID-LIQUID EQUILIBRIUM

$H_2O\ (s) \rightleftharpoons H_2O\ (l)$

(ice) **(water)**

This type of equilibrium is observed by keeping ice and water at 273 K and one atmospheric pressure in a thermos flask. Though some ice molecules escape into water and some water molecules collide to form ice there is no change in mass of ice and water. Thus, the system of ice and water is said to be in solid liquid dynamic equilibrium.

Hence, **the equilibrium existing between the solid and liquid phase of a same substance at freezing or melting point is called solid-liquid equilibrium.**

b. Liquid-vapor equilibrium

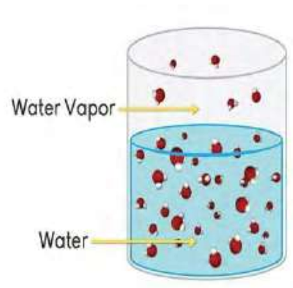


FIGURE: ILLUSTRATION OF LIQUID-VAPOR EQUILIBRIUM

H_2O (s) $\rightleftharpoons H_2O$ (g)
(water) (vapor)

This type of equilibrium is observed by heating distilled water. It is seen that the water level starts decreasing because water is converting into vapor. At the same time some of water vapors continuously collide with each other resulting in the formation of water. After a certain time, we will observe that water level in the vessel becomes constant which implies that, the rate of evaporation (liquid to vapor) and the rate of condensation (vapor to liquid) become equal. Thus, the system of water and vapor is said to be in liquid-vapor dynamic equilibrium.

Hence, **the equilibrium existing between the vapor and liquid phase of a same substance at boiling point is called solid-liquid equilibrium.**

Do you know?

The pressure exerted by the vapors of the liquid at equilibrium is called vapor pressure of the liquid. When vapor pressure of a liquid becomes equal to the atmospheric pressure, then the boiling of the liquid occurs. A liquid with high vapor pressure has lower boiling point and vice versa.

c. Solid-vapor equilibrium

$$I_2(s) \rightleftharpoons I_2(g)$$

Figure: Illustration of solid-vapour equilibrium

Have you observed what happens when a solid iodine piece is placed in a closed jar at room temperature? You will see that the jar will be filled with violet vapors. After a certain time, there will be no change in the color intensity. It is due to the equilibrium established between solid iodine and vapor iodine.

Hence, **the equilibrium existing between the solid and vapor phase of a same substance is called solid-vapor equilibrium.** At this equilibrium, the rate of sublimation is equal to the rate of deposition.

d. Solid-solution equilibrium

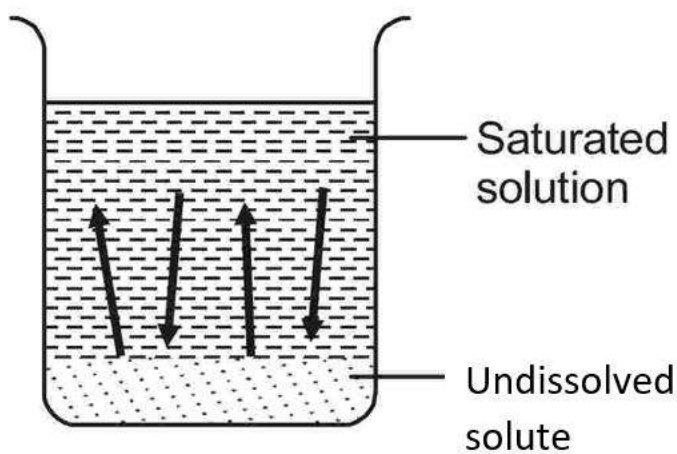


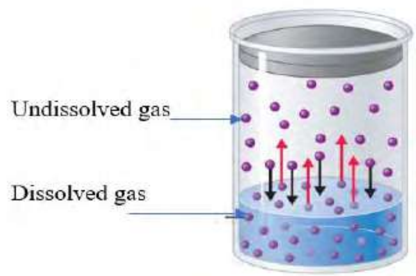
FIGURE: ILLUSTRATION OF SOLID-SOLUTION EQUILIBRIUM

$$\underset{\textbf{(solid)}}{\text{Sugar}} \rightleftharpoons \underset{\textbf{(solution)}}{\text{Sugar}}$$

Have you ever tried dissolving sugar in water? Can you dissolve any amount of sugar? After adding a certain quantity of sugar to water, the added sugar gets deposited. At this point, the solution becomes saturated and no more sugar can be added at a given temperature. Thus, in a saturated solution a dynamic equilibrium exists between solute molecules in solid state and the dissolved solute in solution. In other words, we can say that the rate of dissolution and the rate of deposition (precipitation) become equal.

Hence, **the equilibrium existing between the solid and solution phase of a same substance in a saturated solution at a given temperature is called solid-solution equilibrium.**

e. Gas-solution equilibrium



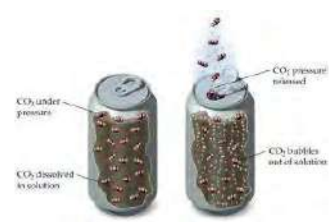


FIGURE: ILLUSTRATION OF GAS-SOLUTION EQUILIBRIUM

$$CO_2\ (gas) \rightleftharpoons CO_2\ (in\ solution)$$

Why do we see fizz and hear a sound when we open a soda bottle? This happens because some of the CO_2 dissolved in it fizzes out rapidly due to the difference in solubility of CO_2 at different pressures. Thus, there establish an equilibrium between the CO_2 molecules in the gaseous state and CO_2 dissolved in liquid under a pressure. At this equilibrium, the rate of dissolution of gas is equal to the rate of escaping of gas from its solution.

Hence, **the equilibrium existing between the gas and solution phase of a same substance under a pressure is called gas-solution equilibrium.**

Why is Carbon dioxide gas added to cold drinks?

The main reason is to create the fizz or bubbles that we enjoy in so many drinks.CO_2 also enhances the flavor and acts as a good preservative by preventing the growth of bacteria.

8.4: Law of Mass Action:

Cato Maximilian Guldberg (11 August 1836 – 14 January 1902) was a Norwegian mathematician and chemist at Royal Frederick University, Oslo, Norway. Guldberg is best known as a pioneer in physical chemistry.

Peter Waage (29 June 1833 – 13 January 1900) was a Norwegian chemist and professor of chemistry at the University of Christiania, Oslo, Norway.

Between 1864 and 1879, Peter waage along with his brother-in-law Cato Maximilian Guldberg,co-discovered and developed the quantitative relationship between rate of reaction and concentration of reactants which is known as **law of mass action**.

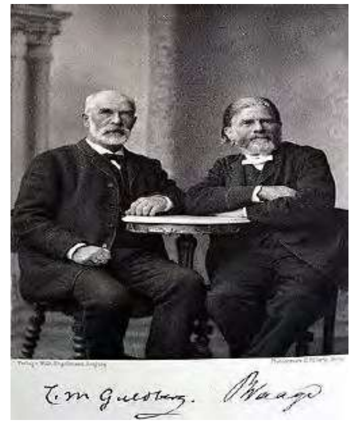
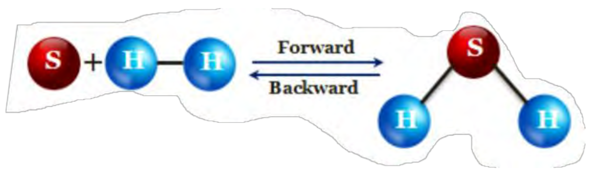


FIGURE: C.M. GULDBERG (LEFT) & P. WAAGE (RIGHT)



What happens to the rate of forward reaction if concentration of Hydrogen and Sulphur increases? Why?

When the number of vehicles increases in the road, the chances of collision increases. In the same way the rate of reaction increases with increasing the concentration of the reacting substances.
The relationship between the rate of reaction and concentration of reactants was given by Norwegian scientists Cato M. Guldberg and Peter Waagein the name of **law of mass action.**

Let us consider the following reaction,
aA + bB → Products
According to law of mass action,
Rate of chemical reaction $\propto [A]^a [B]^b$

$$r = K[A]^a [B]^b$$

Where, r= rate of reaction
[A] = molar concentration of reactant A
[B] = molar concentration of reactant B and
K = proportionality constant called rate constant or velocity constant.
a and b are the stoichiometric coefficients.

Law of mass action states, " **The rate at which a substance reacts is directly proportional to its molar concentration (active mass) and the rate of chemical reaction is directly proportional to the products of molar concentrations of reactants where each reactant term is raised to the power equal to stoichiometric coefficient seen in a balanced chemical equation".**

Do you know?

In chemistry, Active mass and concentration are used to describe the amount of substance present, but they are different concepts. Active mass refers to the amount of a substance that is available to react in a chemical reaction. Concentration, on the other hand, refers to the amount of a substance present in a given volume of solution or mixture.

Expression for equilibrium constant in terms of molar concentration:

Let us consider following reversible reaction at equilibrium,

$$aA + bB \rightleftharpoons cC + dD$$

According to law of mass action,

Rate of chemical reaction in forward direction $(r_f) \propto [A]^a [B]^b$

$$\text{or, } r_f = K_f[A]^a [B]^b \text{ ....(i)}$$

And,

Rate of chemical reaction in backward direction $(r_b) \propto [C]^c [D]^d$

$$\text{or, } r_b = K_b[C]^c [D]^d \text{ ....(i)}$$

At equilibrium,

Rate of forward reaction = Rate of backward reaction

i.e, $r_f = r_b$

or, $K_f[A]^a [B]^b = K_b[C]^c [D]^d$

or, $\frac{Kf}{Kb} = \frac{[C]c\,[D]d}{[A]a\,[B]b}$

or, $K_c = \frac{[C]c\,[D]d}{[A]a\,[B]b}$ ....(iii)

Where, $K_c = \frac{Kf}{Kb}$, a constant called **equilibrium constant**. It is defined as the ratio of forward rate constant to the backward rate constant. The subscript **c** indicates concentration.

Remember that Concentration of perfect solid and liquid =1

Expression for equilibrium constant in terms of partial pressure

In case of gaseous reaction, it is convenient to express the concentration in terms of partial pressure rather than molar concentration. The partial pressure of a gaseous chemical species is directly proportional to its molar concentration.

$PV = nRT$

or, $P = \frac{n}{V} RT$

where,

P= partial pressure

V= volume (in L)

n= no. of moles
T= temperature (in Kelvin scale)
R= Universal gas constant

Since, concentration (C) = $\frac{Number\ of\ moles\ of\ solute\ (n)}{Volumeinlitre\ (V)}$

$\therefore$ P = CRT

i.e, $P \propto C$ ($\because$ R & T are constant)

$\therefore P_A = [A]RT$
$P_B = [B]RT$
$P_C = [C]RT$
$P_D = [D]RT$

Now let us consider the reversible reaction in case 1 occurring in gaseous phase then the molar concentration term will be replaced by partial pressure and equation (iii) becomes,

$Kp= \frac{[PC]c\,[PD]d}{[PA]a\,[PB]b}$ ....(iv)

Where, P_A, P_B, P_C and P_D are partial pressures of species A, B, C and D respectively.

Test yourself!

Write Kp and Kc for the given reactions

$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H2O(g)$

$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

Relation between K_p and K_c

Let usconsider a reversible reaction.

$aA + bB \rightleftharpoons cC + dD$

On applying law of mass action at equilibrium condition,

The equilibrium constant in terms of molar concentration (K_c) = $\frac{[C]c\,[D]d}{[A]a\,[B]b}$ ....(i)

The equilibrium constant in terms of partial pressure (Kp)= $\frac{[PC]c\,[PD]d}{[PA]a\,[PB]b}$ ....(ii)

Also, from ideal gas equation,

PV= nRT

or, P= $\frac{n}{V}$ RT

where,

P= partial pressure
V= volume (in L)
n= no. of moles
T= temperature (in Kelvin scale)
R= Universal gas constant

Since, Concentration (C) = $\frac{Number\ of\ moles\ of\ solute\ (n)}{Volumeinlitre\ (V)}$

$\therefore$ P = CRT ....(iii)

Then, $P_A = C_ART$, $P_B = C_BRT$, $P_C = C_CRT$, $P_D = C_DRT$

Substituting the values of P_A, P_B, P_C, & P_D in equation (ii) we get,

$[C_C\,RT]^c[C_D\,RT]^d$

$$Kp = \frac{\cdot}{[C_A RT]^a [C_B RT]^b}$$

$$Kp = \frac{[C_C]^c[C_D]^d\ [RT]^c[RT]^d}{[C_A]^a\ [C_B]^b\ [RT]^a[RT]^b}$$

$$Kp = \frac{[C]^c[D]^d\ (RT)^{c+d}}{[A]^a[B]^b\ (RT)^{a+b}}$$

C_A, C_B, C_C and C_D are molar concentrations of A, B, C and D and are replaced by designation [A], [B], [C] and [D] respectively.

On using equation (i) above expression now becomes

$K_p = K_c\ (RT)^{(c+d)\ -\ (a+b)}$

or, $K_p = K_c\ (RT)^{\Delta n}$ ....(iv)

Where, Δn = Number of moles of gaseous products − Number of moles of gaseous reactants

For Example,

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3\ (g)$$

Here, Δn = Number of moles of gaseous products − Number of moles of gaseous reactants

$= 2- (1+3)$

$= -2$

$\therefore ,K_p = K_c\ (RT)^{\Delta n}$

$K_p = K_c\ (RT)^{-2}$

or, $K_p = K_c\ (1/RT)^2$

$K_p < K_c$

Therefore, the relationship between Kp and Kc can be summarized as:

Case−I: If, Δn = 0, then $K_p = K_c$ For example $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Case−II: If, Δn >0, then $K_p > K_c$ For example $PCl_5(s) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Case−II: If, Δn <0, then $K_p < K_c$ For example $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$}

Test Yourself!

How are, K_p and K_c related in the given equations?

i. $FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$

ii. $4NO\ (g) + 6H2O\ (g) \rightleftharpoons 4NH_3(g) + 5O_2(g)$

iii. $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

Characteristics of equilibrium constant

The characteristics of equilibrium constant are-

a. It is experimentally determined and depends upon temperature.
b. It does not depend upon the initial concentration of reactants or products.
c. There is no effect of catalyst on the value of equilibrium constant.
d. The equilibrium constant for backward reaction will be equal to the reciprocal of equilibrium constant for forward reaction.
e. It is the measure of extent of rate of reaction. It means larger the value of equilibrium constant indicates faster the rate of reaction.

Importance of equilibrium constant:

The equilibrium constant helps us to understand whether the reaction tends to have higher concentration of products or reactants at equilibrium. A very large value of equilibrium constant ($K_{eq} >> 1$) indicates higher amount of products than that of reactants. Likewise, the low value of equilibrium constant ($K_{eq} << 1$) indicates that forward reaction does not occur at any significant extent i.e, the equilibrium mostly lies to the left.

Test yourself!

Write the expression for the equilibrium constant (Kc and Kp) for the forward and backward reaction to the given equation.

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

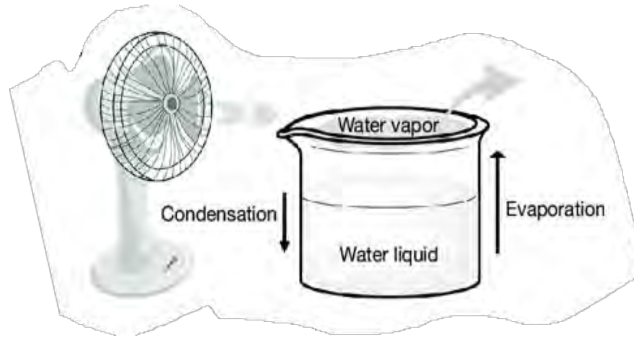
8.5: Le-chatelier's Principle

Activity:

Observe the figure.

a. What happens to this position if only one side is pressed?
b. How can the initial position be restored?

At equilibrium, the rates of condensation and evaporation are equal. However, when a fan blows near the vessel, the equilibrium is disrupted, and the rates of these processes are no longer equal. Identify the process in the figure that increases its rate to counteract the effect of the nearby fan.



What do you mean by disturbing chemical equilibrium? How can you disturb the chemical equilibrium? Well disturbing chemical equilibrium means changing either the rate of forward reaction or rate of backward reaction. We can disturb the chemical equilibrium in following three ways-

a. By changing the concentration of reactants or products.
b. By changing the temperature of the system.
c. By changing the pressure of the system.

This effect of change in equilibrium was first studied by Le-chatlier.

Henry Louis Le Chatelier (8 October 1850 – 17 September 1936) was a French chemist of the late 19th and early 20th centuries. In 1884, he devised Le-chatlier's principle used by chemists and chemical engineers to predict the effect a changing condition has on a system in chemical equilibrium. This principle states, " ***If a chemical system at equilibrium experiences a change in concentration, temperature or total pressure, the equilibrium will shift in order to minimize the effect of change."***

HENRY LOUIS LE CHATLIER

Le-chatlier's principle states that:

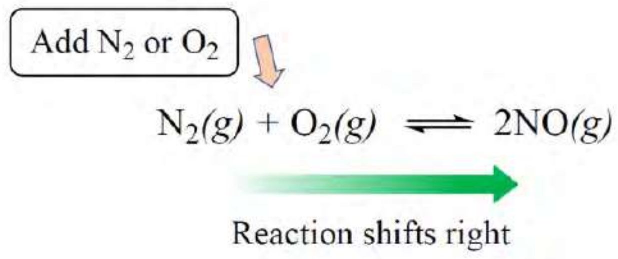
" **When a system in equilibrium is subjected to change in a temperature, concentration or pressure then the equilibrium shifts in the direction which tends to minimize that effect**".

a. Effect of change in concentration on equilibrium:

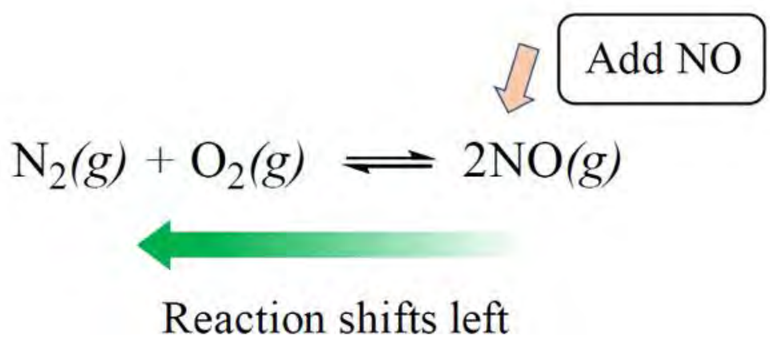
Why is the man removing water from the leaky boat?

Adding a reactant produces more product by consuming some of the reactants so that equilibrium is maintained.

For example:



Likewise, adding a product produces more reactants by consuming some of the product to maintain the equilibrium.



Thus, Le Chatelier's principle implies that increase in concentration of reactants shifts the equilibrium to the right side of the reaction that favors formation of more products, while increase in concentration of products shifts the equilibrium to the left side of the reaction that favors the formation of reactants.

b. Effect of change in temperature on equilibrium

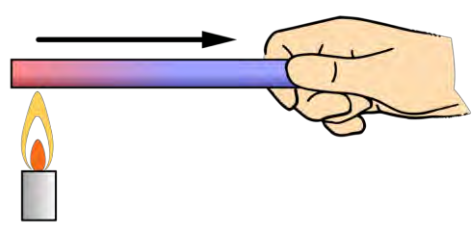


Fig: conduction of heat to maintain the equilibrium

Let us consider an equilibrium reaction which releases heat (exothermic reaction**),**

$$3\,H_2(g) + N_2(g) \rightleftharpoons 2\,NH_3 + \text{heat}$$

Reaction shifts left

Add temperature

Increasing the temperature shifts the equilibrium towards left (backward) so that heat is consumed. Decreasing the temperature shifts the equilibrium towards right(forward) so that heat is released.

$$N_2O_4(g) + \text{heat} \rightleftharpoons 2\,NO_2(g)$$

Reaction shifts left

Decrease temperature

$$N_2O_4(g) + \text{heat} \rightleftharpoons 2\,NO_2(g)$$

Add temperature

Reaction shifts right

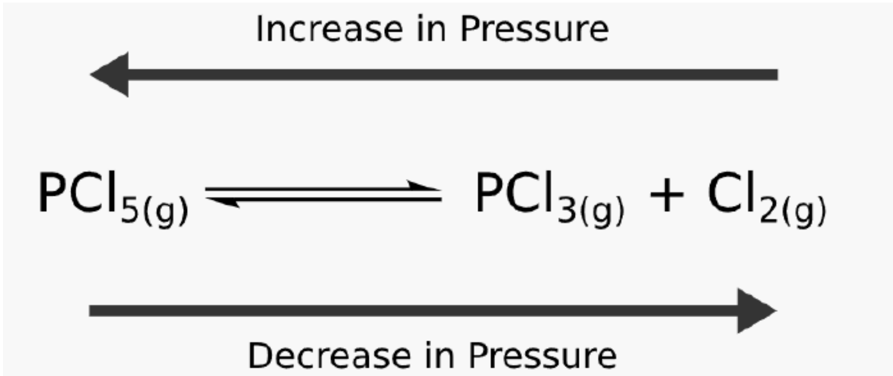
If we consider the case of reaction which proceeds by absorbing heat (endothermic reaction), the effect of temperature is reversed than the above reaction.It means ,when temperature is increased, equilibrium shifts toward right and when temperature is decreased, equilibrium shifts toward left.

Thus, Le Chatelier's principle implies that increase in temperature shifts the equilibrium to that direction where heat is absorbed (i.e, increase in temperature favors endothermic side of reaction) while decrease in temperature shifts the equilibrium where heat is evolved (i.e, decrease in temperature favors exothermic side of reaction).

c. **Effect of change in pressure on equilibrium**

What causes the fizzing phenomenon upon opening bottles of cold drinks?

When we press the balloon filled with air, it changes its shape so that volume is decreased and equilibrium is maintained. Change in pressure has no significant effect in case of solid or liquid phases. However, pressure strongly effects the gas phase as shown below:



Thus, Le Chatelier's principle implies that increase in pressure shifts the equilibrium to the side of the reaction with the fewer number of moles of gas (less volume of gas). In the same way, decrease in pressure shifts the equilibrium to the side of the reaction with the greater number of moles of gas (more volume of gas).

Test yourself!

Birkland-eyed process is commonly used for the synthesis of nitric acid. One of the processes involves the reaction at equilibrium as given.

$$N_{2(g)} + O_{2(g)} + \text{Heat} \underset{\text{Exothermic}}{\overset{\text{Endothermic}}{\rightleftharpoons}} 2NO_{(g)}$$

What will be the effect on the position of equilibrium at

a. high temperature?
b. high pressure?

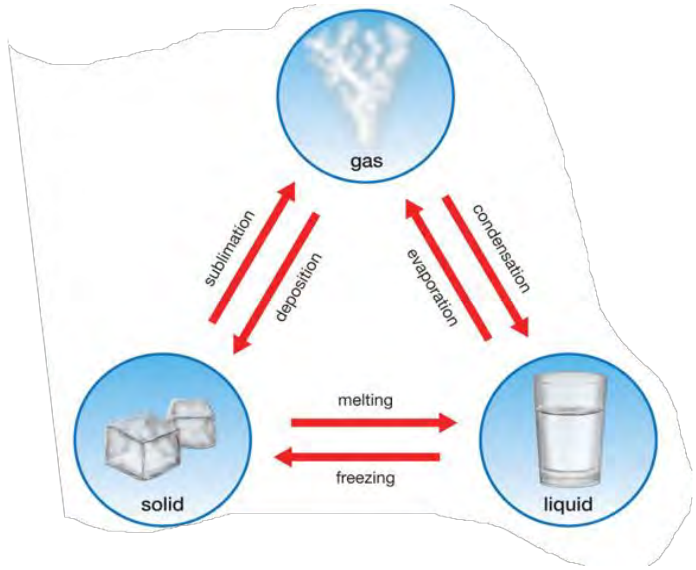
Do you know?

A catalyst in a system that reaches a point of equilibrium increases the rate of both the forward and reverse reactions proportionally but does not change the point of equilibrium. The function of a catalyst is simply to reach the point of equilibrium in less time.

Application of Le-chat liar's principle in physical equilibria

Observe the physical equilibrium among different states. Identify the process:

a. that increases with increasing temperature.
b. that increases with decreasing temperature.



a. Melting of ice

H_2O (ice) $\rightleftharpoons$ H_2O (water)

Melting of ice is an endothermic process. Ice becomes water by absorbing heat.Therefore increasing the temperature increases the rate of melting and more water is formed .

b. Vaporization of water

H_2O (liquid) $\rightleftharpoons$ H_2O (Vapor)

Vaporization of water is an endothermic process. In this phenomena, the rise in temperature will favor vaporization of water. On the other hand, increase in pressure will shift the equilibrium in the direction of less volume i.e. towards water.

Test yourself!

a. What happens to the equilibrium between ice and water when pressure is increased?
b. Why does the boiling point of water increase inside the pressure cooker?

Application of Le-Chatlier's principle in chemical equilibria

a. Synthesis of ammonia by Haber's process

$$\underset{1\text{ vol.}}{N_2\,(g)} + \underset{3\text{ vol.}}{3H_2\,(g)} \rightleftharpoons \underset{2\text{ vol.}}{2NH_3\,(g)} + 22.04\text{ kCal} \quad \text{[exothermic reaction]}$$

Effect of change in temperature

The above reaction is an exothermic reaction in forward direction and endothermic reaction in backward direction. Hence, according to Le Chatelier's principle on decreasing the temperature the equilibrium shifts to right so that heat is released. Therefore, low temperature favors the formation of ammonia.

Effect of change in pressure

In the above reaction it is clear that 4 volumes of total reactants (1 volume of N_2 and 3 volumes of H2) give 2 volumes of product (NH_3). Hence the reaction takes place by decrease in volume. According to Le-chatlier's principle, increase in pressure shifts the equilibrium to the right where volume decreases. Therefore, high pressure favors the formation of ammonia.

Effect of change in concentration

When concentration of N_2 or H_2 is increased then the equilibrium will shift to the right yielding more ammonia. Therefore, high concentration of N_2 or H_2 favors the formation of ammonia.

In conclusion, formation of ammonia is favored by low temperature, high pressure and high concentration of N_2 & H_2.

Do you know?

Without Haber's process at work today, it is estimated that only about 40% of the world's population could be fed.

b. Formation of nitric oxide

.

$$\underset{\text{1 vol.}}{N_2\ (g)} + \underset{\text{1 vol.}}{O_2\ (g)} + 4.32\ \text{kCal} \rightleftharpoons \underset{\text{2 vol.}}{2NO\ (g)}\ \text{[endothermic reaction]}$$

Effect of change in temperature

The above reaction is an endothermic reaction in forward direction and exothermic reaction in backward direction. Hence, according to Le-chatlier's principle, on increasing the temperature the equilibrium shifts to right so that heat is absorbed. Therefore, high temperature favors the formation of nitric oxide in this reaction.

Effect of change in pressure:

In the above reaction it is clear that 2 volumes of total reactants (1 volume of N_2 and 1 volume of O_2) give 2 volumes of product (NO). Hence the reaction takes place without any change in volume i.e, $\Delta n=0$. According to Le-chatlier's principle there is no effect of change in pressure on the equilibrium.

Effect of change in concentration:

When concentration of N_2 or O_2 is increased then the equilibrium will shift in the forward direction. Therefore, high concentration of N_2 or O_2 favors the formation of nitric oxide

Do you know?

A catalyst in a system that reaches a point of equilibrium increases the rate of both the forward and reverse reactions proportionally but does not change the point of equilibrium. The function of a catalyst is simply to reach the point of equilibrium in less time.

Test Yourself!

1. What happens at too low temperature and too high pressure in the formation of ammonia gas by Haber's process?
2. How does increase in temperature and pressure affect the equilibrium of following reactions?

a. $SO_2 (g) + O_2 (g) \rightleftharpoons 2SO_3 (g)\ \Delta H = -46.0$ kCal

b. $PCl_5 (g) \rightleftharpoons PCl_3 (g) + Cl_2(g)$, $\Delta H = 29.52$ kCal

Exercise

A. Multiple Choice Questions

1. To which of the following systems is law of mass action applicable?

a. Irreversible reaction
b. Homogeneous reaction
c. Homogeneous reversible reaction
d. Open system

2. Which of the statements is not true for equilibrium constant?

a. It depends only upon temperature.
b. It is fixed for a reaction at a particular temperature.
c. It can be changed by changing the concentration of reactants.
d. It is independent of the use of the catalyst.

3. Which of the following reactions proceeds only in forward direction?

a. Reversible reaction
b. Rearrangement reaction
c. Irreversible reaction
d. Exothermic reaction

4. What happens when a system attains equilibrium?

a. Rate of forward reaction is faster than backward reaction.
b. Rate of backward reaction is faster than forward reaction.
c. Change in concentration of reactants and products occurs.
d. No change in concentration of reactants and products.

5. For the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ + Heat, which of the following conditions yield maximum ammonia?

a. Low pressure and high temperature
b. High pressure and low temperature
c. Low concentration of N_2 and H_2
d. High pressure and high temperature

6. Which one is correct for the equilibrium constant, for the given reaction?

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g$$

a. $Kc = [CO_2]$ b. $Kp = [CO_2]^2$ c. $Kc = [CaCO_3]/[CaO][CO_2]$ d. $Kc = [CaO]/[CaCO_3]$

7. In the reaction, $X_2(g) + 4Y_2(g) \rightleftharpoons 2XY_4(g)$, $\Delta H < 0$,

8. Which one is favorable condition for product formation?

a. Low temperature and low-pressure
b. High temperature and high pressure
c. Addition of more XY_4
d. Low temperature and high pressure

9. When pressure is applied to the equilibrium, ice $\rightleftharpoons$ water, which of the following will happen?

a. More water will be formed
b. More ice will be formed
c. No change in the system
d. Water will evaporate

10. What do you call a system, when the rate of forward reaction is equal to the rate of backward reaction?

a. Chemical reaction
b. Physical process
c. Chemical constant
d. Chemical equilibrium

11. Choose the correct relation between K_p and K_c.

a. $K_p.K_c = (RT)^{\Delta n}$
b. $K_p = (K_cRT)^{\Delta n}$
c. $K_p = K_c(RT)^{-\Delta n}$
d. $K_p = K_c(RT)^{\Delta n}$

12. Which of the following indicates an equilibrium that strongly favors products?

a. value of K << 1.
b. value of K >> 1.
c. value of K=1
d. value of K=0

13. **For the given system, $N_2(g) + H_2(g) \rightleftharpoons 2NH_3(g)$, what will be the unit of K_c?**

a. atm^{-2} b. $L\ mol^{-2}$ c. $L^2 mol^{-2}$ d. $L\ atm\ mol^{-1}$

14. Consider the equilibrium: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO3(g)$

What is the effect of decreasing the pressure on the position of equilibrium and the value of the equilibrium constant, Kc?

a. The position of equilibrium shifts to the right and Kc increases
b. The position of equilibrium shifts to the left and Kc decreases
c. The position of equilibrium is unchanged and Kc does not change
d. The position of equilibrium shifts to the left and Kc stays the same

B. Short Answer Questions

1. Define the following with an appropriate example.

a. Reversible reaction
b. Irreversible reaction
c. Liquid vapor equilibrium
d. Solid-liquid equilibrium. Gas-solution equilibrium

2. Give reason:

a. Soda water fizz out when the bottle is opened.

b. Active mass of solid is assumed one.

c. A chemical equilibrium is a dynamic equilibrium.

d. Increase in temperature shifts the equilibrium backward for an exothermic reaction.

e. Increase in temperature shifts the equilibrium forward for an endothermic reaction.

3. Equilibrium constant for the chemical reaction at equilibrium is given.

$$K = \frac{(P_{CO})^2 (P_{O_2})}{(P_{CO_2})^2}$$

a. Define equilibrium constant and Write any three characteristics of it.
b. Write the related chemical equation and find the relationship between Kp and Kc for this reaction.

4. Differentiate between:

a. Reversible reaction and irreversible reaction

b. Static and dynamic equilibrium

c. Physical equilibrium and chemical equilibrium

d. Solid-vapor phase equilibrium and liquid- vapor phase equilibrium.

5. What do you mean by reversible reaction? Write an example of a reversible gaseous reaction in which-

i. $K_p > K_c$

ii. $K_p = K_c$

6. Derive the relation between K_p and K_c for an equilibrium reaction.

7. Give an example of:

i. Liquid-vapor equilibrium ii. Solid-solution equilibrium

iiii. Gas- solution equilibrium

iv. Solid-liquid equilibrium v. Solid-vapor equilibrium

8. What do you mean by equilibrium state? "Equilibrium state is dynamic rather than static". Justify the statement.

9. State law of mass action. Derive an expression for equilibrium constant in terms of molar concentration.

10. State law of mass action. Derive an expression for equilibrium constant in terms of partial pressure.

11. State and explain Le-chatlier's principle giving one example.

12. What is the effect of pressure on melting ice? Give your justification on the basis of Le-chatlier's principle.

13. Define equilibrium constant. What information will you get when-

    i. $K_c >> 1$ ii. $K_c << 1$

14. State Le-chatlier's principle. On the basis of this principle, predict the favorable conditions for forward reaction in the following reaction-

$$4NH_3(g) + 5\,O_2\,(g) \rightleftharpoons 4NO_2(g) + 6H_2O\,(g) + \text{Heat}$$

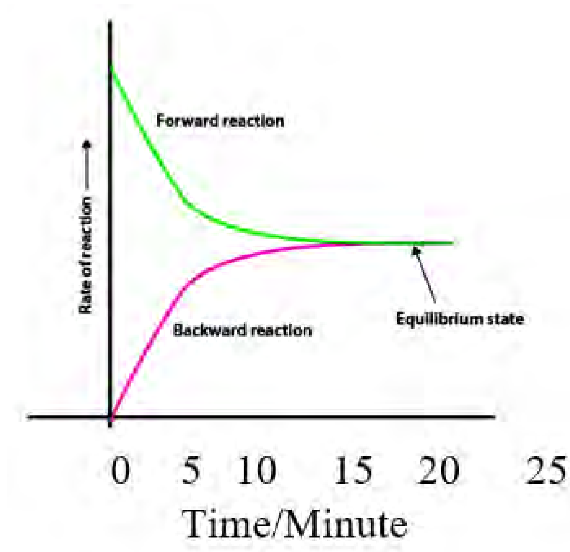
C. Long answer questions

1. Physical equilibrium between solid and liquid is given.

$$\text{Ice} \rightleftharpoons \text{Water} - \text{Heat}$$

   a. Why is this equilibrium called physical equilibrium?
   b. Write the expression for the equilibrium constant.
   c. Name the process that goes toward right and that goes towards left.
   d. What happens to the forward process when temperature is decreased? Why?
   e. What is the application of this process?

2. Rate of reaction versus time for a hypothetical reaction is plotted in the graph.
   a. What is the difference in rate of forward reaction and backward reaction at zero time? Give the reason for such differences.
   b. Find the time at which the rate of forward and backward reaction becomes just equal.
   c. What happens to the rate at 25 minutes? Why?



3. Consider the following reversible reaction,

$$N_2(g) + O_2\,(g) \rightleftharpoons 2NO\,(g) - \text{Heat}$$

   a. What type of reaction is given (exothermic or endothermic) and why?
   b. Write the equilibrium constant in terms of concentration and pressure for this reaction and write their relation.

c. What is the effect of change in temperature on this equilibrium?
d. How do you increase the amount of NO by changing the pressure and why?

4. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ + Heat

a. How does the position of equilibrium shift in the given equilibrium under following conditions. Explain with reason.

i. An inert gas is added at constant volume.
ii. A catalyst is added to the system.
iii. Pressure is increased.

b. Write the expression for the equilibrium constant (kc) for the forward and backward reaction. How are they related?

5. a. Express the equilibrium constant for the following reversible reactions both in terms of concentration and in terms of partial pressure. Write their relationship.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$4NO(g) + 6\,H2O(g) \rightleftharpoons 4NH_3(g) + 5O_2(g)$

b. Which of the given reactions produces more product at high pressure? Why?

6. One of the reactions for the combustion of hydrocarbons is given.

$$C_3H_8(g) + 5O_2(g) \rightleftharpoons 3CO_2(g) + 4H_2O(g)$$

a. Write the expression for the Kp and Kc and write their relationship.
b. Explain the effect of increasing concentration of reactants and increasing pressure on this equilibrium.
c. What is the application of this reaction?
d. What is the value of Kc for forward reaction when pressure is decreased?

7. Nitrogen dioxide is in a gaseous equilibrium forms dinitrogen tetroxide.

Write the balanced equation illustrating this reaction.
Write the expression for Kp and Kc for the given equation.
What is the relationship between these constant?
How does an increase in pressure affect the concentration of nitrogen dioxide at equilibrium?

8. The following equilibrium takes place at high temperatures.
$N_2(g) + 2H_2O(g)$ + heat energy $\rightleftharpoons 2NO(g) + 2H_2(g)$

How will the concentration of NO at equilibrium be affected by these changes?

increasing $[N_2]$

decreasing $[H_2]$

compressing the reaction mixture

decreasing the volume of the reaction container

Project work

Take A4 size paper and draw the column as shown below.

For the following reaction, write how each of the changes will affect the indicated quantity, assuming a container of fixed size. Write "increase", "decrease", or "no change.

$$H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g) \quad \Delta H^o = -103.7kJ$$

Change	**H_2**	**Br_2**	**HBr**	**Value of K**
Add H_2	Increases	Decreases	Increases	Increases
Add Br_2				
Remove H_2				
Remove Br_2				
Add HBr				
Remove HBr				
Increase temperature				
Decrease temperature				
Increase pressure				
Decrease pressure				

UNIT 9: CHEMISTRY OF NON-METALS

9.1 Hydrogen

Activity: 1

Place some pieces of Zn metal in a conical flask and add few ml of dilute HCl, then quickly place the balloon over the mouth of the flask. As soon as the reaction begins, the balloon gets inflated due to the filling of a gas inside it.

a. Which gas is filled inside the ballon?

b. What happens when the mouth of the balloon is tied up with thread and left?

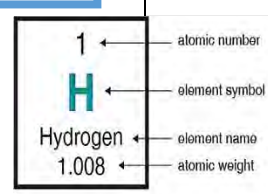
Henry Cavendish (10 October 1731 – 24 February 1810) an English natural philosopher and scientist who was an important experimental and theoretical chemist and physicist. He is noted for his discovery of Hydrogen, which he termed "inflammable air". Antoine Lavoisier later reproduced Cavendish's experiment and gave the element its name.

Henry Cavendish

Antoine Lavoisier (26 August 1743 - 8 May 1794), a prominent French chemist and leading figure in the 18thcentury chemical revolution who developed an experimentally based theory of the chemical reactivity of oxygen and co-authored the modern system for naming chemical substances. He named hydrogen in 1783, after he realized that it makes water when it is burnt in oxygen. Hydrogen means "**water producer**" in Greek.

Antoine Lavoisier

Hydrogen is the first element in the periodic table with atomic number one. It is the simplest and lightest of all chemical elements. It is placed in group 1 as it has certain properties resembling group 1 elements.



Facts

Free hydrogen does not occur in nature. At ordinary temperature hydrogen is colourless, tasteless and odourless non-toxic gas comprising the diatomic molecule dihydrogen (H_2). It is the most abundant element in the universe and 10th most abundant element on the Earth's crust. Water contains about 11.9% by weight of hydrogen. It is a substantial component of more compounds than any other elements. It is a prominent part of petroleum, minerals, fats, oils, carbohydrates and other compounds. It is believed that most of the mass of Jupiter, Saturn, the Sun and other stars is due to hydrogen. The energy of the Sun is due the reaction involving the conversion of hydrogen into helium.

Test yourself!

a. Why is hydrogen kept in group 1?
b. Why is the position of hydrogen still controversial?

9.1.1 Chemistry of Atomic and Nascent Hydrogen

Activity: 2

Observe the following two chemical equations and answer the following question:

$$H\text{-}H \rightarrow H_2 \quad \& \quad H_2 \rightarrow 2H$$

What types of hydrogen are produced as products in the above equation?

Atomic or Active Hydrogen

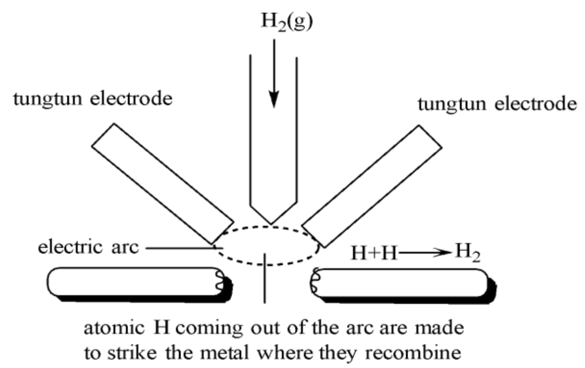


Figure 9.1: Production of atomic hydrogen

When hydrogen gas is passed through an electric arc struck between two tungsten electrodes where it gets dissociated into a new form of hydrogen at temperature of 1000 to 2000 ^{0}C at low pressure. Thus formed hydrogen is very reactive in comparison to hydrogen gas passed.

$$H_2 \xrightarrow{Electric-arc} 2H$$

The hydrogen which is formed by the dissociation of molecular hydrogen is called atomic hydrogen. Atomic hydrogen can be isolated. Since it has a very short half-life period of 0.3 seconds, it quickly reverts back to form molecular hydrogen with the release of a large amount of heat energy.

$$H + H \longrightarrow H_2 + \text{Heat} \quad [\Delta H = -436\ kJmol^{-1}]$$

The above liberated energy can be used to generate a temperature of 4000K, which is ideal for welding and cutting metals. For this reason, atomic hydrogen is used as an atomic hydrogen flame to recombine the metals. Atomic hydrogen is a stronger reducing agent than nascent hydrogen. It reduces oxides, sulphides and chlorides of some metals including Fe, Cu, Ag and Hg.

$FeCl_3 + 3H \rightarrow Fe + 3HCl$

$CuO + 2H \rightarrow Cu + H_2O$

$AgCl + H \rightarrow Ag + HCl$

$CdS + 2H \rightarrow Cd + H_2S$

FIGURE 9.2: ATOMIC

Uses of atomic hydrogen

a. It is used as a reducing agent.
b. It is used for welding metals in the form of atomic hydrogen flame.

Test Yourself!

What do you mean by atomic hydrogen?

Nascent Hydrogen

Activity: 3

Observe the following two chemical equations and answer the following questions:

$$Zn + \text{dil. } H_2SO_4 \rightarrow ZnSO_4 + 2[H]$$

$$Fe + \text{dil. } 2\,HCl \rightarrow FeCl_2 + 2[H]$$

What type of hydrogen is produced in above reaction?
How is this hydrogen different from atomic hydrogen?

Nascent hydrogen is another form of hydrogen that is produced at the time of reaction at ordinary temperature. The meaning of the word '**nascent**' is newly born. Due to its transient existence, it cannot be isolated. It is denoted by [H].

Nascent hydrogen is formed in *situ* during chemical reactions as shown below:

In chemistry, *situ* means, "in the reaction mixture". There are various situations where chemical intermediates are synthesized in *situ*. This may be done because the species is unstable or cannot be isolated.

Activity 4

Take two test tubes marked A and B, each containing acidified $KMnO_4$ solution. Add some Zn pieces in test tube A and leave it for a few minutes. Supply H_2 gas in another test tube marked B and hold it for some time.
a. In which test tube does the colour of the solution disappear, and why?
b. What could be the possible reason for the disappearance of pink colour of $KMnO_4$?

Thus nascent hydrogen is very reactive and is a powerful reducing agent in comparison to molecular hydrogen. It reduces acidified potassium permanganate solution, acidified potassium dichromate solution and ferric chloride solution.

$$\underset{\text{(ink)}}{2KMnO_4} + 3H_2SO_4 + 10[H]\ Zn + H_2SO_4 \rightarrow K_2SO_4 + \underset{\text{(colorless)}}{2MnSO_4} + 8H_2O$$

$$\underset{\text{(orange)}}{2K_2Cr_2O_7} + 4H_2SO_4 + 6[H]\ Zn + H_2SO_4 \rightarrow K_2SO_4 + \underset{\text{(green)}}{Cr_2(SO_4)_3} + 7H_2O$$

$$\underset{\text{(light yellow)}}{FeCl_3} + [H]\ Zn + H_2SO_4 \rightarrow \underset{\text{(light green or almost colorless)}}{FeCl_2} + HCl$$

If molecular hydrogen (H_2 gas) is passed through above acidified solutions, then the colour of the acidified solution does not change which clearly indicates that molecular hydrogen is unable to reduce above chemical species.

Uses of nascent hydrogen

a. It is used as a reducing agent.
b. It is used to reduce elements or compounds that cannot be easily reduced by normal hydrogen.

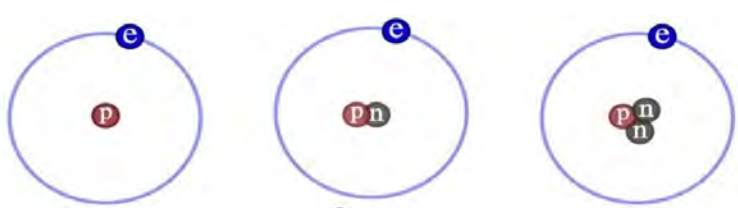
Test yourself!

Write any three differences between atomic and nascent hydrogen.

9.1.2: Isotopes of Hydrogen

Activity: 5

Observe the structure of hydrogen given and complete the table



^1_1H ^2_1H ^3_1H

Symbol	Number of protons	Number of neutrons	Number of electrons

All three atoms in the above activity are hydrogen. Though they are atoms of the same element they differ in their mass number. These atoms with difference in mass number of the same element are isotopes. Thus, isotopes are the atoms of the same element having the same atomic number but different mass number. The difference in mass number is due to the different number of neutrons present in the nucleus of an atom. Hydrogen has three isotopes named as protium, deuterium and tritium. These are commonly called ordinary hydrogen, heavy hydrogen and radioactive hydrogen respectively.

Table-1: Isotopes of hydrogen

Name of the Isotope	Protium	Deuterium	Tritium
Symbol	${}_1H^1$ or H	${}_1H^2$ or D	${}_1H^3$ or T
Atomic No./ No. of electron	1	1	1
Mass number	1	2	3
Atomic mass(amu)	1.008	2.014	3.016
Electronic configuration	$1s^1$	$1s^1$	$1s^1$
Molecular formula	H_2	D_2	T_2
Stability	stable	stable	radioactive

The isotopes of hydrogen differ in their physical properties due to different mass numbers but they all show similar chemical properties as they all have the same electronic configuration.

Facts

It has been found naturally occurring hydrogen consists of 99.986% of protium, 0.014% of deuterium and 7×10^{-16} % of tritium. Hence, properties of hydrogen are the properties of protium (normal hydrogen).

Uses of different isotopes of hydrogen

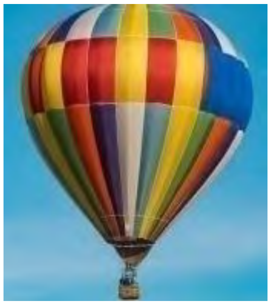


FIGURE 9.3: HYDROGEN BALLOON

FIGURE 9.4: HYDROGEN BOMB

FIGURE 9.5: TRITIUM WATCH

Uses of protium

a. It is used in filling toy balloons.
b. It is used in hydrogenation of vegetable oils to prepare vanaspati ghee.
c. It is used in the manufacture of ammonia gas by Haber's process.
d. It is used as a rocket fuel.

Uses of deuterium

a. Deuterium and its compounds are used as a tracer in the study of reaction mechanism.
b. It is used in preparation of heavy water, D_2O, which is used as a moderator in nuclear reactors to slow down the speed of fast moving neutrons.
c. Compounds of deuterium are used as a solvent in proton NMR spectroscopy.
d. It is used in preparation of hydrogen bombs.

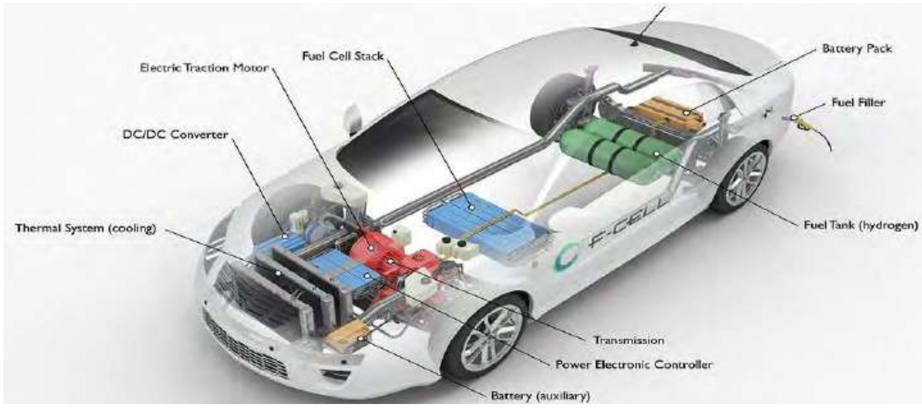
Uses of tritium

a. It is used in producing energy via nuclear fusion reactions.
b. It is used as a radioactive tracer in chemistry, medicine and biology.
c. It is used to make thermonuclear devices and for research into nuclear fusion reactions as a means of energy production.
d. It is used as the energy source in radio luminescent lights for watches, gun sights and other instruments.

Test Yourself!

a. What is meant by isotope effect?
b. Which isotope of hydrogen is radioactive? Why?

9.1.3: Application of Hydrogen as a Fuel



Activity: 6

Observe the vehicle given:

a. Which substance is used as a fuel?

b. What are the advantages and disadvantages of using this fuel?

Hydrogen can be used as an alternative of carbon based fuel for internal combustion engines because of its high inflammability, pollution-less combustion and high enthalpy of combustion. It has already been experimentally used in commercial fuel cell vehicles. The first hydrogen fuel cell bus was demonstrated by Ballard power station in Canada in 1993. It may ultimately be used as fuel for trains, trucks, ships and planes etc. Scientists are trying to completely shift fossil fuel-based economy to the proposed hydrogen-based economy as it is zero emission fuel.

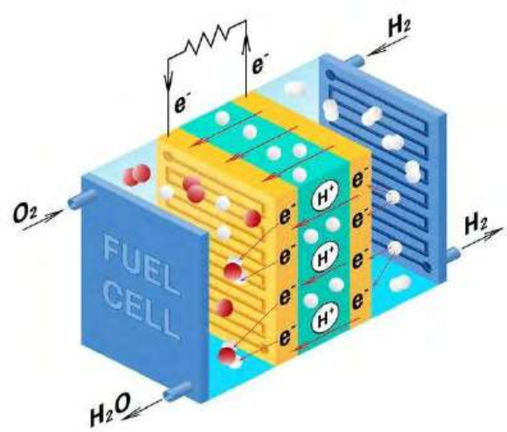


Figure 9.6: Hydrogen fuel cell

Advantages of hydrogen as a fuel

a. Hydrogen fuel cells are very efficient.

b. Unlike petrol and diesel, hydrogen on combustion does not produce greenhouse gas.

c. Hydrogen exists in an unlimited supply.

d. Hydrogen could be an alternative to fossil fuel because it is non-conventional, renewable and pollution free fuel.
e. Hydrogen burning with oxygen produces water as a by-product.
f. Hydrogen fuel is low cost production since water is the richest source of hydrogen which is easily available.

Disadvantages of hydrogen as a fuel

a. Production, storage and transportation of pure hydrogen is difficult.
b. Hydrogen is very flammable, explosive and unsafe.
c. Hydrogen fuel cells are very expensive because materials used in fuel cells like platinum are very expensive.

Test Yourself!

Why hydrogen is called green energy?

9.1.4: Heavy water and its applications

Harold Clayton Urey (April 29, 1893 – January 5, 1981) was an American physical chemist whose pioneering work on isotopes earned him the Nobel Prize in Chemistry in 1934 for the discovery of deuterium. He showed that ordinary water contains fraction of heavy water (1 part out of 6000 parts). He played a significant role in the development of the atom bomb, as well as contributing to theories on the development of organic life from non-living matter.

Harold Clayton Urey

Activity:7

Observe the water given in the glass:

a. Identify the isotopes of hydrogen in each water.
b. Calculate the molecular mass of each and find which one is heavier in mass.

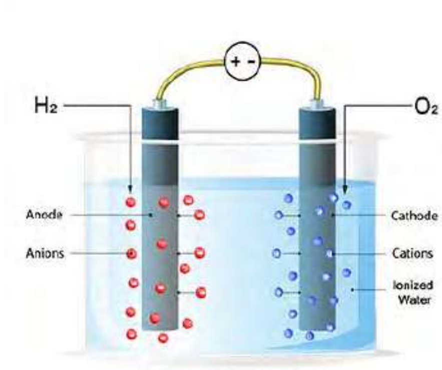


Figure 9.7: Electrolysis of normal water

Heavy water is prepared by prolonged electrolysis of water. Its molecular formula is D_2O and molecular mass is 20 amu which is comparatively heavier than molecular mass of ordinary water (18 amu). So it is called heavy water. About 29000 litres of water must be electrolyzed to obtain one litre of D_2O which is 99% pure. It is colorless, odorless and tasteless just like ordinary water. It shows similar properties to that of ordinary water. The only difference is that D_2O reacts slowly in comparison to H_2O.

Applications of heavy water

Figure 9.8: Nuclear power plant

Some major applications of heavy water are:

a. It is used as coolant and moderator in nuclear reactors.
b. It is used as a tracer compound in determining the reaction mechanism.
c. It is used in the production of heavy hydrogen, D_2.
d. It helps in carrying nuclear fission reactions in a controlled manner by slowing down the speed of fast-moving neutrons.

Test Yourself!

Why is heavy water used as a coolant in nuclear reactors?

Project work

Take a chart paper and draw the structure and mention the applications of:

a. Different isotopes of hydrogen.
b. Fuel cell.

Exercise

A. Multiple choice questions

1. Which of the following is a radioactive form of hydrogen?

a. $_1H^3$ b. $_1H^4$ c. $_1H^2$ d. $_1H^1$

2. Which of the following particles is emitted by tritium?

a. X-ray b. β particle c. α particle d. Neutron

3. How many neutrons are there in protium?

a. 0 b. 1 c. 2 d. 3

4. Which one is the molecular formula of heavy water?

a. H_2O b. D_2O c. T_3O d. H_2O_2

5. Which of the following is the use of heavy water?

a. Filling balloons b. Oxidizing agent

c. Calculating heat content of reaction d. studying reaction mechanism

6. Which of the following is true for hydrogen fuel?

a. Alternative of petroleum b. Highly efficient fuel

c. Pollution free fuel d. All of them

7. When nascent hydrogen comes in contact with acidified $KMnO_4$ solution. Which of the following properties does $KMnO_4$ exhibit?

a. Oxidizing b. Reducing c. Acidic d. Basic

8. An atom X contains one proton and three neutrons. What is the electronic configuration of this element?

a. $1s^2$ b. $1s^1$ c. $1s^2\ 2s^1$ d. $1s^2\ 2s^2$

9. Which of the following is true for nascent hydrogen?

a. It is less reactive than molecular hydrogen.

b. It is more reactive than atomic hydrogen.

c. It cannot be isolated.

d. It can be isolated.

10. Which of the following statements is true for isotopes?

a. They are the different elements with the same mass number.

b. They are the atoms of different elements.

c. They are the elements with different chemical properties.

d. They are the elements with different neutron numbers.

B. Short answer questions

1. **Hydrogen exists in three isotopic forms.**
   a. Define isotope.
   b. What are the isotopes of hydrogen?
   c. List one use of each isotope of hydrogen.
   d. Which isotope of hydrogen is radioactive?

2. **Differentiate between:**
   a. Deuterium and tritium
   b. Nascent hydrogen and atomic hydrogen
   c. Ordinary water and heavy water

3. **'Hydrogen fuel can serve as an alternative to petroleum'**. Justify the statement.

4. What is heavy water? Write any four applications of heavy water.

5. What is nascent hydrogen? Give any three chemical reactions to show that nascent hydrogen is a powerful reducing agent than molecular hydrogen.

6. Mention an important use of each of the following-
a. Atomic hydrogen b. Heavy hydrogen c. Nascent hydrogen d. Heavy water e. Tritium

7. What do you mean by isotopes? Give the main characteristics of isotopes.

Long answer questions

8. H_2, H and [H] are the two different forms of hydrogen. Discuss their properties on following headings:
   a. Production
   b. Reactivity
   c. Stability
   d. Reducing power

9. Some cars use hydrogen-oxygen fuel cells as an alternative source of energy.
   a. Discuss the major advantages of this fuel cell over other cells.
   b. What are the limitations of this fuel?

10. Complete the information regar ding isotopes of hydrogen listed in the given table:

Isotope of hydrogen	No. of neutron	Atomic mass	Uses
Protium			
	2		
		2	

11. Match the items in column I with relevant items in column II.

Column I	Column II
Heavy water is used in	Di-hydrogen
Atomic hydrogen	acidified $KMnO_4$
Electrolysis of water produces	nuclear reactor
Nascent hydrogen reduces	Recombine on metal surface to produce high temperature.
Zn and dil. H_2SO_4 react to produce	hydrogen and oxygen

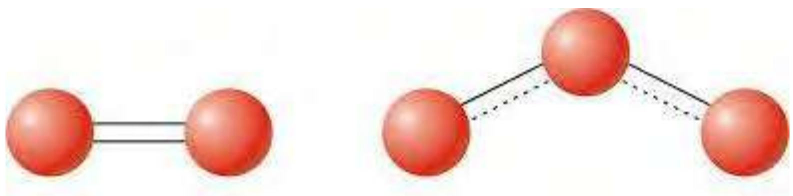
12. Atomic hydrogen combines with almost all elements but molecular hydrogen does not. Explain.

13. How do normal water and heavy water differ? Write their applications.

9.2.1: Allotropes of Oxygen

Activity: 8

Observe the structure of two different molecules containing different numbers of oxygen atoms:



a. What is the molecular formula of these molecules? Calculate the molecular mass of each molecules
b. They both have similar chemical properties. Why?

In the above two forms of carbon, they have the same chemical properties but have different physical properties. Diamond is the hardest substance whereas graphite is soft in nature. These two forms are called allotropes. The word **allotropy** is derived from Greek words, '*allos*' means other and '*tropos*' means forms. Allotropy is defined as the existence of an element in two or more forms which are physically different from each other but chemically identical. As like the carbon given in the above example, other elements like oxygen show the allotropy.

Table: 9.2 List of some elements with their allotropes

Oxygen	i) Dioxygen (O_2) ii) Ozone (O_3) iii) Tetraoxygen (O_4) iv) Octaoxygen (O_8)

Allotropes of oxygen

Among several known allotropes of oxygen, the two most familiar allotropes of oxygen are dioxygen (O_2) and trioxygen (O_3). Dioxygen is commonly known as oxygen or molecular oxygen and trioxygen is commonly known as ozone.
The physical properties of O2 and O3 are given below:

Oxygen (O_2)

i) It is colorless and odorless gas.

ii) It is chemically less reactive than O_3.

iii) It is a weak oxidizing agent.

iv) It cannot absorb UV rays.

v) It is a more stable molecule.

vi) It liquefies at -183^0C.

Oxygen (O_3)

i) It is pale blue gas having a pungent smell.
ii) It is chemically more reactive than O_2.
iii) It is a strong oxidizing agent.
iv) It absorbs UV rays.
v) It is a less stable molecule.
vi) It liquefies at -112 ^{0}C.

9.2.2: Oxygen

Joseph Priestley (24 March 1733 – 6 February 1804) was an English chemist, natural philosopher, separatist theologian, grammarian, multi-subject educator, and liberal political theorist. He published over 150 works, and conducted experiments in several areas of science.

Priestley is credited with his independent discovery of oxygen by the thermal decomposition of mercuric oxide, having isolated it in 1774.

Joseph Priestley

Element	Oxygen
Symbol	O
Atomic Number	8
Atomic Weight	16 amu
Molecular formula	O_2
Electronic configuration	$1s^2\ 2s^2\ 2p^4$
Group	16
Period	2^{nd}
Block	s-block
Isotopes	${}_8O^{16}$ (99.76%), ${}_8O^{17}$ (0.037%) & ${}_8O^{18}$(0.204%)

Oxygen is the most abundant element in the Earth's crust. It constitutes about 21% by volume of the atmosphere. In combined state it constitutes about 88.8% by mass of water and 46.5% by mass of the earth's crust.

Oxides

Activity: 9

Take a small piece of Mg ribbon and burn it over Bunsen flame.
You will notice that Mg burns in air producing a bright white flash.
What is the molecular formula of the white substance thus formed?
Write the chemical equation of the reaction?

When carbon is burnt in air carbon dioxide is formed. Likewise, potassium, sodium, phosphorous, zinc, etc. form compounds with oxygen. These compounds are oxides. Thus, formed binary compounds of oxygen with any other element are called oxides. For example, Na_2O, K_2O, SO_2, P_2O_5, ZnO, SiO_2, Al_2O_3 H_2O_2, Cl_2O_7, etc.

Types of Oxides

Activity: 10

MgO, Al_2O_3, SO_2, CO
Some oxides are given above.
a. Which of the above oxides forms a basic solution when dissolved in water?
b. Which of them forms an acidic solution when dissolved in water?

1. Classification of oxides on the basis of chemical behaviour

On the basis of chemical behaviour like acidic and basic properties, oxides are of four types. They are:

(i) Acidic Oxides

Oxides which produce acids when dissolved in water are called acidic oxides. Acidic oxides when treated with base produce salt and water. Generally, these are the oxides of non-metals. Such oxides are also called anhydrides of corresponding acids. For example, CO_2, SO_2, SO_3, NO_2, N_2O_3, N_2O_5, P_2O_5, SiO_2, ClO_2, CrO_3, V_2O_5, etc.

Formation of acidic oxide-

Non-metal + O_2 → Non-metallic oxide (acidic oxide)

$C + O_2 \rightarrow CO_2$

$S + O_2 \rightarrow SO_2$ etc.

Reaction of oxides to prove the acidic nature are given below.

$CO_2 + H_2O \rightarrow H_2CO_3$(carbonic acid)	$CO_2 + NaOH \rightarrow Na_2CO_3 + H_2O$
$SO_2 + H_2O \rightarrow H_2SO_3$(sulphurous acid)	$SO_2 + KOH \rightarrow K_2SO_3 + H_2O$
$P_2O_5 + 3H_2O \rightarrow 2H_3PO_4$(Phosphoric acid)	$P_2O_5 + 6NaOH \rightarrow 2Na_3PO_4 + 3H_2O$
$N_2O_5 + H_2O \rightarrow 2HNO_3$(Nitric acid)	$N_2O_5 + 2KOH \rightarrow 2KNO_3 + H_2O$
$CrO_3 + H_2O \rightarrow H_2CrO_4$(Chromic acid)	$CrO_3 + 2NaOH \rightarrow Na_2CrO_4 + H_2O$

Facts

Some acidic oxides react with water to produce a mixture of two different acids. For example,

$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$

$P_4O_8 + 6H_2O \rightarrow 2H_3PO_3 + 2H_3PO_4$, etc.

(ii)Basic Oxides

Oxides which produce alkalies when dissolved in water are called basic oxides. Basic oxides when treated with acid produce salt and water. Generally, these are the oxides of metals. For examples,

a) Oxides of group 1(alkali metals) and group 2 (alkaline earth metals except Be)

Li_2O, Na_2O, K_2O, MgO, CaO, BaOetc are basic oxides. They are soluble in water.

b) Oxides of group d-block elements in the lower oxidation states like FeO, MnO, CuO, Ag_2O, etc. They are almost insoluble in water.

Formation of basic oxide

Metal + O_2 → Metallic oxide (basic oxide)

$4Na + O_2 \rightarrow 2Na_2O$

$2Ca + O_2 \rightarrow 2CaO$

$2Cu + O_2 \rightarrow 2CuO$ etc.

Reaction of oxides to prove the basic nature are given below:

$Na_2O + H_2O \rightarrow 2NaOH$(Sodium hydroxide)	$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$
$CaO + H_2O \rightarrow Ca(OH)_2$(Calcium hydroxide)	$CaO + H_2SO_4 \rightarrow CaSO_4 + H_2O$
$BaO + H_2O \rightarrow Ba(OH)_2$(Barium hydroxide)	$BaO + 2HCl \rightarrow BaCl_2 + 3H_2O$
$FeO + H_2O \rightarrow$ Insoluble	$FeO + H_2SO_4 \rightarrow FeSO_4 + H_2O$
$CuO + H_2O \rightarrow$ Insoluble	$CuO + 2HCl \rightarrow CuCl_2 + H_2O$

(iii)Neutral Oxides

Oxides which are neither acidic nor basic in nature are called neutral oxides. These oxides neither react with acid nor base. For example, N_2O, H_2O, NO, CO, etc.

(iv)Amphoteric Oxides

Activity:11

Observe the reactions given below:

$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$

$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$

a. In which of the above reactions ZnO acts as acid?

Oxides which exhibit the properties of both acid and base are called amphoteric oxides. These oxides react with acid as well as base to produce salt and water. For example, ZnO, PbO, SnO, Al_2O_3, Ga_2O_3, Cr_2O_3, BeO, etc.

Some reactions of oxides showing amphoteric nature:

$PbO + H_2SO_4 \rightarrow PbSO_4 + H_2O$

(base)
$PbO + 2NaOH \rightarrow Na_2PbO_2 + H_2O$
(acid) (sodium plumbite)
$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
(base)
$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$
(acid) (sodium aluminate)

Test yourself!

Arrange the given oxides in the given below.

$CO, CO_2, NO, N_2O, N_2O_3, N_2O_4, P_2O_3, P_2O_5, SO_2, SO_3, Cl_2O_7, ZnO, SiO_2, K_2O, ZNO, Al_2O_3, MgO$

Basic oxides	Acidic oxides	Neutral oxides	Amphoteric oxides

2. Classification of oxides on the basis of oxygen content (structural consideration)

On the basis of oxygen content oxides are of following types:

(i) Peroxide

Activity:11
One of the commonly used chemicals for removing ear wax is given in the bottle.
a. Write the molecular formula of this chemical.
b. Why is this chemical called hydrogen peroxide?

Na_2O_2, BaO_2, Li_2O_2, etc. are the examples of polyoxides commonly called peroxides. They contain peroxide unit (O_2^{2-}). In these oxides the oxidation state of oxygen is -1. Peroxides when treated with dilute mineral acids produce hydrogen peroxide (H_2O_2).
$Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$
Alkali metals and alkaline earth metals form peroxides of the type M_2O_2 and MO_2, respectively where M is alkali and alkaline earth metal.

(ii) Mixed oxides
Oxides which are the mixture of two or more oxides of same elements with different oxidation states are called mixed oxides or compound oxides. For example, Fe_3O_4, Mn_3O_4, Pb_3O_4, N_2O_4 etc. Fe_3O_4 is a mixture of FeO and Fe_2O_3 as it gives ferrous chloride and ferric chloride when treated with dilute hydrochloric acid.

$Fe_3O_4 + 8HCl \rightarrow FeCl_2 + 2FeCl_3 + 4H_2O$

Similarly, Pb_3O_4 is a mixture of PbO and PbO_2 as it gives lead nitrate and lead dioxide when treated with dilute nitric acid.

$Pb_3O_4 + 4HNO_3 \rightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

Facts

a. Potassium superoxide (K_2O) is used in space capsule, submarines and breathing masks. This is because K_2O absorbs CO_2 producing O_2.

$4K_2O + 2CO_2 \rightarrow 2K_2CO_3 + O_2$

b. Pb_3O_4 is commonly known as red lead which is called 'Sindoor' in Nepali.

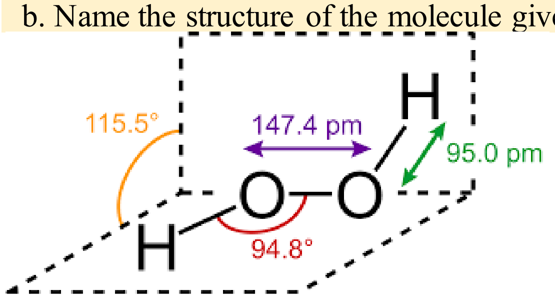
9.2.3: Hydrogen peroxide

Activity:12

Observe the structure of molecule given and

a. Write the molecular formula of compound?

b. Name the structure of the molecule given.



Hydrogen peroxide is the most common peroxide of hydrogen having molecular formula H_2O_2. Pure hydrogen peroxide is a pale blue liquid slightly viscous than water.

Handling of hydrogen peroxide

Hydrogen peroxide is a strong oxidizing and bleaching agent. Since, its high concentration is fatal therefore safety handling of H_2O_2 is required. Some of the safety measures required while using hydrogen peroxide are:

Hydrogen peroxide should be handled below its boiling point under reduced pressure.

Hydrogen peroxide should be stored in an opaque bottle at low temperature and kept away from flammable substances.

In laboratory hydrogen peroxide should be conducted in a properly functioning chemical fume hood whenever possible.

Hydrogen peroxide is corrosive to skin and causes pulmonary irritations therefore its contact with the body should be avoided.

Applications of hydrogen peroxide

Although hydrogen peroxide is hazardous there are some important applications of it. Some of them are listed below:

a. It is used as a bleaching agent.
b. It is used as antiseptic.
c. It is used as an oxidizing agent for various chemical reactions.

d. It is used in wastewater treatment to remove organic impurities.
e. It is used to sterilize surgical tools.
f. It is used to clean ear wax.
g. It is used as rocket propellant.
h. It is used in the preparation of tooth whitening agents and mouthwash.
i. It is used to produce many organic peroxides.
j. It is used to treat acnes.


removing ear wax **removing acne** **cleansing floor**

Figure 9.9: Uses of hydrogen peroxide

Test Yourself!

Why is H_2O_2 called peroxide?

9.2.4: Medical and Industrial Application of Oxygen

Activity: 13

Observe the picture carefully and find the difference in the role of gas in each.


When a person's lungs are unable to provide enough oxygen on their own then s/he requires oxygen therapy. Oxygen therapy helps all the organs and tissues to work properly. It also helps to recover from illness. This oxygen is medical oxygen and there is a big difference between regular oxygen medical oxygen. The highly pure oxygen (99%) used in medical treatment is called medical oxygen and the oxygen that is used for combustion and gasification processes in industries is called industrial oxygen. Some of the applications of these two types of oxygen are listed below:

Medical applications of oxygen

a. It is used to provide life support for artificial ventilated patients.
b. It is used to treat respiratory distress syndrome (RDS) in new-borns.
c. It is used as an antidote for CO poisoning.
d. It is used to treat diseases like chronic obstructive pulmonary disease (COPD), pulmonary embolism, heart attack and pneumonia.
e. It is used for aerobic respiration by living cells in animals.
f. It is used to combust (oxidize) ingested food to produce energy by the cell.

Industrial applications of oxygen

a. It is used with fuel gases in gas welding, gas cutting, oxygen scarfing, flame cleaning, flame hardening, and flame straightening.
b. It is used in sewage treatment and water purification.
c. It is used as a bleaching agent in the production of high quality pulp to produce paper.
d. It is used as an oxidizing agent to burn the fuel in order to launch rockets and missiles.
e. It is used in hydrogen fuel cells to produce energy.

Test yourself!

Why is the 100% pure oxygen harmful to breathe?

Project work

1. Take a chart paper and mention the applications of oxygen in the field of
    a. Medicine
    b. Industry
2. Prepare a chart including various oxides on the basis of their acidic and basic behavior.
3. Prepare a chart paper presentation on the topics:
    Applications and Handlings of Hydrogen Peroxide

Exercise

A. Multiple choice questions

1. What is the correct electronic configuration of an element that belongs to group 16 and 2nd period?

a. $1s^2, 2s^2 2p^1$ b. $1s^2, 2s^2 2p^2$ c. $1s^2, 2s^2 2p^4$ d. $1s^2, 2s^2 2p^6$

2. Which block of the periodic table does oxygen belong to?

a. s b. p c. f d. d

3. Which of the following is not considered as an oxide?

a. Fe_3O_4 b. H_2O c. OF_2 d. ClO_2

4. Pick up the amphoteric oxide.

a. K_2O b. Fe_3O_4 c. BeO d. CO_2

5. Which of the following is acidic oxide?

a. CaO b. SO_3 c. K_2O d. NO

6. How much oxygen is present in the atmosphere?

a. 31% b. 21% c. 19% d. 17%

7. What happens when alkaline earth metals react with oxygen?

a. An acidic oxide is formed b. A neutral oxide is formed
c. A basic oxide if formed d. An amphoteric oxide is formed

8. In peroxide the oxidation state of oxygen atoms is -1. Which of the following belongs to peroxide?

a. BaO_2 b. K_2O c. NO d. ZnO

9. Choose the correct mixed oxide.

a. P_4O_6 b. Na_2O_2 c. P_2O_5 d. Pb_3O_4

10. What are the two forms of oxygen formed in the atmosphere?

a. Oxygen & ozone b. Oxygen & nitric oxide
c. Water vapor & oxygen d. Oxygen & carbon monoxide

11. Which of the following is neutral oxide?

a. BaO b. SO_3 c. Li_2O d. NO

12. Which of the following group 2 elements does not form basic oxide?

a. Ca b. Be c. Mg d. Ba

13. Pick out the strongest acidic oxide?

a. P_2O_5 b. Cl_2O_7 c. ZnO d. SiO_2

14. Which of the following is true about hydrogen peroxide?

a. Hydrogen peroxide is more stable than water.

b. Hydrogen peroxide cannot bleach soft materials like silk, wool, pulp etc.

c. Hydrogen peroxide is stored in a transparent bottle.

d. The term "volume" is used to mention the concentration of hydrogen peroxide.

15. Which of the following acts as an antidote for carbon monoxide poisoning?

a. Pure oxygen b. Pure nitrogen c. Oxygen mixed with propane d. Nitrogen mixed with oxygen

B. Short answer question

1. Define the terms:

a. Allotropy b.Super oxide c. Mixed oxide d. Basic oxide e. Neutral oxide

2. Give reason:

a. N_2O_5 is an acidic oxide whereas NO is neutral.

b. OCl_2 is oxide whereas OF_2 is not.

c. 100% oxygen is harmful to breathe.

d. H_2O is neutral oxide whereas H_2O_2 is peroxide.

e. CO_2 is acidic oxide whereas Na_2O is basic oxide.

3. List some medical and industrial applications of oxygen.

4. What are oxides? Classify the following oxides into acidic, basic and neutral and give reason for your classification.

ZnO	Fe_3O_4	Al_2O_3	CuO	NO	CaO

5. Write a note on:

a. Oxide and its type b. Hydrogen peroxide and its applications

6. Differentiate between:

a. Peroxide and superoxide

b. Acidic oxide and basic oxide

Long answer question

11. Oxygen is the third most abundant element by mass which readily forms oxides with other elements. Some of the oxides are given below:

SO_3	ZnO	CO	K_2O

a. Identify the acidic, basic, amphoteric and neutral oxides from the above table with justification.

b. Give two examples of peroxides.

12. Lewis dot structures of two allotropes of the same element are given.

a. Why are they called allotropes?
b. Write any two differences in their physical properties.
c. Name the shape of the O3 molecule.

13. Different oxides formed by different elements are given in the table.
    a. Identify any two acidic, basic and amphoteric oxides from the table.
    b. Write a chemical reaction to prove that ZnO is amphoteric in nature.
    c. Give reason why Na_2O is called basic oxide
    d. Name the oxide from the table that is most basic and most acidic.

Increasing acidic character →

Increasing base character ↓

IA	IIA	IIIA	IVA	VA	VIA	VIIA
Li_2O	BeO	B_2O_3	CO_2	N_2O_5		OF_2
Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3	Cl_2O_7
K_2O	CaO	Ga_2O_3	GeO_2	As_2O_5	SeO_3	Br_2O_7
Rb_2O	SrO	In_2O_3	SnO_2	Sb_2O_5	TeO_3	I_2O_7
Cs_2O	BaO	Tl_2O_3	PbO_2	Bi_2O_5	PoO_3	At_2O_7

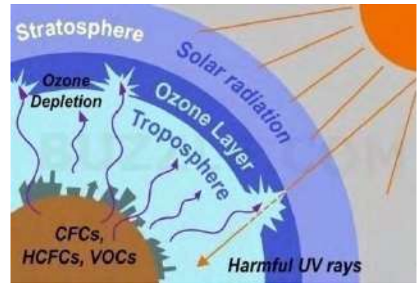
14. Write all possible oxides of following elements and classify them with reason:
    a. N b. H c. C d. K e. Ba f. B

15. X and Y are two series of oxides given below:
    X: SO_3, SiO_2, Cl_2O_7 and P_2O_5
    Y: Al_2O_3, CuO, Na_2O and MgO

Study the above series carefully and answer the following questions:
    a. Which series represents acidic and basic oxides?
    b. Arrange the oxides of each series on the basis of increasing acidic or basic strength with reason.

9.3: Ozone

Activity: 14



Observe carefully the given figure:

a. In which atmospheric layer of the earth ozone layer is found?

b. Why is it called the ozone layer? Why is it important for us?

c. Name the substance that causes the depletion of the ozone layer.

After dioxygen(O_2) the other most common allotrope of oxygen is ozone(O_3). It is also called trioxygen. It is a pale blue gas.

9.3.1: Occurrence of ozone

Ozone naturally occurs in the stratosphere between the altitudes 15 to 30 km shielding the earth from harmful UV radiations coming from the sun. It is naturally formed in the stratosphere by the action of ultraviolet radiations of the sun on oxygen molecules.

$$O_2(g) \xrightarrow{UV} 2O^\bullet(g)$$

$$O^\bullet(g) + O_2(g) \rightarrow O_3(g)]2$$

$$2O_2(g) \xrightarrow{UV} 2O_3(g)$$

Test Yourself!

How is ozone formed naturally?

9.3.2: Preparation of ozone from oxygen

Artificially ozone is prepared by passing silent electric discharge through dry oxygen between two concentric metalized tubes in an apparatus called Siemen's ozonizer. Here the reaction is endothermic.

$$3O_2 \rightarrow 2O_3 \text{ , } \Delta H = +284 \text{ kJ}$$

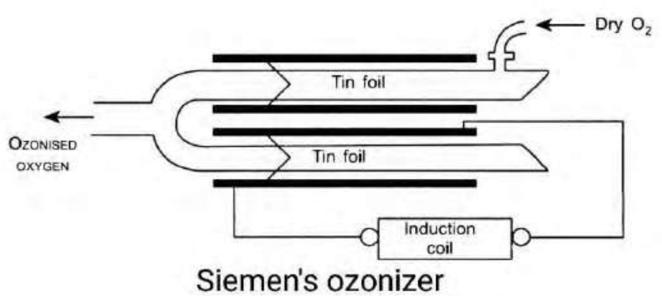


Figure: 9.10: Siemen's ozonizer for the preparation of ozone

9. 3. 3: Structure of ozone

Activity-15

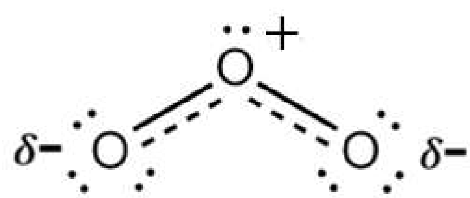
Write Lewis dot symbol of oxygen atoms and try to answer the following questions:

a. What could be the Lewis dot structure of oxygen molecule and ozone molecule?

b. What could be the shape of ozone molecule?

From electron diffraction and microwave spectroscopy it has been found that ozone has a bent structure (i.e, V shape) with bond angle 116.8^0. The oxygen-oxygen bond length is $1.278\ ^0A$, which is the intermediate between the oxygen-oxygen single and double bond. This indicates that the ozone molecule is a resonance hybrid of the following structures.

Figure 9.11: Resonating structures of ozone



Resonance hybrid structure of ozone

9.3.4: Test for Ozone

Test of ozone can be performed on the basis of following properties of ozone:

a. **Starch-Iodide Solution test**

When ozone is passed into potassium iodide (KI) solution, iodine is liberated which can change the color of the starch solution into blue.

$$O_3(g) \longrightarrow O_2 + [O]$$

$$2KI + [O] \longrightarrow K_2O + I_2$$

$$K_2O + H_2O \longrightarrow 2KOH$$

$$2KI + H_2O + O_3 \longrightarrow 2KOH + O_2 + I_2$$

$$I_2 + \text{Starch} \longrightarrow \text{Blue colour}$$

b. **Tailing of Mercury Test**

When mercury is treated with ozone it loses its meniscus and starts sticking to the glass surface due to the formation of mercurous oxide (Hg_2O).

$$2Hg + O_3 \rightarrow Hg_2O + O_2$$

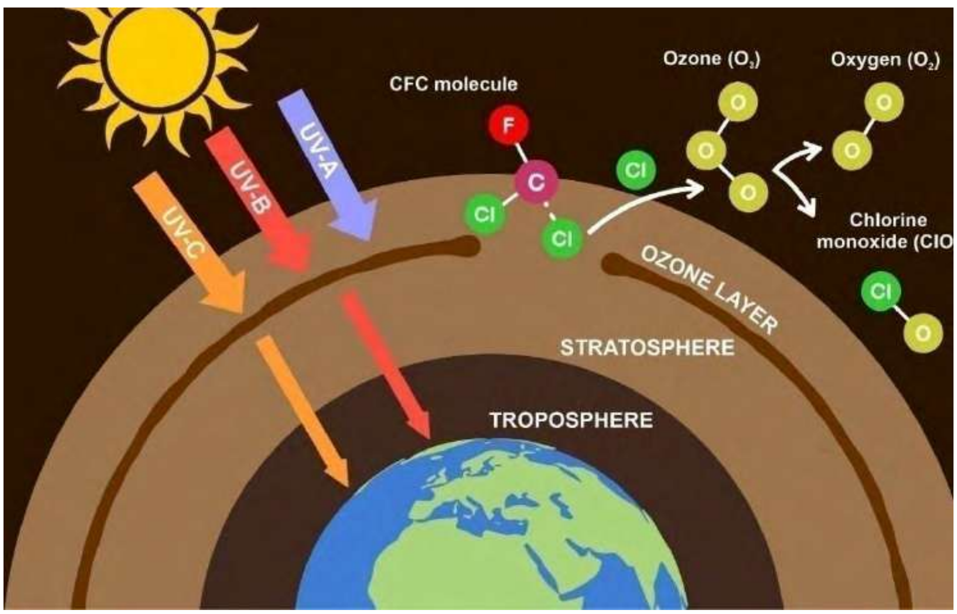
As a result of this mercury starts moving on the surface leaving a tail. So, it is called **tailing of mercury.**

9.3.5: Ozone layer depletion

Activity:16

Observe the given picture carefully and answer the following questions:

a. What are the elements present in CFC?
b. Why is there a hole in the ozone layer?
c. What is the harmful effect of such a hole in the earth?



In 1995 NASA (National Aeronautics and Space Administration) scientists reported the depletion of the ozone layer at an alarming rate over the Antarctic, South Pole of the earth. The main reason for this ozone layer depletion is the release of chlorofluorocarbon (CFCs) compounds into the atmosphere.

Ozone layer is a common term used for high concentration of ozone molecules in the stratosphere around 15-30 Km above the earth's surface. This ozone layer covers the entire planet and protects life on the earth by absorbing harmful UV rays coming from the sun. The gradual thinning of the ozone layer in the upper atmosphere due to release of bromine or chlorine-containing compounds is called **ozone layer depletion**. The compounds that deplete the ozone layer are called ozone depleting substances (ODS). These substances are either released from industries or by other human activities. Among many ozone depleting substances CFCs are the most contributing ozone depleting substances. CFCs are chlorinated and fluorinated methane and ethane derivatives commonly known as Freons. Some common examples of Freons are $CFCl_3$ (Freon-11), CF_2Cl_2(Freon-12), CF_3Cl (Freon-13), $C_2F_2Cl_4$ (Freon-112), etc.

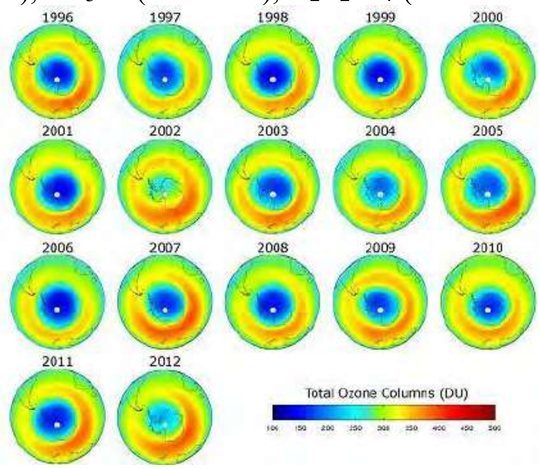
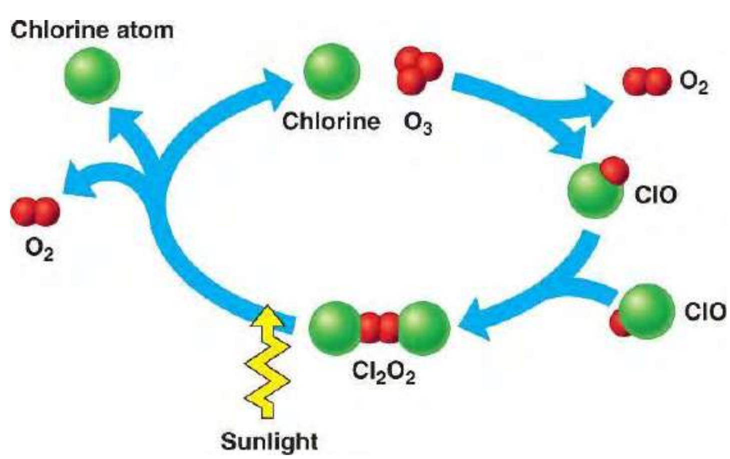


Figure 9.11: Satellites track the health of the ozone layer

> **Note:** ***Freons are extremely stable, non-reactive, non-flammable and non-toxic substances. They can be easily liquefied at room temperature. These special properties of Freons make them to be used as refrigerants in refrigerators and air conditioners. Freon's such asfreon-11 are used in the production of plastic foams.***

Mechanism of ozone layer depletion



Freons produced from the earth's surface slowly rise up in the upper atmosphere where they absorb UV radiations from the sun and then decompose to give free chlorine radicals. This free chlorine radical destroys the ozone layer.

$$CF_2Cl_2 \xrightarrow{UV} CF_2Cl^\bullet + Cl^\bullet \text{ ....(i)}$$

$$Cl^\bullet + O_3 \longrightarrow ClO^\bullet + O_2 \text{ ....(ii)}$$

$$ClO^\bullet + O \longrightarrow Cl^\bullet + O_2 \text{ ....(iii)}$$

The steps (ii) and (iii) are repeated over and over again and O_3 gets destroyed continuously. It has been found that one molecule of CFC can destroy 100,000 molecules of ozone at a time.

Effects of ozone layer depletion

The harmful effects of hole (depletion) in ozone layer are:

a. Humans and other animals will be directly exposed to large amounts of UV radiations that will cause seditious health issues like skin diseases, skin cancer, skin burns, blood cancer, cataract, quick aging and weak immune system etc.
b. It may cause ecological disturbance.
c. It will cause minimal growth, flowering and photosynthesis in plants.
d. It will affect the pond ecosystem.

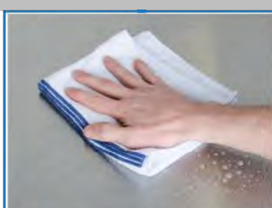
Control measures of ozone layer depletion

Some measures to control ozone layer depletion are listed below:

a. Avoiding or reducing the use of ozone depleting substances.
b. Minimizing the use of vehicles that emit large amounts of greenhouse gases that lead to ozone layer depletion and global warming.
c. Minimizing the use of cleaning products that contain chlorine and bromine. Instead of such products, the use of eco-friendly cleaning products should be praised.
d. Proper rules and laws should be executed by the government to protect the ozone layer.
e. Public awareness should be increased to cut off the chemicals that deplete the ozone layer.

9.3.6: Uses of ozone

It Can Purify Drinking Water



It Can Kill Bacteria On Surfacer

It Can Kill Bacteria In The Mouth

It Can Clean Food

It Can Fight Airborne Contaminats

It Can Remove Odors in Laundry

Some other uses of ozone are listed below:

a. It is used to sterilize surgical tools in hospitals.
b. It is used to purify air in crowded areas like cinema halls, railway tunnels, big marts etc.
c. It is used as a bleaching agent.
d. It is used in preservation of food and raw materials.
e. It is used as an important catalyst in various industrial processes.
f. It is used in the preparation of organic compounds like aldehydes and ketones.

Test yourself!

What are Freons? Give some examples.

Project work

1. Prepare a chart paper presentation on the following headings:
   a. Ozone and its resonating structure
   b. Importance of ozone layer.
   c. Cause, effect and preventive measures of ozone layer depletion.

Exercise

A. Multiple choice questions

1. **In which atmospheric layer ozone is mostly found?**
   a. Thermosphere b. Stratospherec. Troposphere d. Mesosphere

2. **Which of the following gas is detected by starch-iodide solution test?**
   a. N_2 b. O_2 c. O_3 d. Cl_2

3. **What happens when ozone depleting substances are released into the atmosphere?**
   a. They primarily lead to Global warming b. They primarily lead to smog formation
   c. They primarily lead to ozone layer depletion d. They primarily lead to increase rainfall

4. **Ozone depletion from the atmosphere becomes the reason for causing which of the following diseases?**
   a. Brain stroke b. Tuberculosis c. Heart stroke d. Skin cancer

5. **Which of the following synthetic chemicals is responsible for ozone layer depletion?**
   a. Methanol b. Bipolymer c. Chlorofluorocarbon d Ethanol

6. **Ozone is a triatomic form of oxygen. Which of the following properties does it resemble?**
   a. Allotrope b. Isotope c. Isostructural d. Isomer

7. **Which of the following are involved in the artificial preparation of ozone?**
a. UV light on O b. UV light on O_2
c. High pressure on O_2 d. Isomer of O_2

6. **Which of the following statements about ozone is true?**
a. Ozone causes maximum rainfall b. Ozone causes cooling of the earth
c. Ozone blocks the sunlight d. Ozone blocks UV light coming from the sun

7. **Between what altitudes, is the ozone layer found in highest concentration?**
a.20-40 km b. 40-50 km c. 50-60 km d. 60-80km

8. **In which of the following regions, ozone holes are predominantly found?**
a. Tropic of Capricorn b. Equator c. Poles d. Tropic of Cancer

9. **When was the first ozone hole discovered?**
a. 1945 b. 1959 c. 1970 d. 1975

10. **Which of the following represents Freon?**
a. Dichlorodifluoromethane b. Tetrachloroethane
c. Dichlorodiphenyltricholoroethane d. Acetylene tetrachloride

11. **Where was the ozone hole discovered first?**
a. Antartica b. Asia c. Europe d. Arctic

12. **What is the formula of Freon 11?**
a. CCl_3F b. CCl_2F_2 c. $CHCl_2F$ d. $CClF_3$

13. **Which of the following elements is not found in the molecule of CFC?**
a. C b. N c. Cl d. F

14. **What type of chemical substances are regulated by the Montreal Protocol?**
a. Chloropicrin b. Trichloromethane
c. Petrolium products d. Ozone depleting substances

B. Short answer questions

1. **Define the terms-**
a. Freons b. Allotropy c. Ozone layer d. ODS e.

2. What is ozone? Write its importance. How would you prepare ozone artificially?

3. **Differentiate between:**
a. Freon 12 and Freon 21
b. Dioxygen and trioxygen

4. Write any four ozone depleting substances (ODS).

5. What is the primary risk associated with extreme UV rays because of ozone layer depletion?

6. How UV rays coming from the sun are blocked by ozone?

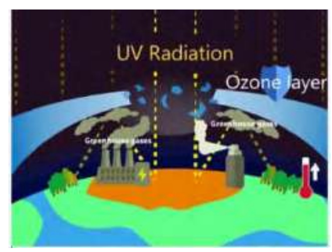
7. "Is ozone a greenhouse gas?" Give your opinion.

8. Write a note on-
a. Mechanism of ozone layer depletion.
b. Preventive measures of ozone layer depletion

9. Write the effects and control measure of ozone layer depletion.

C. Long answer questions

10. Oxygen is the third most abundant element. The allotropes of oxygen are O_2 and O_3.
a. Draw the resonance structure of ozone molecules.
b. Why are both O-O bond distances in ozone molecules equal?
c. Write any two chemicals responsible for destroying the ozone layer.
d. What happens when ozone gas is passed through a starch-iodide solution?



11. Study the given picture and answer the following questions:
a. What does the picture depict?
b. Write the structure of ozone.
c. In which layer of atmosphere does the ozone layer exist?
d. What happens when UV rays are not blocked by the ozone layer?
e. Is ozone a greenhouse gas?

9.4: Nitrogen

Nitrogen

Daniel Rutherford (3 November 1749 – 15 November 1819) was a Scottish physician, chemist and botanist who is known for the isolation of nitrogen in 1772. He postulated the existence of nitrogen gas that does not catalyze the combustion reaction. This inability of nitrogen was tested by conducting an experiment where oxygen and carbon dioxide were eliminated by burning the atmospheric oxygen leaving behind the gas called nitrogen.

Element: Nitrogen	Electronic configuration: $1s^2\ 2s^2\ 2p^3$
Symbol: N	Group: 15
Atomic Number: 7	Period: 3^{rd}
Atomic Weight: 14.007 amu	Block: p block element
Molecular Formula: N_2	Valency: 3

Nitrogen is a non-metallic element. It is a colourless, odourless and tasteless gas. The Earth's atmosphere consists of 75.5% nitrogen by weight and 78% nitrogen by volume.

9.4.1:Reason for inertness of nitrogen

Activity:17

Observe the table with different bond and the bond energy:

a. **Which of the bonds is the strongest bond and why?**
b. **Which of the given bonds can be broken easily?**

Bond	Bond Energy, kJ mol^{-1}
N—N	163
N=N	418
N≡N	946
N—O	222
N=O	590
O—O	142
O=O	498
F—F	159
Cl—Cl	243
Br—Br	193
I—I	151

An electric bulb consists of tungsten filament which quickly burns when exposed to oxygen. Therefore, the bulb is filled with inert gas like argon and nitrogen. Now, the question may arise though nitrogen is not a noble gas why is it used in bulbs? This is because nitrogen is an inert gas. Nitrogen is a diatomic molecule consisting of two atoms of 'N' bonded by triple covalent bonds. This N to N triple bond is very strong and requires very high energy to break and carry the reaction. Thus, the small size of nitrogen atoms and high bond dissociation energy makes N_2 an inert gas. However, at very high temperatures it may react with some metals and non-metals.

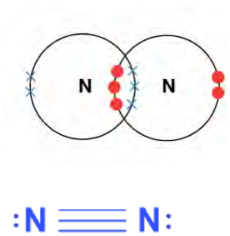


Figure 9.12: Electric bulb is filled with N_2 to make the atmosphere inert

Active nitrogen

Active nitrogen was first reported by Warburg in 1884. When an electric discharge is passed through a nitrogen molecule under very low pressure a brilliant golden yellow luminescence is observed which persists for sometimes after switching off the electric discharge. It is observed that nitrogen after the discharge becomes more active and therefore, this form of nitrogen is called **active nitrogen**. Although this phenomenon was observed by Warburg, later on in 1911 Strut examined thoroughly the physical properties of active nitrogen and found that they differ from molecular nitrogen. It has been suggested that many of the properties of nitrogen could be due to active nitrogen. Thus the energized nitrogen atom which plays an active role in chemical transformation is known as active nitrogen. Excited neutral nitrogen molecules, nitrous oxide, graphitized carbon nitride (g-C_3N_4), etc. are the source of active nitrogen.

Test Yourself!

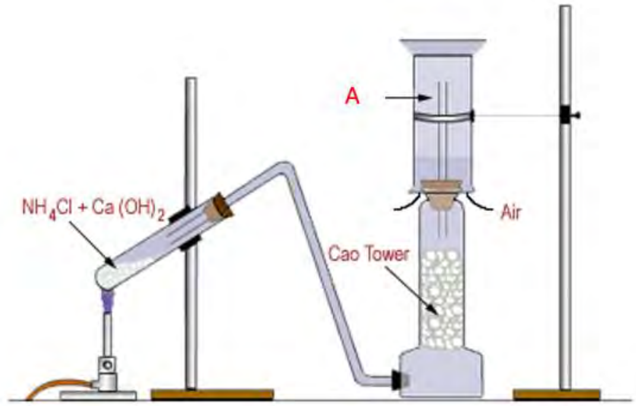
a. Why is nitrogen molecule inert in nature?
b. What is active nitrogen?

Ammonia

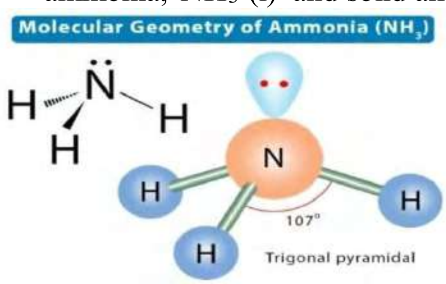
Activity-18

Observe the setup of apparatus as shown in figure and answer the following questions:

a. Identify the gas A collected in the gas jar.
b. Write the balanced chemical equation that occurs in the hard glass test tube.



Ammonia gas is colourless gas with a pungent smell. It is naturally available as well as synthesized on a large scale. It can exist in all three states i.e, ammonia gas, NH_3 (g), liquid ammonia, NH_3 (l) and solid ammonia NH_3 (s).



Facts

a. Ammonia is colourless gas with a characteristically pungent smell that brings tears to eyes.
b. Ammonia is lighter than air and highly soluble in water due to its capacity to form intermolecular hydrogen bonding with water molecules.
c. Ammonia is neither combustible nor a supporter of combustion.
d. Ammonia produces cooling on vaporization. So, it is used as a refrigerant.

9.4.2: Chemical properties of ammonia

1. Action with $CuSO_4$ solution (Formation of complex)

Bluish white ppt. of $Cu(OH)_2$ deep blue color of $[Cu(NH_3)_4]SO_4$

When ammonia gas is passed through copper sulphate solution a bluish white ppt. is obtained which gets dissolved in excess of ammonia producing a deep blue complex called tetraamine copper(II) sulphate. This complex is also known as Schweizer's reagent which is used in purifying cellulose.

$NH_3 + H_2O \rightarrow NH_4OH$

$CuSO_4 + \underset{\text{limited}}{2NH_4OH} \rightarrow \underset{\text{Bluish white ppt}}{Cu(OH)_2\downarrow} + (NH_4)_2SO_4$

$Cu(OH)_2\downarrow + (NH_4)_2SO_4 + \underset{\text{excess}}{2NH_4OH} \rightarrow \underset{\text{Deep blue complex}}{[Cu(NH_3)_4]SO_4} + 4H_2O$

2. Action with water

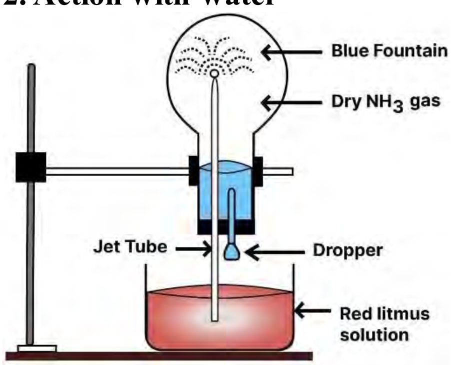


Figure 9.13: solubility of NH_3 in water

Ammonia is highly soluble in water. The solubility of ammonia in water is due to the formation of intermolecular hydrogen between ammonia and water molecules. When ammonia is dissolved in water ammonium hydroxide (also called ammonia solution) is formed.

$$NH_3 + H_2O \rightarrow NH_4OH \leftrightarrow NH_4^+ + OH^-$$

The aqueous solution of ammonia is basic in nature due to the formation of hydroxide ions.

3. Action with Ferric chloride (Ammonia as a precipitant agent)

Aqueous ammonia (or moist ammonia) acts as a precipitant. It precipitates metal ions like Fe^{3+}, Zn^{2+}, etc. as hydroxides from their salt solutions. When ammonia gas is passed into ferric chloride solution, reddish brown ppt. of ferric hydroxide is obtained.

$NH_3 + H_2O \rightarrow NH_4OH$

$FeCl_3 + 3NH_4OH \rightarrow \underset{\text{Reddish brown ppt.}}{Fe(OH)_3\downarrow} + 3NH_4Cl$

This reaction shows ammonia is a precipitating agent.

Figure 9.14: Test tube containing ppt. $Fe(OH)_3$

4. Action with Concentrated HCl

When fumes of concentrated HCl come in contact with ammonia gas then dense white fumes of ammonium chloride are obtained.

NH_3 + conc. HCl $\rightarrow NH_4Cl$ (ammonium chloride)
(Dense white fumes)

Figure 9.15: Formation of dense white fumes of NH_4Cl

5. Action with mercurous nitrate paper

When ammonia gas is passed over mercurous nitrate paper it turns into black precipitate. This black precipitate is a mixture of mercury metal and basic mercuricamido nitrate.

$$\underset{\text{Mercurous nitrate}}{Hg_2(NO_3)_2} + 2NH_3 \rightarrow \underbrace{Hg + Hg(NH_2)NO_3}_{\substack{\text{Black ppt} \\ \text{(mercuricamido nitrate)}}}\downarrow + 3NH_4NO_3$$

6. Action with Oxygen

Although ammonia is neither combustible nor a supporter of combustion it burns with oxygen to produce greenish yellow flame giving nitrogen and water vapour.

$$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$

Here ammonia reduces oxygen to oxide of hydrogen and itself get reduced to nitrogen gas. This proves the reducing nature of ammonia.

If the above reaction is carried out in the presence of platinum at about 850^0C then nitric oxide is formed. This reaction is called catalytic oxidation of ammonia.

$$4NH_3 + 5O_2 \xrightarrow{Pt} 4NO + 6H_2O$$

This reaction forms the basis for the manufacture of nitric acid on a large scale by Ostwald's process.

Test Yourself!

a. Give a reaction to show that ammonia is a reducing agent.
b. Write a balanced chemical equation showing ammonia a precipitating agent.
c. Why does mercurous nitrate paper turns black when treated with NH_3?
d. What happens when ammonia gas is passed through copper sulphate solution till saturation?

9.4.3: Applications of Ammonia

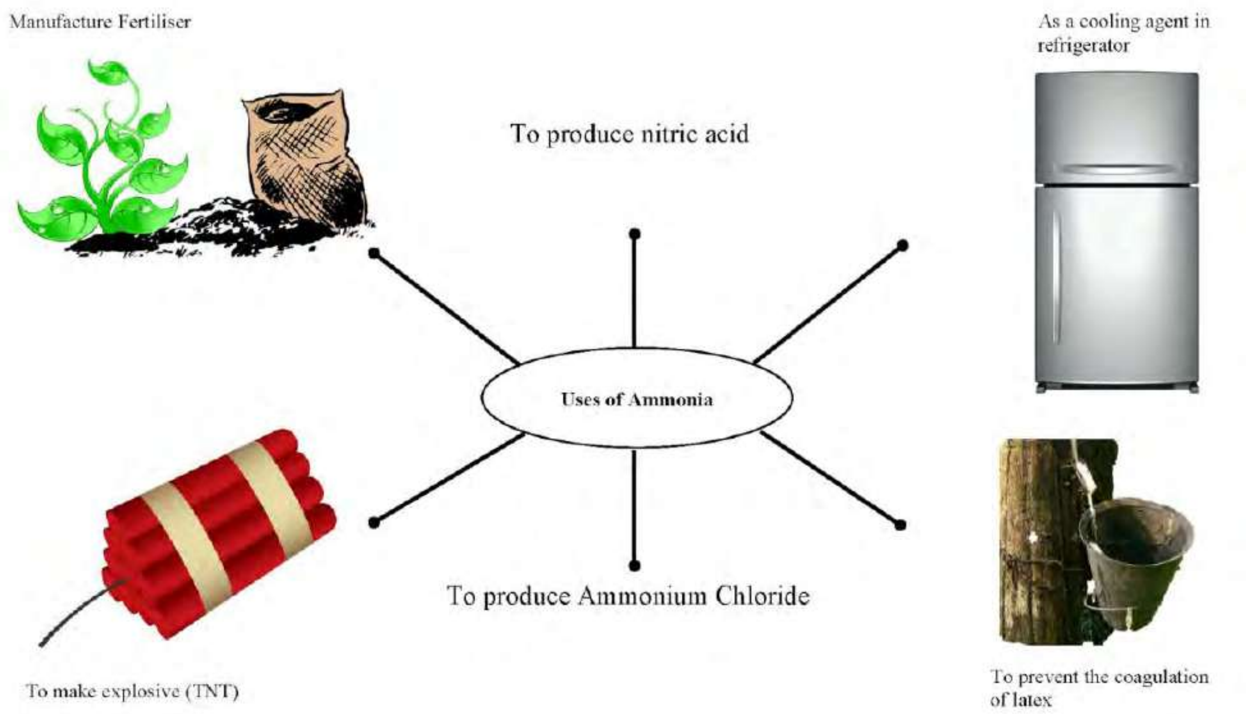


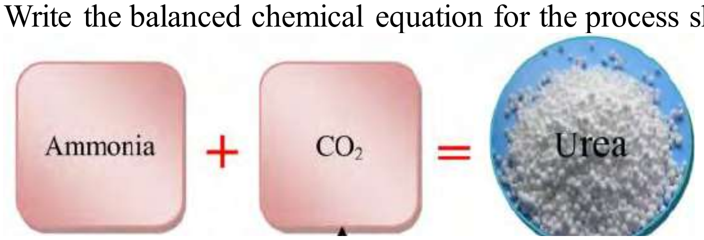
Figure ***9.16:*** Applications of ammonia

Ammonia is the most important building block chemical in the manufacture of products that people make use of everyday. Some of its applications are listed below:

a. It is used in the manufacture of most common fertilizer urea and other nitrogenous fertilizers.
b. It is used in the synthesis of explosives, fibers, plastics, etc.
c. It is used as a laboratory reagent.
d. It is used in the manufacture of washing soda by Solvay process.
e. It is used as a cleansing agent to remove oils and grease.
f. It is used as an analytical reagent in qualitative analysis of salt.

Test yourself!

Write the balanced chemical equation for the process shown.



9.4.4: Harmful effects of ammonia

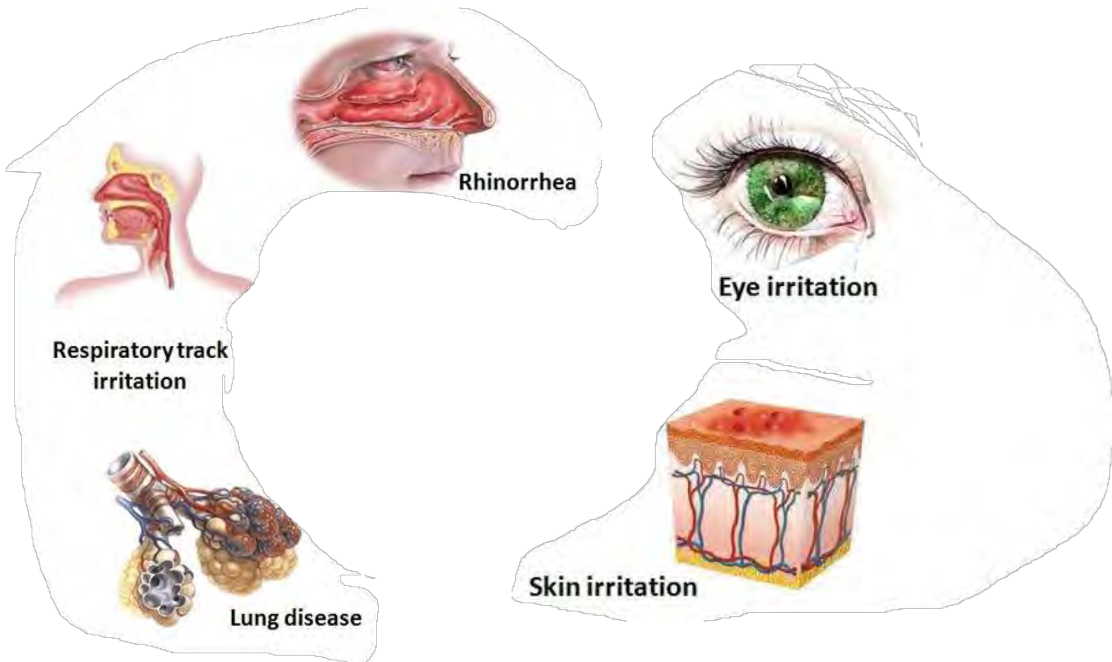


Figure 9.17: Harmful effects of ammonia

Ammonia is one of the most widely produced chemicals worldwide. It is also produced in the human body and is commonly found in nature from decomposition of organic matter, including plants, animals and animal wastes. It is irritating, pungent and corrosive and causes toxicity. Some of the harmful effects of this gas are listed below:

a. It causes nausea, skin irritation and dizziness to humans.
b. It can damage the kidney and cause cataract pneumonia and ulceration etc.
c. It can damage brain cells.
d. Its exposure to pregnant women can affect her new born child.
e. It can harm aquatic life.
f. Its irritating and corrosive nature causes immediate burning and itching of eyes, lungs, throat and respiratory tract.

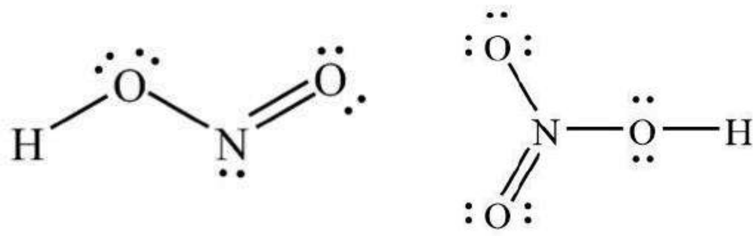
9.4.5: Oxy-acids of Nitrogen

Activity: 19

Observe the Lewis dot structures of two compounds given:

a. Write the molecular formula of the compounds
b. Why are these compounds called oxyacids of nitrogen?

c. **What is the difference in these molecules?**



When an acid contains oxygen, it is regarded as oxyacid. Thus, an **oxyacid** is an acid that contains an oxygen atom bonded to a hydrogen atom and at least one other element, usually non-metals like S, N, P, etc. It is also known as **oxo acids** or **ternary acid**. The relative strength of oxy acids depends upon the electronegativity and oxidation state of the non-metals bonded. Nitrogen forms several oxy acids. Some of them are listed in the table below-

Table 9.3: Oxy-acids of nitrogen

S. N.	Oxy-acids of nitrogen	Molecular formula	Oxidation number	Nature
1.	Hyponitrous acid	$H_2N_2O_2$	+1	white solid, unstable, explosive and oxidizing acid
2.	Nitrous acid	HNO_2	+3	unstable, oxidizing acid
3.	Nitric acid	HNO_3	+5	colorless, fuming liquid, stable, strong acid
4.	Pernitric acid	HNO_4	+5	unstable, explosive, oxidizing agent

Facts

With increase in oxidation state of central atom, the acidic strength of oxy-acids increases as the O-H bond in oxy- acids becomes more polar and release of H^+ ion becomes easier.

Test Yourself!

HNO_3 is always an oxidizing agent but HNO_2 acts as both oxidizing and reducing agent. Why?

Nitric acid

The most commonly used oxyacid of nitrogen is nitric acid that is a highly corrosive mineral acid with molecular formula HNO_3. It is also known as aqua-fortis. The oxidation state of N in this oxyacid is +5.

Facts

Pure nitric is a colorless liquid with a choking smell. On long standing it turns to brown color as it decomposes to produce NO_2 and O_2. Further it turns to yellow color when NO_2 gets dissolved.

$$HNO_3 \xrightarrow{light} 4NO_2 + O_2 + 2H_2O$$

Due to this reason, it is stored in a dark brown bottle.

9.4.6: Chemical properties of nitric acid

1. HNO_3 as an acid

Nitric acid is a very strong acid. The acidic nature of it is due to the availability of replaceable hydrogen atoms.

$HNO_3 \rightarrow H^+ + NO_3^-$

The acidic property of HNO_3 can be tested in following ways:

a. It changes the colour of blue litmus paper into red, light yellow colour of methyl orange in to pink, pink colour of phenolphthalein (added to base) in to colourless, etc.

b. It reacts with various bases and basic oxides to form nitrate salt like:

$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$

$2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + H_2O$

$2HNO_3 + MgO \rightarrow Mg(NO_3)_2 + H_2O$

$HNO_3 + NH_3 \rightarrow NH_4NO_3 + H_2O$

$4HNO_3 + CaCO_3 \rightarrow Ca(NO_3)_2 + CO_2 + H_2O$

c. Nitric acid does not produce hydrogen gas when treated with metals. However, Mg and Mn are two metals that can liberate hydrogen gas when treated with very dilute nitric acid.

$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$
(very dil.)

$Mn + 2HNO_3 \rightarrow Mn(NO_3)_2 + H_2$
(very dil.)

2. HNO_3 as an oxidizing agent

Nitric acid is a strong oxidizing agent because in HNO_3 the oxidation state of N is +5 which is highest. The oxidation state of N in HNO_3 can only be decreased so it acts as an oxidizing agent and oxidizes many metals, metalloids, non-metals and other inorganic compounds etc.

Some oxidizing reactions of nitric acid are:

A. Oxidation of metals

Several metals except inert metals like Pt, Au, Ir, etc. react with nitric acid under different conditions to produce nitrate salts, water, oxides of nitrogen. The nature of the products depends upon the following three main factors:

a. The nature of the metals
b. The concentration of nitric acid and
c. The temperature

The mode of reaction between nitric acid and metals can be categorized as

Action with more electropositive metals

Metals which are more electropositive (like Mg, Mn, Zn, Fe, etc.) than hydrogen react with nitric acid according to nascent hydrogen theory.

$$\text{Metal} + HNO_3 \rightarrow \text{Metal nitrate} + [H]$$

$$HNO_3 + [H] \rightarrow \text{Reduction product} + \text{Water}$$

Some examples-

a. Action on Zinc: Zinc reacts with nitric acid under various conditions to produce different products as explained below-

With hot and conc. HNO_3: Zn reacts with hot and conc. HNO_3 to give nitrogen dioxide along with zinc nitrate and water.

$$Zn + 2HNO_3 \rightarrow Zn(NO_3)_2 + 2[H]$$

$$\{HNO_3 + [H] \rightarrow NO_2 + H_2O\} \times 2$$

$$Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$$

With hot and moderately conc. HNO_3: Zn reacts with hot and moderately conc. HNO_3 to give nitric oxide along with zinc nitrate and water.

$$\{Zn + 2HNO_3 \rightarrow Zn(NO_3)_2 + 2[H]\} \times 3$$

$$\{HNO_3 + [H] \rightarrow NO + 2H_2O\} \times 2$$

$$3Zn + 8HNO_3 \rightarrow 3Zn(NO_3)_2 + 2NO + 4H_2O$$

With cold and dil. HNO_3: Zn reacts with cold and dil. HNO_3 to give nitrous oxide along with zinc nitrate and water.

$$\{Zn + 2HNO_3 \rightarrow Zn(NO_3)_2 + 2[H]\} \times 4$$

$$2HNO_3 + 8[H] \rightarrow N_2O + 5H_2O$$

$$4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$$

With very dil. HNO_3: Zn reacts very dil. HNO_3 to give ammonium nitrate along with zinc nitrate and water.

$$\{Zn + 2HNO_3 \rightarrow Zn(NO_3)_2 + 2[H]\} \times 4$$
$$HNO_3 + 8[H] \rightarrow NH_3 + 3H_2O$$
$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$
$$4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

b. Action on Magnesium: The reaction of Mg with conc. and moderately conc. nitric acid are same as that of zinc.

With hot and conc. HNO_3: Mg reacts with hot and conc. HNO_3 to give nitrogen dioxide along with magnesium nitrate and water.

$$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + 2[H]$$
$$\{HNO_3 + [H] \rightarrow NO_2 + H_2O\} \times 2$$
$$Mg + 4HNO_3 \rightarrow Mg(NO_3)_2 + 2NO_2 + 2H_2O$$

With hot and moderately conc. HNO_3: Mg reacts with hot and moderately conc. HNO_3 to give nitric oxide along with magnesium nitrate and water.

$$\{Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + 2[H]\} \times 3$$
$$\{HNO_3 + [H] \rightarrow NO + 2H_2O\} \times 2$$
$$3Mg + 8HNO_3 \rightarrow 3Mg(NO_3)_2 + 2NO + 4H_2O$$

But, Mg reacts with very dilute nitric acid to produce magnesium nitrate and hydrogen gas.

$$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2 \uparrow$$

Note: ***These above reactions can also be balanced by producing nascent oxygen instead of nascent hydrogen.***

c. Action on Iron: Iron reacts with nitric acid under various conditions to produce different products as explained below-

With hot and conc. HNO_3: When iron reacts with heat and conc. HNO_3, a layer of iron oxide (Fe_3O_4) is formed on the surface of it which stops the reaction. This is called passivation or passivity of iron and such iron is called passive iron.

$$HNO_3 \rightarrow H_2O + 2NO_2 + [O]\} \times 4$$
$$3Fe + 4[O] \rightarrow Fe_3O_4$$
$$3Fe + 8HNO_3 \rightarrow Fe_3O_4 + 8NO_2 + 4H_2O$$

Hence,

$$\underset{\text{(Hot \& Conc.)}}{Fe + HNO_3} \rightarrow \text{No reaction}$$

Facts

Similar behavior is shown by other metals like Al, Cr, Co and Ni.

With hot and moderately conc. HNO_3: Iron reacts with moderately conc. HNO_3 to give nitrogen dioxide along with ferric nitrate and water.

$$Fe + 3HNO_3 \rightarrow Fe(NO_3)_3 + 3[H]$$
$$\{HNO_3 + [H] \rightarrow NO_2 + H_2O\} \times 3$$
$$Fe + 6HNO_3 \rightarrow Fe(NO_3)_3 + 3NO_2 + 3H_2O$$

With cold and dil. HNO_3: Iron reacts with cold and dil. HNO_3 to give nitrous oxide along with ferric nitrate and water.

$$\{Fe + 3HNO_3 \rightarrow Fe(NO_3)_3 + 3[H]\} \times 8$$
$$\{2HNO_3 + 8[H] \rightarrow N_2O + 5H_2O\} \times 3$$
$$8Fe + 30HNO_3 \rightarrow 8Fe(NO_3)_3 + 3N_2O + 15H_2O$$

With cold and very dil. HNO_3: Iron reacts with very dil. HNO_3 to give ammonium nitrate along with ferrous nitrate and water.

$$\{Fe + HNO_3 \rightarrow Fe(NO_3)_2 + 2[H]\} \times 4$$
$$HNO_3 + 8[H] \rightarrow NH_3 + 3H_2O$$
$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$
$$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$$

Action with less electropositive metals

Metals which are less electropositive (like Cu, Bi, Hg, Ag etc) than hydrogen react with nitric acid according to nascent oxygen theory.

$$\underset{\text{conc.}}{HNO_3} \rightarrow NO_2 + H_2O + [O]$$
$$\underset{\text{dil.}}{HNO_3} \rightarrow NO + H_2O + [O]$$

Some examples are:

a. Action on Copper: Copper reacts with nitric acid under various conditions to produce different products as explained below-

With hot and conc. HNO_3: Copper reacts with conc. nitric acid to give nitrogen dioxide along with cupric nitrate and water.

$$2HNO_3 \rightarrow 2NO_2 + H_2O + [O]$$
$$Cu + [O] \rightarrow CuO$$
$$CuO + 2HNO_3 \rightarrow Cu(NO_3)_2 + H_2O$$
$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

With cold and dil. HNO_3: Copper reacts with cold and dil. HNO_3 to give nitric oxide along with cupric nitrate and water.

$$2HNO_3 \rightarrow 2NO + H_2O + 3[O]$$
$$\{Cu + [O] \rightarrow CuO\} \times 3$$
$$\{CuO + 2HNO_3 \rightarrow Cu(NO_3)_2 + H_2O\} \times 3$$
$$Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

Facts

Hg and Ag also react with HNO_3 in the similar manner.

Oxidation of Non-metals

Solid non-metals like carbon, sulphur, phosphorous and iodine react with hot and concentrated nitric acid to produce nitrogen dioxide, water and corresponding oxy-acids or oxides of non-metals.

i. Action with Sulphur: Sulphur reacts with nitric acid to give sulphuric acid.

$$2HNO_3 \rightarrow 2NO_2 + H_2O + [O]\} \times 3$$
$$S + 3[O] \rightarrow SO_3$$
$$SO_3 + H_2O \rightarrow H_2SO_4$$
$$S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 3H_2O$$

ii. Action with Carbon: Carbon reacts with hot and conc. nitric acid to give carbon dioxide.

$$2HNO_3 \rightarrow 2NO_2 + H_2O + [O]\} \times 2$$
$$C + 2[O] \rightarrow CO_2$$
$$C + 4HNO_3 \rightarrow CO_2 + 4NO_2 + 2H_2O$$

Here some of the CO_2 may get dissolved in water to produce carbonic acid.

$$CO_2 + 4H_2O \rightarrow H_2CO_3$$

Oxidation of reducing agents

Reducing agents like H_2S and SO_2 react with conc. nitric acid as:

i. Action with SO_2: When sulphur dioxide is treated with conc. nitric acid it is oxidized to sulphuric acid.

$$2HNO_3 \rightarrow 2NO_2 + H_2O + [O]$$
$$SO_2 + [O] \rightarrow SO_3$$
$$SO_3 + H_2O \rightarrow H_2SO_4$$
$$SO_2 + 2HNO_3 \rightarrow 2NO_2 + H_2SO_4$$

ii. Action with H_2S: When hydrogen sulphide is treated with conc. nitric acid gets oxidised to sulphur.

$$2HNO_3 \rightarrow 2NO_2 + H_2O + [O]$$
$$H_2S + [O] \rightarrow H_2O + S$$
$$H_2S + 2HNO_3 \rightarrow 2NO_2 + 2H_2O + S$$

Test yourself!

a. Name two metals that can liberate hydrogen gas from nitric acid.
b. What happens when a piece of copper turning is kept in hot and conc. nitric acid?
c. Write the action of conc. nitric acid on-
a) S b) H_2S c) Zn d) C

Facts

The mixture of conc. HNO_3 and conc. HCl in the ratio of 1:3 is called aqua regia or king's water. It can dissolve the inert metals like Au, Pt or Pd.

9.4.7: Ring test for nitrate ion

Activity: 20

Take 1 mL of nitrate salt (like ammonium nitrate or potassium nitrate) solution in a test tube and add 1mL of freshly prepared $FeSO_4$ solution. Now, slowly add conc. H_2SO_4 to it followed by cooling, then add freshly prepared $FeSO_4$ solution slowly keeping the test tube inclined and wait for few seconds.
What do you see at the junction of two liquid layers?

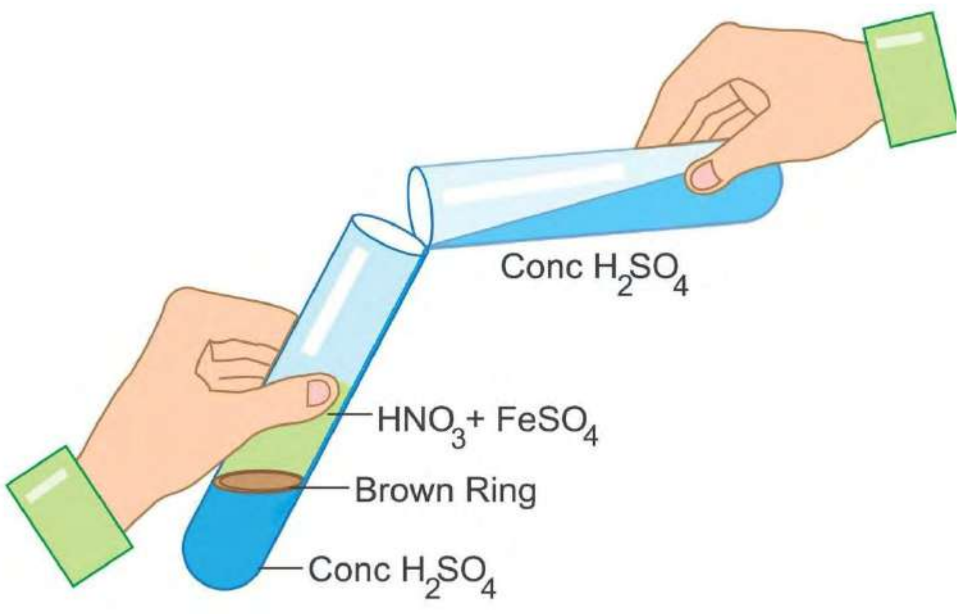


Figure 9.17: Demonstration of ring test

Ring test (or brown ring test) is the test that detects the presence of nitrate ions by the formation of a brown colored ring at the junction of two layers of the solution. This brown ring is formed because of the formation of the brown-colored iron complex, which is called the brown ring complex.

This test is based on the fact that nitrate ion acts as an oxidizing agent. In the reaction mixture, reduction of nitrate ion takes place by Fe^{2+} and itself gets oxidized to Fe^{3+}. Nitric acid is reduced to nitric oxide gas (NO) which is absorbed by a layer of $FeSO_4$ and forms a brown ring at the junction of two layers as shown in the figure.

$\{NO_3^- + H_2SO_4 \rightarrow HNO_3 + HSO_4^-\}\times 2$

$2HNO_3 \rightarrow H_2O + 2NO + 3[O]$

$\{2FeSO_4 + [O] + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + H_2O\}\times 2$

$2NO_3^- + 6FeSO_4 + 5H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + 2HSO_4^- + 4H_2O + 2NO$

$FeSO_4 + NO \rightarrow FeSO_4.NO$ (Nitrosyl ferrous sulphate)

On shaking the test tube the ring disappears.

Facts

Ring test can also be used to determine the presence of nitrate ion in many food samples, soil and water as high quantity of this ion in food will cause food poisoning, make soil more acidic and deteriorate the quality of water. Water containing nitrate ion more than 10 ppm is not safe to drink.

Test Yourself!

Why is freshly prepared ferrous sulphate solution used in ring test?

$3HCl + HNO_3 \longrightarrow NOCl + 2H_2O + 2[Cl]$
(nitrosyl chloride)

$Au + 3[Cl] \longrightarrow AuCl_3$

$AuCl_3 + HCl \longrightarrow HAuCl_4$
(chloroauric acid)

Project work

Prepare a separate chart paper presentation on each the following headings-

a. Oxyacids of nitrogen including oxidation state of N.
b. Applications, harmful effects and safety handlings of ammonia.
c. Various reduced products of concentrated nitric acid with their oxidation state.
d. Laboratory test of nitrate ion.
e. Action of nitric acid on iron and zinc under various conditions.

Exercise

A. Multiple choice questions

1. Why is N_2 relatively non-reactive?

a. It is first element of group 15.

b. Its dissociation energy is high.

c. Its atomic radius is small.

d. It is highly electronegative.

2. Which of the following chemicals is used as a refrigerant?

a. NH_4OH b. NH_4Cl c. NH_4NO_3 d. NH_3

3. Why does ammonia possess higher m.p. and b.p. as compared to other hydride of the same group?

a. Co-ordinate bond b. Hydrogen bond c. Strong ionic bond. d. Strong metallic bond

4. Which of the following combines with Fe^{2+} ion to form a brown complex?

a. N_2O_3 b. NO c. N_2O d. N_2O_5

5. Pick out the metal that liberates hydrogen from dilute HNO_3.

a. Zn b. Cu c. Mg d. Pt

6. What product is expected when Zn is treated with very dilute nitric acid?

a. NH_4NO_3 b. NO c. N_2O_5 d. N_2O

7. Passivity of iron is due to the formation of a thin layer. Which of the following is the formula of this thin layer?

a. $Fe(OH)_2$ b. Fe_3O_4 c. FeO d. $Fe(NO_3)_3$

8. What is the major product formed when P_4 reacts with conc. HNO_3?

a. H_3PO_4 b. H_3PO_2 c. P_2O_5 d. PH_3

9. For ammonia, which of the following is correct?

a. Lewis acid b. Arrhenius acid c. Lewis base d. Arrhenius base

10. What is the molecular formula of hyponitrous acid?

a. HNO_3 b. HNO_4 c. HNO_2 d. $H_2N_2O_2$

11. What is the role of ammonia in the reaction below?

$FeCl_3 + 3NH_4OH \rightarrow Fe(OH)_3 \downarrow + NH_4Cl,$

a. Precipitant b. Oxidizing agent c. Reducing agent d. Dehydrating agent

12. Which of the following chemicals is used in the brown ring test?

a. $FeCl_3$ b. $FeSO_4$ c. $CuSO_4$ d. $CuCl_2$

13. Identify X in the following reaction:

$Zn + X \rightarrow Zn(NO_3)_2 + NO + H_2O$

a. Very dil. HNO_3 b. Dil. HNO_3 c. Moderately conc. HNO_3 d. Hot and conc. HNO_3

14. What is the reduced product of nitric acid when KI is treated with conc. HNO_3?

a. Very dil. HNO_3 b. Very cold & Dil. HNO_3 c. Moderately conc. HNO_4 d. Conc. HNO_3

15. Which of the following mixtures represent aqua-regia?

a. 3 vol. HNO_3 + 1 vol. HCl b. 3 vol. HNO_3 + 1 vol. HCl
c. 3 vol. HNO_3 + 2 vol. HCl d. 1 vol. HNO_3 + 3 vol. HCl

16. Pick out the formula of a brown ring that confirms the presence of nitrate ion?

a. $FeSO_4.NO$ b. $FeSO_4.N_2O$ c. $Fe_2(SO_4)_3.NO$ d. $FeSO_4.NO_2$

17. Which of the following is most acidic?

a. NaOH b. Alk. $KMnO_4$ c. NH_3 d. $Ca(OH)_2$

B. Short answer question

1. Give reason:
   a. Ammonia turns moist red litmus paper into blue.
   b. Concentrated nitric acid turns yellow-brown on long standing.
   c. Nitrogen is filled in a tungsten lamp.
   d. Ammonia is highly soluble in water.
   e. Ammonia is a complexing agent.

2. Differentiate between:
   a. Active nitrogen and molecular nitrogen
   b. Ammonia and nitric acid

3. Give a few large-scale applications of ammonia.

4. Write your observation when ammonia gas is passed through:
   a. Moist red litmus paper.
   b. Aqueous copper sulphate solution.
   c. Aqueous ferric chloride solution.
   d. Concentrated hydrochloric acid.
   e. Mercurous nitrate paper.

5. Write a note on:

a. Harmful effects of ammonia b. Ring test c. Oxidizing nature of nitric acid d. Basic nature of ammonia

6. What happens when-

a. Mercurous nitrate paper comes in contact with ammonia?

b. Very dilute nitric acid is poured on magnesium metal?
c. Very hot and concentrated nitric acid comes in contact with copper?
d. Sulphur dioxide is treated with concentrated nitric acid?
e. Concentrated hydrochloric acid is treated with ammonia?

7. How does copper react with nitric acid under different conditions?

8. Give a reaction to show the acidic nature of nitric acid. How would you perform the laboratory test of nitric acid? Give a balanced chemical equation.

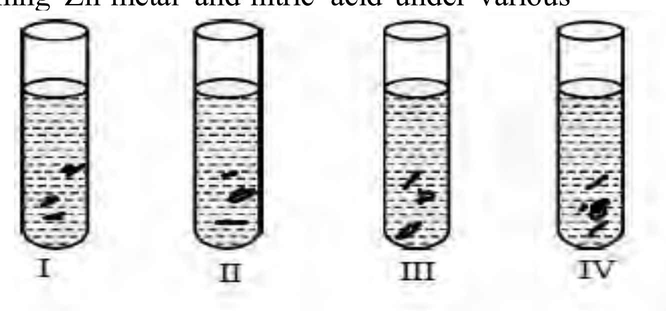
9. In what way does dilute nitric acid differ from other acids when it reacts with metals?

10. Show your acquaintance with:

a. Aqua-fortis b. Aqua-regia

Long answer question

11. Observe the given test tubes (I-IV) each containing Zn metal and nitric acid under various conditions and answer the following questions:



Write balanced reaction for I, II, III and IV when-

a. I contains very hot and conc. HNO_3.

b. II contains moderately conc. HNO_3.

c. III contain cold and dilute HNO_3.

d. IV contains cold and very dilute HNO_3.

12. Mention any three important uses of nitric acid. Also mention the property of nitric acid involved in the use.

13. (A) Dilute nitric acid is generally considered a typical acid except for its reaction with metals. In what way dilute nitric acid differs from other mineral acids like H_2SO_4 and HCl when it reacts with metals?
(B) Write the equation for the reaction of dilute and conc. nitric acid with zinc.

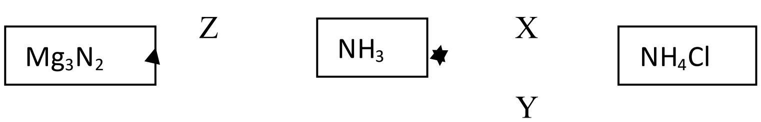
14. (A) Explain with the help of a balanced chemical equation, the brown ring test for nitrate radical.
(B) Why is freshly prepared ferrous sulphate solution used in the above test?

15. An unknown sample of acid is given to you. How do you prove that acid as nitric acid? Write the balanced chemical equation to identify the acid as nitric acid. Draw the Lewis dot structure of nitric acid and write its one use.

16. Complete the given table:

Species	HNO_3	Gas produced	Reaction involved
Mg	Dilute		
Zinc	Conc.		
Cu	Conc.		
H_2S	Dilute		
C	Moderate		

17. Study the given flow chart and write the balanced chemical equation to represent the reaction X, Y and Z.



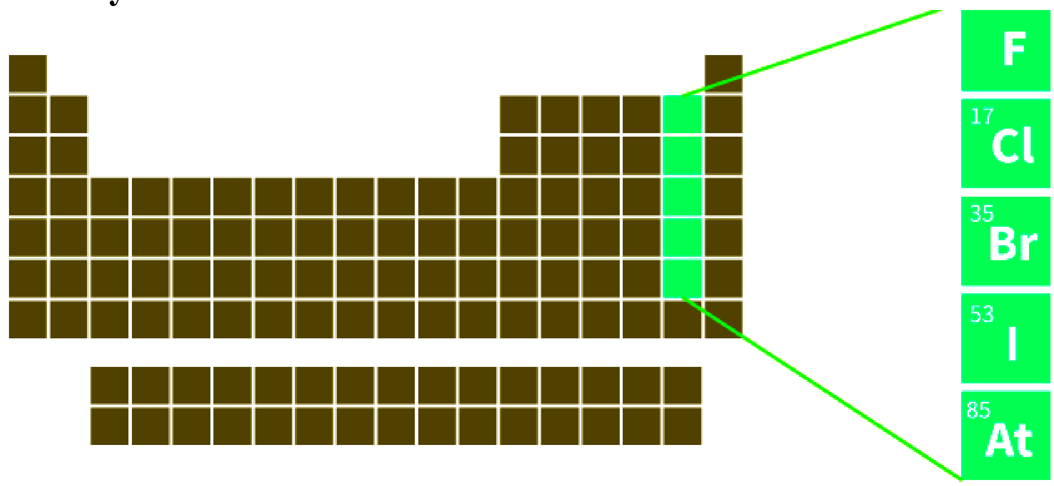
18. Write the balanced chemical equation for each of the following observation:
   a. Addition of ammonia solution to ferric chloride.
   b. Addition of excess of ammonia to copper sulphate solution.
   c. Addition of ammonia to mercurous nitrate solution.
   d. Addition of phosphorous to hot and conc. nitric acid.
   e. Addition of conc. sulphuric acid and freshly prepared ferrous sulphate to nitric acid.

19. Complete the given table:

Metal/ Non-metal	**Concentration and temperature of nitric acid**	**Products**
Zn		$Zn(NO_3)_2 + NH_4NO_3 + H_2O$
	moderate	$Zn(NO_3)_2 + NO + H_2O$
Fe	very dilute	
		$Cu(NO_3)_2 + NO_2 + H_2O$
Mg	very dilute	
		$Cu(NO_3)_2 + N_2O + H_2O$
C	Concentrated	

9.5: Halogens

Activity: 21



a. **What is the group of the elements highlighted in the periodic table?**
b. **Why are these elements kept in the same group?**
c. **Why are these elements called halogens?**

We all are familiar with group 17 (VIIA) of the periodic table. These elements include Fluorine (F), Chlorine (Cl), Bromine (Br), Iodine (I) and Astatine (At). The word halogen is derived from the Greek word, halos =sea, and genus = producing. Thus halogens mean 'salt forming compounds'. Due to this reason they are called halogens because they react rapidly with metals to form wide range of salts like calcium fluoride (CaF_2), sodium chloride (NaCl), potassium iodide (KI), silver bromide (AgBr), etc.

Elements	**Symbol**	**Atomic Number**	**Atomic Mass (amu)**	**Electronic configuration**
Fluorine	F	9	19	[He] $2s^2 2p^5$
Chlorine	Cl	17	35.5	[Ne] $3s^2 3p^5$
Broimine	Br	35	80	[Ar] $3d^{10} 4s^2 4p^5$
Iodine	I	53	127	[Kr] $4d^{10} 5s^2 5p^5$
Astatine	At	85	210	[Xe] $4f^{14}\ 5d^{10} 6s^2 6p^5$

Halogens are very reactive non-metals. Their high reactivity is due to their high value of electronegativity. So they are not found in free-state, rather they occur in a combined state in nature. Some of the sources of halogens are listed below:

Halogens	Source
Fluorine(F)	Fluorspar (CaF_2), Cryolite (Na_3AlF_6), Fluorapatite [$3Ca_3(PO_4)_2$,CaF_2] etc.
Chlorine(Cl)	Sea water or rock salt or common salt (NaCl), Sylvine (KCl), Carnallite ($KCl.MgCl_2.6H_2O$), Horn silver (AgCl) etc.
Bromine(Br)	Sea water, salt lakes like NaBr, KBr, $MgBr_2$, $CaBr_2$ etc.
Iodine(I)	NaIO$_3$)

Yellow gas (F_2) Yellow gas(Cl_2) Brown liquid (Br_2) Brown solid(I_2)

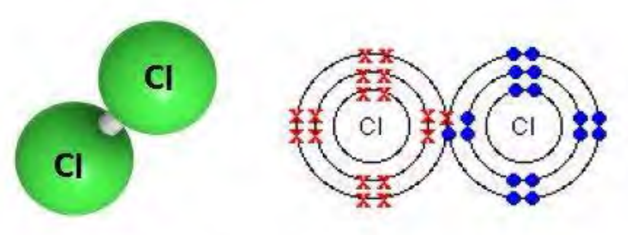
Facts

Fluorine is added in the toothpaste in the form of sodium fluoride which prevents tooth decay. Bromine is used in the swimming pool to prevent germs. Iodine is added to alcohol and used as an antiseptic.

Test Yourself!

Why are halogens not found in free-state?

9.5.1: General characteristics of halogens



Chlorine (Cl_2)

1. **Electronic configuration:** All halogens contain 7 electrons in their valence shell. Their general valence shell electronic configuration is ns^2np^5.
2. **Atomicity:** Halogens exist as non-polar diatomic molecules. F_2, Cl_2, Br_2, I_2
3. **Electro-negativity:** Halogens have high electro-negativity (EN) value in the corresponding period due to their smallest atomic size and higher nuclear charge. Fluorine is the most electronegative element. Their electro-negativity decreases down the group.

Figure-9.18: Elecronegativity trend down the group

4. **Electron Affinity:** Halogens have high electron affinity (EA) value in the corresponding periods due to the smallest atomic size and high nuclear charge. By gaining one electron these elements attain the electronic configuration of noble gas. EA value decreases down the group. However, Cl has a higher EA value than F because of strong inter-electronic repulsion in the relatively compact 2p orbital of F. As a result of which, the incoming electron experiences lesser attraction in F than Cl.

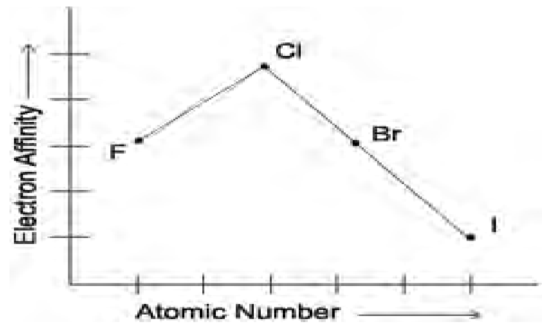


Figure 9.19: Electronaffinity trend -down the group

5. **Oxidation State:** Halogens exhibit -1 oxidation state. F exhibits -1 oxidation state only but other halogens exhibit +1, +3, +5, & +7 oxidation state also. F cannot show any positive oxidation state. It is a second period element and cannot expand its octet due to non-availability of d-orbital. Other halogens have d-orbital and therefore, can expand their octet and exhibit variable positive oxidation state in addition to -1 oxidation state.

Halogens	**Oxidation State**
Fluorine(F)	-1
Chlorine(Cl)	-1, +1, +3, +4,+5,+6 & +7
Bromine(Br)	-1, +1, +3,+4,+5 & +6
Iodine(I)	-1, +1, +3, +5 & +7

6. **Ionization Energy:** Halogens have very high ionization energy value as they have very little tendency to lose electrons. Due to increase in atomic size, ionization energy decreases down the group from F to I.
7. **Atomic Radius:** Halogens have smallest atomic radii in their respective periods because of high effective nuclear charge. The atomic radius increases down the group. Similar is the case for ionic radius too i.e,

$F^- < Cl^- < Br^- < I^-$

8. **Oxidizing nature:** Halogens are powerful oxidizing agents as they have a very high tendency to gain electrons. Fluorine is the most powerful oxidizing agent. The oxidizing power decreases down the group.

$F_2 > Cl_2 > Br_2 > I_2$

9. **Non-metallic character:** Halogens are non-metallic in nature due to high electro-negativity and high ionization energy. However, the non-metallic character of halogens decreases down the group.

10. **Melting point and boiling point**: Both m.p. and b.p. of halogens increase from fluorine to iodine due to increase in Van der waal's force among their molecules.

$F_2 < Cl_2 < Br_2 < I_2$

11. **Reactivity:** Halogens are the most reactive non-metals because of their high tendency to acquire noble gas electronic configuration. Fluorine is the most reactive element known.

Test Yourself!

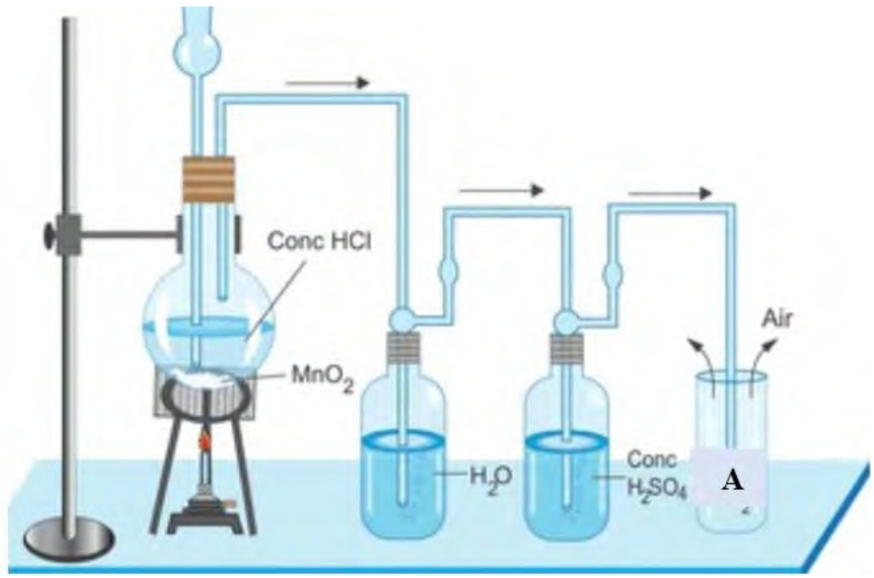
a. Why do halogens possess highest ionization energy?
b. What makes halogens powerful oxidizing agents?

9.5.2: Comparative study on preparation of halogens

Activity: 22

Observe the picture of laboratory preparation of one of the halogens and answer the following questions:

a. **Name the gas A produced in the gas jar.**
b. **Write the balanced chemical equation for the production of gas**
c. **Is the gas lighter or heavier than air? Why?**



1. By chemical oxidation of halides:

Except F_2 other halogens can be prepared by this method. Cl_2, Br_2 and I_2 can be prepared by the oxidation of halides with strong oxidizing agents like MnO_2, $KMnO_4$, $K_2Cr_2O_7$, etc.

$$MnO_2 + 2HX \rightarrow MnX_2 + H_2O + [O]$$
$$2HX + [O] \rightarrow X_2 + H_2$$
$$MnO_2 + 4HX \rightarrow MnX_2 + X_2 + 2H_2O$$

Where, X= Cl, Br and I

Instead of halo acids (HX) a mixture of halide salt and concentrated sulphuric acid can also be taken with MnO_2.

$$NaX + H_2SO_4 \rightarrow NaHSO_4 + HX \}\times 2$$
$$MnO_2 + H_2SO_4 \rightarrow MnSO_4 + H_2O + [O]$$
$$2HX + [O] \rightarrow X_2 + H_2O$$
$$2NaX + 3H_2SO_4 + MnO_2 \rightarrow NaHSO_4 + MnSO_4 + X_2 + 2H_2O$$

Where, X= Cl, Br and I

Although all halogens are prepared by the same general method their method of collection is different.

a. Laboratory preparation reaction of Cl_2

$$\{NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl \}\times 2$$
$$MnO_2 + H_2SO_4 \rightarrow MnSO_4 + H_2O + [O]$$
$$2HCl + [O] \rightarrow Cl_2 + H_2O$$
$$2NaCl + 3H_2SO_4 + MnO_2 \rightarrow 2NaHSO_4 + MnSO_4 + Cl_2 + 2H_2O$$

b. Laboratory preparation reaction of Br_2

$$\{NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr \}\times 2$$
$$MnO_2 + H_2SO_4 \rightarrow MnSO_4 + H_2O + [O]$$

$2HBr + [O] \rightarrow Br_2 + H_2O$

$2NaBr + 3H_2SO_4 + MnO_2 \rightarrow 2NaHSO_4 + MnSO_4 + Br_2 + 2H_2O$

c. Laboratory preparation reaction of I_2

$\{NaI + H_2SO_4 \rightarrow NaHSO_4 + HI\} \times 2$

$MnO_2 + H_2SO_4 \rightarrow MnSO_4 + H_2O + [O]$

$2HI + [O] \rightarrow I_2 + H_2O$

$2NaI + 3H_2SO_4 + MnO_2 \rightarrow 2NaHSO_4 + MnSO_4 + I_2 + 2H_2O$

2. By displacement reaction halogens

More reactive halogens can displace less reactive halides to produce halogens. Example,

Br_2 can be prepared by passing Cl_2 gas through a bromide salt solution. Here, Cl_2 oxidizes bromide to give bromine.

$2KBr + Cl_2 \rightarrow 2KCl + Br_2$

$MgBr_2 + Cl_2 \rightarrow MgCl_2 + Br_2$

I_2 can be prepared by passing Cl_2 and Br_2 in an iodide salt solution. Here, both Cl_2 and Br_2 can oxidize iodide to give iodine.

$2KI + Cl_2 \rightarrow 2KCl + I_2$

$2KI + Br_2 \rightarrow KBr + I_2$

F_2 cannot be prepared by this method as halogens lying below fluorine cannot displace fluoride from its salt.

9.5.2.1: Chemical properties of halogens

All the halogens are most reactive non-metals. Their high reactivity is due to low bond dissociation energy of their molecules, high electro-negativity and electron affinity. Fluorine is the most reactive of the elements known. The reactivity of the halogens follows the order- $F_2 > Cl_2 > Br_2 > I_2$

Some of the important chemical properties of halogens Cl_2, Br_2 and I_2 are discussed below:

1. Action with water: All halogens react with water to produce two acids

Water + Halogens → Acid-1 + Acid-2

$H_2O + X_2 \rightarrow \underset{\text{(Hydrohalic acid)}}{HX} + \underset{\text{(Hypohalous acid)}}{HOX}$

$X_2 = Cl_2$, Br_2 and I_2

HOX being unstable gets decomposed to form oxygen and hydrohalic acid when exposed to light.

$HOX \xrightarrow{light} 2HX + O_2$

Where, X: Cl, Br & I

Example,

$H_2O + Cl_2 \rightarrow \underset{\text{(Hydrochloric acid)}}{HCl} + \underset{\text{(Hypochlorous acid)}}{HOCl}$

$H_2O + Br_2 \rightarrow HBr + HOBr$

(Hydrocbromicacid) (Hypobromous acid)

I_2 is very slightly soluble in water. However, its solubility can be increased in presence of KI as it forms potassium triiodide (KI_3).

$$2KI + I_2 \rightarrow 2KI_3$$

Chlorine dissolved in water is called chlorine water (Cl_2+ H_2O). However, if this chlorine water is left to stand for extended period of time then it begins to lose its yellow colour due to the production of HCl and HOCl. The chlorine water is added to swimming pool, drinking water, etc. as it kills parasites, bacteria and viruses.

2. Action with alkali: Halogens react with alkali KOH or NaOH to give two different products under two different conditions.

(i) Action with cold and dilute alkali: Halogens react with cold and dilute alkali to form metal halides and metal hypohalites.

$$Cl_2 + H_2O \rightarrow HCl + HClO$$

$$NaOH + HCl \rightarrow NaCl + H_2O$$

$$NaOH + HClO \rightarrow NaClO + H_2O$$

$$\underset{\text{(cold \& dil.)}}{Cl_2 + 2NaOH} \rightarrow \underset{\text{(Sodium chloride)}}{NaCl} + \underset{\text{(Sodium hypochlorite)}}{NaClO} + 2H_2O$$

Br_2 and I_2 react in the similar manner.

(ii) Action with hot and conc. alkali: Halogens react with hot and conc. alkali to form metal halide and metal halate.

Figure 9.20: $NaClO_3$ is used as a weedicide

$$\{NaOH + Cl_2 \rightarrow NaCl + NaClO + H_2O\} \times 3$$

$$NaClO + HCl \rightarrow NaClO_3 + 2NaCl$$

$$\underset{\text{(hot \& conc.)}}{3Cl_2 + 6NaOH} \rightarrow \underset{\text{(Sod. chloride)}}{5NaCl} + \underset{\text{(Sod. hypochlorate)}}{NaClO_3} + 3H_2O$$

Br_2 and I_2 react in the similar manner.

Note: In the above reactions one atom of halogen is oxidized and other atom of the same halogen is reduced. Therefore, the reaction of halogen with alkali is called disproportionation (self-oxidation and reduction) reaction.

3. Action with ammonia: Cl_2 and Br_2 react with excess of NH_3 to produce nitrogen gas.

$$\underset{\text{(excess)}}{8NH_3} + 3Cl_2 \rightarrow N_2 + 6NH_4Cl$$

$$\underset{\text{(excess)}}{8NH_3} + 3Br_2 \rightarrow N_2 + 6NH_4Br$$

But with excess of halogens (Cl_2 and Br_2) nitrogen tri-halide is produced.

$$NH_3 + \underset{\text{(excess)}}{3Cl_2} \rightarrow NCl_3 + 3HCl$$

$$NH_3 + \underset{\text{(excess)}}{3Br_2} \rightarrow NBr_3 + 3HBr$$

Iodine (I_2) reacts with ammonia to produce nitrogen tri-iodide ammoniate and ammonium iodide.

$$2NH_3 + 3I_2 \rightarrow NI_3.NH_3 + HI$$
$$\{HI + NH_3 \rightarrow NH_4I\} \times 2$$
$$5NH_3 + 3I_2 \rightarrow NI_3.NH_3 + 3NH_4I$$

Facts

Nitrogen triiodide is an explosive chemical. It is often called **'touch power'** due to its inherent instability which causes it to explode even when touched with a feather.

Figure 9.21: Nitrogen tri-iodide an explosive

4. Oxidizing character

$$F_2(g) + 2e^- \longrightarrow 2F^-(aq)$$
$$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$$
$$Br_2(l) + 2e^- \longrightarrow 2Br^-(aq)$$
$$I_2(s) + 2e^- \longrightarrow 2I^-(aq)$$

All halogens are good oxidizing agents because of their high tendency to accept electrons. The oxidizing power of halogens follows the order, $F_2 > Cl_2 > Br_2 > I_2$. Cl_2, Br_2& I_2 oxidize various substances in water due to the liberating nascent oxygen when dissolved in water.

Some of the reactions showing oxidizing nature of halogens are:

i. Action with H_2S: Hydrogen sulphide is oxidized to sulphur by all halogens.

$$H_2O + Cl_2 \rightarrow HCl + HClO$$
$$HClO \rightarrow HCl + [O]$$
$$H_2S + [O] \rightarrow H_2O + S$$
$$H_2S + Cl_2 \rightarrow 2HCl + S$$

Br_2 and I_2 also react in similar manner.

ii. Action with SO_2: Sulphur dioxide is oxidized to sulphuric acid by all halogens.

$$H_2O + Cl_2 \rightarrow HCl + HClO$$
$$HClO \rightarrow HCl + [O]$$
$$SO_2 + [O] + 2H_2O \rightarrow H_2SO_4$$
$$SO_2 + 2H_2O + Cl_2 \rightarrow 2HCl + H_2SO_4$$

Br_2 and I_2 also react in similar manner.

iii. Action with ferrous salt: Ferrous salts are oxidized to ferric salt by all halogens.

$$H_2O + Cl_2 \rightarrow HCl + HClO$$
$$HClO \rightarrow HCl + [O]$$

$$4FeSO_4 + 2H_2SO_4 + [O] \rightarrow 2Fe_2(SO_4)_3 + H_2O$$

$$4FeSO_4 + 2H_2SO_4 + Cl_2 \rightarrow 2Fe_2(SO_4)_3 + 2HCl + H_2O$$

Br_2 and I_2 also react in similar manner.

iv. Action with thiosulphate ($Na_2S_2O_3$): Cl_2 and Br_2 oxidize sodium thiosulphate to free sulphur.

$$H_2O + Cl_2 \rightarrow HCl + HClO$$

$$HClO \rightarrow HCl + [O]$$

$$Na_2S_2O_3 + [O] \rightarrow Na_2SO_4 + S$$

$$Na_2S_2O_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl + S$$

But I_2 oxidizes $Na_2S_2O_3$ to sodium tetrathionate ($Na_2S_4O_6$).

$$Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$$

5. Bleaching action

Activity: 23

Observe the picture carefully and answer the following questions:

a. What do you mean by bleach?

b. Can you list some examples of bleaching substances?

c. What could be the reason for using chlorine as a bleach?

Halogens being good oxidizing agents show excellent bleaching action by oxidation. Fluorine being most reactive is not used as a bleaching agent as it destroys the substance to be bleached. Iodine is the weakest bleaching agent and is also not preferred for bleaching the substance. Bromine is a mild bleaching agent and is also not generally used to bleach a substance. Chlorine is a strong and excellent bleaching agent among all halogens. It bleaches different substances. Its bleaching action is irreversible and permanent due the production of nascent oxygen.

$$\underset{\text{(moisture)}}{H_2O} + Cl_2 \rightarrow 2HCl + HClO$$

$$HClO \rightarrow HCl + [O]$$

$$\text{Colored Material} + [O] \rightarrow \text{Colorless material}$$

Cl_2 is used to bleach hard substances like textiles, wood, paper and pulp etc.

Test Yourself!

a. Why is chlorine added to swimming pool?

b. Define chlorine water.

c. What happens when iodine reacts with sodium thiosulphate?

Uses of halogens

S. N.	Cl_2	Br_2	I_2
1.	It is used to bleach paper, pulp, textiles.	It is used to prepare ethylenedibromide that is used as insecticides and as a scavanger of lead in petrol.	It is used as an antiseptic. A solution of I_2, KI and rectified spirit called tincture of iodine, is used for disinfecting wounds.
2.	It is used as a sanitation and disinfectant.	It is used in making certain fungicides.	Human body needs traces of iodine to prevent goiter.
3.	It is used in the manufacture of PVC (polyvinylchloride)	It is used in making flame proofing agents, dyes and sanitizers.	AgI is used in cloud seeding (production of artificial rain).
4.	It is used in the extraction of gold.	Its product AgBr is used in preparing photographic films.	HIO_4 is used as an oxidizing agent in organic chemistry.

9.5.4: Comparative study on preparation, properties and uses of haloacids

Activity: 24

Observe the table given:

Reactivity of hydrogen	Decreases from fluorine to iodine
Stability	Decreases from HF to HI
Volatility of the hydrides	HF < HI < HBr < HCl
Thermal stability	HF > HI > HBr > HCl
Boiling point	HCl < HBr < HI
Acid strength	Increases from HF to HI

a. Which of the halides is most difficult to decompose by heating?
b. Identify the compounds that evaporates easily among.

Halogens (X: F, Cl, Br & I) form binary compounds with hydrogen commonly known as hydrogen halides (or haloacids). Their aqueous solution is called hydrohalic acids.

$$\underset{\text{(Hydrogen chloride)}}{HCl(g)} + H_2O \rightarrow \underset{\text{(hydrochloric acid)}}{HCl\ (aq)}$$

Let us discuss some of the methods of preparation haloacids (HX: HCl, HBr & HI)

1. By the action of conc. sulphuric acid on metal halides (Laboratory preparation)
In laboratory HCl gas is prepared by heating common salt with conc. sulphuric acid.

$$NaCl + \text{conc. } H_2SO_4 \rightarrow NaHSO_4 + HCl \text{ (at low temperature)}$$

$NaCl + conc. H_2SO_4 \rightarrow Na_2SO_4 + HCl$ (at high temperature)

$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

This method cannot be applied to prepare HBr and HI. This is because both HBr and HI are powerful reducing agents. Therefore, HBr and HI produced quickly reduce sulphuric acid to sulphur dioxide and themselves get oxidized to Br_2 and I_2 respectively.

$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr\} \times 2$

$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$

$2HBr + [O] \rightarrow H_2O + Br_2$

$2NaBr + 3H_2SO_4 \rightarrow NaHSO_4 + SO_2 + Br_2 + 2H_2O$

Similarly,

$KI + H_2SO_4 \rightarrow KHSO_4 + HI\} \times 2$

$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$

$2HI + [O] \rightarrow H_2O + I_2$

$2KI + 3H_2SO_4 \rightarrow 2KHSO_4 + SO_2 + I_2 + 2H_2O$

Hence, HBr and HI are prepared by heating bromide and iodide salt with non-oxidizing agents like phosphoric acid.

$NaBr + H_3PO_4 \rightarrow NaH_2PO_4 + HBr$

$NaI + H_3PO_4 \rightarrow NaH_2PO_4 + HI$

Test yourself!

a. What happens when a mixture of NaI and conc. H_2SO_4 is heated?
b. What happens when sodium chloride is heated with conc. sulphuric acid?
c. Why does aqueous solution of hadrohalic acid behave as acid?

2. Laboratory Preparation of HBr and HI

In laboratory HBr is prepared by the action of bromine on moist red phosphorous.

$P_4 + 6Br_2 \rightarrow 4PBr_3$

$\{PBr_3 + 3H_2O \rightarrow H_3PO_3 + 3HBr\} \times 4$

$P_4 + 6Br_2 + 12\ H_2O \rightarrow 4H_3PO_3 + 12HBr$

Similarly, HI is preapred by the action of water on a mixture of red phosphorous and iodine.

$P_4 + 6I_2 \rightarrow 4PI_3$

$\{PI_3 + 3H_2O \rightarrow H_3PO_3 + 3HI\} \times 4$

$P_4 + 6I_2 + 12\ H_2O \rightarrow 4H_3PO_3 + 12HI$

Test yourself!

Write the reaction for the preparation of HBr from phosphorous and iodine.

Facts

i. All hydrogen halides are colorless pungent smelling gases.

ii. All hydrogen halides are extremely soluble in water.

iii. All hydrogen halide gases do not change the blue litmus paper into red. However, they turn moist blue litmus paper into red showing acidic nature.

iv. The boiling point of hydrogen halide is in the order, HF> HI >HBr >HCl. HF has abnormally high b.p. due to intermolecular hydrogen bonding between their molecules.

9.4.5 Comparative chemical properties of HX (HCl, HBr and HI):

Some of the chemical properties of hydrogen halides are listed below:

1. Reducing nature

Hydrogen halides behave as a reducing agent. Their reducing power decreases in the order, HI> HBr> HCl >HF. This is because on moving from F to I in the group bond dissociation energy of H-X decreases and reducing strength increases. Some examples that show reducing nature of haloacids are:

HCl reduces MnO_2, $KMnO_4$, $K_2Cr_2O_7$ etc.

$$MnO_2 + 2HCl \rightarrow MnCl_2 + H_2O + [O]$$
$$2HCl + [O] \rightarrow H_2O + Cl_2$$
$$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + H_2O$$

$$2KMnO_4 \rightarrow K_2O + 2MnO + 5[O]$$
$$K_2O + 2HCl \rightarrow 2KCl + H_2O$$
$$\{MnO + 2HCl \rightarrow MnCl_2 + H_2O\}\times 2$$
$$\{2HCl + [O] \rightarrow Cl_2 + H_2O\}\times 5$$
$$2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 5Cl_2 + 8H_2O$$

HBr and HI react in the similar manner as HCl does. HI can even reduce ferric salt to ferrous salt, sulphuric acid to sulphur dioxide, nitric acid to nitrogen dioxide, etc.

$$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$$
$$2HBr + [O] \rightarrow Br_2 + H_2O$$
$$2HBr + H_2SO_4 \rightarrow H_2O + SO_2 + Br_2$$

$$CuSO_4 + 2HI \rightarrow CuI_2 + H_2SO_4$$
$$CuI_2 \rightarrow Cu_2I_2 + I_2$$
$$CuSO_4 + 4HI \rightarrow Cu_2I_2 + I_2 + H_2SO_4$$

$$FeCl_3 + 2HI \rightarrow 2FeCl_2 + 2HCl + I_2$$
$$2HNO_3 + HI \rightarrow 2H_2O + 2NO_2 + I_2$$
$$H_2SO_4 + HI \rightarrow 2H_2O + SO_2 + I_2$$

Test Yourself!

a. What happens when acidified $KMnO_4$ solution comes in contact with HBr?
b. How does HI react with acidified potassium dichromate solution?

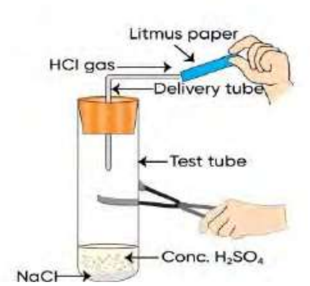
2. Acidic nature

Activity: 25

Take 1 g of solid NaCl in a clean and dry test tube and add few drops of conc. H_2SO_4. Set up the apparatus as shown in the picture and perform the activity.

a. Which gas is produced in the test tube?

b. What happens if you bring moist blue litmus paper near the delivery tube?



Perfectly dry hydrogen halides have no effect on blue litmus paper.

However, their aqueous solution behaves as a strong acid. The aqueous solution of hydrogen halides is called hydrohalic acid or halogen acid. They completely ionize in water-

$HX \rightarrow H^+ + X^-$ where, X= F, Cl, Br and I

The order of acidic strength of halo acids is HI> HBr> HCl> HF

HI is the strongest acid because of the weaker H^+ I^- interaction that facilitates the dissociation of H^+ (proton) more easily from HI. Moreover, I^- is much larger than other halide ions which results in dispersal of negative charge over large space and stabilize I^-.

Being acidic in nature hydrogen halides react with bases, metals, metal oxides, metal carbonates, metal bicarbonates etc. as shown below:

$HCl + NaOH \rightarrow NaCl + H_2O$

$HCl + Na \rightarrow NaCl + H_2\uparrow$

$2HCl + Na_2CO_3 \rightarrow 2NaCl + CO_2\uparrow + H_2O$

$2HCl + MgO \rightarrow MgCl_2 + H_2O$

$HCl + 2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

Test yourself!

a. Give a reaction to show that HI is acidic in nature.

b. HI is stronger acid than HCl, why?

3. Solubility:

All hydrogen halides are extremely soluble in water. This is because they can form intermolecular hydrogen bonds with water molecules. The solubility of HX in water increases from HCl to HBr to HI.

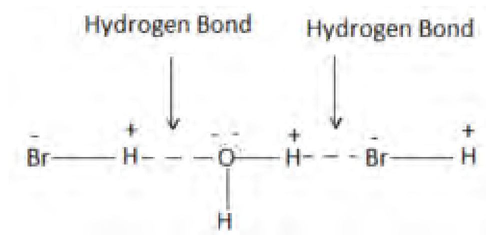


Figure 9.21: H-bonding between H_2O and HBr

Test yourself!

Why does HCl dissolve in water?

9.5.3 Test for halides (X^-: Cl^-, $Br^-$and I^-)

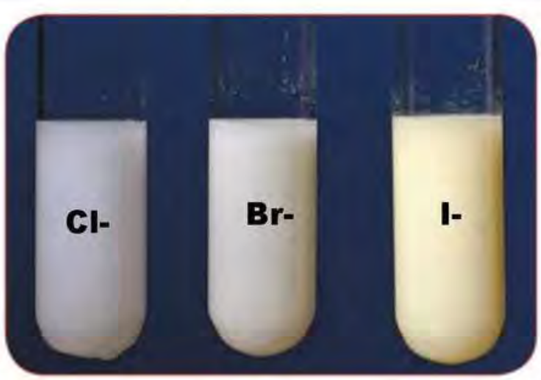
Activity: 26

Take a pinch of chloride, bromide and iodide salt separately in three different test tubes and make their aqueous solution by adding little water. Add a few drops of silver nitrate solution to each test tube and observe the color of the precipitate obtained.

a. Name the color of the chloride, bromide and iodide precipitate.

b. Write the formula of each colored precipitate obtained.

HALIDE ION	COLOUR OF PRECIPITATE FORMED	NAME OF PRECIPITATE FORMED
Chloride ion (Cl^-)	White	Silver chloride
Bromide ion (Br^-)	Cream	Silver bromide
Iodide ion (I^-)	Yellow	Silver iodide



Halides can be tested in laboratory in following ways:

Test	Cl^-	Br^-	I^-
$AgNO_3$ Test	Gives white ppt. with $AgNO_3$ solution which disappears in addition to excess of ammonia solution and ppt. reappears in addition of conc. HNO_3. Reaction Involved: $AgNO_3 + Cl^- \rightarrow AgCl\downarrow + NO_3^-$ (white ppt.) $AgCl + NH_3(aq) \rightarrow Ag(NH_3)_2Cl$ $Ag(NH_3)_2Cl + HNO_3 \rightarrow AgCl\downarrow$ (whiteppt.)	Gives pale yellow ppt. with $AgNO_3$ which is slightly soluble in ammonia solution and ppt. reappears in addition of conc. HNO_3. $AgNO_3 + Cl^- \rightarrow AgBr\downarrow + NO_3^-$ (pale yellow ppt.) $AgCl + NH_3(aq) \rightarrow Ag(NH_3)_2Br$ $Ag(NH_3)_2Br + HNO_3 \rightarrow AgBr\downarrow$ (pale yellow ppt.)	Gives yellow ppt. with $AgNO_3$ solution which is insoluble in ammonia solution. $AgNO_3 + Cl^- \rightarrow AgI\downarrow + NO_3^-$ (yellow ppt.)

Test yourself!

How would you confirm the presence of chloride ion present in rock salt (table salt)?

Uses of haloacids

Some of the uses of haloacids are listed below:

S. N.	HCl	HBr	HI
1.	As a laboratory reagent.	As a laboratory reagent.	As a reducing agent.
2.	Used in the preparation of Cl_2.	Used in the preparation of AgBr&KBr which are used in photography.	Used in medicine.
3.	Used in preparation of aqua-regia, glue, purifying common salt and bleaching textiles.	Used in bromination of organic compounds.	Used to prepare iodide and iodine.

Project work

1. Prepare a chart paper presentation on general characteristics of halogens.
2. Make a chart paper presentation on test of halide ions.
3. Make a chart paper presentation on the use of haloacids.
4. Prepare a list of chemical substances, containing halogen, with their formula that are used in day to day life.

Exercise

A. Multiple choice questions

1. **Which of the following has the highest EA value?**
a. Iodine b. Brominec. Chlorine d. Fluorine

2. **Which of the following is the tincture of iodine?**
a. I_2 and water b. I_2, KI and rectified sprit c. KI and rectified sprit d. I_2 and rectified sprit

3. **Select the correct order of electronegativity of halogens.**
a. Cl> F> Br>I b. Br> I> Cl>F c. F> Cl> Br>I d. I> Br> Cl>F

4. **Which of the following is responsible for the bleaching action of chlorine?**
a. Dehydrating property b. Oxidizing property
c. Reducing property d. Antiseptic property

5. **Pick out the correct order of acidic strength.**
a. HI> HBr >HCl b. HCl> HBr >HI
c. HCl >HBr> HI d. HCl>HBr> HI

6. **Choose the strongest oxidant.**
a. O_2 b. O_3 c. Cl_2 d. F_2

7. **Which of the following shows only one oxidation state?**
a. F b. I c. Cl d. Br

8. **Which of the following can displace bromine from KBr?**
a. I_2 b. Cl_2 c. Br_2 d. HI

9. **The valence shell electronic configuration of an element X is ns^2 np^5. To which group does X belong to?**
a. 1 b. 5 c. 15 d. 17

10. **What product is obtained when Cl_2 reacts with heat and conc. NaOH?**
a. NaCl b. NaCl & $NaClO_3$ c. NaCl &NaOCl d. $NaClO_3$

11. **Name the halogen present in rock salt.**
a. Cl b. At c. F d. Br

12. **What is the molecular formula of Horn silver?**
a. NaCl b. $NaIO_3$ c. AgCl d. CaF_2

13. **Which gas is produced when a mixture of rock salt, manganese dioxide and concentrated sulphuric acid is heated?**
a. O_2 b. Br_2 c. I_2 d. Cl_2

14. **Which of the following is the correct order of oxidizing power of halogens?**
a. $Cl_2>F_2>I_2>Br_2$ b. $F_2>Cl_2>Br_2>I_2$ c. $Cl_2>Br_2>I_2>F_2$ d. $I_2>Br_2>Cl_2>F_2$

15. **Which of the following halogen is generally used in the preparation of non-sticky cooking utensils?**
a. Chlorine b. Bromine c. Fluorine d. Iodine

16. **Pick out the most common bleaching agent.**
a. Cl_2 b. I_2 c. Br_2 d. F_2

17. **Which of the following halogen is commonly added in public water reservoir and swimming pools?**
a. Br_2 b. I_2 c. Cl_2 d. F_2

18. **Identify A^+ and B^- in the following equation:**
$HCl + H_2O \rightarrow A^+ + B^-$
a. A^+ (H) & B^- (Cl^-) b. A^+ (H_3O^+) & B^- (Cl^-)
c. A^+ (HO^-) & B^- (Cl^-) d. A^+ (H^-) & B^- (Cl^+)

19. **What is A in the following reaction?**
$Cl^- + AgNO_3 \rightarrow A + NO_3^-$
a. Ag_2Cl b. $AgCl_2$ c. AgCl d. Ag_2Cl_2

20. **Which reagent is used to test halide ion in the laboratory?**
a. AgCl b. $BaCl_2$ c. DMG d. $AgNO_3$

B. Short answer question

1. Give reason:
a. HCl turns moist blue litmus paper into red.
b. Cl is more electronegative than F.
c. Halogens are never found in free state.
d. HI is a better reducing agent than HCl.
e. HF is soluble in water.

2. Differentiate between:
a. Oxidizing power of Cl_2 and Br_2
b. Reducing power of HCl and HI

3. An element X has electronic configuration [Ne]$3s^2$ $3p^5$.
a. Identify the element X.
b. Write the bleaching action of X_2

c. Write one method of preparing X_2.
d. Write any two uses of X_2.

4. **Give an example each of the following:**
   a. Oxidizing action of chlorine.
   b. Bleaching action of chlorine.
   c. Reducing property of HI.
   d. Acidic nature of hydrohallic acid.
   e. Disproportionation of chlorine.

5. **Write a note on:**
   a. General characteristics of halogens
   b. Uses of haloacids

6. **You are provided three test tube containing chloride, bromide and iodide salt solution separately. Give a test reaction to distinguish them.**

7. **Hydrogen chloride gas can be produced by the action of conc. H_2SO_4 on solid NaCl.**
   a. HBr is not produced by the same method like HCl, why?
   b. What is the difference between hydrogen chloride gas and hydrochloric acid?
   c. Why is hydroiodic acid more acidic than hydrochloric acid?
   d. HCl is a covalent molecule but soluble in water, why?

8. **Give proper justification to the following statement:**
   a. HBr is a better reducing agent than HCl.
   b. HBr cannot be prepared in the same way as HCl.
   c. HF has abnormally higher boiling points than rest haloacids.
   d. HI is stronger acid than HCl
   e. Dehydrating agents like CaO and H_2SO_4 cannot be used to dry HBr and HI.

9. **Identify the observations when items in column I comes in contact with items in column II and complete the table-**

Coulmn I	Column II	Result
HCl	Moist blue litmus paper	
HI	$AgNO_3$	
HBr	Zn	
Moist Cl_2	Red rose	
Cl_2	Hot and conc. NaOH	

10. The addition of $AgNO_3$ solution followed by addition of ammonia solution is used to distinguish Cl^- and Br^- ions. What would you observe for each of these ions when they are tested as stated above? Write the balanced chemical equation involved.

11. (A) Write the balanced chemical equation involved when cold and dilute sodium hydroxide reacts with chlorine.
    (B) Write any two uses of the above resulting major product.

12. (A) State and explain the trend of boiling point of halogens down the group.
    (B) How does electronegativity of halogens vary down the group?

C. Long answer question

1. Compare the properties of HCl, HBr and HI.

2. Cl_2 is prepared by heating sodium chloride with conc. sulphuric acid in presence of MnO_2. Why can't HBr and HI be prepared by heating bromide and iodide salt with conc. sulphuric acid?

3. **What happens when-**
   a. excess of chlorine gas is treated with ammonia?
   b. excess of ammonia is treated with bromine?
   c. iodine is treated with sodium thiosulphate?
   d. bromine is heated with moist red phosphorous?

4. **Elements of group 17 are called halogens which means salt former.**
   a. Write one method of preparation of Cl_2.
   b. Write the action of Cl_2 with excess ammonia.
   c. How can you identify the presence of chloride? Explain with reaction.
   d. Write a bleaching action of chlorine.
   e. Write the oxidizing character of chlorine with suitable examples.

5. **Halogens are very active non-metals.**
   a. Write a silver nitrate test for the distinction between Cl^-, Br^- and I^-.
   b. Why can't we prepare HI by heating KI with conc. H_2SO_4?
   c. Write the action of HI with- i) $CuSO_4$ solution ii) acidified $K_2Cr_2O_7$ iii) $FeCl_3$

6. **In 1774 Scheele discovered chlorine by the action of HCl on MnO_2. While the name chlorine was suggested by Davvy in 1810.**
   a. What is meant by chlorine?
   b. Write the two general reactions for the preparation of X_2 (X: Cl, Br and I).
   c. Write the action of iodine with hypo.

d. Write any two applications of chlorine.
e. How are X_2(X: Cl, Br and I) tests done in the laboratory?

7. A student performed an experiment in chemistry laboratory and made following observations:

Experiment No.	Test	Observation
1	Add iodine to sodium thiosulphate.	Yellowish-brown color of iodine is discharged.
2	Add silver nitrate solution to sodium chloride solution.	Curdy white precipitate is produced.

a. Identify the substance responsible for the discharge of colour of iodine in experiment 1.
b. Write the balanced chemical equation for the reaction that was taken in experiment number 1.
c. Write the formula of the substance responsible for the formation of curdy white precipitate in experiment 2.
d. State what you would observe when excess ammonia solution is added to the white precipitate obtained in experiment 2.
e. Write the balanced chemical equation that has taken place in experiment 2.

8. Write the order of hydrogen halides with proper justification on following headings-
a. Boiling point b. Solubility
c. Reducing strength d. Acidic strength

9.6: Carbon

Why is carbon called an element of life?

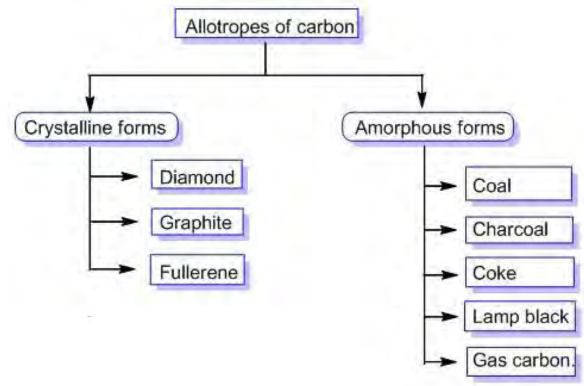
Figure 9.22: Carbon an element of life

Carbon is a non-metallic element of group 14. It is called element of life because of its unique structure and property to form a large number of organic compounds including inorganic compounds.

Element: Carbon	Electronic configuration: $1s^2\ 2s^2\ 2p^2$
Symbol: C	Group: 14
Atomic Number: 6	Period: 2^{nd}
Atomic Weight: 12 amu	Block: p block
Oxidation state: -4 to +4 including 0	Electronegativity: 2.55

9.6.1: Allotropes of carbon

You are quite aware about the meaning of allotrope and allotropy. So let's' discuss various allotropes of carbon. The various allotropes of carbon can be displayed through the following chart.



A: Crystalline allotropes of carbon

Activity: 27

Observe the pictures of diamond and graphite given below and answer the following questions:

a. What common property do they share?

b. Why are these allotropes, called crystalline allotropes?

c. What could be the mode of hybridization of carbon in these allotropes?

d. Which one of them is a good conductor of electricity?

e. Can you name one more crystalline allotrope of carbon besides diamond and graphite?

Figure 9.23: Graphite

Figure 9.24: Diamond

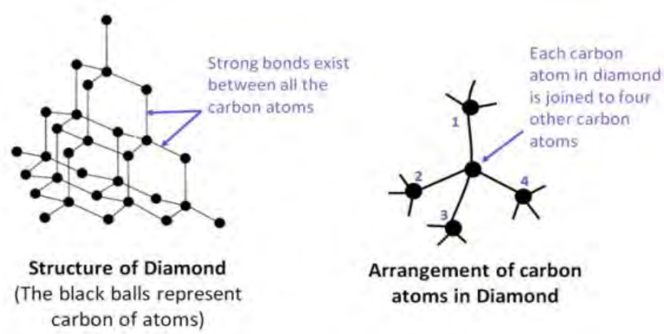
Allotropes of carbon that possess fixed geometrical shape are crystalline allotropes. The crystalline allotropes of carbon are-

1. Diamond

Diamond is one of the purest and hardest substances known. It is believed that diamonds were formed over 3 billion years ago deep within the earth's crust under intense conditions of high temperature and pressure.

Structure of Diamond

In diamond each carbon atom is sp^3 hybridized. Each carbon atom is tetrahedrally bonded to four other carbon atoms through a single strong covalent bond and tetrahedra are extended equally in all directions as shown in figures below. It possesses three dimensional networks in which C-C bond length is 1.54 Å and each bond angle is 109.5^0. Due to the high strength of covalent bonds, numerous carbon atoms are held together strongly that makes diamond very hard with high density.



Since all four valence electrons of each carbon are involved in bond formation there are no free electrons as a result diamond becomes a bad conductor of electricity. Diamond acts like a tiny complicated prism through which the light ray travels at different angles and gives a shiny appearance.

Properties of Diamond:

a. It is the hardest substance ever known.
b. It possesses low critical angle (24^0) which causes total internal reflection as a result of diamond shine.
c. It is transparent to X-ray.
d. Its melting point is about 3930 ^{0}C.
e. It possesses a high refractive index (2.148).
f. It possesses high density (3.5-3.53 g/mL).
g. It is insoluble in water and other organic solvent.
h. It is kinetically stable but thermodynamically less stable in comparison to graphite.
i. Above 1800^0C diamond can be changed into graphite.
j. C (diamond) $\xrightarrow[Heat]{1800\ ^0C}$ C(graphite)
k. It is chemically inert. However, it burns at a very high temperature (900 ^{0}C) and gets oxidized to CO_2.
l. C (diamond) $\xrightarrow[Heat]{900\ ^0C}$ CO_2

Uses of Diamond:

Figure-9.25: Diamond for cutting and polishing

Figure 9.26: Diamond as an ornament

Figure 9.27: Diamond for drilling

a. It is used as a jewellery item because of its ability to reflect light.
b. It is used for drilling, grinding and polishing other hard materials.
c. It is used for cutting glasses and diamond

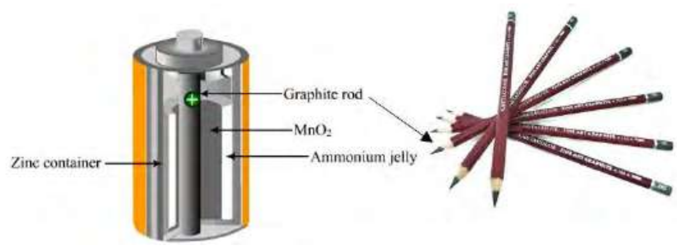
Test Yourself!

a. What is the mode of hybridization of carbon in diamond?
b. Why is diamond bad conductor of electricity?

2. Graphite

Activity: 28

What is the reason behind the use of graphite in batteries and pencils?



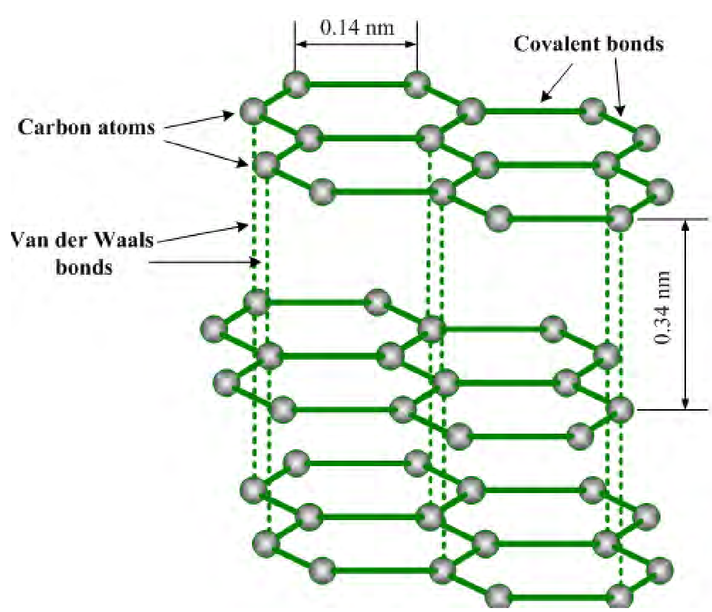
Graphite is a greyish black smooth slippery and opaque substance lighter than diamond. It occurs naturally and can also be prepared artificially as

SiO_2 +3C $\xrightarrow{\text{Electric furnace}}$ SiC + 2CO
(sand) (coke) (Silicon carbide)

SiC → Si + C (graphite)

This is the Acheson process.

Structure of Graphite



In graphite, each carbon atom is sp^2 hybridized. Each carbon atom is covalently bonded to three other carbon atoms forming a hexagonal flat layer. Therefore, unlike diamond, graphite possesses two dimensional structures. The C-C bond length in graphite is 1.42Å and bond angle is 120 ^{0}. Each hexagonal sheet is held together by a weak Van der Waal force which makes graphite slippery. Since one valence electron of each carbon atom is free to move therefore graphite behaves as a good conductor of electricity.

Properties of Graphite

a. It is a good conductor of electricity.
b. It possesses the ability to mark the paper. Therefore, it is used to prepare lead pencils for writing.
c. It is lighter than diamond and is slippery to touch.
d. It is insoluble in water and other solvent.
e. It is non-flammable.
f. It is soft due to the weak Van der Waal's force.
g. Its melting point is 3500 ^{0}C which is higher than diamond.
h. It is chemically inert. However, it burns on strong heating to produce CO_2.
i. It is converted into diamond when heated at about 1600 ^{0}C and 50,000 - 60,000 atmospheric pressure.

$$C(\text{graphite}) \xrightarrow{\text{High T and P}} C\ (\text{diamond})$$

Uses of Graphite

a. It is used to make electrodes as it is a good conductor of electricity.
b. It is used to prepare lead pencils.
c. It is used as lubricants for machines working at high temperature.
d. It is used to make crucibles (melting pots for extremely hot chemical reactions).

FIGURE 9.28: GRAPHITE LEAD

FIGURE 9.29: GRAPHITE LUBRICANT

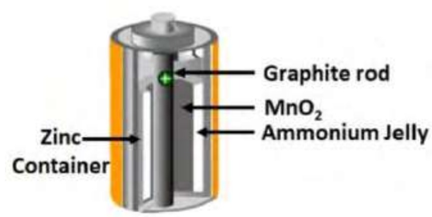
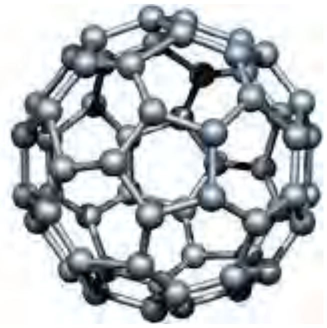


FIGURE 9.30: GRAPHITE ELECTRODE

Test Yourself:

a. Why is graphite good conductor of electricity?
b. Mention any two important use of graphite.

3. Fullerene



What is the shape of a given structure?

In 1985, Harlod W. Kroto with his colleagues Robert E. Curl and Richard E. Smalley at Rice university of USA discovered the third crystalline allotrope of carbon called Fullerene. This fullerene forms a cluster of 60 carbon atoms that looks like a soccer ball. It is called Buckminster Fullerene in the honour of American architect Buckminster Fuller, who designed geodesic domes to which the fullerene molecules resemble. The most common fullerene C_{60}. Other forms of Buckminster fullerene are C_{32}, C_{50}, C_{70}, C_{76}, C_{84} etc. Fullerenes are composed of 5-6 member rings and form a soccer ball shape. Therefore, they are also called "bucky ball".

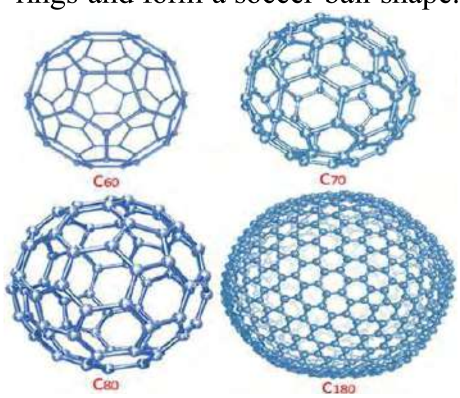


Figure 9.31: Structure of Fullerene C60 and some other examples of fullerene

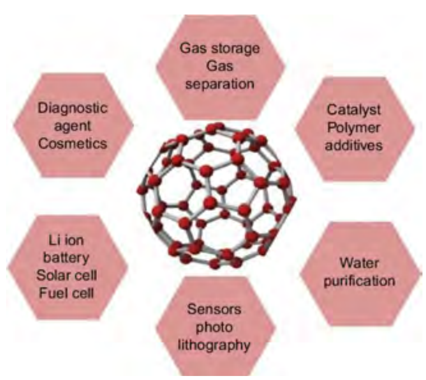


Figure 9.32: Some potential applications of fullerene

Structure of Fullerene

Fullerenes are spherical molecules with three dimensional patterns of soccer balls. They have pentagonal and hexagonal rings. They are often called **bucky ball**. The most common fullerene is 'C_{60}' in which 60 carbon atoms are joined together by single and double covalent bonds to form a hollow sphere with 12 pentagonal and 20 hexagonal rings resembling a soccer ball as shown in figure below. Each carbon in fullerene is sp^2hybridized.The C_{60} molecule has a **truncated icosahedron** structure. An icosahedron is a polygon with 60 vertices and 32 faces. It has been called the most symmetrical molecule.

A Brief History of Fullerene

In 1985 British chemist Sir Harold W. Kroto and the colleagues Richard E. Smalley and Robert F. Curl, Jr., discovered fullerene by using pulsed laser to vaporize graphite rods in an atmosphere of helium gas.

The structure of fullerene was suggested to be like a soccer ball: a spherical shape that can be made using 12 pentagons and 20 hexagons.

C60 was named buckminster fullerene in honor of Buckminster Fuller. The shortened name fullerene is used to refer to the family of fullerene.

In 1996 Curl, Kroto and Smalley was awarded the Nobel Prize in chemistry for the discovery of fullerenes.

Properties of Fullerene:

a. It is insoluble in water but soluble in organic solvents.
b. It is a good conductor of electricity.
c. It reacts with alkali metals to form solid compounds like K_3C_{60} which behaves as a superconductor below 18 ^{0}C.
d. It is prepared by passing electric spark between graphite electrodes in presence of an inert atmosphere of argon.

Uses of Fullerene:

a. It is used as a superconductor.
b. It is used in biomedical field.
c. It is used in making cosmetics related items.
d. It is used as lubricants in micro machines.

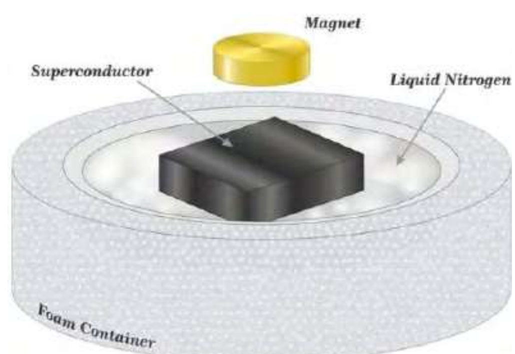


Figure: Structure of Buckminster fullerene

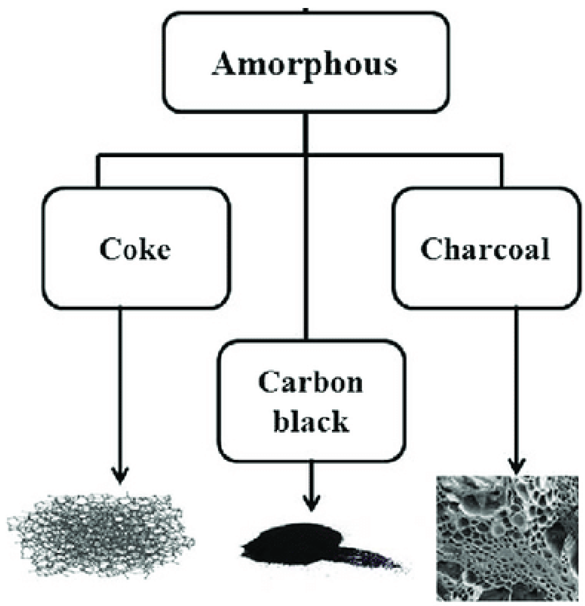
Test Yourself!

> a. What is fullerene?
> b. List the important applications of fullerene.

B. Amorphous allotropes of Carbon

Activity:29

a. **Lit a kerosene lamp and place a white paper over the burning flame to collect the smoke.**
b. **After 5 minutes observe the colour and texture of the substance formed on the paper.**
c. **What differences would you find between the collected smoke and graphite?**



Why are they called amorphous allotropes of carbon?

1. Coal

Coal is a combustible black or brownish black substance formed from the remains of plants that lived millions of years ago.

Uses of Coal

i. It is used as a fuel.
ii. It is used in preparation of graphite electrodes.

2. Coke:

It is a grayish black solid substance obtained by destructive distillation of coal.

Uses of Coke:

i. It is used as a fuel.
ii. It is used as a reducing agent in metallurgy.

3. Gas Carbon

It is a dense form of deposited carbon scraped out from the wall of the retort during the preparation of coal.

Uses of Gas Carbon

i. It is used as a fuel.
ii. It is used in preparation of electrode in dry cell.

4. Lamp black

It is a velvety black powder obtained by burning carbon rich substances like kerosene oil, petroleum, turpentine oil, vegetable oil, and other compounds in a limited supply of oxygen.

Uses of lamp black

i. It is used in the manufacture of ink, paints, shoe polish, printer ink etc.
ii. It is used in preparation of tyres.

5. Charcoal

It is a black porous substance obtained by destructive distillation of either plants or animal bones and blood. It possesses high surface area. Depending upon the source of origin it is of three types:

a. Plant Charcoal:

Plant charcoal or wood charcoal is obtained by destructive distillation of plant/wood.

Uses of Plant Charcoal
i. It is used as adsorbent.
ii. It is used as a fuel.
iii. It is used in preparation of gunpowder**.**

b. Animal Charcoal
Animal charcoal is obtained by destructive distillation of bones or blood of animals.

Uses of Animal Charcoal
i. It is used as adsorbent.
ii. It is used in purification of organic compounds.
iii. It is used in preparation of superphosphate of lime.

c. Activated Charcoal (or Sugar Charcoal)
Activated charcoal is obtained by heating carbon containing compounds in the absence of air at high temperature. It is the purest form of amorphous carbon that contains a larger surface area with highest adsorption capacity.

$$C_{12}H_{22}O_{11} \rightarrow 12\ C + H_2O$$

Figure: Activated charcoal obtained from coconut shell

Uses of Activated Charcoal
i. It is used as adsorbent.
ii. It is used for removing toxins.
iii. It is used in cleaning teeth and whitening skin.

Test Yourself!

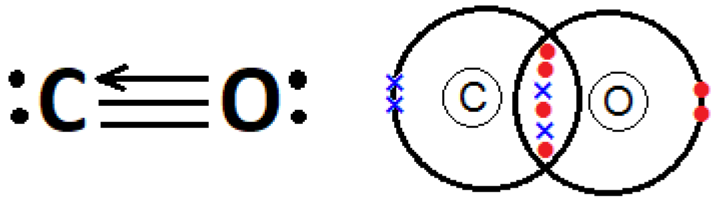
What is the difference between charcoal and activated charcoal?

Facts

A carbon fiber is a polymer of carbon extremely of light weight and super strong. It is five times stronger, two times stiffer and two-third times lesser in weight than so-called steel.

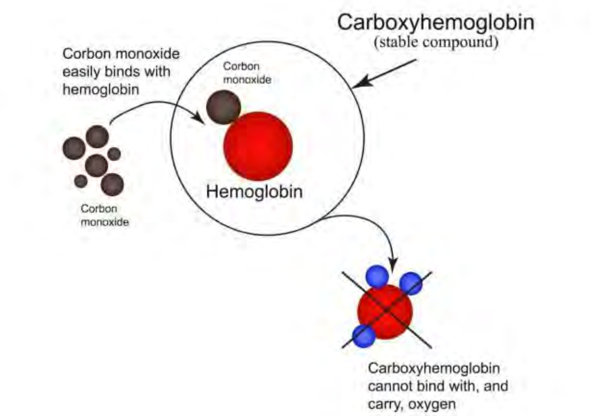
9.6.2: Carbon Monoxide(CO)

Activity: 30
Observe the Lewis structure of oxide of carbon:
a. Name the molecule given.
b. Does it follow octet rule, why?



When a carbon-containing source is burnt incompletely in limited supply of oxygen then carbon monoxide gas (CO) is produced. It is a colourless gas with highly toxic in nature. It is obtained when carbon-containing substances are burnt in insufficient supply of air.

We all know that CO gas is a silent killer. The toxic effect of CO gas is due to the formation of carboxy haemoglobin in blood stream which results suffocation and the person may even die. Therfore, one should avoid sitting in the atmosphere of CO gas.



Let's discuss some of its chemical properties:

Figure 9.32: Incomplete combustion of carbon based fuel produces carbon monoxide

1. Reducing Action

Carbon monoxide is a powerful reducing agent. It reduces many metal oxides to respective metals. For example,

$$ZnO + CO \xrightarrow[Heat]{} Zn + CO_2$$

$$PbO + CO \xrightarrow[Heat]{} Pb + CO_2$$

$$Fe_2O_3 + CO \xrightarrow[Heat]{} 2Fe + CO_2$$

$$CuO + CO \xrightarrow[Heat]{} Cu + CO_2$$

Beside this carbon monoxide also reduces iodine oxide to iodine, Tollen's reagent to metallic silver, Fehling's solution to red ppt. of cuprous oxide.

$$I_2O_5 + CO \xrightarrow[Heat]{} I_2 + CO_2$$

$$2[Ag(NH_3)_2]OH + CO \xrightarrow[Heat]{} Ag\downarrow + 4NH_3 + CO_2 + H_2O$$

(Tollen's reagent)

$$2Cu(OH)_2 + CO \xrightarrow[Heat]{} Cu_2O\downarrow + CO_2 + 2H_2O$$

(From Fehling's solution)

2. Action with metals

Carbon monoxide reacts with many transition metals like nickel, iron, cobalt, etc. to form a variety of metal carbonyl compounds. For example,

$$Ni + 4CO \xrightarrow[Heat]{80\ 0C} Ni(CO)_4 \text{ (Nickel tetracarbonyl)}$$

When nickel tetracarbonyl is heated at 180 ^{0}C then pure nickel is obtained back. This process of purification of nickel metal is called Mond's process.

$$Ni(CO)_4 \xrightarrow[Heat]{180\ 0C} Ni + 4CO$$

$$Fe + 5CO \xrightarrow[Heat]{} Fe(CO)_5 \text{ (Iron pentacarbonyl)}$$

$$Co + 4CO \xrightarrow[Heat]{} Co(CO)_4 \text{ (Cobalt tetracarbonyl)}$$

3. Action with non-metals

Carbon monoxide gas reacts with non-metals like chlorine, sulphur, hydrogen, etc.

a. Action with Cl_2:

In the presence of sunlight carbon monoxide reacts with chlorine gas to produce a poisonous gas called phosgene (Carbonyl chloride) gas.

$$CO + Cl_2 \rightarrow COCl_2$$

b) Action with H_2

Carbon monoxide reacts with hydrogen gas, in presence of catalyst like ($ZnO + CrO_3$) at 300 ^{0}C and 200 atmospheric pressure, to produce methyl alcohol.

$$CO + 2H_2 \xrightarrow[T\&P]{ZnO + CrO3} CH_3OH$$

c) Action with S

Carbon monoxide reacts with sulphur vapor to produce carbonyl sulphide.

$$CO + S \rightarrow COS$$

Test yourself!

a. What happens when CO gas is heated with Ni at 80 ^{0}C?
b. Give a chemical reaction to show reducing nature of CO gas.

Uses of CO

Some of the uses of CO gas are:

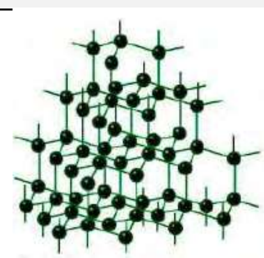
a. It is used as a reducing agent in metallurgy.
b. It is used in manufacture of phosgene gas which is used in warfare.
c. It is used as a fuel with hydrogenic water in the name of water gas.

d. $\underset{\text{(steam)}}{H_2O} + \underset{\text{(coke)}}{C} \xrightarrow[Heat]{} \underset{\text{(water gas)}}{CO + H_2}$

e. It is used in purification of nickel metal by Mond's process.
f. It is used in manufacture of methyl alcohol.

Project work

1. Take a chart paper and draw a table and complete it as shown below:

Allotropes of carbon	Nature of allotropes	Structure	Major uses
Diamond	Crystalline		Cosmetics Jewelry Automotive Medicine

Graphite			
Coal			
Charcoal			

2. Prepare a chart of different allotropic forms of carbon with their properties and uses.

3. Prepare a chart paper presentation on toxicity of carbon monoxide and its preventive measures.

Exercise

A. Multiple choice questions

1. Which of the following is a good conductor of electricity?
a. Coal b. Diamond c. Graphite d. Gas carbon

2. What is the mode of hybridization of carbon in graphite?
a. sp b. sp^2 c. sp^3 d. sp^3d^2

3. What is the nature of bonding in diamond?
a. Covalent b. Metallic c. Ionic d. Co-ordinate covalent

4. Which of the following gas mixture is water gas?
a. $CO + 3N_2$ b. $CO + H_2O$ c. $CO + H_2$ d. $CO + CH_4$

5. CO + Ni →? What is the product of the given reaction?
a. Nickel tetracarbonyl b. Nickel pentacarbonyl
c. Nickel dicarbonyl d. Nickel octacarbonyl

6. Which of the following gases is produced when carbon monoxide reacts with chlorine?
a. O_2 b. $CoCl_2$ c. Br_2 d. CH_3Cl

7. Which of the following is not a crystalline allotrope of carbon?
a. Fullerene b. Diamond c. Graphite d. Charcoal

8. Which of the following properties makes graphite a good conductor?
a. Delocalized electron b. Hydrogen bond c. Soft nature d. sp^3 hybridization

9. Which element is present in lead pencil?
a. Sulphur b. Silicon c. Carbon d. Phosphorous

10. What is the number of pentagon and hexagon rings present in Fullerene?
a. 6 & 10 b. 12 & 20 c. 30 & 30 d. 60 & 0

11. **Pick the substance which is not an allotrope of carbon.**
a. Charcoal b. Diamond
c. Carbon monoxide d. Graphite

12. **Which of the following is called a truncated isohedron?**
a. C_6 b. C_{10} c. C_{20} d. C_{60}

13. **What is the ratio of CO and H_2 in water gas?**
a. 1:1 b. 1:2 c. 2:1 d. 2:3

14. **Which of the following species is formed when CO combines with haemoglobin?**
a. Deoxyhaemoglobin b. Carboxyhaemoglobin
c. Carbhaemoglobin d. Carbinohaemoglobin

15. **Identify A in the following reaction.**

$$CO + 2H_2 \xrightarrow[T\&P]{ZnO+CrO3} A$$

a. C_2H_5OH b. CH_3OH c. C_2H_6 d. CH_4

16. **What is the mode of hybridization of carbon in CO?**
a. sp b. sp^2 c. sp^3 d. sp^3d

17. **Wood charcoal is an allotrope of which element?**
a. Nitrogen b. Oxygen c. Phosphorous d. Carbon

B. Short answer question

1. **Define the following terms:**

a. Allotrope b. Allotropy c. Activated charcoal d. Plant charcoal e. Gas carbon

2. **List crystalline and amorphous allotropes of carbon.**

3. **Discuss the structure, properties and uses of diamond.**

4. **Both carbon and carbon monoxide are used as reducing agents. Which one is a better reducing agent and why?**

5. **Give reason:**
a. Graphite conducts electricity but diamond does not even though both are composed of the same element carbon.

b. Charcoal is good adsorbent than coal.
c. CO gas is called a silent killer.
d. Graphite is called solid lubricant.
e. C_{60} is called truncated isohedron.

6. **Differentiate between:**
   a. Diamond and graphite.
   b. Diamond and fullerene.
   c. Normal charcoal and activated charcoal.
   d. Coke and gas carbon

7. Show your acquaintance with different types of charcoal with their use.

8. Show your familiarity with structure, properties and uses of fullerene.

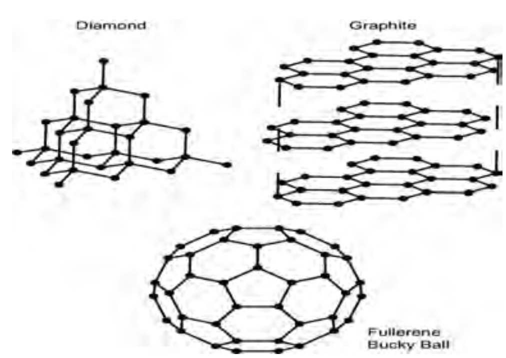
9. At room temperature diamond remains extremely unreactive whereas graphite reacts readily, why?

10. Compare the structure of diamond and graphite.

C. Long answer question

10. (A) Give any two reactions showing the reducing nature of CO gas.
    (B) What happens when CO gas is-
    a. heated with Co?
    b. warmed with Tollen's reagent?
    c. heated with sulphur?

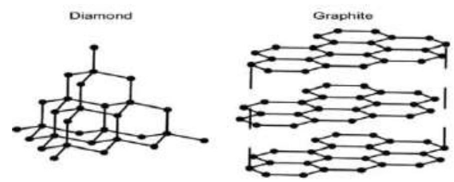
11. Study the given figures are allotropes of carbon - diamond, graphite and fullerene and answer the following questions:



   a. Which allotrope of carbon is the hardest in nature and why?
   b. Which allotrope is the most reactive and why?
   c. Which allotrope is a good conductor of electricity and why?
   d. What is the mode of formation of fullerene?
   e. What is the mode of hybridization in diamond and graphite?

12. Although CO gas is called a silent killer. It does not produce long term ecological damage. It is widely used in different sectors. Can you list some uses of CO gas?

13. Both diamond and graphite are the allotropes of carbon. Study the structure of diamond and graphite and answer the following questions-



a. Why is graphite softer than diamond?
b. What is the mode of hybridization of carbon in graphite and diamond?
c. Graphite is a good conductor of electricity but not diamond, why?
d. Write two uses of each.

9.7: Phosphorous

Activity: 31

Observe the given picture of most commonly used red phosphorous and answer the following questions:

a. To which group and period does phosphorous belong?

b. Can you write a molecular formula for some phosphorous containing compounds?

Figure 9.33: Red phosphorous

The element phosphorous was first discovered by German alchemist Brandit in 1669. It is non-metallic element and does not occur in a free state as it has a higher affinity towards oxygen. It is an essential constituent of plants and animal cells. In combined state it is mainly found in the form of phosphates like-

Phosphorite: $Ca_3(PO_4)_2$

Fluorapatite: $Ca_3(PO_4)_2 . CaF_2$

Chlorapatite: $Ca_3(PO_4)_2 . CaCl_2$, etc.

Element: Phosphorous	Electronic configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
Symbol: P	Group: 15
Atomic Number: 15	Period: 3rd
Atomic weight: 31 amu	Molecular Formula: P_4

9.7.1: Allotropes of Phosphorous

Like carbon, phosphorous also exists in different allotropic forms. The various allotropes of phosphorous are listed below:

i. White or Yellow Phosphorous.
ii. Red Phosphorous
iii. Black Phosphorous
iv. Violet Phosphorous
iv. Scarlet Phosphorous

Facts

The most common allotropes of phosphorous are white phosphorous and red phosphorous. White phosphorous (i.e, yellow) phosphorous is used to make rat poison and red phosphorous is used to make matches in the match industry.

Figure 9.33: Use of white (yellow) phosphorous in rat poison

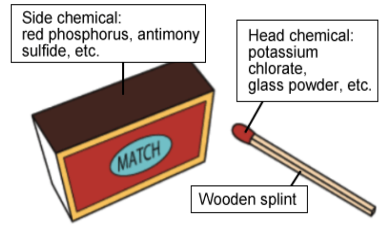


Figure 9.34: Use of red phosphorous in match

Test yourself!

Name the different allotropes of phosphorous.

9.7.2: Phosphine Gas

Phosphine is one of the best hydrides of phosphorous having molecular formula PH_3. It is a colourless and poisonous gas with the smell of rotten fish.

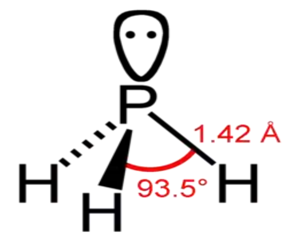


Figure 9.35: Structure of phosphine

Preparation of phosphine (Laboratory Preparation):
In the laboratory, phosphine gas is prepared by heating white phosphorous with aqueous caustic soda or potash in an inert atmosphere of CO_2.

$$P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3\uparrow$$

Test yourself!

How would you obtain phosphine gas in laboratory?

Note: ***There are some other methods too that can be used to prepare phosphine gas. For example,***

$$Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + PH_3\uparrow$$

$$PH_4I + KOH \rightarrow KI + H_2O + PH_3\uparrow$$

Physical properties of phosphine

a. It is colourless gas having the smell of rotten fish.
b. It is slightly soluble in water.
c. It is heavier than air.
d. It is toxic in nature.
e. In its pure form it is non-inflammable but becomes inflammable in presence of P_2H_4 or P_4 vapours.

Chemical properties of phosphine

Some of the chemical properties of phosphine are:

1. Action with air (Oxygen)

Just like ammonia phosphine is non-supporter of combustion. But it burns when heated in the presence of air to give phosphorous pentoxide or phosphoric acid.

$PH_3 + 4O_2 \rightarrow P_2O_5 + 3H_2O$

$PH_3 + 2O_2 \rightarrow H_3PO_4$

2. Basic Nature

Phosphine is a weak base. It does not change the color of moist red litmus paper. Being basic in nature it reacts with halogen acids to give phosphonium compounds.

$PH_3 + HCl \rightarrow PH_4Cl$ (phosphonium chloride)

$PH_3 + HBr \rightarrow PH_4Br$ (phosphonium bromide)

Note: PH_3 is a weaker base than NH_3.

3. Action with Cl_2

Depending upon the reaction condition phosphine reacts with chlorine to give phosphorous trichloride or phosphorous pentachloride.

$PH_3 + 3Cl_2 \rightarrow PCl_3$ (phosphorous trichloride)

$PH_3 + 4Cl_2 \rightarrow PCl_5 + HCl$

(phosphorous pentachloride)

4. Reducing Nature

Phosphine acts as a reducing agent. For example,

a) Phosphine reduces silver nitrate solution to metallic silver.

$PH_3 + AgNO_3 + 3H_2O \rightarrow 6Ag\downarrow + 6HNO_3 + H_3PO_3$

b) Phosphine reduces copper sulphate to black ppt. of copper phosphide.

$PH_3 + 3CuSO_4 \rightarrow Cu_3P_2\downarrow + 3H_2SO_4$

c) Phosphine reduces mercuric chloride to black ppt. of mercuric phosphide.

$PH_3 + 3HgCl_2 \rightarrow Hg_3P_2\downarrow + 6HCl$

Test yourself!

a. How would you confirm that PH_3 is a reducing agent?
b. How does phosphine react with chlorine gas?

Uses of phosphine

Some of the applications of phosphine gas are listed below:

a. It is used as a reducing agent.
b. It is used in preparation of smoke screens which are used in warfare.
c. It is used to prepare various compounds of phosphorous like P_2O_5, PCl_3, PCl_5, etc.
d. It is used as a Holme's signal to guide the ship.

Facts

Holme's signal is a mixture of calcium carbide and calcium phosphide placed in a container. These two substances when react, produce phosphine and acetylene which burns to guide the ship.

Figure 9.36: Use of PH_3 as a Holme's signal to guide the ship

Project work

1. Make a chart list of allotropes of phosphorous.
2. Take a chart paper and write the preparation, properties and uses of phosphine gas.

Exercise

A. Multiple choice questions

1. What is the atomicity of phosphorous?

a. 1 b. 2 c. 3 d. 4

2. $P_4 + Cl_2 \rightarrow$? What is the product of the given reaction?

a. PCl_3 b. PCl_6 c. P_2Cl_5 d. $POCl_3$

3. Which of the following allotrope of phosphorous is used to prepare poison?

a. White phosphorous b. Red phosphorous

c. Violet Phosphorous d. Scarlet phosphorous

4. Which of the following forms Holme's signal?

a. $CaC_2 + C_2H_2$ b. $CaC_2 + Ca_3P_2$

c. $Ca_3P_2 + C_2H_2$ d. $CaCl_2 + Ca(OH)_2$

5. **Which of the following allotrope of phosphorous gives phosphine with caustic alkali?**

a. Scarlet phosphorous b. Red phosphorous

c. Yellow phosphorous d. Black phosphorous

6. **Which of the following is not true about phosphine?**

a. It has rotten fish smell b. It is a colourless gas

c. It is poisonous d. It is inflammable

7. **Which of the following gas is used as Holmes signal?**

a. Phosphine b. Acetylene c. Nitrogen d. Hydrogen

8. **Which of the following compounds reacts with water to give PH_3?**

a. Phosphorous trichloride b. Aluminium phosphide

c. Phosphorous pentachloride d. Black phosphorous

B. Short answer questions

1. Name the allotropes of phosphorous. Which allotrope of phosphorous is used to make matches and rat poison?

2. Why is white phosphorous kept under water? What happens when gas produced by boiling white phosphorous and caustic soda is passed through copper sulphate solution?

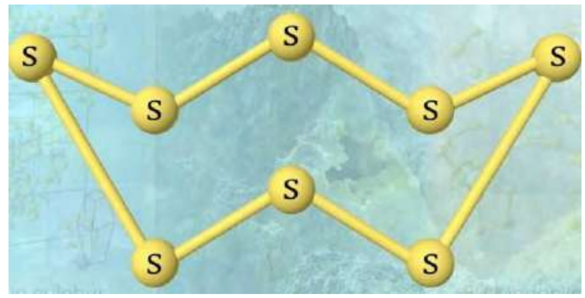
3. Write any two methods of preparation of phosphine gas. Write the uses of it.

4. What is Holme's signal? How is it produced? Write its significance.

9.8: Sulphur

Activity: 32

Observe the structure of Sulphur molecule given:

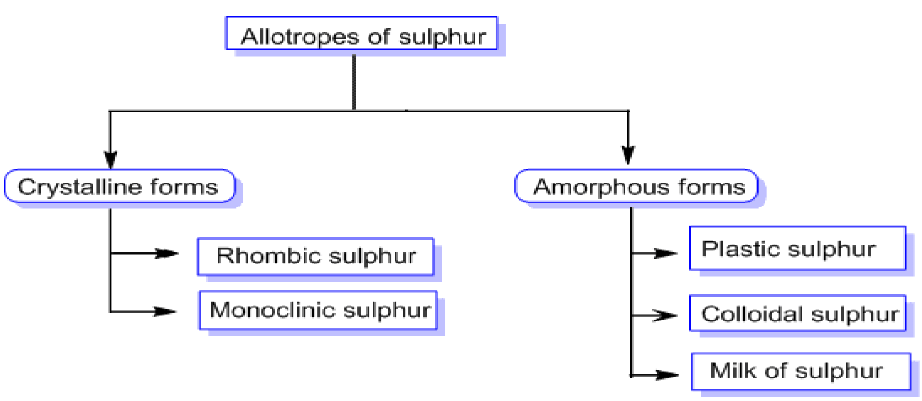


a. What is the atomicity of Sulphur?
b. Name some compounds containing sulphur?

Sulphur is a non-metallic element found in native form as well as combined form. Under normal conditions it is an octatomic molecule with chemical formula S_8. The elemental sulphur is bright yellow crystalline solid as shown in the figure. In combined state sulphur is usually found as sulphide, sulphite and sulphate minerals. It is also present in petroleum and natural gas.

Element: Sulphur	Electronic configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
Symbol: S	Valency: 2, 6
Atomic Number: 16	Group: 16
Atomic Mass: 32.07 amu	Period: 3rd
Molecular Formula: S_8	Electronegativity: 2.5

9.8.1: Allotropes of sulphur and uses of sulphur



a. What do you mean by allotropes?
b. What is the difference between crystalline and amorphous forms?

Facts

The head of a matchstick contains chemicals including an oxidizing agent (typically potassium chlorate), a fuel (such as sulphur or charcoal). Striking the match against a rough surface generates heat through friction, initiating a chemical reaction between the oxidizing agent and the fuel.

Some uses of Sulphur are listed below:

a. It is mainly used in the manufacture of sulphuric acid in a large quantity.
b. It is used to prepare gunpowder and in match industry
c. It is used to provide desirable resilience and toughness to rubber through a process called vulcanization of rubber.
d. It is used as germicides, insecticides and fungicides.
e. It is used in the manufacture of medicine, dyes and explosives.
f. It is used as a bleaching agent in the paper industry in the form sulphides of calcium and magnesium.

Figure 9.37: Sulphur one of the component of gunpowder

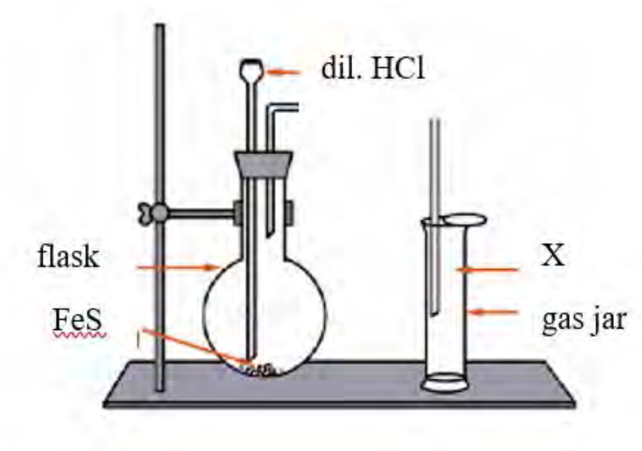
Figure 9.38: Use of sulphur in match stick

9.8.2: Hydrogen sulphide

Activity: 33

Observe the figure below for the preparation of gas X.

a. Identify the gas X.
b. Write the balanced chemical reaction for the production of gas.



When iron sulphide comes in contact with dilute sulphuric acid then hydrogen sulphide gas is obtained. The gas exactly smells like rotten egg. It is invisible, colourless, flammable gas. It is also known as sulphuretted hydrogen or hydrosulphuric acid. It is produced whenever sulphur containing compounds are decomposed.

Preparation of Hydrogen sulphide using Kipp's apparatus

You might have used H_2S gas during qualitative analysis of basic radicals in the laboratory. Have you noticed why Kipp's apparatus is used for the supply of H_2S gas? During the qualitative analysis of salt, we need H_2S in a small quantity frequently. Therefore, for intermittent supply of this gas Kipp's apparatus is used.

Hydrogen sulphide gas can be prepared by the action of dilute sulphuric acid on iron sulphide.

$$FeS + H_2SO_4 \rightarrow H_2S + FeSO_4$$

Now, let's discuss the working principle of Kipp's apparatus.

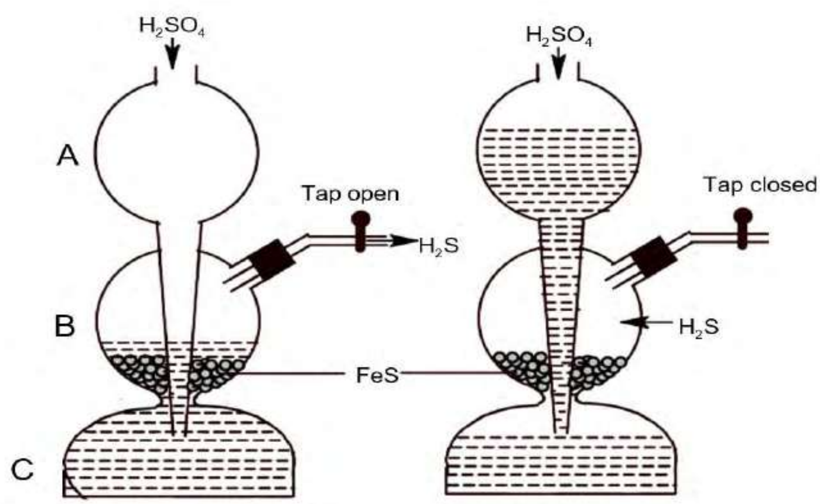


Figure 9.39: Use of Kipp's apparatus for intermittent supply of H_2S gas

Working Principle of Kipp's apparatus

A Kipp's apparatus consists of two parts- Bulb A with a long stem and the base with two connecting bulbs B and C as shown in figure above. The long stem of bulb A reaches to the bulb C. Pieces of iron sulphide are kept in bulb B and dilute sulphuric acid is poured into the bulb A till the iron sulphides are covered with the acid. Bulb B is connected with a small neck for the delivery of hydrogen sulphide gas. Hydrogen sulphide gas is produced as soon as acid comes in contact with iron sulphide. Hydrogen sulphide gas is easily collected by opening the tap. But if the tap remains closed then pressure is developed in the bulb B due to evolution of hydrogen sulphide gas. This causes the rise of acid up in the bulb A and connection between the acid and iron sulphide breaks ceasing the formation of hydrogen sulphide gas. When tap is opened, the gas pressure decreases in the bulb B as a result acid comes in contact with iron sulphide and reaction begins to produce hydrogen sulphide gas.

Facts

a. H_2S is poisonous gas. It may cause nausea, vomiting, dizziness, headache, and even death.
b. It possesses low melting point and boiling point as a result it cannot be easily liquified and solidified.

Chemical properties of Hydrogen sulphide gas

Some of the important chemical properties of hydrogen sulphide gas are-

1. Acidic Nature

Hydrogen sulphide is a diprotic acid. It ionizes to produce two series of radicals as shown below:

$$H_2S \leftrightarrow H^+ + HS^-$$
$$H_2S \leftrightarrow 2H^+ + S^{--}$$

However, H_2S is a weak acid. It can turn moist blue litmus paper into red. Being acidic in nature it reacts with bases to produce salt. For example,

$$\underset{}{NaOH + H_2S \rightarrow} \underset{\text{(Sod. bisulphide)}}{NaHS} + H_2O$$

$$2NaOH + H_2S \rightarrow \underset{\text{(Sod. sulphide)}}{Na_2S} + H_2O$$

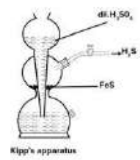
$$CaO + H_2S \rightarrow \underset{\text{(calcium. sulphide)}}{CaS} + H_2O$$

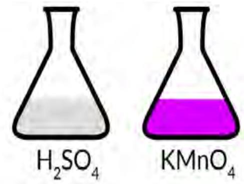
However, H_2S cannot decompose carbonates because it is weaker acid than carbonic acid.

Test yourself!

Give a reaction to show H_2S is acidic?

Activity: 34





Observe the given pictures carefully where gas is produced in Kipp's apparatus and reagents are taken in two conical flasks as shown. Answer the following questions:

a. What happens when H2S gas is passed through the above mixture?

b. What peculiar property of the gas is observed when reaction is carried out between them?

2. Reducing nature

S in H_2S possesses -2 oxidation state which is minimum. Therefore, H_2S in a reaction tends to get oxidized to S. Hence H_2S behaves as a good reducing agent. Some of the reactions showing reducing nature of H_2S are:

a. Action with acidified $KMnO_4$ solution

$KMnO_4$ + dil H_2SO_4 $KMnO_4$ + dil H_2SO_4 + H_2S

Figure 9.40: Action of H_2S on acidified $KMnO_4$ solution

Hydrogen sulphide reduces the pink colour of acidified potassium permanganate solution to colourless manganese sulphate and itself gets oxidized to sulphur.

$$2KMnO_4 + H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$$

$$\underline{\{H_2S + [O] \rightarrow H_2O + S\}\times 5}$$

$$\underset{\text{(pink)}}{2KMnO_4} + 3H_2SO_4 + 5H_2S \rightarrow K_2SO_4 + \underset{\text{(colorless)}}{2MnSO_4} + 8H_2O + 5S$$

b. Action with acidified $K_2Cr_2O_7$ solution

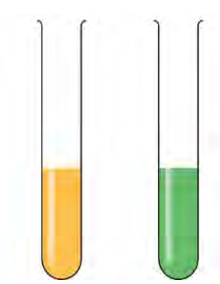


Figure 9.41: acidified $K_2Cr_2O_7$ becomes green by H_2S gas

Hydrogen sulphide reduces the orange colour of acidified potassium dichromate solution to light green colour of chromium sulphate and itself gets oxidized to sulphur.

$$K2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$$
$$\{H_2S + [O] \rightarrow H_2O + S\}\times 3$$
$$K2Cr_2O_7 + 4H_2SO_4 + 3H_2S \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

(orange) (light green)

c. Action with ferric salt

Hydrogen sulphide reduces ferric salt to ferrous salt and itself gets oxidized to sulphur.

$$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$$

d. Action with halogens

Hydrogen sulphide reduces halogens to hydrogen halides and itself gets oxidized to sulphur.

$$H_2S + Cl_2 \rightarrow 2HCl + S$$

e. Action with SO_2

Hydrogen sulphide reduces sulphur dioxide to sulphur.

$$H_2S + SO_2 \rightarrow 2H_2O + 3S$$

f. Action with concentrated HNO_3:

Hydrogen sulphide reduces concentrated nitric acid to nitrogen dioxide and itself gets oxidized to sulphur.

$$2HNO_3 \rightarrow H_2O + 2NO_2 + [O]$$
$$H_2S + [O] \rightarrow H_2O + S$$
$$2HNO_3 + H_2S \rightarrow 2H_2O + 2NO_2 + S$$

g. Action with concentrated H_2SO_4:

Hydrogen sulphide reduces concentrated sulphuric acid to sulphur dioxide and itself gets oxidized to sulphur.

$$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$$
$$H_2S + [O] \rightarrow H_2O + S$$
$$H_2SO_4 + H_2S \rightarrow 2H_2O + SO_2 + S$$

Test yourself!

What happens when H2S gas is passed through acidified $K_2Cr_2O_7$ solution?

3. H_2S as an analytical reagent

Activity: 35

Take a test tube containing a few mL of acidified copper sulphate solution. To this solution add hydrogen sulphide gas prepared in the laboratory and observe the reaction.

a. Write the balanced chemical equation of the reaction carried out in this activity.
b. Identify the colour of the precipitate obtained.

$$Cu^{++} + H_2S \xrightarrow{HCl} CuS \downarrow + 2H^+$$
(black ppt.)

When H_2S gas is passed through the aqueous solution of copper salt in presence of dil. HCl a black ppt. of copper sulphide is obtained. The formation of black ppt. indicates the presence of Cu^{2+} ion as a basic radical in the salt sample.

In a qualitative analysis of salt, we detect basic radicals present in the given salt sample. Hydrogen sulphide has the ability to form sulphides with several metal ions. The difference in the solubility of these sulphides and their characteristic colour are used for identification of metal ions present in the given salt sample. Thus various basic radicals can be detected with the help of hydrogen sulphide gas. Therefore, hydrogen sulphide gas plays an important role in qualitative analysis of salt. Some examples are:

a. Detection of group II basic radicals

Group II basic radicals consist of Cu^{++}, Hg^{++}, Pb^{++} Cd^{++}, Sb^{++} etc. When H_2S gas is passed through an acidified solution of these radicals then they get precipitated in the form of their sulphides. By observing the characteristic colour of these sulphides one can easily identify the metal.

$$Cd^{++} + H_2S \xrightarrow{HCl} CdS \downarrow + 2H^+$$
(yellow ppt.)

$$Sb^{++} + H_2S \xrightarrow{HCl} Sb_2S_3 \downarrow + 2H^+$$
(orange ppt.)

b. Detection of group IIIB basic radicals:

Group III basic radicals consist of Ni^{++}, Co^{++}, Zn^{++} etc. When H_2S gas is passed through the alkaline solution of these radicals then they get precipitated in the form of their sulphides. By observing the characteristic colour of these sulphides one can easily identify the metal.

$$Zn^{++} + H_2S \xrightarrow{Alkalinemedium} ZnS \downarrow + 2H^+$$
(white ppt.)

$$Ni^{++} + H_2S \xrightarrow{Alkalinemedium} NiS \downarrow + 2H^+$$
(black ppt.)

$$Mn^{++} + H_2S \xrightarrow{Alkalinemedium} MnS \downarrow + 2H^+$$
(flesh colored ppt.)

In this way other group basic radicals are detected by the use of hydrogen sulphide gas.

Test Yourself!

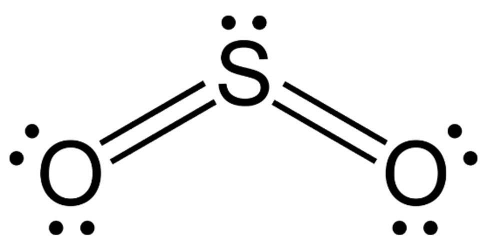
What happens when-
a. H_2S gas is passed through aqueous solution of copper sulphate solution?
b. H_2S gas is passed through aqueous solution of Mn^{2+}?

Uses of H_2S gas

a. It is used as a reducing agent.
b. It is used as analytical reagent in laboratory
c. It is used to prepare metallic sulphides which are used as pigments.
d. It is used in preparation of hydrogen halides in the laboratory.

9.8.5: Sulphur dioxide

Activity: 36



a. **Name the shape of the SO_2 molecule given and give reason for such shape.**
b. **Identify whether it is acidic or basic oxide of suplhur**.

Figure 9.42: Ancient practice of cleaning wine making container by adding SO_2

Sulphur dioxide is regarded as anhydrides of sulphurous acid (H_2SO_3). It is a colourless gas with a harsh odour similar to that of a burning match. It is obtained by the combustion of sulphur containing compounds. Naturally it is found in volcanic gases. It was first prepared by Joseph Priestly in 1774 AD by heating mercury with concentrated sulphuric acid.

Physical properties of SO_2

a. It is a colourless gas with harsh odour of burning match (i. e, possesses chocking smell).
b. It is heavier than air.
c. It is highly soluble in water.
d. Its melting point is -75.5^{0}C and boiling point is -10^{0}C.
e. It is acidic in nature.

Chemical properties of SO_2

Some of the chemical properties of SO_2 gas are discussed below:

1. Acidic Nature: SO_2 dissolves in water to form a sulphurous acid which is a weak diprotic acid. Due to this SO_2 is called an anhydride of sulphurous acid.

$$SO_2 + H_2O \rightarrow H_2SO_3 \text{ (sulphurous acid)}$$

Sulphurous acid exists only in aqueous solution. When it is boiled sulphur dioxide gas gets expelled.

$$H_2SO_3 \xrightarrow[Heat]{} SO_2 + H_2O$$

SO_2 being acidic in nature reacts with bases to produce salts. For examples,

$$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$$

$$SO_2 + Mg(OH)_2 \rightarrow MgSO_3 + H_2O$$

Since sulphurous acid is a diprotic acid, therefore aqueous solution of SO_2 on reaction with bases gives two series of salts. For examples,

$$SO_2 + H_2O \rightarrow H_2SO_3$$

$$H_2SO_3 + NaOH \rightarrow \underset{\text{(sod. bisulphite)}}{NaHSO_3} + H_2O$$

$$H_2SO_3 + 2NaOH \rightarrow \underset{\text{(sod. sulphite)}}{Na_2SO_3} + H_2O$$

Test yourself!

Show that SO_2 behaves as an acid.

2. Reducing Nature

Facts

SO_2 can acts as an oxidizing agent as well as a reducing agent. This is because S in SO_2 has +4 oxidation state and it has tendency to change its oxidation state from +6 (highest oxidation state) to +2(lowest oxidation state).

Some of the reactions showing reducing nature of SO_2 are:

a. Action with acidified $KMnO_4$: Sulphur dioxide reduces pink colour of acidified potassium permanganate solution into colourless manganate sulphate and itself gets oxidized to sulphuric acid.

$$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$$

$$\underline{\{SO_2 + H_2O + [O] \rightarrow H_2SO_4\}\times 5}$$

$$\underset{\text{(pink)}}{2KMnO_4} + 5SO_2 + 2H_2O \rightarrow K_2SO_4 + \underset{\text{(colorless)}}{MnSO_4} + 2H_2SO_4$$

b. Action with acidified $K_2Cr_2O_7$: Sulphur dioxide reduces the orange colour of acidified potassium dichromate solution into light green chromium sulphate and itself gets oxidized to sulphuric acid.

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + 2Cr_2(SO_4)_3 + 4H_2O + 3[O]$$
$$\{SO_2 + H_2O + [O] \rightarrow H_2SO_4\} \times 3$$
$$K_2Cr_2O_7 + 5SO_2 + 2H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

(orange) (light green)

c. Action with ferric salt: Sulphur dioxide reduces yellow colour of ferric salt to light green colour of ferrous salt and it gets oxidized to sulphuric acid.

$$SO_2 + 2H_2O \rightarrow H_2SO_4 + 2[H]$$
$$Fe_2(SO_4)_3 + 2[O] \rightarrow FeSO_4$$
$$Fe_2(SO_4)_3 + 5SO_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$$

(yellow) (light green)

d. Action with halogens: Sulphur dioxide reduces halogens (X_2: Cl_2, Br_2 and I_2) into haloacids (HX: HCl, HBr and HI) and it gets oxidized to sulphuric acid.

$$X_2 + 2H_2O \rightarrow 2HX + [O]$$
$$SO_2 + H_2O + 2[O] \rightarrow H_2SO_4$$
$$SO_2 + 2H_2O + X_2 \rightarrow 2HX + H_2SO_4$$

Where, X : Cl, Br and I

Test yourself!

a. What happens when SO_2 gas is passed through chlorine solution?
b. What happens when SO_2 gas is passed through acidified solution of potassium permanganate?

3. Oxidizing Nature

As discussed earlier, SO_2 also behaves as a good oxidizing agent. Let's see some reactions that depict the oxidizing nature of SO_2.

a. Action with H_2S

Sulphur dioxide oxidizes hydrogen sulphide to sulphur.

$$2H_2S + SO_2 \rightarrow 2H_2O + 3S$$

b. Action with Fe

Sulphur dioxide oxidizes iron to iron oxide and iron sulphide.

$$3Fe + SO_2 \rightarrow FeO + FeS$$

4. Bleaching property of SO_2

Facts

In the figure alongside table, grapes are bleached with SO_2 to have longer shelf life for the grapes including longer storage and long distance transportation. However, if excess of SO_2 is added then the grapes may decolorize.

Figure 9.43: Table grapes bleached with SO_2

Can you guess what is bleach?

When you insert a red rose in a gas jar filled with SO_2, the red colour of the rose disappears. Here SO_2 removes the natural colour of the rose but its action is temporary. When the red rose is shaken in air it restores its original colour. Thus any chemical (or agent) that is used to remove the natural colour of a substance either temporarily or permanently is called a bleach or bleaching agent. Besides this, bleach is also used as disinfectant because of its microbicidal properties. Some examples of bleach SO_2 are chlorine, sodium hypochlorite, calcium hypochlorite, hydrogen peroxide, etc.

The bleaching action SO_2 is due to two factors. They are:

i. Bleaching by reduction

SO_2 reacts with moisture to produce nascent hydrogen which is responsible for the bleaching action of it.

$$\underset{(\text{moisture})}{SO_2 + 2H_2O} \rightarrow H_2SO_4 + 2[H]$$

Vegetable colouring matter + [H] → Colourless reduction product

Here, the bleaching action is temporary. If the colourless reduced product is exposed to the atmosphere, then it will regain its original colour.

Example, if we keep red roses in a gas jar filled with SO_2, then the rose becomes colourless. On shaking the rose in the atmosphere it gains its red colour.

ii. Bleaching by addition reaction

In some cases, the bleaching action of SO_2 is due to an additional reaction.

Vegetable colouring matter + SO_2 → Colourless addition product

If the colourless addition product is treated with dilute sulphuric acid, then the product decomposes liberating SO_2 gas and retaining its original colour.

Thus in both cases the bleaching action of SO_2 is temporary i.e, reversible. SO_2 is a mild bleaching agent and bleaches soft materials like wool, silk, can sugar, flowers, fruits etc.

Test yourself!

How would you explain the bleaching phenomena of SO_2?

Uses of SO_2

Some uses of SO_2 are listed below:

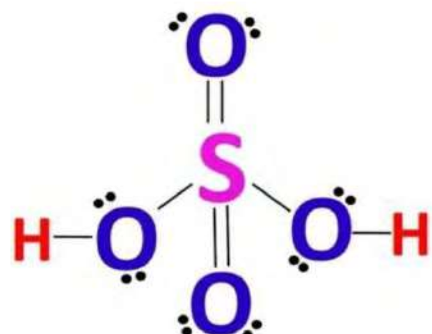
a. It is used in the manufacture of sulphuric acid.
b. It is used as a bleaching agent in the paper and sugar industry.
c. It is used as a preservative in alcohol drinks.
d. It is used as a disinfectant and fungicides.
e. It is used as a non-aqueous solvent in its liquid form.
f. It is used as an antichlor (i.e, removing chlorine traces from a substance bleached by chlorine).
g. It is used in preparation of calcium bisulphate which is an important chemical for the production of paper from wood pulp.
h. It is used to bleach very soft materials like, straw, paper, pulp, wool, silk, flower etc.

9.8.5: Sulphuric acid

Activity: 37

Observe the structure of sulphuric acid given:

a. Why is it called oxyacid of Sulphur?
b. What is the geometry around the central atom Sulphur?



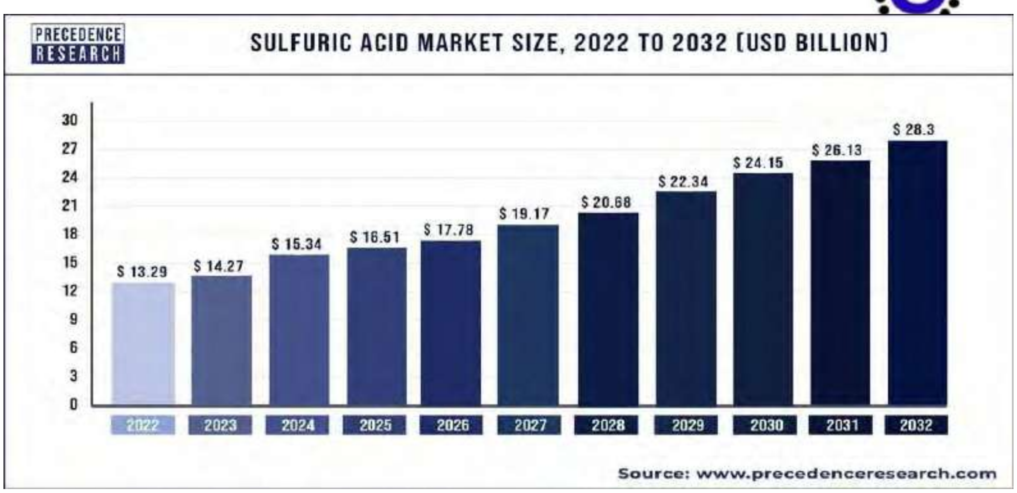


Figure 9.44: Global market graph of Sulphuric acid

Sulfuric acid is one of the most widely used industrial chemicals, with applications in various industries such as fertilizers, chemicals, petroleum refining, metal processing, and wastewater treatment. As industrial activities grow, the demand for sulfuric acid also increases.

In early days, it was prepared from ferrous sulphate crystal (green vitriol), therefore, it is also known as **'oil of vitriol'**. Due to its large production and applications in industries, it is known as **'king of chemicals'**. Nowadays, sulphuric acid is manufactured by **contact process**.

Physical properties of sulphuric acid

a. It is a colourless syrupy liquid without any odour having specific gravity 1.84 at 15^{0}C.
b. It is highly soluble in water. When dissolved in water it produces a large amount of heat.
c. Its melting point is 10.5 ^{0}C and boiling point is 338 ^{0}C.
d. It is hygroscopic in nature and therefore used as a drying agent.
e. It turns blue litmus paper into red in aqueous form.
f. It is highly corrosive in nature and causes severe burns when it comes in contact with skin.

Chemical properties of sulphuric acid

Some of the chemical properties of sulphuric acid are discussed below:

1. Acidic Nature

Sulphuric acid is a strong dibasic acid. It ionizes completely in aqueous solution producing a large concentration of hydrogen ions. It ionizes in two steps:

$$H_2SO_4 \rightarrow H^+ + HSO_4^-$$

$$H_2SO_4 \rightarrow 2H^+ + SO_4^{--}$$

Being dibasic sulphuric acid reacts with bases to produce two series of salts. For example,

$$NaOH + H_2SO_4 \rightarrow \underset{\text{(sod. bisulphate)}}{NaHSO_4} + H_2O$$

$$NaOH + H_2SO_4 \rightarrow \underset{\text{(sod. sulphate)}}{Na_2SO_4} + H_2O$$

Test yourself!

How would you test the acidic nature of sulphuric acid in laboratory?

2. Oxidizing Nature

S in H_2SO_4 is in +6 oxidation state which is highest. Therefore, H_2SO_4 behaves as a good oxidizing agent. In fact, hot and concentrated sulphuric acid is a strong oxidizing agent while dilute H_2SO_4 has very weak oxidizing power. H_2SO_4 decomposes to produce nascent oxygen which oxidizes a number of chemical species.

$$H_2SO_4 \rightarrow H_2O + SO_2 + [O]$$

Some reactions showing oxidizing property of H_2SO_4 are-

a. Action with metals

Concentrated sulphuric acid oxidizes metals to their corresponding metal sulphate along with evolution of sulphur dioxide gas.

$$Zn + \text{conc. } 2H_2SO_4 \rightarrow ZnSO_4 + SO_2 + 2H_2O$$

$$Fe + \text{conc. } 2H_2SO_4 \rightarrow FeSO_4 + SO_2 + 2H_2O$$

$$Cu + \text{conc. } 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$$

Note: Noble metals like platinum (Pt) and gold (Au) do not react with concentrated sulphuric acid.

Metals that lie above hydrogen in electrochemical series react with dilute sulphuric acid to liberate hydrogen gas. For examples,

$$Zn + dil.\ H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$$

$$Mg + dil.\ H_2SO_4 \rightarrow MgSO_4 + H_2\uparrow$$

b. Action with non-metals

Sulphuric acid oxidizes several non-metals like C, S, P, etc. as explained below:

i. Action with C

Sulphuric acid oxidizes carbon to carbon dioxide.

$$\{H_2SO_4 \rightarrow SO_2 + H_2O + [O]\}\times 2$$

$$C + 2[O] \rightarrow CO_2$$

$$2H_2SO_4 + C \rightarrow 2H_2O + 2SO_2 + CO_2$$

ii. Action with S

Sulphuric acid oxidizes sulphur to sulphur dioxide.

$$\{H_2SO_4 \rightarrow SO_2 + H_2O + [O]\}\times 2$$

$$S + 2[O] \rightarrow SO_2$$

$$2H_2SO_4 + S \rightarrow 2H_2O + 3SO_2$$

iii. Action with P

Sulphuric acid oxidizes phosphorous to phosphoric acid.

$$\{H_2SO_4 \rightarrow SO_2 + H_2O + [O]\}\times 2$$

$$2P + 5[O] \rightarrow P_2O_5$$

$$P_2O_5 + 3H_2O \rightarrow 2H_3PO_4$$

$$5H_2SO_4 + 2P \rightarrow 2H_3PO_4 + 5SO_2 + H_2O$$

c. Action with other reducing agents

Sulphuric acid oxidizes other reducing agents like haloacids, hydrogen sulphide etc.

i. Action with halogen acid

Sulphuric acid oxidizes halogen acids to halogens.

$$H_2SO_4 \rightarrow SO_2 + H_2O + [O]$$

$$2HX + [O] \rightarrow H_2O + X_2$$

$$H_2SO_4 + 2HX \rightarrow 2H_2O + SO_2 + X_2$$

Where, X: Cl, Br and I

Test yourself!

What happens when hydrochloric acid comes in contact with concentrated sulphuric acid?

ii. Action with H_2S

Sulphuric acid oxidizes hydrogen sulphide to sulphur.

$H_2SO_4 \rightarrow SO_2 + H_2O + [O]$

$H_2S + [O] \rightarrow H_2O + S$

$H_2SO_4 + H_2S \rightarrow 2H_2O + SO_2 + S$

3. Dehydrating Nature

Activity: 37

Take a beaker containing 5 g of sugar and carefully add concentrated sulphuric acid slowly and observe the reaction.

- **a. What do you observe?**
- **b. What could be the significance of this property of sulphuric acid?**

Concentrated sulphuric acid has greater affinity towards water. Due to this property of the sulphuric acid, it is used as an abstract water molecule form a substance, i.e, it is widely used as a dehydrating agent in many chemical reactions. Some reactions depicting the dehydrating nature of the acid are discussed below:

a. Action with carboxylic acid

Concentrated sulphuric acid dehydrates carboxylic acids like formic acid, oxalic acid etc.

$HCOOH + \text{conc. } H_2SO_4 \rightarrow CO + H_2SO_4.H_2O$

$(COOH)_2 + \text{conc. } H_2SO_4 \rightarrow CO + CO_2 + H_2SO_4.H_2O$

b. Action with hydrated salt

Concentrated sulphuric acid dehydrates hydrated slats like blue vitriol ($CuSO_4.5H_2O$), washing soda ($Na_2CO_3.10H_2O$), etc.

$CuSO_4.5H_2O + \text{conc. } H_2SO_4 \rightarrow CuSO_4 + H_2SO_4.5H_2O$

c. Action with carbohydrates

Concentrated sulphuric acid dehydrates carbohydrates like glucose, fructose, sucrose, etc. producing black charred mass of carbon.

$C_6H_{12}O_6 + \text{conc. } H_2SO_4 \rightarrow 6C + H_2SO_4.6H_2O$

(charred mass)

$C_{12}H_{22}O_{11} + \text{conc. } H_2SO_4 \rightarrow 12C + 6H_2SO_4.11H_2O$

(charred mass)

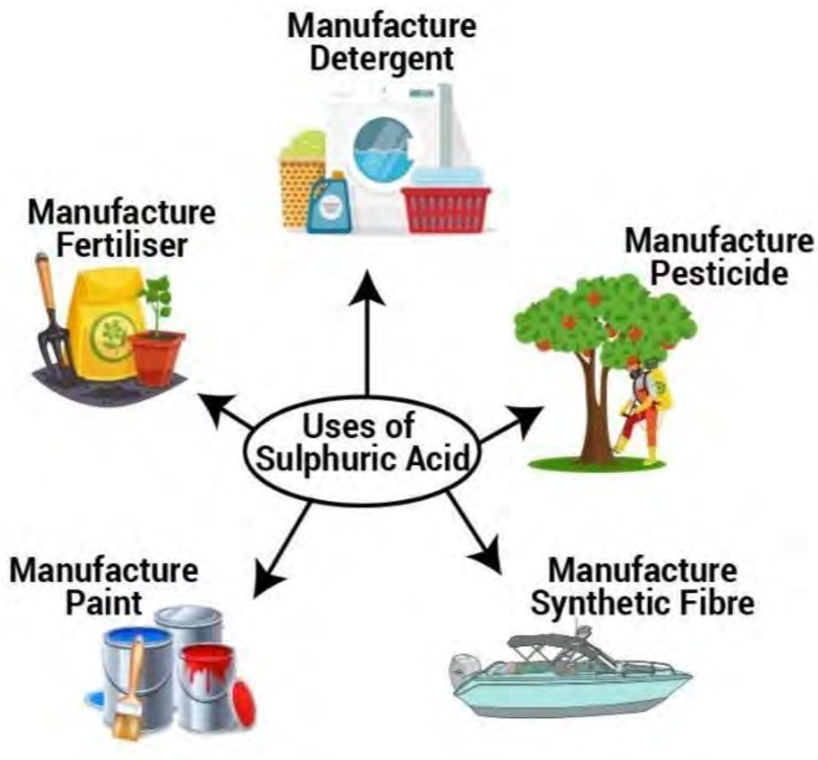
Figure 9.45: Reaction between sugar anc conc.H_2SO_4

Test yourself!

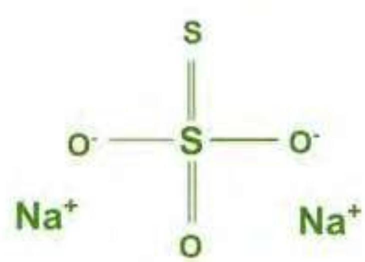
Give a reaction to show that sulphuric acid is a dehydrating agent.

Facts

Dilution of sulphuric acid should be done by adding concentrated acid to water. One should not dilute the acid by adding water to the acid because sulphuric acid has greater affinity towards water, therefore, it reacts violently producing large amounts of heat which may cause accidents.



9.8.6: Sodium thiosulphate and its use



Sodium thiosulphate is a white crystalline hydrated inorganic compound with molecular formula $Na_2S_2O_3.5H_2O$. It is also known as ***hypo*** derived from its original name hyposulphite of soda. It is soluble in water.

a. It is used for iodine titration in volumetric analysis.
b. It is used as anitchlor for removing excess chlorine on fabrics that have been bleached by chlorine.
c. It is used in photography for fixing films and prints in the name of hypo.

Project work

1. Make a list of allotropes of sulphur in a chart paper and write the use of each.
2. Take a chart paper and draw Kipp's apparatus labelling its various parts and write the working principle of it for the preparation of hydrogen sulphide gas.
3. Make a chart paper presentation showing comparison of bleaching action of Cl_2 and SO_2.
4. Take a chart paper and write the acidic, oxidizing and dehydrating properties of sulphuric acid.

Exercise

A. Multiple choice questions

1. Which of the following is used in photography?
a. Na_2SO_4 b. $Na_2S_2O_3$ c. $Na_2S_2O_4$ d. $Na_2S_2O_6$

2. What happens to the later compound, when SO_2 gas is passed through potassium permanganate solution?
a. Oxidized b. Reduced c. Dehydrated d. Decomposed

3. Which of the following behaves as both oxidant and reductant?
a. H_2S b. Cl_2 c. $Na_2S_2O_4$ d. SO_2

4. Which of the following is true for sulphurdioxide gas?
a. Bleaching agent b. Charring agent
c. Precipitating agent d. Complexing agent

5. What happens when dilute H_2SO_4 is treated with zinc metal?
a. O_2 b. Cl_2 c. Br_2 d. H_2

6. What is the oxidation state of S in H_2SO_4?
a. +6 b. +4 c. +2 d. 0

7. In the reaction, $BaCl_2$ + dil. $H_2SO_4 \rightarrow BaSO_4\downarrow + 2HCl$, what is the role of the acid?
a. Precipitate b. Precipitant c. Complexing agent d. Reductant

8. Which of the following statements is wrong for SO_2?
a. It bleaches by reduction b. Its bleaching action is reversible
c. It bleaches hard substances d. It is a mild bleaching agent

9. Which of the following turns lead acetate paper black?
a. H_2S b. SO_2 c. $Na_2S_2O_4$ d. H_2SO_4

10. **SO_2 is an acidic anhydride of which of the following acids?**
a. Sulphurous acid b. Sulphuric acid
c. Pyrosulphuric acid d. Thio-sulphurous acid

11. **Which of the following is known as oil of vitriol?**
a. H_2SO_3 b. H_2SO_4 c. $H_2S_2O_8$ d. H_2SO_5

12. **What is the chemical structure and formula of sulphur?**
a. S & S_3 b. S & S_4 c. S & S_8 d. S & S_5

13. **Who first recognized sulphur as an element?**
a. Louis Pasteur b. Antonie Lavoiser c. John Dalton d. Humphry Davy

14. **What is the geometry of H_2S?**
a. Linear b. Angular c. Trigonal d. Non-linear

15. **Which compound is used as reducing bleach for paper?**
a. SO_2 b. H_2SO_4 c. SO_3 d. H_2S

16. **What is the chemical formula of sodium thiosulphate?**
a. Na_2SO_2 b. $Na_2S_2O_5$ c. $Na_2S_3O_2$ d. $Na_2S_2O_3$

B. Short answer questions

1. Define the term polymorphism and allotropism. List the allotropes of sulphur.
2. Show that sulphuric acid is a diprotic acid. Write the action of pure sulphuric acid towards blue and red litmus paper.
3. Write the formula of hypo. Mention some uses of sulphur.
4. Give reason:
   a. SO_2 acts as oxidizing as well as reducing agent.
   b. H_2S acts as a reducing agent.
   c. H_2SO_4 acts as an oxidizing agent.
   d. H_2SO_4 is a dehydrating agent.
5. Differentiate between:
   a. Bleaching action of Cl_2 and SO_2.
   b. Acidic nature of SO_2 and H_2SO_4
6. Describe the preparation of H_2S gas by Kipp's apparatus. How would you confirm the acidic nature of hydrogen sulphide gas?
7. Explain the working principle of Kipp's apparatus. Write its significance.

8. What happens when H_2S gas is passed through-
   a. acidified $KMnO_4$?
   b. Copper sulphate solution in acidic medium?
   c. SO_2 gas?
   d. concnetrated sulphuric acid?
   e. acidified lead acetate solution?

9. Write a note on:
   a. Use of H_2S as an analytical reagent.
   b. Bleaching action of SO_2.
   c. Dehydrating property of H_2SO_4
   d. SO_2 as an oxidizing and reducing agent.

10. List the uses of SO_2 gas.

11. Why is sulphuric acid called "king of chemicals"? Enlist the uses of sulphuric acid.

12. Write a molecular formula of sodium thiosulphate and mention some of its uses.

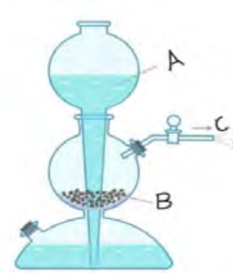
13. Write reactions to justify the sulphuric acid as-
   a. Oxidizing agent b. Dehydrating agent c. Acidic nature.

14. What happens when a moist red petal of a flower is kept in a jar full of SO_2 gas?

15. Write the action of conc. H_2SO_4 on:
   a. Sugar b. P_4 c. S_8 d. Copper turning

16. Write the action of H_2S on:
   a. $KMnO_4$ b. $K_2Cr_2O_7$ c. Moist Cl_2 d. SO_2 e. $FeCl_3$

C. Long answer questions



17. Observe the given set up and answer the following questions:
   a. Identify the apparatus and write its one advantage.
   b. Identify A, B and C.
   c. Write the balanced chemical equation involved in this set up.
   d. Is the nozzle open or closed in the given set up?
   e. Give one application of gas released from C.

18. Complete the given table with appropriate balanced chemical equation.

Reactant A	Reactant B	Products
....	H_2S	$PbS + CH_3COOH$
H_2S	$K_2Cr_2O_7 + H_2SO_4$	....
Cl_2	SO_2	....
SO_2	....	$FeO + FeS$
....	H_2SO_4	$CO + CO_2 + H_2O$
H_2SO_4	HNO_3	....
SO_3	....	$H_2S_2O_7$
Cu^{2+}	H_2S (acidic medium)	....

19. Write the products of the following reactions and balance them.

a. $C_{12}H_{22}O_{11}$ + conc. $H_2SO_4 \rightarrow$?

b. $H_2SO_4 + P \rightarrow$?

c. $H_2S + SO_2 \rightarrow$

d. $Fe_2(SO_4)_3 + SO_2 + H_2O \rightarrow$?

e. $Ni^{++} + H_2S \xrightarrow{\text{Alkaline medium}}$?

f. $FeCl_3 + H_2S \rightarrow$?

g. $KMnO_4 + H_2SO_4 + H_2S \rightarrow$?

h. $FeS + H_2SO_4 \rightarrow$?

20. Concentrated sulphuric acid is both an oxidizing agent and a non-volatile acid.

a. Write an equation each to illustrate the above properties of sulphuric acid.

b. Write the test reaction of this acid in the chemistry laboratory.

c. Write any three uses of this acid.

d. Why is this acid called king of the acid?

e. How would you carry out dilution of this acid?

21. Copy and complete the following table:

Column A **Substance reacted with acid**	**Column B** **Dilute or Concentrated Sulphuric acid**	**Column C** **Gas produced**
		Chlorine
		Carbon monoxide
		Hydrogen
		Carbon dioxide

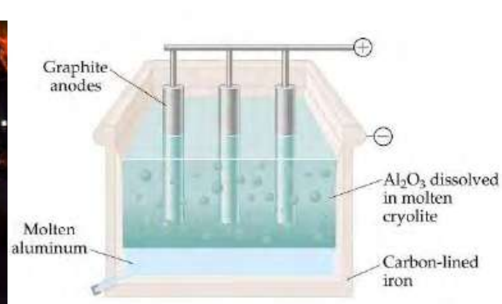
UNIT 10:
CHEMISTRY OF METALS

10.1: Metals and Metallurgical Process

10.1.1: Metallurgy and its types

Activity-1

Which of the given figures is related to pyrometallurgy and electrometallurgy? Why?



Metallurgy is the branch of material science that is concerned with extraction of metals (in pure form) from their respective ores and utilization of them. Actually it is an art of science and engineering where metals from their chief ores are extracted in pure form more conveniently and economically. Several physical and chemical processes are involved during metallurgy among which reduction of metallic compounds to obtain pure metal is a very important step. On the basis of method of reduction applied, metallurgy is classified into three types. They are pyrometallurgy, hydrometallurgy and electrometallurgy.

1. Pyrometallurgy

The branch of metallurgy in which extraction and purification of metal is carried out at higher temperature is called pyrometallurgy. Here, *pyro* means heat. In this type of metallurgy, once the ore becomes free from impurities it is strongly heated with a suitable reducing agent to get metal in molten state. Examples of metals extracted by this process include the oxides of less reactive elements like Cr, Fe, Cu, Zn, Hg, Sn and Mn, etc. Pyrometallurgical process is generally grouped into one or more of the following categories:

i. Calcination

ii. Roasting

iii. Smelting

iv. refining

2. Hydrometallurgy

The branch of metallurgy in which extraction and purification of metal is carried in presence of aqueous solution is called hydrometallurgy. Here, *hydro* means water. In this type of metallurgy, the metallic ore is dissolved in a suitable solvent like sodium or potassium cyanide to produce their respective soluble complex leaving behind the impurities unaffected. Finally, metal is reduced (i.e, recovered) from the complex through precipitation reaction by adding a precipitant like Zn. Hydrometallurgical process is used to extract noble metals like Cu, Ag, Au etc. It is generally divided into three main areas:

i. Leaching
ii. Solution concentration and purification
iii. Metal recovery

Example,

Silver ore (Ag_2S) can be dissolved in 0.3% NaCN solution to form a complex and finally silver can be recovered by precipitation with Zn as shown in the reaction.

$$Ag_2S + 4NaCN \rightarrow 2Na[Ag(CN)_2] + Na_2S$$

$$2Na[Ag(CN)_2] + Zn \rightarrow 2Ag\downarrow + Na_2[Zn(CN)_4]$$

This hydrometallurgy includes leaching process.

3. Electrometallurgy

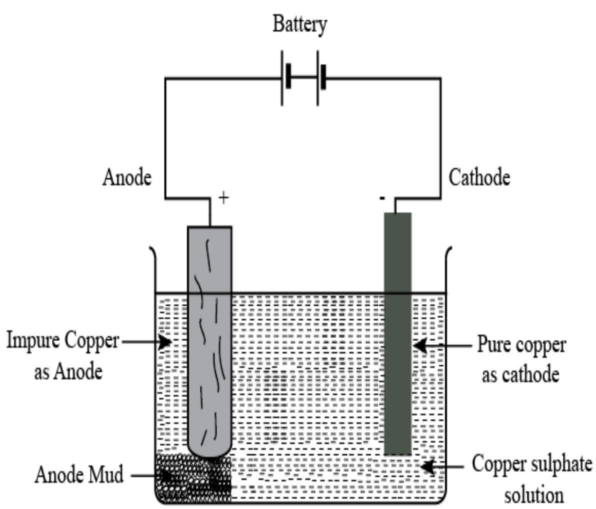


FIGURE 2:

The branch of metallurgy in which extraction and purification of metal is carried out through electrolysis is called electrometallurgy. In this type of metallurgy, more electropositive metals are extracted from their ores. Metals like alkali metals and alkaline earth metals, which cannot be extracted by pyrometallurgy due to higher reactivity and low boiling point, are extracted by this method. It is actually the last step in metal extraction preceded by pyrometallurgical or hydrometallurgical process. In the case when metal ore is not suitable for electrolysis then first it is converted into electrolysable form before electrolysis. In this method metals are collected (i.e, deposited) at cathode.

Test Yourself!

Why is electrometallurgy suitable for the extraction of sodium metal from NaCl?

10.1.2: Introduction of Ores

Activity-2: Observe the ores of iron, zinc and copper given in the figure:

a. Identify one ore of iron, zinc and copper from the figure.
b. Identify the molecular formula of the ores given.

Majority of elements in the periodic table are metals. They include alkali metals, alkaline earth metals, transition metals, lanthanides and actinides. They possess special properties like hardness, malleability, ductility, sonority, conductivity, good reducing nature, metallic lusture, etc. Mercury is the only metal that exists in liquid state at room temperature. Metals generally form basic oxides. They form electropositive radicals as they have a tendency to lose their valence electrons.

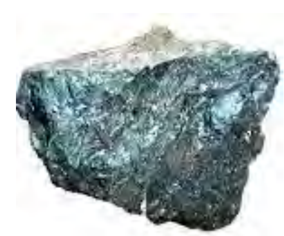
Naturally metals are found either in pure state (i.e, metallic form) or in combined state. Metals that are found in their elemental form (metallic form) in nature are called native metals. Gold and silver are the most well-known metals that are found in their native state. Most of the metals are found in combined state as their oxides, sulphides, silicates, carbonates, sulphates, phosphates, halides, nitrates, etc.

The above figure shows the different forms of copper found in the earth's crust. These forms are minerals of copper. One cannot extract the copper metal more efficiently and cost effectively from all of these forms. You can see the percentage of copper in cuprite is greater among others and therefore extraction of copper from this mineral will be beneficial and cost effective. This cuprite is now called chief ore of copper.

Hence, naturally occurring substances containing the metals along with impurities present in the erath's crust are called **minerals.** The mineral through which the metal can be extracted conveniently and economically is called **ore.** It should be noted that ***all ores are minerals but all minerals are not ores.*** Some common minerals are listed below:

A. Oxide minerals: These include metals like Al, Cr, Fe, Zn, etc.

S. N.	Name of the minerals	Chemical formula
1.	Alumina	Al_2O_3
2.	Bauxite	$Al_2O_3.2H_2O$
3.	Chromeochore	Cr_2O_3
4.	Chromite	$FeO.Cr_2O_3$
5.	Diaspare	$Al_2O_3.H_2O$
6.	Franklinite	$ZnO. Fe_2O_3$
7.	Haematite	Fe_2O_3



pyrolusite

8.	Ilmenite	$FeO.TiO_2$
9.	Limonite	$Fe_2O_3.3H_2O$
10.	Magnetite	Fe_3O_4
11.	Pitch Blende	U_3O_8
12.	Pyrolusite	MnO_2
13.	Rutile	TiO_2
14.	Zincite (Red zinc)	ZnO


Dolomite

B. Carbonate minerals: These include metals like Mg, Ca, Cu, Zn, Pb, etc.

S. N.	Name of the minerals	Chemical formula
1.	Azurite	$2CuCO_3.Cu(OH)_2$
2.	Calamine	$ZnCO_3$
3.	Cerussite	$PbCO_3$
4.	Dolomite	$MgCO_3.CaCO_3$
5.	Limestone	$CaCO_3$
6.	Magnesite	$MgCO_3$
7.	Malachite	$CuCO_3.Cu(OH)_2$
8.	Siderite	$FeCO_3$


Galena

C. Sulphate minerals: These include metals like Na, K, Mg, Ca, Ba, etc.

S. N.	Name of the minerals	Chemical formula
1.	Barytes	$BaSO_4$
2.	Epsom salt	$MgSO_4.7H_2O$
3.	Gypsum	$CaSO_4.2H_2O$
4.	Glauber's salt	$Na_2SO_4.10H_2O$
5.	Kainite	$KCl.MgSO_4.5H_2O$
6.	Kisserite	$MgSO_4.H_2O$
7.	Schonite	$K_2SO_4.MgSO_4.6H_2O$


Malachite

D. Halide minerals: These include metals like alkali metals, Ca, Mg, Ag, etc.

S. N.	Name of the minerals	Chemical formula

1.	Common salt (Rock salt)	$NaCl$
2.	Carnallite	$KCl.MgCl_2.6H_2O$
3.	Cryolite	Na_3AlF_6
4.	Fluorspar	CaF_2
5.	Horn Silver	$AgCl$
6.	Sylvine	KCl

E. Silicate minerals: These include metals like Li, K, Al, etc.

S. N.	Name of the minerals	Chemical formula
1.	Asbestos	$CaMg_3(SiO_3)_4$
2.	China clay	$Al_2O_3.2SiO_2.2H_2O$
3.	Feldspar	$KAlSi_3O_8$
4.	Mica	$K_2O.2Al_2O_3.6SiO_2.2H_2O$
5.	Patalite	$LiAl(Si_2O_3)_2$
6.	Talc	$Mg_2(Si_2O_5)_2.Mg(OH)_2$

Hematite

F. Sulphide minerals: These include metals like Fe, Ni, Cu, Zn, Ag, etc.

S.N.	Name of the minerals	Chemical formula
1.	Cinnabar	HgS
2.	Copper pyrite	$CuFeS_2$
3.	Copper glance	Cu_2S
4.	Iron Pyrite	FeS_2
5.	Lead Galena	PbS
6.	Ruby silver	$3Ag_2S.Sb_2S_3$
7.	Zinc blende	ZnS

10.1.3: Gangue (or matrix), Flux and Slag, Alloys and Amalgams

Activity-3

Observe the given table and answer the following questions:

Impurity	Flux	Slag
SiO_2 +	$CaO \longrightarrow$	$CaSiO_3$
P_2O_5 +	$3CaO \longrightarrow$	$Ca_3(PO_4)_2$
MnO +	$SiO_2 \longrightarrow$	$MnSiO_3$

a. Identify the acidic and basic impurities.
b. Name the acidic and basic flux used to remove the impurities.

Gangue: The unwanted material or earthy impurities present in ores are called **gangue**. They do not melt and do not get volatilized even at high temperatures. They are also known as **refractory materials** or **infusible materials** or **matrixes.** They are associated with ores as rock, stone, sand, mud or any other components. Ores contain silicates, carbonates, aluminates, etc. as impurities.

Flux: A chemical substance added to ore during extraction of metal in order to remove gangue is called **flux**. Actually flux converts infusible materials (i.e, gangue) into fusible (molten) mass called slag. Depending upon the nature of gangue present in the ore, flux is of two types. They are:

i) Acidic Flux: When the impurities present in the ore are basic like MgO, CaO, FeO, etc. then, the flux, like SiO_2, P_2O_5, borax ($Na_2B_4O_7.10H_2O$), etc., added are called **acidic flux.** For example, during the extraction of copper the basic impurity, CaO is removed by adding acidic flux SiO_2.

$$\underset{\text{(Gangue)}}{CaO} + \underset{\text{(Flux)}}{SiO_2} \rightarrow \underset{\text{(Slag)}}{CaSiO_3} \text{ (calcium silicate)}$$

ii) Basic Flux: When the impurities present in the ore are acidic like SiO_2, P_2O_5etc then the flux, like MgO, quick lime (CaO), limestone ($CaCO_3$), magnesite($MgCO_3$), haematite (Fe_3O_4),etc. added are called **basic flux**. For example,

$$\underset{\text{(Gangue)}}{SiO_2} + \underset{\text{(Flux)}}{CaCO_3} \rightarrow \underset{\text{(Slag)}}{CaSiO_3} + CO_2\uparrow$$

Slag: The fusible mass formed by the reaction of gangue present in the ore with suitable flux is called **slag.**

Flux + Gangue → Slag

Example,

$$\underset{\text{(Flux)}}{SiO_2} + \underset{\text{(Gangue)}}{CaO} \rightarrow \underset{\text{(Slag)}}{CaSiO_3} \text{ (calcium silicate)}$$

Slag, being lighter than the molten metal, floats over the surface of metal which can be easily skimmed off and removed.

Test Yourself!

Why is slag a fusible mass?

Alloys



Identify the metal used in the given pictures:

Bell metal is an **alloy** that is primarily used for making bells. The typical composition of bell metal is copper and tin. The composition of stainless steel is iron chromium, carbon and other elements.

Thus, an alloy is a homogeneous mixture of metals or metals with nonmetals.

Now can you say why alloys are prepared? Well, in order to have a desired property by a product which constituent elements do not possess, alloys are made. For example, iron becomes hard when C is added and it becomes hard stainless steel when Ni and Cr are added. Some examples of alloys are:

A: **Ferrous alloys:** In these alloys one of the constituent elements is always iron.

S. N.	Alloy	Composition
1.	Ordinary steel	Fe (98-99%) + C (0.2-2%)
2.	Stainless steel	Fe(74-80%) + Cr((12-18%) + C(0.2-2%) + Ni (1%)
3.	Tungsten steel	Fe(80-85%) + W(14-20%) + C(0.2-2%)
4.	Nickel steel	Fe (≈95%) + Ni (≈5%) + C(0.2%)
5.	Invar	Fe (64%) + Ni(36%)
6.	Nichrome	Ni(≈60%) + Fe(≈25%) + Cr(≈15%) + C(0.2%)

B: **Non-Ferrous alloys:** These alloys do not contain iron.

S. N.	**Alloy**	**Composition**
1.	Brass	Cu (60-80%) + Zn (20-40%)
2.	Bronze	Cu(75-90%) + Sn((10-15%)
3.	German silver	Cu(60%) + Zn(20%) + Ni(20%)
4.	Bell metal	Cu (80%) +Sn (20%)
5.	Gun metal	Cu (90%) + Sn(10%)
6.	Soft solder	Sn(50%) + Pb(50%)
7.	Nichrome	Cr(45%) + Ni(55%)
8.	Pewter	Sn (80%) + Pb(20%)

Amalgam

Figure 3: Example of amalgam (Dental amalgam)

An amalgam is an alloy of two or more metals in which one of the metals is always mercury. For example, dental amalgam is an alloy of Hg, Ag, Cu and Sn along with some Zn. It is used for filling teeth to improve handling characteristics and clinical performance. Some examples of amalgam are- sodium amalgam (Na + Hg), Gold amalgam (Au + Hg), Silver amalgam (Ag + Hg) etc. Mercury does not form an amalgam with iron, platinum, tungsten, etc.

Test yourself!

a. Give an example of alloy and amalgam.
b. Write the composition of- i) Bell metal   ii) Brass

10.1.4: General principle of extraction of metals

Activity-4

Observe the figure given and discuss how iron utensils are made from its ore.

You might have observed iron rods used for construction of houses, buildings, overhead bridges, etc. You are quite aware of the fact that this iron metal is obtained from one of its minerals called hematite. Now, all you need to know is the various steps involved during the extraction of metals from their chief ore. The principle and steps involved in metallurgy may vary due to the nature of metals and other factors. However, some general steps applied in metallurgy are explained below along with flow charts:

1. Crushing and pulverization
2. Concentration of ore
3. Conversion of ore in to oxide
4. Reduction and
5. Purification of the metal

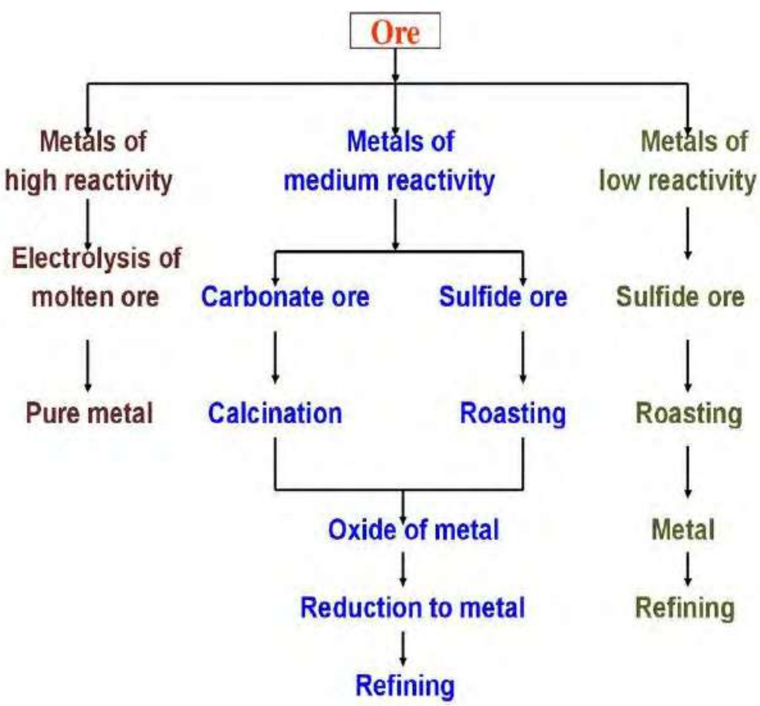
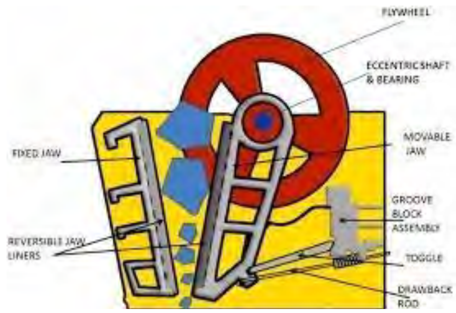


Figure: General steps involved in metallurgy

1. Crushing and pulverization

This is the first step of metallurgy where big lumps of ore mined from the earth's crust are broken down into smaller pieces either with the help of a hammer or jaw crusher. This process is called **crushing**. Finally crushed ore pieces are powdered in the ball mills or stamp mills. This process is called **pulverization.**



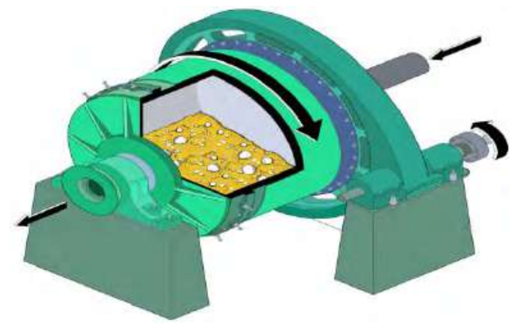


FIGURE-1: JAW

FIGURE-2: BALL MILL

2. Concentration (or Dressing) of ore

The powdered ore obtained in the first step is associated with earthy impurities like mud, sand, lime, rock, stone etc. These impurities are collectively called gangue or matrix. The process applied to remove such impurities is called **concentration or dressing of ores**. The method applied for dressing of ore depends upon the nature of ore, nature of impurities and some type the type of metallurgy employed. Some of the common methods employed for concentration or dressing of ores are explained below:

i. **Hand Picking**

Hand picking can only be used when the impurities are clearly seen and large enough to pick. This method is generally slow and tedious. However, it can be used in the initial stage of concentration.

ii. Gravity separation or hydraulic washing (Levigation) method

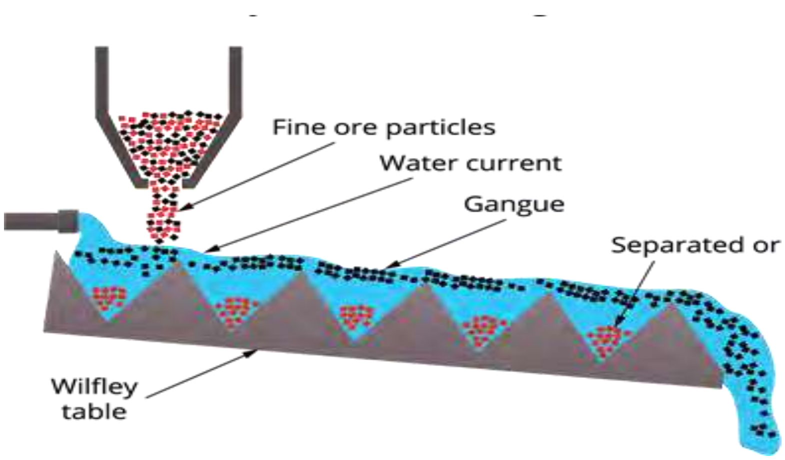


FIGURE: GRAVITY SEPARATION METHOD

This method of concentration of the ore is based on the differences in densities of gangue and ore. It is generally applied for oxygen containing impurities which are comparatively heavier in densities. In this method the concentrated ore is washed with an upward flow of running water where the lighter gangue particles are carried away by a stream of water leaving to settle down the ore particles at the bottom. For example, oxides like SnO, Fe_2O_3etc are concentrated by this method.

ii. Froth Floatation method

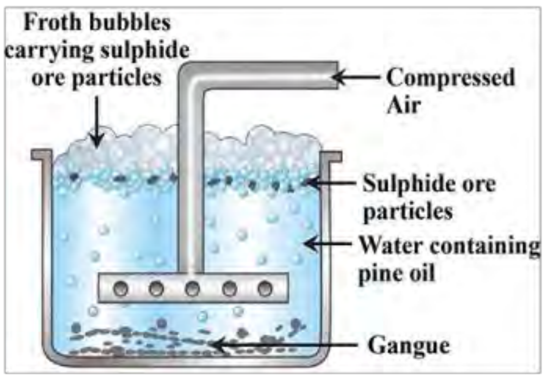


FIGURE: FROTH FLOATATION PROCESS

Froth floatation is applied when the concentrated ore and gangue have differences in the degree of wettability in a particular vegetable oil like pine oil or eucalyptus oil. In other words, froth floatation is a technique of separating ores from their gangue by exploiting differences in their hydrophobicity. This method is suitable for sulphide ores like ZnS, $CuFeS_2$, PbSetc which preferentially get adsorbed into pine oil.

In this method the powdered ore is mixed with pine oil (frother) and water along with some potassium ethyl xanthate as a collecting agent in a big tank. When the mixture is vigorously agitated by passing compressed air then the gangue particles are wetted by water and sink to the bottom of the tank. On the other hand, the ore particles get stuck to the pine oil and float at the top as shown in figure.

Do you know?

In froth floatation, the substances like pine oil, eucalyptus oil etc, that forms stable froth is called frother.
The substance that makes ore particles water repellers when passed through froth oil is called collector. For example, potassium ethyl xanthate, sodium ethyl xanthate, etc.

iii. Magnetic Separation method

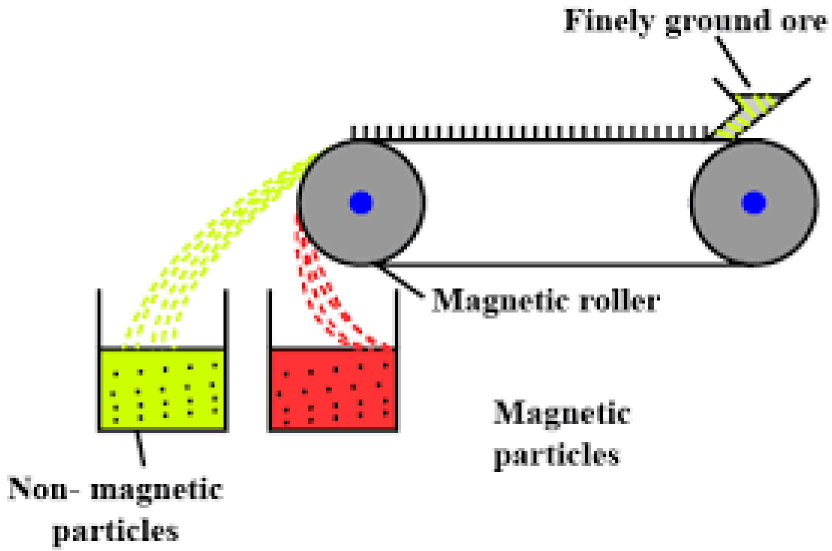
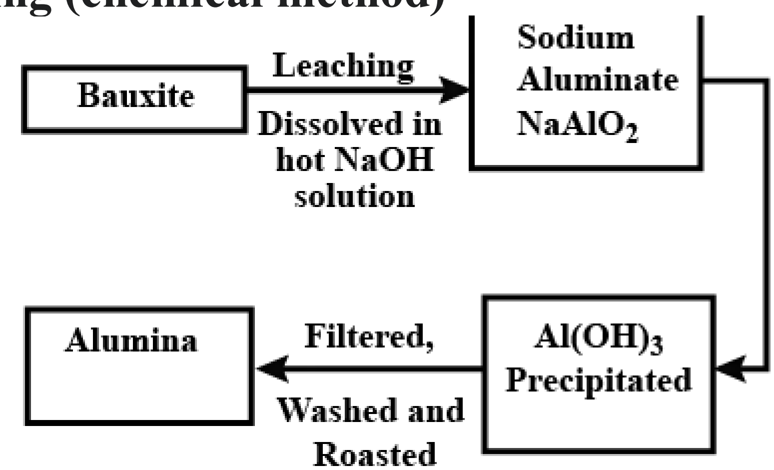


FIGURE: MAGNETIC

Magnetic separation method is applied only when either ore or gangue is magnetic in nature. Non-magnetic ores having magnetic impurities or magnetic ores having non-magnetic impurities are concentrated by this technique. In this method, pulverized ore is passed on a conveyor belt passing over a magnetic roller. The magnetic ore remains on the belt and the gangue falls off the belt as shown in figure. For example, tin stone (SnO_2) having magnetic impurity wolframite ($FeWO_4$) is concentrated by this method.

iv. Leaching (chemical method)



Leaching is a chemical method of concentrating ore where ore is treated with certain chemicals (acid, base or other chemicals) so that metals present in ore get converted into soluble salts while the impurities remain unaffected. Actually it is a part of hydrometallurgy.

For example, in the **metallurgy of Al**, the concentration of aluminum ore is done through leaching. Pure Al_2O_3 can be obtained by leaching bauxite ($Al_2O_3.2H_2O$) with concentrated solution of NaOH. The ore dissolves leaving behind the impurities that are separated by filtration.

$$\underset{\text{(Bauxite)}}{Al_2O_3.2H_2O} + 2NaOH \rightarrow \underset{\text{(sodium aluminate)}}{2NaAlO_2} + 3H_2O$$

This sodium aluminate is further treated with water to aluminum hydroxide which on ignition produces pure alumina.

$$2NaAlO_2 + 2H_2O \rightarrow Al(OH)_3 \downarrow + NaOH$$

$$Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$$

Facts

Leaching technique is used to extract noble metals like gold and silver.

Test Yourself!

a. You are provided with PbS, an ore of lead. Which concentration technique will you use to clean this ore and why?
b. What is leaching?

3. Conversion of ore into suitable reducible form
Calcination and Roasting

Before getting metal in free and pure state some metallurgical operations are still left to be carried. The main aim of this step is to convert the metal into its suitable oxide form from which metal can be easily obtained. In case of pyrometallurgy the concentrated ore is either subjected to calcination or roasting in order to obtain metal oxide so that reduction of metal becomes easier.

Calcination

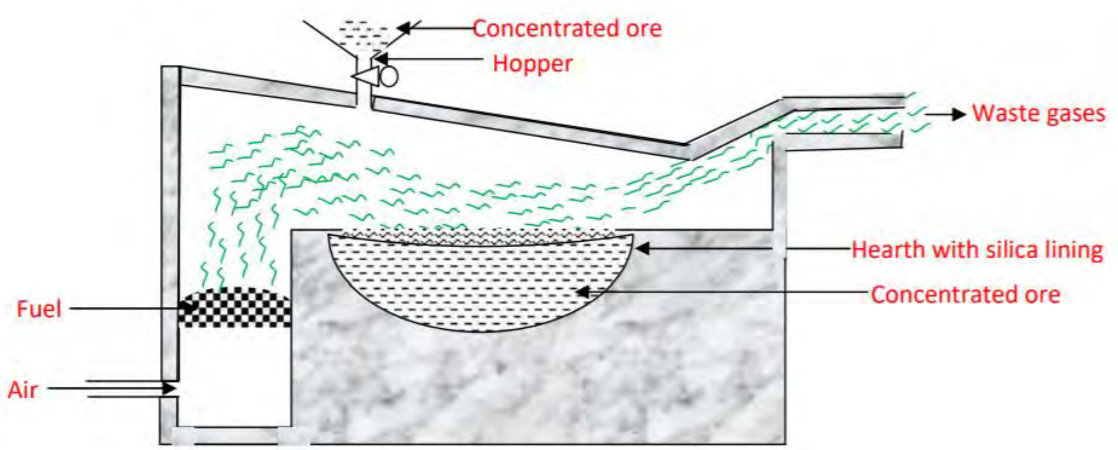


Figure: calcination

The process of heating concentrated ore below its melting point in a limited supply of air or in absence of air to convert ore into dry metallic oxide is called calcination. It is carried out in case of oxide or hydroxide or carbonate ores that generally do not require oxygen to produce metal oxide. This process is carried out in a reverberatory furnace. Some examples are:

When calamine is subjected to calcination it produces zinc oxide and carbon dioxide.

$$ZnCO_3 \xrightarrow[Heat]{} ZnO + CO_2\uparrow$$

When malachite is subjected to calcination it produces copper oxide, carbon dioxide and water.

$$CuCO_3.Cu(OH)_2 \xrightarrow[Heat]{} CuO + CO_2\uparrow + H_2O$$

Advantages of calcination

a. The concentrated ore gets decomposed into suitable metal oxide.

b. Non-metallic impurities like S, P, As, etc. get removed in the form of their volatile oxides.

c. Organic matters, water of crystallization and other impurities are also removed.

d. The ore becomes porous that results in easy handling.

Roasting

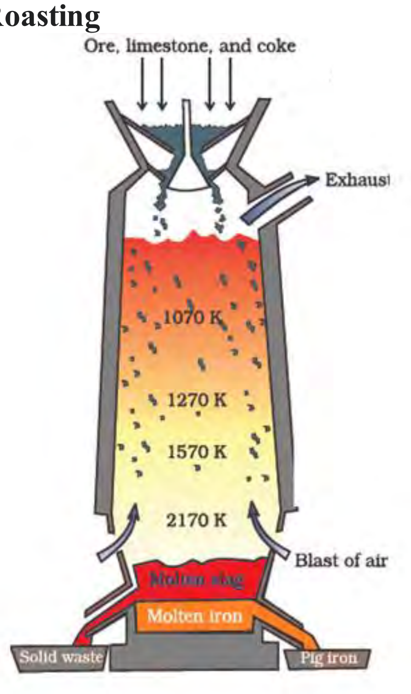


Fig: roasting of ore in blast furnace

The process of heating concentrated ore below its melting point in excess supply of air to convert ore into dry metallic oxide is called roasting. It is carried out for non-oxide ore (especially for sulphide ore) to produce metal oxide. This process is also carried out in a reverberatory furnace or sinterere. Some examples are:

When concentrated ore of zinc blende is subjected to roasting, then zinc oxide is produced.

$$2ZnS + 3O_2 \xrightarrow{Heat} 2ZnO + 2SO_2\uparrow$$

When concentrated ore of galena is subjected to roasting, lead oxide is produced.

$$2PbS + 3O_2 \xrightarrow{Heat} 2PbO + 2SO_2\uparrow$$

Some advantages of roasting:

a. The concentrated ore gets decomposed into suitable metal oxide.

b. Non-metallic impurities like S, P, As, etc. get removed in the form of their volatile oxides.

c. Organic matters, water of crystallization and other impurities are also removed.

d. The ore becomes porous that results in easy handling.

Test Yourself!

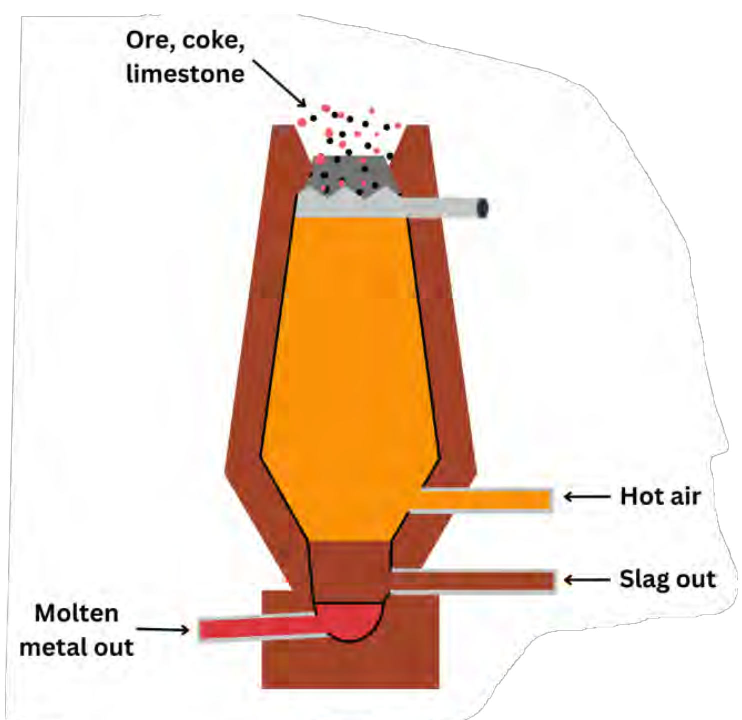
What is the major difference between calcination and

4. Reduction

Reduction is the chemical process where calcined or roasted ore is reduced with suitable reducing agents in order to get free metal. The choice of reducing agent depends upon the nature of metal and principle of metallurgy. Some common reducing agents used in metallurgy are coke(C), CO, H_2, Al, Mg, Zn etc. The calcined or roasted ore consists of some kind of refractory materials (gangue) as impurities which are infusible. These impurities are removed by adding flux in the form slag. Some commonly used methods of reduction are discussed below:

A. Chemical reduction methods

i) Carbon reduction method (Smelting)



The reduction process in which metal oxide is reduced to free metal when strongly heated with a reducing agent like carbon (coke) or carbon monoxide (CO) is called the carbon reduction process. This process is also known as smelting. The carbon reduction process is carried out in a blast furnace.

For example, in the **metallurgy of Fe,**

$$Fe_2O_3 + C \xrightarrow{Heat} 2Fe + 3CO\uparrow$$

$$Fe_2O_3 + CO \xrightarrow{Heat} Fe + CO_2\uparrow$$

FIGURE: BLAST FURNACE

The main aim of smelting is to remove earthy impurities (i.e, gangue) associated with ore. Hence, some suitable flux is always added along with a reducing agent which converts gangue into slag.

Calcined/ Roasted ore + Gangue + Reducing agent + Flux

The molten metal is taken out through the tapping hole of the blast furnace.

Test Yourself!

Why is CO_2 gas not used as a reducing agent in metallurgy?

ii. Thermite Process or Aluminothermic reduction (Goldschmidt process)

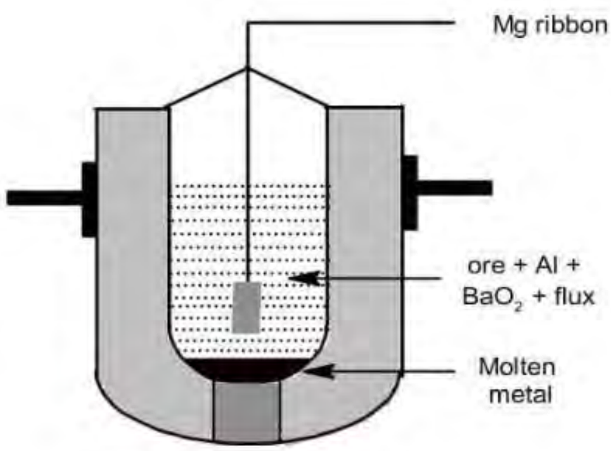


Fig: Alumino-thermite process

Aluminothermic process is employed for reducing certain metal oxides like Cr_2O_3, Mn_3O_4, etc. which cannot be reduced by carbon reduction method. This is because oxygen has higher affinity with these Cr and Mn metals than with carbon. In this case aluminum is used as a reducing agent. Hence, the reduction process in which oxides of metals like Cr, Mn and W are reduced with the help of alumnium as a reducing agent is called **thermite process** or **aluminothermic reduction or Goldschmidt process.** In this method the roasted or calcined ore is mixed with alumnium powder with little barium peroxide (BaO_2) with suitable flux and charged into a large crucible. A piece of burning magnesium wire is introduced in the crucible to ignite the mixture. For example, in the **metallurgy of Cr**,

$$Cr_2O_3 + 2Al \xrightarrow{Heat} 2Cr + Al_2O_3\uparrow$$

Here, BaO_2 acts as an oxidizing agent and hence facilitates the ignition. The above reaction is highly exothermic therefore the heat produced is sufficient to melt the metal.

Test Yourself!

What types of metal oxides are reducedby aluminothermic process?

B. Electrochemical reduction method

More electropositive metals like alkali metals and alkaline earth metal are reduced from their compounds by electrolytic reduction. The reduction of these metals by other chemical means like carbon reduction process is not suitable as they form metal carbides with carbon and at high temperature their vapours become extremely reactive. Moreover, these metals are themselves stronger reducing agents and have greater affinity for oxygen than carbon.

In this method, the fused state of the metal ore is subjected to electrolysis where metal is obtained at cathode via reduction. In certain cases, some other salts are also added so as to lower the melting point of the ore so as to favour the electrolysis process.

For example,

Sodium metal is extracted by this technique from the electrolysis of fused sodium chloride along with addition of little of calcium chloride to lower its melting point.

$NaCl \rightleftharpoons Na^+ + Cl^-$

At anode,	At cathode,
$Cl^- \rightarrow Cl + e^-$	$Na^+ + e^- \rightarrow Na$
$Cl + Cl \rightarrow Cl_2$	$Na + Na \rightarrow 2Na$

Facts

Al is also extracted by the electrolysis of fused alumina by this technique (Hall Heroult Method).

Test Yourself!

a. What is electrochemical reduction in metallurgy?
b. Why is sodium not extracted by pyrometallurgy?

10.1.5: Refining (Purification)

A few impurities may be present in the metal that obtained after different metallurgical processes because of incomplete reduction or the metal may take up certain impurities from the furnace itself. This impure metal is further subjected for purification processes known as refining of metals. The following methods can be used to refine metals:

i. **Poling**

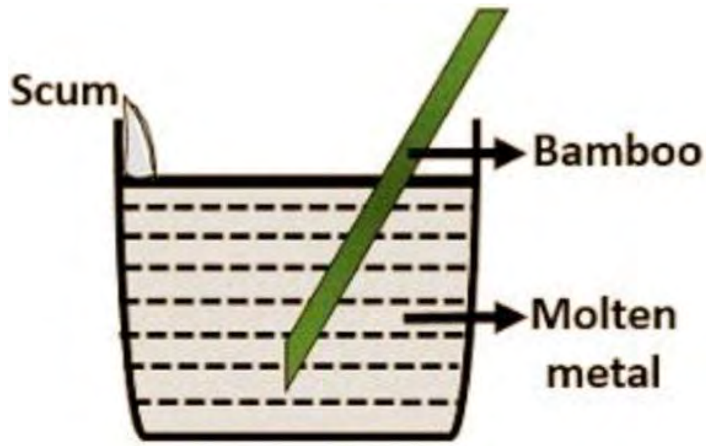


Fig: Poling

This technique of refining is done for those metals which have oxidized impurities. It is especially done for the refining of Cu, Pb and Sn that are in the impure form of cuprous oxide (Cu_2O), lead oxide (PbO) and tin oxide (SnO_2). In this technique the impure molten metal is stirred with a green wooden pole where evolved hydrocarbon gases like methane (CH_4), ethane (C_2H_6), etc. reduce the molten metal into free pure metal. For example,

In the metallurgy of Cu,

Cu_2O + Hydrocarbon → Cu + CO_2 + H_2O

ii. **Electrolytic refining**

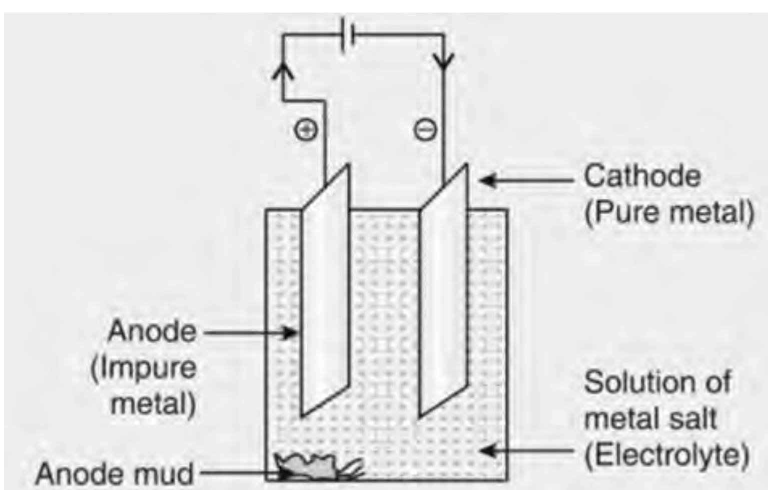
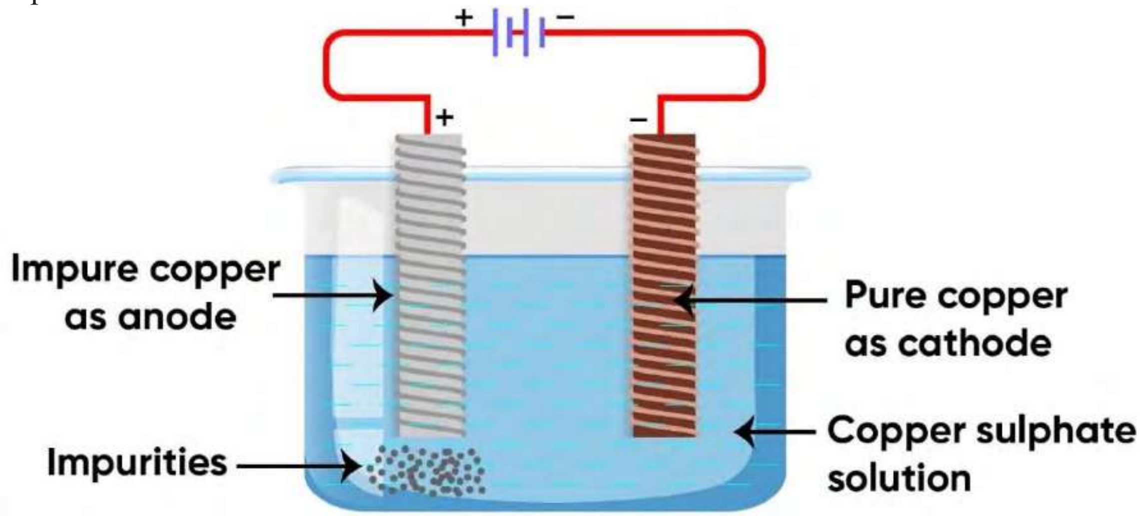


FIGURE: ELECTROLYTIC

This is an important method of purifying almost all metals. In this method, impure metal which is to be purified is kept as anode and thin sheet of pure metal is kept as cathode. Both anode and cathode are dipped in a vessel containing an electrolytic solution of the metal salt. The circuit is completed by connecting the two poles through a conducting wire and a battery. When current is passed through the solution, impure metal at anode gets dissolved into the solution and pure metal gets deposited at the cathode. This is confirmed by observing the weight loss of anode and weight gain cathode. For example, in the metallurgy of copper, impure Cu is made anode and a thin sheet of pure Cu is made cathode. Both poles are dipped in acidified solution of $CuSO_4$ solution and the circuit is completed as shown in figure. When current is passed then impure metal oxidizes from anode and pure metal gets deposited (reduced) at cathode. Less electropositive impurities deposited as anode mud under anode.



$CuSO_4 \rightleftharpoons Cu^{++} + SO_4^{--}$

At anode, At cathode,

$Cu(s) \rightarrow Cu^{++}(aq) + 2e^-$ $Cu^{++}(aq) + 2e^- \rightarrow Cu(s)$

Test Yourself!

a. What is anodic mud?
b. What is electrolysis?

Project work

Take a chart paper and draw a flow sheet diagram showing all the metallurgical processes to convert the impure ore to pure metal.

Exercise

Multiple choice questions

1. **What do you call the impurities associated with minerals?**
a. Slag b. Gangue c. Flux d. Ore

2. **Point out the metal for which aluminothermic process is done?**
a. Cr b. Ni c. Al d. Fe

3. **Which of the following compositions makes brass?**
a. Zn + Sn b. Cu + Sn c. Zn + Pb d. Cu + Zn

4. **Which of the following is acidic flux?**
a. Borax b. CaO c. MgO d. Dolomite

5. **Which of the following is the chemical formula of haematite?**
a. Fe_2O_3 b. Fe_3O_4 c. $FeCO_3$ d. FeO

6. **Name the metallurgy by which silver and gold are extracted?**
a. Pyrometallurgy b. Electrometallurgy c. Hydrometallurgy d. All

7. **Which of the following metal extraction is suited best for the pyrometallurgy principle?**
a. Au b. Na c. Ag d.Cu

8. **Which of the following methods is not applied for dressing of ore?**
a. Froth floatation b. Calcination c. Gravity separation d. Hand picking

9. **Pick out the concentration process in which collectors and activators are used.**
a. Hand picking b. Leaching c. Magnetic separation d. Froth floatation

10. **Which of the following ore is subjected to roasting?**
a. $CaCO_3$ b. ZnS c. $CuCO_3.Cu(OH)_2$ d. $ZnCO_3$

11. **For which of the following ore calcination is carried out?**
a. $ZnCO_3$ b. Ag_2S c. CuS d. ZnS

12. **During, electrochemical refining, at which electrode pure metal is collected?**
a. Cathode b. Anode c. Both pole d. Neither of them

13. **Which of the following is applied for metal refining?**
a. Poling b. Liquation c. Oxiadation d. All

Short answer questions

1. a. Define minerals and ores with examples.
   b. What are alloys? Why are they made?
   c. Show your acquaintances with the term- flux and matrix.
   d. What are metalloids? Give some examples.

2. What is metallurgy? Briefly explain pyrometallurgy and electrometallurgy.

3. What types of metals are extracted by hydrometallurgy?

4. Give reason:
   a. "All ores are minerals but all minerals are not ores".
   b. Alkali metals are not extracted by carbon reduction process.
   c. Sulphide ores are roasted rather than calcinated.

5. Differentiate between:
   a. Calcination and roasting
   b. Electrometallurgy and hydrometallurgy
   c. Pyrometallurgy and hydrometallurgy
   d. Gangue and Flux
   e. Poling and liquation
   f. Metalloids and amalgam
   g. Self-reduction and electrochemical reduction
   h. Carbon reduction process and alumino-thermic process

6. Write a note on - Gangue, flux and salg.

7. Describe the steps involved in metallurgy.

8. Draw a flow sheet diagram showing different steps involved in the extraction of metal from its ore.

9. What are metalloids? Give some examples. Write the general characteristics of metalloids.

10. What do you mean by alloys? Why are they prepared? Give some examples of alloys.

11. What is an amalagam? Write its importance.

12. Write a brief note on:
    a. Pyrometallurgy b. Crushing and pulveriation
    c. Gravity separation technique d. Froth folatation

13. Show your familiarity with:
    a. Distillation b. Poling
    c. Electrolytic refining d. Self-reduction
    e. Thermite process

14. What do you mean by refractory material? Give some examples. How would you remove them during extraction of metal?

15. What is smelting? Write its significance in metallurgy.

16. Give any two chemical formula of ores, in which following metallurgical operations are carried out:
a. Leaching b. Smelting
c. Froth floatation d. Poling
e. Self-reduction f. Liquation
g. Roasting

17. What types of metals use following metallurgical operations?
a. Gravity separation b. Calcination
c. Froth floatation d. Electrochemical reduction

18. What principles are involved during metallurgy in gravity separation and magnetic separation technique?

18. How metals are generally extracted and purified from sulphide ore?

10.2 Alkali Metals

Humphry Davy, (17 December 1778 – 29 May 1829) was a British chemist. Davy was a pioneer in the field of electrolysis using the voltaic pile to split common compounds and thus prepare many new elements. He went on to electrolyze molten salts and discovered several new metals, including sodium and potassium, highly reactive elements known as the alkali metals. Davy discovered potassium in 1807, deriving it from caustic potash (KOH). Before the 19th century, no distinction had been made between potassium and sodium. Potassium was the first metal that was isolated by electrolysis. Davy isolated sodium in the same year by passing an electric current through molten sodium hydroxide.

Activity-1

a. Name the group in which they are kept in the periodic table?
b. Why are these elements called alkali metals?
c. Which of the elements is most reactive and why?

3 Li	lithium	[He]$2s^1$
11 Na	sodium	[Ne]$3s^1$
19 K	potassium	[Ar]$4s^1$
37 Rb	rubidium	[Kr]$5s^1$
55 Cs	cesium	[Xe]$6s^1$
87 Fr	francium	[Rn]$7s^1$

The elements such as Lithium (Li), Sodium (Na), Potassium (K), Rubidium (Rb), Caeseium (Cs) and Francium (Fr) excluding Hydrogen (H) that belong to group 1 (IA) of periodic table are commonly known as **alkali metals.** These metals have only one valence electron entering s orbital. Their valence shell electronic configuration is ns^1. So, they are also called **s block elements.** Since alkali metals have only one valence electron so they all behave in a similar manner. Hydrogen is listed just above alkali metals in group I-A of the periodic table due to its electronic configuration, $1s^1$. It is not considered as alkali metals due to its non-metallic nature. **Activity-2**

Take a porcelain basin containing 15 ml of water added with few drops of phenolphthalein indicator. Add a small piece (pea sized) of sodium metal to basin and observe how metal will scoot around on the surface of water producing bubbles of hydrogen causing phenolphthalein to turn pink.

Finding:

The activity indicates that alkali metals are highly reactive and have produced alkali (base) that is responsible for the color change of the phenolphthalein.

Table No. 1

Atomic Number	Name of Alkali metal	Symbol	Electronic configuration
3	Lithium	Li	$[He]2s^1$
11	Sodium	Na	$[Ne]3s^1$
19	Potassium	K	$[Ar]4s^1$
37	Rubidium	Rb	$[Kr]5s^1$
55	Caesium	Cs	$[Xe]6s^1$

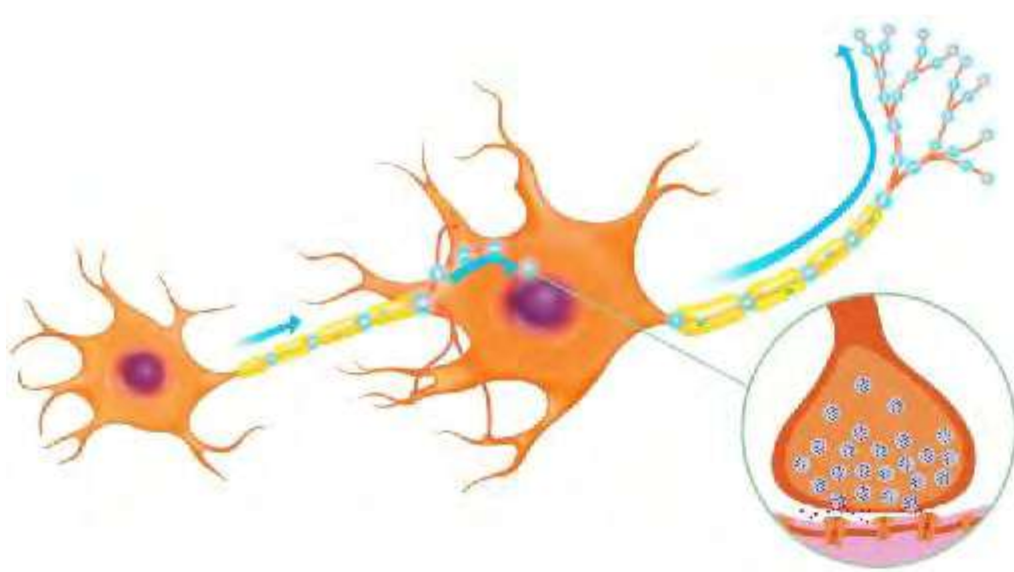
Since alkali metals have only one valence electron, they all behave in a similar manner. Alkali metals being highly reactive react with water to produce alkali with evolution of hydrogen gas. Let us see how the first three alkali metals react with water.

$$Li + H_2O \rightarrow LiOH + H_2$$
$$Na + H_2O \rightarrow NaOH + H_2$$
$$K + H_2O \rightarrow KOH + H_2$$

Similarly remaining alkali metals also react with water to produce alkali and hydrogen. This is the reason why group 1 elements (except H) are called alkali metals.

Facts

Sodium and potassium ions are important in the nerve transmission of the body.



Test Yourself!

Why is hydrogen not considered as alkali metal?

10.2.1: General Characteristics of Alkali Metals

A. Physical Properties

Physical Properties	Characteristics	Reason
Appearance	All alkali metals are silvery white metals possessing metallic lusture. However they lose the metallic lusture when exposed to atmosphere.	
State	Alkali metals are very soft and can be cut even with a cake knife. They are malleable and ductile.	Alkali metals are soft because of the weak metallic bond between the neighboring atoms which decreases as we move from top to bottom in a group. The weak metallic bond in them is due to the presence of only one valence electron.
Atomic and ionic radius	Alkali metals possess the largest atomic as well as ionic radii in their respective period except noble gas. Increasing order of size:	Both atomic and ionic radii increase on moving down the group because a new electron is added to a new shell at each step

<table>
<tr><td></td><td><table><tr><td>$\mathbf{Li<Na<K<Rb<Cs}$</td><td>$\mathbf{Li^+<Na^+<K^+<Rb^+<Cs^+}$</td></tr></table></td><td>which increases the shielding effect and nuclear distance.</td></tr>
<tr><td>Density</td><td>Alkali metals are light metals with low densities. Lithium is the lightest known metal. Increasing order of density:

$\mathbf{Li<Na<K<Rb<Cs}$</td><td>Low density of alkali metals is due to their relatively larger atomic size as a result atoms are less densely packed.</td></tr>
<tr><td>Conductivity</td><td>Alkali metals are good conductors of heat and electricity. Electrical conductivity increases on going down the group.

Increasing order of electrical conductivity:

$\mathbf{Li^+<Na^+<K^+<Rb^+<Cs^+}$</td><td>Excellent conductivity of alkali metals is due to loosely held one valence electron which is free to move throughout the crystal structure.</td></tr>
<tr><td>Melting point and boiling point</td><td>Alkali metals have quite low melting point (m.p.) and boiling point (b.p.). Their mp and b.p. decreases on moving down the group.</td><td>Alkali metals have low m.p. and b.p. because their atoms are held together with weak metallic bond.</td></tr>
<tr><td>Ionization energy(IE)</td><td>Alkali metals possess relatively lowest first ionization energies amongst the elements in their respective periods. On descending the group IE value decreases.

The order of first ionization energies of alkali metals:

<table><tr><th>Element</th><th>Li</th><th>Na</th><th>K</th><th>Rb</th><th>Cs</th></tr><tr><td>IE(KJmol-^{1})</td><td>520</td><td>496</td><td>419</td><td>403</td><td>376</td></tr></table></td><td>Due to the larger atomic size of alkali metals, valence electrons are weakly held by the nucleus. Therefore, low energy is enough to remove the valence electron. On descending the group IE value decreases because atomic size increases and nuclear force decreases.</td></tr>
<tr><td>Electropositive character(Metallic character)</td><td>Alkali metals are highly electropositive (or metallic) in nature. The metallic character increases on moving down the group. Increasing order of metallic character:

$\mathbf{Li<Na<K<Rb<Cs}$</td><td>The strong electropositive character of alkali metals is due to low ionization energy.</td></tr>
<tr><td>Oxidation state</td><td>Alkali metals exhibit +1 oxidation state.

$M \xrightarrow{-e} M^+$ (M= Li, Na, K, Rb, Cs)</td><td>Alkali metals have only one valence electron and can lose this valence electron to form monopositive cation with stable electronic configuration ns^2np^6.</td></tr>
</table>

<table>
<tr>
<td>Reducing nature</td>
<td>Alkali metals are good reducing agents. The reducing character increases from Na to Cs. However Li is the strongest reducing agent as it possesses the highest oxidation potential.</td>
<td>Alkali metals are regarded as powerful reducing agents because they have greater tendency to lose their valence electrons. Li is the strongest reducing agent because it possesses highest oxidation potential</td>
</tr>
<tr>
<td>Flame color</td>
<td>Alkali metals and their salts impart characteristic color in the flame of a gas burner. This is a qualitative test used in chemistry to identify metal in a compound called flame test.

Flame coloration of alkali metals:
<table>
<tr><th>Li</th><th>Na</th><th>K</th><th>Rb</th><th>Cs</th></tr>
<tr><td>Crimson red</td><td>Golden yellow</td><td>Pale violet (Purple)</td><td>Violet

(Purple)</td><td>Sky blue (Bluish violet)</td></tr>
</table>
</td>
<td>Alkali metals give positive flame tests because their electrons get excited to the higher energy levels due to absorption of energy. This excited electron on returning to ground state, emits radiation of particular wavelength belonging to visible region.</td>
</tr>
<tr>
<td>Electronegativity(EN)</td>
<td>Alkali metals have low electronegativity values. The EN value decreases down the group. The decreasing order of EN of alkali metals is

Li>Na>K>Rb>Cs</td>
<td>Alkali metals have low electronegativity value because of large atomic size and low nuclear charge due to which they are not able to attract electrons towards them.</td>
</tr>
</table>

Test Yourself!

a. Why do alkali metals form ionic compounds?
b. "Alkali metals have lowest first IE but very high second IE." Why?
c. What is flame test?
d. "Na shows +1 oxidation state whereas Mg shows +2 oxidation state", Why?

Project work

Clean a nichrome (or platinum) wire by dipping it into concentrated HCl taken in a beaker and then place it in the flame of the burner so that wire does not produce any color in the flame.

a. When the wire is clean, moisten it deep again in conc. HCl and then dip it into small amount of the salt NaCl and KCl and place it over burning flame
b. Write down the color of the flame produced by each salt.

Alkali Metal	Na	K
Flame color		

B. Chemical Properties

Alkali metals have large atomic size and low first ionization energy. They have greater tendency to lose their valence electron forming monovalent cation of stable noble gas configuration. Hence, they are highly reactive and typically form ionic compounds. Their reactivity increases on descending the group because their ionization energy decreases down the group.

1. Action with dry air

Alkali metals react with air to form a layer of caustic metal oxides. They burn vigorously in air to produce oxides.
Lithium burns in air to produce monoxide and some peroxide.

$$2Li + O_2 \longrightarrow Li_2O \text{ (Lithium oxide)}$$

Sodium burns in air to produce peroxide and some monoxide.

$$2Na + O_2 \longrightarrow Na_2O_2 \text{ (Sodium peroxide)}$$

Potassium, Rubidium and Caesium burn in air to produce superoxide.

$$K + O_2 \longrightarrow KO_2 \text{ (Potassium superoxide)}$$

Due to the formation of such layers of oxides alkali metals soon get tarnished when exposed to the atmosphere. Therefore, they are stored in inert solvents like kerosene or paraffin oil.

2. Action with water

Alkali metals react with water to form corresponding hydroxides and hydrogen gas with evolution of heat. The reaction is purely exothermic and may even catch fire. Only lithium reacts gently with water and the rest of the alkali metals react violently with water.

$$2M + 2H_2O \longrightarrow 2MOH + H_2 + \text{Heat (Where, M= alkali metals)}$$

Examples,

$$2Li + 2H_2O \longrightarrow 2LiOH + H_2$$

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

$$2K + 2H_2O \longrightarrow 2KOH + H_2$$

Activity-3

Observe the given pictures carefully and discuss in a group to answer the following questions.

a. Why these metals are kept in inert solvent?

b. What happens if these metals are kept in water rather than inert solvent?

c. Why these metals get tarnished in air?

Alkali metals react with hydrogen at high temperature to form ionic compounds and metal hydrides.

$$2M + H_2 \longrightarrow 2MH \text{ (Metal hydride)}$$

Example,

$$2Li + H_2 \longrightarrow 2LiH \text{ (Lithium hydride)}$$

$$2Na + H_2 \longrightarrow 2NaH \text{ (Sodium hydride)}$$

$$2K + H_2 \longrightarrow 2KH \text{ (Potassium hydride)}$$

4. Action with Non metals

Alkali metals react with non-metals like halogens, sulphur and phosphorous to form corresponding halides, sulphides and phosphides.

With X_2: $2M + X_2 \longrightarrow 2MX$ (Metal halide) Where, M= alkali metals & X= halogens

Example,

$$2Li + Cl_2 \longrightarrow 2LiCl \text{ (Lithium chloride)}$$

$$2Na + Cl_2 \longrightarrow 2NaCl \text{ (Sodium chloride)}$$

$$2K + Br_2 \longrightarrow 2KBr \text{ (Potassium bromide)}$$

With S: $2M + S \longrightarrow M_2S$ (Metal sulphide) Where, M= alkali metal)

Example,

$$2Li + S \longrightarrow Li_2S \text{ (Lithium sulphide)}$$

$$2Na + S \longrightarrow Na_2S \text{ (Sodium sulphide)}$$

With P: $2M + P \longrightarrow M_3P$ (Metal phosphide) Where, M= alkali metal)

Example,

$$2Li + P \longrightarrow Li_3P \text{ (Lithium phosphide)}$$

$$2Na + P \longrightarrow Na_3P \text{ (Sodium phosphide)}$$

4. Action with ammonia

Alkali metals dissolve in liquid ammonia producing deep blue solution having high electrical conductivity, paramagnetic character and reducing properties due to presence of ammoniated metal ion and ammoniated electrons. The reaction is

$$M + (x+y)\,NH_3 \longrightarrow \underset{\text{(Ammoniated metal ion)}}{[M(NH_3)_x]^+} + \underset{\text{(Ammoniated electron)}}{[e(NH_3)_y]^-} \quad (M= \text{alkali metals})$$

The blue color is due to the presence of ammoniated electrons which absorbs energy in the visible region of light and thus imparts blue color.

Note: If the above solution is kept for a long time then metal amide is formed as a product.

$$M^+ + e^- + NH_3 \longrightarrow \underset{\text{(Metal amide)}}{MNH_2} + H_2 \quad (M= \text{alkali metals})$$

$$Na^+ + e^- + NH_3 \longrightarrow NaNH_2 + H_2$$

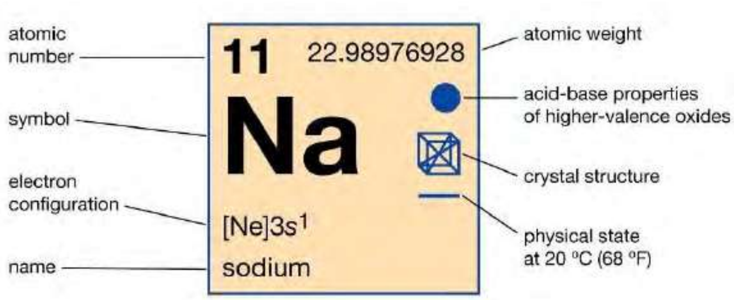
Test Yourself!

How do alkali metals react with ammonia solution?

10.2.2: Sodium (Extraction, properties and uses)

Activity-4

a. Take a small piece of sodium and keep in a beaker containing water.
b. Observe the changes and discuss the cause of such changes.



sodium metal

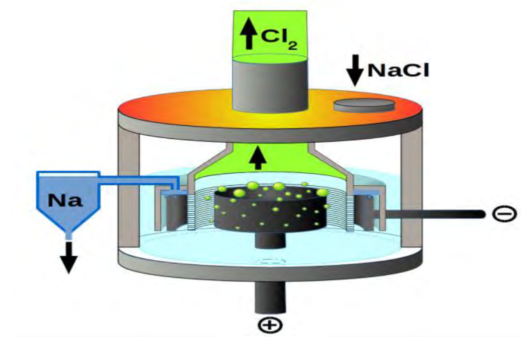
Alkali metals are highly reactive so they do not occur in native states. They are found in a combined state in the earth's crust. Sodium is the most abundant alkali metal found naturally in a combined state in the earth's crust. Some of the important minerals of sodium are listed below:

Mineral of sodium	Molecular formula
Rock salt (Common salt)	$NaCl$
Caliche or Chile saltpetre	$NaNO_3$
Glauber's salt	$Na_2SO_4.10H_2O$
Borax	$Na_2B_4O_7.10H_2O$
Soda feldspar	$NaAlSi_3O_8$

Extraction of sodium metal (Down's process)

Activity-5

What is the raw material used and what are the products obtained in the figure given?



Sodium is a highly reactive alkali metal and occurs in the form of various minerals as shown in the above table. It cannot be obtained from its oxide by carbon reduction method. This is because sodium is highly reactive metal and readily reacts with carbon to form sodium carbide (Na_2C_2) rather than being reduced to its elemental form. Moreover, sodium being less electronegative than carbon has greater affinity for oxygen than carbon does. This means in presence of oxygen sodium forms sodium oxide (Na_2O) rather than being reduced by carbon. Also sodium metal cannot be extracted from aqueous solution of its compounds like NaCl because the metal liberated at once quickly reacts with water.

Now the question arises, 'how sodium metal is extracted?' Actually, sodium metal is extracted by the electrolysis of molten sodium chloride using Down's process. Although sodium chloride is very cheap compound and is easily available extraction of sodium from molten sodium chloride has following difficulties:

- **a.** Sodium chloride melts at 801^0C which is difficult to attain and maintain the high temperature throughout the experiment.
- **b.** Sodium boils at about 883^0C and at this temperature of electrolysis; the sodium metal liberated will vaporize.
- **c.** Molten sodium forms a metallic fog (colloidal solution) with fused sodium chloride. This leads to difficulty in separation of metallic fog and may cause a short circuit.
- **d.** The products of electrolysis i.e, sodium, and chlorine, will corrode the material of the cell at this high temperature.

In 1924 J.C. Down overcame all these difficulties. He observed that the addition of calcium chloride ($CaCl_2$) lowers the melting point of sodium chloride to about 600 0C.
Sodium is now obtained by electrolysis of a fused mixture of sodium chloride and calcium chloride in a Down's cell at 6000C using graphite anode and iron (or steel) cathode.

Working of Down's Cell

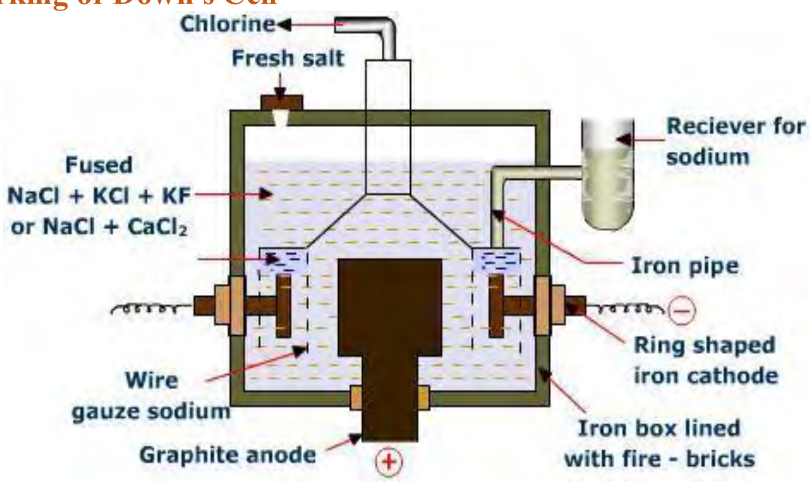


Fig: Down's Cell for the extraction of sodium metal

The Down's cell consists of a central graphite anode surrounded by a cylindrical steel cathode**.** The inner wall of the steel tank is lined with heat resistant bricks as shown in figure. The anode and cathode are separated by steel gauze. Anode is covered by a dome shaped hood provided with an outlet for the escape of Cl_2 gas. Molten sodium is formed at the cathode and collected through another hood around the top of the cathode cylinder in the receiver containing kerosene oil (it is less dense than the sodium chloride). In order to lower the melting point of sodium chloride a small amount of calcium chloride is added. The sodium chloride is melted electrically and kept molten by the flow of current through the cell. More sodium chloride is added as the electrolysis proceeds.

The following electrode reaction takes place,

$$NaCl \rightarrow Na^+ + Cl^-$$

At anode: Oxidation of chloride ion takes place

$2Cl^- \rightarrow$

$Cl_2 + 2e^-$

At cathode: Reduction of sodium ion takes place

$2Na^+ +$ $2e^-$

$\rightarrow 2Na$

Overall cell reaction is

$2Na^+ + 2Cl^- \rightarrow 2Na + Cl_2$

Note: ***Ca^{++} ion does not get reduced in preference to Na^+ ion because its reduction potential (-2.87V) is lower than that of sodium (-2.38 V). Hence only sodium ions are reduced at cathode.***

Advantages of Down's cell

a. Very high purity of sodium metal is obtained (99.5%).
b. Sodium chloride is a very cheap and abundant compound.
c. Long life span of the cell (2000 working hrs).
d. Chlorine is obtained as a valuable by-product.

Test Yourself!

a. Why carbon reduction method cannot be used to obtain sodium metal?
b. Name elements that can be extracted by electrometallurgical process?
c. Calcium ion is not discharged at cathode in place of Na^+, why?

Facts

a. It is soft silvery white metal and can be easily cut with a knife.

b. It appears bright but when it reacts with air it becomes dull.

c. It is less dense than water.

d. It is a good conductor of heat and electricity.

e. Its melting point is 98 ^{0}C and boiling point is 883 ^{0}C.

Properties of sodium metal

a. Why is it easy to cut sodium?

b. What could be the reason for keeping sodium in kerosene oil?

Earlier we have discussed how reactive the alkali metals are. Now let us study some of the properties of sodium metal.

1. Action with air

Sodium metal reacts with oxygen to form sodium oxide and traces of yellowish sodium peroxide. Sodium loses its silvery appearance due to formation of layers of these oxides.

$$4Na + O_2 \longrightarrow 2Na_2O$$

This sodium oxide absorbs moisture to form sodium hydroxide film which further absorbs CO_2 from air to produce sodium carbonate.

$$Na_2O + H_2O \longrightarrow 2NaOH$$

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

Due to this reason sodium metal is stored in an inert solvent like kerosene.

Test yourself!

Why does sodium metal become dull when exposed to atmosphere?

2. Action with Water

Sodium reacts with water vigorously to form a strong base of sodium hydroxide and hydrogen gas. Since, the reaction is exothermic therefore hydrogen gas produced may catch fire or explode.

$$2Na + H_2O \longrightarrow 2NaOH + H_2 + Heat$$

Test Yourself!

What happens when a piece of sodium metal is added to water containing phenolphthalein indicator?

3. Action with mineral acids

Sodium reacts with dilute hydrochloric acid and dilute sulphuric acid to produce corresponding sodium salt and hydrogen gas. The reaction is exothermic, violent and may cause an explosion.

$$2Na + 2HCl \longrightarrow 2NaCl + H_2$$

$$2Na + H_2SO_4 \longrightarrow Na_2SO_4 + H_2$$

Sodium also reacts with phosphoric acid to produce sodium phosphate and hydrogen gas.

$$6Na + 2H_3PO_4 \longrightarrow 2Na_3PO_4 + 3H_2$$

Test Yourself!

Can sodium metal be used in place of zinc to prepare hydrogen gas in laboratory?

4. Action with Non-metals

Sodium reacts with non-metals like halogens, sulphur, phosphorous to produce sodium halides, sodium sulphide and sodium phosphide respectively.

$$2Na + X_2 \longrightarrow 2NaX \qquad \text{(Where, X= F, Cl, Br and I)}$$

(Sodium halide)

Example,

$$2Na + Cl_2 \longrightarrow 2NaCl \text{ (Sodium chloride)}$$

$$2Na + S \longrightarrow Na_2S \text{ (Sodium sulphide)}$$

$$Na + P \longrightarrow Na_3P \text{ (Sodium phoshide)}$$

Test Yourself!

Complete the following reaction:

a) $Na + Br_2$? b) $Na + P_4$?

5. Action with ammonia

Sodium metal dissolves in liquid ammonia producing deep blue solution having high electrical conductivity, paramagnetic character and reducing properties due to presence of ammoniated metal ion and ammoniated electron (solvated electron). The reaction is

$$Na(s) + (x+y)\,NH_3(l) \rightarrow \underset{\text{(Ammoniated metal ion)}}{Na(NH_3)_x^+(aq)} + \underset{\text{(Ammoniated electron)}}{e^-(NH_3)_y(aq)} \text{ (blue)}$$

The blue color is due to the presence of ammoniated electrons which absorbs energy in the visible region of light and thus imparts blue color. As the solution becomes more concentrated, it takes on a bronze color.

Sodium forms sodamide on reaction with ammonia gas on heating.

$$2\,Na + 2\,NH_3 \rightarrow 2\,NaNH_2 + H_2$$

Uses of Sodium

a. It is used in sodium vapour lamps.
b. It is used in preparation of sodium extract to detect the presence of hetero element in organic compounds.
c. It is used in many organic reactions like Wurtz reaction, Fittig reaction, Birtch reduction, etc.
d. It is used as a catalyst in preparation of artificial rubber.
e. It is used in the form of sodium amalgam (Na+Hg) in the preparation of various organic compounds.
f. It is used in the extraction of carbon, magnesium and silicon.
g. It is used as a reducing agent.
h. It is used in the form of sodium chloride which is an important animal nutrient.
i. It is used in production of TEL (Tetraethyl lead), an anti-knocking agent.
j. It is used as a coolant in nuclear reactors.

10.2.3: Sodium hydroxide

We all are quite familiar with the most popular hydroxide base i.e, sodium hydroxide**.** Now in this subunit let us discuss some of the properties of this sodium compound. One of the most valuable compounds of sodium is sodium hydroxide having molecular formula NaOH. It is sometimes called caustic soda or lye as it is a common ingredient of cleansing agents like soap, detergents. It is manufactured on a large scale for its extensive use for various purposes in different areas.

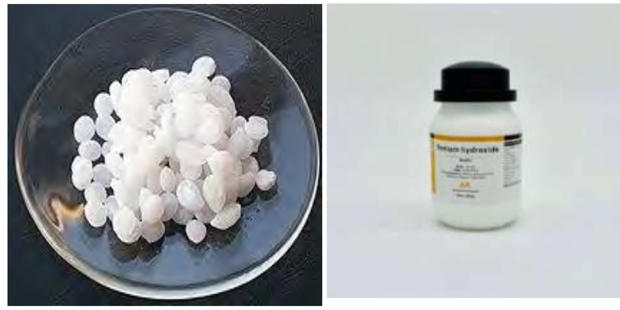


FIGURE: PELLETS OF

Facts

a. It is a white crystalline odourless solid with a soapy touch.
b. It is deliquescent and corrosive in nature.
c. It is highly soluble in water. The dissolution of NaOH in water is an exothermic phenomenon.
d. It is also soluble in alcohol and glycerin.
e. Its aqueous solution is basic in nature.
f. Its melting point is 318 ^{0}C and boiling point is 1388 ^{0}C.

10.2.3: Properties of sodium hydroxide

1. Precipitation of hydroxides

Activity-5

Take two test tubes marked A and B. Take 1 mL of $FeCl_3$ solution in test tube A and 1mL of $FeSO_4$ solution test tube B. Add a few drops of NaOH solution to each test tube and observe the reaction and write the formula and color of the ppt. formed in the test tube A and B.

Sodium hydroxides serve as a precipitant for many metals in qualitative salt analysis. When aqueous salt solutions of metals are treated with dilute sodium hydroxide solution then metals get precipitated as their hydroxides. For examples,

$$\underset{(aq)}{CuSO_4} + \underset{(aq)}{2NaOH} \rightarrow \underset{\text{(bluish white ppt.)}}{Cu(OH)_2\downarrow} + \underset{(aq)}{Na_2SO_4}$$

$$\underset{(aq)}{ZnSO_4} + \underset{(aq)}{2NaOH} \rightarrow \underset{\text{(white ppt.)}}{Zn(OH)_2\downarrow} + \underset{(aq)}{Na_2SO_4}$$

$$FeSO_4 + 2NaOH \rightarrow Fe(OH)_2\downarrow + Na_2SO_4$$

(aq) (aq) (dirty green ppt.) (aq)

$$FeCl_3 + 3NaOH \rightarrow Fe(OH)_3\downarrow + 3NaCl$$

(aq) (aq) (Reddish brown ppt.) (aq)

Bluish white ppt of $Cu(OH)_2$

White ppt of $Zn(OH)_2$

Dirty green ppt of $Fe(OH)_2$

In certain cases, the metal hydroxide formed gets dissolved in excess of sodium hydroxide to produce a soluble salt solution. For example,

$$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2\downarrow + Na_2SO_4$$

(aq) (aq) (white ppt.) (aq)

$$Zn(OH)_2\downarrow + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$$

(white ppt.) (excess) (sodium zincate)

$$AlCl_3 + 3NaOH \rightarrow Al(OH)_3\downarrow + 3NaCl$$

(aq) (aq) (white ppt.) (aq)

$$Al(OH)_3\downarrow + 2NaOH \rightarrow 2\,NaAlO_2 + H_2O$$

(white ppt.) (excess) (sodium aluminate)

2. Action with Carbon monoxide: Under normal conditions sodium hydroxide does not react with carbon monoxide. However, at high temperature (210^0C) and high pressure (5-10 atm) it reacts with carbon monoxide to produce sodium salt of formic acid which on acidification gives formic acid.

$$NaOH + CO \rightarrow HCOONa \text{ (sodium formate)}$$

$$HCOONa + HCl \rightarrow HCOOH + NaCl$$

Facts

Formic acid (HCOOH) is commercially prepared by this method. Formic acid is used in a wide range. It is used in leather tanning, food preservative, pharmaceuticals, etc.

Uses of sodium hydroxide

Some of the common uses of sodium hydroxide are listed below:

a. It is used as a common laboratory reagent.

b. It is used in the manufacture of soaps, detergents, artificial silk, paper and pulp.

c. It is used in the manufacture of washing soda.

d. It is used in the preparation of soda lime.

e. It is used in the purification of aluminum ore from bauxite.

f. It is used as a reagent in many inorganic and organic reactions.

g. It is used for cleaning machinery equipment, waste discharge pipes, etc.

h. It is used in petroleum refining.

Sodium Carbonate:

Activity-6:

Observe the pictures given:

a. Why are they called soda?

b. What is the molecular formula of each substance given?

c. What is the purpose of these substances?

Washing soda

Baking soda

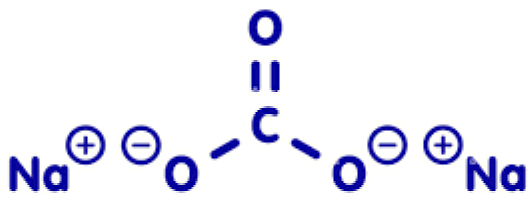
Sodium carbonate is the disodium salt of carbonic acid with an alkalinizing property. It is a white crystalline solid that can exists as anhydrous salt (Na_2CO_3), monohydrate salt ($Na_2CO_3.H_2O$), heptahydrate salt ($Na_2CO_3.7H_2O$) and decahydrate salt ($Na_2CO_3.10H_2O$). The anhydrous sodium carbonate is commonly called ***soda ash*** and sodium carbonate decahydrate is commonly called **washing soda**. When washing soda is exposed to open atmosphere it slowly changes into soda ash as shown below:

$$Na_2CO_3.10H_2O \xrightarrow{air} Na_2CO_3.H_2O + 9\,H_2O$$

$$Na_2CO_3.H_2O \xrightarrow{>100^0C} Na_2CO_3 + H_2O$$

Facts

a. It is a white crystalline solid substance soluble in water.
b. Its aqueous solution is a weak base and turns red litmus paper into blue.
c. It is a colorless and odorless substance.



Structure of Na_2CO_3

10.2.4: Properties of sodium carbonate

1. Action with carbon dioxide

When CO_2 gas is passed through aqueous solution of sodium carbonate, sparingly soluble sodium bicarbonate is obtained.

$Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$

2. Action with SO_2

When sulphur dioxide gas is passed through aqueous solution of sodium carbonate, sodium sulphite (Na_2SO_3) or sodium bisulphite ($NaHSO_3$) is produced.

$Na_2CO_3 + SO_2 \rightarrow Na_2SO_3 + CO_2 \uparrow$

$Na_2CO_3 + 2SO_2 + H_2O \rightarrow NaHSO_3 + CO_2 \uparrow$

This property of sodium carbonate is important to remove SO_2 pollutants from air.

3. Action with water

Sodium carbonate readily dissolves in water. Its aqueous solution is alkaline due to hydrolysis.

$Na_2CO_3 + H_2O \rightarrow 2NaOH + H_2CO_3$ (This process is exothermic)

$Na_2CO_3 \rightleftharpoons 2Na^+ + CO_3^{--}$

$Na+ + H_2O \rightarrow NaOH$

$CO_3^{--} + H_2O \rightarrow H_2CO_3 + 2OH^-$

4. Precipitaion reaction

Sodium carbonate acts as a precipitant. When it is treated with some water soluble metallic salts then the metal gets precipitated as their carbonates. For example,

$MgCl_2 + Na_2CO_3 \rightarrow MgCO_3 \downarrow + 2NaCl$
(aq) (aq) (s) (aq)

$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2NaCl$
(aq) (aq) (s) (aq)

$BaCl_2 + Na_2CO_3 \rightarrow BaCO_3\downarrow + 2NaCl$
(aq) (aq) (s) (aq)

$Pb(NO_3)_2 + Na_2CO_3 \rightarrow PbCO_3\downarrow + 2NaNO_3$
(aq) (aq) (s) (aq)

$MgSO_4 + Na_2CO_3 \rightarrow MgCO_3\downarrow + 2Na_2SO_4$
(aq) (aq) (s) (aq)

Test Yourself!

Give two examples showing sodium carbonate acts as a precipitant.

Facts

The precipitation action of sodium carbonate makes its use to remove hardness of water.

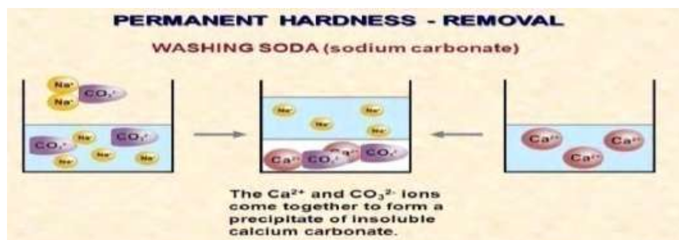


Figure: Use of sodium carbonate to remove hardness of water

Uses of sodium carbonate

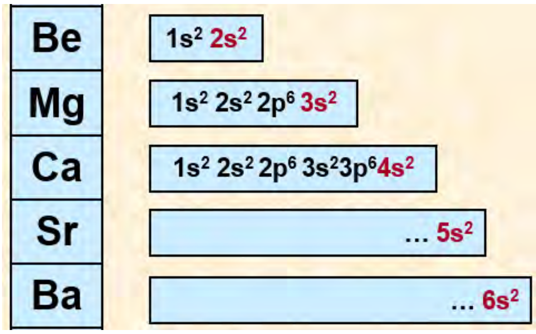
Some uses of sodium carbonate are highlighted below:

a. It is used in the manufacture of glass, caustic soda, soap, detergents, borax, paper and pulp etc.
b. It is used to remove hardness in water.
c. It is used for laundry work.
d. It is used as a laboratory reagent.
e. It is used in textile and petroleum refining.
f. It is used to increase the alkalinity of swimming pools, helping to ensure the proper pH balance of water.
g. It is used as air purifier in the form of soda ash to remove SO2 from air.
h. It is used as a fertilizer in the form of soda ash.

10.3: Alkaline earth metals

Activity-7

a. Complete the electronic configuration for Sr and Ba.
b. Identify the group in which these elements are kept and give reason for your choice.
c. Name two elements that are biologically important for human beings.



The elements such as Beryllium (Be), Magnesium (Mg), Calcium (Ca), Strontium (Sr), Barium (Ba) and Radium (Ra) that belong to group 2 (IIA) of periodic table are commonly known as **alkaline earth metals.** These metals have only two valence electrons entering s orbital. Their valence shell electronic configuration is ns^2. So they are also called **s block elements.** Since alkaline earth metals have only two valence electrons so they all behave in a similar manner.

Atomic Number	Name of Alkali metal	Symbol	Electronic configuration
4	Beryllium	Be	$[He]2s^2$
12	Magnesium	Mg	$[Ne]3s^2$
20	Calcium	Ca	$[Ar]4s^2$
38	Strontium	Sr	$[Kr]5s^2$
56	Barium	Ba	$[Xe]6s^2$
88	Radium	Ra	$[Rn]7s^2$

They are called alkaline earth metals because their oxides are found in the earth's crust and their oxides dissolve in water to produce alkali solution. They also react with water to produce their respective hydroxides.

$MgO + H_2O \rightarrow Mg(OH)_2$ Also, $Mg + H_2O \rightarrow Mg(OH)_2 + H_2$

$CaO + H_2O \rightarrow Ca(OH)_2$ Also, $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$,

Similarly remaining alkaline earth metals also react with water to produce alkali and hydrogen.

Test Yourself!

Why is calcium called alkaline earth metal?

10.3.1: General Characteristics of Alkaline Earth Metals

Berylium

Magnecium

Calcium

A. Physical properties

Physical Properties	Characteristics	Reason
Appearance	All alkaline earth metals are silvery white metals possessing metallic luster. However they lose the metallic luster when exposed to the atmosphere.	
State	Alkaline earth metals are silvery white (grayish) elements harder than alkali metals. They are malleable and ductile possessing metallic luster.	In alkali metals there is a single valence electron due to which metallic bond is too weak but in alkaline earth metals there are two valence electrons making stronger metallic bond. Alkali metals are soft because of weak metallic bond and alkaline earth metals are harder due to strong metallic bond.
Atomic and ionic radius	Alkaline earth metals have large atomic size but smaller than corresponding alkali metals. Atomic size increases down the group. **$Be<Mg<Ca<Sr<Ba<Ra$**	Both atomic and ionic radii increase on moving down the group because new electron is added to new shell at each step which increases the shielding effect and nuclear distance.
Density	Alkaline earth metals have low density but higher than that of alkali metals. Increasing order of density:	Low density of alkaline earth metals is due to their relatively larger atomic size as a result atoms are less densely packed.

<table>
<tr><td>Conductivity</td><td>Alkaline earth metals are good conductors of heat and electricity. Electrical conductivity increases on going down the group.

Increasing order of electrical conductivity:</td><td>Excellent conductivity of alkaline earth metals is due to loosely held two valence electrons which are free to move throughout the crystal structure.</td></tr>
<tr><td>Melting point and boiling point</td><td>Alkaline earth metals have low melting point (m.p.) and boiling point (b.p.) but higher than that of alkali metals. Their m.p. and b.p. decreases on moving down the group.</td><td>Alkaline earth metals have low m.p. and b.p. because their atoms are held together with weak metallic bond.</td></tr>
<tr><td>Ionization energy(IE)</td><td>Alkaline earth metals possess relatively low first ionization energies but higher than that of alkali metals. They have two values of ionization energy. By losing two electrons they attain noble gas electronic configuration. On descending the group IE value decreases.

The order of first ionization energies of alkali metals:

<table>
<tr><td>Element</td><td>Be</td><td>Mg</td><td>Ca</td><td>Sr</td><td>Ba</td></tr>
<tr><td>Ist IE($KJmol^{-1}$)</td><td>899</td><td>737</td><td>590</td><td>549</td><td>503</td></tr>
<tr><td>IInd IE ($KJmol^{-1}$)</td><td>1757</td><td>1450</td><td>1145</td><td>1064</td><td>965</td></tr>
</table></td><td>Ionization energies of alkaline earth metals are low because s- electrons in valence shell are far away from the nucleus as a result valence electrons in them are loosely held by the nucleus. Therefore, low energy is enough to remove the valence electron. On descending the group IE value decreases because atomic size increases and nuclear force decreases. However, the first ionization energies of alkaline earth metals are higher than those of the alkali metals. This is because of the increase in the nuclear charge of the alkaline earth metals.</td></tr>
<tr><td>Oxidation state</td><td>Alkaline earth metals exhibit +2 oxidation state.

$M \xrightarrow{-2e} M^{++}$ (M= Be, Mg, Ca, Sr, Ba)</td><td>Alkaline earth metals have only two valence electrons and can lose these valence electrons to form dipositive cation with stable electronic configuration ns^2np^6.</td></tr>
<tr><td>Reducing nature</td><td>Alkaline earth metals are good reducing agents. The reducing character increases from Be to Ba. Ba is the strongest reducing agent.

[Alkaline earth metals are weaker reducing agents than alkali metals]</td><td>Alkaline earth metals are regarded as powerful reducing agents because they have greater tendency to lose their valence electron. Ba is the strongest reducing agent because on moving from Be to Ba the</td></tr>
</table>

<table>
<tr><td></td><td></td><td>electropositive character increases and ionization enthalpy decreases.</td></tr>
<tr><td>Flame color</td><td>Alkaline earth metals and their salts impart characteristic color in the flame of a gas burner except Be and Mg. This is a qualitative test used in chemistry to identify group II A metals in a compound called flame test.

Flame coloration of alkali metals:
<table><tr><th>Ca</th><th>Sr</th><th>Ba</th><th>Ra</th></tr><tr><td>Brick red</td><td>Crimson</td><td>Apple green</td><td>Carmine red</td></tr></table></td><td>Alkaline earth metals give positive flame test because their electrons get excited to higher energy levels due to absorption of energy. This excited electron on returning to ground state, emit radiation of particular wavelength belonging to the visible region.</td></tr>
</table>

Facts

Animal bones contain calcium in the form of hydrated compound called hydroxyapatite.

B. Chemical properties

Alkaline earth metals lose two valence electrons and form a dipositive cation which has noble gas electronic configuration. Therefore, they are highly reactive and typically form ionic compounds.

Some of the chemical properties of alkaline earth metals are discussed below:

1. **Action with air**

Figure: Burning Magnesium ribbon in air

When alkaline earth metals are exposed to atmosphere, then they become dull as a layer of oxide is formed.

$2Mg + O_2 \rightarrow 2MgO$ (magnesium oxide)
$2Ca + O_2 \rightarrow 2CaO$ (calcium oxide)
Note: Be is much less affected by air.

Test Yourself!

Why does magnesium ribbon get tarnished when exposed in atmosphere for long time?

2. Action with water

Alkaline earth metals react with water to produce alkali and hydrogen gas except Be.

$2Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$
$2Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$
$2Sr + 2H_2O \rightarrow Sr(OH)_2 + H_2$

Mg reacts only on boiling,

$$Mg + 2H_2O \xrightarrow{Boil} Mg(OH)_2 + H_2$$

Figure: calcium metal in water

Test Yourself!

What happens when a piece of magnesium ribbon is kept in boiling water?

3. Action with Hydrogen

Except Be, all alkaline earth metals react with hydrogen to produce their corresponding hydrides.

$Mg + H_2 \rightarrow MgH_2$ (magnesium hydride)

$Ca + H_2 \rightarrow CaH_2$ (calcium hydride)

$Ba + H_2 \rightarrow BaH_2$ (barium hydride)

Test Yourself!

How does Sr react with hydrogen gas?

4. Action with Nitrogen

Alkaline earth metals react with nitrogen at high temperature to form their corresponding nitride.

$3Mg + N_2 \rightarrow Mg_3N_2$ (magnesium nitride)

$3Ca + N_2 \rightarrow Ca_3N_2$ (calcium nitride)

Test Yourself!

What happens when Mg ribbon is heated with nitrogen gas?

5. Action with acids

Activity-8

Take a beaker containing some dilute sulphuric acid and to this acid add a piece of magnesium ribbon. Observe the reaction and answer the following questions.

a. Name the gas evolved.

b. Write the balanced chemical reaction.

Alkaline earth metals react with mineral acids to produce hydrogen gas.

$Mg + HCl \rightarrow MgCl_2 + H_2$

$Ca + H_2SO_4 \rightarrow CaSO_4 + H_2$

Note: Be reacts with acid with most common acids (except HNO_3) more vigorously such that the reacting vessel has to be kept in an ice bath.

6. Action with non-metals: Alkaline earth metals react with non-metals like halogens, sulphur, phosphorous, carbon, etc. as

$Mg + Cl_2 \rightarrow MgCl_2$ (magnesium chloride)

$Ca + P \rightarrow Ca_3P_2$ (calcium phosphide)

$Ba + S \rightarrow BaS$ (barium sulphide)

Test Yourself!

Write the reaction between Mg and P_4.

10.3.2: Molecular formula and uses of some compounds of alkaline earth metals

List of some important compounds of alkaline earth metals are listed below:

1. Quick Lime

Calcium oxide is commonly known as quicklime or burnt lime. Its molecular formula is CaO. In Nepali it is called chuna.

Uses of quick lime

Some uses of quick lime are listed below:

Figure: Quick lime

a. It is used in the manufacture of calcium carbide (CaC_2) and bleaching powder.
b. It is a raw material for the preparation of cement.
c. It is used in the preparation of soda lime (NaOH + CaO).
d. It is used in the manufacture of glass, washing soda etc,
e. It is a basic flux in metallurgy.
f. It is used as a dehydrating agent for drying gases, alcohols and other compounds.
g. It is used as a fertilizer.

2. Bleaching powder

Calcium chlorohypochlorite is commonly known as bleaching powder. Its molecular formula is $CaOCl_2$. It is also known as chloride of lime or chlorinated lime.

Uses of bleaching powder

Some uses of bleaching powder are listed below:

Figure: Bleaching powder

a. It is used in the manufacture of chloroform.
b. It is used to bleach paper, pulp, cotton, linen goods, etc.
c. It is used as a disinfectant. In the presence of moisture it releases chlorine that kills germs and disinfects the area.
d. It is in sterilizing drinking water.
e. It is used as an oxidizing agent.
f. It is used to whiten clothes and remove stains.
g. It is used to clean old items like jewellery items or showpieces.

3. Magnesia

Magnesium oxide is chemically known as magnesia. Its molecular formula is MgO.

Uses of magnesia

Some uses of magnesia are listed below:

a. It is used as refractory material in lining of furnaces, bricks etc due to its high melting point.
b. It is used in the manufacture of cement.
c. It is used to prepare crucible.
d. Its aqueous suspension is called milk of magnesia which is used for treating constipation (laxative).
e. It is used as an antacid for curing gastritis.
f. It is used as a basic flux in metallurgy.

Figure: Milk of magnesia

4. Plaster of paris

Calcium sulphate hemihydrate is chemically known as plaster of paris. Its molecular formula is $CaSO_4.1/2\ H_2O$.

Figure: use of Plaster of paris for setting fractured bones

Uses of plaster of paris

Some uses of plaster of paris are listed below:

a. It is used for setting fractured and broken bones in the body.
b. It is used for making statues, models and various decorative items.
c. It is used as a fireproofing materials and also used for making chalks to write.
d. It is used in making artificial marbles and false-ceiling.
e. It is used in construction of buildings to fill small gaps on the walls and roofs.

5. Epsom salt

Magnesium sulphate heptahydrate is chemically known as plaster of Epsom salt. Its molecular formula is $MgSO_4.7H_2O$.

Figure: Epsom salt

Uses of Epsom salt

Some uses of plaster of Epsom salt are listed below:

a. It is used as a mild purgative (laxative) in medicine.
b. It is used in the manufacture of paints, soaps and fire proof materials.
c. It is used as a mordant in dying and in the tanning industry.
d. It is used as a laboratory reagent.
e. It is used in cosmetic lotions.

Test Yourself!

a. Write the molecular formula of plaster of paris.
b. Write one medical use of quick lime.
c. Write one use of bleaching powder.

10.3.3: Solubility of hydroxides, carbonates and sulphates of alkaline earth metals

Figure-1: $Ba(OH)_2$

Figure SEQ Figure * ARABIC 1: Mg(OH)2

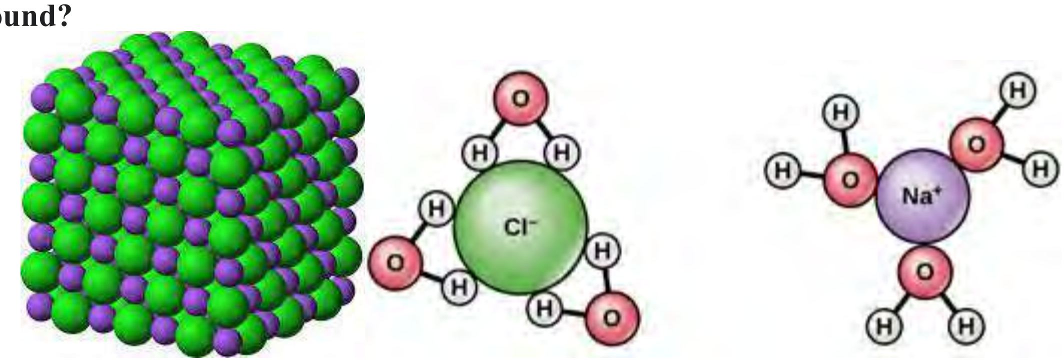
Activity-9

Take two beakers each containing about 50 mL of water. Add about one gram, of each compounds shown in picture-1 and picture-2 and observe their solubility to answer following questions:
a. Do both compounds dissolve in water to same extent?
b. Why do they differ in their solubility?

Facts

The solubility of ionic compounds depends upon two factors i.e, lattice energy and hydration energy. Both the factors are opposing factors. For a compound to be soluble in water its hydration energy must be greater than its lattice energy.

What are lattice energy and hydration energy? How do these energies affect solubility of a compound?



(a) (b)

Fig: NaCl crystal (a) before dissolving in water (b) after dissolving in water

Lattice energy is the energy through which ions of the solids are held together in a crystal. Greater the lattice energy more strongly the ions hold each other and thereby making the compound less soluble.

Hydration energy is the energy evolved when ions in a crystal of the solid held together get separated upon hydration. It is also known as hydration enthalpy. Thus with increase in hydration energy the solubility of the compound increases and vice versa. Hydration energy depends on the charge of the ion and the ionic radius. Higher the charge and smaller the size, greater will be the hydration energy.

Now, let us discuss the solubility of hydroxides, carbonates and sulphates of alkaline earth metals.

Solubility of hydroxides of alkaline earth metals

Activity-10

Use the given structure and answer the questions given

a. Compare the size and charge density of Mg^{++} and Ca^{++} ions.
b. Which of the ions hold the OH^- ions strongly?

In case of alkaline earth metals, on moving down the group from Be to Ba the ionic size of these metals increases and therefore lattice energy decreases. This tends to increase the solubility as it overcomes the counter effect of decrease in hydration energy down the group. Thus the solubility of hydroxides of alkaline earth metals increases from Be to Ba.

$$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$$

Increasing order of solubility

	$MgSO_4$	$CaSO_4$	$SrSO_4$	$BaSO_4$
Solubility g/100cm^3 of water	**3.6×10^{-1}**	**1.1×10^{-3}**	**6.2×10^{-5}**	**9.0×10^{-7}**

Solubility of carbonates and sulphates of alkaline earth metals

a. Identify the compound that is highly soluble in water

Figure: formation of precipitate means salt is insoluble in water

Due to larger anionic size (CO_3^{--} & SO_4^{--}) and minor change in the size of alkaline earth metal ions, the lattice energy does not change by much i.e, lattice energy remains constant as we move down the group. However, hydration energy decreases down the group because the size of alkaline metal ions increases on moving from Be to Ba. Therefore, the solubility of carbonates and sulphates of alkaline earth metals decreases down the group.

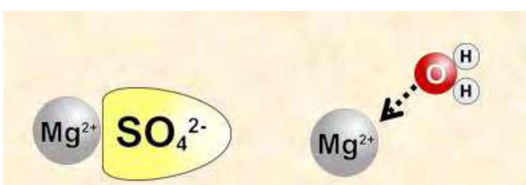
$BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

▼ Decreasing order of solubility

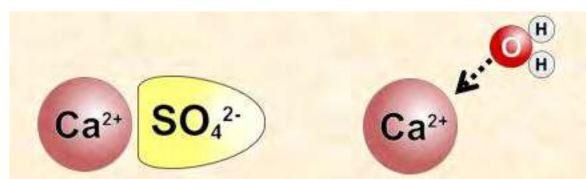
$BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$

▼ Decreasing order of solubility

As the cation gets larger, it has a lower charge density, it becomes less attracted to the polar water molecules and becomes less soluble in water.

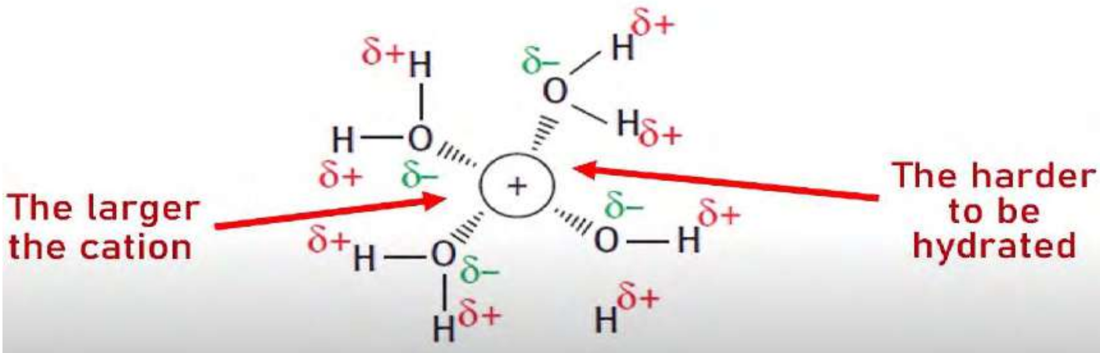


Greater charge density of Mg^{2+} ion means that it is more attracted to water, so, it is more soluble in water.



Lower charge density of larger Ca^{2+} means that it is less attracted to water so it is less soluble in water.

Test



Yourself!

> **a. Barium sulphate is less soluble in water than magnesium sulphate, why?**
> **b. Among, $BeCO_3$ and $SrCO_3$, which one is more soluble and why?**

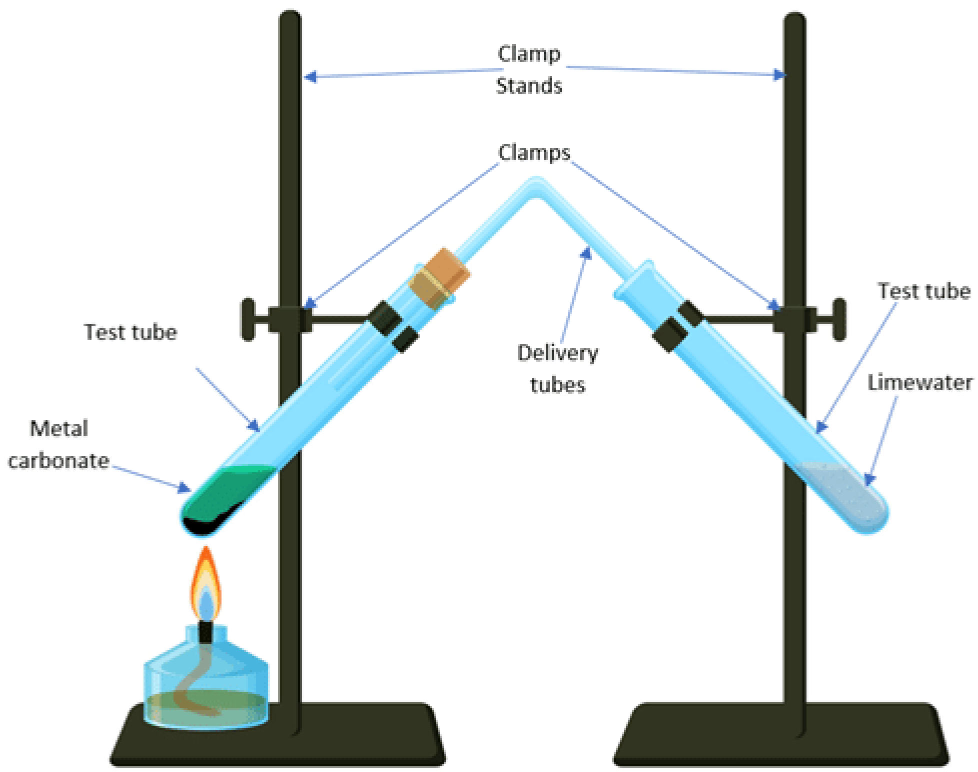
10.3.4: Stability of carbonates and nitrates of alkaline earth metals:

Activity-10

Observe the figure given:

a. What is the cause of the change in colour of lime water?

b. Write the chemical equation for the decomposition of metal carbonate of group 2.



Both carbonates and nitrates of alkaline earth metal decompose on heating. Decomposition of these compounds become harder as we move down the group from Be to Ba. In other words, as we move down the group larger compounds require more heat to decompose than the lighter one.

When carbonates of alkaline earth metals are heated they produce their corresponding oxides and carbon dioxide gas.

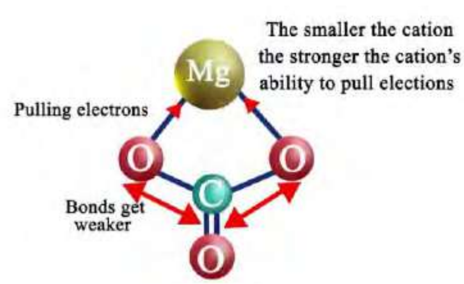
$$MCO_3\,(s) \xrightarrow{Heat} MO(s) + CO_2 \uparrow$$

Where, M: Be, Mg, Ca, Sr, Ba.

Similarly, when nitrates of alkaline earth metals are heated they produce their corresponding oxides and nitrogen dioxide and oxygen.

$$2M(NO_3)_2(s) \xrightarrow{Heat} 2\,MO(s) + 4NO_2 \uparrow + O_2\uparrow$$

The smaller the M^{2+} ion, higher the charge density, and greater the effect it will have on carbonate ion. As we move down the group size of cation increases and their effect on carbonate ion decreases. It means less distortion is caused by the larger cations on carbonate ions than the smaller cations do. The amount of heat required depends on the degree to which ion is polarized. More the polarization lesser the heat required. Hence on moving down the group more heat is required for CO_2 to leave the metal oxide and thereby become thermally more stable. Thus, the thermal stability of carbonates of alkaline earth metals follows the order:



$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$

Increasing order of thermal stability

Note: Similar is the case for stability of nitrates of alkaline earth metals.

$Be(NO_3)_2 < Mg(NO_3)_2 < Ca(NO_3)_2 < Sr(NO_3)_2 < Ba(NO_3)_2$

Increasing order of thermal stability

Test Yourself!

Carbonates	Decomposition temperature/^{0}C
$BeCO_3$	540
$MgCO_3$	900
$CaCO_3$	1280
$SrCO_3$	1360

Which of the given carbonates is easiest to decompose?
Which of the given carbonates is most stable?

Exercise

Multiple choice questions

1. Which of the following is valence shell electronic configuration of alkali metals?
a. ns^2 b. ns^{1-2} c. $ns^2\ np^1$ d. ns^1

2. Pick out the false statement for alkali metals.
a. Alkali metals can be easily cut with a knife.
b. Alkali metals show +1 oxidation state.
c. Alkali metals are stored under water.
d. The I.P. values of alkali metals decrease down the group.

3. Which of the following properties is shown by alkali metals?
a. Electropositive b. Electronegative c. Neutral d. Non-metallic

4. What is the color of the flame produced by caesium metal?
a. Red violet b. Blue c. Yellow d. White

5. Alkali metals dissolving in liquid ammonia give a blue colored solution. This is due to the formation of ammoniated
a. ions b. metal cation & electrons c. metal cation only d. electron only

6. Which of the following alkali metals cannot form superoxide?
a. Potassium b. sodium c. Lithium d. Caesium

7. Which pair of atomic numbers represents an s-block element?
a. 19, 56 b. 18, 56 c. 12, 15 d. 7, 9

8. Which of the following possesses the lowest melting point?
a. K b. Li c. Na d. Cs

9. Name the principle used for extraction of sodium from halide ores like NaCl.
a. Pyrometallurgy b. Hydrometallurgy c. Electrometallurgy d. Magnetic separation

10. Which of the following processes is used to prepare sodium metal commercially by electrolysis of fused NaCl?
a. Down's process b. Solvay process c. Castner& Kellner's cell d. Nelson cell

11. What happens at anode during the electrolysis of fused NaCl?
a. Na^- ions are oxidized b. Cl^- ions are oxidized c. Cl^- ions are reduced d. Na^+ ions are reduced

12. By which of the following sodium metals is extracted?
a. Heating sodium oxide with hydrogen b. Heating sodium oxide with carbon
c. Electrolysis of fused sodium chloride d. Electrolysis of aqueous sodium chloride

13. Which of the following is the most abundant alkali metals in nature?

a. Sodium b. Lithium c. Caesium d. Rubidium

14. Which metal is present in the mineral Chile saltpeter?

a. Potassium b. Calcium c. Lithium d. Sodium

15. What is the molecular formula of Borax?

a. $NaNO_3$ b. $Na_2B_4O_7.10H_2O$ c. NaCl d. $NaAlSiO_3$

16. What is the purpose of adding $CaCl_2$ during extraction of sodium metal in Down's process?

a. To increase the melting point of NaCl. b. To increase the reactivity of chlorine
c. To decrease the melting point of NaCl. d. To stop the formation of Cl_2 gas

17. During electrolysis of concentrated aqueous solution of NaCl, what is the product at cathode?

a. Sodium b. Hydrogen c. Oxygen d. Chlorine

18. Pick out the alkaline earth metal.

a. Sr b. Snc.Pb d. Cs

19. Pick out the false statement for alkaline earth metal.

a. They are softer than alkali metals.
b. They show +2 oxidation states.
c. They are highly reactive.
d. They produce alkali when dissolved in water.

20. Which of the following imparts apple green color to Bunsen flame?

a. Mg b. Ba c. Sr d. Ca

21. What is the molecular formula of bleaching powder?

a. $CaOCl_2.H_2O$ b. $CaOCl.Ca(OH)_2$ c. $CaOCl_2$ d. $CaCl_2$

22. Which of the following oxidation states is shown by alkaline earth metal?

a. 0 b. +1 c. -1 d. +2

23. Which of the following compounds is produced when calcium is exposed to air?

a. Calcium oxalate b. Calcium hydride c. Calcium oxide d. Calcium hydroxide

24. What is the chemical formula of quick lime?

a. CaO b. MgO c. $CaOCl_2$ d. Ag_2O

25. Which one is less soluble in water?

a. $CaCO_3$ b. $MgCO_3$ c. $SrCO_3$ d. $BaCO_3$

26. Which of the following is the most soluble sulphate?

a. $BeSO_4$ b. $MgSO_4$ c. $CaSO_4$ d. $SrSO_4$

27. What is the general trend of thermal stability of nitrate of group II A metal?

a. $Be(NO_3)_2 < Ca(NO_3)_2 < Mg(NO_3)_2 < Ba(NO_3)_2$

b. $Ba(NO_3)_2 < Ca(NO_3)_2 < Mg(NO_3)_2 < Be(NO_3)_2$

c. $Be(NO_3)_2 < Mg(NO_3)_2 < Ca(NO_3)_2 < Ba(NO_3)_2$

d. $Ca(NO_3)_2 < Be(NO_3)_2 < Mg(NO_3)_2 < Ba(NO_3)_2$

28. Which of the following has high thermal stability?

a. $BesO_4$ b. $CaSO_4$ c. $MgSO_4$ d. $BaSO_4$

Short answer questions

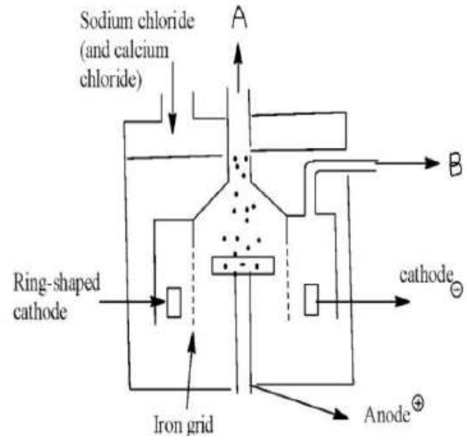
1. a. What are alkali metals? List them. Why are they called so?
   b. How does sodium metal react with ammonia?

2. Give reason:
   a. Alkali metals become dull when exposed to the atmosphere.
   b. Alkali metals are good reducing agents.
   c. Potassium is more reactive than sodium.
   d. Sodium metal stored in paraffin oil.
   e. Sodium metal cannot be extracted by pyrometallurgical process.
3. Define the terms- a) Soda ash b) Caustic soda. Also write their molecular formula.

4. Differentiate between:
   a) Sodium and potassium
   b) Baking soda and washing soda
   c) Caustic soda and soda ash
   d) Mg and Ba
   e) Alkali metal and alkaline earth metals.

5. An element 'X' has valence shell electronic configuration $3s^1$.
   a. Identify X.
   b. How does X react with sulphur?
   c. Why is X stored in kerosene?
   d. What happens when a piece of X is added to water?

6. Sodium is the most abundant alkali metal found naturally in the earth crust. It is extracted by electrolysis of a mixture of molten NaCl and $CaCl_2$ in the ratio 2:3 in Down's process.
   a. Write any two advantages of using $CaCl_2$.
   b. What happens if NaCl is directly subjected to electrolysis?
   c. Write the electrode reaction occurring in Down's cell.
   d. Give any three uses of sodium metal.



7. The given figure is Down's cell which is used for the extraction of the most abundant alkali metal sodium.
   a. Write the importance of Down's cell.
   b. What is the purpose of mixing $CaCl_2$ with NaCl?
   c. Identify A and B with electrode reaction.

8. What happens when:
   a) A pellet of sodium hydroxide is added to $CuSO_4$ solution?
   b) Sodium carbonate is added to aqueous solution of $MgCl_2$?
   c) Caustic soda is heated with CO under high pressure?

9. List the uses of - a) Caustic soda b) Sodium carbonate c) Sodium metal

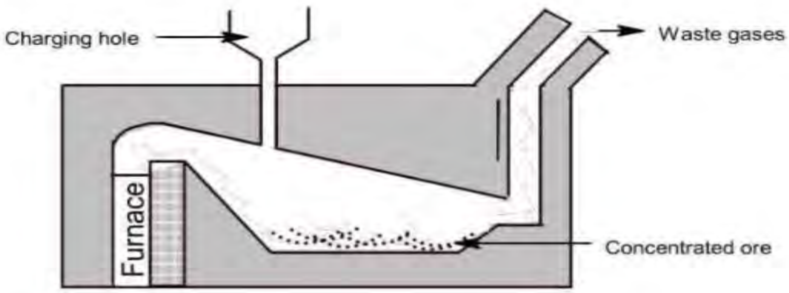
10. What are alkaline earth metals? Why are they called so? List the general characteristics of them.

11. Give reason:
   a) Mg becomes dull when exposed to the atmosphere.
   b) Alkaline earth metals exhibit +2 oxidation states.
   c) Alkaline earth metals do not occur in native state.

12. a) How does magnesium react with boiling water?
   b) What is the use of quick lime in metallurgy?
   c) Write the formula of- Magneisa and Epsom salt.

13. An element 'Y' has valence shell electronic configuration $3s^2$.
   a. Identify Y.
   b. How does Y react with sulphur?
   c. What happens when a piece of Y is added to boiling water?

Long answer questions



14. Calcination is one of the process to convert the ore into its oxides, it is usually done in the furnace given:

  a. Name the furnace given in the figure.
  b. Define the term calcination and write any two examples of ores that can be calcinated.
  c. Write any two reactions that occur in the vessel given.
  d. Name any two waste gases that are released

15. Describe the solubility trend of alkaline earth metal hydroxide.

16. Describe the solubility trend of alkaline earth metal carbonates.

17. Describe the solubility trend of alkaline earth metal sulphates.

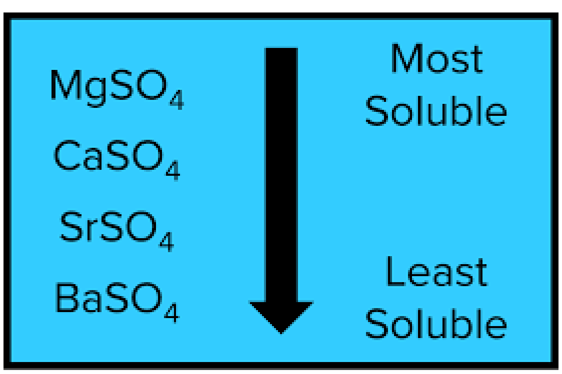
18. Describe the thermal stability trend of alkaline earth metal sulphates.

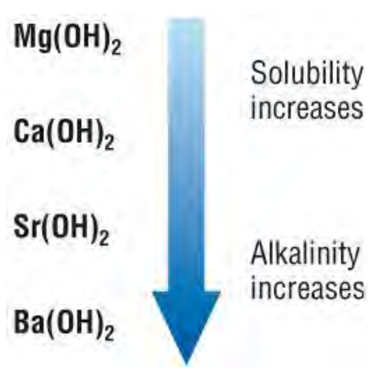
19. Describe the thermal stability trend of alkaline earth metal nitrates.

20. List the uses of - a) Quick lime b) Plaster of paris c) Epsom salt d) Bleaching powder

21. Trend of solubility of sulphates and hydroxides of group 2 is given.

  a. What is the cause of such a trend?
  b. Which of the sulphates precipitated easily and why?
  c. Which of the hydroxide solutions has the highest pH and why?

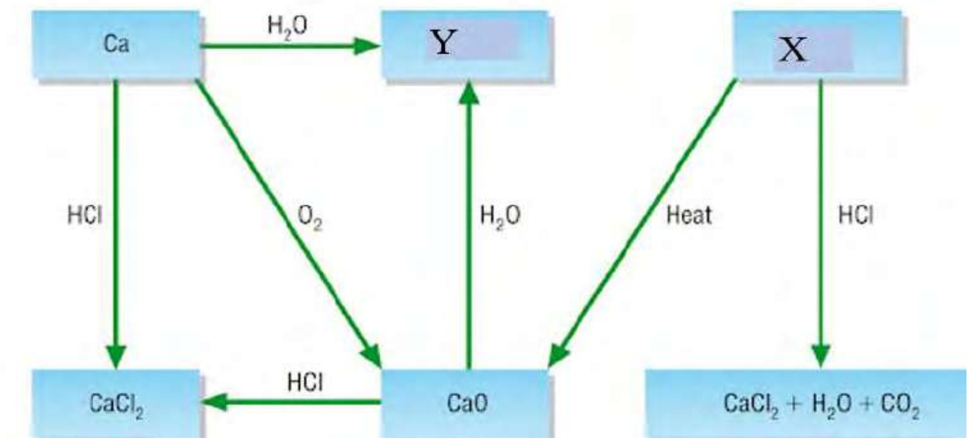




22. Use the given flow sheet diagram to answer the following questions.

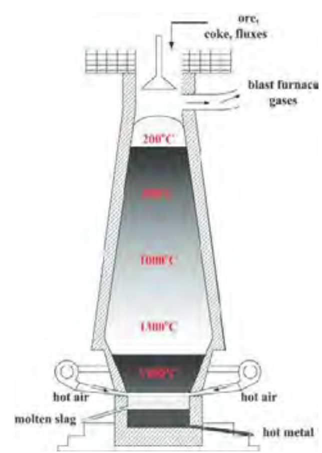
  a. Identify X and Y.

b. Write the balanced chemical equation for the conversion of Ca to Y and CaO to CaCl2
c. Write the balanced chemical equation for the action of heat and HCl on X.
d. What is the common name of X, Y and CaO?



23. Blast furnace for the smelting of ore is given below.

a. Define the term smelting.
b. Write the chemical reactions that occur in the furnce during the process.
c. What is the function of flux in the process?
d. What is the environmental impact of this process?
e. In which region of the furnace slag is formed and why?



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UNIT 11: BIOINORGANIC CHEMISTRY

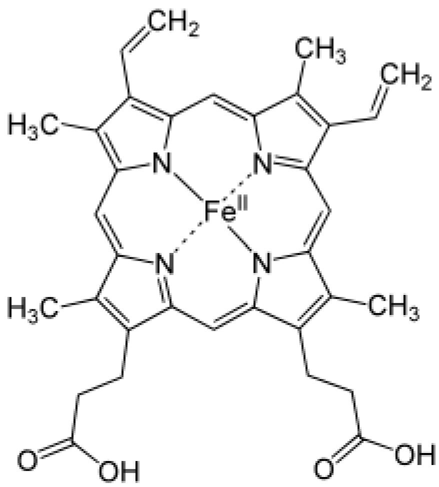


Figure 11.1: Structure of haemoglobin

Activity:

Make four groups in your classroom and discuss.

- Are all the nutrients required in the same amount for the body?
- Write the points that you have made in conclusion.

Figure 11.2: Dry fruits and vegetables

1. ……………………………………………….
2. ……………………………………………….
3. …..

11.1 Introduction

Except carbon, generally metals are important for the proper functioning. Biological processes like respiration, photosynthesis, enzyme actions, transportation of metal ions, oxygenation of blood etc. are possible due to bioinorganic compounds. For example, iron, cobalt, magnesium associated with hemoglobin, vitamin B-12, chlorophyll etc. involve the interaction of cells with metal ions.

Bioinorganic chemistry is an interdisciplinary branch of chemistry which deals with the role of metal or metal containing compounds within a biological system.

Some of the interactions between metal ions and cells are as follows:

a. The study of toxicity of inorganic species
b. Investigation of inorganic elements in nutrition
c. Study of naturally occurring inorganic elements that are related with biochemical systems
d. Study of the structure and function of biomolecules and as drugs to treat diseases
e. Study of the mechanism of photosynthesis and respiration

Test Yourself!

What do you understand by bioinorganic molecules?

What makes enzyme bioinorganic compounds?

11.2 Micro and Macronutrients

Brainstorming

Do we need an equal amount of carbohydrates, protein, vitamins for our body?

On the basis of amount necessity for the body bioinorganic nutrients are classified into following two classes. They are:

i. Micronutrients

The nutrients which are required for the body in lesser amounts for proper functioning of macronutrients to keep the body functioning properly to maintain cellular function, metabolism, physical and mental well-being are called micronutrients. Vitamins, minerals and other trace elements are micronutrients which help in body growth and protection from diseases.

Significances of micronutrients

i. They help proper functioning like metabolism, cellular function etc.
ii. Vitamins are required for production of energy, immune, blood functioning etc.
iii. Some vitamins and minerals act as antioxidants (compounds that inhibit oxidation). They protect against cell damage like cancer, Alzheimer's disease and heart diseases.

iv. These antioxidants are found in vegetables, fruits, eggs, green leafy vegetables etc. their deficiency results in anemia, scurvy, goiter, etc.

ii. Macronutrients

Carbohydrates, protein and fats are required in large amounts as compared to the vitamins and other minerals.

Since these types of nutrients are required by the body in larger quantities for the proper function of the body, they are called macronutrients. These nutrients provide sufficient energy for metabolic activities for the body. Carbohydrates, fats, fiber, protein, etc. are examples of macronutrients. Macronutrients are found in cereals, legumes, meat, fish, potato, nuts, etc. Their deficiency may cause malnutrition, kwashiorkor, etc. Macronutrients provide the body with energy, help prevent disease, and allow the body to function correctly.

Importance of macronutrients

Each type of macronutrient performs an important role in keeping the body healthy. For optimum health, people typically require a balance of macronutrients.

Carbohydrates

Carbohydrates are the preferred and trusted source of energy for several body tissues, and the primary energy source for the brain. The body can break carbohydrates down into glucose, which moves from the bloodstream into the body's cells and allows them to function.

Figure 11.3 Some foods including cereals

Carbohydrates are important for muscle contraction during intense exercise. Even at rest, carbohydrates enable the body to perform vital functions such as maintaining body temperature, keeping the heart beating, and digesting food. Such as rice, pasta, flour, and barley etc. are the sources of carbohydrates.

Protein

Figure 11.4: Proteineous Food

Protein consists of long chains of compounds called amino acids. These play an essential role in the growth, development, repair, and maintenance of body tissues. Protein is present in every body cell, and adequate protein intake is important for keeping the muscles, bones, and tissues healthy. Protein also plays a vital role in many bodily processes, such as aiding the immune system, biochemical reactions, and providing structure and support for cells. For example, meats, fish, nuts, seeds, whole grains, beans, eggs, dairy, soya, etc. are the sources of protein.

Fats

Fats are an important part of the diet that can also provide the body with energy. While some types of dietary fats may be healthier than others, they are an essential part of the diet and play a role in hormone production, cell growth, energy storage, and the absorption of important vitamins. For example: avocados, oily fish, seeds, olive oil, and nuts are the sources of fats.

How much to consume?

It is suggested the following percentages of macronutrients for good health and to provide essential nutrition:

Carbohydrate (in %)	Protein (in %)	Fat (in %)
45-65	20-35	10-35

Test Yourself!

Is there any possibility of human life without carbohydrates?

What are the sources of carbohydrates and proteins?

11.3 Importance of metal ions in biological systems

Metal ions are the auxiliary constituent of organic catalysts in the body which are called enzymes. Enzymes are biological catalysts (also known as biocatalysts) that speed up biochemical reactions in living organisms. They can also be extracted from cells and then used to catalyze a wide range of commercially important processes. For example, they have important roles in the production of sweetening agents and the modification of antibiotics, they are used in washing powders and various cleaning products, and they play a key role in analytical devices and assays that have clinical, forensic and environmental applications. Metal ions have the following functions:

i. Regulation of intracellular activities
ii. Plays role as cofactors of enzymes in the control of metabolic pathways and other mechanisms
iii. To make structural entities as carrier of oxygen and other required materials
iv. For the maintenance of osmotic pressure in the body
v. For balancing the charge, stabilizing structure, replication etc.
vi. For balancing vitamin B-12 etc.

Importance of metal ions is discussed individually as follows:

a. Biological importance of sodium(Na^+)

i. Regulates flow of water across membranes and nerve functions
ii. Controls the buffer system in the body
iii. Balances the osmotic pressure in the body
iv. Helps in transport sugars and amino acids.

b. Biological importance of Potassium(K^+)

i. Controls nerve actions in animals
ii. Regulates opening and closing of stomata
iii. Maintains turgor pressure in plants
iv. Controls kidney stone problem by lowering calcium level in urine
v. Helps in synthesis of ribosomes
vi. Helps in proper heart functioning in which low potassium causes hypertension
vii. Helps in proper functioning in skeleton and muscle contractions.

c. Biological importance of magnesium (Mg^{2+})

i. Forms pyrophosphate linkage which controls the various biological systems
ii. Helps in photosynthesis in green plants since it is present in chlorophyll which helps in metabolic activity of nitrogen in plants
iii. With the adequate supply of magnesium ions, ATP (adenosine triphosphate) is changed into ADP (adenosine diphosphate) which releases excess amounts of energy.
iv. It helps in the regulation of cholesterol in the body. It helps in the activation of enzymes and an enzyme co-factor.
v. It helps in synthesis and proper functioning of DNA.
vi. Maintains electrolyte balance in the body because it is present in exoskeleton and endoskeletons.
vii. Provides strengths and rigidity in bones and teeth
viii. Controls blood glucose level and blood pressure.

d. Biological importance of calcium ion(Ca^{2+})

i. Helps in neuromuscular functioning and cell signaling in the body.
ii. Helps in activation of enzymes, chromosome movement in cell division.
iii. Helps in blood coagulation.
iv. Regulates the functioning of proteins.
v. One of the components of the cell wall which is present in the middle lamella in the form of calcium pectate.
vi. Regulates muscle functioning like contraction and relaxation.

e. Biological importance of irons (Fe^{2+} and Fe^{3+})

i. It helps in oxygen carrying functions in the body and helps in myoglobin for the storage of oxygen in muscle tissues.
ii. It functions as an electron carrier in plants, animals, etc.
iii. It helps in the creation of hormones in the body.
iv. Helps to convert molecular nitrogen into ammonia by nitrogen fixation.

f. Biological importance of copper ions (Cu^{+} and Cu^{2+})

i. Helps in transport of electrons and storage of oxygen in plants.
ii. It helps in the formation of melanin pigment in the skin.
iii. It helps in synthesis of phospholipids.
iv. It helps developing fetuses in pregnant women.
v. Helps in brain functioning and is required for connective tissue formation.

g. Biological importance of zinc ion (Zn^{2+})

i. Regulates functions of insulin secreted by pancreas in the body
ii. Plays an important role in sexual maturation and reproduction in the human body
iii. Helps in metabolism of the body, synthesis of protein and DNA
iv. Helps for the production of growth hormone (auxin) in plants
v. Helps in healing of wounds
vi. Helps to maintain vitamin A in plasma
vii. Helps in genetic transcription and stabilization of proteins.

h. Biological importance of nickel(Ni^{2+})

i. Helps in iron absorption, glucose metabolism and formation of various types of hormones
ii. Helps in the production of RBCs
iii. Helps in electron transfer during production of ATP
iv. Helps in enzyme activities like hydrogenase and hydrolases.

i. Biological importance of cobalt (Co^{2+})

i. Helps in functioning of the brain and nervous system because cobalt is the constituent of vitamin B-12 in the form of cobalamine
ii. Helps in enzyme activity of the body
iii. Helps in the formation of RBCs.

j. Biological importance of chromium (Cr^{3+})

i. Helps in metabolism of fats and carbohydrates
ii. Helps in stimulation of fatty acid and cholesterol synthesis
iii. Helps in brain functioning
iv. Helps in functioning of insulin.

Test Yourself!

- What are the micronutrients that are required for the human body?
- Give any four examples of macronutrients and macronutrients.

11.4 Ion pumps (Sodium potassium and sodium- glucose pump)

a. Sodium potassium pump

The sodium-potassium pump (Na, K-ATPase) was discovered in 1957. It plays an important role in contracting the cardiac muscle, kidney function, and nerve signaling. The purpose of the sodium-potassium pump is to maintain the proper concentration of potassium ions K^+ and sodium ions Na^+ inside and outside of the cell. In the image below, you can see the pump is found in the plasma membrane of cells. Keep in mind a cell does not have one Na, K-ATPase pump. There can be hundreds in each cell.

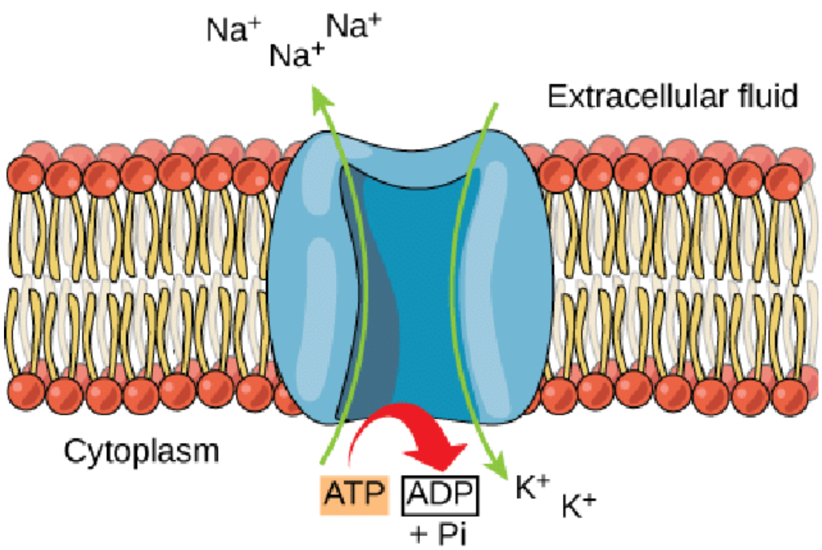


Figure 11.5: Sodium Potassium Pump

The Na, K-ATPase pump is a protein in the cell used to regulate Na^+ and K^+ gradients across the membrane. As gradients change, cells can produce electrical signals. The pump pushes three Na+ to the outside of the cell for every two K+ that are pushed inside the cell. This results in a higher concentration of Na^+ outside the cell and a larger concentration of K^+ inside the cell. The role of Na, K-ATPase is to regulate the Na^+ and K^+ levels across the cell membrane. When the cell is at resting membrane potential, the concentration of Na^+ is higher on the outside of the cell and the concentration of K+ is higher on the inside of the cell. A concentration gradient is the term used to refer to a difference in concentration of molecules. When molecules move from a high to a low concentration, they move down the concentration gradient. It does not require energy and is referred to as **passive transport**. When molecules move from a lower concentration to a higher concentration, they are said to be moving against the concentration gradient. This requires energy and is referred to as **active transport**.

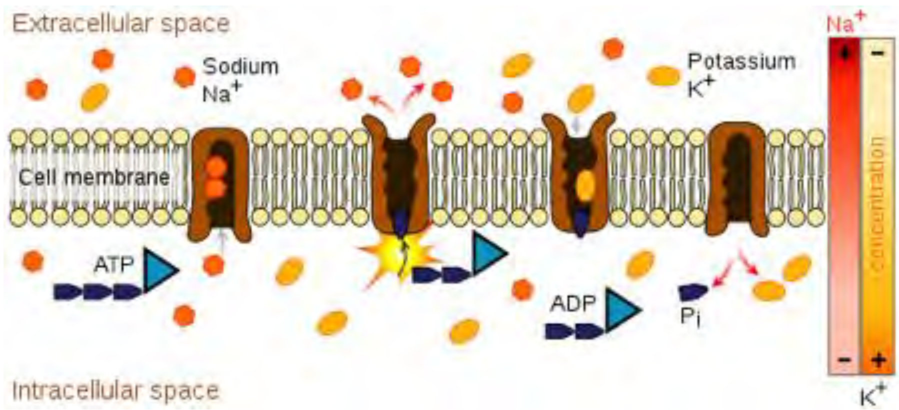


Figure 11.6 : Process showing sodium potassium pump

b. Sodium glucose pump

Sodium-glucose co-transporter (SGLT) activity mediates apical sodium and glucose transport across cell membranes. Co-transport is driven by active sodium extrusion by the baso-lateral sodium/potassium-ATPase, thus facilitating glucose uptake against an intracellular up-hill gradient. Baso-laterally, glucose exits the cell through facilitative glucose transporter 2 (GLUT2). In humans, six SGLT isoforms have been identified but SGLT1 and SGLT2 are the focus of this review.

SGLT1 is responsible for glucose absorption in the small intestine, and for reabsorbing nearly 3% of the filtered glucose load in the renal proximal tubule segment 3 (S3). Until recently, tissue localization studies of SGLT1 were hampered by the lack of well-functioning antibodies. In the intestine, examination of various species showed that SGLT1 is expressed in the luminal brush border of enterocytes, which are responsible for nutrient absorption, and in entero-endocrine L-cells and K-cells, which produce glucagon-like peptide-1 (GLP-1) and glucose-dependent insulin-tropic peptide (GIP). SGLT1 knockout (Sglt1−−) mice show glucose-galactose mal-absorption and do not survive unless kept on a glucose-free diet. However, Sglt1−/− mice have a comparable blood pressure compared with their wild-type littermates.

SGLT2 is responsible for glucose reabsorption in the proximal tubule segment 1 and 2 (S1/2), wherein it reabsorbs more than 90% of the filtered glucose load. In SGLT2 knockout (Sglt2) mice, however, SGLT1 may compensate and reabsorb up to 35% of the filtered glucose load. Sodium homeostasis and blood pressure are unaffected in Sglt2 mice.

The human blood glucose concentration under normal conditions is nearly 5.5 mmol/l, and less than 1% of filtered glucose is excreted in the urine. However, increasing amounts of glucose appear in the urine if blood levels exceed 10–12 mm.mol/L, assuming that glomerular filtration rate is unaffected. This may become evident in patients with insufficiently controlled diabetes mellitus. Under diabetic conditions, proximal tubule glucose reabsorption is enhanced via increased expression of SGLT2, possibly contributing to overt hyperglycemia. The increased proximal tubule sodium reabsorption may reduce the downstream sodium and chloride concentration at the macula, thus activating the tubule-glomerular feedback mechanism. This may lead to early diabetic hyperfiltration and stimulation of the rennin–angiotensin–aldosterone system (RAAS). In patients living with diabetes mellitus for 10–20 years, approximately 20% develop diabetic nephropathy, which is currently the leading cause of end-stage renal disease.

This review focuses on the regulation of intestinal and renal SGLTs. The majority of studies presented in this review comprise investigations in cell cultures, isolated epithelia, purified membranes, isolated transport proteins or intact animals. Consequently, they reveal a rather complex regulatory pattern of SGLTs and may vary depending on the experimental system or species studied. Each regulatory system may contribute to a different extent, resulting in differences in the overall final response.

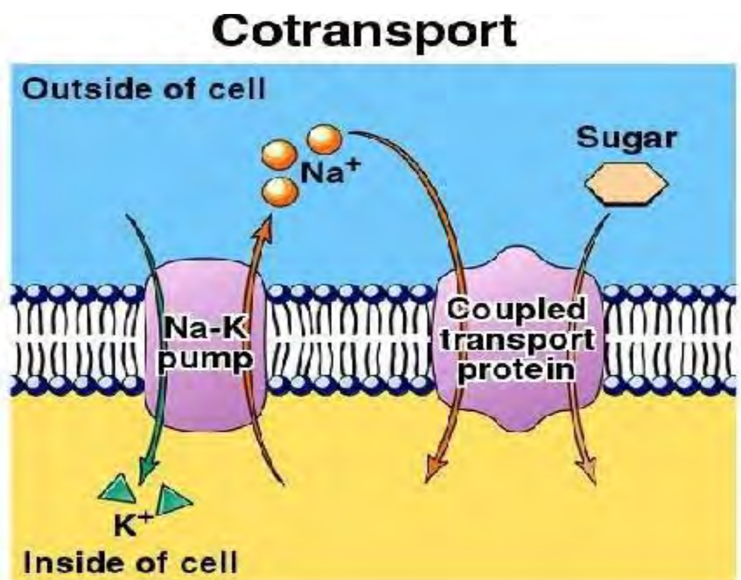


Figure 11.7: Contransport

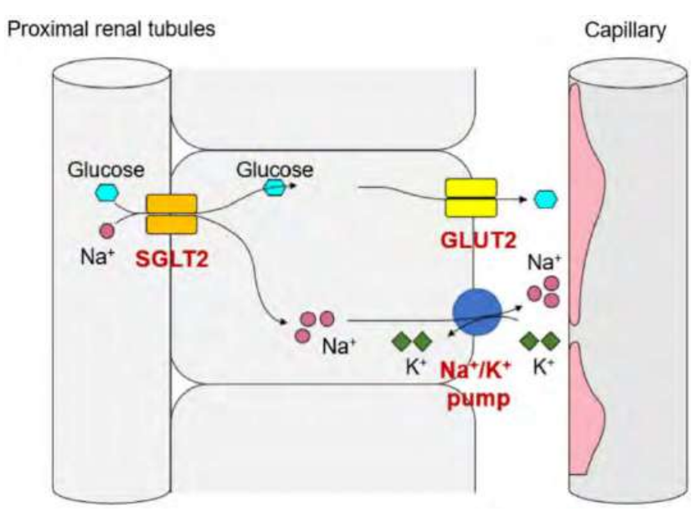


Figure 11.8: Process of sodium potassium pump in renal tubules

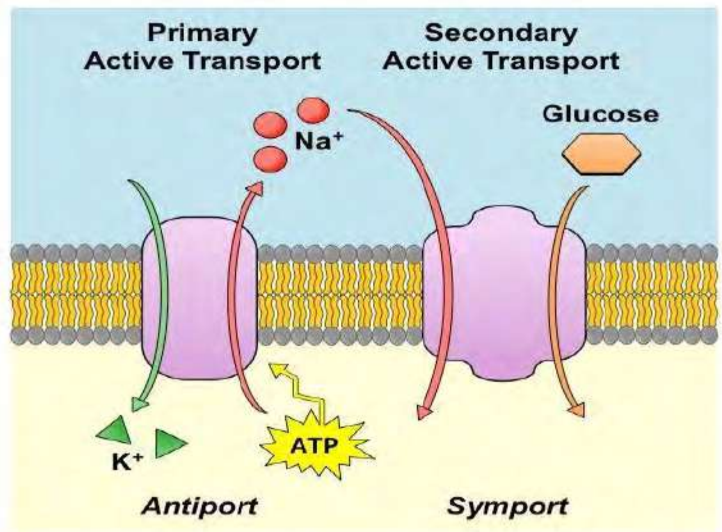


Figure 11.9: Process showing primary and secondary transport

Test Yourself!
Is there any difference between sodium- potassium pump and sodium glucose pump?
What is sodium?

11.5 Metal toxicity

Accumulation of heavy metals which causes problems in soft tissues of the body is called metal toxicity. Metals like lead, mercury, arsenic, cadmium, iron, etc. cause poisoning in human health. Due to free radicals generated by the metals cause oxidative stresses in the enzymes, proteins, lipids, DNA, nucleic acids which disturbs their functioning which are carcinogenic as well as neurotoxic in nature. Due to toxicity of metals functioning of kidney, liver, composition of blood, brain, lungs, etc. Metal toxicity depends on factors like dose, route of exposure, chemical nature(species), age, gender, genetics, time of exposure, etc. Among many metals iron, arsenic, mercury, lead and cadmium have higher degrees of toxicity. So, we are talking about these metals' toxicity hereafter:

a. Iron

i. Except Fe^{2+}, other forms of iron are toxic to human health.
ii. The excess iron causes formation of gastrointestinal ulcers.
iii. Excess iron in the body may cause cancer. So the excess intake of meat may cause cancer.
iv. Asbestos consists of 30% iron. So, the person who works in asbestos is at high risk of lung cancer.
v. Iron may cause damage to DNA which leads to cancer.

b. Arsenic

i. Due to mining and processing of ores, arsenic can cause poisoning.

ii. Generally in paints, soaps, metals, semiconductors and drugs may contain arsenic, in addition certain pesticides, fertilizers and animal feedings may release arsenic in the environment.
iii. Inorganic salts in the form of arsenate and arsenite are dangerous to human health.
iv. Trivalent arsenic deactivates the –SH bond of protein. So, the enzyme is deactivated.
v. Contamination of it also can cause cancer of lungs, skin, gallbladder, liver, etc.
vi. It may be present in drinking water due to herbicides, pesticides, fungicides, wood preservatives, etc.
vii. Toxicity of arsenic can be either acute or chronic. The chronic toxicity of arsenic is called arsenicosis.
viii. Long term exposure in arsenic can cause keratosis, internal cancer, pulmonary diseases, peripheral vascular disease, hypertension, diabetes, cardiovascular diseases, etc.

c. Mercury

i. Generally mercury is used in thermometers, barometers, pyrometers, hydrometers, mercury arc lamps, fluorescent lamps as well as catalysts. It is also used in paper and pulp industries, amalgams, etc.
ii. It can be inhaled in three different forms i.e. metallic elements, inorganic salts and organic salts. These forms of mercury are widely in water resources. It may affect both aquatic creatures; microorganisms and human beings too. Consumption of contaminated aquatic animals is the main route of human exposure. It is neurotoxic.
iii. Poisoning can be caused due to agricultural wastes, wastewater discharges, mining, incineration, contaminated water, contaminated fish, etc.
iv. Mercury can affect lungs, kidneys, skin, brain, etc.
v. Symptoms of mercury poisoning are depression, fatigue, sluggishness, irritability, headache, etc.
vi. The toxicity of mercury causes pneumonia, abnormal formation of fibrous tissues, abnormal formation of fluid in lungs, etc.
vii. Minamata Bay is the waste product dumping site of Japan. Mercury contaminated fish were consumed by the people which caused large casualties. It is reported that about 2000 people died and over thousands were crippling from the mercury poisoning.
viii. Brain is the main target of mercury poisoning in human beings.

d. Lead

i. Lead is poisoned from herbal medicines, toys and candles, stained glasses, leaded glasswares, tetraethyl lead, storage batteries, paints, etc.
ii. Lead poisoning happens when your child is affected by high levels of lead exposure. Lead poisoning is usually caused by eating or drinking (ingesting) lead, but touching or breathing

in the toxic metal can also cause it. Lead poisoning is when any detectable amount of lead is found in your child's blood. Lead can affect many parts of your child's body, including their brain, nerves, blood, digestive organs and more. While lead poisoning can affect anyone, it's especially dangerous in children. It can damage your child's nervous system, brain and other organs.

iii. Lead poisoning can also lead to severe health, learning and behavioral problems, including sudden brain damage and long-term intellectual deficits.

iv. Overexposure of lead to the children may cause less playful, clumsier, irritable, sluggish, etc. but in adults it may cause high blood pressure, damage to the reproductive organs, nervous systems, etc. Some may suffer from kidney diseases, wrist drop, etc.

e. Cadmium

i. Contamination can occur due to industrial production of alloys, vapor lamps, storage batteries, pigments, etc. In addition to this, sewage sludge fertilizers, ground wate,r etc. are the sources that accumulate cadmium.

ii. It damages bone (osteoporosis) and lungs mainly. Once in the bloodstream, cadmium binds to alpha-2-macroglobulin and albumin and gets distributed to the liver and kidneys. Aside from these two main organs, cadmium also concentrates in the pancreas, spleen, heart, lung, and testes. Once in the liver, cadmium binds with metallothionein and this complex is slowly released from the liver. Because of the slow release, its biologic half-life can be at least ten years. Cadmium then travels to the glomerulus, and large amounts concentrate in the proximal tubule–conferring its renal toxic effects.

iii. Within six hours of soldering with cadmium alloys, patients may experience fever, chills, cough, and respiratory distress. Patients may appear well on initial presentation but can progress from pneumonitis to acute respiratory distress syndrome (ARDS). Death can occur in 3-5 days.

iv. Chronic poisoning of cadmium may cause damage in nephrone which is called nephrotoxicity, pulmotoxicity, musculoskeletal toxicity, hepatotoxicity, hypertension, immunosuppresions, etc.

v. Acute poisoning of cadmium has significant risk to the lungs.

Exercises

Multiple choice questions

1. Some of the nutrients and their common functions are given in the following table:

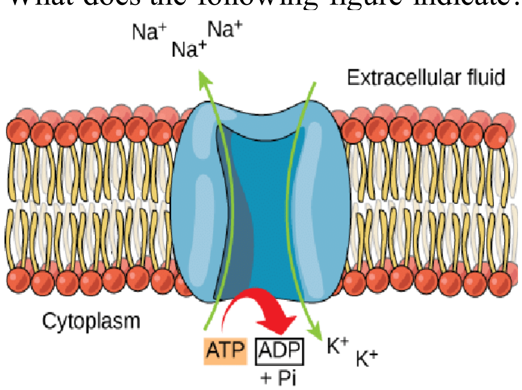
Nutrients	Functions
A. Rice	1. Helps in oxygenation process of blood
B. Iron	2. Gives energy
C. Copper	3. Balances the osmotic pressure in the body
D. Na	4. It helps developing fetus in pregnant women.

Which of the pairs is correct between two columns?

a. A-2, B-1, C-4, D-3
b. A-3, B-4, C-1, D-2
c. A-1, B-2, C-3, D-4
d. A-4, B-3, C-2, D-1

2. Which of the following is about sodium potassium pump?
   a. Primary active transport protein
   b. Primary passive transport protein
   c. Secondary active transport protein
   d. Secondary passive transport protein

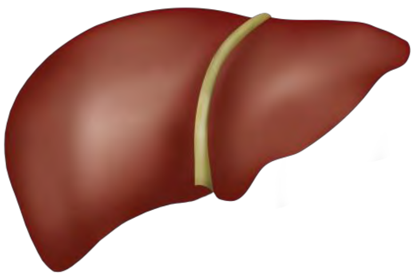
3. What does the following figure indicate?



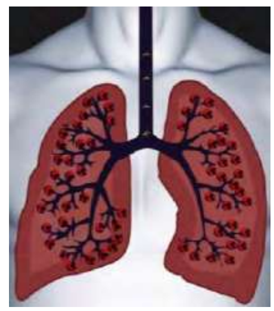
   a. Sodium potassium pump
   b. Sodium glucose pump
   c. Potassium glucose pump
   d. Potassium ion pump
4. Which of the following elements is most abundant in the human body?
   a. Zinc
   b. Titanium

c. Iron
d. copper

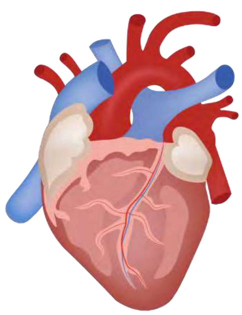
5. Which of the following is the main role of zinc in bio-inorganic chemistry?
   a. It plays catalytic role in many proteins
   b. It forms an octahedral geometry in enzymes
   c. It facilitates acid-base catalysis by coordinating to a water molecule
   d. Carbonic anhydrase does not contain a zinc center.
6. Which of the following is macronutrient?
   a. Magnesium
   b. Iron
   c. Sodium
   d. glucose
7. Which part of the body does mercury affect mostly?



a



b



c

d

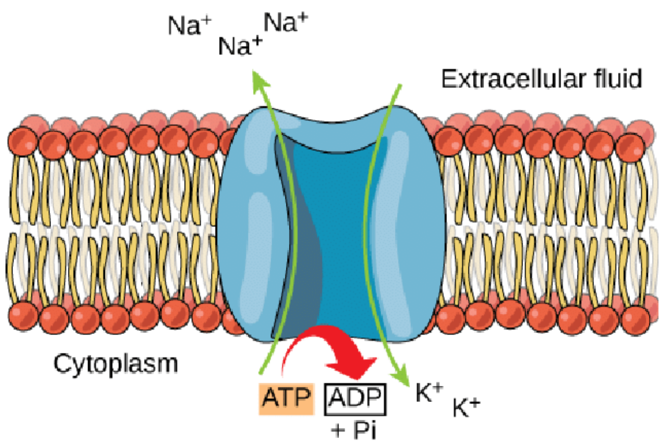
8. Body of a baby is not growing well. Doctor suggested having body building food frequently. Which nutrients should the baby eat frequently?
   a. Vitamin
   b. Carbohydrates
   c. Protein
   d. Minerals

9. Which of the following is correct for the essential biological elements?
   a. Fe, Ru, Zn, Mg
   b. Cu, Mn, Zn, Ag
   c. Fe, Cu, Co, Ru
   d. Fe, Mo, Cu, Zn

10. Which of the following metals is present in chlorophyll?
    a. Mg
    b. Cu
    c. Fe
    d. Ca
11. Which part of the body does sodium metal present?
    a. Heart cells
    b. Liver cells
    c. Nerve cells
    d. Lungs cells

Short answer questions

1. Differentiate between sodium-potassium pump and sodium glucose pump (any five).
2. What is shown in the following diagram? Explain about the process.



3. Draw mechanistic diagram of sodium glucose pump and explain about the procedure.
4. Make a list of macronutrients and write at least two main functions of each of them.
5. What is metal ion poisoning? Explain about mercury poisoning in human beings in detail.
6. Some of the symptoms seen in a person are given below.
    - Retarded growth, skin damage, retarded maturation
    - Anemia and immune disorder
    - Causes osteopenia; retarded skeletal growth and can lead to decreased bone mineral density
    - Shows the symptoms of diabetes
    - It helps in synthesis of ribosomes.

    On the basis of the above symptoms and functions, write the name of the related minerals.
7. What are enzymes? How can we supply enzymes in adequate amounts to our body?
8. A person is suffering from a weak bone and has Alzheimer symptoms. What type of nutrients does the person have deficiency of? Explain with reasons.
9. How is cadmium different from mercury in its toxicity? Explain
10. What are the biological importance of Calcium and magnesium? Write any five of each.

11. What are the major diseases caused by the following metal?

Cadmium	Mercury	Arsenic	Lead	Iron

12. Mention the biological importance of sodium and potassium in the human body.
13. Imagine that the sodium-potassium pump is not functioning properly in a person, what could be its effects on him/her?

Project Work:

Make a list of micronutrients and macronutrients included in the meal in your house in a week and submit to your teacher next week as given in the following format:

	Micronutrients	Macronutrients
Sunday		
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		

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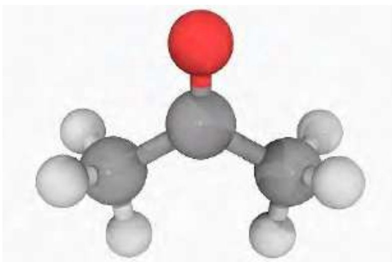
UNIT 12: BASIC CONCEPT OF ORGANIC CHEMISTRY



Activity:

Discuss the following structure of the compound. Observe them about the formation of bonds and fill in the blanks in the given table.

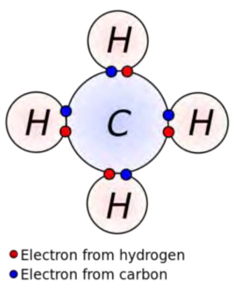


Figure 12.1

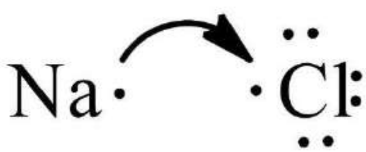


Figure 12.2

Figure No.	How is bond formed?	What is the name of the bond ?
1	………………………	…………………………………
2	……………………….	……………………………….

12.1 Introduction to Organic Chemistry and Organic Compounds

What are the constituents of NaCl and CH_4?

Organic compounds are those compounds which are formed by covalent bonds. Covalent bonds are formed by sharing the electrons within the atoms. The chemistry which deals with the study of those organic compounds is called organic chemistry.

In the early days, it was considered that all the organic compounds could have originated from plants and animals only through some mysterious forces. This force was called 'vital force' and the hypothesis that explains the formation of organic compounds only under the influence of some mysterious force existing in the living organism is known as 'vital force theory'. This theory was first proposed by Berzileus in 1815. The word 'vita' means 'life' in Latin. In 1828, the German chemist, Friendrich Wohler obtained urea by heating ammonium cyanate. After this, Kolbe prepared acetic acid from its respective elements (C, H and O) in 1845, Bertholet synthesized methane in 1856. Hence, the old concept of 'vital force theory' was discarded due to synthesis of organic compounds in the laboratory from inorganic compounds.

Friendrich Wohler

$$NH_4Cl + KCNO \xrightarrow{heat} NH_4CNO + KCl$$

Ammonium cyanate

$$NH_4CNO \xrightarrow{heat} NH_2CONH_2$$

Carbohydrates, proteins, fats, vitamins, hormones, petroleum products, drugs, dyes, plastics, gums, rubber, etc. are some of the compounds that are not considered as organic compounds even though they contain carbon. For example, carbon monoxide (CO), carbondioxide (CO_2), carbondisulphide (CS_2), metal carbonates (MCO_3), metal bicarbonates ($MHCO_3$), metal cyanides (MCN), cyanates (-CNO), isocyanates (RNC), etc. The study of carbon compounds is called organic chemistry.

Later, the new concept of organic compounds has been defined widely considered as modern concept. The binary compounds of carbon and hydrogen and their derivatives which directly or indirectly involve living matters are called organic compounds. Carbon is an essential element of the organic compound and is covalently bonded with other elements like hydrogen, oxygen, nitrogen, sulphur, phosphorous, halogen, etc. The compounds containing only carbon and hydrogen in their molecules are called hydrocarbons and these are regarded as parent organic compounds. All other compounds are derived from hydrocarbons by the replacement of one or more hydrogen atoms by other atoms or group of atoms are called derivatives of hydrocarbons. Organic chemistry is the branch of chemistry which deals with the hydrocarbon and their derivatives.

Test Yourself!

What is vital force theory?

What is the difference between the old and modern definition of organic chemistry?

12.2 Reason for the separate study of organic compounds

Figure 12.3 Molecular structure of Sulphuric acid

Do you find any difference in the nature of above molecular structure in their bonding and constituent elements?

Due to the large number of compounds

Organic compounds are found in large numbers with various chains. Approximately the number of inorganic compounds is 1,00,000 but there are millions of organic compounds. Several thousands of new organic compounds are synthesized every year. The above molecules show more possibility of number of molecules of organic compounds but from the molecule of sulphuric acid no other molecules are possible from above structures.

Due to unique chemical and physical properties

Due to the formation of covalent bond by carbon atom, these compounds have significantly low melting and boiling point, lower solubility in water, they are inflammable in nature, non-conductor of electricity, exhibit isomerism, low rate of chemical reaction, they have high molecular weights since they form complex structure. The inorganic compounds do not show these properties.

Due to unique character of carbon

In the structure of polyethylene and polypropylene above 'n' has not been defined. So, the chains are longer than that of sulphuric acid. Carbon forms rings and long chains with varying lengths and shapes. This property is known as catenation property. Other atoms do not have such properties. The structure of polyethylene and polypropylene shown above have this property.

Organic compounds have some more different nature than inorganic compounds

Property	Organic Compounds	Inorganic Compounds
Composition	Carbon is essential.	Composed of any elements
Occurrence	Most of the compounds are obtained from living beings.	Most of the compounds are obtained from non-living beings.

Bonding	Possess covalent bond	Possess covalent, ionic and coordinate covalent bonds
Isomerism	Exhibit isomerism	They do not exhibit isomerism
Reactivity	Show molecular reactions and are generally slow	Show ionic reactions which are very fast and instantaneous.
Conductivity	Solutions are non-conductor of electricity	Solutions are good conductor of electricity
Catenation	Forms long chain and ring.	Don't form a chain and ring.
Melting point and boiling point	Low melting and boiling point and are volatile in nature.	Comparatively high melting and boiling points but generally non-volatile in nature.
Combustibility	Undergo combustion and produces CO_2, H_2O and energy	Generally non-combustible and give varieties of the products.

Do you know?

Sources of organic compounds are as follows but sources of inorganic compounds are non-living things.

Plants and animals
Natural gas and petroleum
Coal
Fermentation
Synthesis

Test Yourself!

a. Why is it necessary to study organic compounds separately?
b. What is isomerism?

12.3 Tetra covalency and catenation nature of carbon

Tetracovalency (tetra =four, covalency=forms covalent bond)

Activity: Study the following steps of formation of covalent bonds and discuss in groups in your classroom.

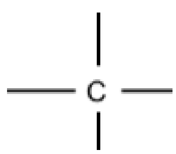
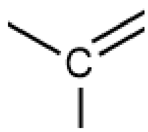
a. Configure the electrons of carbon on the basis of subshell.
b. Draw a table of one electron in a clockwise and anticlockwise pattern.
c. Count the total number of unpaired electrons in excited state.

In the ground state the number of unpaired electrons is only 2 but in the excited state it shows 4 unpaired electrons. It gains 4 electrons to be stable i.e. to reach in octet state.

What is the conclusion of this activity?

Carbon needs to acquire four electrons to be completely filled i.e. to attain octet state. Acquiring four bonds to be octet states is called tetra covalency. The property of the carbon atom by which it forms four covalent bonds is generally known as tetracovalency of carbon.

Tetra- covalent bond is formed by sharing the electrons. Carbon can form sigma bond(σ) and pi-bonds(π) by sharing electrons. Sigma bond(σ) is formed when it forms a single bond. But pi-bond (π) is formed when it forms double and tripple bonds.

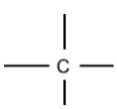
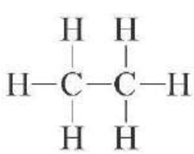
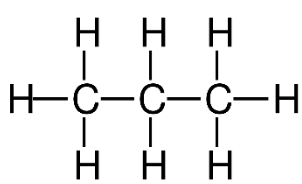
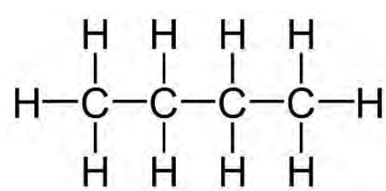
Four σ bonds	**three σ and one π bond**	**two σ and two π** bonds
		$\equiv C -$ =C=

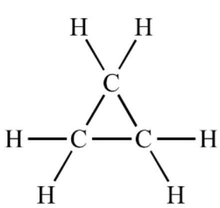
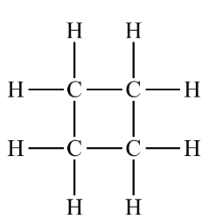
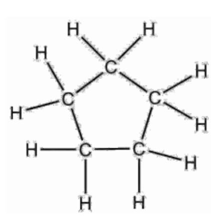
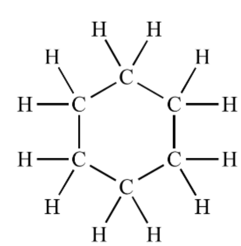
b. Catenation nature

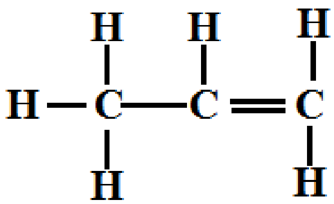
Activity:

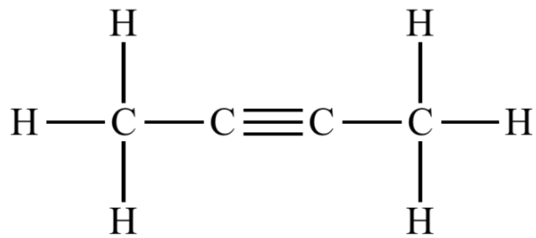
a. Write one carbon atom and make four bonds around it in a chart paper.
b. Add hydrogen at the end of each bond.
c. Replace one hydrogen by another carbon and draw three other bonds around the newly added carbon.

How did you find the new compound from the first compound?



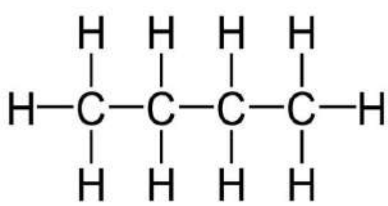


Carbon has the unique ability to bond successively to other carbon atoms to form chains of varying lengths and shapes (straight, branched and closed). Carbon atoms may form single, double or tripple bonds among other carbon atoms. This property is responsible for the existence of large numbers and varieties of organic compounds. Carbon forms stronger bonds between C-C atoms. Therefore, existence of self-linkage is possible. Making a chain or ring of carbon atoms by self-linking is called catenation property.

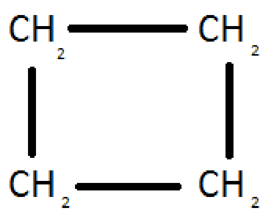
Test Yourself!

a. What is the message of tetracovalency?
b. What is catenation?

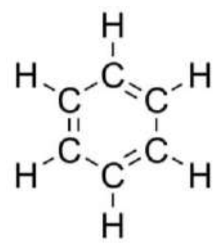
12.4 Classification of Organic Compounds



P



Q



R

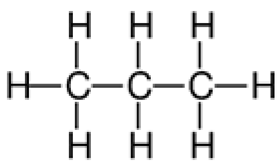
What are the differences in the nature of chains in the structures of compounds P, Q and R given above?

There are millions of organic compounds. The parent organic compounds are hydrocarbons made up of carbon and hydrogen. Other compounds are considered as their derivatives formed by replacing hydrogen by other atoms or functional groups into parent hydrocarbons. There are many ways to classify organic compounds. Broadly organic compounds are classified into two classes. They are:

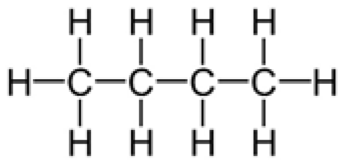
A. Open chain or aliphatic organic compounds
B. Closed chain or cyclic organic compounds

A. Open chain or aliphatic hydrocarbons

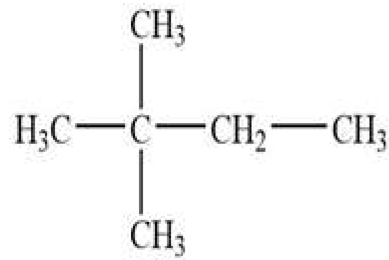
The organic compounds which do not have a bond between terminal carbon atoms are called open chain or aliphatic hydrocarbons. The term was coined by the Greek word '**aleipha**' which means fats. They may be of straight chain or branched chain.



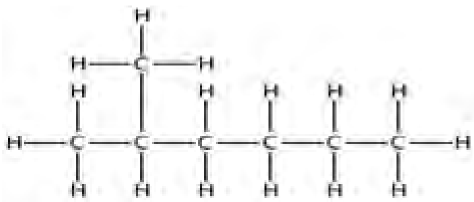
Propane



Butane



2,2-dimethyl butane



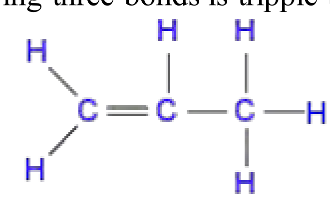
2-methyl hexane

Test Yourself!

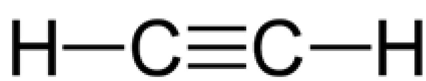
Do you know which one is a straight chain and which one is a branched chain hydrocarbon? Write the name of the compounds in the following table from the given structures.

Straight chain hydrocarbons	**Branched chain hydrocarbons**
……………………………………..	…………………………………….

The chain may be single bonded or double bonded or tripple bonded. You can easily identify the double bonded and tripple bonded hydrocarbons i.e. having two bonds is double bonded and having three bonds is tripple bonded hydrocarbons.

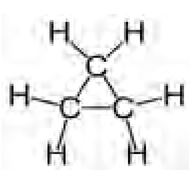


Double

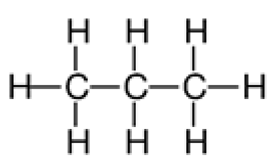


Triple bonded

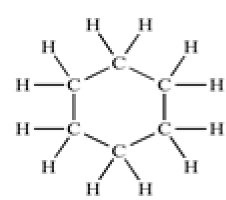
B. Closed chain or cyclic hydrocarbon



Cyclopropane



Propane



Cyclohexane

The organic compounds which are formed with a closed chain or have rings of carbon atoms are called closed chain hydrocarbon or cyclic hydrocarbons. In the compounds given above propane does not have cyclic structure but cyclopropane and cyclohexane have cyclic structure. So they are cyclic compounds.

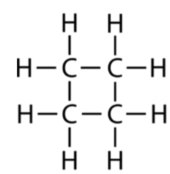
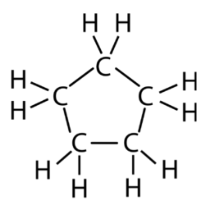
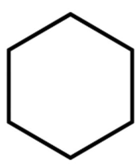
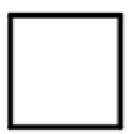
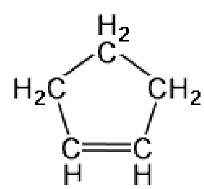
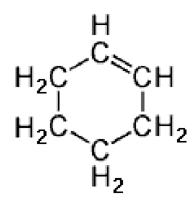
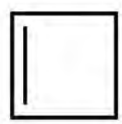
C. Homocyclic or carbocyclic compounds

The compounds which have only carbon atoms in the carbon ring are called homocyclic or carbocyclic compounds. These compounds are further divided into two classes;

i. Alicyclic compounds
ii. Aromatic compounds

i. Alicyclic compounds

The cyclic compounds which show the character of aliphatic compounds is called alicyclic compounds or cyclic aliphatic compounds. They may contain 3 or more carbon atoms. Such compounds may be saturated or unsaturated. They are similar with open chain compounds in their properties.

Saturated alicyclic compounds	Unsaturated alicyclic compounds
 cyclobutane  cyclopentane  	 Cyclopentene  Cyclohexene  

ii. Aromatic compounds

Figure 12.4

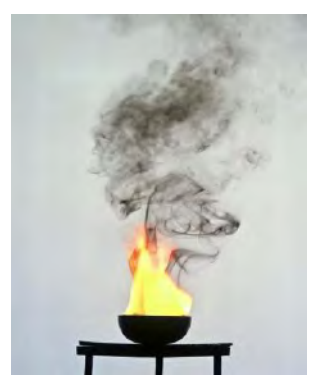
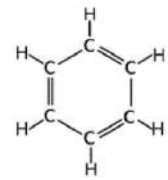


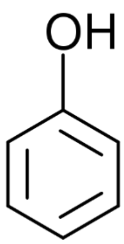
Figure 12.5

1. Have you ever felt the sweet smell of the flowers of figure A yourself?
2. Have you ever seen the flame with dense smoke as of figure B?

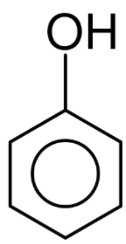
The homocyclic compounds which possess aromatic characters are called homocyclic aromatic compounds. The word aroma refers to ‘sweet smell’. The organic compounds which possess the following characteristics are aromatic compounds. Most of these compounds contain at least one benzene ring. Hence benzene and benzene like compounds are aromatic compounds. Benzene, naphthalene, anthracene, etc.

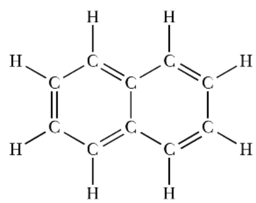


or



or



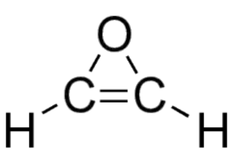


Test Yourself!

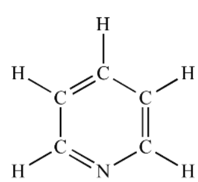
Why is benzene considered an aromatic compound?

b. Heterocyclic Compounds

Some compounds are given as follows:

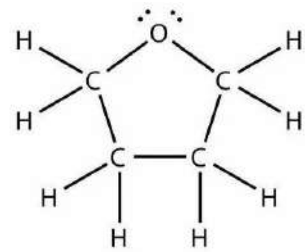


oxirane

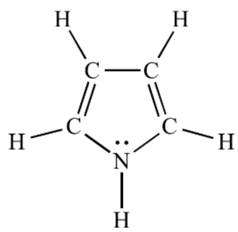


pyridine

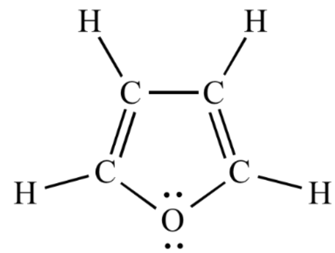
thiophene



tetrahydrofuran(THF)



pyrrole



furan

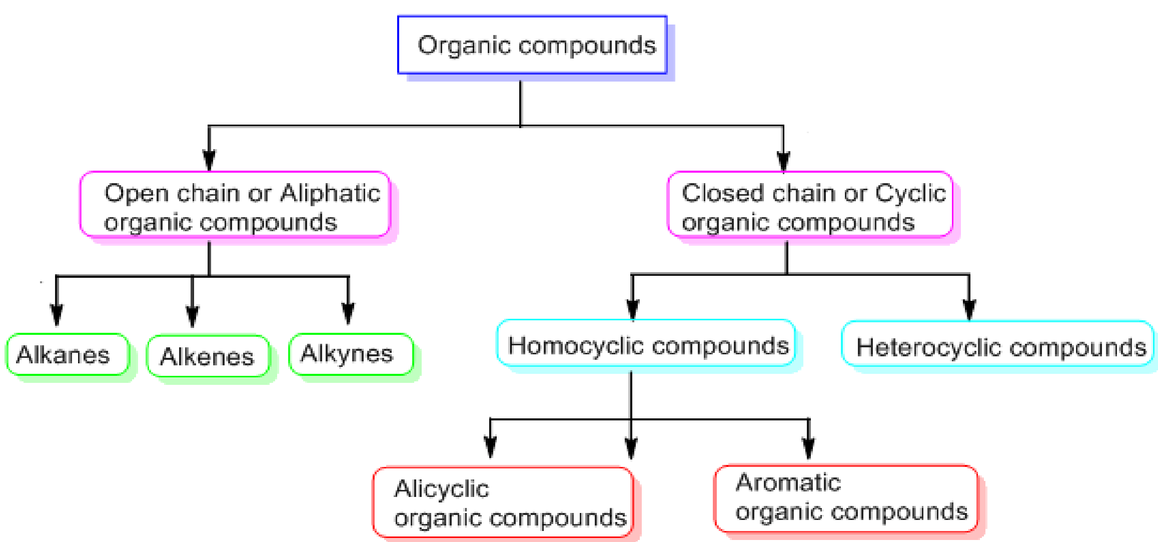
What are the common features in all of the compounds given above?

As you have thought, they have the following common features.

All are cyclic compounds. They are saturated and unsaturated. The major point is; one of the atoms in the ring is different (hetero atom). This makes the compound heterocyclic. The heteroatoms may be either nitrogen (N), sulphur(S), oxygen(O), etc. These compounds may also be aromatic or aliphatic in nature.

Test Yourself!

What is the difference between homocyclic compound and heterocyclic compound?



12.5 Alkyl groups, functional groups and homologous series: Alkyl groups

Activity:

Observe the following chemical equations, discuss in groups in the classroom and draw conclusions what is happening in these equations???

R- H $\xrightarrow{-H}$ R- (alkyl group)
Alkane
CH_3- H $\xrightarrow{-H}$ CH_3- (methyl group)
Methane
CH_3-CH_2- H $\xrightarrow{-H}$ CH_3-CH_2- (ethyl group)
Ethane
CH_3-CH_2- CH_2-H $\xrightarrow{-H}$ CH_3-CH_2-CH_2- (n- propyl group)
Propane
CH_3-CH(H)-CH_3 $\xrightarrow{-H}$ CH_3-CH-CH_3
Propane isopropyl

You may have made some thoughts in your mind. In these equations, one hydrogen atom has been lost from each compound. So, the remaining part is not a complete compound. So, they are called respective groups like methyl group which is derived from methane, ethyl group which is derived from ethane and propyl group which is derived from propane and so on. The group or the radical which is obtained after the removal of a hydrogen atom from an alkane is called alkyl group or alkyl radical. It is denoted by –R or R-. The name of alkyl groups is given by replacing the suffix -ane by –yl in the name of parent alkane.

Functional group

Study the following compounds. All of them have 2 carbon atoms i.e. same carbon number. What makes them different?

CH_3CH_2-OH
CH_3-COOH
CH_3-O-CH_3
CH_3CH_2-NH_2
CH_3-$CONH_2$
CH_3CH_2-NO_2
CH_3-CHO etc.

They do have the same carbon number but differences are –OH, -COOH, -O-, $-NH_2$, -$CONH_2$, $-NO_2$, -CHO. Due to the presence of these groups in these compounds, the properties of the compounds are different.

An atom or group of atoms bonded to the carbon atom of the organic compounds which largely determines the chemical properties of that organic compound is called a functional group.

Organic compounds have two parts. They are carbon-hydrogen skeleton parts (alkyl radical) and functional groups.

Molecular Body $\boxed{R}$ — $\boxed{G}$ Functional group

In methanol, CH_3- is a molecular body and –OH is a functional group.

Molecular Body $\boxed{CH3}$ — $\boxed{OH}$ Functional group

Table of functional group:

Structure	**Name**	**Class or family**	**General representation**
-X	Halo	Haloalkanes /alkyl halides	R-X
-OH	Hydroxyl	Alcohols	R-OH
-O-	Ether	Ethers	R-O-R
-CHO	Formyl/aldo	Aldehydes	RCHO
-CO-	Carbonyl/keto/oxo	Ketones	R-CO-R’
-COOH	Carboxyl	Carboxylic acids	R-COOH
-COX	Acyl halides	Acid halides	R-COX
$-CONH_2$	Amide	Acid amides	R-CO-NH2
-CO.O.CO-	Acid anhydride	Acid anhydrides	R-CO.O.CO-R
-COOR	Ester	Esters	R-COOR’
-N=N-	Azo /diazo	Diazo compounds	R-N=N-R’
$-NO_2$	Nitro	Nitro compounds	R-NO2
$-NH_2$	Amino	Amines	R-NH2
-N=O	Nitroso	Nitroso compounds	R-N=O
$-SO_3H$	Sulphonic	Sulphonic acids	R-SO3H

-C≡N	Cyano/nitrile	Nitriles	R-C≡N
-SCN	Thiocyano	Thiocyanide compounds	R-SCN

Significances of the functional group

1. They are the basis of classification of organic compounds.
2. Nomenclature of the compounds can be done on the basis of functional groups.
3. They possess the characteristic property of the compounds and are the reactive site of the compounds.

Test Yourself!

Write the molecular body and functional group in compounds given above.

Homologous series

Study the following four organic compounds. What is the difference from one compound to another compound in the sequential order?

CH_4	Methane
CH_3-CH_3	Ethane
CH_3-CH_2-CH_3	Propane
CH_3-CH_2-CH_2-CH_3	Butane

What difference did you find?

One member is different by –CH_2 group. Which has the molecular mass of 14 amu. So, ***a regular series of organic compounds having the same functional group and mass difference is 14 amu from one compound to consecutive another compound is called homologous series***. The individual member of a homologous series is called a homologue and the phenomenon is called homology.

Similarly,

Homologous series of alkyl halides:

CH_3Cl	Chloromethane
CH_3CH_2Cl	Chloroethane
$CH_3CH_2CH_2Cl$	Chloropropane
$CH_3CH_2CH_2CH_2Cl$	Chlorobutane

Homologous series of alcohols:

CH_3-OH	Methanol
CH_3CH_2-OH	Ethananol
$CH_3CH_2CH_2$-OH	Propanol
$CH_3CH_2CH_2CH_2$-OH	Butanol

Characteristics of homologous series

1. All the members of a homologous series have similar chemical properties due to the presence of the same functional group.
2. Two consecutive members of the series differ by $-CH_2$ unit. Which is 14 amu.
3. Homologues can be represented by the same general formula. For example, $C_nH_{2n+1}X$ is the general formula of alkyl halides.
4. They can be prepared by the same general methods.
5. The physical properties such as melting point, boiling point, density etc. change gradually from lower member to higher members due to increase in molecular mass.
6. In general, the first member in a series shows unusual behavior due to its smallest size.

Test yourself!

What are the differences between functional groups and homologous series?

12.6 Idea of structural formula, contracted formula and bond line structural formula

Empirical formula

It gives the least details of the compound. It tells about the simplest ratio of the different types of atoms present in the molecules. For example, an organic compound called ethene has the molecular formula CH_2. It means the number of hydrogen is twice the number of carbon atoms i.e. the ratio is 1:2, $C_6H_{12}O_6$ empirical formula of this compound is CH_2O, the ratio is 1:2:1.

Molecular formula

It gives the actual numbers of each type of atom in a molecule. To find the actual molecular formula of the compound we can use the formula

Molecular mass = Empirical formula x n, where n is whole number which can be obtained by dividing molecular mass by empirical formula mass

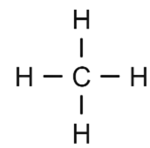
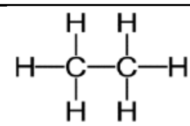
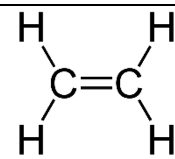
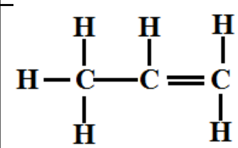
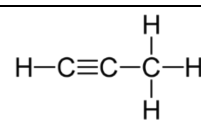
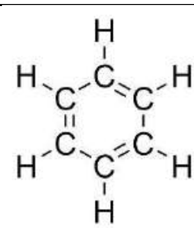
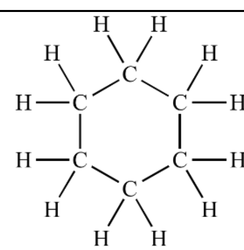
i.e. $n = \frac{molecular\ mass}{empirical\ formula\ mass}$

For example, the molecular formula of propane is C_3H_6 and its empirical formula is CH_2. Molecular mass = 42.

$n = \frac{42}{14} = 3$

Thus, molecular formula = $(CH_2)_3 = C_3H_6$

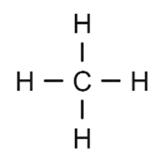
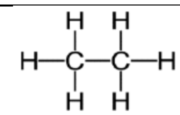
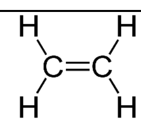
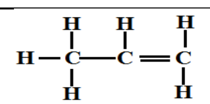
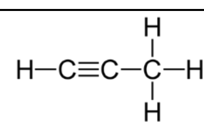
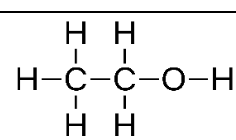
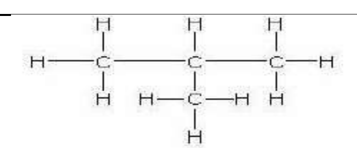
Structural formula

Compounds	Molecular Formula	Molecular Structure
Methane	CH_4	
Ethane	C_2H_6	
Ethene	C_2H_4	
Propene	C_3H_6	
Ethyne	C_2H_2	
Propyne	C_3H_4	
Benzene	C_6H_6	
Cyclohexane	C_6H_{12}	

The structural formula of an organic compound shows every bond between every atom in the molecule. Each bond is represented by a dash line (-). Structural formula is also known as graphic formula.

Elemental analysis is the tool to determine the composition of a compound. It is used for the determination of empirical formulas and then on knowing the molecular mass of the compound. It tells all about the structure of the molecule like the nature of bond, number of atoms bonded, number of carbon atoms.

Contracted formula

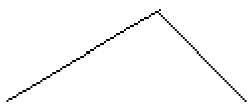
Compounds	Molecular Formula	Molecular Structure	Contracted formula
Methane	CH_4		CH_4
Ethane	C_2H_6		CH_3CH_3
Ethene	C_2H_4		$CH_2 = CH_2$
Propene	C_3H_6		$CH_2 = CHCH_3$
Ethyne	C_2H_2	$H—C \equiv C—H$	$HC \equiv CH$
Propyne	C_3H_4		$HC \equiv CCH_3$
Ethanol	C_2H_6O		CH_3CH_2OH
2-methyl propane	C_4H_{10}		CH3-CH-CH3 CH3

To write structural formulas in chemical equations is impracticable. When these structures are condensed into simpler forms, it is quite convenient. It is also possible to represent an organic molecule without showing any bonds between atoms at all. This is called a condensed or contracted formula. The carbon atoms are grouped with –H atoms bonded directly to it. The bond between these groups are not shown. The branched or substituent groups are shown in brackets after the carbon atoms to which they are attached.

Bond Line Structural Formula

Do you believe both of the following compounds are the same?

CH_3-CH_2-CH_3 Propane

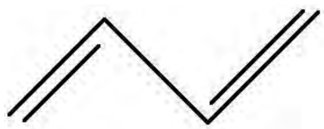


Propane

In the second structure the bond is represented by lines and each vertex and end of the line represents the carbon atoms but no hydrogen atoms are shown. Any unfilled valences on carbon atoms are considered to be filled by hydrogen atoms.

Some examples are shown below:

Butane

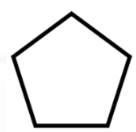


Bond line formation of butane-1,3-diene

Butanoic acid

Octane

Hexan-1,6-dial



Cyclopentane

Test Yourself!

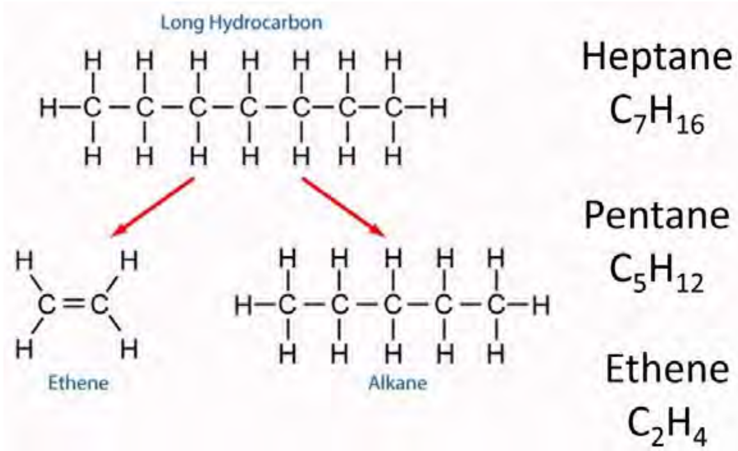
What is the difference between contracted formula and bond line formula?

12.7 Preliminary idea of cracking and reforming, quality of gasoline, octane number, cetane number and gasoline additive

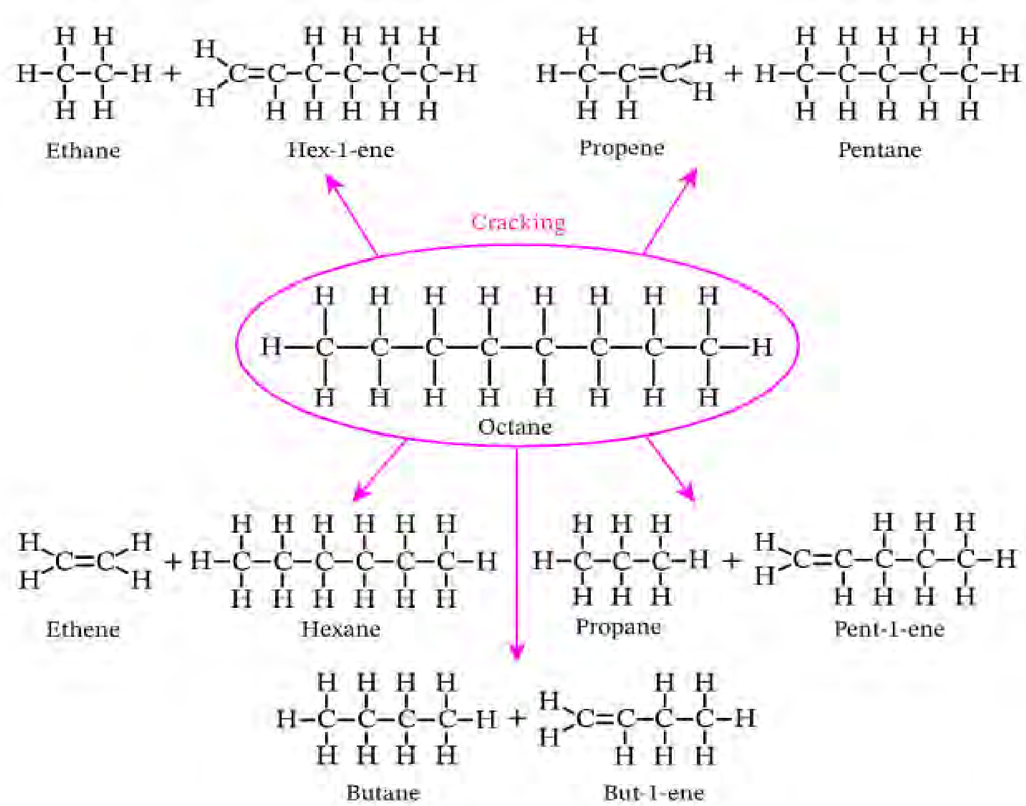
Cracking

Study the following two reactions.

Reaction :1



Reaction: 2



- How do the products form?
- What information did you get from these two reactions?

In above reactions, one larger chain bearing hydrocarbon (petroleum) compound is broken into simpler molecules. This reaction is possible when higher hydrocarbons of petroleum fractions are strongly heated in the absence of air. The reactions occur due to the cleavage of the C-C and C-H bonds.

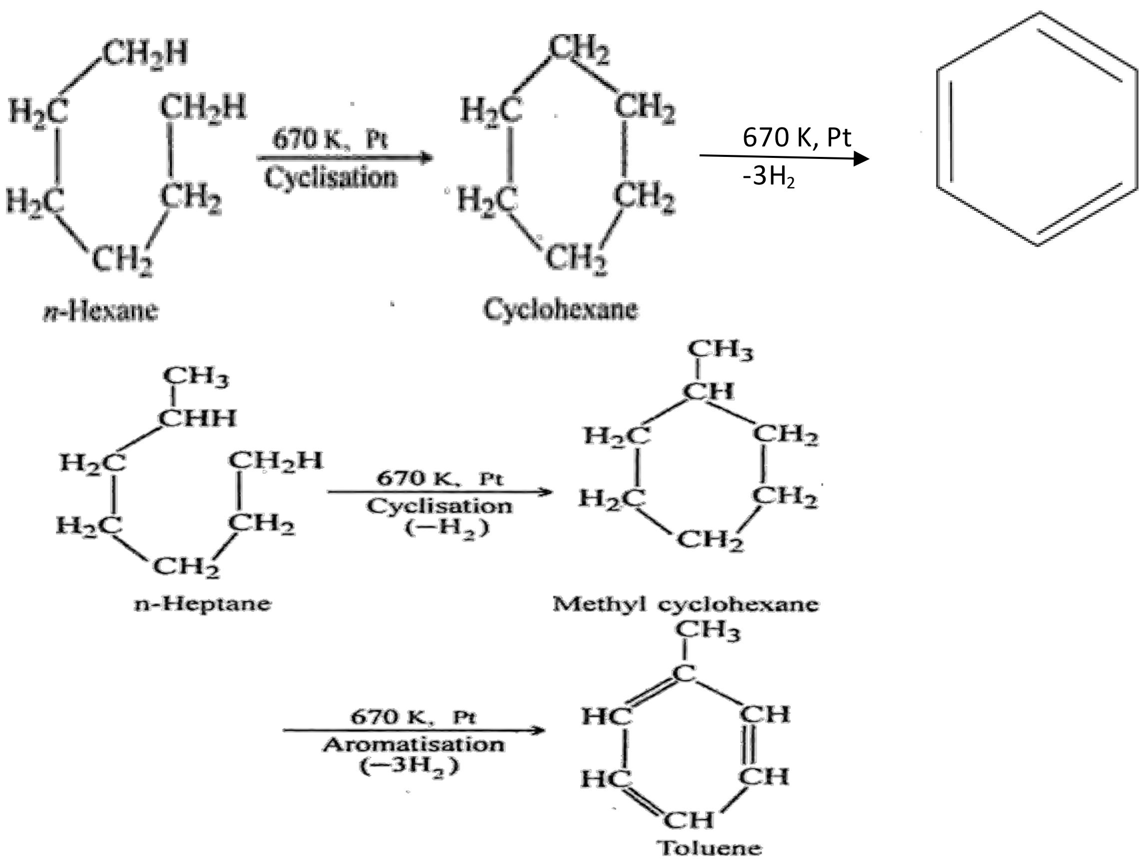
Thermal cracking

The cracking process which is carried out in the presence of heat alone without any catalyst is called thermal cracking or pyrolysis. After this the mixture is cooled immediately. Lower alkenes can be prepared by this method.

Catalytic cracking

It involves the cracking of hydrocarbons in lower temperatures in presence of suitable catalysts. For example, kerosene can be changed into gasoline by catalytic cracking. The mixture of silica and alumina in the ratio 4:1 acts as catalyst during cracking of petroleum.

Reforming or aromatization



How is open chain hydrocarbon changed into closed chain hydrocarbon?

What change has occurred in the following reaction during conversion of open chain hydrocarbon to closed chain hydrocarbon?

The reaction is carried by the bond formed between the terminal carbons by loss of a molecule of hydrogen. Generally, aliphatic hydrocarbons of petroleum (C-6 to C-8) are changed into aromatic hydrocarbons which are used as starting materials for the synthesis of organic compounds. So, the reforming process is defined as a process by which straight chain alkanes and cyclic alkanes are converted into the corresponding aromatic hydrocarbons under suitable condition and catalyst. Platinum is the best catalyst for this process. So, the process is also known as platforming.

Test Yourself!

What is cracking?

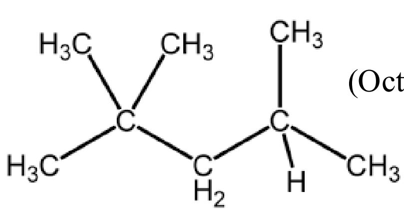
Quality of Gasoline

Have you ever heard unwanted sounds produced from bikes, buses, scooters, etc.?

The unwanted sound produced from these vehicles is call knocking sound. The efficiency of the engine depends upon the extent to which the fuel-air mixture is compressed in the time of ignition. The greater the compression, greater the efficiency. Increase in compression of gasoline beyond a certain point results in the sudden and irregular burning of fuel mixture which causes jerk against the piston resulting in an unwanted sound called knocking sound. Knocking reduces the efficiency of engine. Such gasolines are considered low quality gasolines. The maximum compression of fuel that can be attained without any knocking depends on the quality of gasoline. This is known as the anti-knocking property of the fuel. To enhance the quality of fuel, compounds having anti knocking properties should be mixed.

Octane number (n)

CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3 (octane number =0)



(Octane number =100)

Octane number is the measurement of quality of the fuel. By the experiment it is found that the octane number of n-heptane is 0 i.e. it knocks maximum when used as fuel. On the other hand, iso-octane (2, 2, 4 -trimethyl pentane) is considered the maximum anti knocking property i.e. octane number of it is 100.

Rest of the gasoline is graded at between these two values (0-100). Octane number is an arbitrary scale based on n-heptane and isooctane. **It is defined as the percentage by volume of isooctane in the mixture of isooctane and n-heptane which has the same anti-knocking property as the fuel under examination.** For example, if a petrol has octane number 85, it means that the anti-knocking property of that petrol is equal to a sample which is a mixture of 85% isooctane and 15% n-heptane.

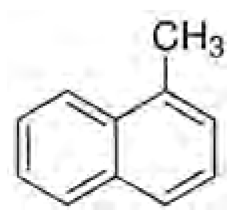
Octane number depends on the structure of the compound;

a. Octane number of straight chain hydrocarbons is low.
b. Cyclo compounds have comparatively higher octane numbers as compared to the corresponding straight chain hydrocarbons.
c. Branching of the chain increases the octane number.
d. Aromatic compounds have a very high value of octane.
e. The compounds having multiple bonds like double or triple have greater octane number.

Cetane number:

$CH_3(CH_2)_{14}CH_3$

Cetane (hexadecane)
(Cetane No. = 100)



methyl naphthalene
(Cetane No. = 0)

As like octane number, it also represents the quality of diesel fuels. It is the expression of quality of diesel fuel as compared with the percentage of the cetane in the mixture of cetane and alpha methyl naphthalene which has the same ignition quality of the diesel sample under examination. Diesel with cetane number 70 has the same ignition properties as a mixture of 70% cetane and 30% alpha methyl naphthalene. It is also known as cetane rating.

Gasoline additives

$(C_2H_5)_4Pb \rightarrow Pb\bullet + 4C_2H_5\bullet$

$CH_2Br\text{-}CH_2Br + Pb\bullet \rightarrow CH_3\text{-}CH{=}CH_2 + PbBr_2$ (volatile)

Knocking sound of the gasoline can be reduced by adding some reagents. These reagents which help to improve the quality of gasoline are called anti-knocking agents or gasoline additives. In the cylinder of the internal combustion engine TEL is decomposed to ethyl radical which is combined with the straight chain hydrocarbon of gasoline to form branched chain hydrocarbons.

Due to which octane number is increased. For example, tetraethyl lead (TEL) [$(C_2H_5)_4Pb$]. TEL is a commonly used anti-knocking agent which is mixed very small amounts (0.01%) with gasoline or petrol. The petrol with TEL is called leaded petrol. To exhaust the lead in the engine ethylene dibromide is used which makes lead dibromide which is volatile and can be escaped from the engine. The use of leaded petrol is banned due to the toxic nature of lead which causes environmental pollution.

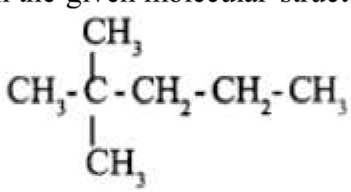
Test Yourself!

Is octane number similar to cetane number?

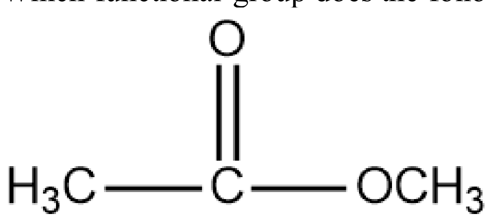
Exercise

Multiple choice questions

1. How many electrons of the valence shell of carbon participate to make the bond?
   a. 2
   b. 3
   c. 4
   d. 5
2. Two methyl groups make a molecule of ethane. What type of property does the condition explain?
   a. Combination
   b. Catenation
   c. Alkylation
   d. methylation
3. In the given molecular structure, how many secondary carbons are there?



   a. 2
   b. 3
   c. 4
   d. 5
4. Which functional group does the following compound consist of?



a. Carboxylic acid
b. Ketone
c. Aldehyde
d. Ester

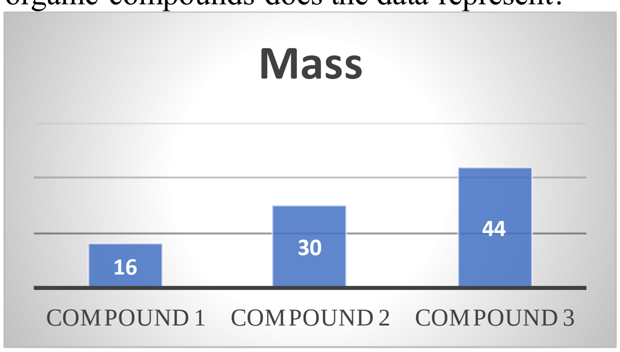
5. What is the relation between these two compounds?

$CH_3—O—CH_3$ $CH_3CH_2\text{-}O\text{-}CH_3$

Compound 1 Compound 2

a. Both of them have same alkyl groups.
b. Compound 1 has molecular mass 46 but compound 2 has molecular mass 58 amu.
c. Compound 1 has molecular mass 46 but compound 2 has molecular mass 58 amu.
d. Compound 1 and compound 2 are homologues.

6. Molecular mass of three organic compounds are given as follows. Which property of organic compounds does the data represent?

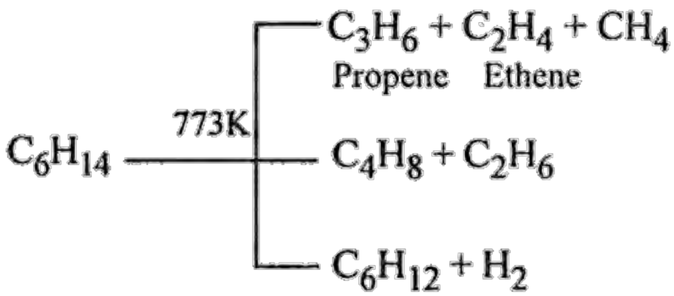


a. Compound 1 is alkane compound 2 and 3 are alcohols.
b. Compound 1 is alcohol and compound 2 and 3 are alkane.
c. Compound 2 is alkane, compound 1 and 3 are alkenes.
d. Compound 2 and 3 are homologues of compound 1.

7. What do you understand by octane number 75?
a. The organic compound having 75 carbon atoms
b. The organic compound having perfect quality of iso-octane
c. The organic compound has the mixture of 75% iso-octane and 25% n-heptane.
d. The organic compound having 75% pure

8. Which of the following is used as an anti-knocking agent?
a. Lead acetate
b. Ethyl acetate
c. Tetraethyl lead
d. Platinum tetrachloride

9. What does the following reaction indicate?



a. Decomposition reaction
b. Fusion reaction
c. Reforming reaction
d. Cracking reaction

10. What is the octane number of n-heptane?
a. 0
b. 70
c. 80
d. 100

11. Who proposed the vital force theory?
a. Berzelius
b. Wohler
c. Kolbe
d. Kekule

12. Which one is the first synthesized organic compound in the laboratory?
a. Methane
b. Formaldehyde
c. Formic acid
d. urea

13. What will be the molecular formula of the organic compound having empirical formula CH_2O and molecular weight 90?
a. $C_3H_6O_3$
b. $C_6H_{12}O_6$
c. $C_4H_8O_3$
d. $C_2H_6O_2$

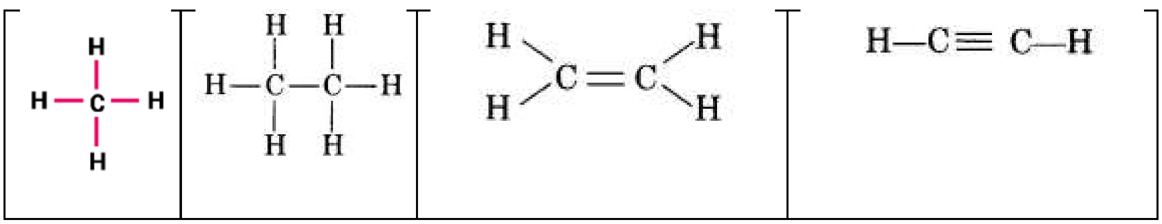
14. A motorbike is producing some explosive sound while running on the road. What could be the reason for such a sound?
a. Due to rough road condition
b. Due to high speed of the bike
c. Due to high volume of petrol in the tank
d. Due to low quality petrol

15. Which of the following compounds is used as an anti-knocking reagent?
a. Iso-octane
b. Iso-heptane

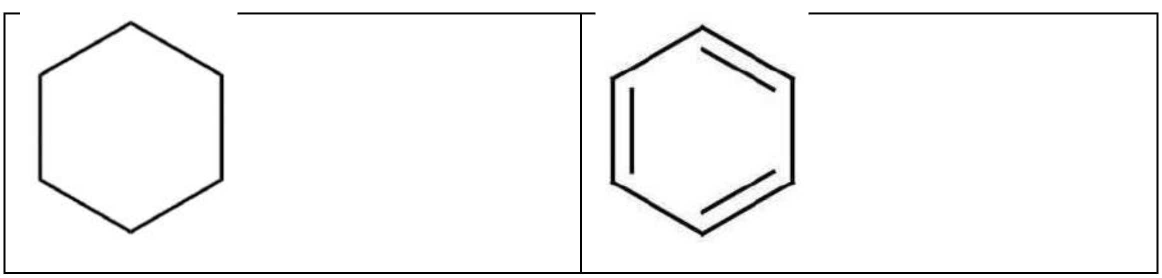
c. Iso-pentane
d. Iso-butane

Short answer questions

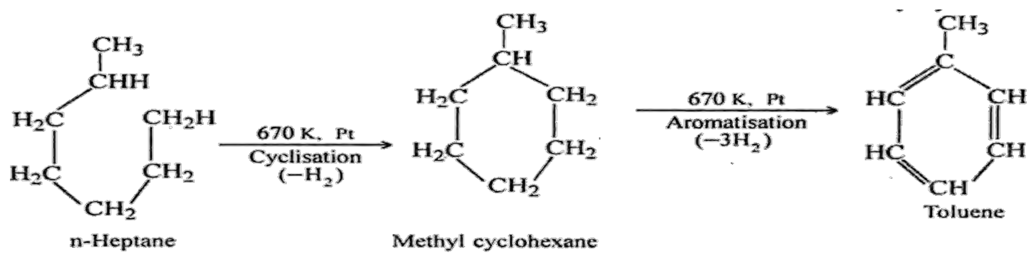
1. Why is it necessary to study organic compounds separately? Explain with five suitable reasons.
2. Write the characteristics of homologous series.
3. Differentiate between;
   a. Ethene and ethane
   b. Propyne and propane
4. How is organic compound originated? Explain with the help of vital force theory.
5. How was urea obtained for the first time?
6. Write any five differences between organic and inorganic compounds.
7. The following compounds have a common characteristic of carbon. Which characteristic of carbon do these compounds represent? Explain about the bonding of carbon in each compound of the following:



8. How do we classify organic compounds? Explain the classification process with the help of a chart.
9. Structures of two compounds have been given below. Are these two compounds different? If so, in which respect these compounds are different? Find any four differences in them.



10. Write first five members of alcohol. Also find the difference in mass of two consecutive members.
11. Why does HCOOH have a unique property in comparison with other members of carboxylic acid? Explain. Write any three characteristics of homologous series.
12. Write the structural formula of the C_6H_6, C_6H_{12}, C_3H_7OH. How is this formula different from the contracted formula? Explain.
13. Which process is represented by the following reaction? How is it different from cracking?



14. Explain about octane number and cetane number with examples. How is knocking property different from anti-knocking property?
15. How has the octane number of fuel increased? A mixture of n-heptane and iso-octane contains 40% n-heptane. What is the octane number of the mixture? Write any two differences between cracking and reforming.
16. What is vital force theory? How was this theory discarded? Explain.

Long answer question

1. Write the answer to the following questions.
   A. Explain about tetracovalency of carbon with an example.
   B. In which respect knocking sound is different from antiknocking sound?
   C. $HO-CH_3$, is an organic compound which part is an active site?
2. An example of homologous series is given below.

HCOOH
X
CH_3CH_2COOH

   a. Write the contracted formula of X.
   b. Which of them has the highest melting point and boiling point? Give a reason.
   c. What is the molecular mass and IUPAC name of X?
   d. What would be the IUPAC name of the fourth compound of the series?

3. A motorbike rider is hearing a more explosive sound nowadays. What could be the reason for it? What suggestions do you give for him to minimize the sound that he is hearing from the silencer of the motorbike? Is there any problem with the engine of my motorbike? Explain. How does tetraethyl lead work to reduce explosive sound?

4. Write the compounds according to the information given in the table.

Number of carbon atoms	Compounds of -OH	Compounds of $-NH_2$
2	………………	…………………

3	………………….	……………….
4	…………………..	………………….
5	…………………….	…………………….

5. Are octane numbers and cetane numbers comparable? Write the similarities and dissimilarities of them in any three points of each.

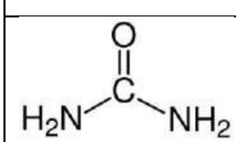
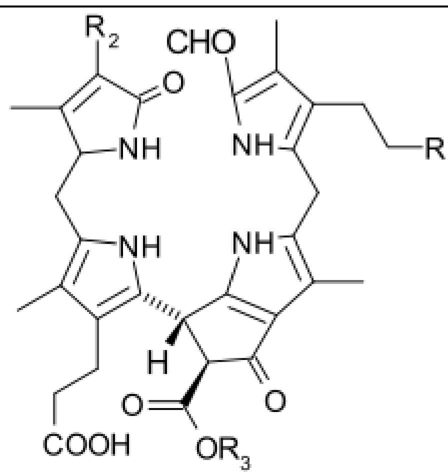
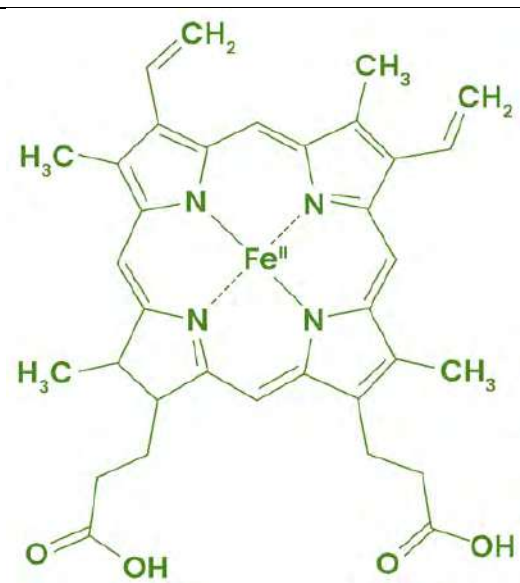
6. An example of homologous series is given below.

HCHO
X
CH_3CH_2CHO

   a. Write the contracted formula of X.
   b. Which of them has the highest melting point and boiling point? Give a reason.
   c. Why is the HCHO has a unique nature than that of the second and third compound?
   d. What is the molecular mass and IUPAC name of X?

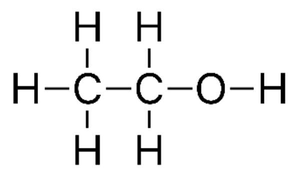
7. Make a chart of classification of organic compounds and give an example of each class.

8. Write the functional group(s) in the following compounds?

Urea	Chlorophyll	Hemoglobin
		

Project work

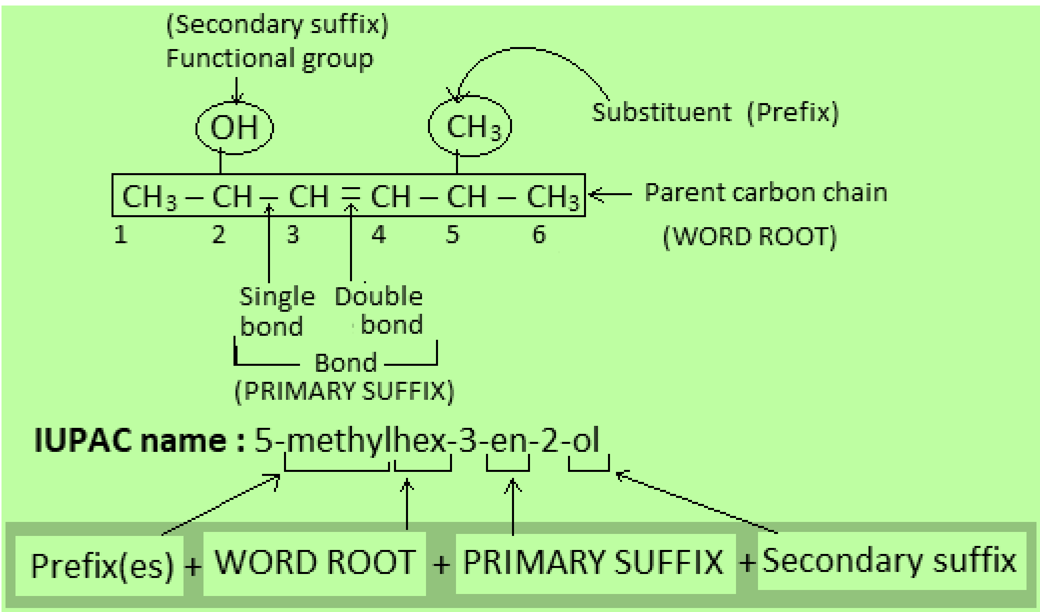
Take a chart paper. Make a list of any 7 functional groups. Add ethyl groups in each functional group. Write their structural formula as given in the example and submit it to your teacher.

SN	Functional Group	Compounds
1.	-OH	
2.		
3.		
4.		
5.		
6.		
7.		
8.		

UNIT 13: FUNDAMENTAL PRINCIPLE OF ORGANIC CHEMISTRY

IUPAC Nomenclature of organic compounds (upto chain having 6-carbon atoms)

Study the different parts of an organic compound in the box given below. There are different parts in the organic compounds. You may find prefix, word root, primary suffix and secondary suffix in the compounds.



To make the same understanding for the same compound in the world it was necessary to develop a common way of writing the name of an organic compound. With the same motif, in 1957 a system of writing the name was evolved. Which is called the IUPAC naming system i.e. commonly called IUPAC nomenclature of the organic compound.

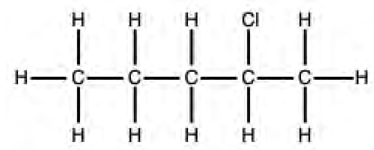
In this system the whole organic compound is divided into following parts:

Word root
Suffix
Prefix

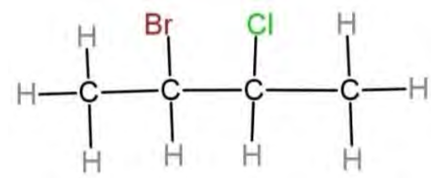
a. Word root

Activity:

Two compounds are given below. Count the number of carbon atoms in these two compounds and write in the following table.



Compound A



Compound B

	Compounds	Compound A	Compound B
	No. of carbon atoms	……..	……

Various chains have different word roots and different names.

The representative name of the main chain (longest chain) of the organic compound is called word root. It is also known as the parent chain of the organic compound. ***Longest chain having the maximum number of functional groups or multiple bonds is called parent chain or word root.*** Different chains of the organic compounds have different word roots. On the basis of the number of carbon atoms word roots are named. The word to be used in the different word roots are given below.

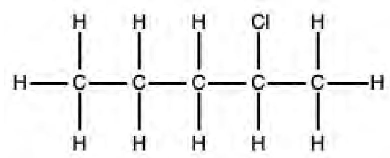
Number of the carbon atoms	Name of word root
C1	Meth
C2	Eth
C3	Prop
C4	But
C5	Pent
C6	Hex
C7	Hept
C8	Oct
C9	Non
C10	Dec
C11	Undec
C12	Dodec

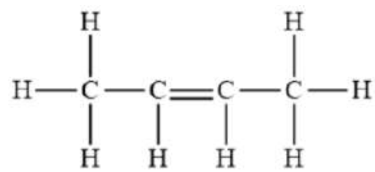
b. Suffix

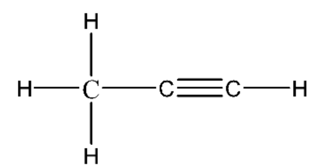
The word root is linked to the suffix which may be primary or secondary or both.

i. Primary suffix

Observe the nature of the bonds between **carbon and carbon atoms** in these three compounds.







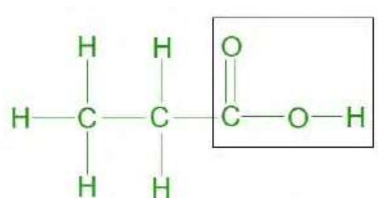
- First compound has a single bond between all carbon atoms.
- Second compound has one double bond between carbon atoms.
- Third compound has one triple bond between carbon atoms.

Primary suffix helps to identify whether the compounds are either saturated or unsaturated.

Carbon chain	Primary suffix
C- C	-ane (saturated)
C=C	-ene (unsaturated)
C≡ C	-yne (unsaturated)

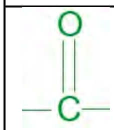
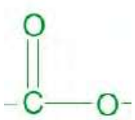
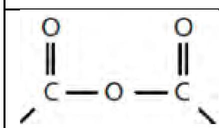
So, the first compound is saturated and the second and third compounds are unsaturated compounds. So, the primary suffix for the first compound is -ane and the second compound is -ene and the third compound is -yne. Primary suffix is written immediately after the word root. It indicates the nature of linkage in the carbon chain.

ii. Secondary suffix



What does the bounded part indicate in the structural formula given above?

In this compound the enclosed part is the group of atoms that determines the properties of the compound. It means secondary suffix in the IUPAC name indicates the functional group which is written after the primary suffix. It is written after the primary suffix. Some secondary suffixes i.e. functional groups are given below:

Functional groups	**Secondary suffix**
-CHO	-al
-OH	-ol
 or –CO-	-one
-COOH	-oic acid
 Or –COO-	-oate
$-CONH_2$	-amide
-CN	-nitrile
-COX	-oyl halide
	- Oic anhydride

Examples:

CH_4

There is no substituent, so no prefix. There is one carbon atom word root is meth. There is only one carbon atom which means single bond at carbon, primary suffix is ane. In this compound there are no functional groups.

So IUPAC name of the compound is meth+ ane = methane.

CH_3-CH_2-COOH

Primary suffix.- ane; all bonds are single. Secondary suffix- oic acid because the functional group is carboxylic acid.

So IUPAC name of the compound is; 3-chloro+prop+ane+oic acid= 3-chloropropanoic acid

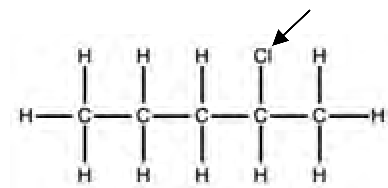
IUPAC names of some compounds are given below.

Compound	Word root	Primary suffix	Secondary suffix	IUPAC name
CH_3-CH_3	eth	ane	-	Ethane

CH_3-CH_2-CH_3	prop	ane	-	Propane
CH_3-CH_2-COOH	prop	ane	oic acid	Propanoic acid
CH_3-CH_2-NH_2	eth	ane	amine	Ethanamine
CH_3-CH_2-OH	eth	ane	ol	2-bromoethanol
CH_3-$(CH_2)_4$-CHO	hex	ane	al	hexanal

a. Prefix

Where do we write for the chlorine atom in the given compound?

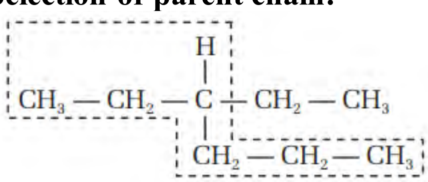


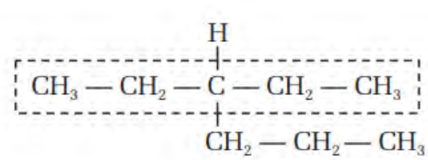
The word which represents the atom or the group of atoms bonded with the main chain of the organic compound is called prefix. In the example given above, chloro group is a substituent and written in the form of prefix i.e. 2- chloro. Some of the substituents are given as follows. In the following compound, chlrorine is substituent because it is bonded with the carbon of the main chain.

Substituents /functional group	**Prefix**
-X(Cl, Br, I)	Halo (chloro, bromo, iodo)
-CN	Cyano
-OR	Alkoxy
=N-	azo
-N=N-	diazo
-NO_2	Nitro
-NH_2	Amino

A. General rules for IUPAC nomenclature of hydrocarbons (alkanes)

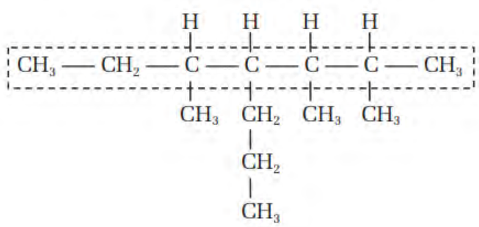
1. Selection of parent chain:

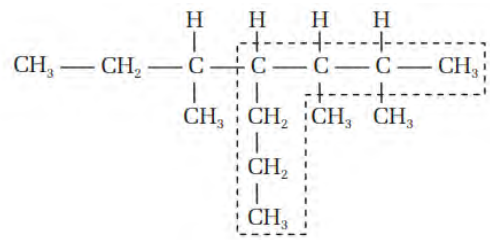


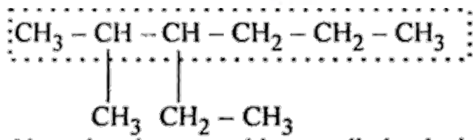


Which one compound has the longest chain?

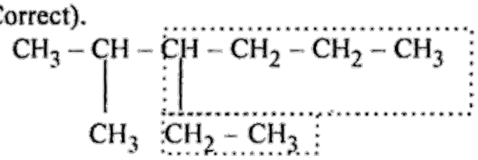
While writing IUPAC name of alkanes, select the longest continuous chain of carbon atoms in the molecule, containing the large number of carbon atoms as parent chain or principal chain or main chain. Carbon atoms which are not included in the parent chain are identified as substituents or branched chains. Longest chain may or may not be straight but must be continuous.





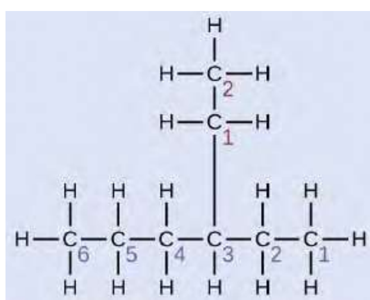


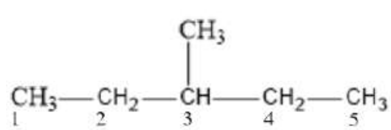
Named as hexane with two alkyl substituents (Correct).



If two different chains of equal length are possible, the chain with maximum number of side chains of alkyl substituent is selected as the parent chain.

2. **Lowest Locant Rule:**

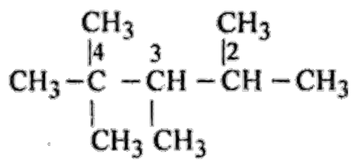




What is the difference in the numbering process in these two compounds?

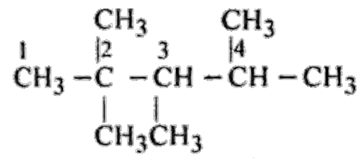
If the given alkane contains only one branch or substituents in the parent chain, the numbering of carbon atoms is done from the nearest terminal carbon of the substituent. Locant is a carbon number that locates the position of substituent.

3. **Lowest sum rule:**



Sum of locants = 2+3+4+4=13

Wrong



Sum of locants = 2+2+3+4=11

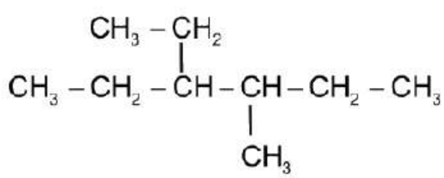
Correct

2, 2, 3, 4-tetramethyl pentane

If the parent chain has two or more than two substituents, numbering is done from the end of the parent chain by which the sum of the locants is minimum.

While naming the compound, the position of substituents or side chain is indicated by the number of the carbon atom to which it is attached.

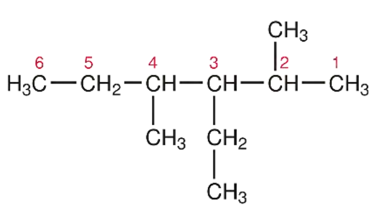
4. Alphabetical order of the substituenet

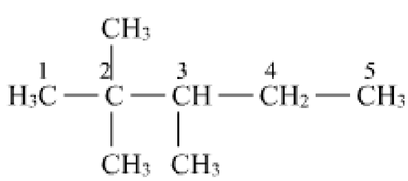


3-ethyl-4-methyl hexane

If two different substituents are located in the same positional number (equivalent position) from the terminal ends of parent chain, the lower number is given to the substituent which comes first in alphabetical order.

5. Similar group in multiple places



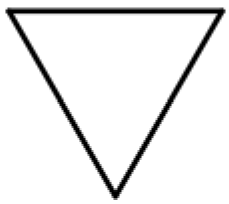


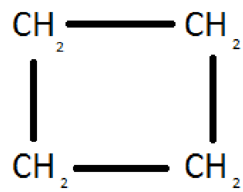
What would be the correct IUPAC name of the compounds given above?

If the same substituent or side chain comes more than once, the prefixes di, tri tetra are added to the name of the substituents. The position of the substituents is written separately, and positional numbers are separated by commas.

If the same substituent or side chain comes more than once, the prefixes di, tri tetra are added to the name of the substituents. The position of the substituents is written separately, and positional numbers are separated by commas.

6. **Cyclic hydrocarbons:**





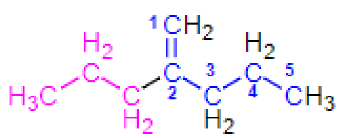
What could be the IUPAC name of these compounds? Any guesses?

Saturated hydrocarbons with rings of carbon atoms in the molecule are called cycloalkanes.

For cyclic compounds, nomenclature is done by prefixing cyclo to the name of the corresponding straight chain alkanes. So, the IUPAC names of the first and second compounds are cyclopropane and cyclobutane respectively.

B. IUPAC name of alkenes and alkynes

1.



Select the longest continuous carbon chain containing the carbon atoms involved in the multiple bond (double and triple) as parent chain. While writing the name of the alkene and alkyne, the suffix 'ene' and 'yne' respectively, are used.

2. The numbering of C-atoms in the parent chain is done in such a way that the carbon atom carrying the double or triple bond gets the lowest possible number. The lower number is assigned to indicate the position of double or triple bond before the suffix -ene or –yne.

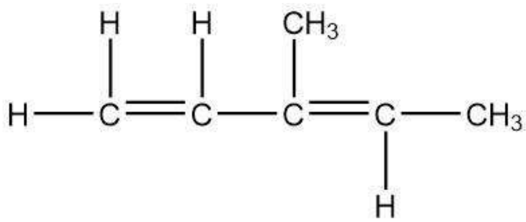
3. If multiple bonds are in equal distance from the either ends of the chain, numbering is done from the ends by which other substituent get the lowest possible number.

4. If the parent chain has more than one double or triple bonds, numbering is done from the ends of the carbon chain by which the sum of the locant has the lowest possible number (Lowest sum rule).

5. If both the double bond and triple bond are located in the same locant number from either end, double bond gets priority over the triple bond. Double bond receives the lowest number.

6. If the compound contains both double and triple bonds in the parent chain, the final name of the compound is given as the derivative of alkynes rather than alkene i.e., the last suffix ends at -yne.

7. If the multiple bond comes two or more times in the chain, then it is named as diene or diyne, triene or triyne, etc. The extra letter 'a' is written at the end of the word root if the primary suffix begins with a consonant as – di, tri, – tetra, etc.

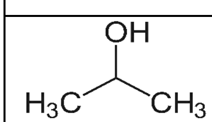
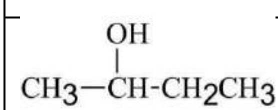


C. IUPAC name of organic compounds having one functional group (mono-functional compounds

Alcohol

Functional group –OH
General formula $C_nH_{(2n+1)}$ OH
General representation: ROH
IUPAC name: Alkane-e+ol = Alkanol
Note: select the longest chain containing –OH group as parent chain and counting of carbon is done from the nearest terminal of the –OH group.
The position of –OH group is indicated by Arabic numerals before the suffix –ol.
If two or three –OH groups are present in the same molecule diol or triol is written along with their position.

Example:

Compounds	Common Name	IUPAC Name
CH_3OH	Methyl alcohol	Methanol
CH_3CH_2OH	Ethyl alcohol	Ethanol
$CH_3CH_2CH_2OH$	Propyl alcohol	Propanol
	Isopropyl alcohol	Propan-2-ol
$CH_3CH_2CH_2CH_2OH$	Butyl alcohol	Butan-1-ol
	Isobutyl alcohol	Butan-2-ol

Aldehyde

Functional group –CHO

General formula $C_nH_{2n}O$

General representation: RCHO

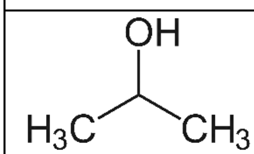
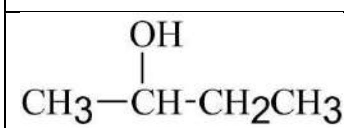
IUPAC name: Alkane-e+al= Alkanal

Note: select the longest chain containing –CHO group as parent chain and counting of carbon is done from the nearest terminal of the –CHO group.

The position of –CHO group is indicated by Arabic numerals before the suffix –al.

If two –CHO groups are present in the same molecule, the dial is written along with their position.

Example:

Compounds	Common Name	IUPAC Name
HCHO	Formaldehyde	Methanal
CH_3CHO	Acetaldehyde	Ethanal
CH_3CH_2CHO	Propyl aldehyde	Propanal
	Isopropyl alcohol	Propan-2-ol
$CH_3CH_2CH_2CH_2OH$	Butyl alcohol	Butan-1-ol
	Isobutyl alcohol	Butan-2-ol

Ketone

Functional group –CO-

General formula $C_nH_{2n}O$

General representation: RCOR’

IUPAC name: Alkane-e+one= Alkanone

Note: select the longest chain containing –CO- group as parent chain and counting of carbon is done from the nearest terminal of the –CO- group.

The position of –CO- group is indicated by Arabic numerals before the suffix –one.

If two or three – CO- groups are present in the same molecule dione or trione is written along with their position.

Example:

Compounds	**Common Name**	**IUPAC Name**
CH_3COCH_3	Acetone	Propan-2-one
$CH_3COCH_2CH_3$	Butyl ketone	Butan-2-one
$CH_3CH_2COCH_2CH_3$	Pentyl ketone	Pentan-3-one

Carboxylic acid

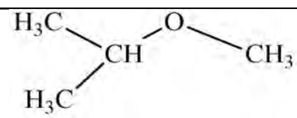
Functional group –COOH
General formula: $C_nH_{(2n+1)}$ COOH
General representation: RCOOH
IUPAC name: Alkane-e+oic acid= Alkanoic acid
Note: select the longest chain containing –COOH group as parent chain and counting of carbon is done from the nearest terminal of the –COOH group.
The position of –COOH group is indicated by Arabic numerals before the suffix –oic acid.
If two or three –COOH groups are present in the same molecule, dioic or trioic is written along with their position.
Example:

Compounds	**Common Name**	**IUPAC Name**
HCOOH	Formic acid	Methanoic acid
CH_3COOH	Acetic acid	Ethanoic acid
CH_3CH_2COOH	Propanoic acid	Propanoic acid
$CH_3CH_2CH_2COOH$	Butyric acid	Butanoic acid
$\begin{array}{c}COOH\\ \vert \\ COOH\end{array}$	Oxalic acid	Ethan-1,2-dioic acid

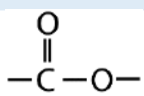
Ether

Functional group –O-
General formula $C_nH_{(2n+2)}O$
General representation: ROR'
IUPAC name: Alkoxy+alkane= Alkoxy alkane

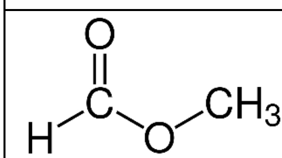
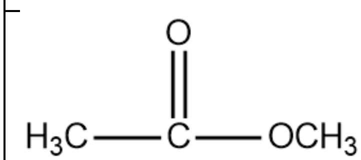
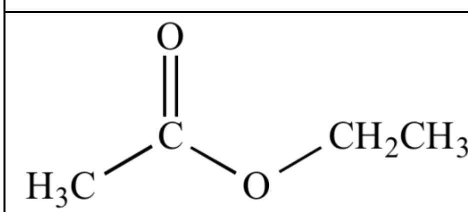
Note: select the longest chain containing –O- group as parent chain and counting of carbon is done from the nearest terminal of the –O- group.
The position of –OH group is indicated by Arabic numerals before the suffix –ol.
If two or three –OH groups are present in the same molecule diol or triol is written along with their position.
Example:

Compounds	**Common Name**	**IUPAC Name**
CH3-O-CH3	Dimethyl ether	Methoxy methane
CH_3-O-CH2CH3	Ethyl methyl ether	Methoxy ethane
CH_3CH_2-O-CH2CH3	Diethyl ether	Ethoxy ethane
	Isopropyl methyl ether	2-methoxy propane

Ester

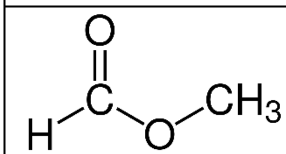
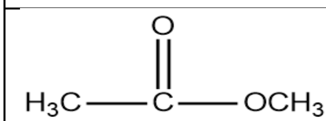
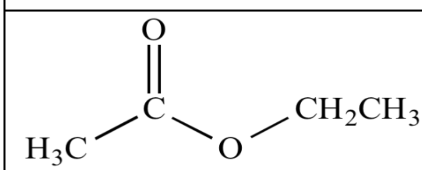
Functional group: 
General formula $C_nH_{(2n)}O_2$
General representation: ROH
IUPAC name: Alkyl alkane-e+oate= Alkyl alkanoate

Example:

Compounds	**Common Name**	**IUPAC Name**
	Dimethyl ester	Methyl methanoate
	Ethyl methyl ether	Methoxy ethane
	Diethyl ether	Ethoxy ethane

Amide

Functional group –OH
General formula $C_nH_{(2n+1)}OH$
General representation: ROH
IUPAC name: Alkane-e+ol= Alkanol
Note: select the longest chain containing –OH group as parent chain and counting of carbon is done from the nearest terminal of the –OH group.
The position of –OH group is indicated by Arabic numerals before the suffix –ol.
If two or three –OH groups are present in the same molecule diol or triol is written along with their position.
Example:

Compounds	**Common Name**	**IUPAC Name**
	Dimethyl ester	Methyl methanoate
	Ethyl methyl ester	Methyl ethanoate
	Diethyl ester	Ethyl ethanoate

Acyl halides (acid halides)

Functional group: –COX
General formula: RCOX
IUPAC name: Alkane-e+oyl halide = Alkanoyl halide
Example:

Compounds	**Common Name**	**IUPAC Name**
HCOCl	Formyl chloride	Methanoyl chloride
CH_3COCl	Acetyl chloride	Ethanoyl chloride
CH_3CH_2COCl	Propanoyl chloride	Propanoyl chloride
C6H5COCl	Benzoyl chloride	

Amine

Functional group $-NH_2$

General formula: $R\text{-}NH_2$ or $C_6H_5\text{-}NH_2$

IUPAC name: Substituent + Alkane-e + amine= Alkanamine or aryl amine

Note: select the longest chain containing $-NH_2$ group as parent chain and counting of carbon is done from the nearest terminal of the $-NH_2$ group.

The position of $-NH_2$ group is indicated by Arabic numerals before the suffix –amine.

Example:

Compounds	Common Name	IUPAC Name
$CH_3\text{-}NH_2$	Methyl amine	Methanamine/aminomethane
$CH_3\text{-}CH_2\text{-}NH_2$	Ethyl amine	Ethanamine/aminoethane
$CH_3\text{-}CH_2\text{-}CH_2\text{-}NH_2$	Propyl amine	Propanamine/aminopropane
$CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2$	Butyl amine	Butanamine/aminobutane

Acid anhydride

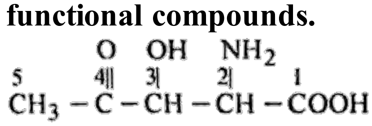
Functional group: –CO.O.CO-

General formula: R-CO.O.CO-R

IUPAC name: Alkane- e + oic anhydride = Alkanoic anhydride

Compounds	Common Name	IUPAC Name
H-CO.O.CO-H	Formic anhydride	Methanoic anhydride
$CH_3CO.O.CO.CH_3$	Acetic anhydride	Ethanoic anhydride
$CH_3\text{-}CH_2\text{-}CO.O.CO.CH_2CH_3$	Propanoic anhydride	Propanoic anhydride

D. IUPAC name of organic compounds having two or more functional group (poly-functional compounds.



What are the functional groups present in the given organic compound?

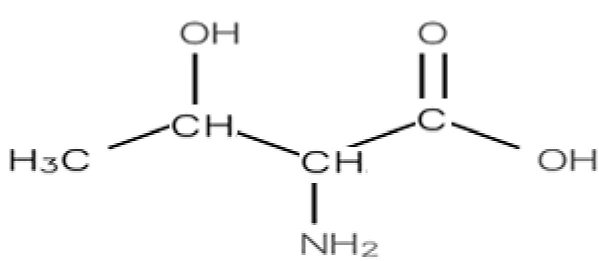
In poly-functional compounds, one of the functional groups is regarded as principal functional group and the remaining group are considered as subsidiary/ auxiliary functional groups (substituents) which are denoted by prefixes. The principal functional group is selected based on the following priority order.

Carboxylic (-COOH) > sulphonic (-SO_3H) > acid anhydride (-CO.CO.CO-) > Ester (-COOR) > acid halide (-COX) > amide (-$CONH_2$) > nitriles (-CN) > aldehyde (-CHO) > ketone (-CO-) > alcohol (-OH) > amine (–NH_2) > aryl (-Ar) > alkyne (-C≡ C-) > alkene (-CH=CH-)

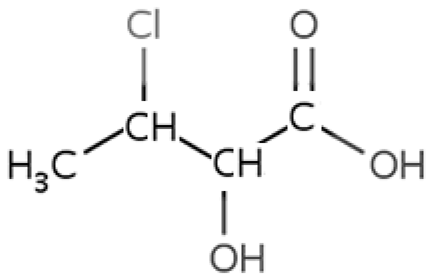
Prefix name of subsidiary/ auxiliary functional group:

Secondary functional group	Prefix name
-OH	Hydroxy
-NH_2	Amino
-NHR	Alkyl amino
-NR_2	Dialkyl amino
-CHO	Formyl
-CO-	Keto or oxo
-COOH	Carboxy
-COOR	Alkoxy carbonyl
-CN	Cyano
-$CONH_2$	Carboxamido
-SO_3H	Sulphonic acid

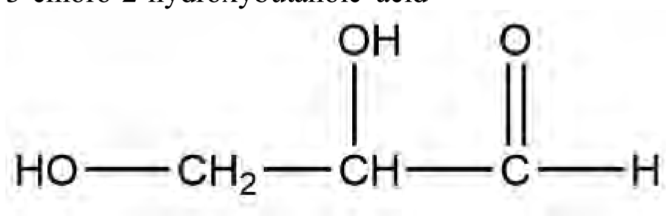
Examples:



3-hydroxy-2-aminobutanoic acid



3-chloro-2-hydroxybutanoic acid



2,3-dihydroxypropanal (Glyceraldehyde)

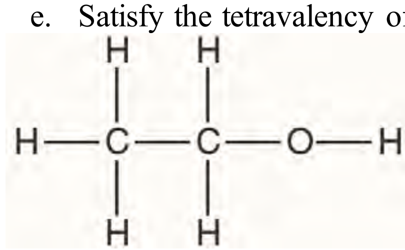
E. Structural formula of organic compounds from IUPAC name

Activity

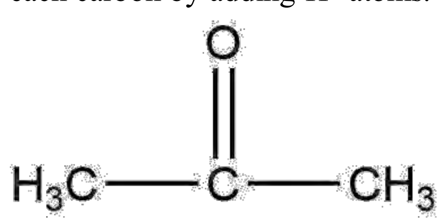
IUPAC name of an organic compound is given below. On the basis of the guidelines draw the structural formula.

2,3-dimethylbut-2-ene-1-ol

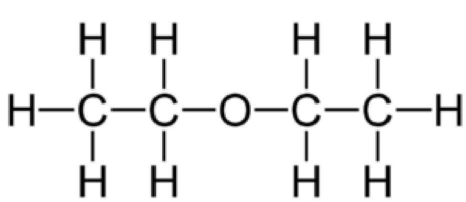
a. Construct the C-C chain (parent chain) from the word root of the given compound.
b. Insert multiple bonds (double or triple by using primary suffix).
c. Add a functional group in the parent chain in the appropriate position.
d. Add substituents given in parent chain.
e. Satisfy the tetravalency of each carbon by adding H- atoms.



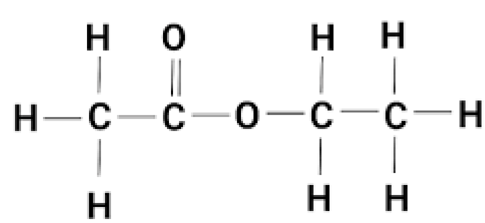
Ethanol



Propan-2-one



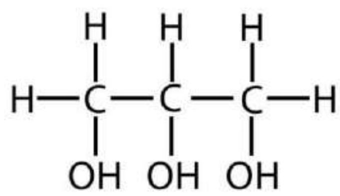
Ethoxy Ethane

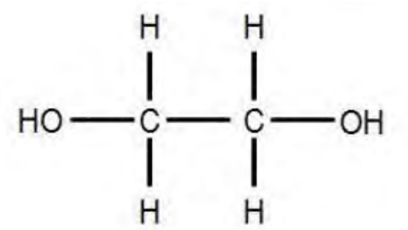


Ethyl ethanoate

Similar multiple functional groups

How do we write the IUPAC name of such compounds, in which you are looking for the same functional groups?



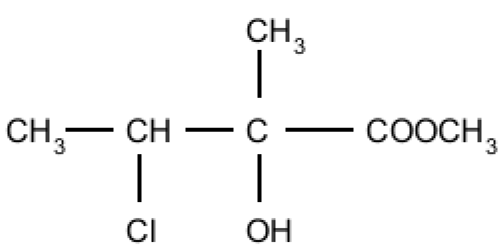


When multiple functional groups are present in the same compound, di, tri, tetra, penta, etc. are put before the name of that functional group. So, IUPAC name of first compound is propan-1,2,3-tri-ol and IUPAC name of second compound is ethan-1,2-di-ol.

Different multiple functional groups

If various functional groups are present in the main chain of the compound they should be written in the following priority order.

What would be the name of the compound having a poly-functional group?



After finding the primary functional group, other functional groups are considered as the substituents. These substituents are written in the alphabetical order.

So, the name of the compound given above is: methyl-3-chloro-2-hydroxy-2-methylbutanoate.

Test Yourself!

What are the steps of writing IUPAC names for organic compounds?

Write the IUPAC name of the first three members of alcohol.

Qualitative analysis of organic compounds (detection of N, S and X by Lassaigne's test)

The qualitative analysis gives the idea of the presence of various types of elements in organic compounds. It gives the idea of the presence of different types of elements. The fundamental elements of the organic compounds are carbon, hydrogen and oxygen. In addition to these elements nitrogen, sulphur and halogen are less commonly present. In some compounds phosphorous is also present as well in some cases metals too. The less commonly present elements N, S and X besides C and H are called foreign elements or hetero elements. Nitrogen, sulphur and halogen may present individually or combined in the same compound. Lassaigne's test is the most convenient method to detect foreign elements present in the organic compounds. Foreign elements are bonded with covalent bonds in organic compounds. So, they do not react with ionic or inorganic reagents. They should be converted into ionic compounds. To convert covalent bond into ionic bond sodium extract solution is prepared. When the organic compound is fused with highly reactive sodium metal, the covalent bond between carbon and foreign element is broken to form water soluble ionic compound (NaCN, Na_2S, NaX).

Preparation of sodium extract solution

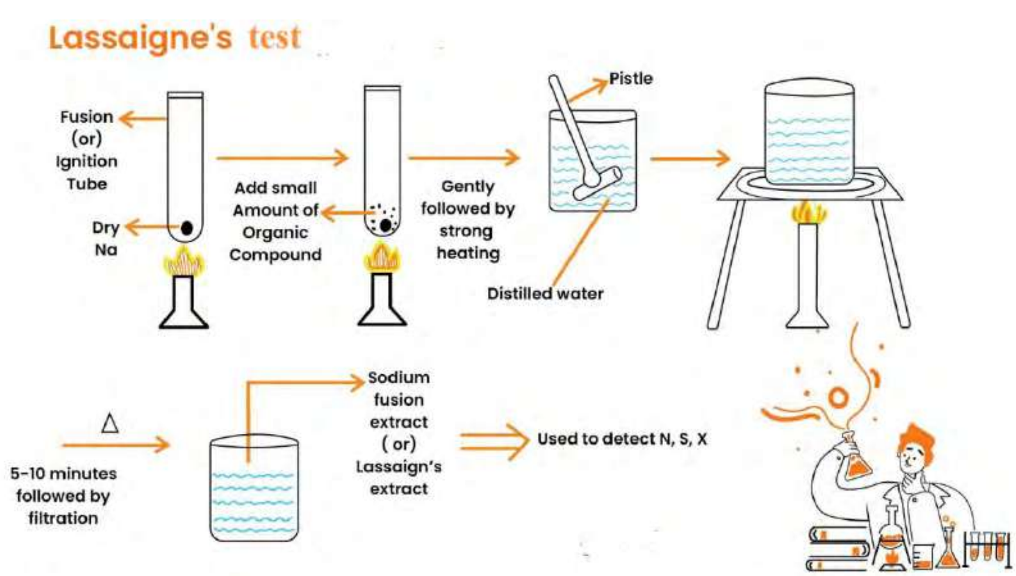


Figure 13.1 Process of preparation of sodium extract

Ignition tube (fusion tube) is taken with the help of fire tongs and a small piece of sodium metal is kept in it and heated gently. Heating is done to make the sodium metal just fused (turns into a small spherical shape). It is then cool in the air. Small amount of organic compound is mixed with the sodium metal. The mixture is heated gently first then to red hot. The red hot ignition tube is kept in the mortar containing about 10 ml of distilled water and ground. Then the solution is filtered. The filtrate is called sodium extract or Lasssaigne's extract solution, which is the test solution for the analysis of N, S and X.

If nitrogen is present: $Na + C + N \xrightarrow{fusion} NaCN$

If sulphur is present : $Na + S \xrightarrow{fusion} Na_2S$

If halogen is present: $Na + X \xrightarrow{fusion} NaX$ ($X = Cl^-$, Br^- , I^-)

Ignition tube → sodium metal → heat → sod. metal fused → organic compound → heat to red hot → keep into mortar containing about 10 ml water → grind with pestle → filter with filter paper → filtrate(Sod. extract solution)

a. Detection of Nitrogen

Activity

About 1 ml of sodium extract solution is taken in a test tube. The solution is boiled with freshly prepared ferrous sulphate. Add a few drops of sodium hydroxide to make the solution distinctly alkaline. It is then cooled. After cooling the mixture, a few drops of ferric chloride solution is added followed by addition of concentrated hydrochloric acid which dissolves excess ferrous hydroxide. A Prussian blue or green coloration confirms the presence of nitrogen in the given organic compound.

$$FeSO_4 + 2NaOH \rightarrow \underset{\text{Ferrous hydroxide (dirty green)}}{Fe(OH)_2} + Na_2SO_4$$

$$Fe(OH)_2 + 2NaCN \rightarrow Fe(CN)_2 + 2NaOH$$

$$Fe(CN)_2 + 4NaCN \rightarrow \underset{\text{Sodium ferrocyanide}}{Na_4[Fe(CN)_6]}$$

$$3Na_4[Fe(CN)_6] + 4\,FeCl_3 \rightarrow \underset{\text{Ferric ferrocyanide (Prussian blue)}}{Fe_4[Fe(CN)_6]_3} + 12NaCl$$

Conc. HCl dissolves excess ferrous hydroxide which otherwise may create confusion with green coloration of ferric ferrocyanide

$$Fe(OH)_2 \rightarrow FeCl_2 + 2H_2O$$

b. Detection of sulphur

Activity

i. Sodium Nitroprusside Test

About 1 ml of sodium extract solution is taken in a test tube. Few drops of freshly prepared sodium nitroprusside are added into it. Formation of violet coloration confirms the presence of sulphur as a foreign element.

$$Na_2S + \underset{\text{Sodium nitroprusside}}{Na_2[Fe(CN)_5.NO]} \rightarrow \underset{\text{sodium sulphonitroprusside (violet color)}}{Na_4[Fe(CN)_5.NOS]}$$

ii. Lead acetate test

About 1 ml of sodium extract solution is taken in a test tube and a few drops of acetic acid followed by lead acetate solution is added into it. A black precipitate confirms the presence of sulphur.

$$Na_2S + Pb(CH_3COO)_2 \xrightarrow{CH_3COOH} \underset{\text{Black ppt.}}{PbS\downarrow} + 2CH_3COONa$$

During the test of nitrogen, blood red coloration appears after adding ferric chloride solution. The appearance of blood red coloration indicates the presence of both N and S in the organic compound.

$$Na + C + N + S \rightarrow \underset{\text{Sodium sulpho(thio)cyanide}}{NaCNS}$$

$$3NaCNS + FeCl_3 \rightarrow \underset{\text{Ferric sulpho(thio)cyanide (Blood red color)}}{Fe(CNS)_3} + 3NaCl$$

c. Detection of halogen

Activity

About 2 ml of sodium extract solution is taken in a test tube and is boiled with dilute nitric acid to remove the sulphides, cyanides, or phosphides if present. After cooling, silver nitrate solution is added into it. If white precipitate is obtained it is due to the presence of chlorine as a foreign element in the form of silver chloride. With the addition of ammonium hydroxide white precipitate gets dissolved. When dilute nitric acid is added into it, white precipitate reappears.

$$AgNO_3 + X^- \rightarrow AgX\downarrow + NO_3^- \text{(general reaction)}$$

For presence of chlorine

$$AgNO_3 + NaCl \rightarrow \underset{\text{White ppt.}}{AgCl\downarrow} + NaNO_3$$

$$AgCl + 2NH_4OH \rightarrow \underset{\text{colorless}}{[Ag(NH_3)_2]Cl} + H_2O$$

$$[Ag(NH_3)_2]Cl + \text{dil. } HNO_3 \rightarrow \underset{\text{White ppt.}}{AgCl} + NH_4NO_3$$

For presence of bromine

With silver nitrate solution bromide gives pale yellow precipitate. Which further partially dissoves with ammonium hydroxide.

$$AgNO_3 + NaBr \rightarrow \underset{\text{Pale yellow ppt.}}{AgBr\downarrow} + NaNO_3$$

$$AgBr + 2NH_4OH \rightarrow [Ag(NH_3)_2]Br + H_2O$$

For presence of iodine

If iodine is present as foreign element, it forms light yellow precipitate of silver iodide. It cannot be dissolved by ammonium hydroxide solution i.e. completely insoluble in liquor ammonia.

$NaI + AgNO_3 \rightarrow AgI\downarrow + NaNO_3$

If N and S are present along with halogen in the same organic compound they interfere with the detection of halogens by forming the precipitate of silver sulphide and silver cyanide and give false test for presence of halogen. So the sodium extract solution should be heated with nitric acid before treatment with silver nitrate solution for the test of halogen. HNO_3 converts cyanide and sulphides compounds into volatile HCN and H_2S.

$Na_2S + 2AgNO_3 \rightarrow Ag_2S\downarrow + 2NaNO_3$

White ppt. of silver sulphide

$NaCN + AgNO_3 \rightarrow AgCN\downarrow + NaNO_3$

White ppt of silver cyanide

$Na_2S + 2HNO_3 \rightarrow H_2S + 2NaNO_3$

$NaCN + HNO_3 \rightarrow HCN + NaNO_3$

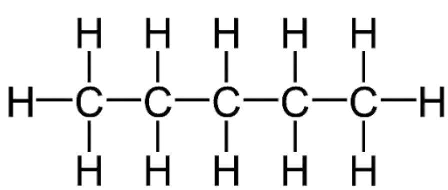
Test Yourself!

Can we detect foreign elements without preparing a sodium extract solution?

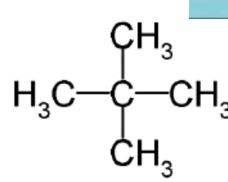
Isomerism in Organic Compounds

Activity:

What are the similarities and differences found in the following compounds? Study the following compounds that fill the table as guided.



Pentane



2,2-dimethyl propane

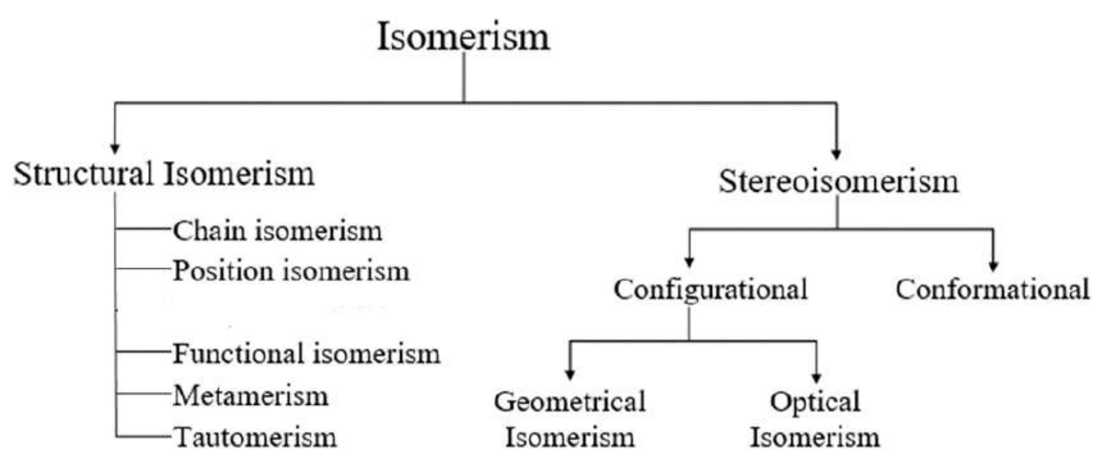
Compounds	No. of carbon atoms	No. of hydrogen atoms	Molecular formula
Pentane	………………..	…………………..	…………….
2,2-dimethyl propane	………………	………………….	…………….

What would be the conclusion of the above activity?
……………………………………………………………………………………………
……………………………………………………………………………………………
……………………………………………………………………………………………

13.4 Definition and Classification of isomerism

Definition:
Two or more organic compounds having the same molecular formula but different physical or chemical properties are called isomers and this phenomenon is called isomerism. The term was proposed by Berzelius.

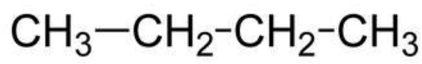
On the basis of arrangement of the atoms within molecules there are two broad classes of isomerism. They are i) structural isomerism and ii) stereoisomerism.

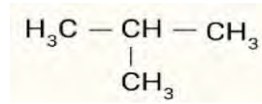


13.5Structural isomerism and its type

The property of the compounds having the same molecular formula but different molecular structures is called structural isomerism. The compounds that exist with this phenomenon are called structural isomers. It is further classified into following types:

a. Chain Isomerism





Butane 2-methyl propane

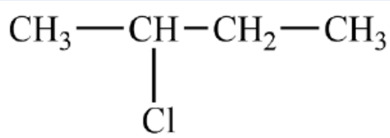
In these two compounds you can see both of them have the same molecular formula but they have differences in parent chain. First compound has a but chain and the second compound has a prop chain.

Two or more organic compounds which have the same molecular formula but different arrangement of the parent chain of carbon atoms are called chain isomers and the phenomenon is called chain isomerism.

Test Yourself!

Make various compounds formed by the molecular formula C_5H_{12} and C_5H_{12} individually with different chains and write their IUPAC name.

b. Position isomerism



2-chlorobutane

CH_3-CH_2-CH_2-CH_2-Cl (1-Chloro butane)

Among the two compounds given above, what difference do you see?

Yes, as you have thought that these two compounds have same molecular formula i.e. C_4H_9Cl, but have difference in the position of chlorine. Such compounds having the same molecular formula but different position of functional group or substituent are called position isomers. The property shown by these compounds is called position isomerism.

CH_3-CH_2-CH_2-NH_2 and CH_3-$CH(NH_2)$-CH_3

are two compounds having the same molecular formula, C_3H_9N. These two compounds are different in the position of the amino group. So, they are position isomers to each other.

c. Functional isomerism

Find the difference in the following two compounds.

CH_3-O- CH_3 and CH_3CH_2OH

Compounds	No. of carbon atoms	No. of hydrogen atoms	Molecular formula	Functional group
CH_3-O- CH_3	……………… ..	………………… ..	…………….	…….
CH_3CH_2OH	………………	………………… .	…………….	……..

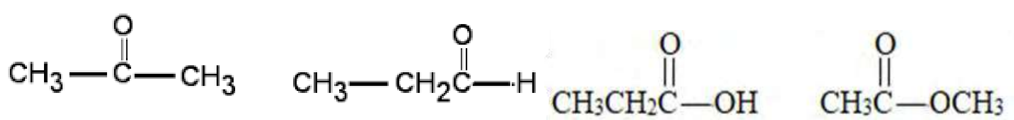
Of course, you may have found that except functional group, these two compounds have similarity in other respects. Two or more compounds having same molecular formula, but have different functional groups are called functional isomers. The property shown by these compounds is called functional isomerism. Functional isomers have different physical as well as chemical properties due to the presence of different functional groups.

The following pairs of compounds represent functional isomers.

a. Alcohols and ethers ($C_nH_{2n+2}O$)

b. Aldehydes and ketones ($C_nH_{2n}O$)

c. Carboxylic acid and esters ($C_nH_{2n}O_2$)



Propanone Propanal Propanoic acid Methyl ethanoate

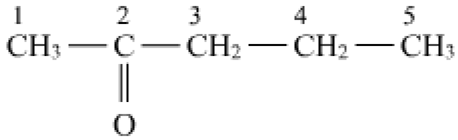
(C_3H_6O)

d. Metamerism

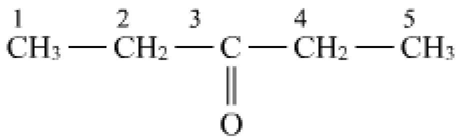
Study the following examples and make a conclusion.

Molecular formula	Compound 1	Compound 2
$C_4H_{10}O$	CH_3-O-CH_2-CH_2-CH_3 Methoxy propane	CH_3-CH_2-O-CH_2-CH_3 Ethoxy ethane
$C_5H_{10}O$	P	
$C_4H_{11}N$	CH_3CH_2NH-CH_2CH_3 N-ethyl ethanamine	$CH_3CH_2CH_2NH$-CH_3 N-methyl propanamine

What is the difference between compound 1 and 2 of each molecular formula?



Pentan-2-one



Pentan-3-one

From a molecular formula given above two compounds are formed and they are different through the alkyl groups substituted in either sides of the functional groups. Such compounds are called metamers. Methoxy propane and ethoxy ethane are metamers to each other as well as N-ethyl ethanamine and N-methyl propanamine, another pair of metamers. Pentan-2-one and pentan-3-one they are also metamers to each other.

Examples:

$CH_3 - CH_2 - O - CH_2 - CH_3$

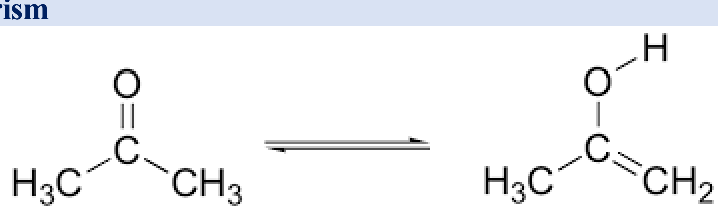
Ethoxyethane

$CH_3 - O - CH_2 - CH_2 - CH_3$

1-Methoxypropane

CH_3-CH_2-NH-CH_2-CH_3	CH_3-NH-CH_2-CH_2-CH_3
N-ethyl ethanamine	N-methyl methanamine

e. Tautomerism

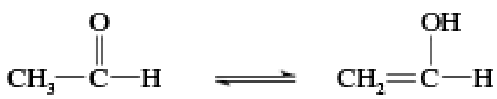


a. What are the functional groups present on these compounds?
b. What is the difference between these two structures?

These compounds are formed by shifting hydrogen from one position to another group then a new functional group appears with the formation of new compounds. The shifting is possible due to the presence of labile (can shift easily) hydrogen atoms from one position to another. Finally, another new compound is formed with a new functional group.

Tautomerism is the special type of functional isomerism, in which isomers differ in arrangement of atoms within the molecule. These isomers exist in dynamic equilibrium with each other.

Some other examples:



acetaldehyde (keto form) ⇌ acetaldehyde (enol form)

This taumerism is also called keto enol tautomerism.

amino enol ethanamide

c. Stereoisomerism

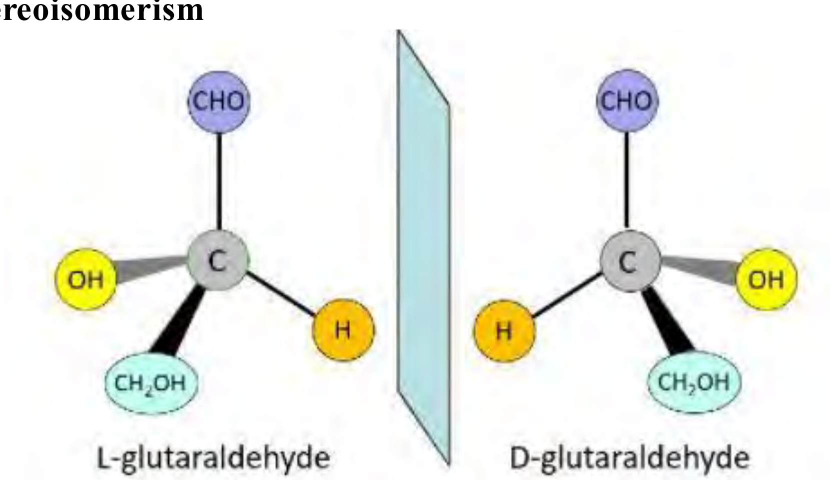


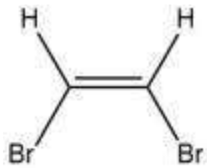
Figure 13.2 Stereoisomers

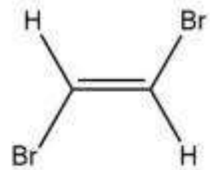
What is the difference between these two figures?

The isomers which are formed due to spatial arrangement of atoms or group of atoms in the space are called stereoisomerism. They are different in configuration of particular atom or group but have the same molecular formula.

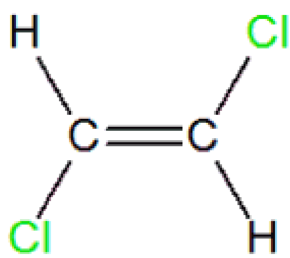
13.6Concept of Geometrical isomerism

Observe the alignment of the atoms in the pair of organic compounds.

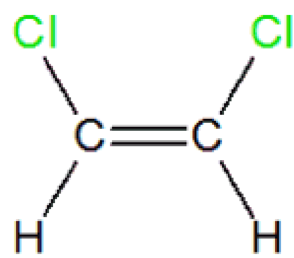




a. What is the molecular formula of these two compounds?
b. What is the difference between these two structures?
c. Do they have similar physical properties?



trans-1,2-dichloroethene



cis-1,2-dichloroethene

If there is a single bond between C-C these carbons can easily be rotated but in such compounds there is a double bond between C=C. So, these carbons cannot be rotated. In *cis*-1,2-dibromoethene both the Br atoms remain fixed on the same side of the C C double bond. On the other hand, in *trans*-1,2-dibromoethene, the Br atoms are positioned across the C C double bond. (Remember that the prefix 'trans-' means across.) These two stereoisomers have different arrangements of the atoms in space so they are different compounds with different physical properties and some different chemical properties, e.g. differing rates for the same reaction.

The main conditions for showing geometrical isomerism are:
a. The molecule must have double bond.
b. The two atoms or groups attached to the same carbon must be different.

Cis: on this side and trans: on the other side
Write the cis and trans structure of the following compounds?
Examples: cis but-2-ene and trans but-2-ene or give compound.

a. Optical isomerism

Compound

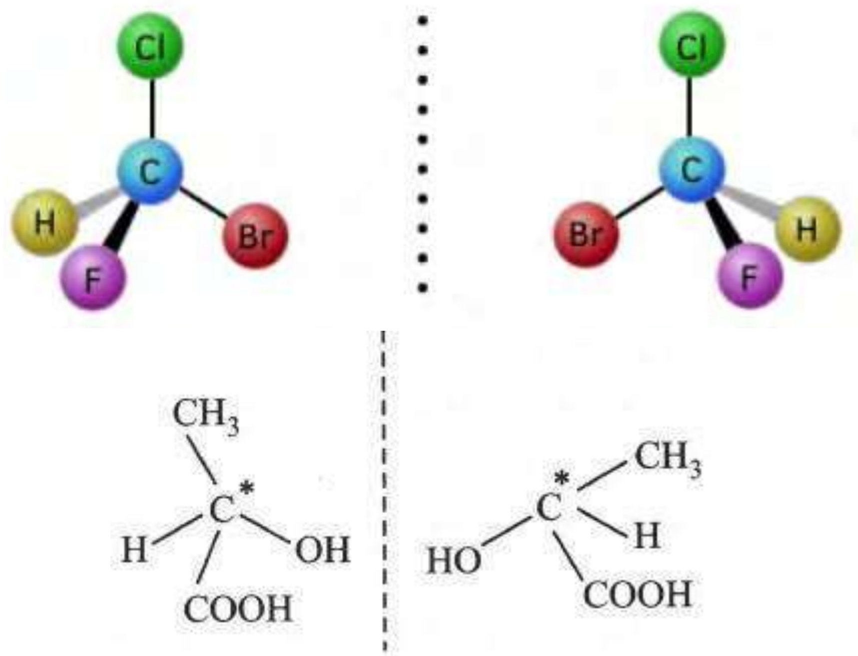


Figure 13.3 Optical isomers

If a molecule contains a carbon atom that is bonded to four different atoms or groups of atoms, it can form two optical isomers. The two different molecules are mirror images of each other and cannot be superimposable. The carbon atom with the four different groups attached is called the **chiral carbon** of the molecule. In the compounds d-isomers which rotate the plane polarized light towards the right or clockwise also known as dextrorotatory or (+) compound. As well as l-isomers are those compounds which rotates the plane polarized light towards left or anticlockwise also known as laevorotatory compound or (-) compound. If the compounds have achiral carbon they do not show optical activity.

Plane polarized light

Observe the following figure and try to distinguish unpolarized and polarized light.

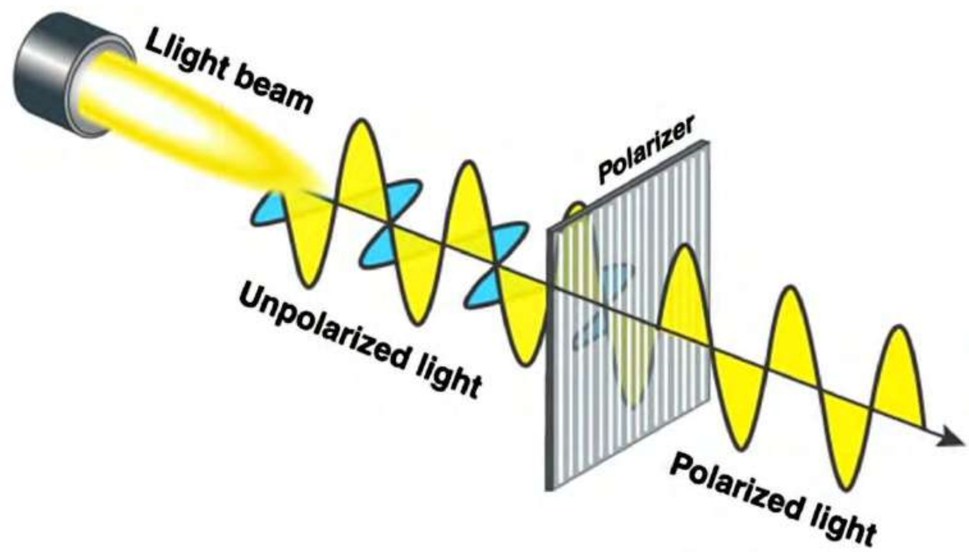
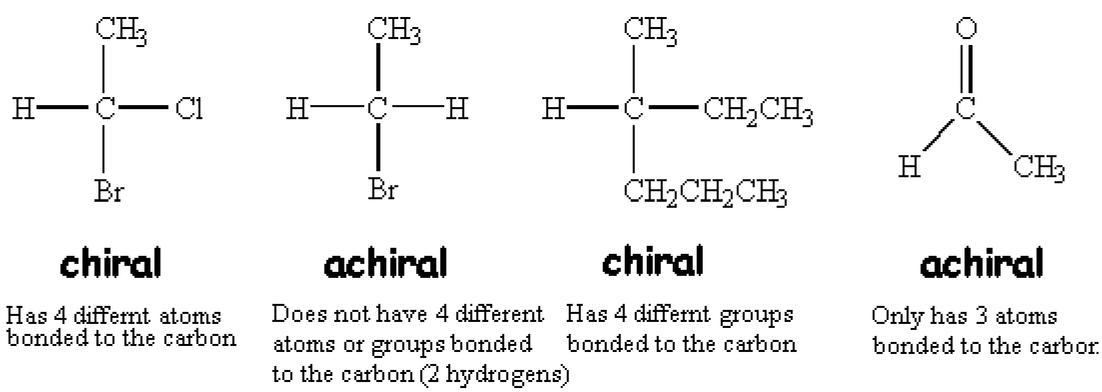


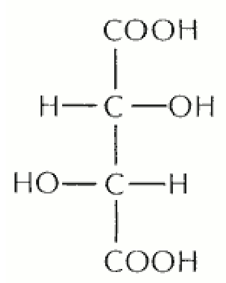
Figure 13.4 showing plane polarized light

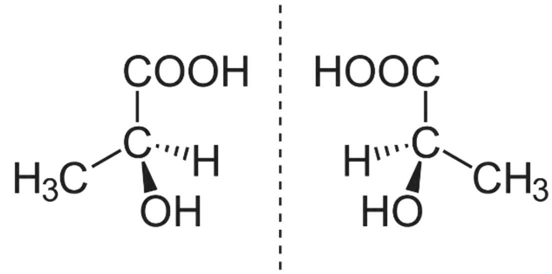
On the unpolarized side the light waves are in different planes but in polarized size the light waves are fixed in certain plain.
The light wave that vibrates in only one plane is called plane polarized light and vice versa.



Enantiomers
Two compounds that are mirror images of each other but that are not identical to each other are called enantiomers. The more common definition of an enantiomer is that it is not superimposable on its mirror image. Enantiomers are the Isomers which are not superimposable to each other.

Example:
Count the chiral carbon of tartaric acid.





Test Yourself!

a. How is position isomerism different from chain isomerism?
b. Give any two examples of functional isomers.

13.7 Preliminary idea of reaction mechanism

Reagent + Substrate $\textit{required condition} \rightarrow$ intermediate product Products.

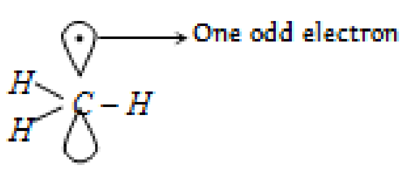
Generally, in chemical reactions substrate is attached by reagent in required condition. An intermediate product is formed before a stable final product. Reaction mechanism is an actual path followed by the reactants to give products with intermediate products. Some components of the reaction mechanism are given below:

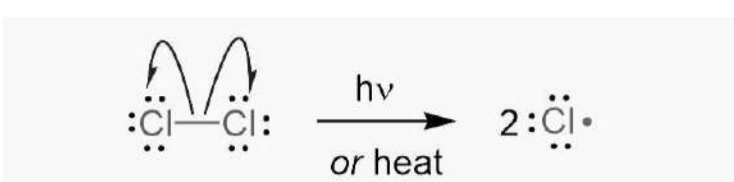
13.7.1 Homolytic and heterolytic fission

a. Homolytic fission

$$CH_3 - CH_3 \xrightarrow[\text{bond fission}]{\text{Homolytic}} \underbrace{CH_3^{\bullet} + CH_3^{\bullet}}_{\text{Methylfreeradicals}}$$

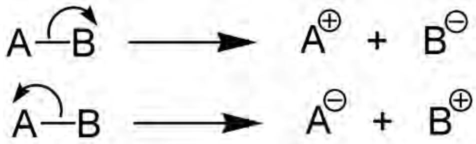
Free radical is formed which is sp^2-hybridized.





In homolytic bond fission, the bond breaks in such a way that each of the fragments will acquire one of the bonding pairs of electrons. After homolytic fission the reactive particles are formed which are called free radicals. This is most common in the vapor phase. This type of fission can be carried by heat, light or organic peroxides, etc.

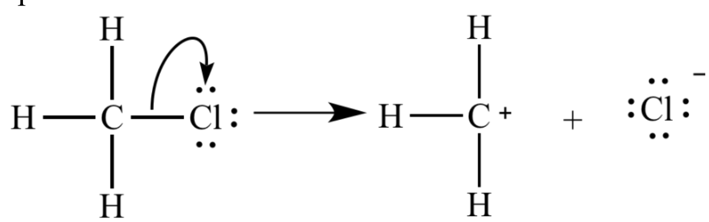
b. Heterolytic fission



B is more electronegative than A in first fission, A is more electronegative than B in second fission.

The breaking of a covalent bond in which both the electrons of a bond will be shifted towards a more electronegative atom. Heterolytic bond fission occurs in such a way that one of the fragments gains both of the bonding pair of electron i.e. It acquires negative charge and another atom acquires positive charge. Ions or charged species formed by heterolytic bond fission. It is the bond Polar compounds in polar solvents show such reactions.

For example:



Methyl carbocation

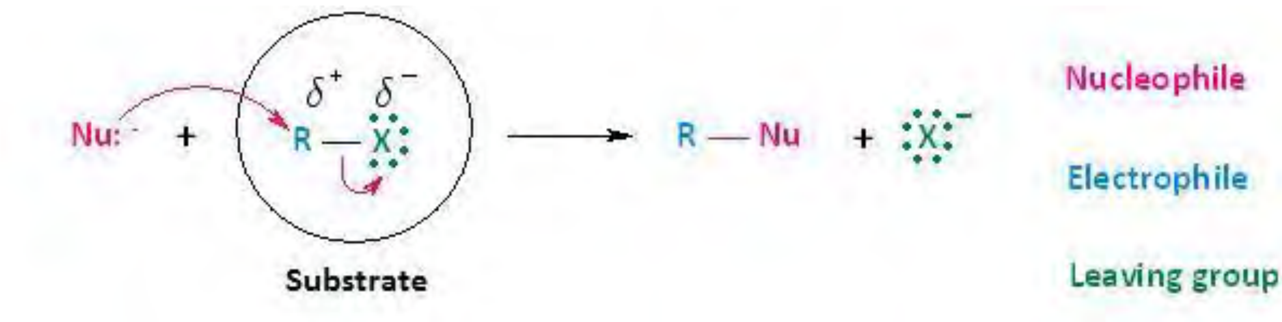
$$CH_3\text{-}CH_2\text{-}Cl \xrightarrow{Energy} CH_3\text{-}CH_2^+ + Cl^-$$

Ethyl carbocation

13.7.2 Electrophile, nucleophiles and free radicals

NO_2^+ and CH_3^+

What type of charges do these chemical species have? Do these charged chemical species get attracted to positively charged chemical species?



They are substrate and reagent. The reagents are those chemical species which attack the substrate and product is obtained. They are as follows.

a. Electrophiles

The word refers to *electro* = electron and *phile* = loving. The atoms or group of atoms which are able to be attracted towards electron are generally called electrophiles. They may either be positively charged species or neutral molecules having electron deficient centres. They have the tendency to attack at the electron rich part of the substrate. Some examples of electrophiles are as follows. NO_2^+, CH_3^+, Br^+, I^+, H_3O^+, BF_3, Cl^+ , SO_3, $FeCl_3$,$AlCl_3$, etc.

b. Nucleophiles

The word refers to *nucleo* = nucleus and *phile* = loving. The atoms or group of atoms which are able to be attracted towards the electron deficient centre. They are also called nucleus loving groups. They may either be negatively charged species or neutral molecules having lone pairs of electrons. They have the tendency to attack at the electron deficient part of the substrate. Some examples of electrophiles are as follows. CH_3O^-, CN^-, NH_2^-, Br^-, I^- , OH^-, NH_3, H_2O, RNH_2, ROH, R-O-R, etc.

c. Free radicals

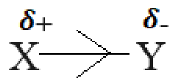
The chemical species which contain an odd (unpaired) number of electrons and are reactive in nature are called free radicals. In organic compounds, if an odd number of electrons is present at carbon it is called carbon free radical. Free radicals are formed by hemolytic bond fission. They do have a neutral nature. In writing symbols of free radical, we generally include a dot to represent an odd electron.

$CH_3 : CH_3 \ Energy \rightarrow CH_3^{\bullet} + CH_3^{\bullet}$ (methyl free radicals)
$H : H \ Energy \rightarrow H^{\bullet} + H^{\bullet}$(hydrogen free radicals)

Test Yourself!

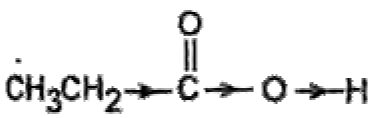
a. In which respect hemolytic fission is different from heterolytic fission?
b. Give an example of hemolytic fission and heterolytic fission of each.

13.7.3 Inductive Effect



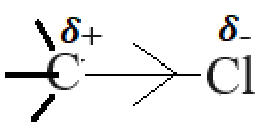
As we see X is donating electrons and the electron is being accepted by the Y in the condition given above. It is possible due to electronegativity differences between X and Y. It seems Y is more electronegative than X. The Sigma bond has become polar due to differences in electronegativity of X and Y. The electron of the covalent bond has been shifted towards Y from X which causes polarization. Thus, the permanent polarization of the bond due to differences in electronegativity of atoms in covalent bond is called inductive effect.

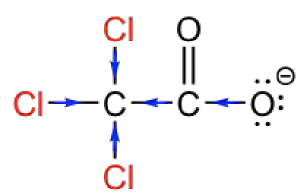
a. Positive inductive effect (+I effect)



In the examples given above, the methyl group is releasing electrons towards the carbon atom. In the second compound also you can see the ethyl group is releasing the electron towards the carbon atom which is the positive effect. The group which releases electrons towards the carbon atom is called electron releasing group. Such groups show positive inductive effects. For example, -R: $-CH_3$, CH_3-CH_2- , $(CH_3)_2CH-$, $(CH_3)_3C-$ etc.

b. Negative inductive effect (- I effect)





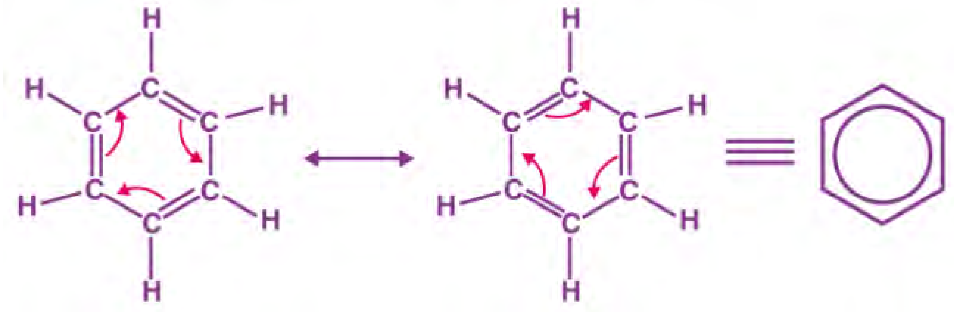
In the first examples chloride has withdrawn electrons from carbon atoms. In the second compound also you can see three chlorine atoms are withdrawing electrons towards them from carbon atoms. Which is the negative effect. The group which withdraws electrons from carbon atoms is called electron withdrawing group. Such groups show negative inductive effects. For example, $-NO_2$, -CN, F, -COOH, Cl, -Br, -I, $-C_6H_5$ etc.

Test Yourself!

NO_2-CH_2-COOH which type of inductive effect does it show?

13.7.4 Resonance Effect:(+R and –R)

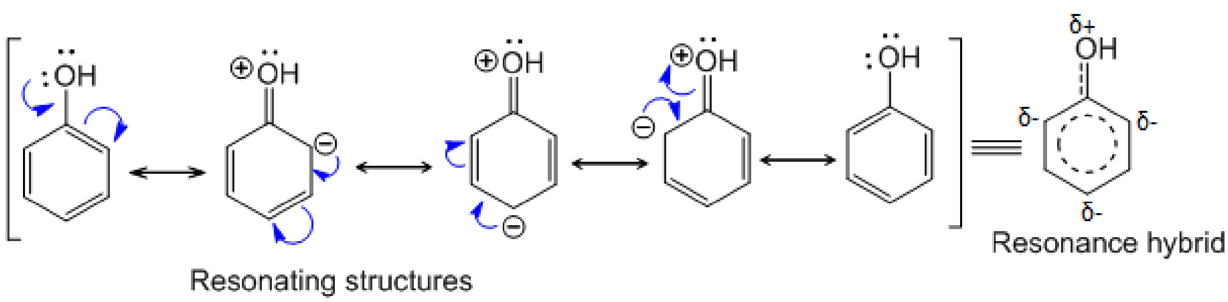
Observe the molecular structures of the benzene given below. Do you find them similar?



In the above, two structures are equivalent. They do have similar structure but position of double bond is different. This differences are due to the interaction of the pi-bonds of the benzene ring. Due to delocalization of electrons resonating (mesomeric) structures are possible. Electron of one atom delocalized and forms bond with electron of another atom. So, the double bond is shifted from one to another. Resonance is seen in open chain hydrocarbon as well as closed chain hydrocarbons. The structures of the compound having same energy state but have different position of the pi-bonds are called resonating structures. It is of two types. They are:

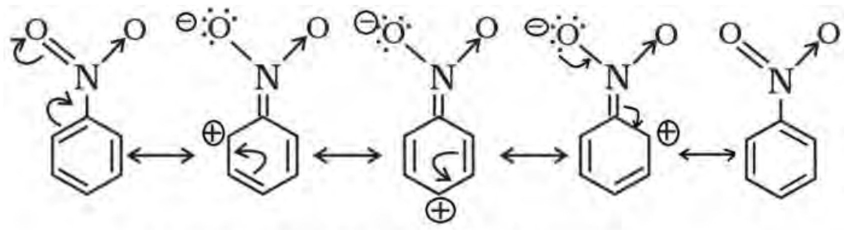
i. Positive resonance effect (+R)

If a group or atom releases electrons to other atoms by delocalization process, then the effect is known as positive resonance effect. In this process, electron density increases in the carbon atom. For example, -OR, -SH, -OH, -NH_2, etc. show this type of positive resonance.



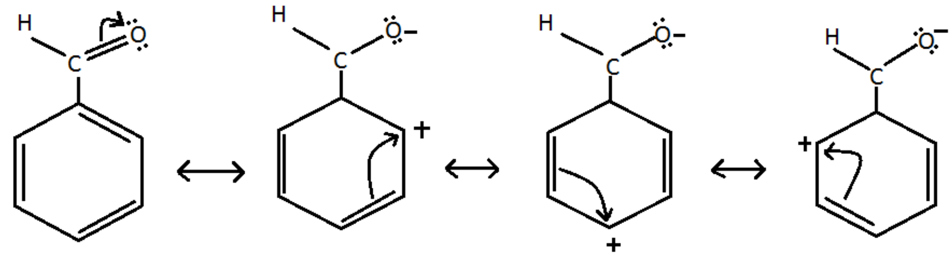
ii. Negative resonance effect (-R)

If an electron withdraws from the carbon of the functional group or main molecule of the organic compound by the process of delocalization, it is considered as negative resonance effect (-R). Consequently, electron density of the molecule decreases. The groups that show negative resonance effect (-R) are –NO_2, >C=O, -CHO, -COOH, -C≡N. In aromatic hydrocarbons, due to –R effect, electron density of ortho and para carbons is decreased.



Resonance structures of nitrobenzene.

In benzaldehyde also, it shows the negative resonance which is shown as follows:



Test Yourself!

What are resonating structures? Give any two examples.

Exercise

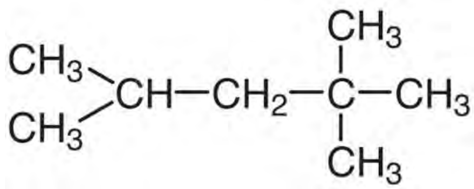
A. Multiple choice questions

1. What does the Z represent in the following skeleton about IUPAC nomenclature represent?
   W + X + Y + secondary suffix
   a. W is prefix, X is primary suffix and Y is word root.
   b. W is word root, X is primary suffix and Y is prefix.
   c. W is prefix, Y is primary suffix and X is word root.
   d. W is prefix, X is primary suffix and Y is word root.
2. Which of the following is correct according to the information given in the following table?

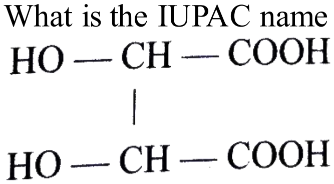
A. –CHO	1. Amide
B. –CONH2	2. Carboxylic acid
C. –OH	3. Aldehyde
D. –COOH	4. alcohol

   a. A-1, B-2, C-3, D- 4
   b. A-3, B- 1, C-4, D-2
   c. A-2, B-3, C-1, D-4
   d. A-4, B-3, C-2, D-1

3. Which of the following is an electrophile?
   a. CH_3-CH_2^-
   b. NH_3
   c. -OR
   d. NO_2^+
4. What is the longest chain of the following compound?



   a. 4
   b. 5
   c. 6
   d. 7
5. What is the IUPAC name of the following compound?



   a. 1,2-dihydoxy butanoic acid
   b. 1,2-dihydroxy butan-dioic acid
   c. 2,4-dihydroxy butan-di-oic acid
   d. 2,3-dihydroxy butan-di-oic acid
6. Na^+ and CNS^- is found in a sodium extract solution. Which foreign element does the organic compound contain?
   a. Sodium only
   b. Sulphur only
   c. Nitrogen and sulphur only
   d. Nitrogen only
7. Which formula is used to calculate the total number of isomers of alkane?
   a. $2^{n-3}+1$
   b. $2^{n-4}+1$
   c. $2^{n-3}-1$
   d. $2^{n-4}-1$
8. What happens in the Lassaigne’s test?
   a. Oxidation of the organic compound
   b. Reduction of the organic compound
   c. Rearrangement of the foreign element within the compound
   d. Formation of ionic compounds
9. An organic compound has nitrogen and sulphur. What would be the product when sodium extract solution is treated with ferric chloride?
   e. $FeSO_4$

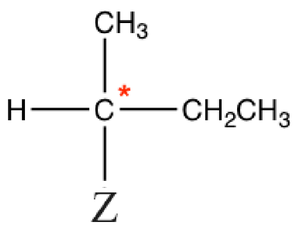
f. $FeCl_2$
g. FeCNS
h. $Fe(CNS)_3$

10. Analyze the pairs of compounds given in the following table and choose the correct option.

CH_3-O-CH_2-CH_3	CH_3CH_2CHO	CH_3COOCH_3	CH_3CH_2CN
$CH_3CH_2CH_2OH$	CH_3COCH_3	CH_3CH_2COOH	CH_3CH_2NC

a. All the pairs have same molecular formula.
b. All the pairs are functional isomers.
c. All the pairs are metamers.
d. All the pairs are position isomers.

11. Which of the following groups is suitable at the position of 'Z' to make the compound non-superimposable?



a. -H
b. CH_3CH_2-
c. $-CH_3$
d. -Br

12. An organic compound has the empirical formula CH2O and molecular weight 90. What would be the molecular formula of the compound?
a. $C_3H_6O_3$
b. $C_4H_8O_4$
c. $C_5H_{10}O_5$
d. $C_6H_{12}O_6$

13. Which of the following statements is true about the following two compounds?

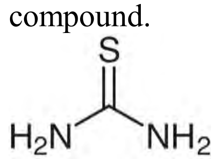
A-Cl-CH_2-CH_2-COOH	B-CH_3-CH_2-CH_2-COOH

a. Compound A is stronger acid than compound B due to negative inductive effect.
b. Compound B is stronger acid than compound A due to negative inductive effect.
c. Compound A is stronger acid than compound B due to positive inductive effect.
d. Compound B is stronger acid than compound A due to positive inductive effect.

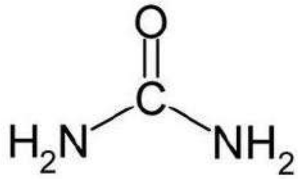
14. Which of the following is a nucleophile?
a. $AlCl_3$
b. BF_3
c. CN^-
d. H_3O^+

15. Which of the following is the general molecular formula of C_3H_8?
   a. C_nH_{2n-2}
   b. C_nH_{2n}
   c. C_nH_{2n-1}
   d. C_nH_{2n+2}

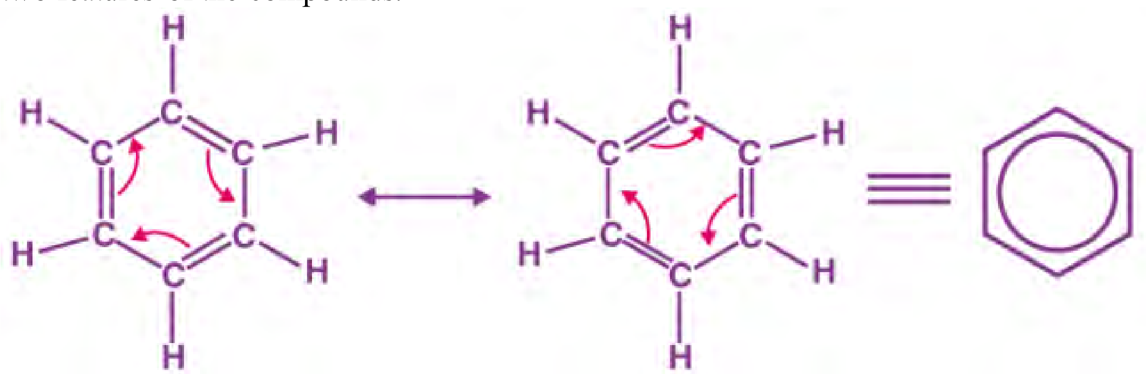
B. Short answer questions

1. Explain about the following terms:
   a. Prefix
   b. Word root
   c. Primary suffix
   d. Secondary suffix
   e. substituent
2. Write an example of the following functional groups:
   a. –NHCO-
   b. -COOH
   c. –CHO
   d. –O-
   e. –COOCO-
3. Write the chemistry of detection of nitrogen present in urea.
4. Write the differences between nucleophile and electrophile. Also write the differences between chain isomer and position isomers of the organic compound.
5. Explain the process and importance of preparation of sodium extract solution. Write the complete reaction for the detection of nitrogen as a foreign element.
6. Explain the process and reaction of the detection of foreign elements present in the given compound.
   
7. How many isomers are possible from the molecular formula C_5H_{12}? Write structure and their IUPAC names of the compounds.
8. Explain about structural isomerism with an example of each.
9. What are the optical isomers? Chiral carbon is necessary to show optical activity, explain with an example.
10. Explain about the following terms with an example of each:
    Hemolytic fission, nucleophiles, free radicals, inductive effect, electrophiles
11. What is Lassaigne's test? Explain about the detection of chlorine as a foreign element in the given organic compound.
12. What are geometrical isomers? Explain about E and Z nomenclature of geometrical isomers.

13. Write the name of the elements present in the given organic compound. Choose the element which is not the fundamental element of the organic compound. How do you confirm the presence of the element by Lassaigne's test? Explain.



14. Which property of the organic compound does the following figures indicate? Write any two features of the compounds.

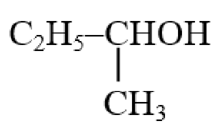


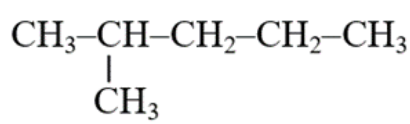
Write any two differences between +R and –R.

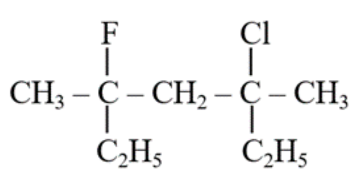
15. Why is the IUPAC system superior to the common system of naming of organic compounds?

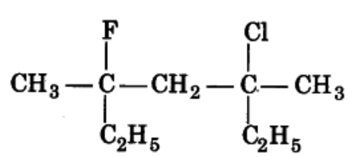
C. Long answer questions

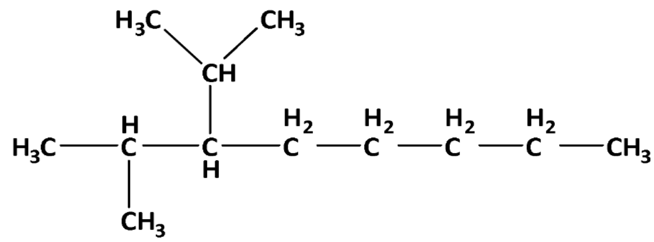
1. Write the IUPAC names of the following compound.

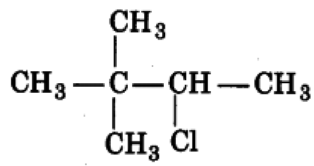
a. 

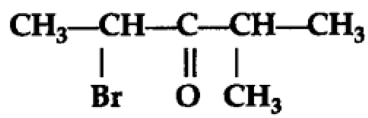
b. 

c. 

d. 

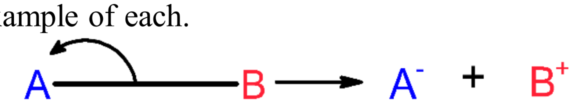
e. 

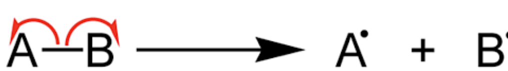
f. 

g. 

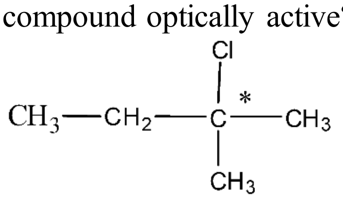
h. $(CH_3)_2CHCOOH$.

2. What is Lassaigne's test? If you have given three organic compounds without labelling. One of them consists of sulphur and another one consists of bromine as a foreign element. How do you identify these elements in the organic compounds? Prepare a laboratory work report mentioning all the procedures with conclusion.
3. Who coined the term isomer? Explain about types of isomerism with subtypes with an example of each.
4. Differentiate between optically active organic compounds and optically inactive organic compounds with any two points. In which respect positive inductive effect is different from negative inductive effect? Explain with examples. Two types of fissions are given below. What makes the products different even if they have similar reagents? Explain. Give an example of each.

a.  b.



5. Draw the resonating structures of benzene. In which respect geometrical isomerism is different from optical isomerism? Explain with a suitable example of each. Is the following compound optically active? Explain.



6. Write rules for the nomenclature of mono-functional organic compounds with suitable examples.
7. Write all the steps to perform Lassaigne's test of sulphur.

Project work

1. Make a chart of prefix, word root, primary suffix and secondary suffix of various types of 15 organic compounds as given in the following table and submit them to your teacher.

Compounds	Prefix	Word root	Primary suffix	Secondary suffix
$Cl\text{-}CH_2\text{-}CH_2\text{-}OH$	Chloro	eth	ane	ol

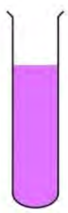
2. Discuss in groups about isomerism of organic compounds. Prepare a chart of various types of isomerism of organic compounds in a chart paper with an example of each and submit to your teacher.

UNIT 14: HYDROCARBONS

Saturated Hydrocarbons(Alkanes; paraffins, C_nH_{2n+2})

Activity:

Observe the following pictures.

 A	 B
 C	 D
 E	 F

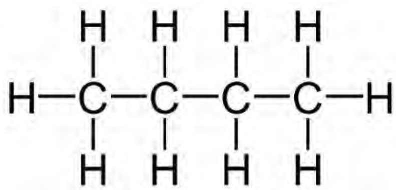
Observation table:

Figure No.	Observation	Inference
A	Pink color of $KMnO4$ remains same	………………..
B	Pink color of $KMnO4$ discharged	…………………
C	Due to unpleasant smell	…………………..

D	Sweet smell(fragrance)	…………………..
E	Non sooty flame(without smoke)	…………………..
F	Sooty flame (with smoke)	……………………

Aliphatic compounds are those compounds which have an open chain. If the compounds are still if they show the reaction as like that of an open chain, they are aliphatic compounds. You can see the differences between aliphatic and aromatic hydrocarbons by observing the above pictures. Alkanes, alkenes and alkynes are aliphatic compounds. In this chapter we are discussing alkanes.

Observe the following given structure of a compound and think. Can we add any atoms in this compound?



In the given compound, all carbons have made 4 covalent bonds. There are no vacant spaces to add any other groups. Such hydrocarbons are saturated hydrocarbons. Alkane has the following features.

They are generally called paraffin (less reactive in Latin).
They have a single covalent bond (sigma bond).
They can be represented by C_nH_{2n+2}.

For example: CH_3-CH_3 (ethane), CH_3-CH_2-CH_3 (propane)

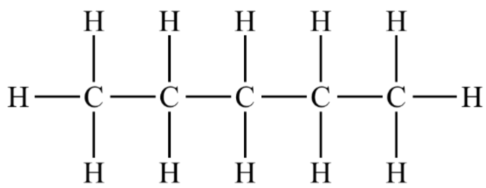
Alkanes are the saturated hydrocarbons. So, they do not give additional reactions. They give only substitution reactions.

Nomenclature

Word root + ane

Compounds	Common name	IUPAC Name
CH_4	Methane	Methane
C_2H_6	n-ethane	Ethane
C_3H_8	n-propane	Propane
C_4H_{10}	n- butane	Butane
C_5H_{12}	n-pentane	Pentane
C_6H_{14}	n-hexane	Hexane

Can you justify?



Why is the compound given above saturated?

a. …………………

b. ………………..

c. ………………….

Classification of alkanes

a. Open chain alkanes

The alkanes having carbon and hydrogen consisting of open chains are called open chain alkanes. They can be straight chain alkanes or branched chain alkanes.

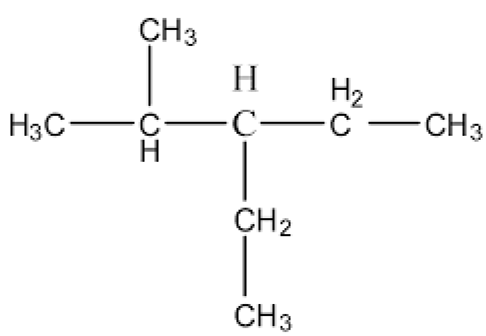
Straight chain alkanes

The alkanes which do not have any groups bonded side wise with the main chain of the hydrocarbon are called straight chain alkanes.

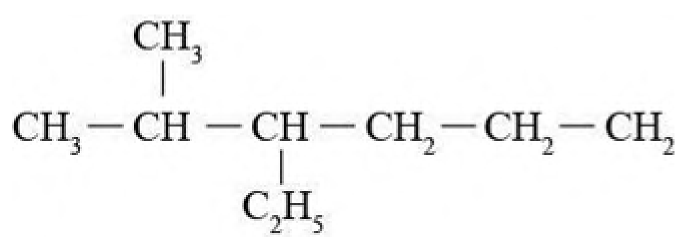
CH_3-CH_3 (ethane), CH_3-CH_2-CH_3 (propane) etc.

Branched chain alkanes

Some groups are bonded in the carbon of main chain are from the main chain of alkanes are called branched chain alkanes.



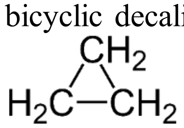
2-methyl-3-ethyl pentane

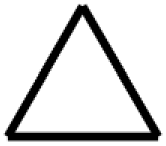
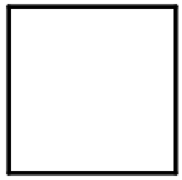
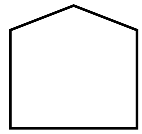
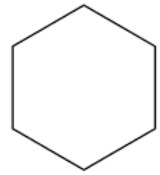
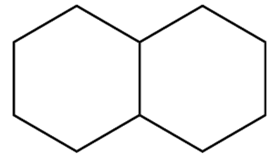


2-methyl-3-ethylhexane

b. Closed chain alkanes(cycloalkanes)

The alkanes in which terminal carbons are bonded with each other and cyclic structure is formed is called cyclic or closed structure. These compounds may be of monocyclic, dicyclic or tricyclic and so on. Cyclopropane, cyclobutane, cyclopentane, cyclohexane and bicyclic decalin have been given in the following table:



 Cyclopropane	 cyclobutane	 cyclopentane	 Cyclohexane	 Bicyclic (decalin)

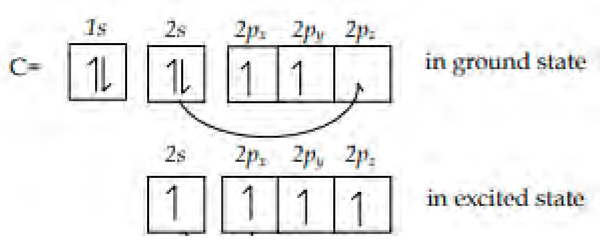
Structure of alkanes

Carbon makes sp^3 hybridization while forming alkanes. Due to the following electronic configuration of carbon, methane has following types of molecular orbital and spatial pictures.

C= $1s^2, 2s^2\ 2p^2$: in ground state

$1s^2, 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^1$: in excited state

With electron spin it can be denoted by;



In an excited state, four orbitals are half filled i.e. one s and three p orbitals. So, carbon of alkanes always has the same case. Hence it forms sp^3 hybridization.

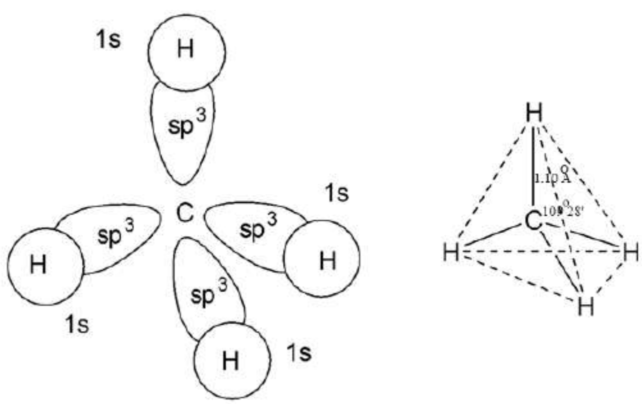


Figure 14.2: Orbital structure of Methane

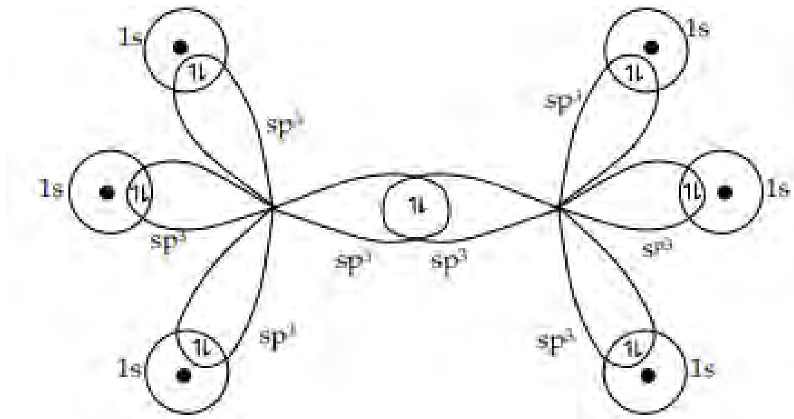


Figure 14.3 Orbital Structure of ethane

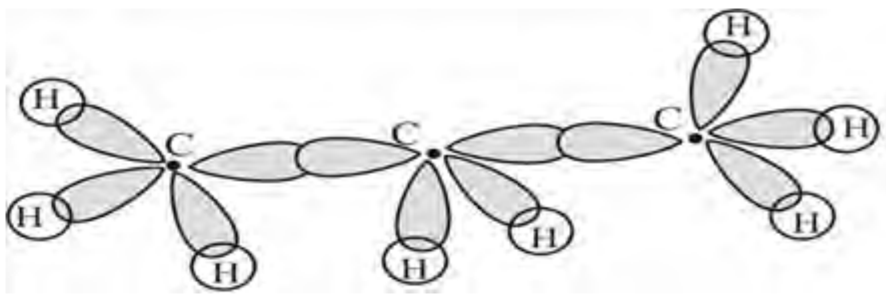


Figure 14.4 : Orbital structure of propane

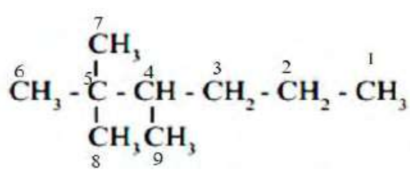
Try yourself!

Which type of hybridization do alkanes have?

Classification of carbons

Activity:

Structure of a hydrocarbon is given below. Study the structure and fill in the blanks on the basis of information given in the table:



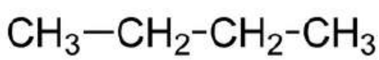
Features	**No. of carbon atoms**
Carbon atoms which is/are bonded with only one other carbon atom	………………..
Carbon atoms which is/are bonded with two other carbon atoms	………………..
Carbon atoms which is/are bonded with three other carbon atoms	………………..
Carbon atoms which is/are bonded with four other carbon atoms	………………..

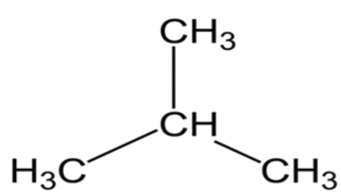
As you have filled in the blanks of the table, they are the basis of classification of carbon atoms.

a. **Primary carbon atom (1^0):** The carbon which is bonded with only one other carbon atom is called primary carbon. Carbon numbers C-1, C-6, C-7, C-8, C-9 are the primary carbons in the given structure of hydrocarbons. 1^o carbon bearing hydrogen atoms are called 1^o hydrogen.
b. **Secondary carbon atom (2^0):** The carbon which is bonded with two other carbon atoms is called secondary carbon. Carbon numbers C-2, C-3 are the secondary carbons in the given structure of hydrocarbons.
c. **Tertiary carbon atom (3^0):** the carbon which is bonded with three other carbon atoms is called tertiary carbon. Carbon number C-4 is the tertiary carbon in the given structure of hydrocarbon.
d. **Quaternary carbon atom (4^0):** the carbon which is bonded with four other carbon atoms is called quaternary carbon. Carbon number C-5 is the quaternary carbon in the given structure of hydrocarbon.

Isomerism in alkanes

Observe the molecular structures and write the answers that follow.

A= 

B= 

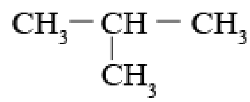
- What is the molecular formula of compound A and B?
  A ………………………. B………………………..
- Do they have the same molecular formula?
  Yes / No

The compounds which have the same molecular formula but different structures (chain) are called isomers. First three members of alkanes do not have isomers but other alkanes show chain isomerism.

The number of chain isomers can be calculated by the formula: $2^{n-4} + 1$, where n represents the number of carbon atoms.

a. Butane (C_4H_{10})
   From this molecular formula total number of isomers formed are,
   $2^{n-4} + 1$
   $2^{4-4} + 1$
   $2^0 + 1$
   $1+1=2$

$CH_3-CH_2-CH_2-CH_3$



2 - methyl propane

b. Pentane (C_5H_{12})

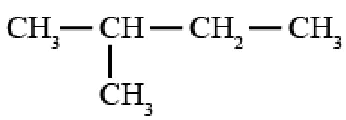
Total number of isomers that can form are,

$2^{n-4} + 1$

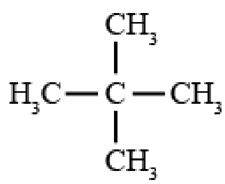
$2^{5-4} + 1$

$2^{1} + 1$

$2+1=3$



2-methylbutane



2,2-dimethylpropane

$H_3C-CH_2-CH_2-CH_2-CH_3$

n-pentane

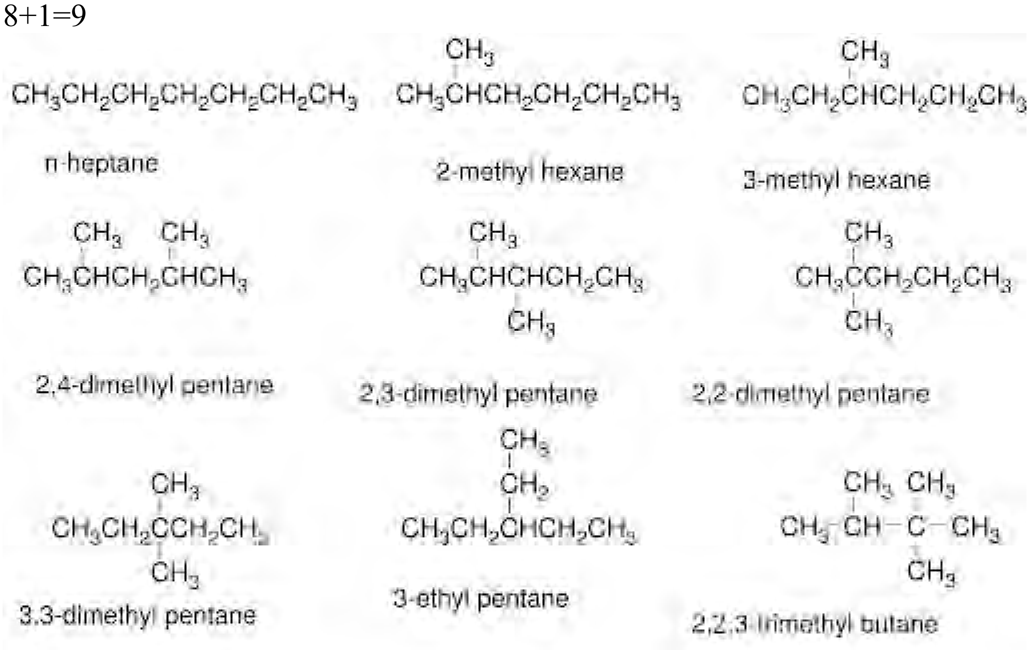
c. Heptane (C_7H_{16})

$2^{n-4} + 1$

$2^{7-4} + 1$

$2^{3} + 1$

$8+1=9$



Preparation of Alkanes:(C_nH_{2n+2})

a. By the reduction of haloalkanes

R-X + 2H $\quad Zn + HCl \rightarrow$ R-X + HX

Zinc and hydrochloric acid produce nascent hydrogen. Nascent hydrogen reduces alkyl halide into alkanes. By the presence of $LiAlH_4$, Ni or Pd/H_2 also reduces alkyl halide into alkanes having the same carbon number.

For example,

CH_3-I + 2H $\xrightarrow{Zn + HCl}$ CH_4 + HI

Iodomethane methane

CH_3-CH_2-Cl + 2H $\xrightarrow{Zn + HCl}$ CH_3-CH_3 + HCl

Chloroethane Ethane

CH_3-CH_2-CH_2-Cl + 2H $\xrightarrow{Zn + HCl}$ CH_3-CH_2-CH_3 + HCl

Chloropropane Propane

b. By Wurtz's reaction

When alkyl halide is heated with sodium metal in presence of dry ether as solvent, a symmetrical alkane containing double the number of carbon atoms on haloalkane is formed. This is called Wurtz's reaction. Two molecules of haloalkane give an alkane. So the product is even carbon numbered.

R-$\boxed{X + 2Na + X}$-R $\xrightarrow{Dry\ ether}$ R –R + 2NaX

For example:

CH_3-$\boxed{Cl + 2Na + Cl}$-CH_3 $\xrightarrow{Dry\ ether}$ CH_3-CH_3 + 2NaCl

Chloro methane ethane

CH_3-CH_2-$\boxed{Cl + 2Na + Cl}$- CH_2-CH_3 $\xrightarrow{Dry\ ether}$ CH_3- CH_2- CH_2-CH_3 + 2NaCl

Chloroethane Butane

Remember:

Odd carbon numbered alkanes cannot be prepared by this method.

c. By decarboxylation method:

Observe the following reaction and make a conclusion.

R-$\boxed{COONa + NaO}$ H $\xrightarrow{CaO/\Delta}$ R-H + Na_2CO_3

Sodium alkanoate alkane

CH_3COONa + NaOH $\xrightarrow{CaO/\Delta}$ CH_3-H + Na_2CO_3

Sodium ethanoate methane

CH_3CH_2COONa + NaOH $\xrightarrow{CaO/\Delta}$ CH_3CH_2-H + Na_2CO_3

Sodium Propanoate Ethane

The reaction occurs between sodium salt of carboxylic acid and soda-lime (3:1) in the presence of heat. If you have analyzed these reactions minutely, you may have found that in all reactions one carbon number is decreased. This reaction is called the decarboxylation reaction, since the carboxyl group is removed.

d. By catalytic hydrogenation of alkenes and alkynes

The word catalytic hydrogenation refers for addition of hydrogen in presence of catalysts. When alkenes and alkynes are heated with hydrogen in presence of catalysts nickel or palladium or platinum, it gives alkanes. (Nickel works in high temperature but palladium and plantinum reacts a bit low temperature i.e at room temperature).

R-CH=CH_2 + H_2 $\xrightarrow{Ni/Pd/Pt/\Delta}$ R-CH_2CH_3
Alkene alkane
For example:
CH_3-CH=CH_2 + H_2 $\xrightarrow{Ni/Pd/Pt/\Delta}$ CH_3-CH_2CH_3
Propene propane
R-CH ≡ CH_2 + H_2 $\xrightarrow{Ni/Pd/Pt/\Delta}$ R-CH=CH_2 + H_2 $\xrightarrow{Ni/Pd/Pt/\Delta}$ R-CH_2CH_3
Alkyne alkene alkane
For example:
CH_3-CH ≡CH_2 + H_2 $\xrightarrow{Ni/Pd/Pt/\Delta}$ CH_3-CH=CH_2 + H_2 $\xrightarrow{Ni/Pd/Pt/\Delta}$ CH_3-CH_2CH_3
Propyne propene propane

Note: If the reduction is carried out in the presence of Lindlar's reagent, single step reduction of alkyne occurs, i.e. when alkyne is reduced in Lindlar's reagent it stops at alkene, further reduction does not take place.

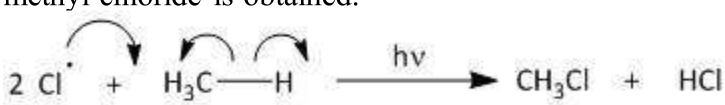
14.1.2 Chemical Properties of alkanes

Each carbon of alkanes has a sigma bond i.e. single bond. So, an additional reaction is impossible. Still the substitution reaction is yet to be discussed. It undergoes with strong reagents and in presence of high energy only.

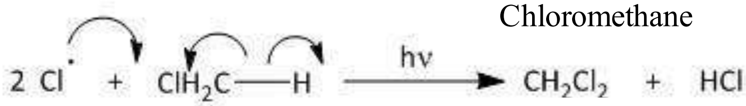
1. **Substitution reaction**

   a. **Halogenation**

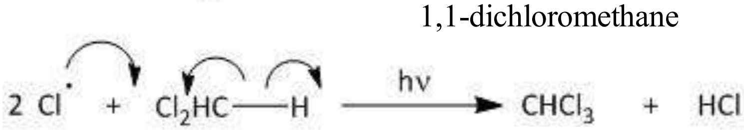
   Chlorine molecules get dissociated into reactive chlorine atoms. The atom gets substituted by replacing hydrogen with methane. After the multiple substitution carbon tetra chloride is obtained as follows. The products depend upon the ratio of chlorine to methane taken. With excess chlorine carbon tetrachloride is obtained as well as with excess of methane, methyl chloride is obtained.



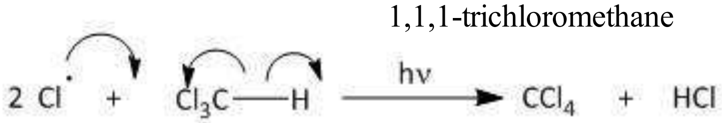
Chloromethane



1,1-dichloromethane



1,1,1-trichloromethane



1,1,1,1-tetrachloromethane

By the same way substitution in other compounds also occurs.

$CH_3\text{-}CH_3 + Cl_2 \xrightarrow{UV\ light} CH_3CH_2Cl + HCl.$

Ethane Chloroethane

$CH_3\text{-}CH_2\text{-}CH_3 + Cl_2 \xrightarrow{Ni/Pd/Pt/\Delta} CH_3\text{-}CH_2\text{-}CH_2\text{-}Cl + HCl$

Propane 1-chloropropane

b. Nitration

The substitution of hydrogen by the nitro ($-NO_2$) group in alkane/other hydrocarbons is called nitration. This reaction is carried out in presence of fuming of concentrated nitric acid or in presence of high temperature i.e at 400- 500^0 C with alkanes. The major products are nitroalkanes.

$R\text{-}H + HNO_3 \xrightarrow{\Delta} R\text{-}NO_2 + H_2O$

Alkane nitroalkane

$CH_3\text{-}CH_3 + HNO_3 \xrightarrow{\Delta} CH_3\text{-}CH_2NO_2 + H_2O$

Ethane nitroethane

$CH_3\text{-}H + HNO_3 \xrightarrow{\Delta} CH_3\text{-}NO_2 + H_2O$

Mehane nitromethane

c. Sulphonation

Study the reactions given below

$R\text{-}H + \text{conc. } H_2SO_4 \xrightarrow{\Delta} R\text{-}SO_3H + H_2O$

Alkane Alkane sulphonic acid

$CH_3CH_2\text{-}H + \text{conc. } H_2SO_4 \xrightarrow{\Delta} CH_3\text{-}CH_2\text{-}SO_3H + H_2O$

Ethane ethane sulphonic acid

In these reactions, alkanes are heated with concentrated sulphuric acid. Fuming occurs. The fuming of SO_3, it dissolves in conc. H_2SO_4 and sulphonic acid is obtained by the substitution of one hydrogen atom by sulphonic group ($-SO_3H$).

2. Oxidation of ethane

a. Combustion

Due to the highly combustible nature of alkanes they burn readily in the presence of oxygen. So alkanes are used as good fuels in different forms, like LPGs. It shows that the reaction is exothermic. When alkanes are burnt in the presence of oxygen, carbondioxide and water are formed with an excess amount of energy.

Figure 14.5 : Fueling in the engine

Reactions

$CH_4 + 2O_2 \xrightarrow{\Delta} CO_2 + 2H_2O + \text{energy}$

$CH_3\text{-}CH_3 + O_2 \xrightarrow{\Delta} CO_2 + 2H_2O + \text{energy}$

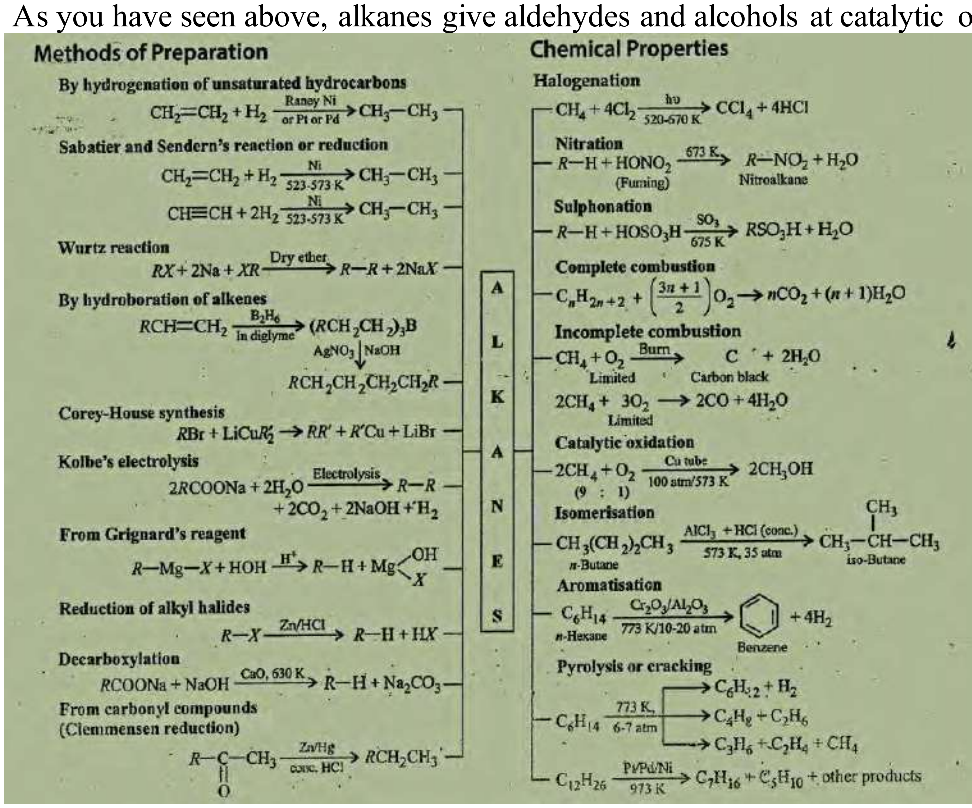
Note: Generally, in the combustion of hydrocarbons, mainly energy is released and byproducts are carbon dioxide and water.

b. Catalytic oxidation

$2CH_4 + O_2 \xrightarrow{Cu,\ 100\ atm\ /\Delta} 2CH_3OH$

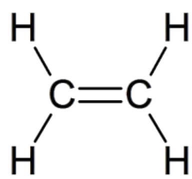
$CH_4 + O_2 \xrightarrow{MoO, 200atm\ /\Delta} HCHO + H_2O$

As you have seen above, alkanes give aldehydes and alcohols at catalytic oxidation.

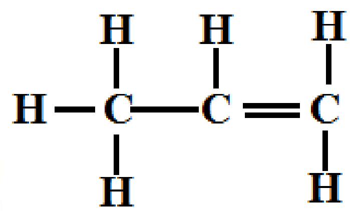


Unsaturated hydrocarbons (Alkenes) or olefins C_nH_{2n}

Analyze the following structure and fill the following table with the information found in these compound



Ethene



Propene

No. of carbon atoms	compounds	No. of Bonds between carbon atoms	Saturated/unsaturated
2	Ethene	Double	……………
3	Propene	Single & double	………………

Orbital Structure of ethene (C_2H_4)
Count the total number of bonds in between C-C and C-H and their nature.

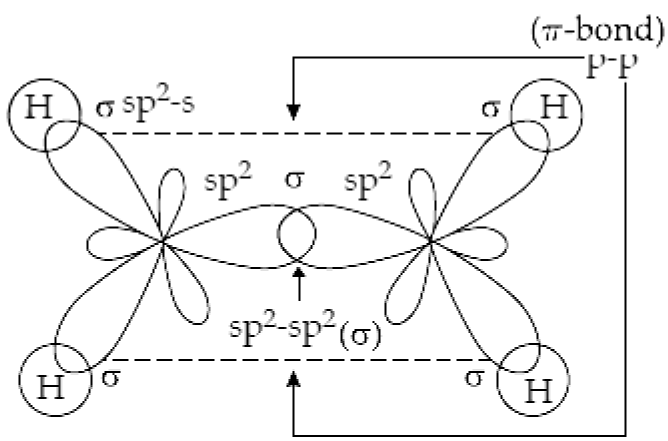


Figure 14.6: Orbital Structure of Ethene

You may have found that there are two bonds between C-C i.e. one is a sigma bond and another one is a pi-bond. Between C- H, there are four sigma bonds.

Preparation of alkene

a. By dehydration of alcohol

When alcohol is heated to about 400°C in presence of aluminium oxide –OH group of alcohol and hydrogen from the adjacent carbon forms a water molecule and is removed by dehydration.

R-CH_2-CH_2OH $\xrightarrow{Al2O3/400}$ RCH =CH_2 + H_2O
For example:
CH_3-CH_2OH $\xrightarrow{Al2O3/400}$ CH2 =CH_2 + H_2O
Alcohol can be dehydrated in the presence of concentrated sulphuric acid also.

Study the following chemical equation and conclude whether the reaction has dehydration of alcohol or not?

CH_3CH_2-OH + $HHSO_4$ $\xrightarrow{Heat/170}$ $CH_3CH_2HSO_4$ + H_2O
Ethanol ethyl hydrogen sulphate
$CH_3CH_2HSO_4$ $\xrightarrow{Heat/170}$ CH_2=CH_2 + H_2SO_4
Ethyl hydrogen sulphate ethene
CH_3-CH(OH)-CH_3 + conc H_2SO_4 $\xrightarrow{Heat/170}$ CH_3-CH=CH_2 + H_2O
Propan-2-ol propene

Saytzeff's rule:

In the dehydration of secondary and tertiary alcohols, there is a possibility of the formation of two types of alkenes, the major product will be the alkene which has more alkyl substituted at doubly bonded carbon atoms called Saytzeff's rule.

For example:
CH_3-CH_2-CH(OH)-CH_3 $\xrightarrow{Heat/170}$ CH_3CH=CH-CH_3 + CH_3-CH_2-CH=CH_2 + H_2O
Butan-2-ol but-2-ene (major) but-1-ene

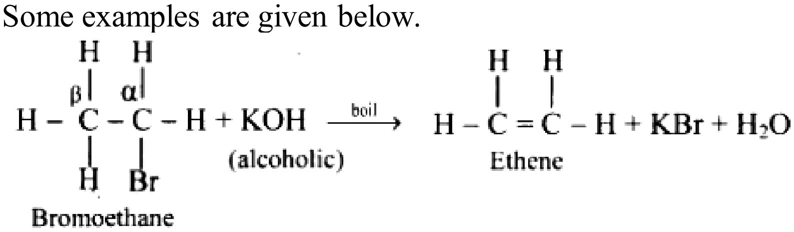
CH_3-CH_2-CH_2-CH(OH)-CH_3 $\xrightarrow{Heat/170}$ CH_3-CH_2-CH=CH-CH_3 + CH_3-CH_2-CH_2-CH=CH_2+ H_2O

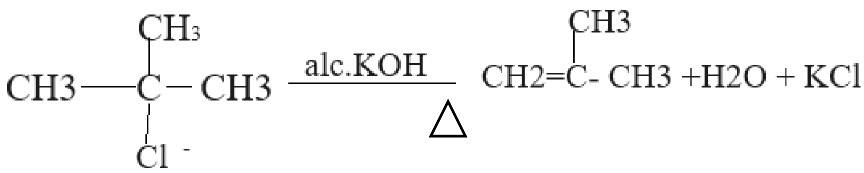
Pentan-2-ol pent-2-ene pent-1-ene

b. By dehydrohalognenation of alkyl halides:

When alkyl halide is heated with alcoholic potassium hydroxide, hydrogen and halogen are eliminated from adjacent carbon atoms. In this chemical reaction, hydrogen is lost from β-carbon, so it is called β -elimination reaction. The ease of the dehydrohalogenation of alkyl halide is in the order of $3^0 > 2^0 > 1^0$.

Some examples are given below.





2-chloro-2-methyl propane 2-methyl-propene

c. **Catalytic hydrogenation of alkynes:**

Hydrogenation refers to the addition of hydrogen. Here, the reduction should be controlled when it occurs in a single step i.e. when it forms alkene. For this controlled hydrogenation, reaction reagents are Pd and calcium carbonate along with quinoline or Lindlar's catalyst($Pd/BaSO_4$) is used.

$R\text{-}C \equiv CH + H_2 \; Pd/BaSO4\,/quinoline \rightarrow \; R\text{-}CH{=}CH_2$

Alkyne alkene

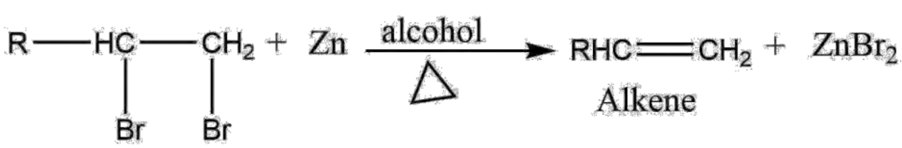
$H\text{-}C \equiv CH + H_2 \; Pd/BaSO4\,/quinoline \rightarrow \; CH_2{=}CH_2$

Ethyne Ethene

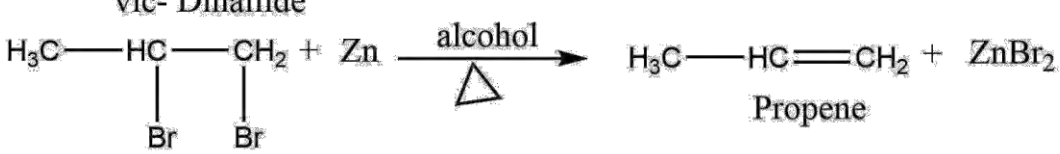
$CH_3\text{-}C \equiv CH + H_2 \; Pd/BaSO4\,/quinoline \rightarrow \; CH_3\text{-}CH{=}CH_2$

Propyne Propene

Note: alkenes can also be prepared by heating vicinal dihalides (two halogen atoms on adjacent (neighboring) carbon atoms) with zinc in presence of alcohol.



vic- Dihallide

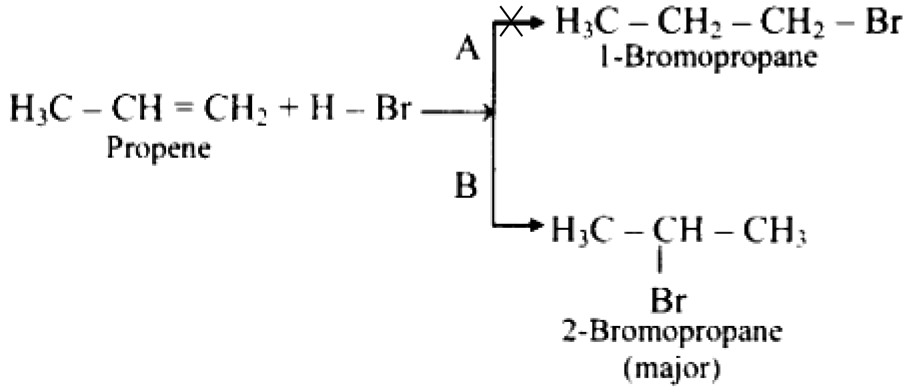


1,2-Dibromopropane

Chemical properties of alkene

a. Addition reaction with HX (Markonikov's rule)

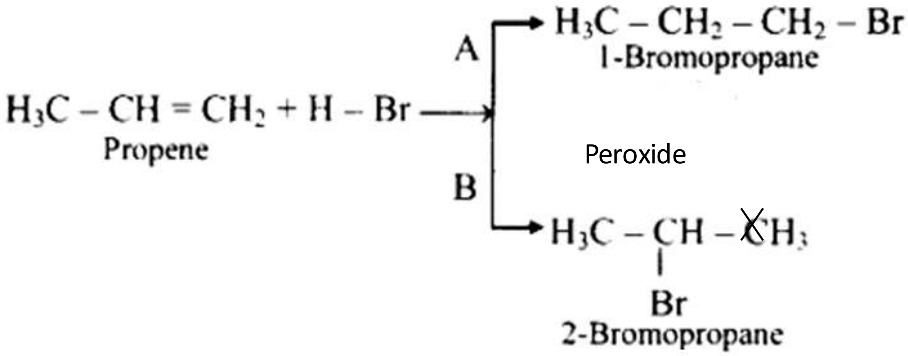
Observe the following chemical equations.



In the reaction given above, HBr is added in unsymmetrical alkene. In this reaction addition of hydrogen has occurred in the carbon of 2 hydrogen atoms rather than 1 hydrogen atom. It is according to Markovnikov' rule. A Russian chemist Vladimir Markovnikov studied a large number of addition reactions of unsymmetrical olefins and he observed two types of isomeric products.

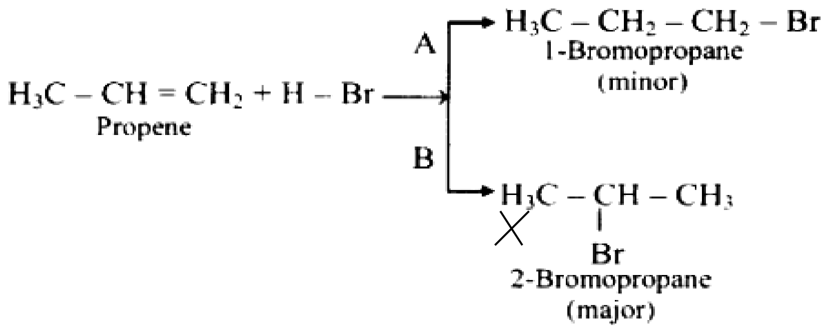
It states that, *the addition of hydrogen halide or other polar reagent occurs in such a way that the positive part of the reagent goes towards the carbon where there is greater number of hydrogen already in unsymmetrical alkenes or alkynes*.

Peroxide effect or Kharash effect (Anti-Markovnikov's rule):

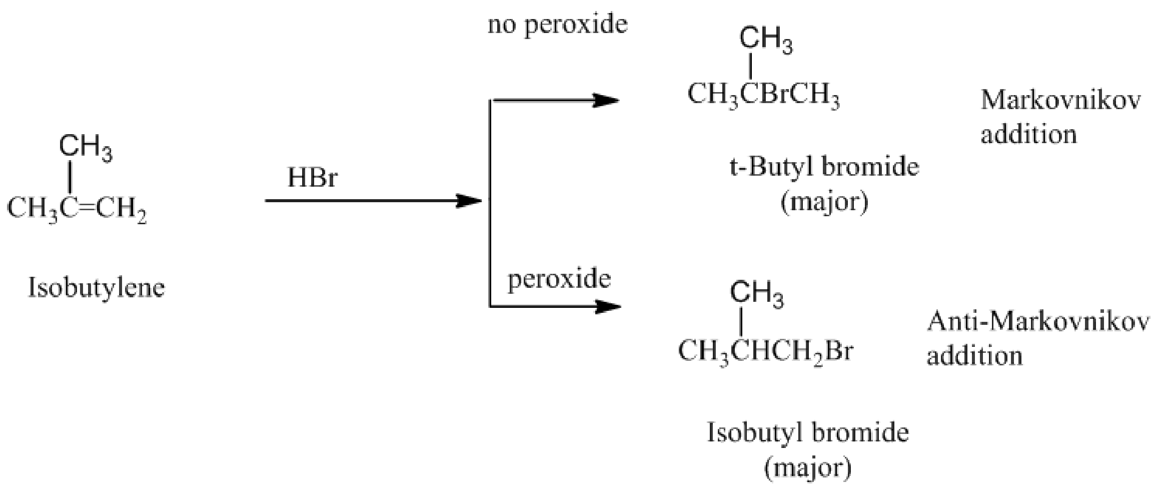


If you see this example, you will see that reverse type of reaction as you have seen in Markovnikov's rule. Addition of only HBr but not HI and HCl to an unsymmetrical alkene in presence of organic peroxide (R – O – O – R) takes place just opposite of Markovnikov's rule. It is due to the peroxide effect.

It states that, ***the addition of hydrogen bromide occurs in such a way that the positive part goes towards the carbon where there is lesser number of hydrogen already in unsymmetrical alkenes or alkynes.***

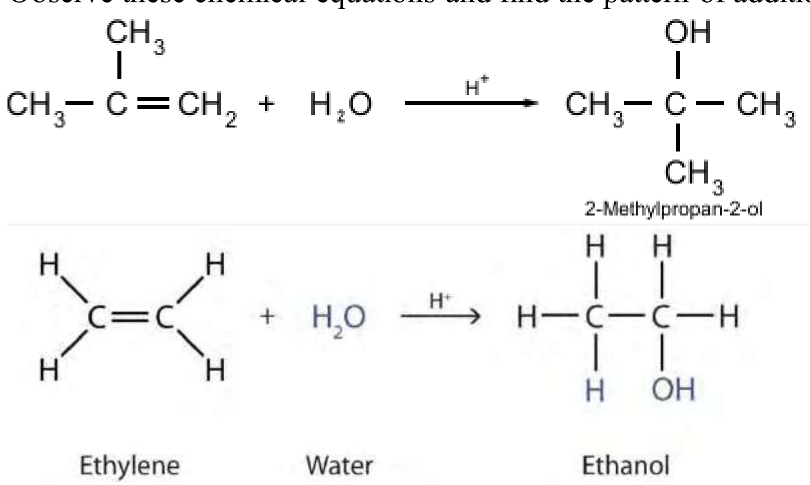


Summary of the reactions is:



b. Addition of water

Observe these chemical equations and find the pattern of addition of water into alkenes given.



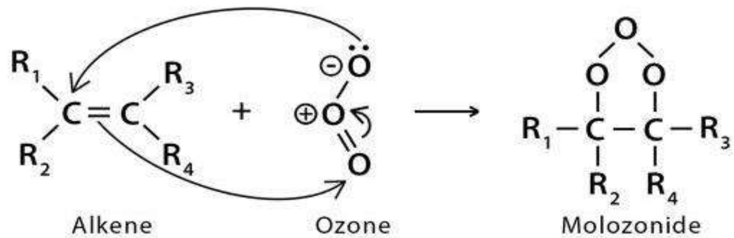
As, there are two ionic parts in water i.e. H^+ and OH^-. These ionic parts are added to the unsaturated group by following Markovnikov's rule in unsymmetrical alkenes.

$CH_3CH{=}CH_2 + H_2O \xrightarrow{H_2SO_4} CH_3\text{-}\underset{\displaystyle OH}{\underset{|}{C}H}\text{-}CH_3$

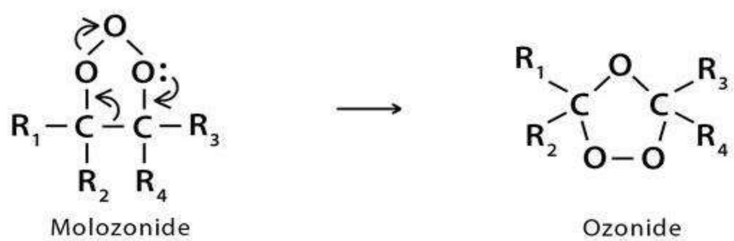
c. Addition of ozone

Alkene reacts with ozone in the presence of inert conditions in the presence of carbon tetrachloride or in the presence of ethers. First it forms ozonide then by hydrolysis it gives carbonyl compounds i.e. aldehydes and ketones.

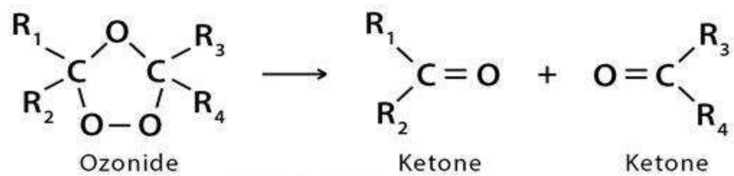
Step 1: Nucleophilic C=C bond attacks the positive O of ozone while a second nucleophilic attack by the negative O takes place at the other end of the C=C bond



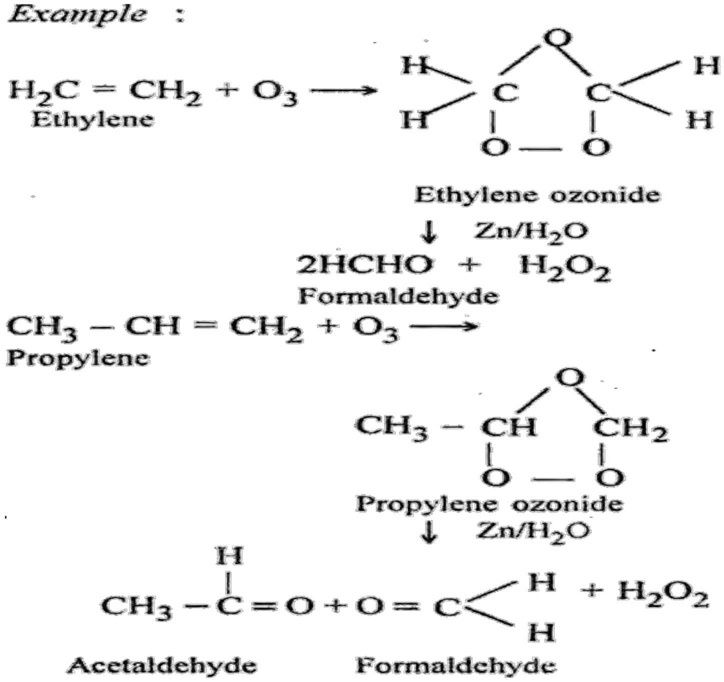
Step 2: The cyclic molozonide rearranges itself to ozonide.



Step 3: Decomposition of the ozonide after the reductive workup resulting in two carbonyl groups.



Example :



d. Addition of sulphuric acid (Addition of H_2SO_4)

As with other reactions, sulphuric acid gives two ions H^+ and OSO_3H^-. These groups are added in a double bond of alkene. In unsymmetrical alkenes addition is occurred by Markovnikov's rule.

$$-\overset{|}{C}=\overset{|}{C}- + H_2SO_4 \longrightarrow -\underset{H}{\overset{|}{\underset{|}{C}}}-\underset{SO_3H}{\overset{|}{\underset{|}{C}}}-$$

Alkyl hydrogen sulphate

$$H_2C=CH_2 + H_2SO_4 \longrightarrow CH_3-\underset{OSO_3H}{\underset{|}{CH_2}}$$

Ethene Ethyl hydrogen sulphate

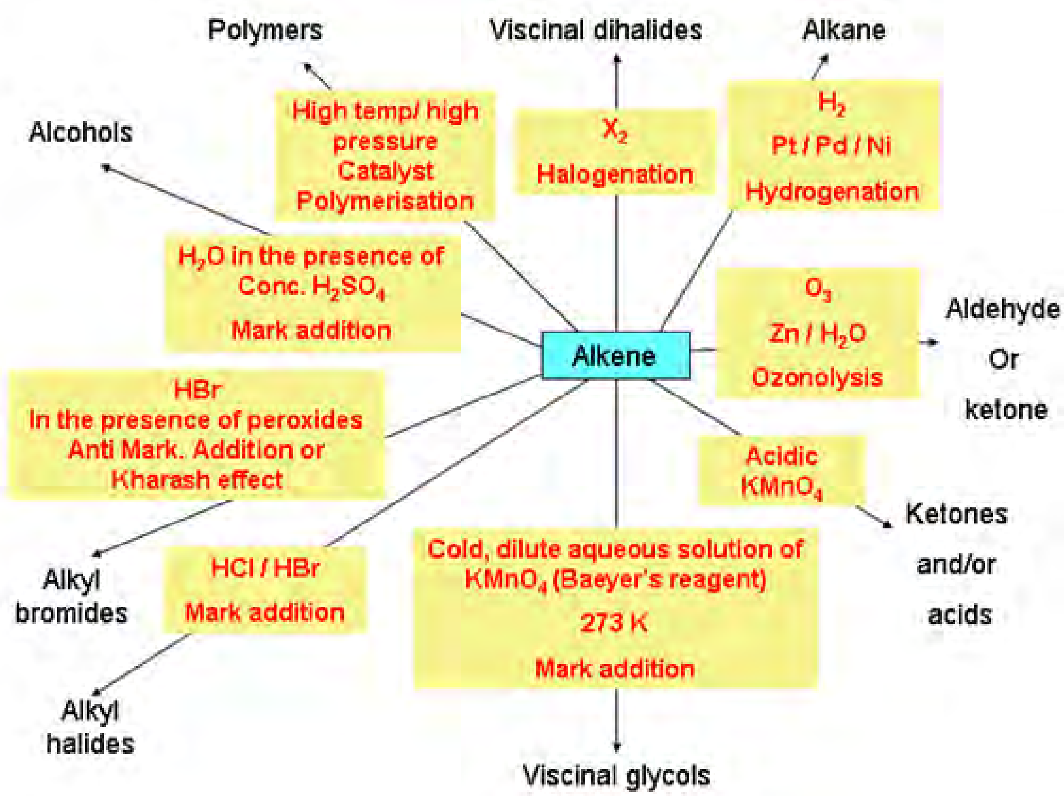
$$CH_3-CH=CH_2 + H_2SO_4 \longrightarrow$$

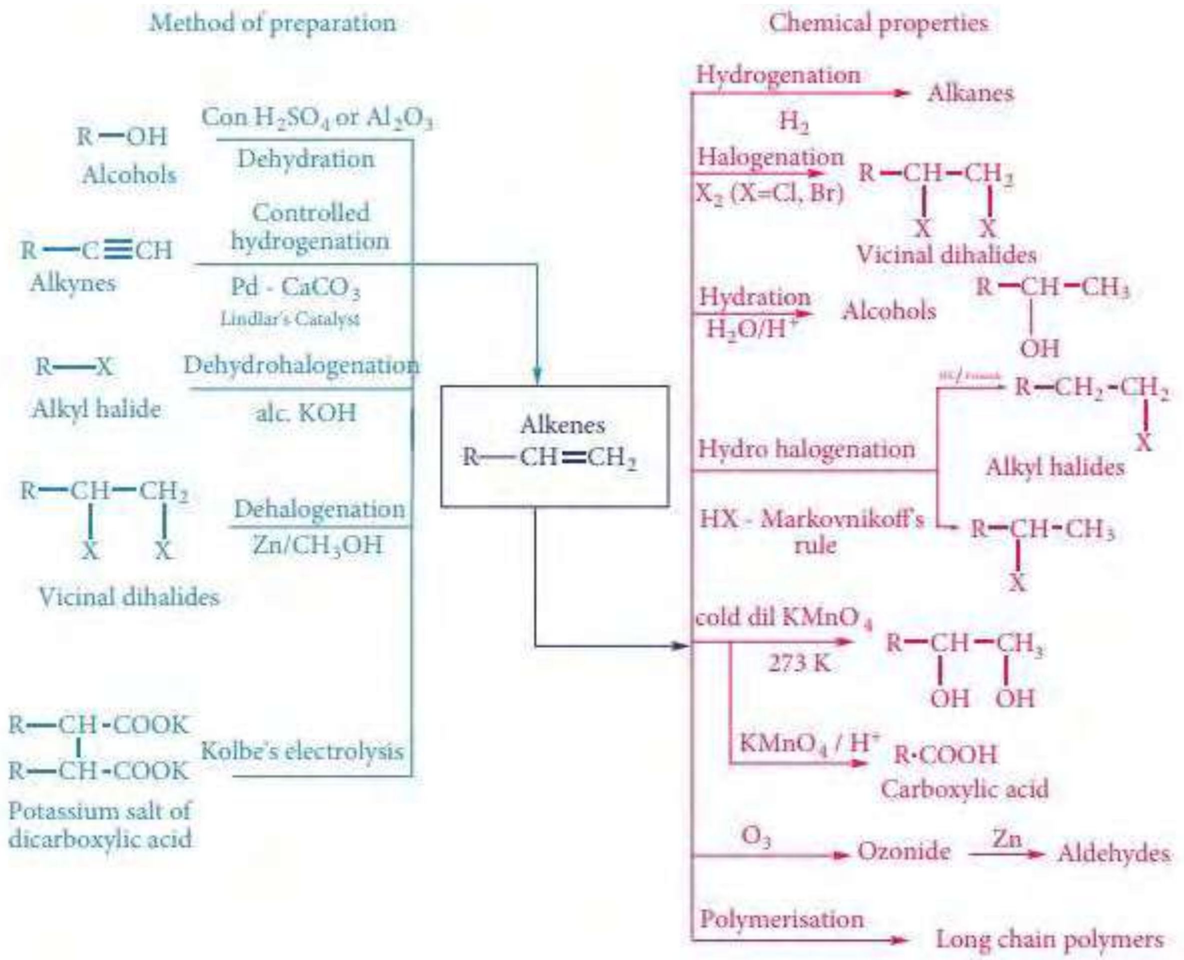
Propylene

$$CH_3-\underset{OSO_3H}{\underset{|}{CH}}-CH_2$$

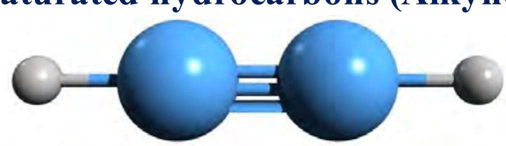
Isopropyl hydrogen sulphate

A summary of the reactions

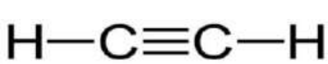




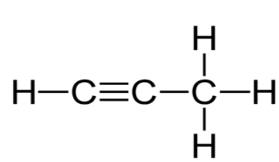
Unsaturated hydrocarbons (Alkynes)[C_nH_{2n-2}]



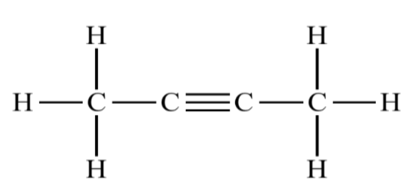
Study the following compounds and fill in the blanks of the following table.



Ethyne



Propyne

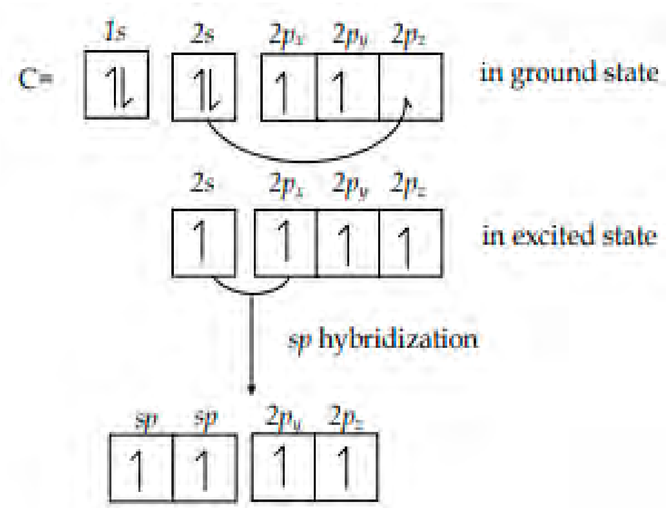


But-2-yne

Compounds	No. of carbon atoms	No. of hydrogen atoms	C_nH_{2n-2}(Yes/NO)
Ethyne	………….	……………	…………….
Propyne	…………..	……………	…………..
But-2-yne	…………..	…………….	………….

Structure of alkynes

Alkynes have a triple bond between two carbon atoms with sp hybridization. They have one sigma bond and two pi bonds. Sigma bond is formed by head to head overlapping of sp hybridized orbitals but pi bond is formed by sidewise overlapping of the p orbitals of two adjacent carbon atoms. The scheme of hybridization is as follows:



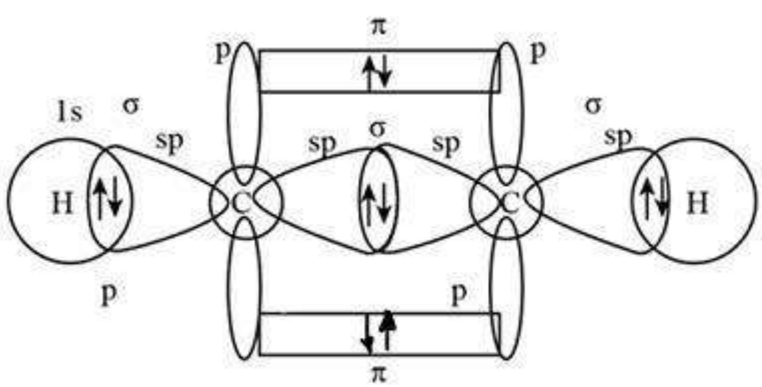


Fig 14.7: Orbital structure of acetylene molecule

Preparation

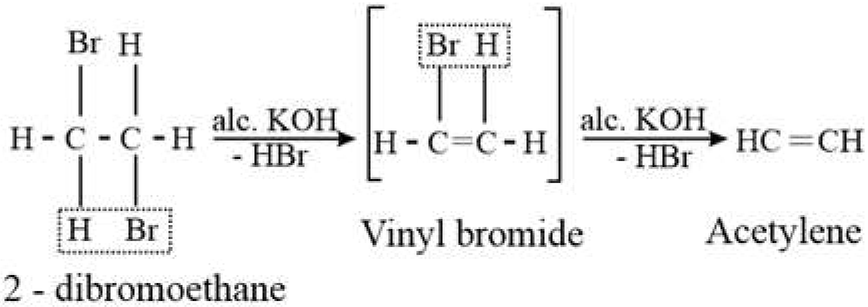
a. From carbon and hydrogen

For the preparation of ethyne, hydrogen gas is passed to make a hydrogen atmosphere in the tube fitting a carbon electrode fused with silica. This synthesis was carried out by Berthelot. Hence, called Berthelot's synthesis of ethyne.

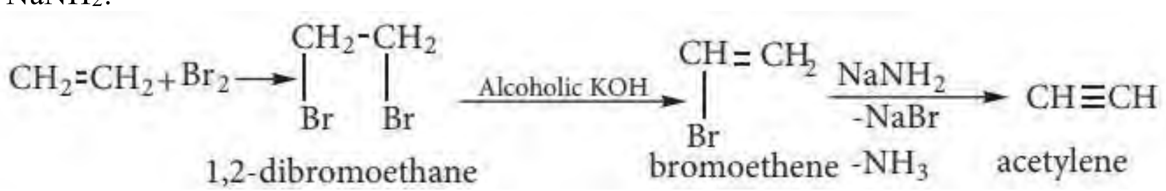
$$2C + H_2 \xrightarrow{\text{electric spark}} \underset{\substack{\text{Ethyne} \\ \text{(Acetylene)}}}{H\text{-}C{\equiv}C\text{-}H}$$

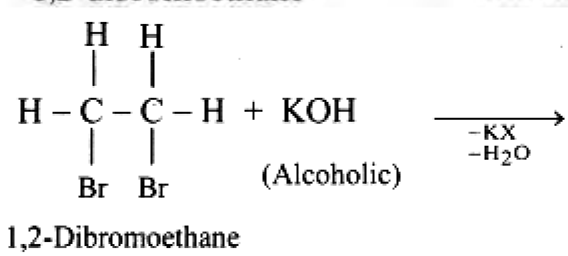
b. 1,2-dibromoethane (By dehydrohalogenation of vicinal dihalides)

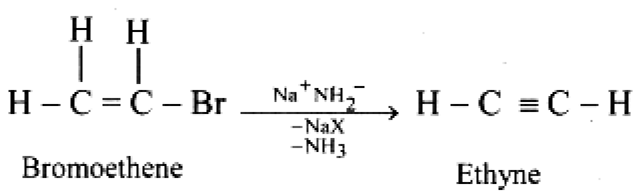
When vicinal dihalides are reacted with alcoholic potassium hydroxide elimination reaction is obtained by the removal of two halogens making the byproducts potassium bromide and water.

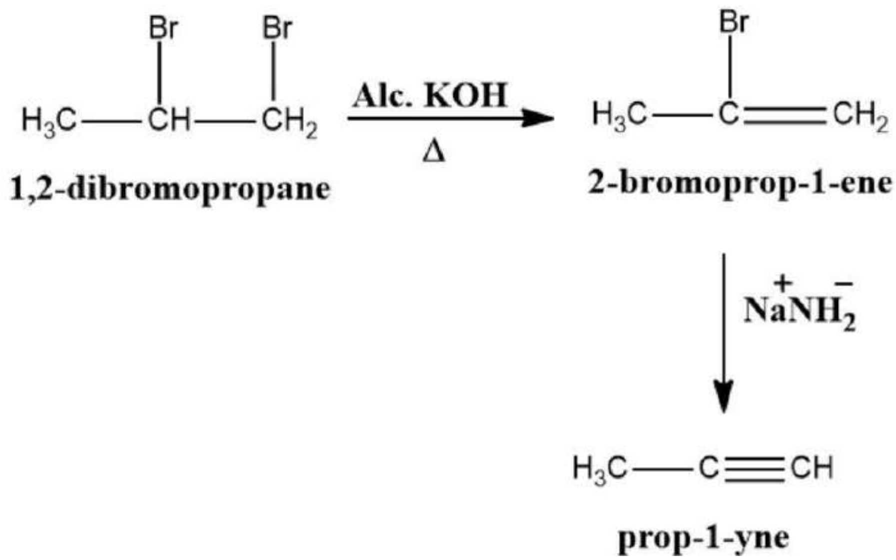


Note: alkynes can be obtained by the addition of alcoholic KOH followed by addition of $NaNH_2$.



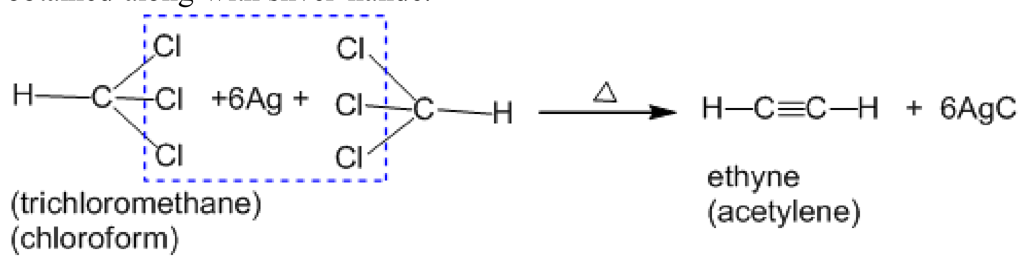


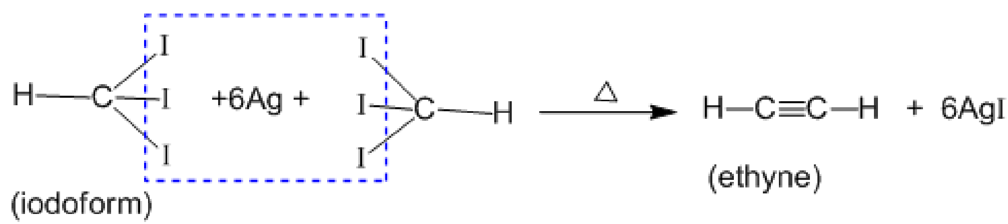




c. Chloroform/iodoform

When two molecules of chloroform are heated with silver powder, acetylene is obtained along with silver halide.





Chemical Properties of alkynes

Due to the presence of triple bonds in alkynes, they are very reactive as compared with other hydrocarbons. So, alkynes undergo an additional reaction easily.

a. Addition of hydrogen

What is happening in the following reaction?

$R\text{- }C\equiv CH + H_2 \quad Ni/\Delta \quad \rightarrow RCH{=}CH_2 + H_2 \quad Ni/\Delta \quad \rightarrow RCH_2CH_3$

Alkynes Alkene alkane

$HC\equiv CH + H_2 \quad Ni/\Delta \quad \rightarrow CH_2{=}CH_2 + H_2 \quad Ni/\Delta \quad \rightarrow CH_3CH_3$

Ethynes Ethene Ethane

$CH_3\text{ -}C\equiv C\text{-}CH_3 + H_2 \quad Ni/\Delta \quad \rightarrow CH_3CH{=}CH\text{-}CH_3 + H_2 \quad Ni/\Delta \quad \rightarrow CH_3CH_2CH_2CH_3$

But-2-yne But-2-ene Butane

Addition of hydrogen molecules occurs in the presence of nickel catalyst and heat. It gives alkene first, then the alkene is again reduced to alkane.

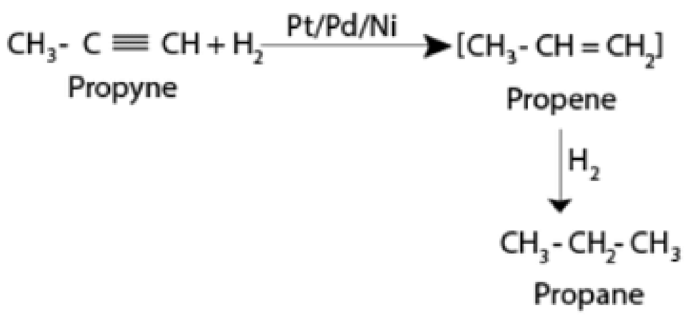
But when special catalysts like palladium and barium sulphate in presence of quinoline are used the reduction occurs single step i.e. alkene is obtained.

$R-C\equiv CH + H_2 \quad Pd/BaSO4/Quinoline \quad \rightarrow RCH{=}CH_2$

Alkynes   Alkene

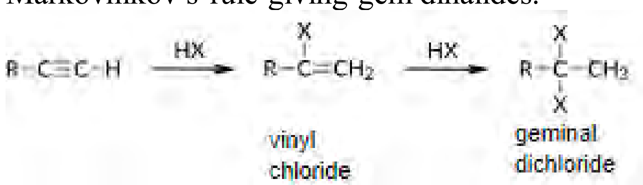
$CH_3-C\equiv C-CH_3 + H_2 \quad Pd/BaSO4/Quinoline \quad \rightarrow CH_3CH{=}CH-CH_3$

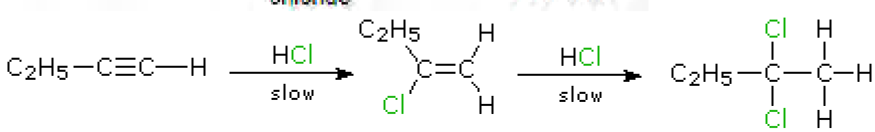
But-2-yne   But-2-ene

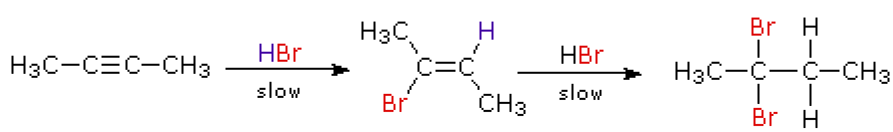


b. Addition of HX (HCl, HBr, HI)

Addition of hydrogen halide occurs by following two steps, i.e. in the first step haloalkene is obtained, in the second step the same molecule is added by following Markovnikov's rule giving gem dihalides.







Note: In unsymmetrical alkene addition of hydrogen halide follows Markovnikov's rule.

c. Addition of water (H_2O):

Alkynes react with water in presence of 1% mercuric sulphate and dilute sulphuric acid. The products thus obtained would be aldehyde or ketone by tautomerization of enols. The process is generally called hydration of akynes.

$RC\equiv CH + H_2O \xrightarrow{1\% HgSO4/dil.H2SO4} RC(OH){=}CH_2$

Ethyne ethene-ol(enol)

$RC(OH){=}CH_2 \xrightarrow{keto-enol\ tautomerization} RCO\text{-}CH3$

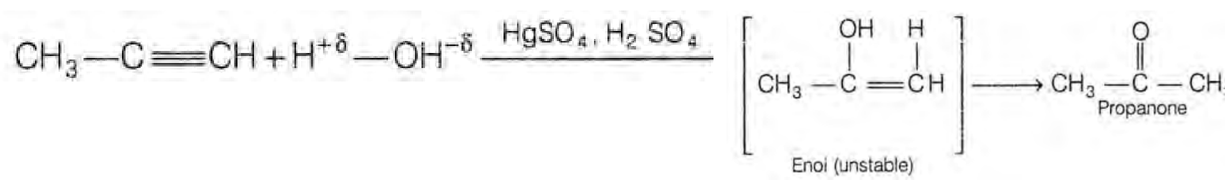
ethene-ol(enol) Alkanone

$HC\equiv CH + H_2O \xrightarrow{1\% HgSO4/dil.H2SO4} CH_2{=}CH\text{-}OH$

Ethyne ethene-ol(enol)

$CH_2{=}CH\text{-}OH \xrightarrow{Aldo-enol\ tautomerization} CH_3CHO$

ethene-ol(enol) ethanal



d. Acidic nature

Compounds	hybridization	% s character
$CH_3\text{-}CH_3$	sp3	25%
$CH_2{=}CH_2$	sp2	33%
$CH\equiv CH$	sp	50%

Higher the % s- character greater the positivity of the compound i.e. such compounds can lose proton or hydrogen (H^+) more easily. So, ethyne is most acidic among ethane, ethene and ethyne. Here are some examples to show the acidic nature of alkynes.

i. **Action with sodium**

$2CH\equiv CH + 2Na \xrightarrow{Liq.NH3} 2H\text{-}C\equiv C\text{-}Na + H_2$

Ethyne monosodium acetylide

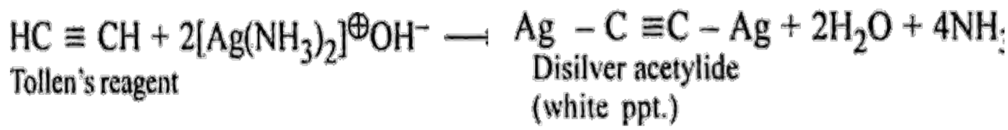
$2H\text{-}C\equiv C\text{-}Na + Na \xrightarrow{Liq.NH3} Na\text{-}C\equiv C\text{-}Na + H_2$

Monosodium acetylide di-sodium acetylide

When alkynes are treated with sodium metal it gives sodium acetylide and hydrogen gas. In this reaction sodium metal is base and acetylene is acid so salt and hydrogen gas are obtained.

ii. **Action with ammoniacal $AgNO_3$**

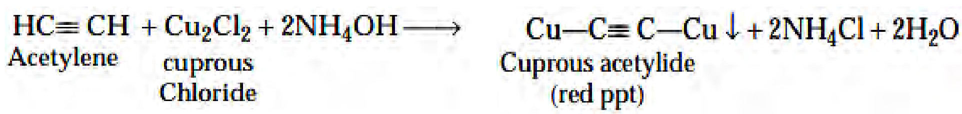
When alkyne is treated with ammoniacal silver nitrate solution (Tollen’s reagent) it gives disilver acetylide and water.



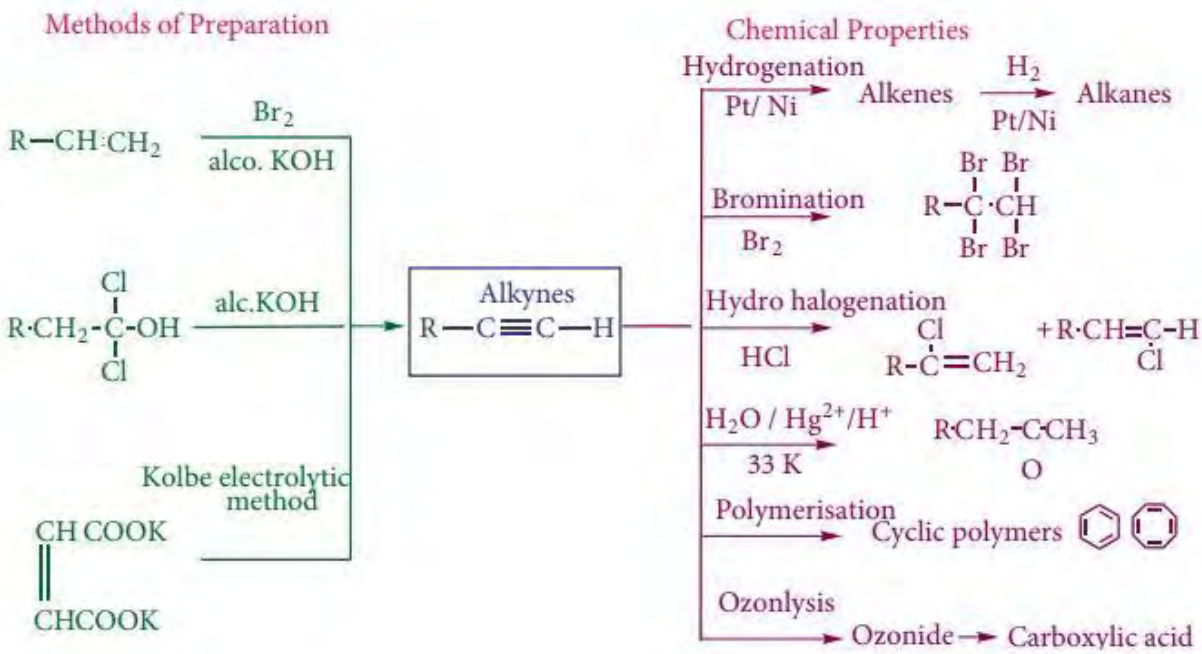
Tollen's reagent: $AgNO_3 + NaOH \rightarrow AgOH + NaNO_3$
Grey ppt.

$AgOH + NH_4OH \rightarrow [Ag(NH_3)_2]OH$(Colorless)
Tollen's reagent

iii. Action with ammoniacal Cu_2Cl_2

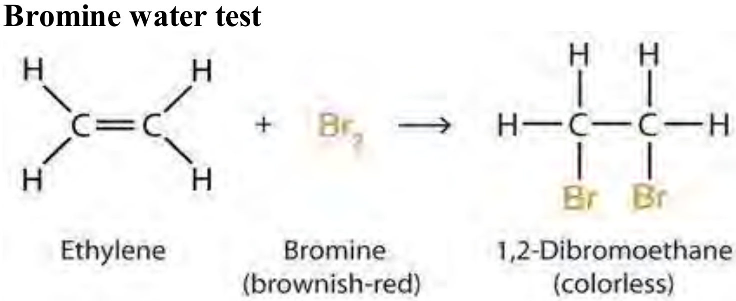


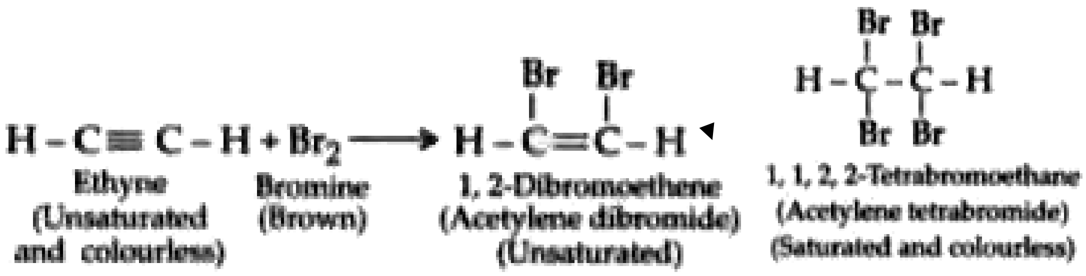
In this reaction acetylene is treated with cuprous chloride, which gives cuprous acetylide and water.



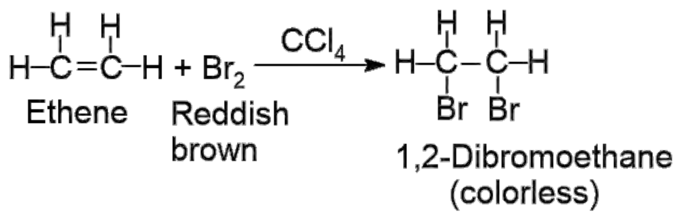
Test of unsaturation (Ethene and Ethyne)

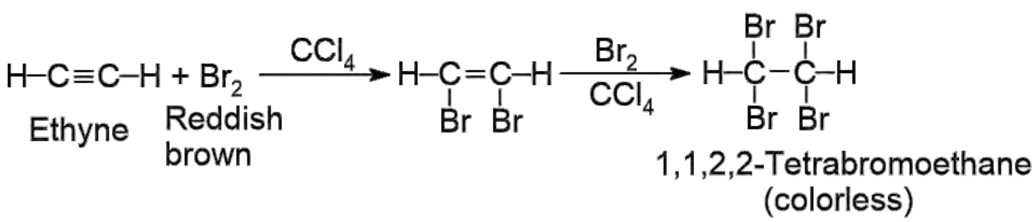
a. Bromine water test





b. Bayer's reagent test





Comparative Study of physical properties of alkane, alkene and alkynes

a. **State**

Alkanes: The alkanes containing up to carbon atoms four are in gaseous state, then upto carbon atoms 17 are in liquid state and higher alkanes are in solid state.

Alkenes: The alkenes upto carbon atoms 4 are in gaseous state, then up to carbon atoms fifteen are in liquid state the rest of the higher alkenes are in solid state.

Alkynes: The first three members of alkynes are in gaseous state then next eight members are in liquid state and other higher alkynes are in solid state.

b. Solubility

Alkanes: Insoluble in polar solvent like water and alcohol but soluble in non-polar solvents like petrol, ether, benzene, carbon tetra chloride, etc.

Alkenes: Insoluble in polar solvent like water and alcohol but soluble in non-polar solvents like petrol, ether, benzene, carbon tetra chloride, etc.

Alkynes: Slightly soluble in water but highly soluble in organic solvents like benzene, acetone, chloroform. etc.

c. Nature

Alkanes: Non acidic

Alkenes: Non acidic

Alkynes: Acidic

d. Hybridization

Alkanes: sp^3

Alkenes: sp^2
Alkynes: sp

e. **Density**

Alkanes: lighter than water their density increases with increase in molecular weight. Melting point boiling point is smaller when alkanes have branched chains.

Alknes: melting point, boiling points and specific gravity rises with increase in molecular weight.

Alkynes: melting point, boiling points and specific gravity rises with increase in molecular weight.

f. **Polarity**

Alkanes: non polar
Alkenes: slightly polar due to sp^2 hybridization
Alkynes: slightly polar due to sp hybridization

Kolbe's Electrolysis method of preparation of alkane, alkene and alkynes

a. **For alkanes**

When sodium salt of fatty acid or potassium salt of fatty acid is electrolyzed, two ions are formed. The carboxylate ion goes towards anode to give anodic reaction where it loses electron and carbon dioxide. Ultimately it gives alkane at anode.

$RCOONa + H_2O \xrightarrow{Electrolysis} RCOO^- + Na^+$

At cathode:

$Na^+ + e^- \rightarrow Na$

At Anode:

$RCOO^- - 1e \rightarrow RCOO$

$RCOO \rightarrow R. + CO_2$

$R. + R. \rightarrow R\text{-}R$

For example,

$$\underset{\text{(sodium acetate)}}{2\,CH_3COONa} \rightleftharpoons \underset{\text{(acetate ion)}}{2\,CH_3COO^-} + 2\,Na^+$$

At anode :

$$2\,CH_3COO^- - 2e^- \longrightarrow \underset{\text{(unstable)}}{2\,CH_3COO} \longrightarrow \underset{\text{(ethane)}}{CH_3\text{-}CH_3} + 2CO_2\uparrow$$

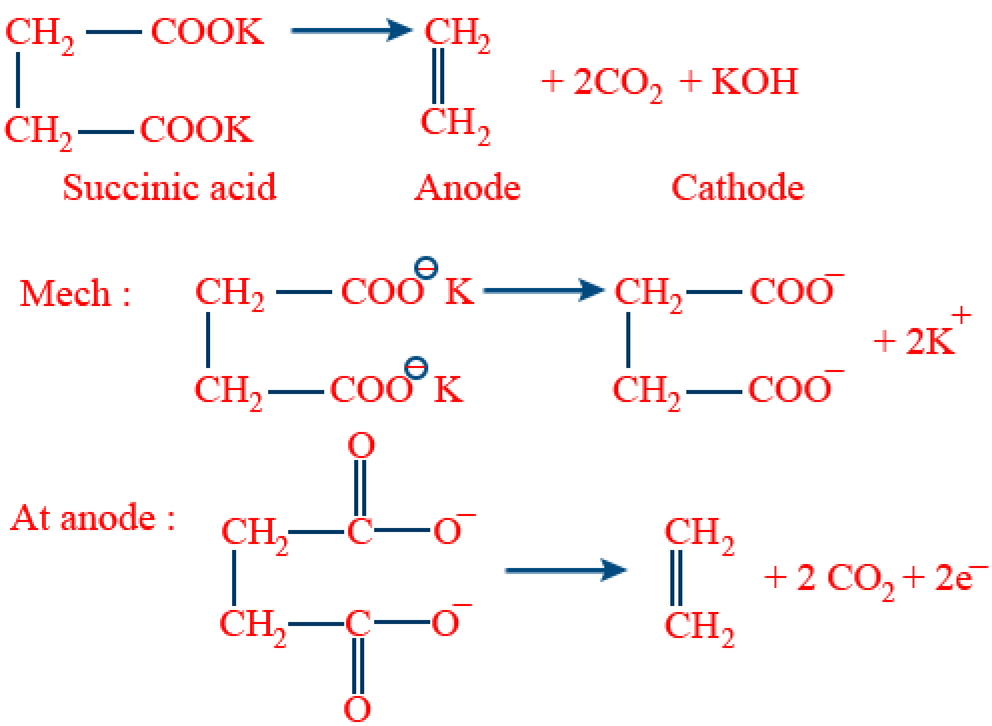
At cathode :

$$2\,Na^+ + 2e^- \longrightarrow 2Na$$

$$2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2\uparrow$$

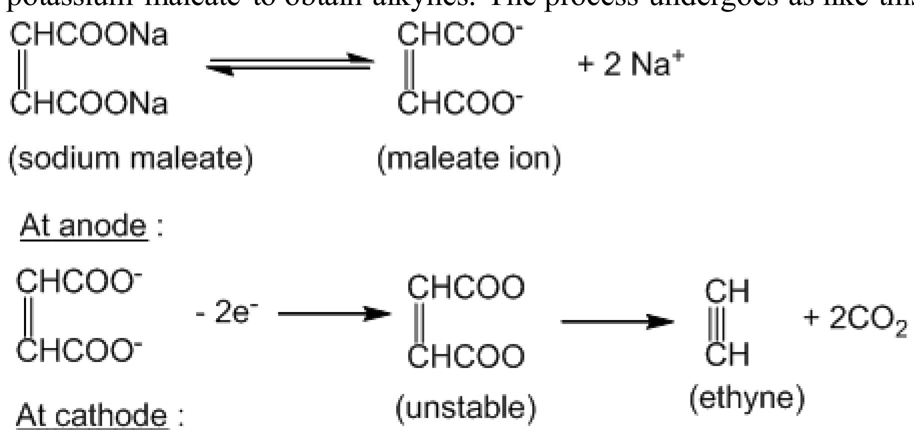
b. **For alkenes**

When potassium or sodium succinate is electrolyzed ethene is obtained as anodic reaction.



c. For alkynes

As we did electrolysis to prepare alkanes and alkenes, we can do electrolysis of sodium or potassium maleate to obtain alkynes. The process undergoes as like this.



At cathode :

$2\ Na^+ + 2e^- \longrightarrow 2Na$

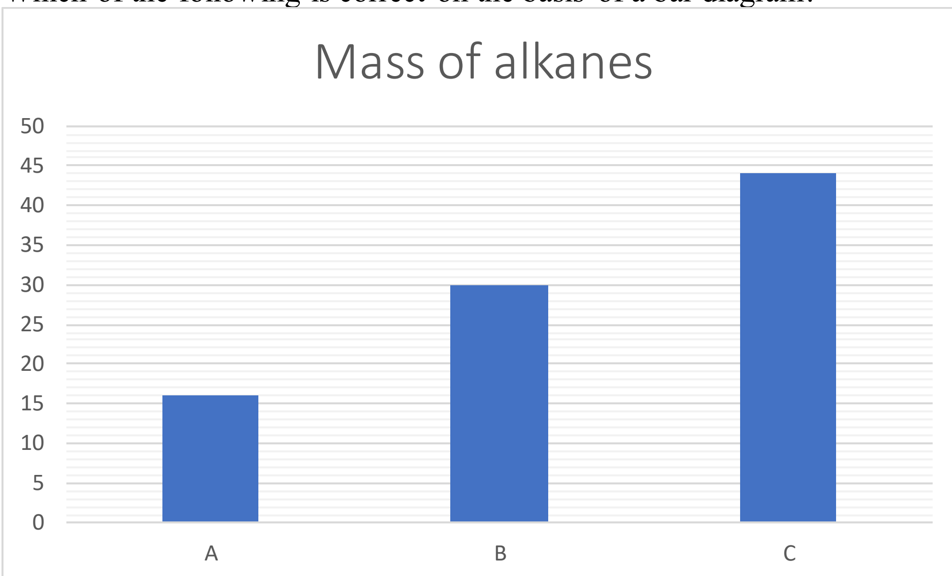
$2\ Na + 2\ H_2O \longrightarrow 2\ NaOH + H_2\uparrow$

Exercise

A. Multiple choice questions

1. Which type of hybridization does saturated hydrocarbon form?
   a. s^3p b. Sp c. sp^2 d. sp^3

2. Which of the following is correct on the basis of a bar diagram?



   a. A is ethane, B is propane and C is butane.
   b. A is methane, B is propane and C is butane.
   c. A is methane, B is ethane and C is propane.
   d. A is propane, B is butane and C is pentane.

3. Which of the following is alkane?
   a. C_nH_{2n+2} b. C_nH_{2n-2} c. C_nH_{2n} d. C_nH_{2n+1}

4. What is B in the following chemical equation?
   $CH_3COONa \quad B \quad \rightarrow CH_4 + Na_2CO_3$
   a. Sodium hydroxide b. Calcium hydroxide c. Soda-lime d. Soda-lime+heat

5. Which of the following chemical equations is a perfect example of Wurtz's reaction?
   a. $CH_3Cl + 2Na + Cl\text{-}CH_3 \; heat \rightarrow CH_3\text{-}CH_3 + 2NaCl$
   b. $CH_3Cl + Na + Cl\text{-}CH_3 \; heat \rightarrow CH_3\text{-}CH_3 + NaCl$
   c. $CH_3Cl + 2Na + Cl\text{-}CH_3 \; heat/dry\ ether \rightarrow CH_3\text{-}CH_2\text{-}CH_3 + 2NaCl$
   d. $2CH_3Cl + 2Na \; heat/dry\ ether \rightarrow CH_3\text{-}CH_3 + 2NaCl$

6. What type of hybridization is formed in alkene?
   a. Sp3d b. sp3 c. sp2 d. sp

7. What would be the product Y in the following sequence of the equation?
$CH_3CH_2Cl \xrightarrow{alc.KOH} Y + KCl + H_2O$
a. Ethanol b. Ethanal c. 2-chloro ethane d. Ethene

8. Choose the correct option in between the relation of following two columns.

A. Markovnikov' rule	1. $KMnO_4$/KOH
B. Wurtz's reaction	2. Ethanal and methanal
C. Ozonolysis	3. Butane
D. Baeyer's test	4. 2-chloro propane

a. A-1, B-2, C-3, D-4
b. A-4, B-3, C-2, D-1
c. A-3, B-1, C-4, D-2
d. A-2, B-4, C-1, D-3

9. Which of the following is correct related to the table?

Reactant(s)	**Condition(s)**	**Product(s)**
P	Alc. KOH	Ethene
But-2-ene	Q	Ethanal
Sodium ethanoate	NaOH+CaO + heat	R
Ethyne	S	Ethanal

a. P-chloroethane, Q- Zn/H_2O, R-Methane, S- dil H_2SO_4+$HgSO_4$
b. Q-chloroethane, R- Zn/H_2O, S-Methane, P- dil H_2SO_4+$HgSO_4$
c. R-chloroethane, S- Zn/H_2O, P-Methane, Q- dil H_2SO_4+$HgSO_4$
d. S-chloroethane, P- Zn/H_2O, Q-Methane, R- dil H_2SO_4+$HgSO_4$

10. Which of the following is a paraphrase of alkane?
a. Olefins
b. ethylene
c. acetylene d. paraffin

Short answer questions

1. How many isomers are there of the hydrocarbon C_4H_{10}? Write all possible isomers and write their IUPAC names.

2. Suggest a possible reason why the peroxide effect is observed for HBr, but not for HCl or HI. Why is Wurtz reaction not a good method for preparing alkanes containing an odd number of carbon atoms? Explain with examples.
3. Give an example of each of the following.
   a. Wurtz's reaction
   b. Kolbe's electrolysis
   c. Decarboxylation
   d. Markovnikov's rule
   e. Dehydrohalogenation
4. Identify the following compounds P, Q, R, S and T in the following conditions in which reactants, conditions and products lie in the same row.

Reactant(s)	**Condition(s)**	**Product(s)**
P	Alc. KOH	Propene
But-1-ene	Q	Propanal and methanal
Sodium propanoate	NaOH+CaO + heat	R
Propyne	S	Propan-2-one
T	Peroxide	1-chloro propane

5. Convert methane into ethyne.
6. Two containers contain but-2-ene and octane. How do you recognize the presence of these compounds by a chemical test? Explain with a chemical equation.
7. Differentiate between examples of each.
   a. Wurtz's reaction and decarboxylation reaction
   b. Markovnikov's rule and anti-markovnikov's rule
   c. Olefins and paraffin
8. What product would you obtain by the ozonolysis of following compounds?
   a. Ethene
   b. Propene
   c. But-2-ene
   d. 2-methyl but-2-ene
   e. 2,3-dimethyl but-2-ene
9. Addition reaction is impossible in alkanes and it is possible in alkenes and alkynes, but substitution reaction is possible in alkanes, explain with an example of each.
10. CH_3CH_3, $CH_2{=}CH_2$ and $CH{\equiv}CH$, among these three compounds which one is most acidic? Explain with examples.
11. Draw the orbital structures of methane, ethene and ethyne. Also mention the number and nature of bonds formed between carbon-carbon atoms on them.

Long answer Questions

1. Alkanes saturated hydrocarbons. How do we confirm this? Explain with examples. Alkanes are less acidic than that of alkynes, how do you elaborate this statement? What product would you obtain when ethane is under nitration and sulphonation individually? Explain with suitable examples.

2. Compare the any three physical and any five chemical properties of alkane, alkene and alkynes.

3. Study the following table in which three requirements for a complete chemical reaction are given. Write the structure and name of P, Q, R, S, T, U, V, W.

Reactants	**Conditions**	**Products**
P	Na/dry ether/heat	$CH_3CH_2CH_2CH_3$
$CH_2=CH_2$	Q	CH_3-CH_3
CH_3-CH_2Cl	alc.KOH	R
S	$HgSO_4/H^+$	CH_3CHO
CH_3CH_2OH	T	$CH_2=CH_2$
$CH\equiv CH$	Na/NH_4OH	U
V	HCl	$CH_3CHClCH_3$
W	NaOH/CaO	CH_4

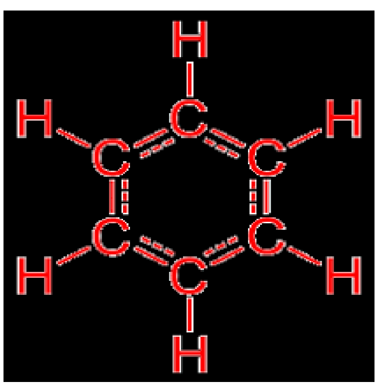
4. Convert methane into ethene?

5. What happens when:
   a. Ethane is treated with iodine?
   b. Ethane is treated with concentrated nitric acid?
   c. Ethane is treated with concentrated sulphuric acid?
   d. Chloroethane is heated with sodium metal in presence of dry ether?
   e. Ethylene is heated with hydrogen in presence of nickel?
   f. Ethane is combusted with air?
   g. Sodium acetate is heated with soda-lime?
   h. Chloroethane is heated with zinc and HCl?

Project work

Make a list of alkane, alkene and alkyne used in our daily life by the help of Google or from the internet and submit it to your teacher by the next week. The format is given below.

SN	Name of the compound	Alkane/alkene/alkyne	Uses
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			



UNIT 15: AROMATIC HYDROCARBONS

15.1 Introduction and characteristics of aromatic compounds

Activity: 1 Observe the nature of flames produced in the following two figures. What is the difference that you find? Note down in the given box.

Friedrich August Kekule, later Friedrich August Kekule von Stradonitz, was a German organic chemist. From the 1850s until his death, Kekulé was one of the most prominent chemists in Europe, especially in theoretical chemistry.

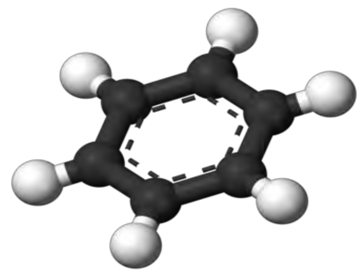
Born: September 7, 1829, Darmstadt, Germany

Died: July 13, 1896, Bonn, Germany

Education: University of Giessen (1847–1852)

Full name: Friedrich August Kekule von Stradonitz

Nationality: German



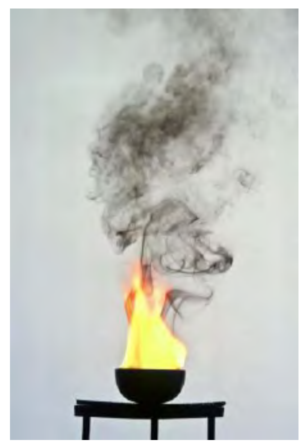


Figure 15.1 Bluish flame

Figure 15.2 Dense smoke

Findings:

Flame of figure 1:…………………………………

Flame of figure 2:…………………………………

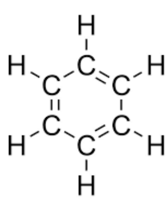
Activity: 2 Take a little amount of benzene in a test tube. Take the smell of it. Does it have a different smell than other solutions available in chemistry laboratories; NaOH, NH_4OH, water, etc.?

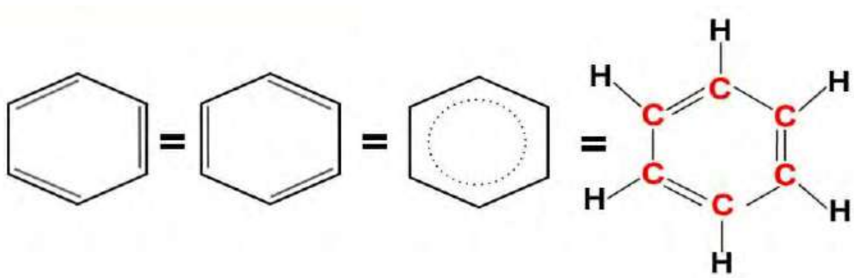
Figure 15.3: Sweet smelling flower

Atomatic compounds give a sooty flame (yellowish white flame with dense smoke) due to high degree of unsaturation in the ring and have a fragrance smell (physical test). So these compounds are called aromatic. i.e. aroma- sweet smell.

Aromatic hydrocarbons are also known as arenes. The hydrocarbon having a ring of carbon atoms with alternate single and double bonds and have delocalized π-electrons (π-electrons moving in the ring continuously) within the ring are called arenes. It is difficult to cause an addition reaction in arenes due to delocalized π-electrons.

Out of many aromatic compounds, the most important aromatic compound is benzene (C_6H_6). Benzene has the structure as given in the following figure.





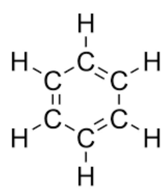
15.2 Huckel's rule of aromaticity

At first, theories of aromaticity were Robinson's aromatic sextet and valence bond theory. Now, the most explicit theory is evolved as Huckel's theory given by German Chemist and Physist Erich Huckel. According to this theory, a planar cyclic system with alternate single bond and multiple bond satisfies the equation;

$4n+2=\pi$ electrons. From this equation, the value of should be in the whole number i.e. n=0,1,2,3,4….. In the cyclic system, all the carbon atoms should be sp^2 or sp hybridized.

For example,

1. Benzene



All the carbon atoms are sp^2 hybridized. Each double bond has 2 π electrons. So, in benzene there are 6 π electrons. Substituting the value in the given equation.

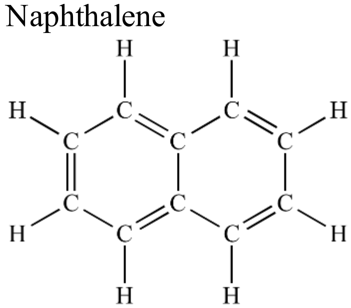
$4n+2=\pi$ electrons.

Or, $4n+2=6$

Or, $4n=4$

Or, n=1, so benzene is aromatic in nature.

2. Naphthalene



All the carbon atoms of naphthalene are sp^2 hybridized. Each double bond has 2 π electrons. So, in naphthalene there are 10 π electrons. Substituting the value in the given equation.

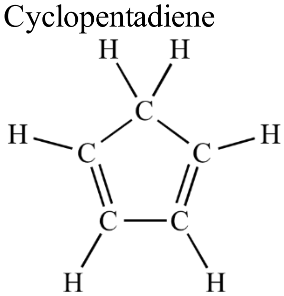
$4n+2=\pi$ electrons.

Or, $4n+2=10$

Or, $4n=8$

Or, n=2, It is whole number. So, naphthalene is aromatic

3. Cyclopentadiene



Cyclopentadiene has 4 carbon atoms are sp^2 hybridized. Each double bond has 2 π electrons. So, it has 4 π electrons. Substituting the value in the given equation.
$4n+2=\pi$ electrons.
Or, $4n+2=4$
Or, $4n=2$
Or, n=1/2, it is rational number. So, cyclopentadiene is not aromatic. in nature.

<u>Key points of aromatic hydrocarbons:</u>

- Should have cyclic structure
- Should have single bond and multiple bonds (double or triple) in alternate position. In some cases, lone pair of electrons are sufficient to delocalize electrons instead of double or tripple bond.
- Should follow Huckel's rule of aromaticity

15.3 Kekule's Structure of Benzene

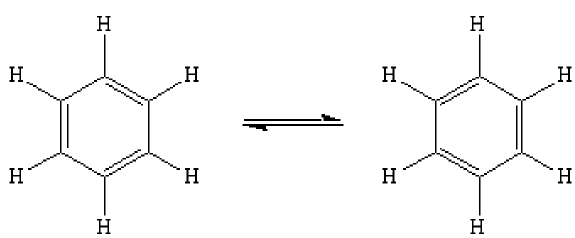
Benzene is an aromatic compound which took a long time to for confirmation of its structure. **August Kekule von Stradonitz**, original name **Friedrich August Kekulé**, (born Sept. 7, 1829, Darmstadt, Hesse-died July 13, 1896, Bonn, Ger.), German chemist who established the foundation for the structural theory in organic chemistry.

The structure of benzene has been established on the basis of following facts.

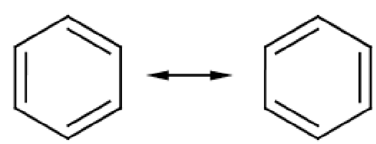
a. Determination of molecular weight and elemental analysis indicate that the molecular formula of benzene is C_6H_6.

b. Comparing the molecular formula of benzene with saturated hydrocarbon, hexane (C_6H_{14}) indicates that there is deficiency of eight hydrogen atoms which indicates the unsaturation in benzene i.e. presence of double or tripple bond.

   So, he proposed the structure of benzene as a ring with six carbon atoms and six hydrogen atoms.



Or

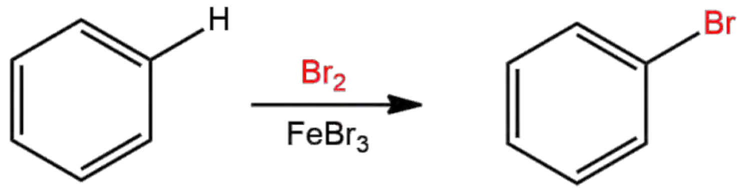


Kekule's Structure of benzene

Facts to support Kekule's structure of benzene

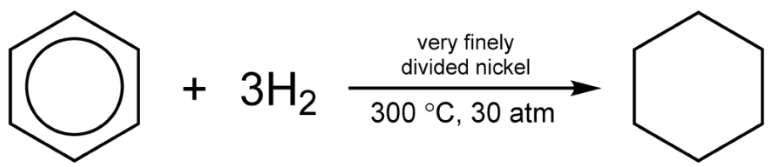
1. Kekule's structure of benzene has six equivalent hydrogen atoms. It should give only one monosubstituted product. In actual practice too it gives mono substituted products.

Think a while:



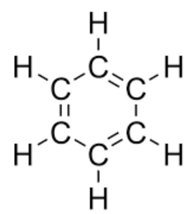
Benzene *Bromobenzene*

2. During hydrogenation(reduction) of benzene, three molecules (six atoms) of hydrogen are added into it and give cyclohexane. It infers benzene consists of six hydrogen atoms only.



3. Benzene does not give unsaturation tests like Baeyer's test and bromine water test due to delocalization of electrons in the ring. But aliphatic unsaturated hydrocarbons give such tests.

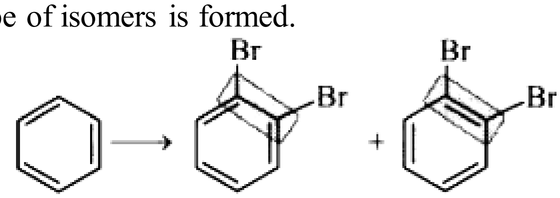
$CH\equiv C\text{-} CH_2\text{-}CH_2\text{-}C\equiv CH$
Open chain structure(I)



Ring Structure(II)

Drawbacks (objections) of Kekule Structure of benzene

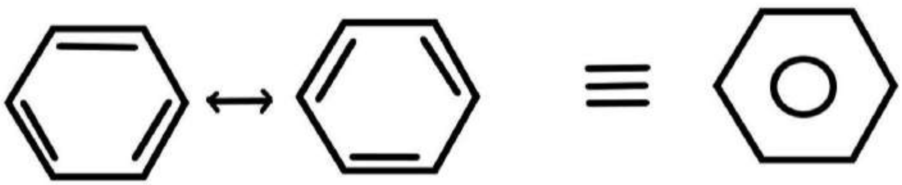
1. Benzene is stable towards an additional reaction. If structure II is correct, it should give an additional reaction easily.
2. If there are single bonds ($1.54A^0$) and double bonds ($1.34A^0$) in the ring, there should be two types of bond lengths in benzene but it is not found so. It is found only intermediate of them i.e. $1.39A^0$.
3. If the structure is correct, there should be two types of disubstituted isomers i.e. one should be two halogen substitutions at single bond and another at double bond. But in practice only one type of isomers is formed.



15.4 Resonance and Isomerism

Resonance Theory

Activity: 3 Are these first two structures of benzene similar? Observe and find, in which respect they are similar or different.



Two equivalent resonating structures of benzene. Resonance hybrid

Finding 1: ……………………………………………

Finding 2……………………………………………..

If you shift the double bond of the first structure in an adjacent position, the second structure of benzene is obtained. This is due to delocalization of electrons in the benzene ring. So, these two structures are similar.

a. Resonance hybrids are more stable than resonating structures. Since both the first and second structures are exactly equivalent. Benzene is quite stable towards an additional reaction due to resonance.
b. In both structures, each structure has equal contribution in resonance hybrid. Therefore, in benzene all carbon-carbon bonds are identical and are intermediate in length between double and single bonds.

Molecular Orbital Structure of benzene to support Kekule's Structure

Carbon has two half-filled orbitals in ground state. $1s^2$, $2s^2$ $2p^2$

In an excited state, carbon has four half-filled orbitals. $1s^2$, $2s^1$ $2p_x^1$ $2p_y^1$ $2p_z^1$.

Now, 2s and 2p orbitals undergo hybridization to form sp2 hybridization. Each sp2 hybridized orbitals are equivalent in energy and identical in shape and size with trigonal planar geometry having bond angle 120^0. Other information on benzene is given in following orbital structure.

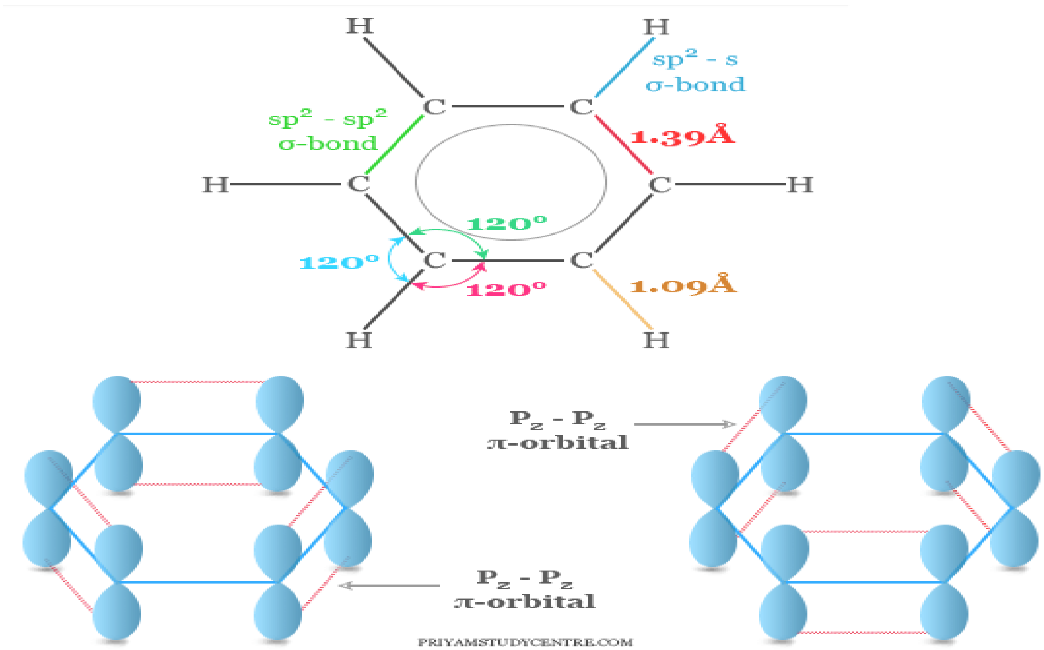


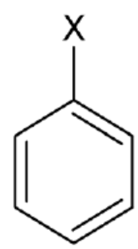
Figure 15. 4 Orbital Structure of benzene

For detail watch the video in the link:
https://www.youtube.com/watch?v=EgQAOSRfz3E

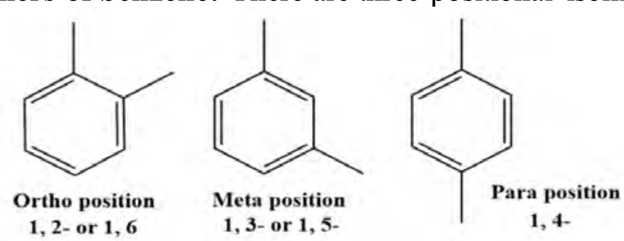
Isomerism in benzene

The organic compounds having the same molecular formula but of different structure are called isomers. The property shown by the compound is called isomerism.

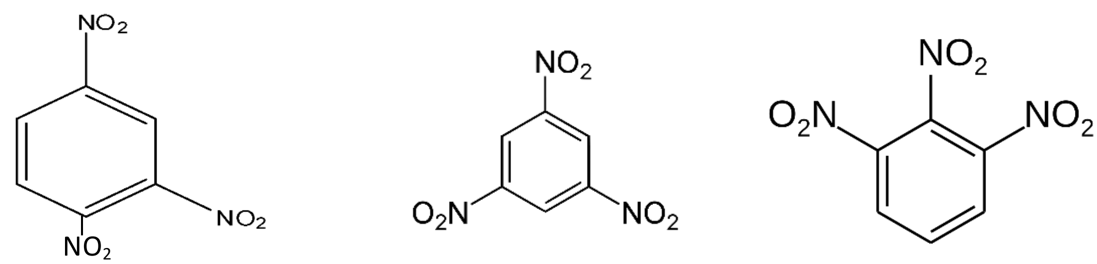
1. **Monosubstituted isomers of benzene:** all six carbon atoms of benzene are equivalent. So, replacement of any of six hydrogen atoms gives only one mono-substituted benzene.



2. **Disubstituted isomers of benzene:** There are three positional isomers of benzene.



3. **Tri-substituted isomers of benzene:** Number of isomers depends on the nature of substituents. If all three substituents are the same, three isomers are possible. They are vicinal, asymmetrical and symmetrical.

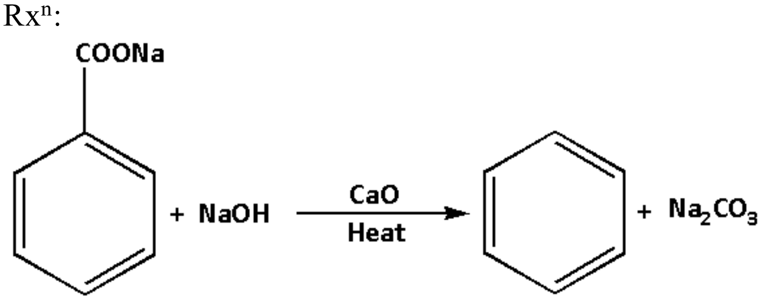


(1,2,4-trinitrobenzene unsymmetrical) (1,3,5-trinitrobenzene, symmtrical) (1,2,3-trinitrobenzene, vicinal)

15.5 Preparation of benzene

a. **By decarboxylation of sodium benzoate:** When sodium benzoate is heated with soda-lime (NaOH+CaO) benzene is obtained by removing –COONa. In this reaction, one carbon atom is lost from the substrate reagent. So, it is called a decarboxylation reaction.

Rx^n:



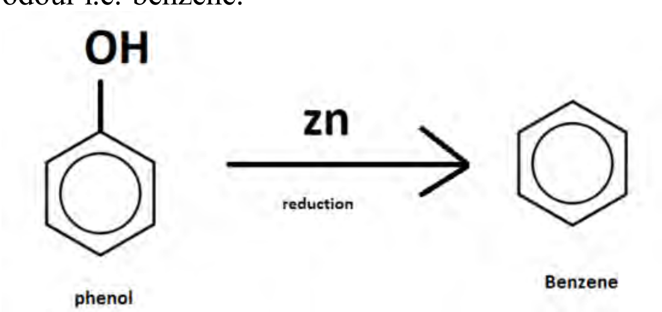
Requirements:

Sod. Benzoate
Soda-lime
heat

b. **By heating phenol with zinc dust**

Activity: 4

Take about 5 ml of phenol in a test tube. Add a piece of zinc in it. Heat the mixture. After sometime you will see pale yellow liquid with pleasant gasoline like petroleum odour i.e. benzene.



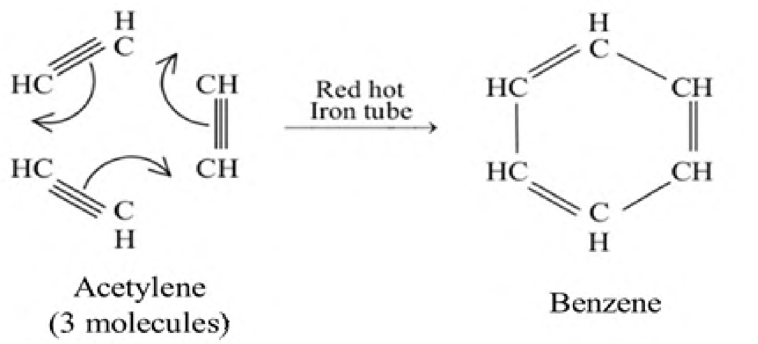
Requirements:

Phenol
Zinc-dust
heat

c. **By aromatization or polymerization of ethyne**

When three molecules of ethyne or acetylene are passed into iron tube or cu-tube, at about 500^0C, under pressure of about 25 atm, benzene is obtained by polymerization. Aliphatic compounds are cyclized. So, this reaction is also known as cyclization or aromatization reaction.

$3CH\equiv CH \ \ Cu/Fe - tube\ 500^0C/Pressure \rightarrow \qquad C_6H_6$



<u>Requirements:</u>

Acetylene
Fe/Cu-tube
Temperature
about 500^0C
Pressure
about 25atm.

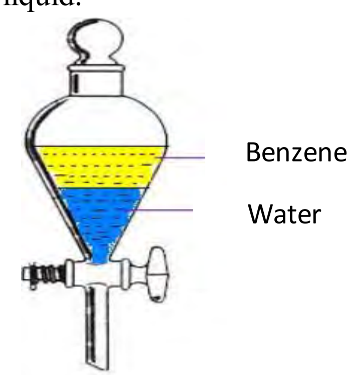
Question:

a. How many molecules of ethyne are taking part to complete the reaction?

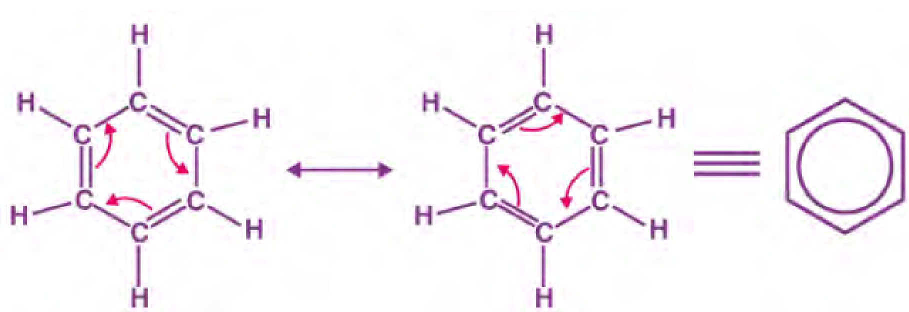
b. What is the process to obtain benzene from acetylene called?

15.6 Physical Properties of Benzene

1. Benzene is a colorless, aromatic odour and volatile liquid.
2. Benzene is immiscible in water but soluble in organic solvents.
3. Boiling point of benzene is 80.09^0C i.e. ~ 80.1^0C and melting point is 5.53^0C.
4. It has a density of $0.87 g\ cm^{-3}$.



5. Benzene shows resonance.



6. It is highly inflammable and burns with a sooty flame.

Test Yourself!

Have you ever seen such flames? If you have seen, where have seen?

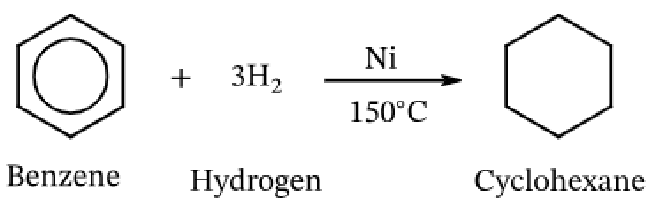
15.7 Chemical Properties of benzene

A. Addition reaction

Benzene consists of double bonds in 3 places in the ring. Due to delocalization of π-electron in the benzene ring, it does not give an additional reaction. In vigorous conditions it gives an additional reaction.

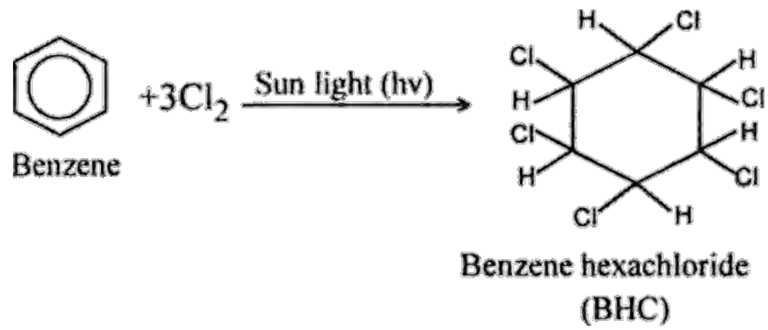
i. **Addition of hydrogen:** When vapor of benzene is mixed with hydrogen in presence of Nickel or palladium catalyst at about 250^0C under pressure, three molecules of hydrogen get added to the benzene molecule to give cyclohexane. In condensed formula the reaction can be written as:

$$C_6H_6 + 3H_2 \xrightarrow{Ni\ or\ Pt, 250^0C} C_6H_{12}$$



What are the differences in the structure of benzene and cyclohexane? Find any two.

ii. **Addition of halogen:** When benzene is treated with halogen in the presence of sunlight, three molecules of chlorine get added to the benzene ring and benzene hexachloride is obtained. It is an insecticide.

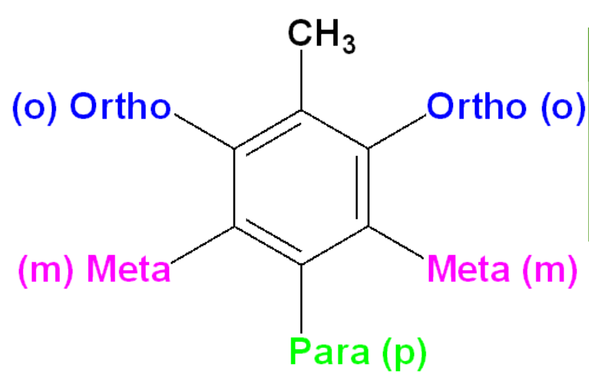


In condensed form the reaction is written as;
$C_6H_6 + 3Cl_2 \ Sunlight(hv) \rightarrow \ C_6H_6Cl_6$

B. Electrophilic substitution reaction

i. Orientation of benzene derivatives (o, p, m)

The position o, m, and p are labelled in reference with the particular group substituted in the benzene ring. For eg. In toulene, with reference to the methyl group, ortho(o), para(p) and meta(m) have been labelled. The adjacent carbon or C-2 from the reference carbon, at which substituent group substituted is called ortho carbon and the position is called ortho position. C-3 from the reference carbon is called meta carbon as well as C-4 is called para carbon. The orientation means, the site for substitution of incoming groups (charged ions or radicals) in the benzene ring.



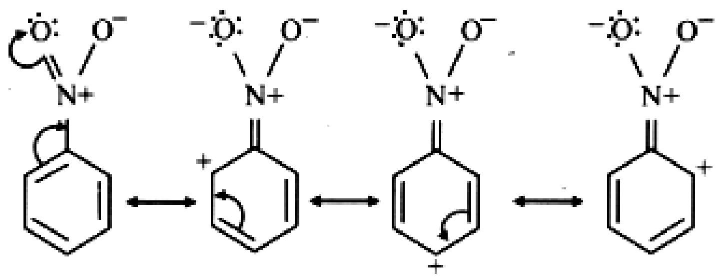
Electrophilic substitution:

It is the replacement of an atom (for eg. H) by another group of atoms or atom by attacking electron deficient species

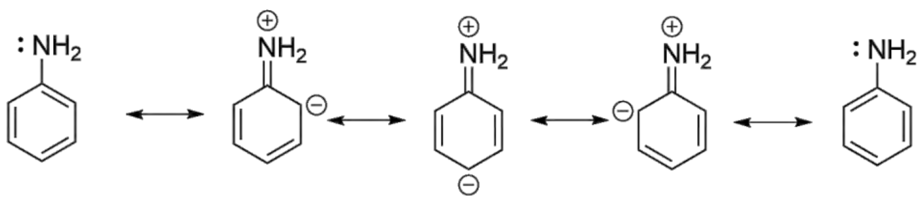
o and p directors(ring activator)	m-directors(ing deactivator)
$-NH_2$, -OH, -OR, -O-, -NHR, $-NR_2$, -NHCOR etc.	$-NO_2$, -COOH, -CN, $-SO_3H$, -CHO, -COR, -COOR etc.

Activity

Observe the following resonating structures and identify ring activator and ring deactivator group which is substituted in benzene ring.



Resonance structures of nitrobenzene

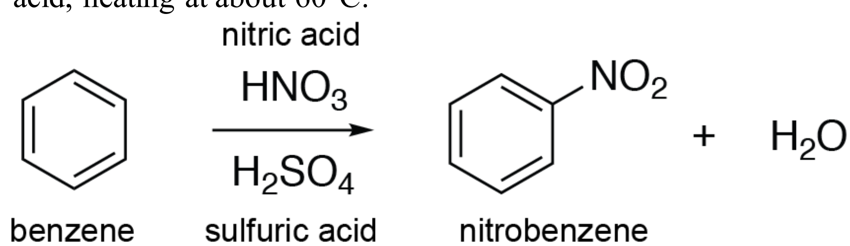


1 2 3 4 5

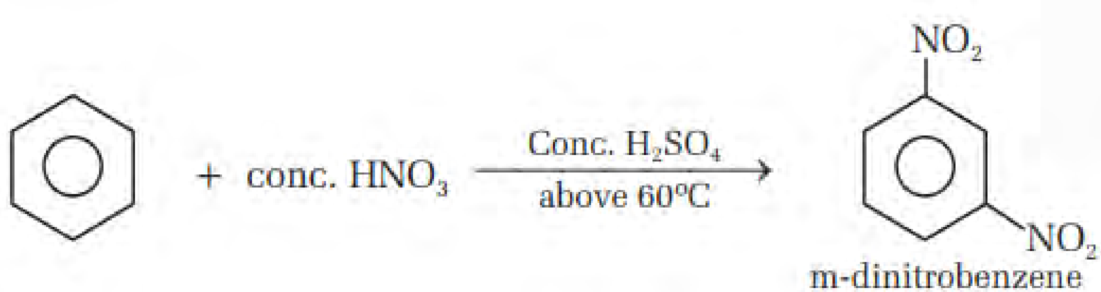
Resonating Structures of Aniline

ii. Nitration of benzene

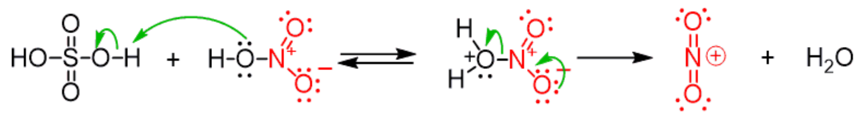
Introduction of $-NO_2$ group into benzene is called nitration of benzene. In this reaction the electrophile is the NO_2^+ ion, known as the nitronium ion (nitryl cation). This ion is produced from the mixture of concentration nitric acid and concentrated sulphuric acid, heating at about 60^0C.



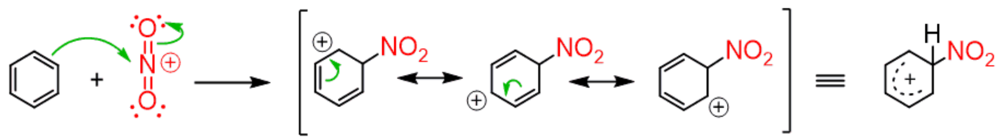
Note: If heated above 60ºC (i.e., 80–100ºC), meta-dinitrobenzene (1,3-dinitrobenzene) is formed.



Mechanism:(for concept; optional)

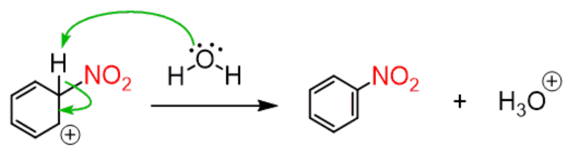


Nitronium ion



Addition of the electrophile   Sigma Complex   hybrid

Loss of a proton - restoring the aromaticity

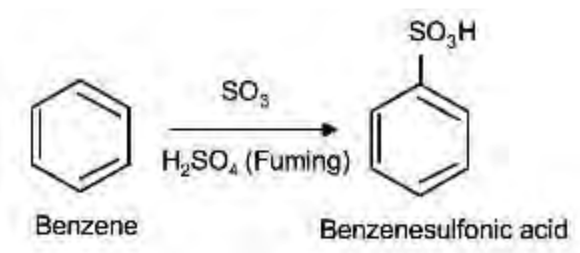


nitrobenzene

iii. Sulphonation

As nitration sulphonation also goes through the same process. SO_3 acts as an elctrophile and gets substituted in the benzene ring.

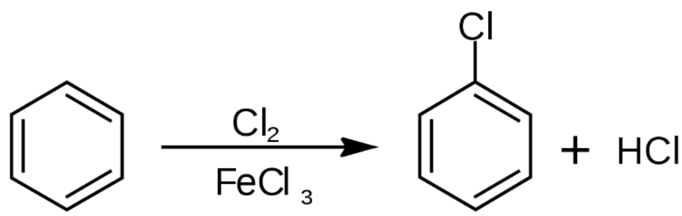
What happened in the following reaction? Do you know?



+ H_2O

iv. Halogenation

When benzene is treated with halogen (Cl_2 or Br_2) in presence of halogen carriers such as Lewis acid of iron ($FeCl_3$ or $FeBr_3$) at dark condition, an atom of halogen is substituted at the benzene ring. This process is known as halogenation. For example: chlorination of benzene.



v. **Friedel- Craft's reactions**

The reaction involves introduction of an ***alkyl group*** or an ***acyl group*** in the benzene ring in presence of anhydrous aluminium chloride as a Lewis acid. The reaction was proposed by Charles Friedel and Jame Mason Craft in 1877. The reactions are studied in two different types.

Charles Friedel James Mason Craft

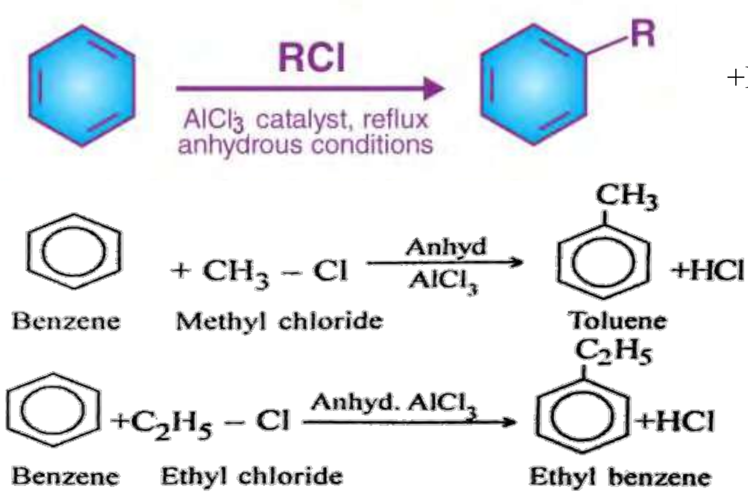
They are:

a. **Friedel-Craft's alkylation**

b. When benzene is heated with alkyl halide in presence of anhydrous $AlCl_3$, BF_3, $FeCl_3$, etc. ***alkyl group(-R)*** is added to the benzene ring. So, alkyl benzene is obtained. If methyl halide is used, methyl benzene is obtained, as well as if ethyl halide is used ethyl benzene is obtained and so on. Some examples are as follows.

Requirements:

An. $AlCl_3$
Alkyl halide
Benzene
Heat



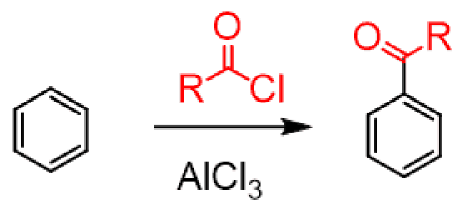
+HCl

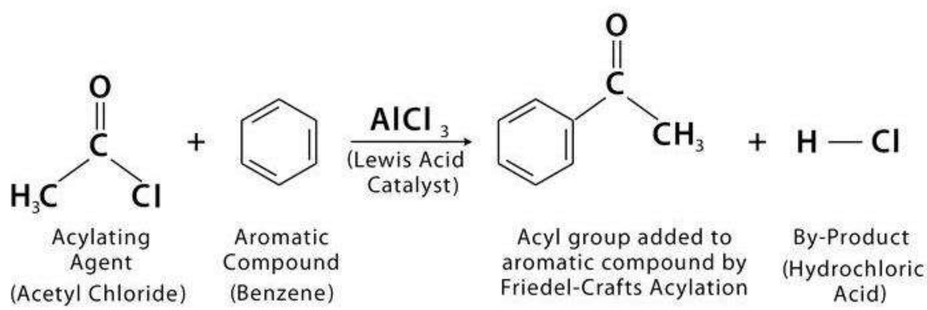
c. **Friedel-Craft's acylation**

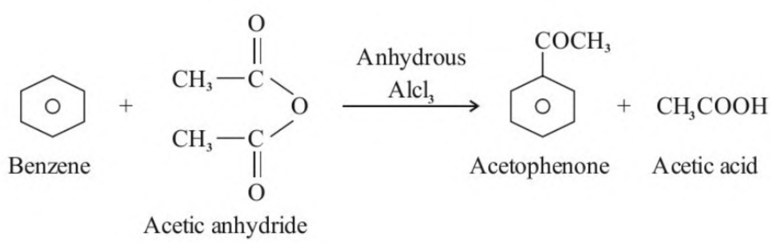
When benzene is heated with ac
yl halide in presence of anhydrous $AlCl_3$ and ether ***acyl group(RCO-)*** is added to the benzene ring. So, acyl benzene is obtained. If acetyl halide is used, acetophenone is obtained, as well as if acetic anyhydride is used acetophenone is obtained as a major product and so on. Some examples are as follows.

<u>Requirements:</u>

An. $AlCl_3$
Acyl halide
Benzene







vi. **Combustion of Benzene (free combustion)**

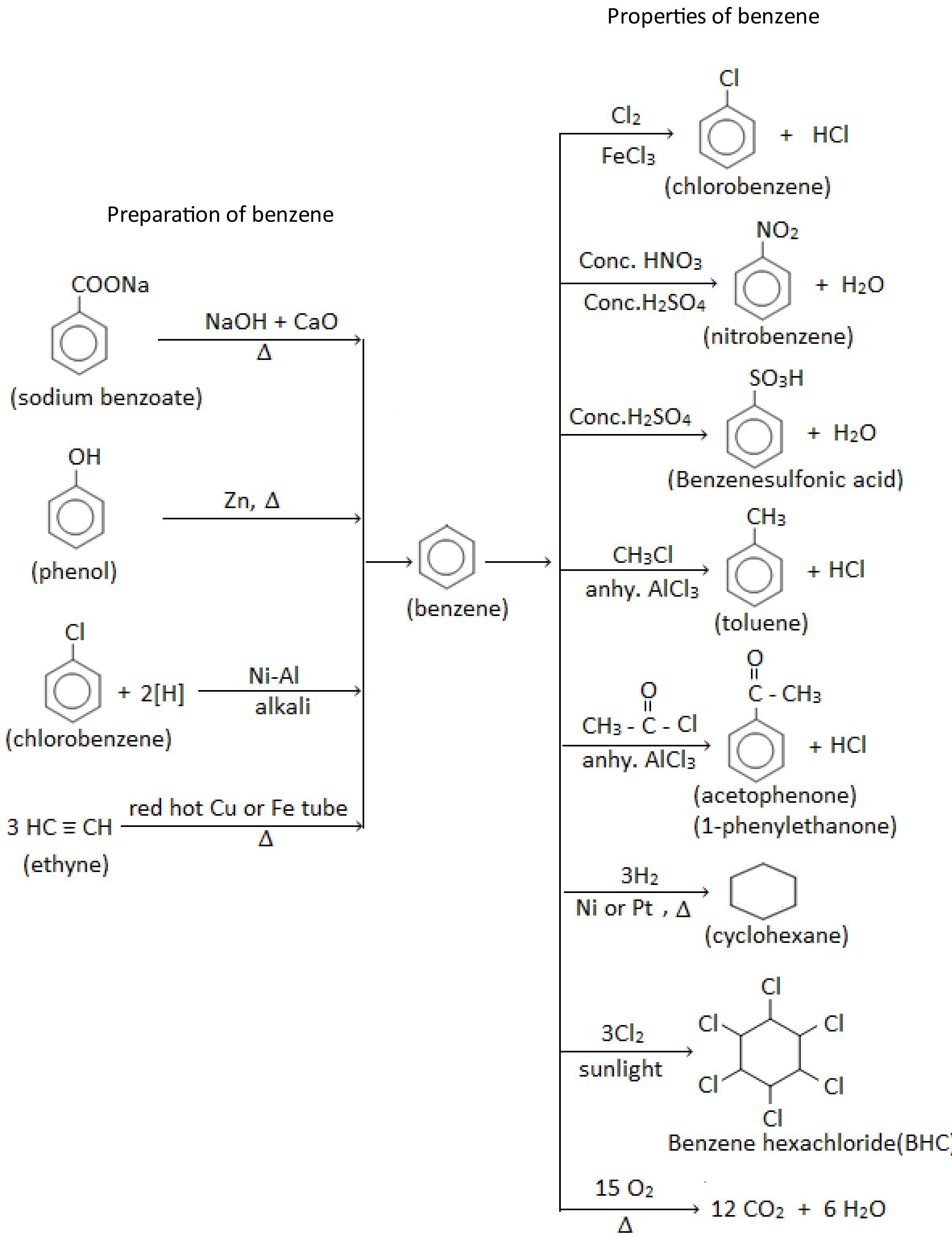
When benzene is combusted in the presence of air, it burns with a yellow luminous sooty flame and is oxidized into carbondi-oxide and water.

$$2C_6H_6 + 15O_2 \xrightarrow{Combustion} 12CO_2 + 6H_2O$$

Uses of Benzene

a. Used as organic solvent.
b. Used as solvent for the extraction of fats and oils.
c. Used as fuel in the name of benzol
d. Starting material in the synthesis of dyes, drugs, plastics, perfumes, etc.
e. Used in dry cleaning agents.
f. Component of anti-knocking gaseline.

Flow chart of preparation and properties of benzene.


Properties of benzene
Preparation of benzene
COONa
NaOH + CaO
Δ
(sodium benzoate)
OH
Zn, Δ
(phenol)
Cl
+ 2[H]
Ni-Al
alkali
(chlorobenzene)
3 HC ≡ CH
red hot Cu or Fe tube
Δ
(ethyne)
(benzene)
Cl
Cl2
FeCl3
+ HCl
(chlorobenzene)
NO2
Conc. HNO3
Conc.H2SO4
+ H2O
(nitrobenzene)
SO3H
Conc.H2SO4
+ H2O
(Benzenesulfonic acid)
CH3
CH3Cl
anhy. AlCl3
+ HCl
(toluene)
O
C - CH3
O
CH3 - C - Cl
anhy. AlCl3
+ HCl
(acetophenone)
(1-phenylethanone)
3H2
Ni or Pt , Δ
(cyclohexane)
Cl
Cl
Cl
3Cl2
sunlight
Cl
Cl
Cl
Benzene hexachloride(BHC)
15 O2
Δ
12 CO2 + 6 H2O

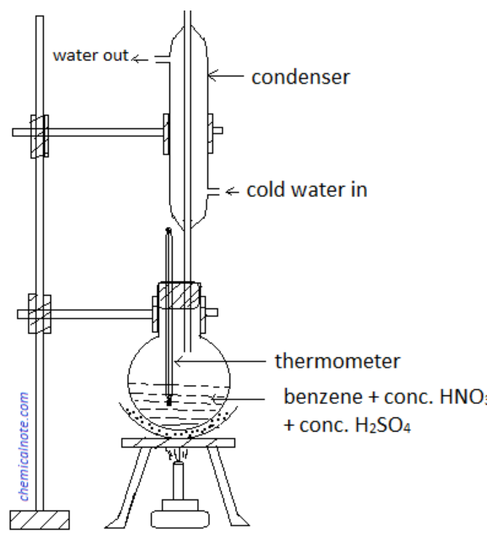
Project work

Visit a chemical industry where benzene is used as a component material to produce some daily life products. Prepare chart how is it used to produce that material in the industry and submit it to your teacher.

Exercise

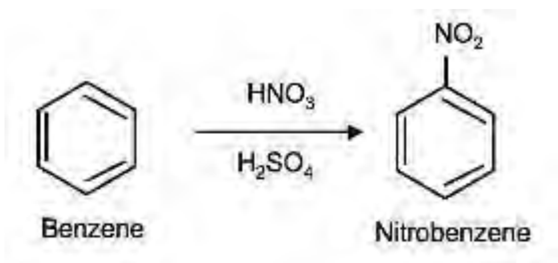
A. Multiple choice questions.

1. What is the molecular mass of benzene?
   a. 66 amu
   b. 68 amu
   c. 72 amu
   d. 78 amu
2. Study the given figure and identify the compound that would be produced in the given round bottomed flask?

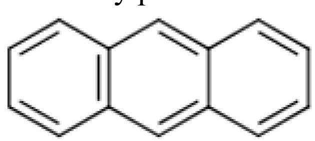


   a. Benzene
   b. Benzene sulphonic acid
   c. Nitrobenzene
   d. Phenol
3. Which of the following chemical species is suitable in the blank?
   C_6H_5OH+ Zn-dust $\xrightarrow{\Delta}$ ………… + Zn
   a. C_6H_5O
   b. C_6H_5-
   c. $CO_2 + H_2O$
   d. C_6H_6
4. Which of the following is a catalyst for Friedel-Crafts' reaction?
   a. Conc. H_2SO_4
   b. CH_3Cl
   c. $AlCl_3$
   d. Conc. HNO_3

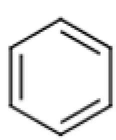
5. Which of the following is correct about nitration of benzene?
   a. It is nucleophhilic substitution reaction.
   b. It is electrophilic addition reaction.
   c. It is nucleophilic addition reaction.
   d. It is electrophilic substitution reaction.
6. Which gas is evolved when sodium benzoate is heated with sodalime?
   a. Carbon monoxide
   b. Vapor of water
   c. Carbondioxide
   d. Sodium carbonate
7. What is the common name of aromatic hydrocarbons?
   a. Olefins
   b. Arene
   c. Ethylene
   d. Parafin
8. Which type of reaction does the following example indicate?



   a. Electrophilic substitution
   b. Electrophilic addition
   c. Nucleophilic substitution
   d. Nucleophilic addition
9. How many pi-electrons does the following compound contain?



   a. 7
   b. 10
   c. 12
   d. 14
10. What type of hybridization does the following carbon atoms contain?



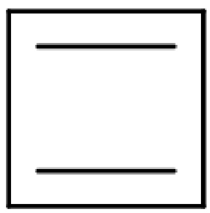
   a. sp^3

b. sp^2
c. sp
d. sp^2d

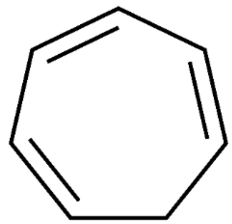
11.

B. **Short answer questions**

1. Write physical and chemical characteristics of benzene.
2. Arrange the following compounds in justifiable order of reactant and products with suitable conditions.
   C_6H_6, $C_6H_5NO_2$, $CH{\equiv}CH$
3. Check whether these compounds are aromatic or not?



i.



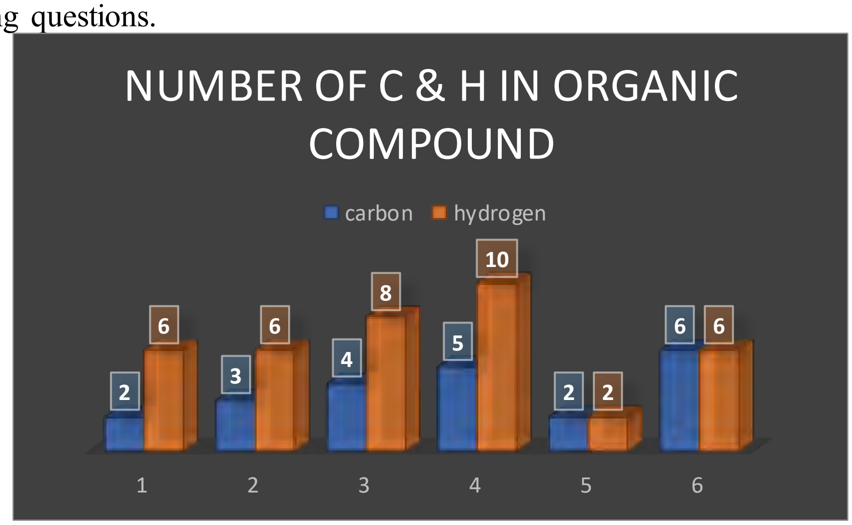
ii.

iii.

4. What would be the product when:
   i. Benzene is treated with chlorine in presence of sunlight?
   ii. Sodium benzoate is heated with soda-lime?
   iii. A compound which is obtained by passing acetylene into a red-hot iron tube at about 500^0C is heated ethanoyl chloride in presence of anhydrous $AlCl_3$?
5. Differentiate between:
   i. Aliphatic compounds and aromatic compounds
   ii. Friedel-Crafts's alkylation and acylation
   iii. Sooty and non-sooty flame

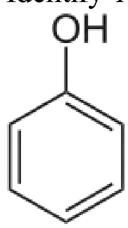
C. **Long answer questions**

1. Answer the following questions.

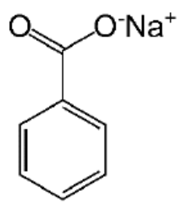


a. The histogram shown above represents the six compounds (1,2,3,4,5,6 at the bottom) and the bar consists of a number of hydrogen and carbon atoms as a pair bars.
   i. Which one of the bar pairs can represent aromatic hydrocarbons?
   ii. In which respect compound 1 and compound 6 are different? Write any two points.
   iii. Find the product when the compound of question i is treated with chlorine in dark medium.
   iv. What happens when compound 5 is heated in a copper tube?

b. Identify X and Y in the following reaction sequence:

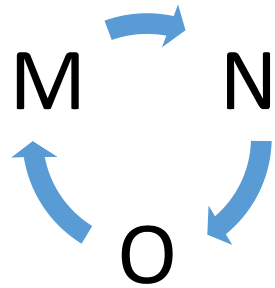


$Zn/\Delta \rightarrow$ X $H2/Ni/\Delta \rightarrow$ Y

2. Write the answer to the following questions:
   a. Write the main differences between addition reaction and electrophilic substitution reaction of benzene.
   b.  $+ NaOH$ $CaO/\Delta \rightarrow$ A $CH3Cl/AlCl3/\Delta \rightarrow$ B

3. An aromatic hydrocarbon having molecular mass 78 amu is heated with conc.nitric acid and conc. Sulphuric acid gives compound B. A reacts acetic anhydride to give C. Identify A, B and C in the reaction sequence.
4. A compound 'A' of carboxylic acid having molecular formula $C_7H_6O_2$ reacts with NaOH solution to give B. The compound B is heated with soda-lime and gives compound C. C is again reacted with chloromethane and toluene is obtained. Identify A, B and C in the given reaction sequence.
5. MNO are organic compounds. M is the aromatic compound of molecular mass 78. N is the product when M is under alkylation. From the compound O, M is obtained.

Identify the compounds M, N and O. Which factor is responsible to direct incoming electrophile at o, p and m position? Explain with resonating structures.



6. A. Identify S, T, U, V on the basis of information given in the following table.

Compounds	Conditions	Type of reaction/products
Benzene	T	Combustion
S	Red-hot iron tube	C_6H_6
Benzene	Chlorine/ sunlight	V
U	Zn-dust	Benzene

B. Explain about resonating structures of benzene. What would be the product when benzene is combusted in the presence of air? Explain.

Project work:

1. **Take slime of different color or clay and some colored sticks. Prepare a model of benzene and submit it to your teacher.**
2. **Prepare a list of 5 aromatic and 5 aliphatic organic compounds consulting the internet and submit to your teacher. Also mention, whey the compounds that you have chosen such properties as of the following table.**

SN	**Aliphatic**	**Reason**	**Aromatic**	**Reason**
1				
2				
3				
4				
5				

UNIT-16: FUNDAMENTALS OF APPLIED CHEMISTRY

16.1: Fundamental of applied chemistry

Activity:

Observe the given picture and answer the following questions:

a. What are the people doing?
b. Make a list of advantages and disadvantages of this process.

The branch of chemistry that deals only with theoretical knowledge is called pure chemistry. It is also known as theoretical chemistry or fundamental chemistry. It involves focusing on research to advance one's knowledge and understanding of the chemistry principles or process. On the other hand, the branch of chemistry that deals with the applications of the principles and theories of chemistry to solve real world problems using existing information effectively, efficiently, and economically and in an eco-friendly manner is called **applied chemistry**.

The main aspect of applied chemistry is to develop chemical industries which produce varieties of daily use products like cement, oil, detergents, soaps, food, medicines, paints, fertilizers, pesticides, textiles, paper and pulp etc. Therefore, we can say applied chemistry plays a key role for the betterment of human life.

Pure chemistry can be considered the "first step" towards **applied chemistry**

A student goes to school and learns about chemistry
(Pure Chemistry)

↓

The student uses what they learn to cure the common cold
(Applied Chemistry)

Test Yourself!

How does applied chemistry help in the betterment of human life?

16.1.2: Chemical industry and its importance

Activity:

a. Identify the chemical industry mentioned and its everyday applications.
b. Outline the raw materials utilized in each industry.
c. Evaluate the environmental consequences related to their production.

You might have heard the name of some chemical industries like Surya Paints and Chemicals (Nepal), Asian Pharmaceutical Pvt. Ltd. (Nepal), Gujarat Alkali and Chemicals (India), TATA Chemicals (India), Dow Chemical Cooperation (USA), Saudi Kyan Petrochemical Company (Saudi Arabia), etc. There are many more chemical industries all over the world which produce different products used in daily life.

A chemical industry is a manufacturing company that converts the raw materials into commodity chemicals for industrial and consumer products.

Various professionals like chemists, researchers, scientists, chemical engineers and technicians are involved in chemical industries.

Importance of chemical industries

Chemical industries are the prime factors to convert the raw materials into a large number of desired products. Chemicals play a vital role in our daily life. Today one cannot imagine human existence without chemicals. Some importance of chemical industries is listed below:

i. It provides job opportunities as many professionals like chemical engineers, chemists; technicians etc. are involved in chemical industries.

ii. It manufactures fertilizers, pesticides and insecticides widely used in agriculture.

iii. It manufactures various drugs, vaccines, and many newer medicines to cure different diseases.

iv. It produces various day to day life products like soap, detergents, shampoo, perfumes, petroleum products, etc.

v. It manufactures polymers, plastics for regular use like packing, wiring, clothing, home decor, water tanks, piping, storage containers, furniture and electronics.

vi. It has a significant direct impact on the economy through its contribution to the gross domestic product (GDP) of a nation.

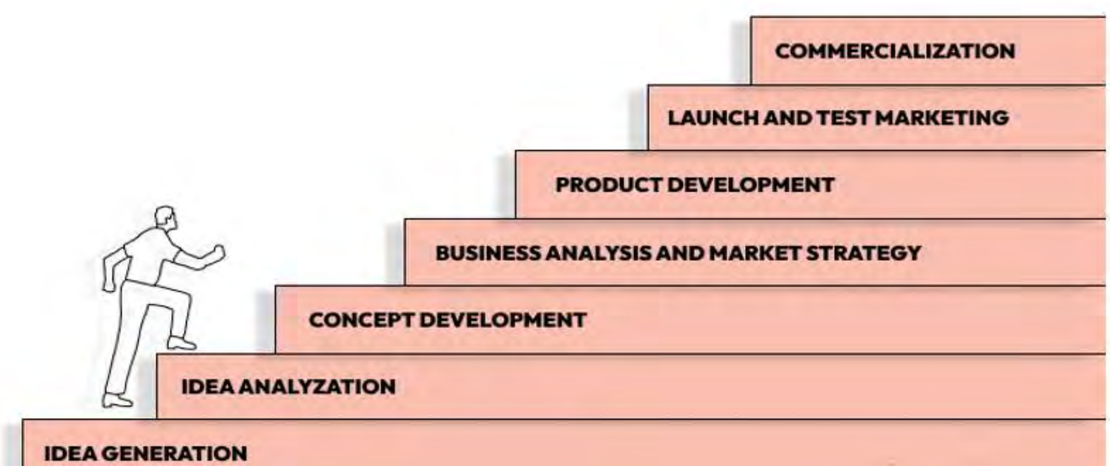
vii. It manufactures chemicals that preserve, maintain the quality and enhance the taste of food items.

viii. It acts as a center for research in the field of bioengineering, mutation, artificial organ production, and many more.

Test Yourself!

Why are number of cement industries comparatively more in Nepal than other chemical industries?

16.1.3: Stages in producing new products



Activity:

Observe the different stages of producing new product and identify the stages:

a. In which stages do companies launch the product in complete with full-scale production?
b. Which will outline product pricing, distribution, promotion, budget, etc.?

A chemical industry can survive only when it makes good business. If a company does not improve its product in quality and as per the need, it cannot exist. Therefore, it is important for a company to produce its product in a smart way through new technology in order to sustain in a competitive market. There are certain stages that a company should follow while producing a new product to sustain in the market.

Step-1: Idea Generation

This is the first step in production of a new product. The purpose of this stage is to focus on the need of the product and possible profit that can be generated. This idea may be generated by internal sources such as research and development, market research or from external sources such as customers, suppliers, distributors, or even competitors.

New ideas can be generated by:

i. Identifying gaps and trends in the market.
ii. Analyzing and researching customer needs.
iii. Reviewing feedback and complaints of the products and services offered by competitors.
iv. Brainstorming suggestions for new products.

Step-2: Idea Screening

This is the process of filtering the ideas collected from different sources. It is done by picking the one of the most likely to turn a profit. There is an "Idea Committee" in a company. This committee selects or rejects the idea through questionnaire like-

i. Is it necessary to introduce a new product and do the customer need it?

ii. Is there any competition regarding the product?

iii. Is the company able to produce this product and can sell the product?

iv. What impact will the idea have on stakeholders, organization and customers?

The idea for new product development is selected if all the answers to the above questions are positive otherwise it is rejected. Idea screening is an effective risk management tactic.

Step-3: Concept development and testing

It is conducted to find out the customer's reaction towards the new product. For this a small group of people is selected and questions regarding the new product are asked. The main motto of this communication is to find the problem and its significant solution. If the people like the product, then business analysis is done.

Step-4: Marketing strategy development

Actually, it is a business's game plan for reaching potential consumers and turning them into customers. It will outline product positioning, pricing, distribution, promotion, budget, and profit. The ultimate goal of this step is to achieve and communicate sustainable competitive advantages over rival companies by understanding the needs and wants of its customers.

Step-5: Business analysis

This is an important step as it will decide whether the project is beneficial from a financial as well as marketing point of view. Here the company tries to find out:

i. Whether the new product will be profitable or not?
ii. Whether the demand of the product will be regular or seasonal?
iii. Who are the competitors of this product?
iv. What will be the total expense during the lunch of the product?
v. How much profit will the product make?

If the company finds the new product profitable then the idea of developing a new product will be selected otherwise, rejected.

Step-6: Product development

In this step, the company will decide to introduce the new product on a small scale to test whether the project will be successful or not. There are three departments in a company for this step. Financial department provides money to introduce the product. Production department takes all necessary steps in producing and distributing the new product. Advertising department will make plans to advertise the new product.

Step-7: Test marketing

In this step, the company aims to explore consumer response to a product on a very small scale in a small market. This step will help the company to understand how the product and marketing strategy is going to work on consumers and market. If the test marketing fails, the company can make necessary changes to introduce the product again. Therefore, test marketing is helpful to reduce the risk of large-scale marketing especially for very costly products.

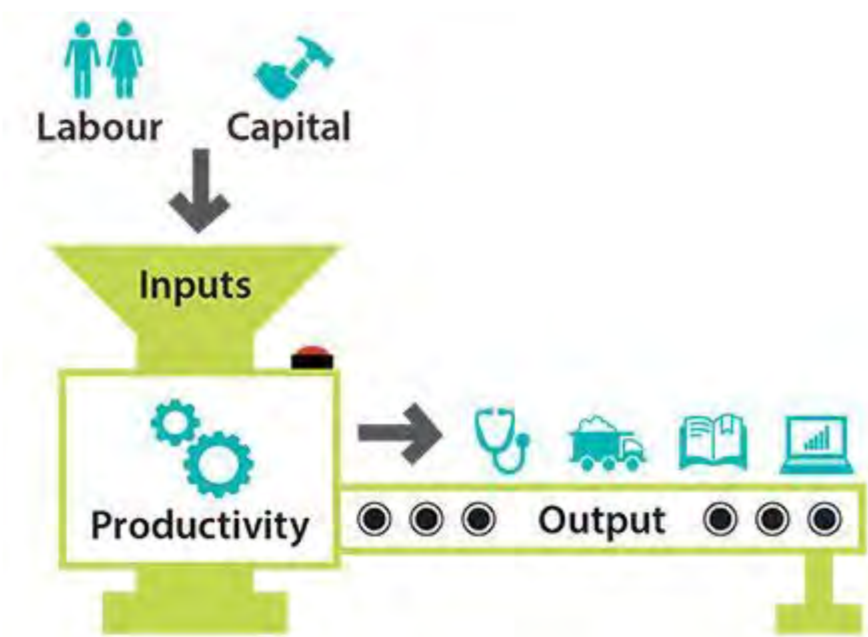
Step-8: Product launch

This is the crucial step in the product road map and is carried out only when test marketing is successful. While launching new product successfully, the company considers timing, pricing, packaging, product name (branding), promotion, target market competition, testing and reviews.

Test yourself!

a. What is the product produced in this industry and what are the raw materials used?
b. What are the challenges of this industry?

16.1.4: Economics of production



Activity:

How can we increase the output in the company with a given set of inputs?

Production refers to how much output can be produced with a given set of inputs. Output refers to a quantity of goods and services produced in a given time period. Productivity increases when more output is produced with the same amount of inputs, i.e., labor and capital.

All companies attempt to combine their inputs in a way to maximize their profits. Based on the operational area, companies can choose **labor intensive factories** or **capital-intensive factories** that will help them to maximize their profit. The company also considers the cost of input and technology in the production of new products. Hence, production is a crucial component of economics and is essential for any economy to function. Let us look at an example of production:

Suppose you are an owner of a food packaging company and want to expand your business in a city where labor charges are low. Then, you will choose to use **a labor-intensive factory** in order to maximize your profit.

On the other hand, if you want to expand your business in a city where labor charges are high, then you will choose to use **capital-intensive factories** in order to maximize your profit.

There are three types of production:

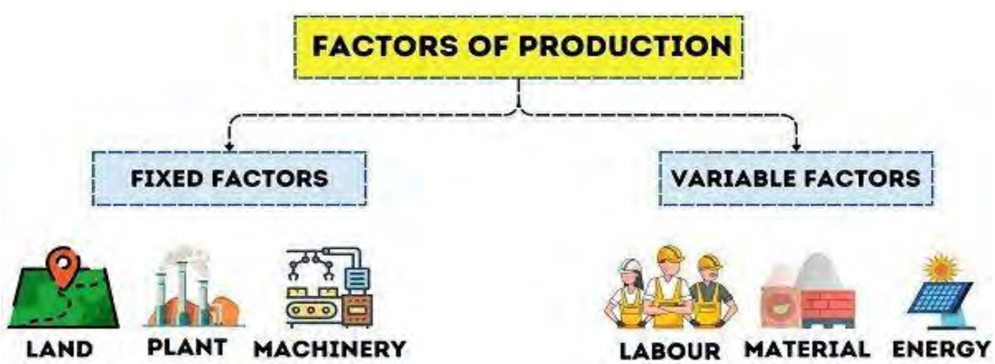
i. Primary production: It is the process in which raw materials are produced which are required by secondary industries like agriculture, mining, fishing and others.

ii. Secondary production: It is the process in which raw materials are converted into finished goods. This includes manufacturing plants, construction companies and others.

iii. Tertiary production: It is the process in which finished goods produced by secondary industries are sold by wholesalers, retailers, communication services and others.

Factors of production economics

The efficiency of a company is measured on the basis of how the company utilizes its factors of production. There are four factors of production- **land, labor, capital and entrepreneurship**. All these factors of production are economically important. They are further categorized as fixed factors and variable factors as shown below:



A fixed factor of production is one whose quantity cannot be readily changed. For example: plant, machinery, heavy equipment, land, factory building etc. A variable factor of production is one which can be changed. For example, labor, raw material, fuel, electricity, etc.

Productivity growth is important for maintaining the economic welfare and prosperity of the company as well as nation.

Fig: benefits of increasing productivity

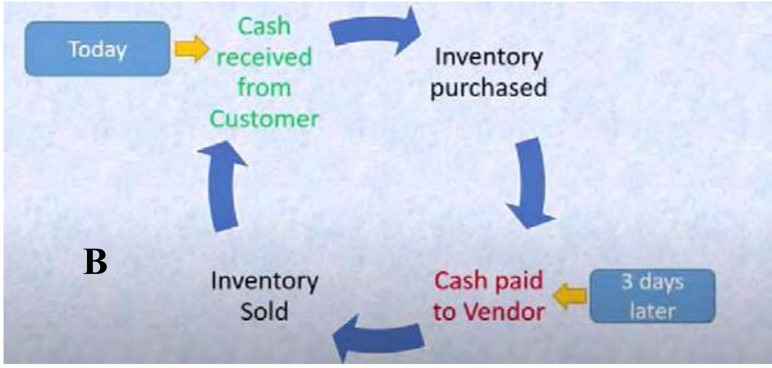
Test Yourself!

How can the increasing productivity lower the prices of the goods or services?

16.1.5: Cash flow in production cycle

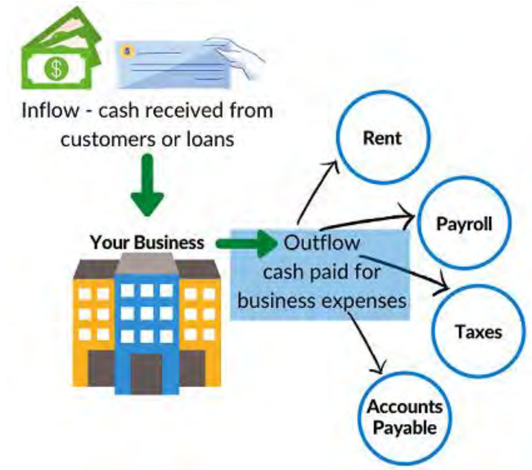
Activity:

Compare the inflow and outflow of the cash in figures A and B and answer the following question:



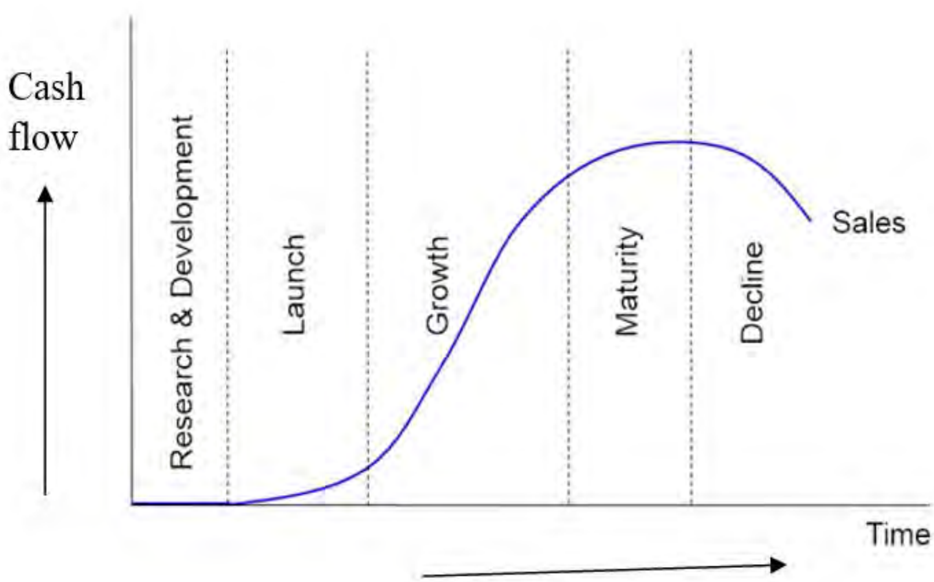
Which figure indicates more flow of cash and why?

Once the company is set to produce a new product, it requires cash to turn the raw materials into useful products and supply to the market. Hence, cash flow is the net cash transferred in and out of a company.



The cash flow cycle is also called cash conversion cycle is a measure of the time taken to convert a company's investments into cash.

A shorter cash conversion cycle is always better than a longer one. This is because a shorter cash conversion cycle represents that company is running successfully. However, the cash flow cycle varies from company to company. At the beginning the cash flow is less even negative due to high development cost. When the product moves through the growth into maturity the cash flow starts to increase as shown in the graph below.



Company can shorten their cash conversion cycles by reducing their inventory or by collecting customer payments quickly.

Test Yourself!

Why is cash flow maximum at the maturity of the company and start to decline after?

16.1.6: Running a chemical plant

Fig: Haber-Bosch plant

What is a chemical plant? Have you ever visited any chemical plant?

A chemical plant is an industrial plant that manufactures or processes chemicals on a large scale. These products can range from basic chemicals like ammonia and sulfuric acid to more complex compounds such as pharmaceuticals, plastics, etc.

Chemical plants consist of highly specialized equipment, units and technology that convert the raw materials into desired products.

Chemical plants can be categorized into different types based on the types of raw materials they use and the products they produce. Some common types of chemical plants are:

a. **Petrochemical plants**
These plants primarily use petroleum-based raw materials such as crude oil, natural gas to produce a variety of chemicals and products. These plants produce chemicals like ethylene, propylene, benzene, xylene, etc. These are the essential chemicals for the production of plastics, synthetic fibers, rubber, and various other materials.

b. **Pharmaceutical plants**
Pharmaceutical plants are designed for the synthesis and purification of pharmaceutical compounds and drugs. They use various organic and inorganic chemicals as raw materials to produce medications, vitamins, vaccines, and other pharmaceutical products.

c. **Fertilizer plants**
Fertilizer plants produce fertilizers and agricultural chemicals using raw materials such as ammonia, urea, phosphates, and potash. These chemicals are essential for enhancing soil fertility and promoting plant growth in agriculture.

d. **Chlor-Alkali plants**
Chlor-alkali plants produce chlorine, sodium hydroxide (caustic soda), and hydrogen gas through the electrolysis of brine (sodium chloride solution). These chemicals have various industrial applications, like production of PVC plastics, detergents, paper products etc.

e. **Agrochemical plants**
Agrochemical plants produce pesticides, herbicides, fungicides, and other chemicals used in agriculture to protect crops from pests, weeds, and diseases. These plants utilize a variety of chemical synthesis and processes to produce effective and environmentally safe agrochemical products.

Whatever may be the operation process, special precautions and safety measures should be taken while running a chemical plant. This is because the manufacturing process involves high temperature, high pressure, toxic and corrosive substances. Therefore, varieties of workers like plant managers, engineers, plant operators, highly skilled technicians and laborers are involved to run a chemical plant. Nowadays smart technology has been used in the chemical plant called **automation**. Due to the use of advanced sensors, control systems, artificial intelligence, and other technologies, a large number of chemical plants are operated with limited skilled manpower.

In addition, regular inspection of chemical plants is required to avoid leakage of toxic substances, explosion of chemicals and other accidents.

Test Yourself!

What happens if a chemical plant is not inspected frequently?

16.1.7: Designing a chemical plant

Fig: Chemical plant design

Chemical plant is a complex combination which includes both physical structure and chemical processes. Designing a chemical plant involves use of engineering principles and chemical processes to convert raw materials into valuable products efficiently, safely, and cost-effectively. Designing of chemical plant includes:

i. Objectives and scope
ii. Process and place selection
iii. Equipment selection and sizing

iv. Piping and instrumentation diagram
v. Safety and environmental considerations
vi. Cost estimation
vii. Risk assessment and management
viii. Operations and maintenance

There are various challenges while designing a chemical plant. Some of these challenges are:

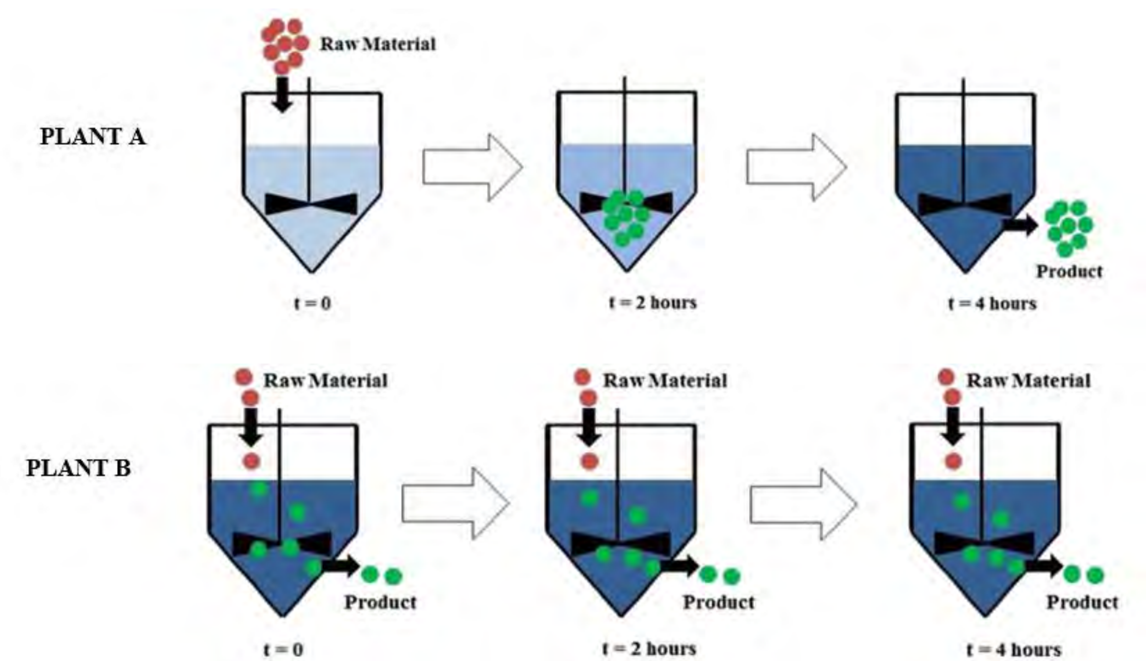
i. Ensuring the safety of both workers and the surrounding environment.
ii. Developing efficient and cost-effective chemical processes
iii. Materials used in the plant are suitable with the chemicals and reaction conditions.
iv. The plant can operate continuously with minimal stoppage.
v. Minimizing emissions, waste generation and resource consumption to reduce their environmental impact.

Hence, design, operation and safety considerations of chemical plants are essential for ensuring efficient production, maintaining product quality, safety measures of workers and protection of the environment.

Test Yourself!

Why do we need planning and designing before starting the project or work?

16.1.8: Continuous and Batch processing



Let us take an example of processing plant A and B. In plant **B** production is continuous without interruptions whereas in plant **A**, production occurs in separate cycles or **batches**. There is a stable flow of materials and constant operating conditions in plant **B**. So that it can result in products with uniform quality than that of **A**. The approach of plant B is called **Continuous processing** whereas the approach in plant A is called **Batch processing**.

Batch processing refers to a technique that consists of a sequence of one or more steps that should be performed in a defined order. At the end of the sequence a finite quantity of product is produced. This is again repeated in order to produce another product batch. Batch processing is a commonly working technique in the manufacture of chemicals, pharmaceuticals and food-based products on a large scale.

Continuous processing refers to a technique where the production of a product is continuous without any interruptions or pause between batches. This process is traditionally used for manufacturing high-volume products which don't require customization. Such products include paper and pulp manufacturing, petrochemicals, or oil and gas refineries, electricity production and water treatment plants. For the majority of applications, continuous flow saves costs, energy and time. For example, cement produced by **Hetauda Cement Industry**, Nepal is a continuous process.

Some examples of these processing are:

Figure: An introduction of batch

Type of Industry	Batch processing	Continuous processing
Chemical industry	special chemicals, cosmetics, and pharmaceuticals	Petrochemical refining
Food and Beverage Industry	Baking, brewing, and confectionery production	Bottling and packaging
Pharmaceutical Industry	drug formulation, blending, and packaging	tablet manufacturing
Textile Industry	Dyeing and printing processes	Spinning and weaving processes
Water Treatment Industry	Treatment of small volumes of water	Large-scale municipal water treatment plants

Test yourself!

Complete the table given:

	Batch Process	Continuous process
What		
Conditions		
Feeding of raw materials		
Number of products		
Cost required		
Nature of product produced		
Control system		
Life span		
Work force		
Examples		

16.1.9: Environmental impact of chemical industry

Activity:

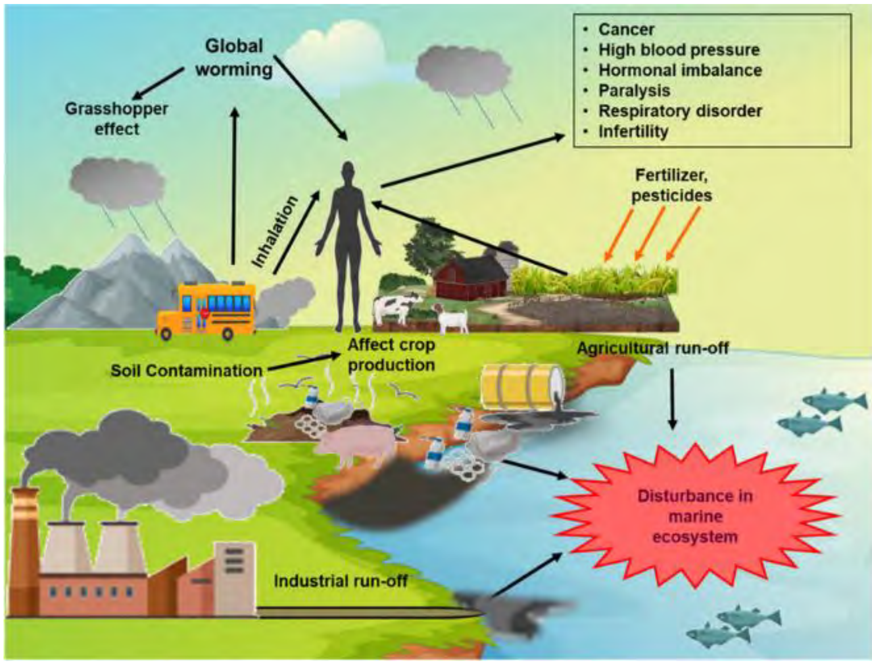
A

B

Identify the types of pollution in the given figure A and B and their harmful effects in the environment.

Activity:

Observe the picture and discuss the impact of the chemical industry in air, water and land quality.



Chemical industry plays a significant role in modern society, providing essential products for various sectors such as agriculture, healthcare, manufacturing, and consumer goods. However, it also poses environmental challenges due to its extensive use of energy, water, and raw materials, as well as the generation of waste and emissions. Some of the key environmental impacts of the chemical industry include:

Air pollution

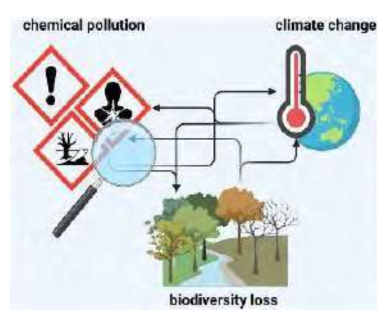
Chemical industries cause air pollution through the release of various pollutants into the atmosphere. These pollutants can include volatile organic compounds, nitrogen oxides, and sulfur dioxide, particulate matter, etc. These pollutants can contribute to smog formation, acid rain, respiratory problems, and damage to ecosystems

Greenhouse gas emissions

Most of the chemical industries and chemical processes use fossil fuels as a source of energy. These processes therefore release greenhouse gases like carbon dioxide, methane, nitrous oxide, etc. These emissions contribute to climate change including global warming, rising of sea level, etc.

Water pollution

The chemical industry adds pollutants in the water resources through the discharge of untreated or incompetently treated chemicals. These pollutants can include toxic chemicals, heavy metals, and other contaminants. These contaminations can harm aquatic ecosystems, disturb aquatic life, and affect drinking water quality.



Soil contamination

Improper disposal of chemical waste or leaks from the factory can cause soil contamination. These pollutants can harm soil quality and reduce soil productivity, which can have negative impacts on agriculture and other land uses.

Depletion of natural resources

Chemical manufacturing can contribute to the depletion of natural resources through the use of non-renewable resources, such as fossil fuels and minerals, in the production of chemicals. This can have negative impacts on the availability and quality of these resources for future generations.

Preventive measures

The chemical pollution can be reduced or prevented by the combined effort of related company, society, government and other stakeholder. Some of the preventive measures are:

i. Governments can implement strict regulations to control the production, use, and disposal of chemicals.
ii. Encouraging the use of less hazardous or non-toxic alternatives to harmful chemicals (called green chemistry)
iii. Proper handling, storage, and transportation of chemicals are essential to prevent spills, leaks, and accidental releases into the environment.
iv. Effective waste management practices, such as recycling, treatment, and disposal of hazardous waste, are best practices for preventing chemical pollution.
v. Regular monitoring of air, water, and soil quality can detect and prevent the possible incidents from pollution.
vi. Raising public awareness about the possible risks, individuals and communities may take the initiative to prevent pollution.

Facts

There are lots of multilateral agreements to improve the global impacts of hazardous chemicals. Some of them are:

a. Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal
b. Montreal Protocol on Substances that Deplete the Ozone Layer
c. Stockholm Convention on Persistent Organic Pollutants
d. The Minamata Convention on Mercury

Test Yourself!

How does chemical industry loss the biodiversity?

Exercise

Multiple choice questions

1. What is the name of a product concept that is developed during the new product development process?

a. concept
b. model
c. prototype
d. precedent

2. If you have gathered customers in the target market and presented them with a description of your product. What is the name of this process?

a. prototyping
b. concept testing
c. test-marketing
d. crowdsourcing

3. At what stage in the new product development process would you launch the product, complete with full-scale production, distribution, advertising, and sales promotion?

a. Test-marketing
b. Product development
c. Business analysis
d. Commercialization

4. What is the first stage in the new product development process?

a. Idea screening and evaluation
b. Market strategy development
c. Commercialization
d. Idea generation

5. What is a distinguishing characteristic of batch processing than that of continuous processing?
a. Continuous flow of materials
b. Fixed processing time for each batch
c. Simultaneous processing of multiple batches
d. Minimal need for equipment cleaning

6. Which of the following is an advantage of continuous processing over batch processing?
a. Greater flexibility in production scheduling
b. Lower initial capital investment
c. Reduced risk of contamination
d. Easier monitoring and control of individual batches

7. Which is true for continuous processes?

   a. Low cost equipment
   b. Moderate variety in the process
   c. Same product with different version
   d. Identical product

8. Who is the designer of a chemical plant?

   a. Chemist b. Plant manager
   c. Chemical Engineer d. Worker

9. Which of the following best describes cash flow in a chemical industry?

   a. The total revenue generated by the industry
   b. The net profit after deducting expenses from revenue
   c. The movement of money in and out of the industry over a specific period
   d. The value of moneys owned by the industry

10. What is the primary source of cash inflow in an industry?

    a. Sales of products
    b. Borrowing from financial institutions
    c. Government subsidies
    d. Purchase of raw materials

11. Which factor can increase cash outflow in an industry?

    a. Efficient energy utilization
    b. Decreased maintenance costs
    c. Expansion of production capacity
    d. Reduction in workforce

12. How does depreciation affect cash flow in a chemical plant?

    a. Increases cash inflow
    b. Decreases cash inflow
    c. No impact on cash flow
    d. Increases cash outflow

13. Which one is true at the beginning of the cash flow in the life of a product?

    a. Positive b. Negative
    c. Sometimes positive sometimes negative d. Not defined

B. Short answer questions

1. Define the following terms:
    a. Applied chemistry
    b. Cash flow
    c. Economics of production
    d. Batch processing

2. Differentiate between:
    a. Pure chemistry and applied chemistry
    b. idea generation and idea screening
    c. Batch process and continuous process.
    d. Fixed factor of production and variable factors of production
    e. Cash inflow and cash outflow
    f. Concept development and product development
    g. Designing a chemical plant and running a chemical plant

3. Give reasons:
    a. Market testing helps in the commercialization of the new product.
    b. Cash flow is maximum at the maturity of the company compared to the initial stage.
    c. Material and energy balance should be considered during the designing of chemical plants.
    d. Chemical pollution can be minimized using green solvents.

4. What do you mean by chemical industry? Write any three importance of the chemical industry?

5. What is applied chemistry? Why is it called applied? Give one application of it.

6. Define cash flow in the production cycle. What are the key factors that influence cash flow in the company?

7. What do you mean by new product? What are the different stages of development of a new product?

8. What do you mean by running a chemical plant? What should safety measures be considered while running a chemical plant?

C. Long answer questions

9. One of the stages of the development of new product is research and development.

a. What are the other stages in the development of a new product?
b. What is the role of research and development in this process?

c. Write any three challenges commonly encountered during the development of a new product and give suggestions to overcome these challenges.

10. Urea is primarily produced in a chemical plant through a process known as the Haber-Bosch process,

a. Define the term chemical plant.
b. What are the raw materials used in the production of urea?
c. What is the application of urea in daily life?
d. How does it pollute the environment and what are the preventive measures?

11. Batch processing and continuous processing are the two major types of processing done in the production industry

a. Define batch processing and continuous processing, focusing on key characteristics and operational differences.
b. Write the advantages and disadvantages of these processes based on flexibility, control, and resource utilization.
c. Provide an example of chemical processes or industries where batch processing is preferred over continuous processing.

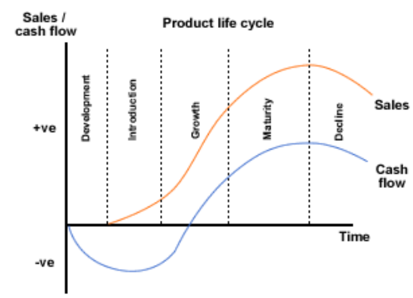
12. Let us consider the case of a chocolate manufacturing company, which produces different brands of chocolate bars. Each brand has a different shape, size and list of ingredients but the process and raw materials involved are identical for the most part with minor alterations.

a. Identify the type of processing (batch or continuous) in the company and give reason for your choice.
b. Write any three characteristics of this processing.

13. The production of sulfuric acid is primarily done through the Contact Process in a plant. It is the most widely used method globally.

a. Why is this plant called a chemical plant?
b. What are the uses of sulfuric acid?
c. Write any three environmental impacts of this plant.
d. How can we solve the chemical pollution caused by this process?

14. Use the plot which shows the relationship between cash flow and sales with time.



   a. What do you mean by cash flow in the production cycle?

   b. Identify the stages in which cash flow is negative whereas sales are positive.

   c. Why does cash flow start to decrease after maturity of the company?

   d. Find the relationship between sales and cash flow in the growth of the company.

Project work

Make a list of industries and factories in Nepal and write the raw materials used and the product produced as shown in the table below.

Industry	Raw materials used	Product produced

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UNIT 17: MODERN CHEMICAL MANUFACTURES

(Principle and flow sheet diagram only)

Figure 17.1: An industry

17.1.1 Manufacture of Ammonia by Haber's process

A German chemist and Nobel Laureates Dr. Fritz Haber (1868-1934) manufactured ammonia from a direct combination of nitrogen and hydrogen. Ammonia can be manufactured by Haber's process, cyanamide process and Serpeck's process. Out of them Haber's process is considered more economic and developed.

He was born on 9th December, 1868 and died in 1934. He was awarded the Nobel Prize in chemistry in 1918 for the development of the Haber-Bosch process for high scale synthesis of ammonia, fertilizer and explosives. The method was adopted for commercial use after 1930 by German Chemist Karl Bosch. Then it is used to say the Haber-Bosch process.

Principle

When dry nitrogen and hydrogen in the ratio of 1:3 is mixed by volume, then heated in presence of iron as catalyst containing metal oxides as a promoter at a temperature of 450^0C-500^0C and atmospheric pressure of 200-900 atm, ammonia is obtained.

$N_2(g)+ H_2(g)\ Fe/Al2O3,450-500degree,200-900atm \rightarrow NH_3(g) + 22KCal$

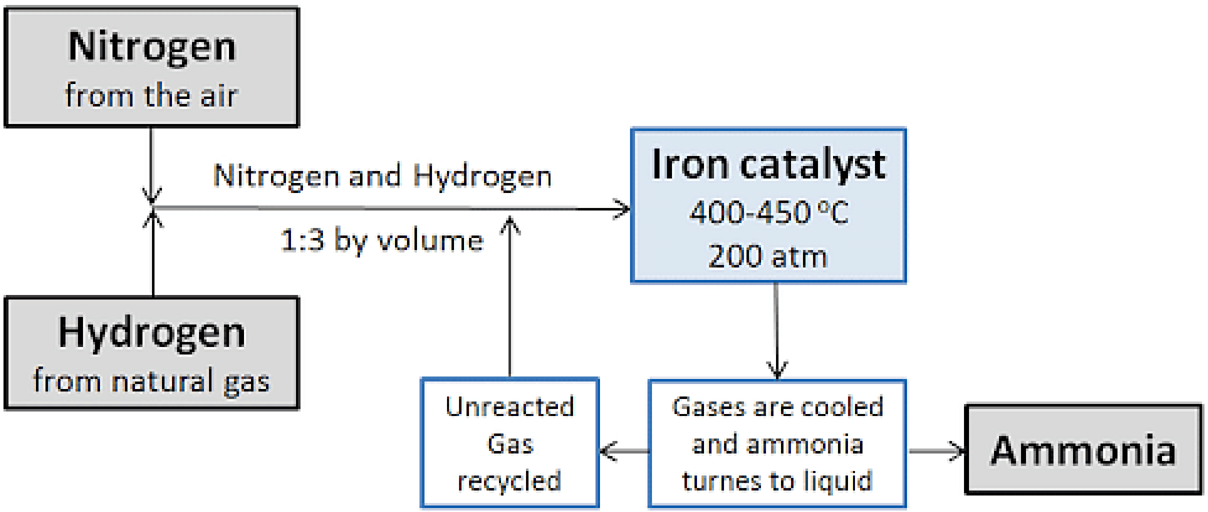


Figure 17.2: Summary chart of manufacture of ammonia by Haber's process

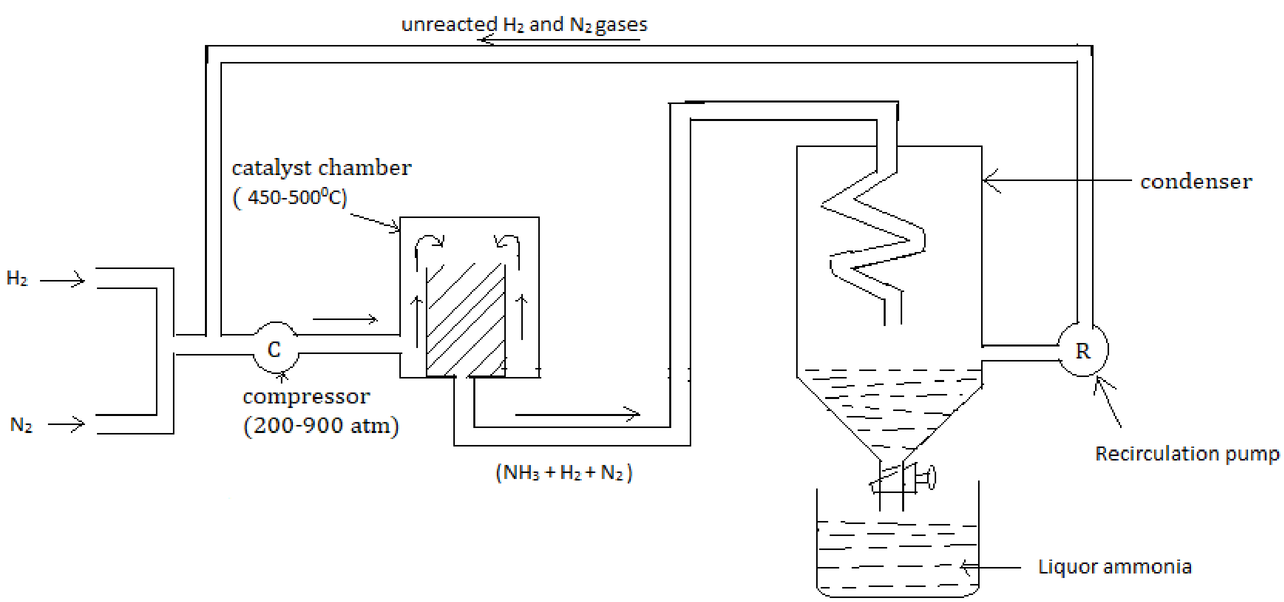


Figure 17.3 Diagrammatic sketch of manufacture of ammonia

Conditions for high yielding of ammonia

a. Low temperature

Low temperature is favourable for the formation of ammonia. Below 450^0C, reaction would be very slow as well as at higher than 500^0C, prepared ammonia may decompose. So, temperature should be low i.e. according to Le-Chatelier's principle at low temperature forward reaction is favoured.

b. High pressure

High pressure i.e. 200 to 900 atmospheres favours the formation of ammonia i.e. according to Le-Chatelier's principle forward reaction is favourable.

c. High concentration of hydrogen and nitrogen

With high concentration of hydrogen and nitrogen forward reaction is occurred. The high concentration of hydrogen or nitrogen or both should be used for high yield of ammonia.

d. Proper use of catalyst

The suitable catalyst has the proper alignment of the reactants for collision and proper use of catalyst and promoter enhances the rate of formation of product i.e. ammonia.

e. Purity of nitrogen and hydrogen

To prevent from the chemical poisoning of catalyst and reactants the gases- hydrogen and nitrogen used should be pure.

Process

Mainly there are six units for the manufacture of ammonia by Haber's process.

a. Production of hydrogen and nitrogen

Hydrogen gas is produced by electrolysis of water or from water gas and nitrogen is produced by fractional distillation of liquid air.

b. Compressor unit

The gases are compressed by the help of air compressor maintaining about 200-900 atmosphere. For proper collision of hydrogen and nitrogen compressor is used.

c. Purifying unit

The mixture of gas may contain impurities like carbondioxide. To remove such impurities, the gas is passed into a purifying unit. In this unit a mixture of sodalime is kept which absorbs carbondioxide gas.

$NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

$CaO + H_2O \rightarrow Ca(OH)_2$

d. Catalyst chamber

The vertical cylindrical steel chamber with thick wall is packed with finely divided iron promoted with Al_2O_3, K_2O and ZnO is kept. It is heated to about 500^0C, which helps to initiate the chemical reaction. The reaction is exothermic which helps to continue the reaction for a long time. Only about 15% of the gases are converted into ammonia.

e. Condenser

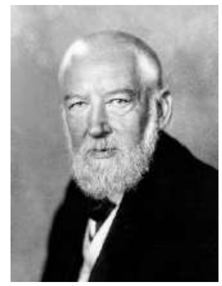
The ammonia is condensed at the condenser chamber in liquid form which is called liquor ammonia. The unreacted nitrogen and hydrogen gases are recycled to the compressor again.

f. Recirculation

All the nitrogen and hydrogen gases are not converted into ammonia. So, the unreacted gases are recycled for the same process to the condenser to yield more ammonia gas.

17.1.2 Manufacture of nitric acid by Ostwald's process

Introduction of Freidrich Wilhelm Ostwald



A German chemist, Freidrich Wilhelm Ostwald was a person who developed the process of manufacture of nitric acid by dilution method. He was awarded the Nobel prize for the work on catalysis, chemical equilibrium and reaction velocities in the year 1909. He has a special contribution on dilution theory for weak electrolytes and was named as Ostwald's dilution law. He was born in 1853 and died in 1932.

Principle

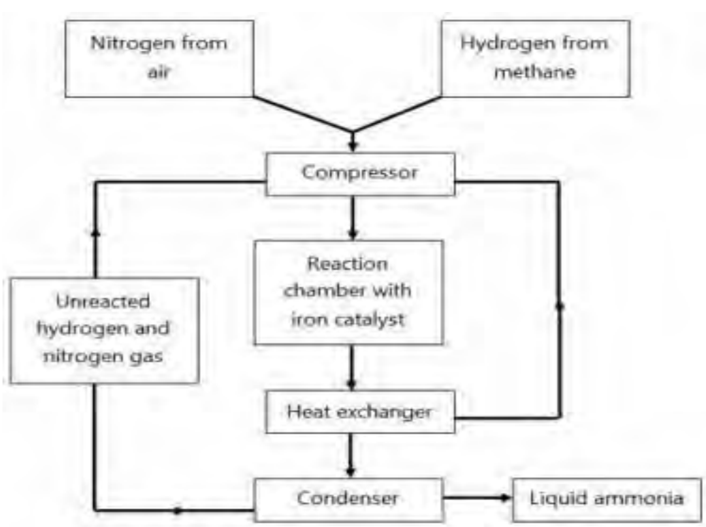
In the manufacture of nitric acid, ammonia is catalytically oxidized into nitric oxide and is further oxidized to nitrogen dioxide. Finally, nitrogen dioxide is absorbed by water in presence of oxygen to produce nitric acid.

$4NH_3 + 5O_2 \xrightarrow{90\% Pt + 10\% Rh,\ (973-1073)K} 6H_2O + 4NO + heat$

$2NO + O_2 \xrightarrow{173\ K} 2NO_2$

$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$

A diagrammatic representation of the process is as follows:



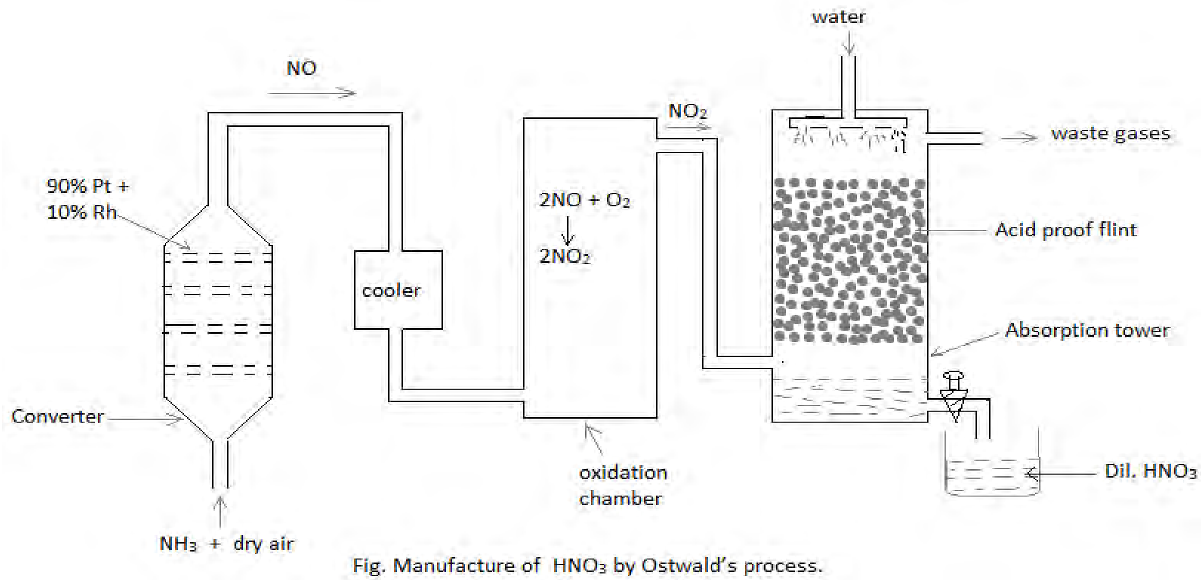


Figure 17.4 Diagrammatic sketch of manufacture of HNO3 by Ostwald's process

Process

a. Oxidation of ammonia

$4NH_3 + 5O_2 \xrightarrow{90\% Pt + 10\% Rh,\ 1073K} 6H_2O + 4NO + \text{heat}$

Can you say in the given reaction ammonia is oxidized?

Ans ……………………………………………………….

In this reaction ammonia manufactured from Haber's process is oxidized in a converter chamber in which platinum or rhodium are kept in the form of gauze or seive or bar as a catalyst. The temperature of the converter chamber is maintained about 700^0C to $800\ ^0C$ (973 to 1073 K). In this condition ammonia is oxidized to nitric oxide. Due to the exothermic nature of the reaction, the process continues for continuous oxidation of ammonia.

b. Oxidation of nitric oxide

What is the difference between NO and NO_2? How is NO converted to NO_2?

$2NO + O_2 \xrightarrow{173\ K} 2NO_2$

Nitric oxide obtained from the converter is passed into the cooler and is passed into the oxidation chamber. The nitric oxide is oxidized in the presence of air which gives nitrogen dioxide. Oxidation of nitric oxide is feasible at low temperature.

c. Absorption of nitrogen dioxide

$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$

Nitrogen dioxide obtained from the above step is passed into an absorption chamber in which water is sprayed from the top and acid proof flint or quartz is kept at the middle of the tower. In this chamber nitrogen dioxide is absorbed by water forming nitric acid. Thus obtained dilute nitric acid is collected at the receiver. To obtain pure nitric acid it is distilled in presence of reduced pressure and concentrated sulphuric acid. To make the nitric acid anydrous, thus obtained acid is treated with phosphorus pentoxide.

This process is more advantageous than other methods because it is cheap and can be controlled easily. Moreover, it is economical to convert dilute nitric acid to concentrated nitric acid.

17.1.3 Manufacture of sulphuric acid by contact process

Principle

This method is called contact process because the oxidation of sulphurdioxide occurs by contact process at contact chamber on the surface of catalyst V_2O_5. Principle of this process undergoes in the following steps-

Step I: Production of sulphur dioxide

$S + O_2 \xrightarrow{\Delta} SO_2$

$4FeS_2 + 11O_2 \xrightarrow{\Delta} 2Fe_2O_3 + 8SO_2$

To produce sulphurdioxide sulphur is burnt or iron pyrite is burnt as shown in the above chemical equation.

Step II: Oxidation of sulphurdioxide

$2SO2 + O2 \ \ V2O5/500C \rightarrow \Leftrightarrow \ \ 2SO3 + Heat$

This reaction has the key role to obtain the quantity of sulphuric acid produced since the production of oleum depends upon the quantity of sulphurtrioxide formed.

Step III: Formation of oleum

Sulphurtrioxide is treated with pre-existed sulphuric acid. It forms oleum($H_2S_2O_7$)

$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$(oleum)

Step IV: Preparation of sulphuric acid of desired concentration

The oleum thus formed is further treated with a calculated amount of water for desired concentration.

$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

Process

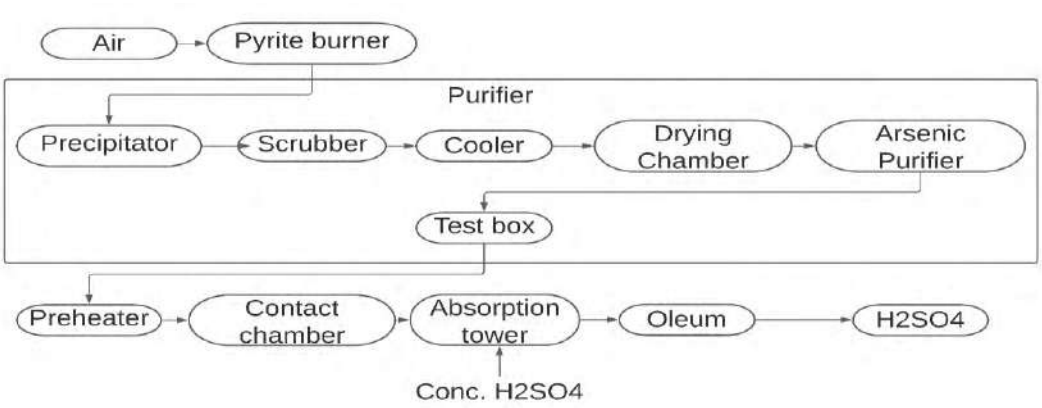


Figure 17.5: Flowsheet diagram of manufacture of H_2SO_4 by contact process

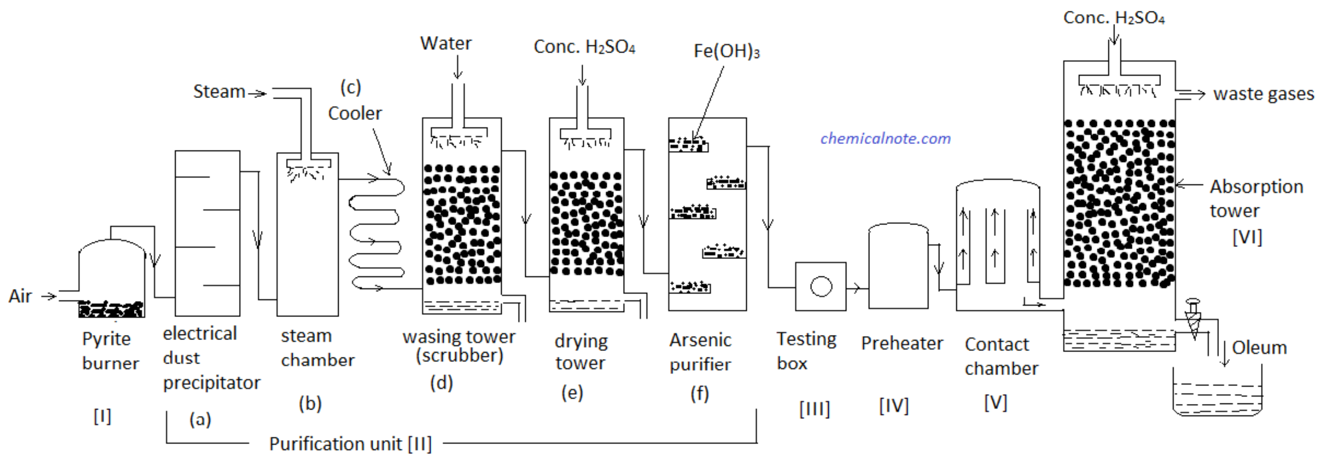


Figure 17.6: Flowsheet diagram of manufacture of H_2SO_4 by contact process

a. Sulphur or pyrite burner

In this burner sulphur or iron pyrite is burnt in the presence of air. Air is blown through the inlet and sulphur or iron pyrite is kept inside in the form of fire bricks. It produces suphurdioxide.

b. Purification units

SO_2 obtained in pyrite burner may contain dust, water soluble impurities, arsenic oxides, moisture, unoxidized suphur etc. To remove these impurities SO_2 should be passed through various chambers in which different types of impurities are removed. Which are described below.

i. Dust precipitator

If dust is present in SO_2 prepared in the previous chamber, this dust should be precipitated out. The dust remover works under the influence of a high potential difference fitted in the precipitator.

ii. Steam purification chamber

The chamber is fitted for the removal of precipitates of dust. In this chamber steam is passed through the top which helps to settle down the dust at the bottom.

iii. Washing tower

The water soluble impurities, acids, mists etc. are washed in this chamber. Water is sprayed through the top which consists of a scrubber packed with quartz which helps to increase contact time of water and impurities.

iv. Drying tower

The gas obtained from the washing tower may contain moisture. To make SO2 dry, concentrated sulphuric acid is sprayed through the top of the tower. It absorbs moisture and SO_2 gas becomes dry.

v. **Arsenic purifier**
Sulphurdioxide may consist of arsenic oxide as impurity. To remove arsenic oxide, ferric hydroxide is kept in shelves, which forms ferric arsenate. If arsenic is present in the sulphuridioxide it reduces the catalytic activity of the vanadium petoxide.

vi. **Testing box**
A strong ray of light is passed into the test box, which identifies whether the gas is pure or not. If gas is free from the impurities it is passed into the contact chamber, if not recycled to make the gas pure.

c. **Preheater:**
The mixture of gas is preheated from 450^0C to 500^0C, and is passed into the contact chamber. If the preheater is not kept, the heating system can be maintained in the contact chamber.

d. **Contact chamber**
In this chamber, vanadium pentoxide is coated in the inner wall. It catalyses for the oxidation of sulphuridoxide to sulphurtrioxide.

e. **Absorption chamber**
Pre-existed concentrated sulphuric acid is used to absorb sulphurtrioxide. After the absorption of sulphurtrioxide, sulphuric acid forms oleum. Oleum is further treated with a calculated amount of water to obtain desired concentration of sulphuric acid.

Conditions of high yielding of sulphuric acid

a. **High pressure**
Higher the pressure of gases, higher the possibility of production of products because pressure creates lower distance between the reactants.

b. **Low temperature**
Temperature should be maintained to about 450 to 500 ^{0} C

c. **Purity of the gases i.e. sulphurdioxide and oxygen**
Purity of the gases gives more product i.e. purity of sulphurdioxide and oxygen gives more oxidized products i.e. sulphurtrioxide.

d. **High concentration of gases**
More the concentration of gases, the more is the product. Since there will be more collisions between the reactants to give product.

e. **Use of proper catalyst i.e. vanadium pentoxide**
Vanadium pentoxide is the catalyst which catalyzes to oxidize sulphurdioxide into sulhurtrioxide.

17.1.4 Manufacture of sodium hydroxide by Diaphragm Cell

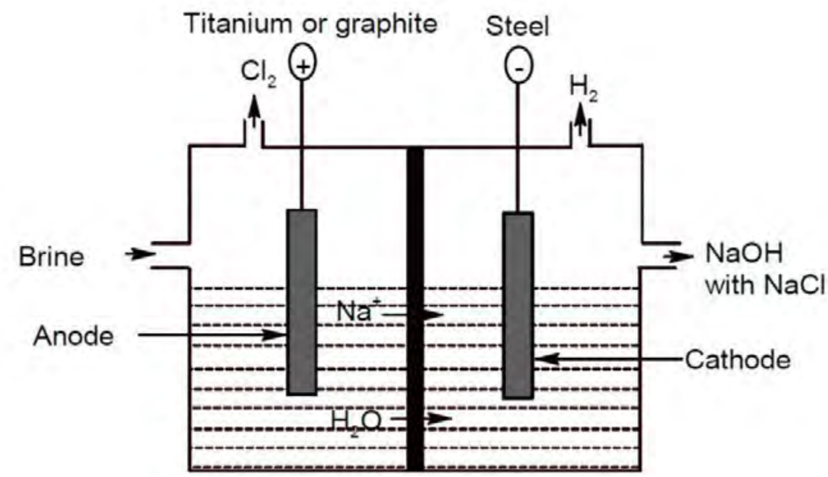


Figure 17.7: Manufacture of sodium hydroxide by Diaphragm cell

Principle and procedure

In the diaphragm cell, electrolysis of brine solution is performed in order to obtain sodium hydroxide in presence of coating of titanium in Ru-Ti oxide anode and steel cathode. In this process chlorine is discharged through anodic outlet and hydrogen is discharged through cathodic outlet.

$$NaCl \rightarrow Na^+ + Cl^-$$

$$H_2O \rightarrow H^+ + OH^-$$

At anode:

Between Cl^- and OH^-, OH^- remains unaffected but Cl_2 gets discharged from the anode in the diaphragm cell.

$$2Cl^- \rightarrow Cl_2 + 2e^-$$

At cathode:

Between H^+ and Na^+, Na^+ has more discharge potential. So, it remains unaffected but H^+ forms hydrogen molecules. Molecular hydrogen is discharged from the outlet of the cathode.

$$2H^+ + 2e^- \rightarrow H_2 \quad \text{or}$$

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$$

When electrolysis is over, the hydrogen and chlorine gases are removed separately. Sodium ion passes through the diaphragm and combines with hydroxide and forms sodium hydroxide since the membrane is selectively permeable. The sodium hydroxide solution is formed at the cathode compartment.

$$Na^+ + OH^- \rightarrow NaOH$$

The hydroxide ions are prevented from reacting with chlorine gas by diaphragm. During electrolysis, the solution from the cathode is periodically removed and concentrated by the evaporation process. Dilute solution of NaOH is obtained which is about 10-12 % (w/w) and about 16% of NaCl. Sodium chloride is crystallized out and can be recycled for the same process.

Advantages of the membrane cell

a. It is economical.
b. Consumption of very low amounts of electricity.
c. No hazardous waste.
d. Diaphragm cell is selectively permeable which allows Na+ and water to flow to the cathode compartment.

Disadvantages of the membrane cell

a. Purity of caustic soda is low
b. Poor quality of chlorine
c. High steam consumption to make the concentration about 50%.
d. Use of asbestos, which may create problems frequently for replacement.

17.1.5 Manufacture of sodium carbonate by ammonia soda or solvay process

Flow sheet diagram of manufacture of sodium carbonate by ammonia soda or solvay process.

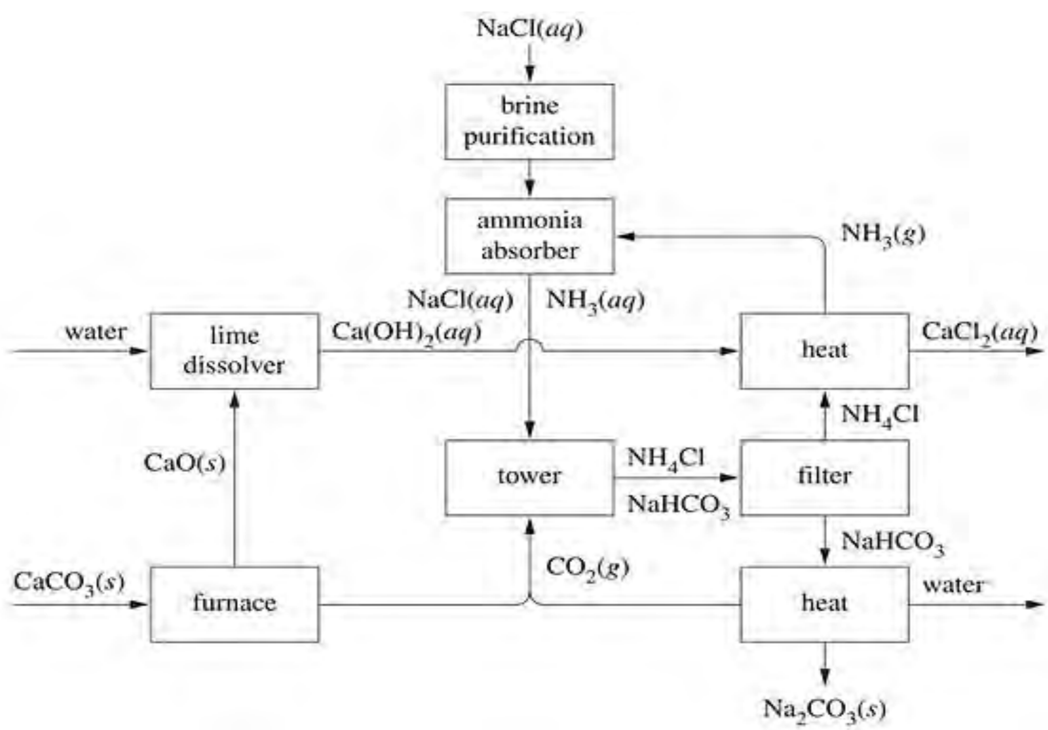


Figure: 17.8 Diagrammatic sketch of Sodium carbonate by Soda-Solvay process

Solvay Process

Principle

When carbon dioxide is passed into a concentrated solution of brine saturated with ammonia, ammonium bicarbonate is produced.

$NH_3 + H_2O \rightarrow NH_4OH$

$CO_2 + H_2O \rightarrow H_2CO_3$

$NH_4OH + H_2CO_3 \rightarrow NH_4HCO_3 + H_2O$

Again, ammonium bicarbonate is treated with brine solution which forms sodium bicarbonate. Due to the partial soluble nature of sodium bicarbonate in water it dissolves to form sodium carbonate. Finally, sodium carbonate is crystallized out from its solution.

$NH_4HCO_3 + NaCl \rightarrow NaHCO_3 + NH_4Cl$

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

$Na_2CO_3 + H_2O \rightarrow Na_2CO_3.10H_2O$

Process

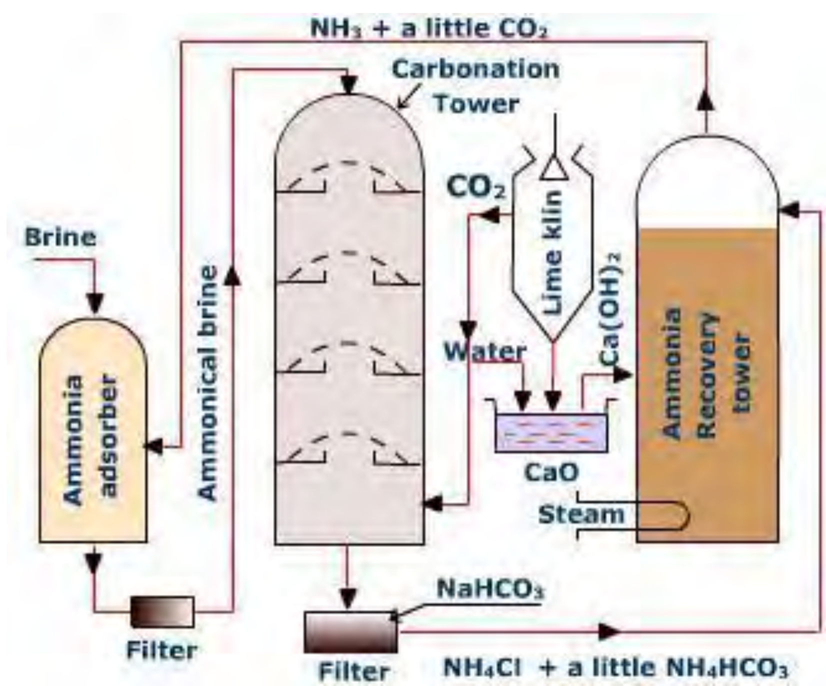


Figure 17.9 Process of manufacture of Sodium carbonate by Soda-Solvay process

The entire process consists of five steps.

1. In the first step, an aqueous solution of NaCl or brine solution is prepared.
2. In the second step, the solution is saturated by passing NH_3 gas into the saturated brine solution.
3. In the third step, CO_2 gas is passed through the ammonia saturated brine solution. As a result, NH_4HCO_3 is obtained. The reaction takes place under 30-35°C temperature.
   The producing ammonium bicarbonate reacts with NaCl resulting in the formation of $NaHCO_3$ precipitate and NH_4Cl. Both are reversible reactions.
   $NH_3 + CO_2 + H_2O \longrightarrow NH_4HCO_3$
   $NH_4HCO_3 + NaCl \longrightarrow NaHCO_3 + NH_4Cl$
   The $NaHCO_3$ precipitate thus obtained is filtered and washed well with water. It is then dried thoroughly.
4. In the fourth step, the dry $NaHCO_3$ precipitate is heated to 180°C to dissociate to produce $\mathbf{Na_2CO_3}$**.**
   $2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 \uparrow + H_2O$
5. In the last step, NH_3 is produced from NH4Cl present in the filtrate solution. In this case, CaO, a stronger alkali than NH_3, is used. Again, the by-product CaO is used to produce CO_2 from limestone.

   $2NH_4Cl + CaO = 2NH_3 + CaCl_2 + H_2O$

   $CaCO_3 \rightarrow CaO + CO_2 \uparrow$

Ammonia produced in the **Solvay process** is recycled. This process only takes brine and limestone. Calcium chloride is its only waste product. The process is substantially more economical than the LeBlanc process.

The LeBlanc process produces two waste products, calcium sulfide and hydrogen chloride. The Solvay process dominates the production of large quantities of sodium carbonate worldwide.

17.2 Fertilizers

Study the following figures and make some your opinion on the use of these products.

Figure: 17.10 Some common chemical fertilizers

Why do we use these things in the farms? Give your opinion.

a. ………………………..
b. ……………………….
c. ………………………

Have you ever seen using of such things by the farmers? What do these substances do on the farm that farmers use? What are they called?

Figure 17.11: Using urea in plants

The substances that you see above are the fertilizers. The substances which are used to enhance the quality of soil are called fertilizers. Crop production is increased due to proper use of fertilizers consequently economic development also occurs. These chemicals are used to supply the nutrients for the plants which are not available in the soil. On the basis of presence of nutrients, fertilizers are of two types:

a. Organic fertilizer
b. Chemical fertilizer

We are discussing only chemical fertilizer in this chapter.

Chemical fertilizers are those fertilizers which are synthesized by chemical process. These substances consist of various types of nutrients that can be used by the plants. On the basis of need for the plants these nutrients are classified into following groups.

i. Primary nutrients/ macronutrients: Nitrogen, Phosphorus and Potassium
ii. Secondary nutrients: Calcium, Magnesium, Sulphur
iii. Micronutrients: Copper, Chlorine, Cobalt, Nickel, Manganese, Zinc, Boron, Molybdenum etc.

The nutrients which help in growth of the plants and are required for the plants in significant quantities are called primary nutrients. Secondary nutrients are those which have lesser role in the growth of the plants and the micronutrients are used to maintain the pH of the soil.

In the context of Nepal, due to the sloppy nature of land nutrients are being washed away. To supply adequate nutrients to the plants chemical fertilizer is necessary. Generally, three types of fertilizers are used in the farm. They are

a. Urea

b. Diammonium phosphate (DAP)

c. Potash

Nitrogenous fertilizer

a. **Nitrate fertilizer:** Sodium nitrate or chile saltpetre ($NaNO_3$), calcium nitrate(Ca(NO3)2, potassium nitrate(KNO3) etc. are used as nitrogenous fertilizer.

b. **Ammoniacal fertilizer:** ammonium sulphate$(NH_4)_2SO_4$, ammonium chloride(NH_4Cl), diammonium phosphate$[(NH_4)_2.(HPO_4)]$(DAP)
c. **Ammonium and nitrate:** ammonium nitrate (NH_4NO_3), calcium ammonium nitrate[**$5Ca(NO_3)_2 \cdot NH_4NO_3 \cdot 10H_2O$.]**
d. **Amide fertilizer:** urea [NH_2-CO-NH_2]

Question: calculate the percentage of nitrogen present in these compounds.

Functions of nitrogen in plants:

a. It helps in rapid growth of the plants.
b. Helps in formation of chlorophyll.
c. Helps in protein synthesis.
d. Entire development of the plant.
e. Mainly urea is used for the supply of nitrogen in plants.

Phosphorous fertilizer

a. Single superphosphate(SSP) $3Ca(H_2PO_4)_2$
b. Double superphosphate(DSP)$6[CaH_2(HPO_4)_2]$
c. Triple superphosphate(TSP) $10[Ca(H_2PO_4)_2$

Functions of phosphorous in plants:

a. It helps for growth of roots.
b. Helps in formation of fruits and seeds.
c. Helps in maturity of the plants.

<u>Potash(potassium)</u>

a. Potassium sulphate(K_2SO_4)
b. Muriate of potash(KCl)

Functions of potassium in plants

a. It controls the opening of stomata in leaves.
b. It helps to resist drought and disease.
c. It helps to increase stalk and increase the quality of seed and grains.
d. It helps in the production of protein and sugars.

A good fertilizer should have following characteristics:

a. Should be affordable by farmers.
b. Should be water soluble, so that plants can get the nutrients easily.
c. Should have adequate quality of nutrients.
d. Should be safe for plants.

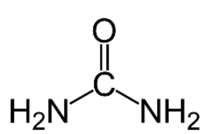
e. Should be stable; hence the plants absorb when they need.
f. Should have the quality to check the acidity of soil.
g. Should be easily absorbed by plants etc.

Note: **The fertilizer which consists of all three macronutrients is called NPK fertilizer.**

Effects on environment due to chemical fertilizers

a. **Water eutrophication:** When nitrogen containing fertilizer is reached to the ocean/ sea or any water resources, it reduces the fauna because nitrogen uses dissolved oxygen. Ultimately all the fauna will be lost but flora covers all the aquatic area. This condition is called eutrophication.
b. Change in soil acidity: Urea and nitrate fertilizers make soil acidic. Acidic soil is loose. Due to which there are high chances of landslides.
c. **Effects in health:** Nitrate containing food materials can cause cancers, mental retardations, diabetes. Nitrate reduces oxygen carrying capacity of blood and ultimately reduces hemoglobin in the blood.
d. **Effects in atmosphere:** Excessive use of ammonium fertilizer causes global climate change as it depletes the ozone layer.
e. **Biological disturbance of soil:** Due to accumulation of toxic elements like fluorides, radioactive elements like lead, arsenic, chromium, nickel etc. disrupts the quality of soil.

Production of urea with flow-sheet diagram




A nitrogenous fertilizer having molecular formula NH2-CO-NH2 with molecular mass 60.05 amu is urea. It is an amide. So, it is also called carbamide. It was discovered by ***Friedrich Wohler in 1828***. The granular urea is chemically called prilled urea. It is nitrogen rich fertilizer with 46.67% nitrogen. Specific gravity of the fertilizer is 1.33 and highly soluble in water.

Principle

a. **Formation of ammonium carbamate**: When ammonia and carbon dioxide in the ratio of 2:1 is heated to about (170- 200)^{0}C and at high pressure of 100-200 atm, ammonium carbamate is formed. This is an exothermic reaction which is a fast reaction.
$2NH_3 + CO_2 \leftrightarrow NH_2COONH_4$, $\Delta H = -37.4$ kcal
b. **Dehydration of ammonium carbamate:** On dehydration of ammonium carbamate at low pressure gives urea solution
NH_2COONH_4, $\leftrightarrow NH_2CONH_2 + H_2O$

Net reaction is:

$2NH_3 + CO_2 \leftrightarrow NH_2CONH_2 + H_2O$

For the high yield of urea:

a. Carbon dioxide should be pure.
b. Proper ratio of ammonia and carbon dioxide should be maintained.
c. Ammonia should be preheated.
d. Temperature and pressure should be maintained i.e. suitable for the reaction.

Problems during the prills formation:

Formation of dimer of urea i.e. biuret ($NH_2CONHCONH_2$).

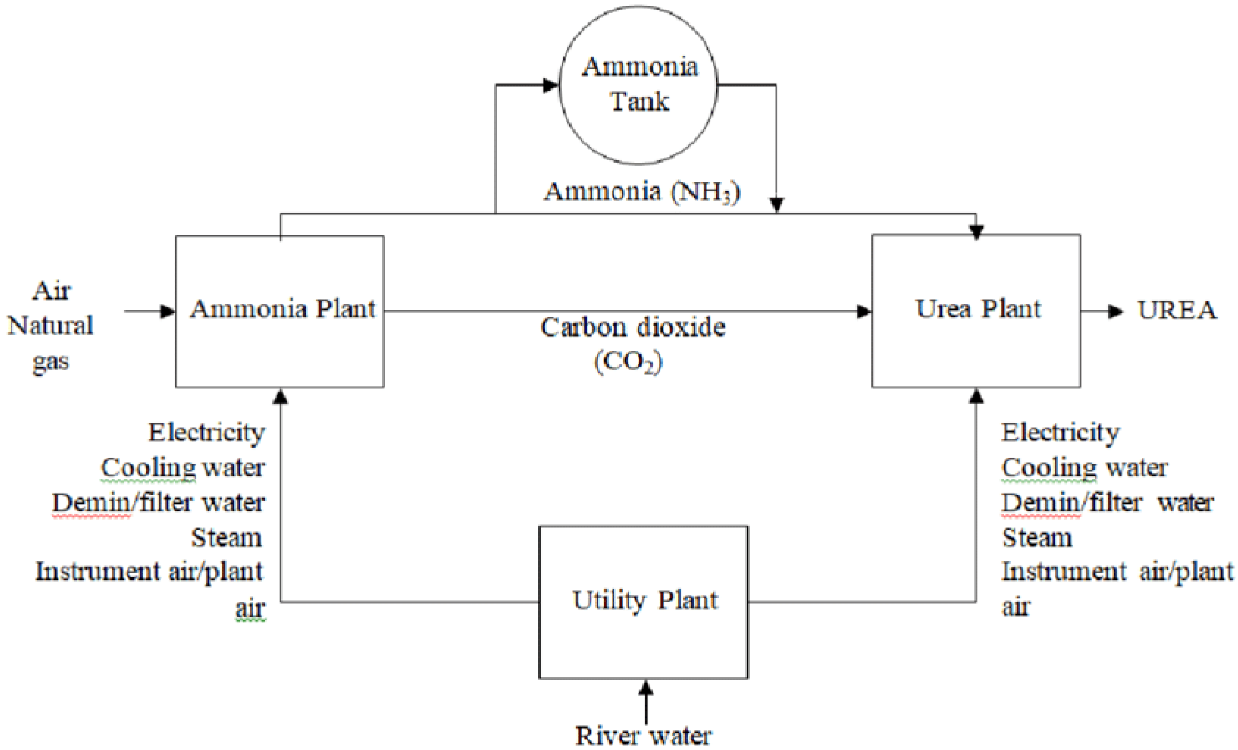


Figure 17.12: Chart showing prill formation

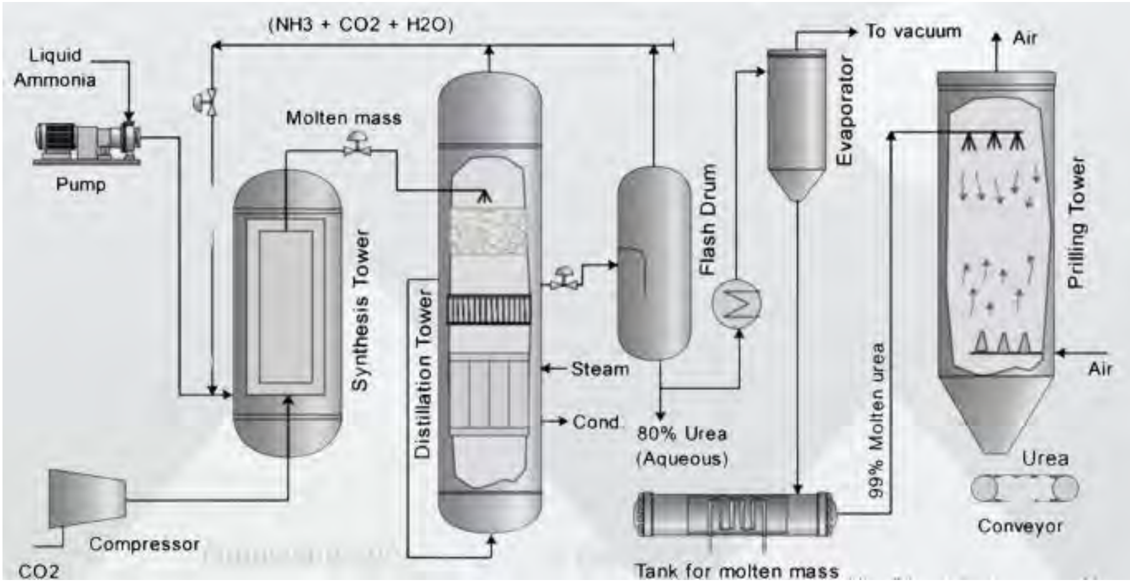


Figure 17.13: Urea manufacturing plant

Process of manufacture of urea in detail

1. **Ammonia pumping**
   Ammonia obtained from Haber's process is pumped with high pressure to the urea synthesis section called autoclave of the plant
2. **Compression of carbon dioxide**
   The carbon dioxide obtained from the decomposition of calcium carbonate is compressed under high pressure.
3. **Autoclave tower (Urea synthesis tower)**
   Inner wall of the tower is lined with refractory oxides which protects the tower from corrosion. The temperature of the tower is maintained to about 170-200^{0}C and pressure is maintained to about 180 atm at these conditions carbon dioxide and ammonia form ammonium carbamate. Thus obtained ammonium carbamate is dehydrated which forms urea. The reaction is exothermic; the evolved heat is controlled by a cooling jacket system of the tower. The running time of the autoclave is about 2 hours. The urea thus obtained is passed into the distillation tower.
4. **Distillation tower**
   When urea is passed into the distillation tower, the slurry is flashed under low pressure which contains a gas liquid separator and condenser at about 130^{0}C. In this chamber undecomposed ammonium carbamate is decomposed into ammonia, carbon dioxide and water which are again recycled to the urea synthesis tower. The liquid urea of about 50-60% concentration is passed into the flash drum for further process.
5. **Flash drum**
   The slurry of urea obtained from the distillation tower is concentrated in a flash drum and is distilled to get about 85% concentrated aqueous by removing excess ammonia, water vapor and decomposed ammonium carbamate. These components can be recycled to get more urea. This can also be used as fertilizer directly.

6. **Vacuum evaporator**
   The urea obtained from former steps is further heated and passed into a vacuum evaporator. It makes molten urea about 99% pure by evaporating water molecules and other impurities.
7. **Prilling tower**
   Actually this is a dryer. In this tower molten slurry obtained from vacuum evaporator is fed from the top of the tower into a bucket which rotates and sprinkles the slurry and air is passed from the bottom of the tower. Due to this air passed from the bottom of the tower helps to dry and changes into prills or granules of the urea. In this process biuret also formed. The urea should contain impurities less than 1%.

Advantages of urea

a. Highest nitrogen containing nitrogenous fertilizer
b. Production cost is low as compared to other fertilizers
c. Suitable to use all types of crops.
d. Easy to store and transport.
e. Environment friendly as compared to other fertilizers.
f. Can be used to manufacture nitro urea.

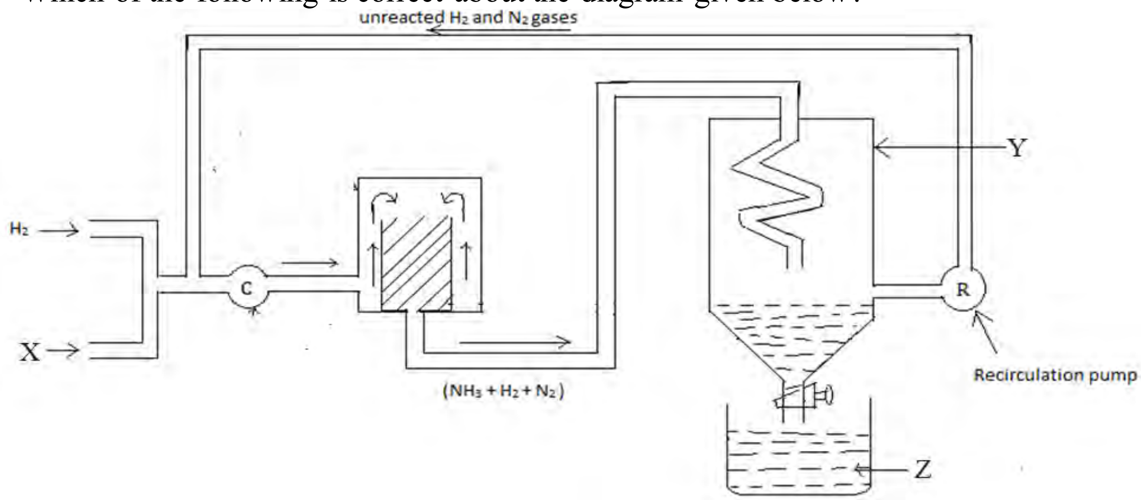
Disadvantages of urea

a. It is unstable so hard to store for long time
b. Decomposes at room temperature too, so may be high economic loss
c. Highly soluble in water so storage and transportation is challenging
d. If more impurities, it should not be used which may cause serious problems in the soil and environment.

Exercise

Multiple choice questions

1. What is the catalyst used in Haber's process?
   a. Co
   b. Fe
   c. Fe with promoter
   d. Co with promoter
2. Which of the following is correct about the diagram given below?



   a. X-NH_3, Y- catalyst chamber, Z-Catalyst chamber
   b. Y-NH_3, Z- catalyst chamber, X-Catalyst chamber
   c. Z-NH_3, Y- catalyst chamber, X-Nitrogen
   d. Z-NH_3, Y- condenser, X-nitrogen

3. By Ostwald's process nitric acid is prepared by which of the following methods?
   a. By catalytic reduction of ammonia
   b. By catalytic oxidation of ammonia
   c. By liquefaction of ammonia
   d. By condensation of ammonia
4. Which of the following is the suitable criteria to manufacture high yielding of ammonia by Haber's process?
   a. High temperature and high pressure
   b. High temperature and low pressure
   c. Low temperature and high pressure
   d. Low temperature and low pressure
5. What are the required conditions for high yielding of sulphuric acid?
   a. Purity of oleum, catalyst V_2O_5, high temperature, low pressure
   b. Purity of gases, catalyst V_2O_5, high temperature, low pressure
   c. Purity of gases, catalyst Fe, low temperature, high pressure
   d. Purity of SO_2 and O_2, catalyst V_2O_5, low temperature, high pressure
6. Which reaction occurs in the contact chamber of the contact process?
   a. $S+O_2 \xrightarrow{Heat} SO_2$

b. $SO_2+O_2\ V2O5 + Heat \rightarrow$ SO_3
c. $H_2S_2O_7 + H_2O$ $2H_2SO_4$
d. $SO_3 + H_2SO_4$ $H_2S_2O_7$

7. Which of the following is the molecular formula of oleum?
   a. H_2SO_7
   b. HS_2O_7
   c. $H_2S_2O_7$
   d. $H_2S_3O_7$
8. In which vessel does the following reaction occur in the carbonation tower?
   $NH_3+ CO_2 + H_2O$ NH_4HCO_3
   $NH_4HCO_3 + NaCl$ $NaHCO_3 + NH_4Cl$
   a. Ammonia absorber tower
   b. lime kiln
   c. ammonia recovery chamber
   d. carbonation tower

9. Which of the following by-products are reused in the Solvay ammonia process?
   a. Sodium chloride and carbon dioxide
   b. Sodium chloride and calcium chloride
   c. Sodium chloride and ammonia
   d. Calcium oxide and ammonium chloride
10. Which of the following is the method of manufacture of sulphuric acid?
    a. Haber's process
    b. Ostwald's process
    c. Nelson's process
    d. Contact process
11. What would be the product when ammonium chloride is treated with slaked lime?
    a. Calcium chloride, ammonia and water is obtained
    b. Potassium chloride, ammonia and water is obtained
    c. Calcium carbonate, ammonia and water is obtained
    d. Potassium carbonate, ammonia and water is obtained
12. Plants are being wilted and leaves are turning yellow due to which farmers are sad. What could be your best suggestion to minimize this issue of the farmer?
    a. To use fertilizers in the farm
    b. To use nitrogenous fertilizer in the farm
    c. To use phosphatic fertilizer in the farm
    d. To use green fertilizer in the farm
13. For what purpose ammonium carbamate is used?
    a. For manufacture of sodium carbonate
    b. For the manufacture of nitric acid
    c. For the manufacture of urea
    d. For the manufacture of sulphuric acid

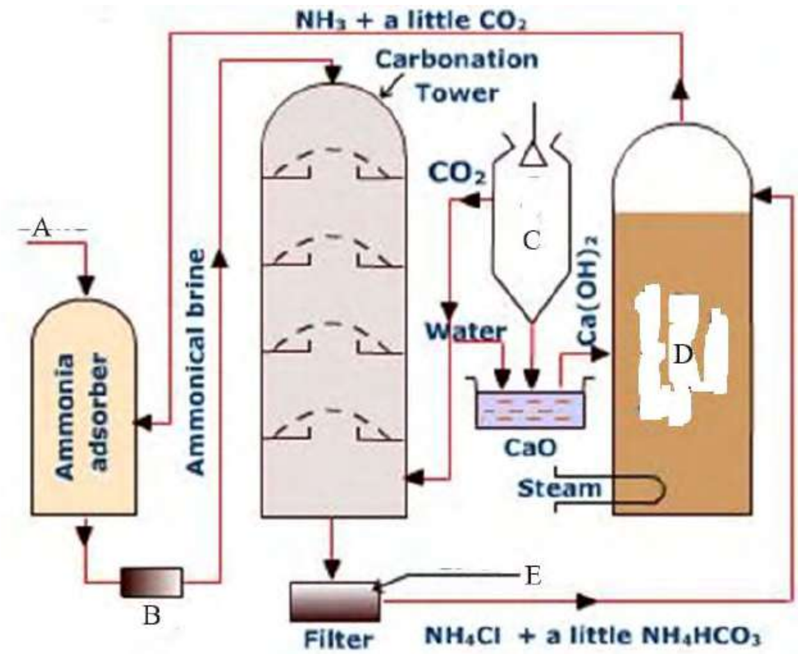
14. A urea manufacturing company has the following labelling in the sack. What is the meaning of it?
    a. It consists 46.67% hydrogen
    b. It consists 46.67% carbon
    c. It consists 46.67% oxygen
    d. It consists 46.67% nitrogen

Short answer questions

1. What are the conditions for high yielding of ammonia? Explain all the conditions with suitable examples.
2. A chemist wants to produce a gas in high scale that is basic in nature and obtains from hydrogen and nitrogen. What process should he/she follow? Explain the process with a suitable flow sheet diagram.
3. Draw a well labelled diagram of manufacture of nitric acid by Ostwald's dilution process.
4. Draw a well labelled diagram of contact process for the manufacture of sulphuric acid.
5. How do you manufacture a basic hydroxide of sodium? Explain.
6. Explain the process of manufacture of sodium carbonate. (no figure is required)
7. Identify P, Q, R, S and T from the information about compounds and their reaction at the same rows as given in the following table.

Compounds	**Reaction at**
Sodium carbonate	Carbonation tower………Q
HNO_3	Oxidation chamber…………R
P	Contact chamber…………S
NaOH	At cathode……….T

8. What are fertilizers? What are any four qualities of good fertilizer?
9. What would happen when the washing tower and drying towers are not kept in contact with the process of manufacture of sulphuric acid? Explain. How are oleum and sulphuric acid formed? Explain with suitable reactions.
10. What are A, B, C, D and E in the following figure?



11. Write M, N, O, P, Q, from the following table.

Elements	Fertilizer	Functions in plants
Nitrogen	M	N
Q	P	Helps in the growth of root and fruit
Potassium	Potassium nitrate	O

Long answer questions

1. Explain the principle, and detail process of manufacture of urea by ammonium carbamate with the help of flow sheet diagram.
2. How is sulphuric acid manufactured by contact process? Explain all the procedures with the help of a diagram. What are the conditions for high yielding of sulphuric acid?
3. What is NPK fertilizer? Write any two functions of all these nutrients in the plants of each. Why do we use fertilizer in the soil? Explain.
4. Write the principles of manufacture of sulphuric acid by contact process, manufacture of nitric acid by Ostwald's dilution method. Explain about manufacture of sodium hydroxide by diaphragm cell method.
5. Show a flow sheet diagram of manufacture of sulphuric acid by contact process. Write any three advantages of diaphragm cells for the manufacture of sodium hydroxide.

Project work

1. Visit an industry of manufacturing sulphuric acid (otherwise, in virtual mode on YouTube). Make a chart of advantages and disadvantages of sulphuric acid in society. (Social, economic, health, etc.) and present in your classroom.
2. Make a chart of the manufacture of ammonia by Haber's process and sodium hydroxide by diaphragm cell method and submit it to your teacher.
3. Make a table as given below and discuss in classroom

Compounds	**Principle**	**Procedure**	**Health hazards**
Ammonia			
Nitric acid			
Sulphuric acid			
Sodium carbonate			
Sodium hydroxide			

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